
OPERATION MANUAL (DATA)

Please read this manual carefully before starting operation.

brother

Introduction

Thank you for purchasing this machine.

Always be sure to read this manual carefully first, in order to use the machine functions properly and safely.

This machine manual is divided into the following sections.

- **Operation Manual**
This manual describes the operation procedure for the machine.
- **Installation Manual**
This manual describes the machine's installation procedure and inspections.
- **Programming Manual**
This manual provides a program description.

Brother is not responsible or liable for accidents that occur during special machine use or handling that does not follow the general safety usage guidelines.

Keep this manual for future reference.

Attach this manual to the machine if it is resold.

Contact the nearest Brother sales office or Brother approved service dealer if this manual or the safety labels are damaged, lost or missing. (Charges apply)

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This manual is divided into the following sections.

(1) Overview ----- A summary of the content is provided for the corresponding section.

(2) Warning----- A warning is provided for any hazards that could potentially cause serious bodily injury, death or damage to the machine.

The hazards are described in the following order.

(2-1) Hazard level

(2-2) Type of hazard

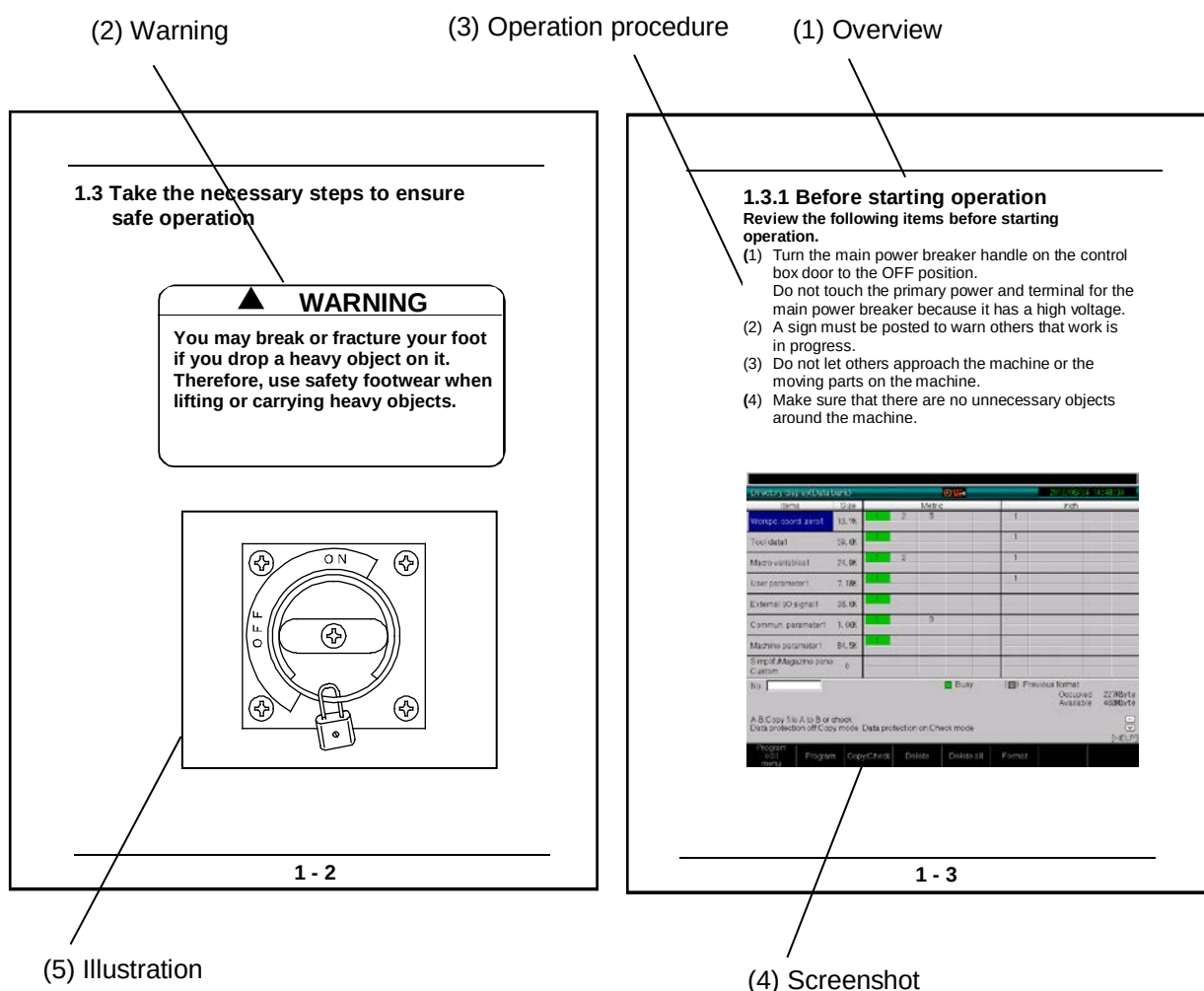
(2-3) Potential damage

(2-4) Safety directions to avoid danger

(3) Operation procedure --- The procedure describes how to operate each function.

(4) Screenshot --- A screenshot is inserted into places to highlight certain points in the operation procedure. The screenshot messages are shown at an approximate position and may differ slightly from the actual position of the line or column. The same applies to the font.

(5) Illustration ---- Illustrations, such as explanatory drawings, diagrams that show the dimensions, positioning, ranges, figures or configurations, are used in certain places where a written explanation alone may be hard to understand.



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CHAPTER 1	PROGRAM EDIT	1-1
1.1	Outline	1-2
1.2	Program edit (NC)	1-3
1.2.1	Program edit	1-3
1.2.1.1	Basic operations	1-7
1.2.1.2	Search	1-8
1.2.1.3	Replacement	1-9
1.2.1.4	Jump	1-10
1.2.1.5	Copy	1-11
1.2.1.6	Paste	1-11
1.2.1.7	Cut	1-12
1.2.1.8	Insertion of automatic sequence number	1-13
1.2.1.9	Search for block	1-13
1.2.1.10	Menu programming	1-14
1.2.1.11	Edit finish	1-14
1.2.2	Program edit in memory operation	1-15
1.2.2.1	To edit a program not in operation or a new program	1-15
1.2.2.2	To edit a Program in execution	1-15
1.2.3	Menu programming	1-16
1.3	Program edit (Conversation)	1-18
1.4	Directory display	1-18
1.4.1	Outline	1-18
1.4.2	Directory display (Program)	1-19
1.4.3	Directory display (Data bank)	1-25
1.4.4	Format	1-29
1.4.4.1	Delete all conversation files	1-31
1.5	Folder operation	1-32
1.5.1	Operation between folders	1-32
1.5.1.1	Moving a current folder	1-34
1.5.1.2	Moving a copy destination folder	1-34
1.5.1.3	Program selection screen	1-35
1.5.1.4	Copying/checking a program between folders	1-35
1.5.1.5	Program movement between folders	1-38
1.5.2	Folder operation	1-40
1.5.2.1	To create a folder	1-41
1.5.2.2	To delete a folder	1-41
1.5.2.3	To change a folder name	1-43
1.5.2.4	To copy/check a file in a folder	1-43
1.6	How to convert G codes	1-45
1.6.1	Operational procedure	1-45
1.6.1.1	Conversational parameters	1-48
1.6.2	Conversion results	1-49
1.6.2.1	Tool list conversion	1-49
1.6.2.2	Conversion of coordinate system settings	1-49
1.6.2.3	Comment output	1-49
1.6.2.4	Notes	1-49
1.6.2.5	Conversion example	1-51
CHAPTER 2	DATA BANK	2-1
2.1	Overview of data bank	2-2
2.1.1	Outline	2-2
2.1.1.1	Data bank for NC language mode	2-2
2.1.1.2	Data bank for conversational language mode	2-3
2.1.2	How to edit	2-4
2.1.2.1	To start editing	2-4
2.1.2.2	To finish editing	2-5
2.2	Workpiece coordinate zero point (NC)	2-6
2.2.1	Outline	2-6
2.2.1.1	Edit screen	2-6
2.2.2	How to edit	2-8

2.2.2.1	To move the cursor	2-8
2.2.2.2	Direct input of data	2-8
2.2.2.3	Automatic setting of data	2-8
2.2.2.4	To add data	2-13
2.2.2.5	Data set by reading a parameter	2-13
2.2.2.6	To finish editing	2-13
2.2.3	Editing during operation	2-13
2.3	Tool data (NC)	2-14
2.3.1	Outline	2-14
2.3.2	To set tool list	2-14
2.3.2.1	List screen	2-15
2.3.2.2	Extension 1 screen	2-16
2.3.2.3	Expansion 2 screen	2-18
2.3.2.4	Movement of cursor	2-19
2.3.2.5	Direct input of data	2-19
2.3.2.6	Automatic setting	2-20
2.3.2.7	To add data	2-21
2.3.2.8	To copy data	2-21
2.3.2.9	To delete data	2-21
2.3.2.10	To paste data	2-21
2.3.2.11	To finish editing	2-21
2.3.3	To set tool group	2-22
2.3.3.1	Tool group list	2-22
2.3.3.2	To edit tool group	2-22
2.3.3.3	To finish editing	2-23
2.3.4	To set tool data range	2-23
2.3.4.1	To edit tool data range	2-23
2.3.4.2	To finish editing	2-25
2.3.5	Edit during operation	2-25
2.4	Macro variables (NC)	2-25
2.4.1	Outline	2-25
2.4.1.1	Edit screen	2-25
2.4.2	How to edit	2-26
2.4.2.1	Direct input of data	2-26
2.4.2.2	To add data	2-26
2.4.2.3	To delete data	2-26
2.4.2.4	To finish editing	2-26
2.4.3	Editing during operation	2-26
2.5	User parameter	2-27
2.5.1	Outline	2-27
2.5.1.1	Edit screen	2-27
2.5.2	How to edit	2-28
2.5.2.1	Input of data	2-28
2.5.2.2	To delete data	2-28
2.5.2.3	To finish editing	2-28
2.5.3	User parameter 1 (Switch 1)	2-29
2.5.4	User parameter 2 (Switch 2)	2-51
2.5.5	User parameter 3 (Switch 3)	2-57
2.5.6	User parameter 4 (Ref. position/Conversation zero point)	2-63
2.5.7	User parameter 5 (User registered auxiliary functions)	2-66
2.5.8	User parameter 6 (Tool breakage detection)	2-66
2.5.9	User parameter 7 (Automatic workpiece measurement)	2-68
2.5.10	User parameter 8 (Automatic thermal distortion compensation)	2-69
2.5.11	User parameter 9 (Quick table)	2-70
2.5.12	User parameter 10 (High accuracy)	2-71
2.5.13	User parameter 11 (PLC)	2-74
2.5.14	User parameter 12 (PROFIBUS DP)	2-75
2.5.15	User parameter 13 (DeviceNet)	2-75
2.5.16	User parameter 14 (CC-Link Slave)	2-75
2.5.17	User Parameter 15 (Rotary Fixture Offset)	2-75
2.5.18	User Parameter 17 (Feature Coordinate Setting)	2-75
2.5.19	User Parameter 18 (CC-Link Master)	2-75

2.5.20	List of Default Values for User Parameter-----	2-76
2.5.20.1	Metric(S500X1, S700X1, S300X1, S1000X1, M140X1, R450X1)-----	2-76
2.5.20.2	Inch(S500X1, S700X1, S300X1, S1000X1, M140X1, R450X1)-----	2-99
2.5.20.3	Metric(R650X1, M140X2, S500X2, S700X2, S300X2, F600X1)-----	2-122
2.5.20.4	Inch(R650X1, M140X2, S500X2, S700X2, S300X2, F600X1)-----	2-144
2.5.20.5	Metric(R650X2_40MG, R450X2, R650X2, M200X3, M300X3)-----	2-166
2.5.20.6	Inch(R650X2_40MG, R450X2, R650X2, M200X3, M300X3)-----	2-187
2.6	External I/O Signal -----	2-208
2.6.1	Outline-----	2-208
2.6.1.1	Edit screen -----	2-209
2.6.2	How to edit-----	2-210
2.6.2.1	Locating the cursor -----	2-210
2.6.2.2	Data input -----	2-210
2.6.2.3	Data deletion-----	2-214
2.6.2.4	Overlapping input terminal check-----	2-214
2.6.2.5	Ending editing-----	2-215
2.6.3	External input signals -----	2-216
2.6.3.1	List of input signals-----	2-216
2.6.3.2	Detailed description of external input signals-----	2-227
2.6.4	External output signals-----	2-231
2.6.4.1	List of output signals -----	2-231
2.6.4.2	Detailed description of external output signals -----	2-242
2.6.4.3	Pallet indexing complete PFIN1, PFIN2-----	2-247
2.7	Communication parameter -----	2-249
2.7.1	Outline-----	2-249
2.7.1.1	Edit screen -----	2-249
2.7.2	How to edit-----	2-249
2.7.2.1	Data input/output -----	2-249
2.7.2.2	Data deletion-----	2-249
2.7.2.3	Ending editing-----	2-249
2.7.3	Common-----	2-250
2.7.4	Serial port -----	2-250
2.7.5	Ethernet/FTP-----	2-253
2.7.6	Memory card -----	2-257
2.7.7	Auto notification function -----	2-258
2.7.8	List of Communication Parameter Default Values-----	2-260
2.8	G/M Code Macro Settings -----	2-262
2.8.1	Overview -----	2-262
2.8.1.1	Edit screen -----	2-262
2.8.2	Edit method -----	2-263
2.8.2.1	Enter data directly -----	2-263
2.8.2.2	Delete data -----	2-263
2.8.2.3	Exit editing-----	2-263
2.8.3	List of G/M code macro parameters -----	2-263
2.8.3.1	Common parameter-----	2-263
2.8.3.2	G code macro-----	2-264
2.8.3.3	M code macro -----	2-264
2.9	Tool list (Conversation) -----	2-265
2.9.1	Outline-----	2-265
2.9.1.1	Edit screen -----	2-265
2.9.1.2	Setting items-----	2-266
2.9.2	How to edit-----	2-273
2.9.2.1	Locating cursor-----	2-273
2.9.2.2	Direct data input -----	2-273
2.9.2.3	Setting of tool length-----	2-273
2.9.2.4	Data Addition-----	2-274
2.9.2.5	Data copy-----	2-274
2.9.2.6	Data deletion-----	2-274
2.9.2.7	Data paste -----	2-275
2.9.2.8	Registering tools with auxiliary code -----	2-275
2.9.2.9	End of editing -----	2-275

2.9.3	Editing during operation-----	2-275
2.10	Tool pattern (Conversation)-----	2-276
2.10.1	Outline-----	2-276
2.10.1.1	Edit screen-----	2-276
2.10.2	How to edit-----	2-277
2.10.2.1	Locating cursor-----	2-277
2.10.2.2	Data change-----	2-277
2.10.2.3	Data insertion-----	2-277
2.10.2.4	Data deletion-----	2-277
2.10.2.5	Ending editing-----	2-277
2.10.3	List of Default values-----	2-278
2.11	Tapping drill diameter (Conversation)-----	2-279
2.11.1	Outline-----	2-279
2.11.1.1	Edit screen-----	2-279
2.11.2	How to edit-----	2-279
2.11.2.1	Data change-----	2-279
2.11.2.2	Ending an Edit-----	2-279
2.11.3	List of default values-----	2-280
2.11.3.1	Drill diameters for tapping primary hole-----	2-280
2.11.3.2	Drill diameters for tapping primary hole by rolled tap-----	2-284
2.12	Cutting condition (Conversation)-----	2-286
2.12.1	Outline-----	2-286
2.12.1.1	Edit screen-----	2-286
2.12.2	How to edit-----	2-287
2.12.2.1	Locating cursor-----	2-287
2.12.2.2	Data change-----	2-287
2.12.2.3	Ending an Edit-----	2-287
2.12.3	List of default values-----	2-287
2.12.3.1	Cutting conditions 1 (drill)-----	2-287
2.12.3.2	Cutting conditions 2 (milling tools)-----	2-293
2.12.3.3	Cutting conditions (other than tool conditions)-----	2-298
CHAPTER 3	GRAPHIC-----	3-1
3.1	Outline-----	3-2
3.1.1	Outline-----	3-2
3.1.2	Changing between operation drawing and tool path simulation-----	3-3
3.1.3	Changing between 2D and 3D-----	3-3
3.1.4	Drawing size and coordinate display-----	3-3
3.1.5	Display of drawing alarm-----	3-6
3.2	Operation drawing-----	3-7
3.2.1	Outline-----	3-7
3.2.2	Drawing during operation-----	3-8
3.2.3	Cycle drawing-----	3-8
3.2.4	Single drawing-----	3-9
3.2.5	Zoom-----	3-9
3.2.6	Drawing setting-----	3-11
3.2.7	To stop drawing-----	3-15
3.2.8	To cancel drawing-----	3-15
3.2.9	To delete graphic-----	3-15
3.3	Tool path simulation-----	3-16
3.3.1	Outline-----	3-16
3.3.2	Cycle drawing-----	3-17
3.3.3	Single drawing-----	3-17
3.3.4	Zoom-----	3-17
3.3.5	Drawing setting-----	3-20
3.3.6	To stop drawing-----	3-23
3.3.7	To cancel drawing-----	3-24
3.3.8	To delete graphic-----	3-24

CHAPTER 4	TAPE OPERATION MODE	4-1
4.1	How to perform tape run	4-2
4.1.1	Tape run (General COMM device)	4-2
4.1.2	Tape run (Memory card)	4-4
4.2	Configuration of tape run program	4-6
4.2.1	General communication device	4-6
4.2.2	Memory card	4-7
4.3	Notices	4-8
CHAPTER 5	COMMUNICATION	5-1
5.1	Outline	5-2
5.1.1	Special Notes Before Using Communication Function	5-2
5.1.2	Master station and slave station communication	5-3
5.1.3	File Name	5-3
5.2	Master station communication	5-4
5.2.1	Screen	5-4
5.2.1.1	External program I/O menu	5-4
5.2.1.2	Screen flow	5-4
5.2.1.3	Program (external I/O)	5-5
5.2.1.4	Data bank (external program I/O)	5-8
5.2.2	Change connection device	5-10
5.2.2.1	Connection device changing procedure	5-10
5.2.2.2	Memory card	5-11
5.2.3	Folder operation	5-12
5.2.3.1	Folder operation screen	5-12
5.2.3.2	Moving the folder	5-13
5.2.3.3	Creating a folder	5-13
5.2.3.4	Deleting a folder	5-14
5.2.3.5	Changing the folder name	5-14
5.2.3.6	Sort	5-15
5.2.3.7	Latest directory	5-15
5.2.4	Parameters	5-16
5.2.5	Selecting programs to communicate	5-16
5.2.5.1	Program selection screen	5-16
5.2.5.2	Selecting programs to be input	5-17
5.2.5.3	Selecting programs to output	5-18
5.2.5.4	Selecting programs to delete	5-18
5.2.6	Input/output of programs	5-19
5.2.6.1	Program input method (from connected device to machine)	5-19
5.2.6.2	Program output method (from machine to connecting directory)	5-22
5.2.6.3	Program deletion method (from connected device)	5-24
5.2.7	Data bank I/O	5-26
5.2.7.1	Data bank input method (from connected device to machine)	5-26
5.2.7.2	Data bank output method (from machine to connected device)	5-29
5.2.7.3	Data bank deletion method (deleting data bank from the connected device)	5-30
5.2.7.4	I/O in the unit of record (User parameter, machine parameter, and ladder program)	5-31
5.3	Slave station communication	5-33
5.3.1	Outline	5-33
5.3.1.1	Slave station communication from "NC COMM software"	5-33
5.3.1.2	Slave station communication from FTP client	5-33
5.3.1.3	Slave station communication from browser	5-34
5.3.2	Restrictions	5-34
5.4	Maintenance communication	5-36
5.4.1	What is maintenance communication?	5-36
5.4.1.1	Maintenance menu	5-36
5.4.1.2	Screen flow	5-37
5.4.2	Output all data	5-38
5.4.3	Input all data	5-39
5.4.4	Output all data banks	5-42
5.4.5	Output survey data	5-42

5.4.6	Delete all data	5-42
5.4.7	Disk format	5-42
5.4.8	External input signal key	5-44
5.4.8.1	External input signal key screen	5-44
5.4.8.2	Creation of external input signal key	5-45
5.4.8.3	Adding external input signals	5-45
5.4.8.4	Deleting external input signals	5-45
5.4.8.5	Cancellation of external input signal key	5-46
5.4.9	Request code output	5-46
5.5	Communication protocol	5-47
5.5.1	Character code table (NC language mode)	5-47
5.5.2	Character code table (Conversation mode)	5-51
5.5.3	Internal data code list	5-51
5.5.4	Ethernet Communication	5-52
5.5.4.1	Slave station communication from "NC communication software"	5-52
5.5.4.2	FTP communication	5-52
5.5.4.3	Slave station communication from browser	5-54
5.5.5	General communication device protocol	5-54
5.5.5.1	Tape format (NC language mode)	5-54
5.5.5.2	Tape format (conversation mode)	5-54
5.5.6	Communication methods of general communication device	5-55
5.5.7	NC communication device protocol type 1 (NC language mode)	5-57
5.5.7.1	Data output request	5-57
5.5.7.2	Data input request	5-57
5.5.7.3	Data delete request	5-57
5.5.7.4	Request for inputting information on slave station directory	5-58
5.5.8	NC communication device protocol type 1 (conversation mode)	5-59
5.5.8.1	Data output request	5-59
5.5.8.2	Data input request	5-59
5.5.8.3	Data delete request	5-59
5.5.8.4	Request for slave station directory information	5-60
5.5.9	NC communication device protocol type 2	5-61
5.5.9.1	Control format	5-61
5.5.9.2	Completion code list	5-62
5.5.9.3	Details of commands	5-64
5.5.10	Communication methods of NC communication device	5-88
5.6	Details of communication data	5-91
5.6.1	Data name list	5-91
5.6.2	Operation range	5-93
5.6.2.1	Operation from external devices to NC (NC: slave)	5-93
5.6.2.2	Operation from NC to external devices (NC: Master) (NC language mode)	5-96
5.6.2.3	Operation from NC to external devices (NC: master) (Conversation mode)	5-99
5.6.3	Data compatibility with other models	5-102
5.6.3.1	Compatibility of NC language data	5-102
5.6.3.2	Compatibility of conversational language data	5-103
5.6.4	File format	5-106
5.6.4.1	Workpiece counter	5-106
5.6.4.2	Production data 2 (Time display)	5-106
5.6.4.3	Measurement results	5-108
5.6.4.4	Automatic thermal distortion compensation	5-109
5.6.4.5	Maintenance notice	5-111
5.6.4.6	Production data 3 (Status history)	5-112
5.6.4.7	Position data	5-112
5.6.4.8	Memory operation	5-115
5.6.4.9	ATC tool	5-115
5.6.4.10	Operation panel data	5-115
5.6.4.11	Version data	5-116
5.6.4.12	Current alarm	5-116
5.6.4.13	Alarm history	5-117
5.6.4.14	Operation history	5-119
5.6.4.15	Data for Machine Monitor (Production Data 4)	5-120
5.6.4.16	Request code	5-120

5.6.4.17 Workpiece coordinate zero (Type 1) -----	5-120
5.6.4.18 Workpiece coordinate zero (Type 2) -----	5-121
5.6.4.19 Tool data (Type 1) -----	5-121
5.6.4.20 Tool data (Type 2) -----	5-123
5.6.4.21 1Macro variables (Type 1) -----	5-123
5.6.4.22 Macro variables (Type 2) -----	5-123
5.6.4.23 Machining load monitor -----	5-123
CHAPTER 6 DATA PROTECTION -----	6-1
6.1 Outline -----	6-2
6.2 Set operation level -----	6-3
6.2.1 Setting by password -----	6-3
6.2.2 Setting by external input signals -----	6-4
6.2.3 Precautions -----	6-4
6.3 Change password -----	6-5
6.3.1 Changing password -----	6-5
6.3.2 Changing password to no-set -----	6-6
6.3.3 Reset password to default value -----	6-6
6.4 Edit detailed settings -----	6-7
6.4.1 Changing settings for detailed settings -----	6-7
6.4.2 Item List -----	6-7
CHAPTER 7 SUPPLEMENT -----	7-1
7.1 Help -----	7-2
7.1.1 Help screen -----	7-2
7.1.1.1 Outline -----	7-2
7.2 Display in different languages -----	7-3
7.2.1 Changing display language -----	7-3
7.2.2 Selecting temporary display language -----	7-4
7.3 HTTP (Hyper Text Transfer Protocol) -----	7-5
7.3.1 Outline -----	7-5
7.3.2 Monitor display -----	7-5
7.3.2.1 Workpiece counter -----	7-6
7.3.2.2 Time display -----	7-6
7.3.2.3 Measurement results -----	7-7
7.3.2.4 Status log -----	7-7
7.3.2.5 Maintenance notice -----	7-8
7.3.2.6 Alarm -----	7-8
7.3.2.7 ATC tool -----	7-10
7.3.3 Data I/O -----	7-11
7.3.4 3D smoothing viewer -----	7-11

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CHAPTER 1	PROGRAM EDIT	1
CHAPTER 2	DATA BANK	2
CHAPTER 3	GRAPHIC	3
CHAPTER 4	TAPE OPERATION MODE	4
CHAPTER 5	COMMUNICATION	5
CHAPTER 6	DATA PROTECTION	6
CHAPTER 7	SUPPLEMENT	7

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CHAPTER 1

PROGRAM EDIT

- 1.1** **Outline**
- 1.2** **Program Edit (NC)**
- 1.3** **Program Edit (Conversation)**
- 1.4** **Directory Display**
- 1.5** **Folder Operation**
- 1.6** **How to Convert G Codes**

1.1 Outline

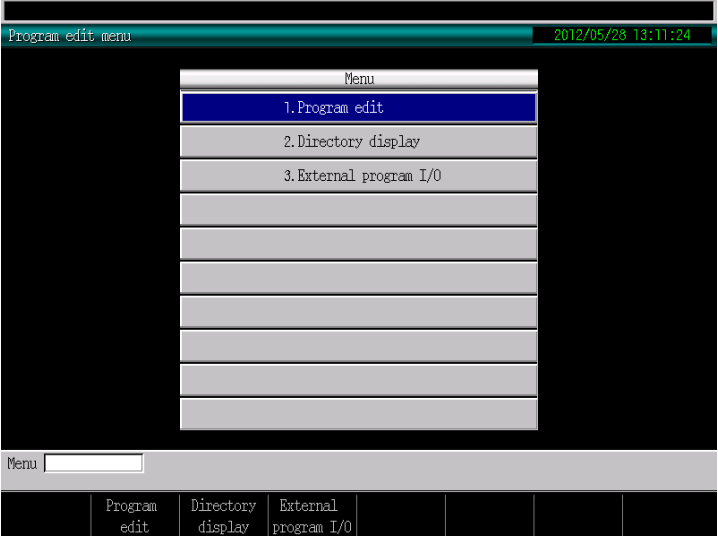
The following four functions are available in program edit mode.

1. Creation and modification of NC type program
2. Creation and modification of conversation type programs
3. Display of registered program No., deletion, copy and check of the programs.
4. I/O of programs and data to/from an external unit
5. Preprocessing for NC program

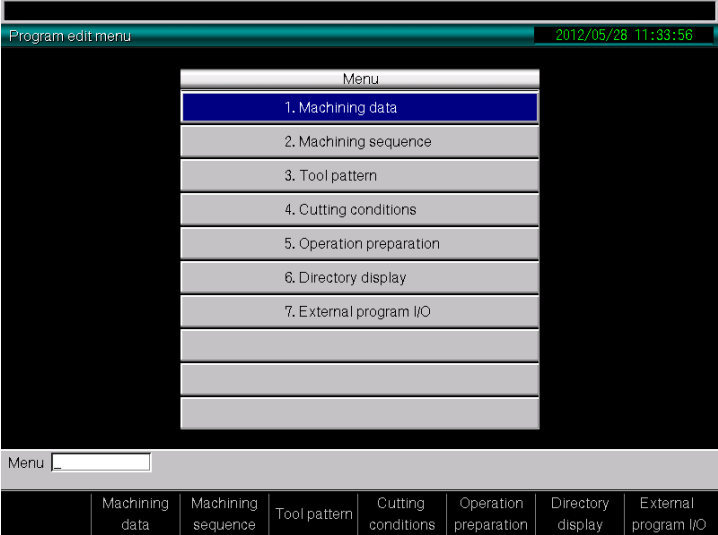
Operation method is as follows.

- (1) Press the **[EDIT]** key (mode selection key).
- (2) Press the menu No. and the **[ENT]** key, or move the cursor to the menu No. and press the **[ENT]** key or press a function key. Each operation screen is displayed.

<Program edit menu> screen (NC language)



<Program edit menu>screen (Conversational language)



When the **[EDIT]** key is pressed while in memory operation, the operation temporarily stops or pauses (block stop). To edit or reference a program without stopping operation, press the **[EDIT]** key while pressing the **[MEM]** key.

1.2 Program Edit (NC)

1.2.1 Program Edit

This function is available only for the NC language mode.

Creation and modification of NC program are available.

In addition, you can move a program to a folder or create a folder.

- * For creation, correction, or storage of a program and creation of a folder, there are following capacity limits:
 - Total capacity of usable memory is about 128M bytes.
 - The total number of usable files and folders including NC programs, conversational programs, databanks, and folders is 4096 files.

Press **[1]** and then the **[ENT]** key on the <Program edit menu> screen, or point the cursor at the menu No. 1 and press the **[ENT]** key to display the <Program list> screen. When the **[F1]** (list/detail) key is pressed, the list display and detail display are switched. When the **[F2]** (Sort) key is pressed, the order of the programs are sorted.

- | | |
|---|--|
| [F0] (Date(ascending)): | Displays programs in the order from the oldest to the newest. |
| [F1] (Date(descending)): | Displays programs in the order from the newest to the oldest. |
| [F2] (Program No. (ascending)): | Displays in the order from the smallest to the biggest.
Displays the folders in the alphabetical order of the folder names. |
| [F3] (Program No. (descending)): | Displays in the order from the biggest to the smallest.
Displays the folders in the reverse alphabetical order of the folder names. |
| [F4] (Comment(ascending)): | Displays in the alphabetical order of the character strings of the comment.
Displays the folders in the alphabetical order of the folder names. |
| [F5] (Comment(descending)): | Displays in the reverse alphabetical order of the character strings of the comment.
Displays the folders in the reverse alphabetical order of the folder names. |

Press the **[F6]** key (Create folder) to display the screen for creating folders. The key cannot be pressed when the screen is already inside the folder.

Example of the <Detailed program list> screen

Detailed program list			2012/05/28 11:40:59	
Program(0001)				
Current folder/				
No.	Size	Comment	Date	
0001	1.27K	(*****)	12/05/24	14:05:10
0002	144	(PRG2)	11/11/22	13:20:28
0004	74	(PRG3)	11/11/22	13:20:28
0005	4	M99	11/11/22	13:20:29
0010	64	M00	11/11/22	13:20:29
0011	24	M450	11/11/22	13:20:30
0012	14	T901	11/11/22	13:20:30
0013	9	M400	11/11/22	13:20:31
0014	14	M450	11/11/22	13:20:39
0016	21	M08	11/11/22	13:20:46
0017	8	M11	11/11/22	13:20:47
0018	8	M12	11/11/22	13:20:47
0019	8	M13	11/11/22	13:20:48
0020	8	M14	11/11/22	13:20:49
0021	8	M18	11/11/22	13:20:49
0022	8	M21	11/11/22	13:20:50
0023	8	M41	11/11/22	13:20:50
Program No.			Available 430Byte	
No. 0001 1.27K Byte 12/05/24 14:05:10 (*****)				
☒ Running ☐ Protected			☒ ☐	
Program edit menu	List	Sort	Create folder	

The details screen shows the program size, a description of block 1 in the program, the date and time the program was last updated, and the information is displayed according to the selected order.

Seventeen programs can be displayed on one screen. Other programs can be displayed by pressing the **[PAGE DOWN]** or **[PAGE UP]** key.

In addition, if there is a folder, a folder name is added with “/” after the name and displayed in the program number area and the program size area.

Example of the <Program list> screen

Program list

2012/05/28 11:43:07

Program(0001)

Current folder:/

No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
0001	1.27K	12/05/24	0024	8	11/11/22	0041	16	11/11/22	0058	9	11/11/22
0002	144	11/11/22	0025	8	11/11/22	0042	28	11/11/22	0059	42	11/11/22
0004	74	11/11/22	0026	9	11/11/22	0043	16	11/11/22	0060	43	11/11/22
0005	4	11/11/22	0027	12	11/11/22	0044	12	11/11/22	0061	67	11/11/22
0010	64	11/11/22	0028	8	11/11/22	0045	21	12/04/24	0068	58	11/11/22
0011	24	11/11/22	0029	14	11/11/22	0046	1.90M	11/11/22	0069	4	11/11/22
0012	14	11/11/22	0030	14	11/11/22	0047	14	11/11/22	0070	246	11/11/22
0013	9	11/11/22	0031	14	11/11/22	0048	14	11/11/22	0071	41	11/11/22
0014	14	11/11/22	0032	14	11/11/22	0049	12	11/11/22	0072	14	11/11/22
0016	21	11/11/22	0033	44	11/11/22	0050	12	11/11/22	0073	14	11/11/22
0017	8	11/11/22	0034	24	11/11/22	0051	159	11/11/22	0074	14	11/11/22
0018	8	11/11/22	0035	68	11/11/22	0052	41	11/11/22	0075	14	11/11/22
0019	8	11/11/22	0036	9	11/11/22	0053	9	11/11/22	0076	14	11/11/22
0020	8	11/11/22	0037	9	11/11/22	0054	155	11/11/22	0077	25	11/11/22
0021	8	11/11/22	0038	9	11/11/22	0055	2.15M	11/11/22	0078	37	11/11/22
0022	8	11/11/22	0039	9	11/11/22	0056	19	11/11/22	0079	50	11/11/22
0023	8	11/11/22	0040	25	11/11/22	0057	57	11/11/22	0080	57	11/11/22

Program No.

Available 430Byte

No. 0001

1.27K Byte 12/05/24 14:05:10 (***** Running Protected)

Running

Protected

☒
☒

Program edit menu	Details	Sort			Create folder	
-------------------	---------	------	--	--	---------------	--

On the <Program list> screen, program number, program size, and up-date dates are displayed in the selected sorted order.

Sixty eight programs can be displayed on one screen. Other programs can be displayed by pressing the **[PAGE DOWN]** or **[PAGE UP]** key.

In addition, if there is a folder, a folder name is added with “/” after the name and displayed in the program number area and the program size area.

Program number at the center below the title is the program edited last time. On the line below that, a folder name (current folder) displayed in the present list is displayed.

Enter the program No., or point the cursor at the program No. to be edited, and press the **[ENT]** key.

If a program number which is not in the NC is input and the **[ENT]** key is pressed, it is an edit of a new program.

After moving the cursor to the folder and pressing the **[ENT]** key, the screen changes to the selected folder and shows a list of items in that folder.

(Note) Even if folder name is entered and the [ENT] key is pressed, the screen cannot change to the folder screen.

During memory operation, the program that is running is highlighted in green.

If a program is protected and cannot be edited, that program is highlighted in grey.

* Refer to the “1.4 Directory display” for further details on currently running programs and protected programs that cannot be edited.

The edit program number and the memory operation program number are independent numbers. But if the <Set same number to edit/memory operation programs> of the user parameter is set <1: Yes>, they can be accorded including the folder.

For instance, if the memory operation mode is set when the edit program number is 1000, the memory operation program number becomes 1000. In addition, if the edit mode is set when the memory program number is 2000, the edit program number becomes 2000.

- (Note 1) The program numbers 9000 - 9999 cannot be used.
- (Note 2) A description of block 1 in the program is displayed on the <Detailed program list> screen. The user can insert comments using the control-out and control-in codes ("(", ")"). This makes the overview helpful because the user does not have to look inside the program to check the description. Note, there is a 60 character limit for the description that is displayed in the <Detailed program list> screen.

1

Folder creation screen (Example of a list screen)

Program list folder creation 2012/05/28 11:51:46

Program(0001)

Current folder:

No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
0001	1.27K	12/05/24	0024	8	11/11/22	0041	16	11/11/22	0058	9	11/11/22
0002	144	11/11/22	0025	8	11/11/22	0042	28	11/11/22	0059	42	11/11/22
0004	74	11/11/22	0026	9	11/11/22	0043	16	11/11/22	0060	43	11/11/22
0005	4	11/11/22	0027	12	11/11/22	0044	12	11/11/22	0061	67	11/11/22
0010	64	11/11/22	0028	8	11/11/22	0045	21	12/04/24	0068	58	11/11/22
0011	24	11/11/22	0029	14	11/11/22	0046	1.90M	11/11/22	0069	4	11/11/22
0012	14	11/11/22	0030	14	11/11/22	0047	14	11/11/22	0070	246	11/11/22
0013	9	11/11/22	0031	14	11/11/22	0048	14	11/11/22	0071	41	11/11/22
0014	14	11/11/22	0032	14	11/11/22	0049	12	11/11/22	0072	14	11/11/22
0016	21	11/11/22	0033	44	11/11/22	0050	12	11/11/22	0073	14	11/11/22
0017	8	11/11/22	0034	24	11/11/22	0051	159	11/11/22	0074	14	11/11/22
0018	8	11/11/22	0035	68	11/11/22	0052	41	11/11/22	0075	14	11/11/22
0019	8	11/11/22	0036	9	11/11/22	0053	9	11/11/22	0076	14	11/11/22
0020	8	11/11/22	0037	9	11/11/22	0054	155	11/11/22	0077	25	11/11/22
0021	8	11/11/22	0038	9	11/11/22	0055	2.15M	11/11/22	0078	37	11/11/22
0022	8	11/11/22	0039	9	11/11/22	0056	19	11/11/22	0079	50	11/11/22
0023	8	11/11/22	0040	25	11/11/22	0057	57	11/11/22	0080	57	11/11/22

New folder name: Available 430Byte

No. 0001 1.27K Byte 12/05/24 14:05:10 (*****)

Back Create

If the <Program list> screen or the <Detailed program list> screen displays a root folder, press the [F6] (Create folder) key to display the Folder creation screen.

If a folder name to create is input and the [F1] (Create) key is pressed, a new folder is created and returns to the previous screen.

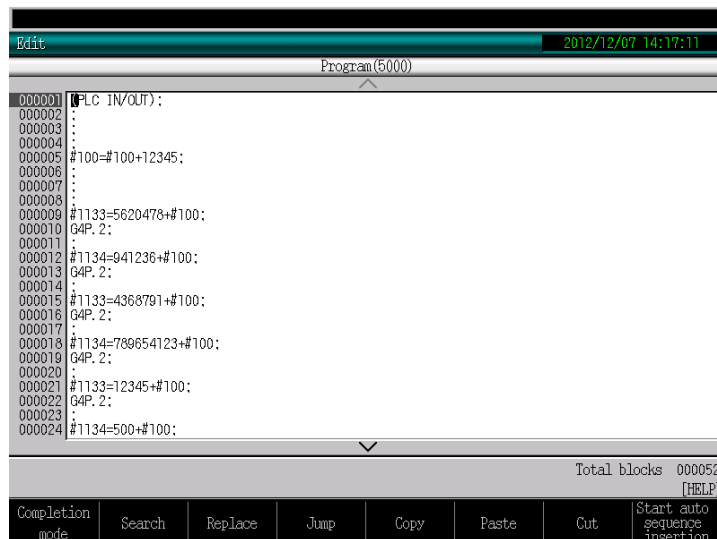
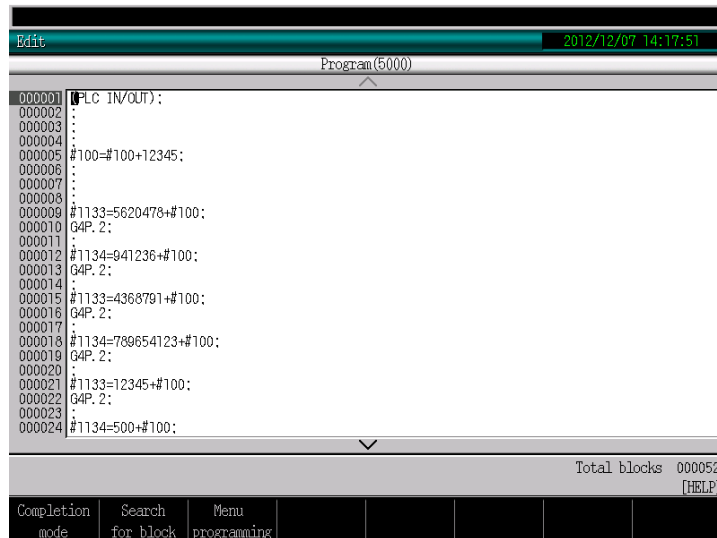
Press the [F0] (Back) key to return to the previous screen without creating a folder.

A folder name can have up to 8 alphanumerical characters and a period.

In addition, the folders that can be created at the machine can extend to 2 levels including the root directory. (You cannot create another folder in a folder which has already been created.)

You cannot create a folder name like an already existing folder, names (O0000-O9999) in an NC program, or a name in a conversational program. If it is tried to create, the operator message <<This is an unusable folder name>> appears.

<Edit> screen



- Edit can be performed in the screen mode (mode which displays the editing programs to modify directly).
- The end of block code (EOB) is indicated as “;”.
- The cursor indicates the position to start editing.
- One screen can display max. 24 lines.
- The block number in a block where the cursor exists is highlighted in a gray background.

The maximum size of an editable program is 20M bytes.

- 163840 blocks if a full characters are input in a block (127 characters + EOB).
- 20971518 blocks if all the blocks are filled with only EOB.

If a program over 20M bytes is tried to be edited, the operator message <<Memory overflow>> appears to disable editing.

In addition, if it exceeds 20M bytes during editing, the operator message <<Editing area exceeded>> appears to disable addition of a character. Divide the program.

When the program size exceeds the program load size, or when the program size including all sub programs being called exceeds the program load size, the program runs in extended memory operation mode during operation.

When the program load size is the size that is selected in the user parameter (switch 1) <Program load size>.

1.2.1.1 Basic Operations

1. Deletion of character

Press the **[DEL]** key.

The character pointed by the cursor is deleted by one character. The remaining of the block after the cursor is moved to the left by one character.

If the cursor is positioned on the end of block (EOB) code “;”, then that block and the next one merge and become one block. The entire next block shifts to the cursor position.

However, if the number of characters of the connected block exceeds 127 characters, the operator message <<Too many characters>> appears to disable deletion.

Push the **[BACKSPACE]** key.

One character right before the cursor position is deleted. All content for that block including the cursor shifts 1 character space to the left.

If the cursor is at the head of a block, the block and the next block are connected into one block and the entire next block moves to the end of the before block.

However, if the number of the characters in a block after connection exceeds 127 characters, the operator message <<Too many characters>> appears to disable deletion.

2. Insertion of character

Insertion is available as long as the **[INS]** LED is lit on.

When any of the alphabetical keys, numerical keys or symbol keys is pressed at this time, the character specified by the keys is inserted before the character pointed by the cursor. The cursor and the rest of the characters are moved to the right by one character.

* The insertion input mode is the prompt when a program edit starts.

3. Overwriting of character

Press the **[INS]** key and turn off the LED. The overwriting mode is valid as long as the LED is lit off.

When any of the alphabetical keys, numerical keys or symbol keys is pressed, the character pointed by the cursor is replaced by a new character.

When the cursor is at “;” the characters specified by the pressed key are inserted before this symbol as in the insertion mode.

After overwriting is finished, press the **[INS]** key again. The LED is lit on and the insertion mode becomes valid.

4. Cursor shift

The cursor is shifted by pressing the **[↑]**, **[↓]**, **[←]** and **[→]** keys.

The cursor cannot move where there is no character. When the **[←]** key is pressed at the beginning of a block, the cursor moves over the “;” of the precedent block.

When the cursor is at the bottom line of the screen, pressing the **[↓]** key scrolls the screen and moves the cursor to the next line or the next block.

If the **[↑]** key pressed at the top of a line, the screen scrolls down.

If the **[←]** key is pressed while pressing the **[SHIFT]** key, the cursor moves to the head character of the line.

If the **[→]** key is pressed while pressing the **[SHIFT]** key, the cursor moves to the last character of the line (or “;” of the line).

5. Page change

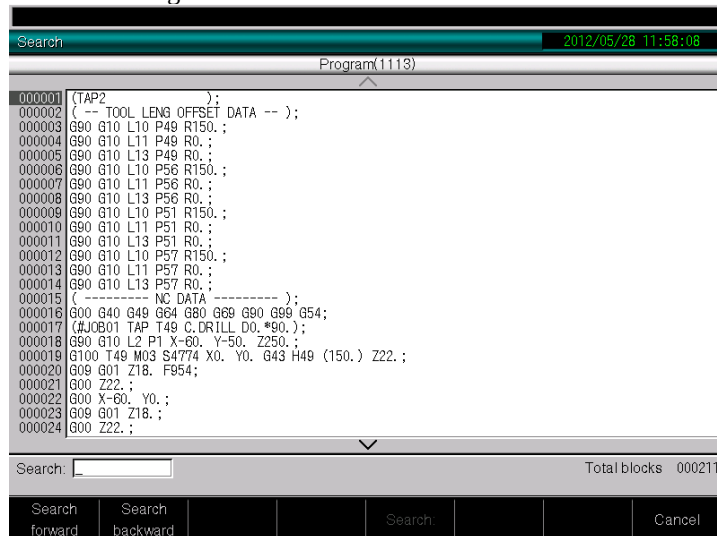
Screen changes by one page by pressing the **[PAGE DOWN]** or **[PAGE UP]** key.

After changing the pages, the cursor moves to the first character of the newly displayed page.

The last line of the precedent page is displayed at the top of the next page before changing the pages. The head line of the precedent page is displayed at the bottom of the precedent page before changing the pages.

1.2.1.2 Search

Character string of maximum 12 characters can be searched from the cursor position.



[F0].....Searches forward.

[F1].....Searches backward.

[F4].....Enter character string and search

[F7].....Exit search mode

Input a search character into the search character string input area. The character string cannot include “;” (EOB).

Search by pressing the **[F0]** (Search forward) key or **[F1]** (Search backward) key.

The cursor moves to the head of the searched character string.

If no designated character string is found, the operator message <<Data not found>> appears but the cursor does not move.

Under this condition, you can move the cursor and edit the program.

After searching a character string, the specified character string is saved until turning off the machine.

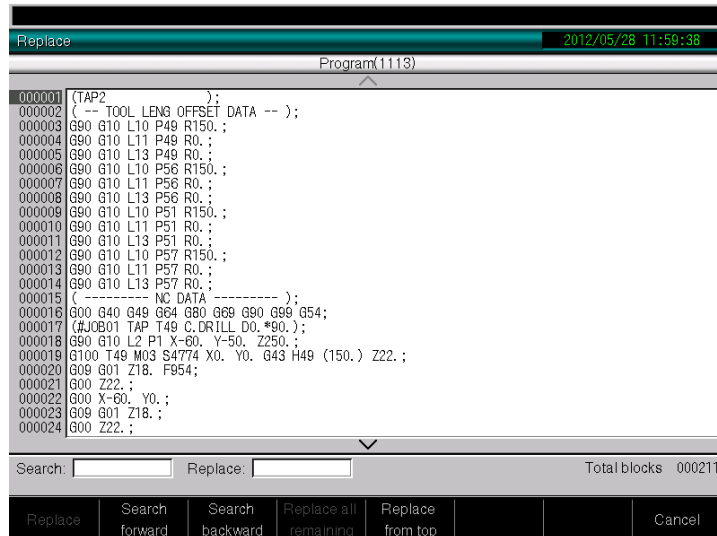
When you want to search the same character string continuously, press **[F0]** (Search forward) or **[F1]** (Search backward).

When you want to change character strings to be searched, press **[F4]** (Search character string) and you can input a desired character string.

Press the **[F7]** (Cancel) key to finish the search mode and return to the <Edit> screen.

1.2.1.3 Replacement

Replace a character string that is 12 characters or less with another character string that is 12 characters or less.



[F0].....Replace

[F1].....To search a downward candidate

[F2].....To search an upward candidate

[F3].....Enable or disable the replacement operation in a specific direction.

[F4].....Replace all from the beginning of the program

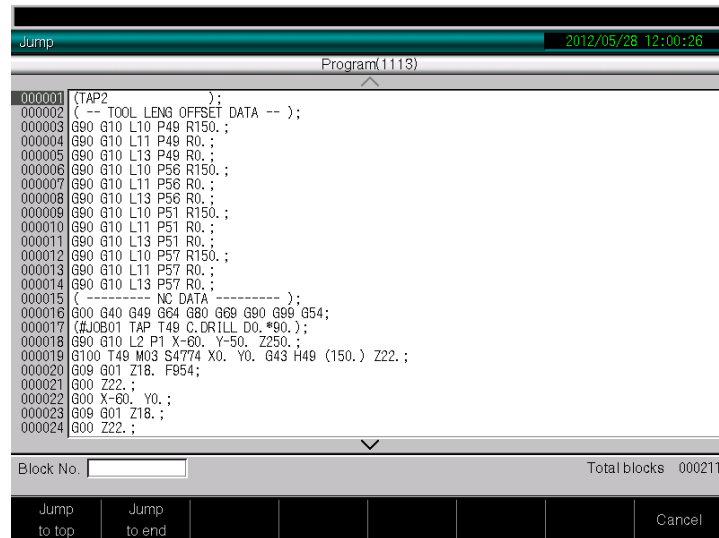
1. Input a character string to be replaced and a new string character.
When entering a character string for a <Search>, after pressing the **[ENT]** key, the cursor moves to the data input field to enter data to <Replace>.
After entering the character string to <Replace> and pressing the **[ENT]** key, the cursor moves to the data input field for a <Search>.
2. When pressing **[F1]** (Search forward) or **[F2]** (Search backward), searching for the character string to be replaced begins from the cursor position forward or backward. When pressing the **[F4]** (Replace from top) key, searching downward from the head of the program is performed. The cursor moves to the beginning of the searched character string. If there is no character string to be replaced, the operator message <<Data not found>> appears to return to the <Edit> screen.
3. When pressing **[F0]** (Replace) key, the character string is replaced with the new one.
When pressing **[F1]** (Search forward) key, replacement is not performed and the cursor moves to the beginning of the next candidate character string. If there is no character string to be replaced, the operator message <<Data not found>> appears to return to the <Edit> screen.
When pressing the **[F2]** (Search backward) key, replacement is not performed and the cursor moves to the beginning of the next candidate character string. If there is no character string to be replaced, the operator message <<Data not found>> appears to return to the <Edit> screen.
If the **[F3]** (Replace all remaining) key is pressed and a downward candidate is pressed, replacement is carried out downward from the cursor position. If an upward candidate is pressed, replacement is carried out upward in one go without checking.
However, if replacement frequency exceeds 1000 times, the operator message <<Replacement performed too many times>> appears and the replacement finishes to return to the <Edit> screen.

The current character string (character string to be replaced) and the new character string are saved until these are changed or turning off the power.

If the size exceeds 20 M bytes due to replacements or if the number of characters in a block exceeds 127 characters, the operator message <<Editing area exceeded>> or <<Too many characters>> appears and the replacement process is held to return to the <Edit> screen.

1.2.1.4 Jump

When the **[F0]** (Jump to top) key is pressed, the cursor jumps to the head of a program. If the **[F1]** (Jump to end) key is pressed, the the cursor jumps to the end of the program. When entering a block number and pressing the **[ENT]** key, the cursor jumps to the first character of the specified block.



[F0] Jumps to the head of a program

[F1] Jumps to the end of a program

[F7] Returns to the <Edit> screen without jumping anywhere

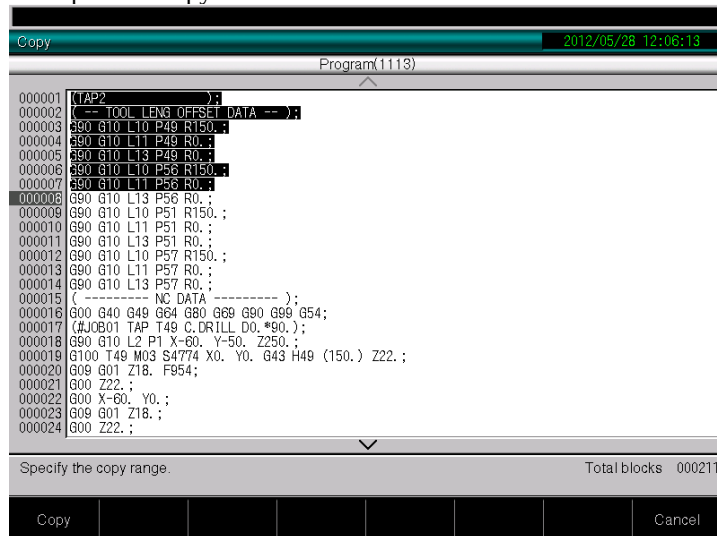
If a block number larger than the total number of the blocks is input, the cursor jumps to the end of a program.

1.2.1.5 Copy

This is a function to copy a part of program to a different part.

1. Point the cursor at the first character you want to copy.
2. Press the **[F4]** (Copy) key. The title displays <Copy>.
3. Move the cursor to the character that comes after the last character to be copied. All characters to be copied will be displayed in reverse.
4. Press the **[F0]** (Copy) key. The part that is displayed in reverse is saved. The title returns to <Edit>.
5. Point the cursor at the part you want to copy.
6. Press the **[F5]** (Paste) key. The taken characters are inserted before the character specified by the cursor.

Example of <Copy> screen



The maximum size of memory for copying or cutting-off is 20 M bytes.

1.2.1.6 Paste

The group of characters that are selected for copy or cut operations is saved. That content is saved until a new copy or cut operation is performed or the power to the machine is turned OFF.

The content can be called for and inserted in the character before the cursor position for any number of times by pressing the **[F5]** (Paste) key.

This function is used in the following cases:

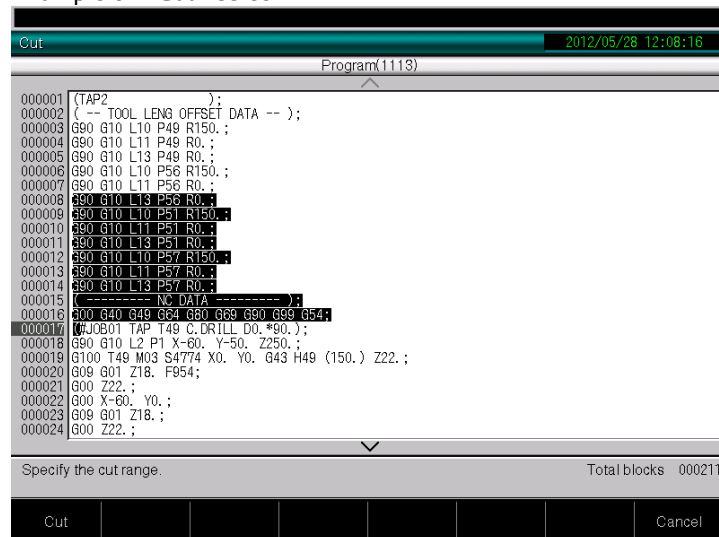
- Restoration after cutting
- Moving a specified character group to another place
- Copying a specified character group to several places
- Copying or moving a specified part of a program to another program

1.2.1.7 Cut

This is a function to cut a part of a program per character. The cut characters can be saved and pasted to another place.

1. Point the cursor at the first character to be cut.
2. Press the **[F6]** (Cut) key. The screen displays <Cut>.
3. Point the cursor at the precedent character next to the last character to be cut. All the characters to be cut are highlighted.
4. Press the **[F0]** (Cut) key. The part that is displayed in reverse is cut and saved. The title returns to <Edit>.
5. When pasting the cut character to another place, move the cursor to the place to be pasted.
6. Press the **[F5]** (Paste) key. The cut character is inserted to the character before the cursor position.

Example of <Cut> screen



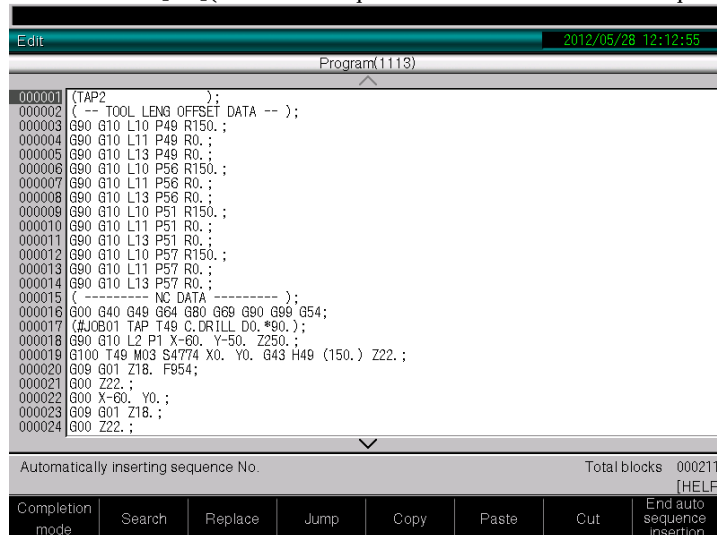
The maximum size of memory for copying or cutting-off is 20 M bytes.

1.2.1.8 Insertion of Automatic Sequence Number

If EOB is input, the sequence number is automatically inserted.

The value of the sequence number is set with the sequence number of the block, where EOB is input, added with the user parameter <Automatic sequence insertion increment>. When the block where EOB is input does not have a sequence number, the value of the sequence number is set with the user parameter <Automatic sequence insertion increment>.

To switch the [F7](Start auto sequence insertion / End auto sequence insertion) key

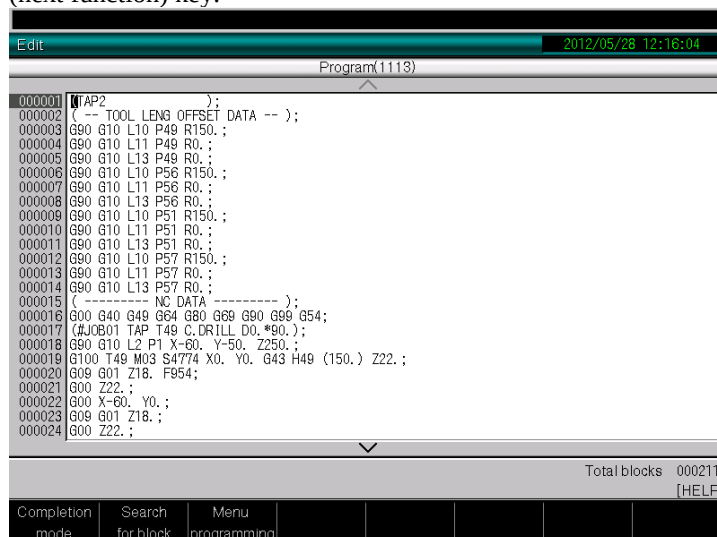


1.2.1.9 Search for Block

The cursor jumps to the currently operating block by pressing the [F1] (Search for block) key when the editing program is the currently operating program on the <Edit> screen of the operating program.

If the program in editing differs from the program in operation or is not in operation, the operator message <<Data not found>> appears and the cursor does not move.

If there is no “Search for block” in the function display, change the functions by pressing [>>] (next function) key.



The block number pointed by the operation pointer is indicated in a black background.

1.2.1.10 Menu Programming

Menu programming is the function enable to make NC program by inputting numerical values according to the menu. You don't need to remember G/M codes.

For details, refer to "1.2.3 Menu programming".

1.2.1.11 Edit Finish

When the **[F0]** (Completion mode) key is pressed on the program edit screen, the edit finish check mode is called and the screen to select a condition for finishing is called.

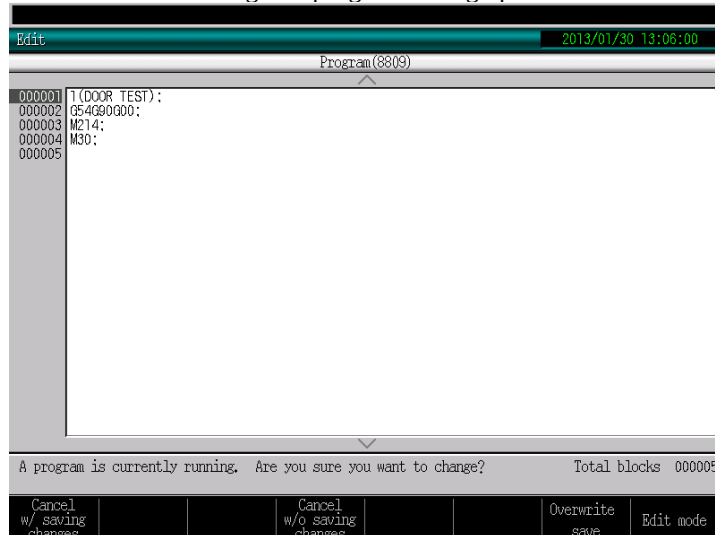
You can save a program for finishing or continue editing.

If an edited program is in operation, the message <A program is currently running. Are you sure you want to change?> appears.

(Note) If the edited program exceeds 2 M bytes and is in extension memory operation, the message <End editing after operation is completed. > appears. Consequently, neither storing nor overwriting becomes available.

In addition, if the edited program is in drawing and the program in drawing exceeds 2 M bytes, the message <Quit editing after drawing is completed.> appears. Consequently, neither storing nor overwriting becomes available.

Screen to check editing of a program during operation is finished



1. **[F0]** (Cancel w/ saving changes) key
Editing finishes when the edited content is saved.
The screen returns to the <Program list> screen.
However, if a program in extension memory operation is edited or a program larger than 2 M bytes in drawing is edited, you cannot push the **[F0]** (Cancel w/ saving changes) key.
2. **[F3]** (Cancel w/o saving changes) key
Edited contents are invalidated.
The screen returns to the <Program list> screen.
3. **[F6]** (Overwrite save) key
The contents ever edited are overwritten in a file and the screen returns to the edit status.
However, if a program in extension memory operation is edited or a program larger than 2 M bytes in drawing is edited, you cannot push the **[F6]** (Overwrite save) key.
4. **[F7]** (Edit mode) key
Returns to the editing status.

1.2.2 Program Edit in Memory Operation (Edit During Operation)

Programs are editable during a memory operation without stopping the operation.

1.2.2.1 To Edit a Program not in Operation or a New Program

1. Press the **[EDIT]** key while pressing the **[MEM]** key.
The LEDs of the **[MEM]** and **[EDIT]** keys go lit.
2. Set a program number other than the one currently executed to create a new program or reedit a current program.

(Note) Unless the **[MEM]** key is pressed in 1 above, the mode becomes the program edit mode and the memory operation stops.

1.2.2.2 To Edit a Program in Execution

1. Press the **[EDIT]** key while pressing the **[MEM]** key. The LEDs of the **[MEM]** and **[EDIT]** keys go lit.
2. Set a program number in operation to enable editing. Here, the message <<Running program>> appears.

If the user parameter <Program edit start position during operation > is set as <1: Pointer>, the cursor moves to the operation pointer position when the editing started.
The block number of the operation pointer is displayed in a black background.

If the user parameter <Save when mode of current program is changed> is set <1: Overwrite save>, the message <Perform overwrite save by mode switching.> appears in the teaching area.

3. If the **[F0]** (Completion mode) key is pressed after editing a program, the message <A program is currently running. Are you sure you want to change?> appears for confirmation.
4. Press the **[F0]** (Cancel w/ saving changes) key to save the edit content.

(Note) Unless the **[MEM]** key is pressed in 1 above, the mode becomes the program edit mode and the memory operation stops.

The saved program in operation is reflected in memory operation after the operation finishes.

If any of the **[MANU]**, **[MDI]**, **[MEM]**, and **[EDIT]** keys is pressed while a program in operation is being edited and the modes are changed, the user parameter <Save when mode of current program is changed> is set <1: Overwrite save>, the edited content is overwritten and saved in a file.

1.2.3 Menu Programming

Menu programming is the function enable to make NC program by inputting numerical values according to the menu. You don't need to remember G/M codes.

If there is no menu programming in the function display, press the next menu key [>>](next function) to switch the function.

1. Press **[F2]** (Menu programming) key to call the Menu programming screen.
2. Press **[F1]** (G code) key or **[F2]** (M code) key.
3. Move the cursor to the G/M code to be made and press **[ENT]** key.
4. Set values on the selected G/M code setting screen. While deleting setting values, press the **[DEL]** key.
5. When returning to the <Edit> screen with registering setting contents, press the **[F0]** (End registration) key. When returning to the Menu programming input menu screen with registering setting contents, press the **[F2]** (Continue registration) key. When returning to the <Edit> screen with invalidating setting contents, press the **[F1]** (Cancel) key.

(Note) The G code or M code that is registered as a <G/M code macro> in the data bank carries out the corresponding G code or M code macro call operation.

Example of Menu programming input menu (G/M code) screen

Menu programming input menu(G code)		2012/05/28 13:16:14	
Program(1113)			
G code	Function	G code	Function
G00	Positioning	G13	Circular cutting CCW
G01	Linear interpolation	G17	XY plane selection
G02	Circular/helical interpolation CW	G18	ZX plane selection
G02	Spiral/conical interpolation CW	G19	YZ plane selection
G03	Circular/helical interpolation CCW	G22	Programmable stroke limit valid
G03	Spiral/conical interpolation CCW	G23	Programmable stroke limit invalid
G04	Dwell	G28	Reference position return
G09	Exact stop check	G29	Return from reference position
G10	Workpiece coord. zero setting	G30	2nd to 6th reference position return
G10	Workpiece coord. zero setting(measurement)	G31	Skip function
G10	Extended workpiece coord. zero setting	G31	Continuous skip function
G10	Extended workpc. coord. zero setting(meas.)	G36	Coordinate calculation function(bolt hole)
G10	Tool length offset setting	G37	Coordinate calculation function(line(angle))
G10	Cutter compensation setting	G38	Coordinate calculation function(line(X, Y))
G10	Tool length wear offset setting	G39	Coordinate calculation function(line(grid))
G10	Cutter wear offset setting	G40	Cutter compensation cancel
G10		G41	Cutter compensation left
G12	Circular cutting CW	G42	Cutter compensation right

G code Total blocks 000211

Back G code M code

Menu programming input menu(M code)		2012/05/28 13:17:13	
Program(1113)			
M code	Function	M code	Function
M00	Program stop	M200	Tool breakage detection
M01	Optional stop	M201	Tool breakage detection
M02	End of program	M203	Tool breakage detection program call
M03	Spindle normal rotation	M211	Workpiece counter 1 on
M04	Spindle reverse rotation	M212	Workpiece counter 2 on
M05	Spindle stop	M213	Workpiece counter 3 on
M06	Tool change	M214	Workpiece counter 4 on
M08	Coolant on	M221	Workpiece counter 1 off
M09	Coolant off	M222	Workpiece counter 2 off
M19	Spindle orientation(0°)	M223	Workpiece counter 3 off
M30	End of program	M224	Workpiece counter 4 off
M96	Interrupt type macro	M230	Tool life on
M97	Interrupt type macro cancel	M231	Tool life off
M98	Sub program call	M238	Waveform display Measurement start
M99	Return from sub program	M239	Waveform display Measurement end
M111	Spindle orientation(180°)	M260	High accuracy mode A on(level 1)
M120	Tool breakage error check(on)	M261	High accuracy mode A on(level 2)
M121	Tool breakage error check(off)	M262	High accuracy mode A on(level 3)

M code Total blocks 000211

Back G code M code

Example of Menu programming input menu (G/M code) screen

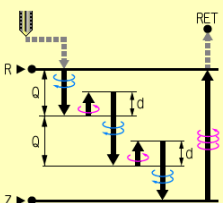
Menu programming input menu(G code)

2012/05/28 13:21:39

Program(1113)

Synchronized tapping cycle(G77)

Sequence No. (N)	
Comment	
Drilling position X (X)	
Drilling position Y (Y)	
Bottom of hole (Z)	
R-point(R)	
Pitch (I)	
Threads (J)	
Infeed amount(Q)	
1st infeed amount(W)	
Min. infeed amount(V)	
Spindle speed(S)	
SP return speed (L)	
No. of repeats (K)	



RET

Rapid feed

Cutting feed

Normal rotation

Reverse rotation

Set the pitch (I) or the threads (J).
Set the relief amount (d) for the user parameter.
When specifying fixed step, set the infeed amount (Q).
When specifying variable step, set the 1st infeed amount (W) and the minimum infeed amount (V).

Sequence No.

Total blocks 000211

End registration

Cancel

Continue registration

Menu programming input menu(M code)

2012/05/28 13:23:43

Program(1113)

Tool change(M06)

Sequence No. (N)	
Comment	
Tool No. (T)	
X-axis (X)	
Y-axis (Y)	
Z-axis (Z)	
R-point(R)	
A-axis (A)	
B-axis (B)	
C-axis (C)	
Next tool No. (L)	
T-len. offset No. (G43 H)	

Set 1** (** indicates magazine or pot No.) when specifying the magazine or pot number or g** (** indicates group No.) when specifying the group number for tool No. (T) or next tool No. (L).
Set the tool No. or group No. for tool length offset No. (G43 H).

Sequence No.

Total blocks 000211

End registration

Cancel

Continue registration

1.3 Program Edit (Conversation)

For creation of a conversational program, refer to the “Programming Manual; Conversational.”

1.4 Directory Display

1.4.1 Outline

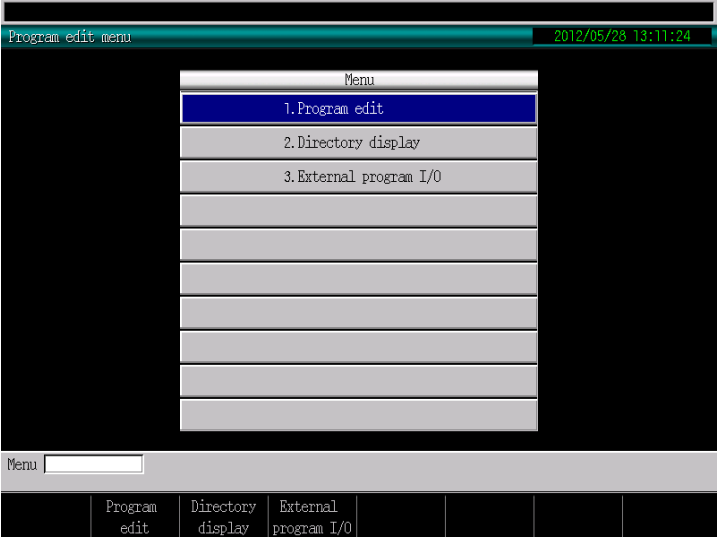
The directory display is to display the amount of memory used by a program or data bank.

Usable amount amount is as follows:

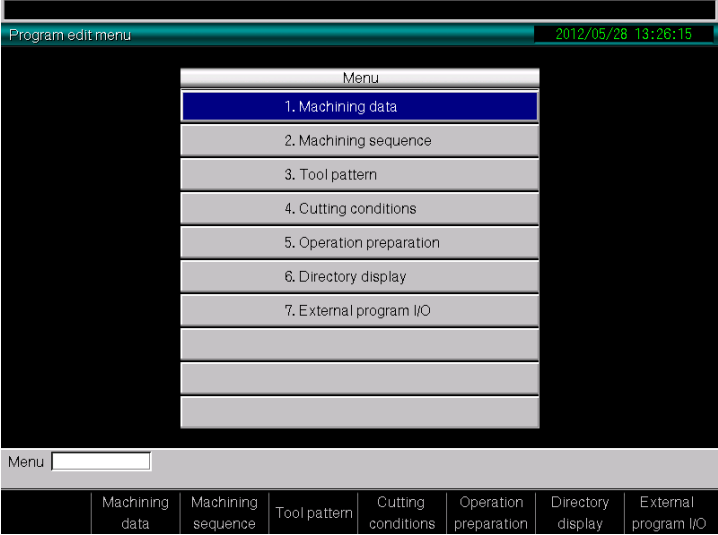
- Total amount of usable memory is about 128 M bytes.
- The total number of usable files and folders including NC programs, conversational programs, databanks, and folders is 4096 files.

1. Set the [DATA PROTECTION] switch to the [OFF] position.
2. Push the [EDIT] key.

<Program edit menu> screen (NC language)



<Program edit menu> screen (Conversational language)



3. NC language mode

Push **[2]** and push the **[ENT]** key.

You can also choose the menu as described below.

Use the cursor to select <2. Directory display> and push the **[ENT]** key or the **[F2]** (Directory display) key.

Conversational language mode

Press **[6]** and then the **[ENT]** key. You can also choose the menu as described below.

Use the cursor to select <6. Directory display> and press the **[ENT]** key or **[F6]** (Directory display) key.

4. Set the **[DATA PROTECTION]** switch to the **[ON]** position.

Every time when the **[F1]** (Program/Data bank) key is pressed on the <Directory display> screen, the <Directory display (Program)> screen and the <Directory display (Data bank)> screen are switched.

1.4.2 Directory Display (Program)

1. Displaying directory of program

Select the directory display from the <Program edit menu> screen.

The <Directory display (Program)> screen or the <Directory display (Data bank)> screen appears.

When the <Directory display (Data bank)> screen appears, press the **[F1]** (Program) key to switch to the <Directory display (Program)> screen.

Example of <Directory display (Program)> screen (list)

Directory display(Program)										2012/05/28 13:33:05	
Program(1113)											
Current folder./											
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
AAA/		12/05/11	SSSPP/		12/05/14	0019	8	11/11/22	0033	44	11/11/22
BBB/		12/04/26	WWW/		12/04/26	0020	8	11/11/22	0034	24	11/11/22
CCC/		12/04/26	0001	1.27K	12/05/24	0021	8	11/11/22	0035	68	11/11/22
DDD/		12/04/26	0002	144	11/11/22	0022	8	11/11/22	0036	9	11/11/22
EEE/		12/04/26	0004	74	11/11/22	0023	8	11/11/22	0037	9	11/11/22
FFF/		12/04/26	0005	4	11/11/22	0024	8	11/11/22	0038	9	11/11/22
GGG/		12/04/26	0010	64	11/11/22	0025	8	11/11/22	0039	9	11/11/22
HHH/		12/05/23	0011	24	11/11/22	0026	9	11/11/22	0040	25	11/11/22
III/		12/04/26	0012	14	11/11/22	0027	12	11/11/22	0041	16	11/11/22
JJJ/		12/04/26	0013	9	11/11/22	0028	8	11/11/22	0042	28	11/11/22
KKK/		12/04/26	0014	14	11/11/22	0029	14	11/11/22	0043	16	11/11/22
LLL/		12/04/26	0016	21	11/11/22	0030	14	11/11/22	0044	12	11/11/22
OOO/		12/05/25	0017	8	11/11/22	0031	14	11/11/22	0045	21	12/04/24
SSS/		12/04/26	0018	8	11/11/22	0032	14	11/11/22	0046	1.90M	11/11/22
Program No.								Occupied 17.1MByte		Available 430MByte	
AAA/ 12/05/11 12:16:17											
A-B: Copy file A to B or check.											
A/B: Delete files A to B.											
Data protection off: Copy mode Data protection on: Check mode											
☒ Running ☐ Protected											
Program edit menu	Data bank	Details	Sort	Copy/Check	Delete	Protect ON	Format				

Example of <Directory display (Program details)> screen

Directory display(Program details)				2012/05/28 13:34:35
Program(1113)				
Current folder:/				
No.	Size	Comment	Date	
KKK/			12/04/26 18:53:48	
LLL/			12/04/26 18:53:48	
OOO/			12/05/25 20:39:02	
SSS/			12/04/26 18:53:48	
SSPPP/			12/05/14 21:58:44	
VVV/			12/04/26 18:53:48	
0001	1.27K	(*****)	12/05/24 14:05:10	
0002	144	(PRG2)	11/11/22 13:20:28	
0004	74	(PRG3)	11/11/22 13:20:28	
0005	4	M99	11/11/22 13:20:29	
0010	64	M00	11/11/22 13:20:29	
0011	24	M450	11/11/22 13:20:30	
0012	14	T901	11/11/22 13:20:30	
0013	9	M400	11/11/22 13:20:31	
Program No.			Occupied	17.1MByte
			Available	430MByte
No. 0001 1.27K Byte 12/05/24 14:05:10 (*****)				
A-B:Copy file A to B or check.				
A/B:Delete files A to B.				
Data protection off:Copy mode Data protection on:Check mode				
☒ Running ☐ Protected				
Program edit menu	Data bank	List	Sort	Copy/Check
			Delete	Protect ON
				Format

- <Current folder> : Displays the name of a folder currently being displayed.
 <No.> : Displays a program number and the name of a folder.
 <Size> : Displays the memory size of a program used for the program.
 <Comment> : Displays the first block of a program for detailed display.
 <Date> : Displays the date and time when a program is finally saved in NC.

In the teaching area, the program number, size, updated day and time, and comment information of program selected by the cursor are displayed.

If a program is in memory operation (in operation / in block stop / in temporary stop), it is displayed in a green background.

NC language

- Main program
- Sub program called from a main program
- Sub program called from a sub program

Conversational language

- Main program
- Schedule program
- Program registered in a schedule program

Uneditable programs are displayed in the gray background.

With respect to the program number in the conversational language:

- The program created in the unit system (metric system or inch system) opposite to the one displayed on the screen is displayed with "M" or "I" in front of the program number.

If the [ENT] is pressed while the cursor is selecting a folder, the cursor moves into the selected folder to display the directory again. The current folder is the selected folder.

(Note) The character input in an input column is ignored even if the [ENT] key is pressed.

- [F2] : To switch displays
 <Details> : To display a comment or update time
 <List> : To list programs on a screen
- [F3] (Sort) : To sort the programs per update time / program number (folder in the alphabetical order) / comment (folder in the alphabetical order) in the ascending or descending order

2. Copying/Checking of programs

A designated program can be copied to another program.

The copy destination and copy source must be in the current folder. Carry out the “Operation between folders” for copying to a different folder.

Copying a program:

Before operation, invalidate the **[DATA PROTECTION]** switch.

(1) Designate the program for an input area.

Method to input to an input area (program number)

“Program number in copy source”, [—], “Program number in copy destination”

Example) When the 100th program number is copied to the 99th
100—99

(2) Press the **[F4]** (Copy/Check) key.

(3) If there is a new program yet, the Overwrite check screen appears.

Press the **[F0]** (Yes) key for overwriting.

Press the **[F1]** (No) key for not overwriting.

The following cases show the following operator messages and disable copying:

- If there is no program designated by the “program number in copy source”: <<No copy source data>>
- If an uneditable program is designated to the “program number in copy destination” (However, checking is possible): <<Program is write-protected>>
- If there is a program designated by the “Program number in copy destination” and the program is in extension memory operation: <<Extended memory operation mode>>
- If a program is designated by the “Program number in copy source” or “Program number in copy destination” and is being input/output by communication: <<Communicating>>

Checking a program

If the **[DATA PROTECTION]** switch is validated and the program is operated, the program is checked.

If the program has the same content, the characters <Check OK> appears in the teaching area.

The following case will display an operator message but will not check the program.

- If there are both programs to be designated: <<Required data not found>>

Copying/checking is unavailable between a schedule program and a program in the conversational language.

3. Deletion of program

A designated program can be deleted.

Carry out the “Operation between folders” for deleting the whole folder.

To delete a program

Before operation, invalidate the **[DATA PROTECTION]** switch.

(1) Designate a desired program in the input area.

How to input a program number into an input area

“Program number”

- * If this is omitted, the program at the cursor position becomes the subject for deletion.
(However, if the cursor position is at a folder, the **[DEL]** key after this is ignored.)

(2) Press the **[F5]** (Delete) key.

Directory display(Program)						2012/05/28 13:49:31					
Program(1113)											
Current folder:/											
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
0001	1.27K	12/05/24	0021	8	11/11/22	0035	68	11/11/22	0049	12	11/11/22
0002	144	11/11/22	0022	8	11/11/22	0036	9	11/11/22	0050	12	11/11/22
0004	74	11/11/22	0023	8	11/11/22	0037	9	11/11/22	0051	159	11/11/22
0005	4	11/11/22	0024	8	11/11/22	0038	9	11/11/22	0052	41	11/11/22
0010	64	11/11/22	0025	8	11/11/22	0039	9	11/11/22	0053	9	11/11/22
0011	24	11/11/22	0026	9	11/11/22	0040	25	11/11/22	0054	155	11/11/22
0012	14	11/11/22	0027	12	11/11/22	0041	16	11/11/22	0055	2.15M	11/11/22
0013	9	11/11/22	0028	8	11/11/22	0042	28	11/11/22	0056	19	11/11/22
0014	14	11/11/22	0029	14	11/11/22	0043	16	11/11/22	0057	57	11/11/22
0016	21	11/11/22	0030	14	11/11/22	0044	12	11/11/22	0058	9	11/11/22
0017	8	11/11/22	0031	14	11/11/22	0045	21	12/04/24	0059	42	11/11/22
0018	8	11/11/22	0032	14	11/11/22	0046	1.90M	11/11/22	0060	43	11/11/22
0019	8	11/11/22	0033	44	11/11/22	0047	14	11/11/22	0061	67	11/11/22
0020	8	11/11/22	0034	24	11/11/22	0048	14	11/11/22	0068	58	11/11/22
Program 0001 (*****)										Occupied 17.1MByte	
Are you sure you want to delete?										Available 439MByte	
YesNo											

Delete the program with the Program No.0001 on this screen.

(3) To delete the program, press the **[F0]** (Yes) key. To cancel the operation, press the **[F1]** (No) key.

The following cases show operator messages and disable deletion.

- If an uneditable program is designated: <<Program is write-protected>>
- If a program you want to delete is in memory operation: <<Running program>>
- If a program you want to delete is in communication: <<Communicating>>

To designate an area and delete multiple programs

Before operation, invalidate the **[DATA PROTECTION]** switch.

(1) Designate a desired program in the input area.

“Program number at the head”, [/], “Program number at the end”

Example) When the programs from 1 to 8999 are deleted

1/8999

- * If the “Program number at the head” is omitted, the programs from the 1st one are deleted.
- * If the “Program number at the end” is omitted, the programs up to the 9999th are deleted.

The conversational language includes the schedule programs from 9900 to 9999.

(2) Press the **[F5]** (Delete) key.

Directory display(Program) 2012/05/28 13:51:35

Program(1113)

Current folder/

No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
0001	1.27K	12/05/24	0021	8	11/11/22	0035	68	11/11/22	0049	12	11/11/22
0002	144	11/11/22	0022	8	11/11/22	0036	9	11/11/22	0050	12	11/11/22
0004	74	11/11/22	0023	8	11/11/22	0037	9	11/11/22	0051	159	11/11/22
0005	4	11/11/22	0024	8	11/11/22	0038	9	11/11/22	0052	41	11/11/22
0010	64	11/11/22	0025	8	11/11/22	0039	9	11/11/22	0053	9	11/11/22
0011	24	11/11/22	0026	9	11/11/22	0040	25	11/11/22	0054	155	11/11/22
0012	14	11/11/22	0027	12	11/11/22	0041	16	11/11/22	0055	2.15M	11/11/22
0013	9	11/11/22	0028	8	11/11/22	0042	28	11/11/22	0056	19	11/11/22
0014	14	11/11/22	0029	14	11/11/22	0043	16	11/11/22	0057	57	11/11/22
0016	21	11/11/22	0030	14	11/11/22	0044	12	11/11/22	0058	9	11/11/22
0017	8	11/11/22	0031	14	11/11/22	0045	21	12/04/24	0059	42	11/11/22
0018	8	11/11/22	0032	14	11/11/22	0046	1.90M	11/11/22	0060	43	11/11/22
0019	8	11/11/22	0033	44	11/11/22	0047	14	11/11/22	0061	67	11/11/22
0020	8	11/11/22	0034	24	11/11/22	0048	14	11/11/22	0068	58	11/11/22

Program 0001 (*****
Are you sure you want to delete?
(Delete range:0001 - 8999)

Occupied 17.1MByte
Available 430MByte

Yes No Delete all remaining Cancel

Program numbers from 0001 to 8999 are deleted from the screen above.

- (3) To deleted the program, press the **[F0]** (Yes) key or **[F4]** (Delete all remaining) key.
To cancel operation, press the **[F1]** (No) key.

[F0] (Yes):

Delete the program.

When the **[F0]** (Yes) key is pressed, <Deleting> appears on the screen.

If the program has been deleted, the next program check screen appears.

If there is no next program, the display returns to the original screen.

[F1] (No):

Cancel the delete operation.

When the **[F1]** (No) key is pressed, the next program check screen appears.

If there is no next program, the display returns to the original screen.

[F4] (Delete all remaining):

Delete all programs in the specified range.

When pressing **[F4]** (Delete all remaining) key, <Deleting> appears on the screen.

[F7] (Cancel):

Cancels deletion and the screen returns to the previous screen.

The following cases display operator messages and disable deletion.

- If an uneditable program is designated: <<Program is write-protected>>
- If a program you want to delete is in memory operation: <<Running program>>
- If a program you want to delete is in communication: <<Communicating>>

4. Protection of programs
The programs from 8000 to 8999 can be set “uneditable.”
The programs set uneditable are the programs from 8000 to 8999 not only in a current folder but also in different folders.
ON: Changes are prohibited.
OFF: Changes are permitted.
5. G-code conversion
This function is only available in a conversational mode.
The conversational program is converted into an NC program.
Programs in other folders can also be converted.

Directory display(Program) 2012/05/28 13:59:29

Program(0001)

Current folder:/

No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
AAA/	12/05/25	12/05/25	SSSPP/	12/05/25	0017	1.02K	11/11/22	0031	1.19K	11/11/22	
BBB/	12/05/25	12/05/25	WWW/	12/05/25	0018	954	11/11/22	0040	25.6K	12/01/31	
CCC/	12/05/25	12/05/25	0001	2.47K	12/05/24	0019	1.02K	11/11/22	0041	2.53M	11/11/22
DDD/	12/05/25	12/05/25	0002	12.1K	11/12/22	0020	1.19K	11/11/22	0050	2.53M	11/11/22
EEE/	12/05/25	12/05/25	0003	1.82K	12/01/23	0021	962	11/11/22	0051	2.53M	11/11/22
FFF/	12/05/25	12/05/25	0004	1.31K	12/04/23	0022	1.14K	11/11/22	0060	4.83K	11/11/22
GGG/	12/05/25	12/05/25	0005	1.22K	12/01/23	0023	1.13K	11/11/22	0103	2.27M	12/04/02
HHH/	12/05/25	12/05/25	0010	4.96K	11/11/22	0024	1.28K	11/11/22	0201	13.6K	11/11/22
III/	12/05/25	12/05/25	0011	4.83K	11/11/22	0025	1.42K	11/11/22	0202	12.6K	11/11/22
JJJ/	12/05/25	12/05/25	0012	1.89K	12/01/30	0026	806	11/11/22	0204	23.7K	11/11/22
KKK/	12/05/25	12/05/25	0013	2.26M	11/11/22	10027	927	11/11/22	1111	2.03K	11/11/22
LLL/	12/05/25	12/05/25	0014	2.25M	12/01/31	0028	1.22K	11/11/22	1112	6.32K	11/11/22
OOO/	12/05/25	12/05/25	0015	1.64K	12/01/31	0029	1.07K	11/11/22	1113	7.73K	11/11/22
SSS/	12/05/25	12/05/25	0016	931	11/11/22	0030	1.51K	11/11/22	1114	15.6K	11/11/22

Program No. [] Occupied 25.1MByte
Available 430MByte

No. 0001 2.47K Byte 12/05/24 14:04:00 *****
A-B:Copy file A to B or check.
A/B:Delete files A to B
Data protection off:Copy mode Data protection on:Check mode
Running Protected I-Inch

Program edit menu Convert G code Change program number Operate between folders

Before operation, invalidate the **[DATA PROTECTION]** switch.

- (1) Press the [$\gg$] (next function) key to have the function display the G code conversion.
- (2) Press the **[F1]** (Convert G code) key.

* For details, refer to “1.6 How to convert G codes”.

6. Change of program numbers
A designated program can be changed to another program number.
Carry out the “Operation between folders” for moving a program into other folder.

Example in NC language

Directory display(Program) 2012/05/28 13:59:29

Program(0001)

Current folder:/

No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
AAA/	12/05/25	12/05/25	SSSPP/	12/05/25	0017	1.02K	11/11/22	0031	1.19K	11/11/22	
BBB/	12/05/25	12/05/25	WWW/	12/05/25	0018	954	11/11/22	0040	25.6K	12/01/31	
CCC/	12/05/25	12/05/25	0001	2.47K	12/05/24	0019	1.02K	11/11/22	0041	2.53M	11/11/22
DDD/	12/05/25	12/05/25	0002	12.1K	11/12/22	0020	1.19K	11/11/22	0050	2.53M	11/11/22
EEE/	12/05/25	12/05/25	0003	1.82K	12/01/23	0021	962	11/11/22	0051	2.53M	11/11/22
FFF/	12/05/25	12/05/25	0004	1.31K	12/04/23	0022	1.14K	11/11/22	0060	4.83K	11/11/22
GGG/	12/05/25	12/05/25	0005	1.22K	12/01/23	0023	1.13K	11/11/22	0103	2.27M	12/04/02
HHH/	12/05/25	12/05/25	0010	4.96K	11/11/22	0024	1.28K	11/11/22	0201	13.6K	11/11/22
III/	12/05/25	12/05/25	0011	4.83K	11/11/22	0025	1.42K	11/11/22	0202	12.6K	11/11/22
JJJ/	12/05/25	12/05/25	0012	1.89K	12/01/30	0026	806	11/11/22	0204	23.7K	11/11/22
KKK/	12/05/25	12/05/25	0013	2.26M	11/11/22	10027	927	11/11/22	1111	2.03K	11/11/22
LLL/	12/05/25	12/05/25	0014	2.25M	12/01/31	0028	1.22K	11/11/22	1112	6.32K	11/11/22
OOO/	12/05/25	12/05/25	0015	1.64K	12/01/31	0029	1.07K	11/11/22	1113	7.73K	11/11/22
SSS/	12/05/25	12/05/25	0016	931	11/11/22	0030	1.51K	11/11/22	1114	15.6K	11/11/22

Program No. [] Occupied 25.1MByte
Available 430MByte

No. 0001 2.47K Byte 12/05/24 14:04:00 *****
A-B:Copy file A to B or check.
A/B:Delete files A to B
Data protection off:Copy mode Data protection on:Check mode
Running Protected I-Inch

Program edit menu Convert G code Change program number Operate between folders

Before operation, invalidate the **[DATA PROTECTION]** switch.

- (1) Designate a desired program in the input area.
How to input to the input area (program number)
“Program number to be changed”, [-] , “New program number”
- (2) Press the [>>] (next function) key to make the function display “Program number change” and press the [F1] ([F2] for conversational mode)] (Change program number) key.

The following cases display operator messages to disable movements.

- If the data protection is validated: <<Data is write-protected.>>
- If the program, whose program number to be changed is made uneditable, is designated: <<Program is write-protected>>
- If the “new program number” is uneditable: <<Program is write-protected>>
- If the “new program number” is already existing: <<A program is already existing>>
- If the program designated by the “Program number to be changed” is in operation: <<Running program>>
- If any of the designated programs is in communication: <<Communicating>>

7. Operation between folders

Copying/checking and moving of a file between folders and copying/checking of an entire folder are available.

In addition, creation and deletion of a folder and changes of a folder name are available.

For details, refer to “1.5.1 Operation between folders”.

1.4.3 Directory Display (Data bank)

1. Displaying directory of data bank

Select <Directory display> on the <Program edit menu> screen.

The <Directory display (Program)> screen or <Directory display (Data bank)> screen appears.

When the <Directory display (Program)> screen appears, press the [F1] (Data bank) key to switch to the <Directory display (Data bank)> screen.

<Directory display (Data bank)> screen (NC language)

Directory display(Data bank)										2012/05/26 14:07:35	
Items	Size	Metric				Inch					
Workpc. coord. zero1	17.1K	1	2	3	5	1					
Tool data1	119K	1	2	3		1					
Macro variables1	24.9K	1	2			1					
User parameter1	14.3K	1		3	4	1					
External I/O signal1	36.6K	1									
Commun. parameter1	544	1									
Machine parameter1	84.5K	1									
Simplif.Magazine pane Custom	0										
No.		☒ Busy				☐ Previous format					
						Occupied		297KByte			
						Available		436KByte			
A-B: Copy file A to B or check.											
Data protection off: Copy mode Data protection on: Check mode											
☒ ☐											
[HELP]											
Program edit	Program	Copy/Check	Delete	Delete all	Format						

<Directory display (Data bank)> screen (Conversational language)

Directory display(Data bank)										2012/05/28 14:12:15	
Items	Size	Metric					Inch				
Tool list1	52.2K	1	2	3	4		1				
					9						
Tool pattern1	1.01K	1	2	3							
						10					
Tapping drill dia.1	15.4K	1		3			1	2			
Cutting conditions1	19.9K	1		3			1				
User parameter1	14.3K	1		3	4		1				
External I/O signal1	36.6K	1									
Commun. parameter1	544	1									
Machine parameter1	84.5K	1									
No.											
Busy ☐ Previous format											
										Occupied	224KByte
										Available	436KByte
A-B: Copy file A to B or check											
Data protection off: Copy mode Data protection on: Check mode											
[HELP]											
Program edit	Program	Copy/Check	Delete	Delete all	Format						

(NC language)

File	Notes
Workpiece coordinate zero	Metric/Inch
Tool data	Metric/Inch
Macro variable	Metric/Inch
User parameter	Metric/Inch
External I/O signal	
Communication parameter	
Machine parameter	Only one
G/M code macro	
Special settings	
Machining load monitor	
Simplified/Magazine panel custom	(Reserved)
Teaching pendant custom	Only one

(Conversational)

File	Notes
Tool list	Metric/Inch
Tool pattern	
Tapping drill diameter	Metric/Inch
Cutting conditions	Metric/Inch
User parameter	Metric/Inch
External I/O signal	
Communication parameter	
Machine parameter	Only one
Special settings	
Machining load monitor	
Simplified/Magazine panel custom	(Reserved)
Teaching pendant custom	Only one

The data bank can register up to 10 files of metric system data and inch system data respectively per item.

However, <External I/O signal>, <Commun. parameter>, <G/M code macro>, <Special setting> and <Load monitor> are used for both meters and inch data and a maximum of 10 files can be registered. <Machine parameter> and <Teaching pendant custom> are used for both meters and inch data and only one file can be registered.

The numerical character on the right side of the data name is an effective data number.

The data number in the green background shows the currently used data bank.

The data number indicated in the dark gray background in () shows an abnormal data bank like an old format or a parity error.

Data other than <Workpc. coord. zero>, <Tool data>, <Macro variables>, <G/M code macro>, <Tool list>, <Tool pattern>, <Tapping drill dia>, and <Cutting conditions> are the data common to the NC language mode and the conversational mode.

2. Changing the data number currently enabled

Items	Size	Metric	Inch
Workpc. coord. zero1	17.1K	1 2 3 5	1
Tool data1	119K	1 2 3	1
Macro variables1	24.9K	1 2	1
User parameter1	14.3K	1 3 4	1
External I/O signal1	36.6K	1	
Commun. parameter1	544	1	
Machine parameter1	84.5K	1	
Simplif./Magazine pane Custom	0		

No. ☒ Busy ☐ Previous format

Occupied 297KByte
Available 430KByte

A-B: Copy file A to B or check.
Data protection off: Copy mode Data protection on: Check mode

[HELP]

Program edit menu Program Copy/Check Delete Delete all Format

Move the cursor to the desired item and enter the data number to be changed on the <Directory display (Data bank)> screen.

For example, when [2] and then [ENT] are pressed, the data number enabled for [Workpc. coord. zero] is changed from [1] to [2].

Limitations when changing valid data numbers:

- * Changes are disabled when the operator message <<Operating>> appears in memory operation.
- * Changes are disabled when the data of the number before changing are being edited in the data bank or input or output by communication while the operator message <<Editing>> or <<Communicating>> is appearing.
- * Changes are disabled when the data of the number after changes are input or output by communication while the operator message <<Communicating>> is appearing.
- * Changes are disabled when the <Life initialization> screen of the <ATC tool> screen is displayed while the tool data (NC language) or the tool list (conversational language) is being tried to be changed and the operator message <<Initializing tool life>> is appearing.
- * Changes of the machine parameter and teaching pendant customization are disabled.

3. Copying/Checking of data bank

An already registered data bank can be copied to a different data number.

Copying of data bank

Before operation, invalidate the [DATA PROTECTION] switch.

- (1) Move the cursor to the data you want to copy.
- (2) Input the "Bank number you want to copy", [-], and "New bank number" into the input area (number) in this order.
- (3) Press the [F2] (Copy/Check) key.
The Overwrite check screen appears if a new data number exists. Press the [F0] (Yes) key on the Overwrite check screen to overwrite the data. Press the [F1] (No) key to return to the previous screen without overwriting.

Checking of data bank

When the [DATA PROTECTION] switch is validated for operation, the data bank is checked. If two data banks have the same contents, the characters <Check OK> appears in the teaching area.

If a nonexistent data number is designated, the operator message <<Required data not found>> appears to disable checking.

Limitations when copying/checking

- * Currently valid numbers cannot be copied.
- * If the operator message <<Operating>> appears in memory operation, no copying is available.
- * If the operator message <<Editing>> or <<Communicating>> appears when the data you want to copy or check is input or output during data bank editing or communication, copying/checking is unavailable.
- * If the <Life initialization> screen of the <ATC tool> screen appears when a validated tool data (NC language) or a tool list (conversational language) is tried to copy/check, the operator message <<Initializing tool life>> appears to disable copying/checking.

4. Deleting a data bank

One data bank is deleted.

Before operation, invalidate the **[DATA PROTECTION]** switch.

- (1) Bring the cursor to the item you want to delete.
- (2) Input a bank number you want to delete in the input area (number).
- (3) Press the **[F3]** (Delete) key.

Directory display(Data bank)									
2012/05/28 14:16:08									
Items	Size	Metric					Inch		
Workpc. coord. zero1	20.6K	1	2	3	4	5	1		
Tool data1	119K	1	2	3			1		
Macro variables1	24.9K	1	2				1		
User parameter1	14.3K	1		3	4		1		
External I/O signal1	36.6K	1							
Commun. parameter1	544	1							
Machine parameter1	84.5K	1							
Simplif./Magazine pane Custom	0								
Workpc. coord. zero4 Are you sure you want to delete?							Occupied	300KByte	
							Available	430KByte	
Yes	No								

- (4) Press the **[F0]** (Yes) key to delete the selected data bank.

If a nonexistent data number is designated, the operator message <<Required data not found>> appears to disable deleting.

Deleting all data banks

All data banks are deleted.

Before operation, invalidate the **[DATA PROTECTION]** switch.

Directory display(Data bank)									
2012/05/28 14:16:55									
Items	Size	Metric					Inch		
Workpc. coord. zero1	20.6K	1	2	3	4	5	1		
Tool data1	119K	1	2	3			1		
Macro variables1	24.9K	1	2				1		
User parameter1	14.3K	1		3	4		1		
External I/O signal1	36.6K	1							
Commun. parameter1	544	1							
Machine parameter1	84.5K	1							
Simplif./Magazine pane Custom	0								
Workpc. coord. zero User parameter Machine parameter Simplified/Magazine panel custom Are you sure you want to delete?							Occupied	300KByte	
							Available	430KByte	
Yes	No								

Press the **[F4]** (Delete all) key on the <Directory display (Data bank)> screen.
The confirmation screen shown above appears to confirm that you really want to delete all data banks.

Press the **[F0]** (Yes) key to delete all data banks.

Limitations when deleting one or all data banks

- * Deletion is disabled when the operator message <<Operating>> appears during memory operation.
- * Changes are disabled when the operator message <<Editing>> or <<Communicating>> is appearing while the data you want to delete is input or output during data bank editing or communication.
- * Changes are disabled when the <Life initialization> screen of the <ATC tool> screen appears while the valid tool data (NC language) or tool list (conversational language) is tried to be deleted and the operator message <<The life is being initialized>> is appearing.
- * Deletion is disabled when the <Parameter change> is set <0: Write-protected> while the machine parameter is deleted and the alarm <<*Machine parameter was not deleted>> is appearing.

1.4.4 Format

When the memory is formatted, all saved data including the followings are lost.

- 1 program
- 2 data banks (NC language)
- 6 data banks (conversational language)

In addition, the following data are deleted or cleared:

Automatic centering history, ATC tool registration data, MDI operation history, production monitoring, measurement results, machine status history, graph parameters, maintenance notifications, waveform displays and teaching pendant customization

Memory can be formatted from the <Directory display (Program)> screen or <Directory display (Data bank)> screen.

Set the **[DATA PROTECTION]** switch to the **[OFF]** position.

Select <1: Writable> for <Parameter change> on the <Input/Output> screen.

Example of the <Directory display (Program)> screen

Directory display(Program details)				2012/05/28 14:22:55
Program(1113)				
Current folder:/				
No.	Size	Comment	Date	
JKJ/			12/04/26 18:53:48	
KKK/			12/04/26 18:53:48	
LLL/			12/04/26 18:53:48	
DDD/			12/05/25 20:39:02	
SSS/			12/04/26 18:53:48	
SSSPPP/			12/05/14 21:59:44	
VVV/			12/04/26 18:53:48	
0001	1.27K	(*****)	12/05/24 14:05:10	
0002	144	(PRG2)	11/11/22 13:20:28	
0004	74	(PRG3)	11/11/22 13:20:28	
0005	4	M99	11/11/22 13:20:29	
0010	64	M00	11/11/22 13:20:29	
0011	24	M450	11/11/22 13:20:30	
0012	14	T901	11/11/22 13:20:30	
Program No.			Occupied	17.1MByte
No. 0001			Available	430MByte
A-B Copy file A to B or check.				
A/B Delete files A to B.				
Data protection off.Copy mode Data protection on:Check mode				
☒ Running ☐ Protected				
Program edit menu	Data bank	List	Sort	Copy/Check
		Delete	Protect ON	Format

Example of the <Directory display (Data bank)> screen

Directory display(Data bank)									
2012/05/28 14:25:20									
Items	Size	Metric					Inch		
Workpc. coord. zero1	20.6K	1	2	3	4	5	1		
Tool data1	119K	1	2	3			1		
Macro variables1	24.9K	1	2				1		
User parameter1	14.3K	1		3	4		1		
External I/O signal1	36.6K	1							
Commun. parameter1	544	1							
Machine parameter1	84.5K	1							
Simplif./Magazine pane Custom	0								
No. ☒ Busy ☐ Previous format									
							Occupied	300KByte	
							Available	430KByte	
A-B:Copy file A to B or check. Data protection off:Copy mode Data protection on:Check mode									
[HELP]									
Program edit menu	Program	Copy/Check	Delete	Delete all	Format				

Input **[FMT]** and then press the **[F7]** (Format) key on the <Directory display (Program)> screen.
Input “FMT” on the <Directory display (Data bank)> screen and then press the **[F5]** (Format) key.

Directory display(Format)									
2012/05/28 14:26:36									
Formatting the memory will delete all programs and data banks.									
Are you sure you want to format?									
Machine parameters are write-protected.									
Yes	No								

Pressing the **[F0]** (Yes) key formats the memory.

Limitations when formatting

- * Formatting is disabled when the operator message <<Operating>> appears in memory operation.
- * Formatting is disabled when the operator message <<Editing>> or <<Communicating>> appears while editing a data bank or program.
- * Formatting is disabled when the operator message <<Initializing tool life>> appears on the <Life initialization> screen of the <ATC tool> screen.
- * Formatting is disabled when the <Parameter change> is set <0: Write-protected> and the operator message <<Machine param. change disabled>> appears.
- * When the inside of the sub folder is displayed on the following screen, the operator message <<Sub folder is being displayed>> appears and formatting is not possible.
 - List of programs for tool path simulation graph setting
 - List of memory operation programs (When changing to editing (background programming) mode during operation while the sub folder is displayed)

1.4.4.1 Delete All Conversation Files

When the following operations are performed, all the conversation files (data banks and machining programs) are deleted.

- Directory display program screen:
Enter **[T]**, **[C]**, **[F]**, **[M]** and **[T]**, and then press the **[F7]** (format) key.
- Directory display data bank screen:
Enter **[T]**, **[C]**, **[F]**, **[M]** and **[T]**, and then press the **[F5]** (format) key.

Conversation files that are deleted

- Tool list
- Tool pattern
- Tapping drill diameter
- Cutting conditions
- Machining data
- Schedule program

(Note) This operation is only valid when the “Conversation/NC language change” is set to “NC”.

1.5 Folder Operation

1.5.1 Operation between Folders

Programs are copied, checked, or moved between folders.

In addition, creation and deletion of a folder, changing of a folder name, and copying/checking of an entire folder are available.

On the <Directory display> screen of a program, the following operation calls a screen:

1. Press the [>>] (next function) key to make the function display the <Operate between folders>.

Directory display(Program)										2012/12/07 14:18:44	
Program(5000)											
Current folder: /											
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
VERUPBK/		12/08/30	0013	298	12/06/04	0046	104	12/07/26	0081	397	12/05/18
YAM/		12/08/31	0014	293	12/06/04	0047	100	12/07/26	0082	397	12/05/18
0001	701	12/06/04	0015	11	12/06/04	0053	305	12/05/18	0083	444	12/05/18
0002	92	12/07/24	0017	57	12/06/04	0055	2.57K	12/05/21	0091	367	12/05/18
0003	73	12/06/04	0018	32	12/06/04	0057	118	12/07/13	0092	367	12/05/18
0004	73	12/06/04	0020	50	12/06/04	0061	84	12/05/18	0093	414	12/05/18
0005	9	12/07/24	0021	163	12/06/04	0062	84	12/05/18	0094	169	12/08/14
0006	29	12/06/04	0022	74	12/06/04	0063	87	12/05/18	0095	210	12/08/18
0007	170	12/06/04	0023	44	12/08/10	0067	421	12/05/18	0100	4	12/12/03
0008	141	12/06/04	0030	63.8K	12/06/04	0068	449	12/05/18	0101	13	12/08/21
0009	148	12/06/04	0031	63	12/06/04	0069	520	12/05/18	0102	13	12/08/21
0010	48	12/06/04	0035	93	12/07/17	0071	425	12/05/18	0103	13	12/08/21
0011	17	12/06/04	0039	94	12/05/18	0072	424	12/05/18	0104	14	12/08/21
0012	247	12/06/04	0045	80	12/07/26	0073	495	12/05/18	0105	5	12/08/21
Program No.										Occupied 17.2MByte Available 454MByte	
VERUPBK/ 12/08/30 14:31:49											
A-B:Copy file A to B or check. [HELP]											
A/B:Delete files A to B.											
Data protection off:Copy mode Data protection on:Check mode											
☒ Running ☐ Protected											
Program edit menu	Change program number	Operate between folders									

2. Press the [F2] ([F3] for conversational mode)] (Operate between folders) key.
3. The <Operate between folders> screen or <Operate between folders detail> screen appears. The <Operate between folders> screen appears from the <Directory display (Program)> screen. The <Operate between folders details> screen appears from the <Directory display (Program detail)> screen.

Example of the <Operate between folders> screen (when the cursor is on the current folder side)

Display directory Operate between holders										2012/05/28 14:38:31	
Program(1113)											
Current folder./						Copy destination folderAAA/					
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
AAA/		12/05/11	SSSPP/		12/05/14	../master folder					
BBB/		12/04/26	WWW/		12/04/26	0001	138	11/11/22			
CCC/		12/04/26	0001	1.27K	12/05/24	0002	6.16K	12/05/11			
DDD/		12/04/26	0002	144	11/11/22	0004	267	12/05/11			
EEE/		12/04/26	0004	74	11/11/22	0005	267	12/05/11			
FFF/		12/04/26	0005	4	11/11/22	0010	267	12/05/11			
GGG/		12/04/26	0010	64	11/11/22	0011	267	12/05/11			
HHH/		12/05/23	0011	24	11/11/22	0025	474	12/05/11			
III/		12/04/26	0012	14	11/11/22	0045	2.10M	12/05/11			
JJJ/		12/04/26	0013	9	11/11/22	0202	6.32K	12/05/11			
KKK/		12/04/26	0014	14	11/11/22						
LLL/		12/04/26	0016	21	11/11/22						
MMM/		12/05/23	0017	8	11/11/22						
SSS/		12/04/26	0018	8	11/11/22						
Program No.						Occupied 17.1MByte		Available 430MByte			
No. 0001 1.27K Byte 12/05/24 14:05:10 (*****)											
A-B: Copy number A to B/Check, Move folder											
A/B: Specify from number A to number B in a lump A B C: Number A and B and C											
Data protection off: Copy mode Data protection on: Check mode											
Running Protected											
Back	Details	Sort	Select program	Copy/Check	Move folder	Operate folder	Switch cursor left and right				

Example of the <Operate between folders> screen (when the cursor is on the copy destination folder side)

Display directory Operate between holders										2012/05/28 14:42:37	
Program(1113)											
Current folder /						Copy destination folderAAA/					
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
AAA/		12/05/11	SSSPP/		12/05/14	../(master folder)					
BBB/		12/04/26	VVV/		12/04/26	0001	138	11/11/22			
CCC/		12/04/26		0001	1.27K	12/05/24	0002	6.16K	12/05/11		
DDD/		12/04/26		0002	144	11/11/22	0004	267	12/05/11		
EEE/		12/04/26		0004	74	11/11/22	0005	267	12/05/11		
FFF/		12/04/26		0005	4	11/11/22	0010	267	12/05/11		
GGG/		12/04/26		0010	64	11/11/22	0011	267	12/05/11		
HHH/		12/05/23		0011	24	11/11/22	0025	474	12/05/11		
III/		12/04/26		0012	14	11/11/22	0045	2.10M	12/05/11		
JJJ/		12/04/26		0013	9	11/11/22	0202	6.32K	12/05/11		
KKK/		12/04/26		0014	14	11/11/22					
LLL/		12/04/26		0016	21	11/11/22					
OOO/		12/05/25		0017	8	11/11/22					
SSS/		12/04/26		0018	8	11/11/22					
Program No.								Occupied 17.1MByte		Available 430MByte	
No. 0002 6.16K Byte 12/05/11 12:16:08 (RENZOKU UNTEN 14T)											
A-B: Copy number A to B/Check, Move folder											
A/B: Specify from number A to number B in a lump A B C: Number A and B and C											
Data protection off:Copy mode Data protection on:Check mode											
☒ Running ☐ Protected											
Back	Details	Sort	Select program	Copy/Check	Move folder	Operate folder	Switch cursor left and right				

1

Example of the <Operate between foldersdetails> screen

Display directory

Operate between folders detail

2012/05/28 14:44:31

Program(1113)

Current folder/

Copy destination folderAAA/

No.	Size	Comment	Date	No.	Size	Comment	Date
SSS/			12/04/26 18:53:48	../(master folder)			
SSSPP/			12/05/14 21:59:48	0001	138	(PRG1)	11/11/22 13:20:27
VVV/			12/04/26 18:53:48	0002	6.16K	(RENZOKU UNTEN)	12/05/11 12:16:08
0001	1.27K	(*****)	12/05/24 14:05:10	0004	267	(*****)	12/05/11 12:16:09
0002	144	(PRG2)	11/11/22 13:20:28	0005	267	(*****)	12/05/11 12:16:09
0004	74	(PRG3)	11/11/22 13:20:28	0010	267	(*****)	12/05/11 12:16:10
0005	4	M99	11/11/22 13:20:29	0011	267	(*****)	12/05/11 12:16:11
0010	64	M00	11/11/22 13:20:29	0025	474	(*****)	12/05/11 12:16:11
0011	24	M450	11/11/22 13:20:30	0045	2.10M	(2.1M)	12/05/11 12:16:14
0012	14	T901	11/11/22 13:20:30	0202	6.32K	(RENZOKU UNTEN)	12/05/11 12:16:17
0013	9	M400	11/11/22 13:20:31				
0014	14	M450	11/11/22 13:20:39				
0016	21	M08	11/11/22 13:20:46				
0017	8	M11	11/11/22 13:20:47				

Program No.

Occupied 17.1MByte

Available 430MByte

No. 0001

1.27K Byte 12/05/24 14:05:10 (*****)

A-B: Copy number A to B/Check, Move folder

A/B: Specify from number A to number B in a lump A B C: Number A and B and C

Data protection off.Copy mode Data protection on:Check mode

☒ Running

☐ Protected

Back

List

Sort

Select program

Copy/Check

Move folder

Operate folder

Switch cursor left and right

On the list screen, the program number, program size, and program update date are displayed in the order of the selected sort.

On the detail screen, the program size, content of the first block of the program, program updated date are displayed in the order of the selected sort. Content of the first block of a program is displayed in 15 one-byte characters.

- * For the uneditable programs, programs in operation, conversational language program displays, and folder displays, refer to “1.4 Directory display”.

If there are folders (including parent folders), both the list screen and detail screen display folder names added with “/” after the name in the program number area and the program size area.

Current folder

Displays the folder name at the source of copying or moving as same as the current folder on the Directory list screen.

If a current folder is moved, the current folder when it returns to the Directory list screen becomes the same folder.

Copy destination folder

Displays the folder name at the destination of copying or moving. It becomes a root folder when the <Operate between folders> screen is displayed.

[F0] (Back) : Returns to the Directory list screen or Directory list/detail screen.

[F1] (Details/list) : Switches the displays.

<Details>: Comment and update time are displayed.

<List>: Many programs are displayed on one screen.

If this is displayed from the Directory list screen, it is displayed in the status (detail/list) displayed by the Directory list screen.

[F2] (Sort) : Sorts the folders per update time, program number (folders in the alphabetical order), or comment (folders in the alphabetical order) in the ascending order or descending order respectively.

Displaying orders on both the current folder side and copy destination folder side is changable.

* For sorting, refer to “1.2.1 Program edit”

[F7] (Switch cursor left and right): Switches the cursor between the current folder and the copy destination folder.

When the cursor is on the current folder side, the following function keys become valid:

[F3] (Select program) : Designates multiple programs to copy or move.

[F4] (Copy/Check) : Copies a designated program to a copy destination folder or crosschecks the program when the **[DATA PROTECTION]** switch is validated.

[F5] (Move folder) : Moves a designated program to a copy destination folder.

[F6] (Operate folder) : Creates a new folder, deletes a folder, or changes names.

1.5.1.1 Moving a Current Folder

Press the **[F7]** (Switch cursor left and right) key to move the cursor to the current folder side.

Point the cursor at the folder to be moved and press the **[ENT]** key.

The current folder changes to a designated folder to display the program in the folder. If it is moved to a root folder, the folder itself is displayed.

* When the current folder is moved, the current folder on the Directory list screen becomes the same folder.

1.5.1.2 Moving a Copy Destination Folder

Press the **[F7]** (Switch cursor left and right) key to move the cursor to the copy destination folder side.

Point the cursor at the folder to be moved and press the **[ENT]** key.

The copy destination folder changes to a designated folder to display the program in the folder. If it is moved to a root folder, the folder itself is displayed.

1.5.1.3 Program Selection Screen

Multiple programs to be copied or moved are selected.

[F3] (Select program): When this key is pressed, the Program selection screen appears.

Move the cursor to the desired program position and operate the following:

* No folder is selected.

[F1] (Select/Cancel): Point the cursor at the desired number and press the key.

It is selected if it has not been selected. It is cancelled if selected.

[F2] (Start range selection): The area is designated and multiple programs are selected.

Point the cursor at the desired number and press the key. Then the cursor moves to the last number. Press the key again.

[F3] (Select all): All the programs are selected.

[F4] (Cancel all): All the programs are selected and cancelled.

Display directory Select program 2012/05/28 15:05:31											
Program(1113)											
Current folder/						Copy destination folderAAA/					
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
1000	72	12/01/18	3210	147	11/12/05	./master folder					
1111	987	11/11/22	6030	485	12/04/16	0001	138	11/11/22			
1112	3.41K	11/11/22	7000	495	12/04/18	0002	6.16K	12/05/11			
1113	4.35K	11/11/22				0004	267	12/05/11			
1114	7.82K	11/11/22				0005	267	12/05/11			
1115	870	11/11/22				0010	267	12/05/11			
1116	550	11/11/22				0011	267	12/05/11			
2000	102	11/11/22				0025	474	12/05/11			
2111	1007	11/11/22				0045	2.10M	12/05/11			
2112	3.49K	11/11/22				0202	6.32K	12/05/11			
2113	4.56K	11/11/22									
2114	8.01K	11/11/22									
2115	648	11/11/22									
2116	536	11/11/22									
With the cursor positioned in a program to be selected, press the [F1] key.											
No.	2111		1007 Byte	11/11/22	13:22:35	(HOLE	)		Occupied	17.1MByte	
									Available	430Byte	
[HELP]											
Back Select/Cancel Start range selection Select all Cancel all											

Press the **[F0]** (Back) key to return to the <Operate between folders> screen.

1.5.1.4 Copying/Checking a Program between Folders

Program can be copied between different folders including the root folder. If the data protection is valid, the program is crosschecked.

* To copy/check the entire folder, carry out the folder operation.

- Call the current folder (copy source folder) and the copy destination folder.
Press the **[F7]** (Switch cursor left and right) key and move the respective folders to the folder which has the copy source program and to the folder which has the copy destination.
- Designate the program you want to copy.
 - To select on the screen
Refer to the aforementioned "Program selection screen".
 - To input a program number
Input a program number in the area at the lower part of the screen.

Display directory Copy between folders 2012/05/28 15:14:20

Program(1113)

Current folder:/			Copy destination folderAAA/		
No.	Size	Date	No.	Size	Date
0802	64	12/04/18	2115	648	11/11/22
0803	21	12/04/18	2116	536	11/11/22
1000	72	12/01/18	3210	147	11/12/05
1111	987	11/11/22	6030	485	12/04/16
1112	3.41K	11/11/22	7000	495	12/04/18
1113	4.35K	11/11/22			
1114	7.82K	11/11/22			
1115	670	11/11/22			
1116	550	11/11/22			
2000	102	11/11/22			
2111	1007	11/11/22			
2112	3.49K	11/11/22			
2113	4.56K	11/11/22			
2114	8.01K	11/11/22			

Copy source folder "I"
1114 (SUB PROGRAM)
Copy destination folder "AAA/"
1114 (SUB PROGRAM)
Are you sure you want to overwrite?

Occupied 17.1MByte
Available 430MByte

Yes No Hence, copy all Cancel

- [F0] (Yes)** : Overwrites and copies the program in inquiry. When the **[F0]** (Yes) key is pressed, the screen display changes to <Transferring>.
- [F1] (No)** : Does not copy the program in inquiry. When the **[F1]** (No) key is pressed, confirmation to copy the next program starts. If there is no next program, the screen returns to the previous screen.
- [F4] (Hence, copy all)** : Displays when multiple programs are designated to copy. All the copy-designated programs after the program in inquiry are copied. When the **[F4]** (Hence, copy all) key is pressed, the screen display changes to <Transferring>.
- [F7] (Cancel)** : Cancels copying and returns to the <Operate between folders> screen.

When the data protection is valid
Programs can be crosschecked.

For checking, if both programs to be designated do not exist, the operator message <<Required data not found>> appears to disable checking.

If multiple programs are selected from the Program selection screen, an area is designated by “/”, or multiple programs are designated by spaces, respective programs are checked.

When the data does not match, the message <Check error> is displayed in the instructions area and the verification check process is discontinued.

SL7508 Check error

Display directory Copy between folders 2012/05/28 15:34:46

Program(1113)

Current folder:/			Copy destination folderAAA/		
No.	Size	Date	No.	Size	Date
AAA/		12/05/11	SSPP/		12/05/14
BBB/		12/04/26	WWW/		12/04/26
CCZ/		12/04/26	0001	1.27K	12/05/24
DDZ/		12/04/26	0002	144	11/11/22
EEE/		12/04/26	0004	74	11/11/22
FFF/		12/04/26	0005	4	11/11/22
GGG/		12/04/26	0010	64	11/11/22
HHI/		12/05/23	0011	24	11/11/22
III/		12/04/26	0012	14	11/11/22
JJJ/		12/04/26	0013	9	11/11/22
KKK/		12/04/26	0014	14	11/11/22
LLL/		12/04/26	0016	21	11/11/22
OOO/		12/05/25	0017	8	11/11/22
SSS/		12/04/26	0018	8	11/11/22

Check source folder "I"
0001 (*****)
Check destination folder "AAA/"
0001 (PRG1)
Check error

Occupied 17.1MByte
Available 430MByte

Cancel

4. When the copying/checking finishes, the <Operate between folders> screen is called back.

1.5.1.5 Program Movement between Folders

Programs can be moved between folders including root folders.

* If a program is moved in the same folder, use the <Directory display (Program)> screen.

1. Invalidate the data protection.

(Note) If the data protection is validated, the operator message <<Data is write-protected>> appears to disable pressing the [F5] (Move Folder) key. In addition, if the data protection is validated during a movement, the operator message <<Data is write-protected>> appears to cancel the process halfway.

2. Call the current folder (folder before movement) and the movement destination folder. Press the [F7] (Switch cursor left and right) key and move each folder to the folder where a desired program exists and to the movement destination folder.

3. Designate the program to be moved.

- (1) To select from a screen

Refer to the aforementioned "Program selection screen".

- (2) To input a program number

Input a program number in the area at the lower part of the screen.

Entry example	Machine operation
1234	1234 on the current folder side → 1234 on the movement destination folder side
1234-5678 (Divided by a minus "-")	1234 on the current folder side → 5678 on the movement destination folder side
1000/1999 (Divided by a slash "/")	All of 1000-1999 on the current folder side → All of 1000-1999 on the movement destination folder side
1000 2000 3000 (Divided by a space, up to 24 characters)	1000, 2000, & 3000 on the current folder side → 1000, 2000, & 3000 on the movement destination folder side

* If the current folder and the movement destination folder are the same folder, the message <<Transfer to the same folder is unavailable.>> appears to disable movement.

4. Press the [F5] (Move folder) key.

File movement starts.

If a same program number is already in a movement destination, the Overwrite check screen appears.

If multiple programs are selected from the Program selection screen, an area is designated by "/", or multiple programs are designated by spaces, the Overwrite check screen appears per program.

Display directory Move between folders 2012/05/28 15:38:03											
Program(1113)											
Current folder:/						Copy destination folderAAA/					
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
SSSP/F/	12/05/14		0019	8	11/11/22	..(master folder)					
vvv/	12/04/28		0020	8	11/11/22	0001	138	11/11/22			
0001 1.27K	12/05/24		0021	8	11/11/22	0002	6.16K	12/05/11			
0002 144	11/11/22		0022	8	11/11/22	0004	267	12/05/11			
0004 74	11/11/22		0023	8	11/11/22	0005	267	12/05/11			
0005 4	11/11/22		0024	8	11/11/22	0010	267	12/05/11			
0010 64	11/11/22		0025	8	11/11/22	0011	267	12/05/11			
0011 24	11/11/22		0026	9	11/11/22	0025	474	12/05/11			
0012 14	11/11/22		0027	12	11/11/22	0045	2.10M	12/05/11			
0013 9	11/11/22		0028	8	11/11/22	0202	6.32K	12/05/11			
0014 14	11/11/22		0029	14	11/11/22	1114	7.82K	11/11/22			
0016 21	11/11/22		0030	14	11/11/22						
0017 8	11/11/22		0031	14	11/11/22						
0018 8	11/11/22		0032	14	11/11/22						
Move source folder "/"						Occupied 17.1MByte					
0001 (*****)						Available 430MByte					
Move destination folder "AAA/"											
0001 (PRG1)											
Are you sure to move?											
Yes		No				Hence, all are moved				Cancel	

- [F0]** (Yes) : Moves the program in inquiry. When the **[F0]** (Yes) key is pressed, the screen display changes to <Moving>.
- [F1]** (No) : Does not move the program in inquiry. When the **[F1]** (No) key is pressed, confirmation of the next program movement starts. If there is no next program, the previous screen is called back.
- [F4]** (Hence, all are moved) : Displays when multiple programs are designated to move. All the movement-designated programs after the program in inquiry are moved. When the **[F4]** (Hence, all are moved) key is pressed, the screen display changes to <Moving>.
- [F7]** (Cancel) : Cancels the movement and calls back the <Operate between folders> screen.

5. When the movement finishes, the <Operate between folders> screen is called back.

If a program unable of edit movement is designated, the operator message <<Program is write-protected>> appears to disable movement.

In addition, if a movement destination number is designated but is uneditable, the operator message <<Program is write-protected>> appears to disable movement.

If a program before movement is in operation, the operator message <<Running program>> appears to disable movement.

If a program before movement or after movement is in communication, the operator message <<Communicating>> appears to disable movement.

1.5.2 Folder Operation

If the **[F6]** (Operate folder) key is pressed on the <Operate between folders> screen while the root folder is appearing on the current folder side, the <Operate folder> screen appears.

1

Display directory Operate folder												2012/05/28 15:49:50	
Program(1113)													
Current folder:/													
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date		
AAA/		12/05/11	SSSP/		12/05/14	0019	8	11/11/22	0033	44	11/11/22		
BBB/		12/04/26	VVV/		12/04/26	0020	8	11/11/22	0034	24	11/11/22		
CCC/		12/04/26	0001	1.27K	12/05/24	0021	8	11/11/22	0035	68	11/11/22		
DDD/		12/04/26	0002	144	11/11/22	0022	8	11/11/22	0036	9	11/11/22		
EEE/		12/04/26	0004	74	11/11/22	0023	8	11/11/22	0037	9	11/11/22		
FFF/		12/04/26	0005	4	11/11/22	0024	8	11/11/22	0038	9	11/11/22		
GGG/		12/04/26	0010	64	11/11/22	0025	8	11/11/22	0039	9	11/11/22		
HHH/		12/05/23	0011	24	11/11/22	0026	9	11/11/22	0040	25	11/11/22		
III/		12/04/26	0012	14	11/11/22	0027	12	11/11/22	0041	16	11/11/22		
JJJ/		12/04/26	0013	9	11/11/22	0028	8	11/11/22	0042	28	11/11/22		
KKK/		12/04/26	0014	14	11/11/22	0029	14	11/11/22	0043	16	11/11/22		
LLL/		12/04/26	0016	21	11/11/22	0030	14	11/11/22	0044	12	11/11/22		
OOO/		12/05/25	0017	8	11/11/22	0031	14	11/11/22	0045	21	12/04/24		
SSS/		12/04/26	0018	8	11/11/22	0032	14	11/11/22	0046	1.90M	11/11/22		

AAA/

12/05/11

12:16:17

Occupied

17.1MByte

Available

430MByte

[HELP]

Running

Protected

⏏

⏏

Back

Details

Sort

Create

Delete

Rename

Copy/Check

- [F0]** (Back) When this key is pressed, the <Operate between folders> screen is called back.
The current folder and the copy destination folder on the <Operate between folders> screen become a root folder.
- [F1]** (Details/list) When this key is pressed, the detail display screen and the list display screen are switched.
- [F2]** (Sort) When this key is pressed, the display order can be changed.
*Refer to section “1.2.1 Program edit” further details on the sort function.
- [F3]** (Create) When this key is pressed, a new folder can be created.
- [F4]** (Delete) When this key is pressed, the folder selected by the cursor is deleted.
- [F5]** (Rename) When this key is pressed, the folder name selected by the cursor is changed.
- [F6]** (Copy/Check) When this key is pressed, the folder selected by the cursor is copied to a different folder and checked.

If the data protection is validated when the **[F3]** (Create), **[F4]** (Delete), or **[F5]** (Rename) key is pressed, the operator message <<Data is write-protected>> appears to disable operation.

If the cursor is pointed at a folder and the **[ENT]** key is pressed, the current folder changes to a designated folder and the program in the folder is displayed. In such a case, the **[F3]** (Create), **[F4]** (Delete), **[F5]** (Rename), or **[F6]** (Copy/Check) key is invalidated to disable pressing.

1.5.2.1 To Create a Folder

A new folder is created under a root folder.

1. Invalidate the data protection.
2. Press the **[F3]** (Create) key. The <Create folde> screen appears.



3. Input a new folder name in the lower input area of the screen and press the **[F1]** (Create folder) key.
 - A folder is named in maximum 8 alphanumerical characters and period.
 - The folder including a root folder is layered in two. (A folder cannot be created in a created folder.)
 - If an undesignable folder name like already existing folder names, names (O0000-O9999) in NC programs, names in conversational programs, etc. is selected, the operator message <<This is an unusable folder name>> appears to disable creation.
 - If the data protection is validated, the operator message <<Data is write-protected>> appears to disable creation.

1.5.2.2 To Delete a Folder

A folder and a file in the folder can be deleted. Only if all the files are deleted from a folder, the folder itself is also deleted.

(Note) A folder may contain files of the current language mode (NC language/conversational language) and of the opposite side. Be cautious that all the files are deleted from the folder if deletion is continued.

If an uneditable program is in a folder, the operator message <<Program is write-protected>> appears to disable deletion of the program and the folder.

If an inoperable program is in a folder you want to delete, the operator message <<Running program>> appears to disable deletion of the program and the folder.

If a communicating program is in a folder you want to delete, the operator message <<Communicating>> appears to disable deletion of the program and the folder.

If the data protection is validated, the operator message <<Data is write-protected>> appears to disable deletion of the program and the folder.

1. Invalidate the data protection.
2. Point the cursor at the desired folder and press the **[F4]** (Delete) key. The Folder deletion check screen appears.

1

Display directory Clear folder										2012/05/28 15:55:54	
Program(1113)											
Current folder:/											
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
AAA/		12/05/11	SSSPPP/		12/05/14	0019	8	11/11/22	0033	44	11/11/22
BBB/		12/04/26	WWW/		12/04/26	0020	8	11/11/22	0034	24	11/11/22
CCC/		12/04/26	0001	1.27K	12/05/24	0021	8	11/11/22	0035	68	11/11/22
DDD/		12/04/26	0002	144	11/11/22	0022	8	11/11/22	0036	9	11/11/22
EEE/		12/04/26	0004	74	11/11/22	0023	8	11/11/22	0037	9	11/11/22
FFF/		12/04/26	0005	4	11/11/22	0024	8	11/11/22	0038	9	11/11/22
GGG/		12/04/26	0010	64	11/11/22	0025	8	11/11/22	0039	9	11/11/22
HHH/		12/05/23	0011	24	11/11/22	0026	9	11/11/22	0040	25	11/11/22
III/		12/04/26	0012	14	11/11/22	0027	12	11/11/22	0041	16	11/11/22
JJJ/		12/04/26	0013	9	11/11/22	0028	8	11/11/22	0042	28	11/11/22
KKK/		12/04/26	0014	14	11/11/22	0029	14	11/11/22	0043	16	11/11/22
LLL/		12/04/26	0016	21	11/11/22	0030	14	11/11/22	0044	12	11/11/22
OOO/		12/05/25	0017	8	11/11/22	0031	14	11/11/22	0045	21	12/04/24
SSS/		12/04/26	0018	8	11/11/22	0032	14	11/11/22	0046	1.90M	11/11/22

Folder "AAA" is cleared including programs in the folder.

Occupied 17.1MByte
 Available 430MByte

 10 files of NC programs exist.
 9 files of interactive programs exist.
 All are cleared. Are you sure?

■ Running ■ Protected

Yes	No						
-----	----	--	--	--	--	--	--

3. Press the **[F0]** (Yes) key to delete the folder. If not to delete the folder, press the **[F1]** (No) key.
4. Confirm the deletion per file.

Display directory Clear folder										2012/05/28 15:58:22	
Program(1113)											
Current folder:/											
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
AAA/		12/05/11	SSSPPP/		12/05/14	0019	8	11/11/22	0033	44	11/11/22
BBB/		12/04/26	WWW/		12/04/26	0020	8	11/11/22	0034	24	11/11/22
CCC/		12/04/26	0001	1.27K	12/05/24	0021	8	11/11/22	0035	68	11/11/22
DDD/		12/04/26	0002	144	11/11/22	0022	8	11/11/22	0036	9	11/11/22
EEE/		12/04/26	0004	74	11/11/22	0023	8	11/11/22	0037	9	11/11/22
FFF/		12/04/26	0005	4	11/11/22	0024	8	11/11/22	0038	9	11/11/22
GGG/		12/04/26	0010	64	11/11/22	0025	8	11/11/22	0039	9	11/11/22
HHH/		12/05/23	0011	24	11/11/22	0026	9	11/11/22	0040	25	11/11/22
III/		12/04/26	0012	14	11/11/22	0027	12	11/11/22	0041	16	11/11/22
JJJ/		12/04/26	0013	9	11/11/22	0028	8	11/11/22	0042	28	11/11/22
KKK/		12/04/26	0014	14	11/11/22	0029	14	11/11/22	0043	16	11/11/22
LLL/		12/04/26	0016	21	11/11/22	0030	14	11/11/22	0044	12	11/11/22
OOO/		12/05/25	0017	8	11/11/22	0031	14	11/11/22	0045	21	12/04/24
SSS/		12/04/26	0018	8	11/11/22	0032	14	11/11/22	0046	1.90M	11/11/22

Delete folder "AAA"
 NC program
 0001 (PRG1)
 Delete status log?

■ Running ■ Protected

Yes	No			Delete all remaining			Cancel
-----	----	--	--	----------------------	--	--	--------

- [F0]** (Yes) : Deletes the program in inquiry. When this **[F0]** (Yes) key is pressed, the screen display changes to <Deleting>.
- [F1]** (No) : Does not delete the program in inquiry. When this **[F1]** (No) key is pressed, confirmation to delete the next program starts. If there is no next program, the <Operate folder> screen is called back.
- [F4]** (Delete all remaining) : Deletes all the programs after the program in inquiry. When this **[F4]** (Delete all remaining) key is pressed, the screen display changes to <Deleting>.
- [F7]** (Cancel) : Cancels the deletion and the <Operate folder> screen is called back.

5. When the deletion finishes, the <Operate folder> screen is called back.

1.5.2.3 To Change a Folder Name

Folder names can be changed.

If a folder has an uneditable program, the operator message <<Program is write-protected>> appears to disable changing folder names.

If a folder has a program in operation, the operator message <<Running program>> appears to disable folder name changes.

If a folder has a program in communication, the operator message <<Communicating>> appears to disable changing folder names.

1. Invalidate the data protection.
2. Point the cursor at the folder name to be changed and press the [F5] (Rename) key. The Rename folder screen appears.

No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
AAA/		12/05/11	SSSP/		12/05/14	0019	8	11/11/22	0033	44	11/11/22
BBB/		12/04/26	WWW/		12/04/26	0020	8	11/11/22	0034	24	11/11/22
CCC/		12/04/26				0021	8	11/11/22	0035	68	11/11/22
DDD/		12/04/26	0001	1.27K	12/05/24	0022	8	11/11/22	0036	9	11/11/22
EEE/		12/04/26	0002	144	11/11/22	0023	8	11/11/22	0037	9	11/11/22
FFF/		12/04/26	0004	74	11/11/22	0024	8	11/11/22	0038	9	11/11/22
GGG/		12/04/26	0005	4	11/11/22	0025	8	11/11/22	0039	9	11/11/22
HHH/		12/05/23	0010	64	11/11/22	0026	9	11/11/22	0040	25	11/11/22
III/		12/04/26	0011	24	11/11/22	0027	12	11/11/22	0041	16	11/11/22
JJJ/		12/04/26	0012	14	11/11/22	0028	8	11/11/22	0042	28	11/11/22
KKK/		12/04/26	0013	9	11/11/22	0029	14	11/11/22	0043	16	11/11/22
LLL/		12/04/26	0014	14	11/11/22	0030	14	11/11/22	0044	12	11/11/22
MMM/		12/05/25	0016	21	11/11/22	0031	14	11/11/22	0045	21	12/04/24
SSS/		12/04/26	0017	8	11/11/22	0032	14	11/11/22	0046	1.90M	11/11/22

w folder name

AAA/ 12/05/11 12:16:17

Occupied 17.1MByte
Available 430MByte

Running Protected

Back Change

3. Input a new folder name in the input area at the lower part of the screen and press the [F1] (Change) key.

1.5.2.4 To Copy/Check a File in a Folder

A file in a designated folder can be copied to a new or different folder. If the data protection is validated, the files in a folder and the other files in a different folder are crosschecked.

(Note 1) When specifying a new folder name to the copy destination while data protection is enabled, the message <<Required data not found>> is displayed and a check cannot be performed.

(Note 2) The folder may contain files that are opposite from current language mode (NC language/conversation language). In this situation, the copy/check that is performed includes those files that are opposite.

1. Move the cursor to the folder you want to copy or check and press the **[F6]** (Copy/Check) key. The screen to select a folder copy destination appears.

No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
AAA/	12/05/11	SSSPPP/	12/05/14	0019	8	11/11/22	0033	44	11/11/22		
BBB/	12/04/26	WWW/	12/04/26	0020	8	11/11/22	0034	24	11/11/22		
CCC/	12/04/26	0001	1.27K	12/05/24	0021	8	11/11/22	0035	68	11/11/22	
DDD/	12/04/26	0002	144	11/11/22	0022	8	11/11/22	0036	9	11/11/22	
EEE/	12/04/26	0004	74	11/11/22	0023	8	11/11/22	0037	9	11/11/22	
FFF/	12/04/26	0005	4	11/11/22	0024	8	11/11/22	0038	9	11/11/22	
GGG/	12/04/26	0010	64	11/11/22	0025	8	11/11/22	0039	9	11/11/22	
HHH/	12/05/23	0011	24	11/11/22	0026	9	11/11/22	0040	25	11/11/22	
III/	12/04/26	0012	14	11/11/22	0027	12	11/11/22	0041	16	11/11/22	
JJJ/	12/04/26	0013	9	11/11/22	0028	8	11/11/22	0042	28	11/11/22	
KKK/	12/04/26	0014	14	11/11/22	0029	14	11/11/22	0043	16	11/11/22	
LLL/	12/04/26	0016	21	11/11/22	0030	14	11/11/22	0044	12	11/11/22	
OOO/	12/05/25	0017	8	11/11/22	0031	14	11/11/22	0045	21	12/04/24	
SSS/	12/04/26	0018	8	11/11/22	0032	14	11/11/22	0046	1.90M	11/11/22	

Folder name:

AAA/ 12/05/11 12:16:17

Enter new folder name or select existing folder with the cursor.

Data protection off: Copy mode Data protection on: Check mode

☒ Running ☐ Protected

Occupied 17.1MByte Available 430MByte

Back Enter

2. Input a new folder name and press the **[F7]** (Enter) key. (Only for copying)
Alternatively, select a folder you want to copy or check by the cursor and press the **[F7]** (Enter).

* When the folder same as the folder you want to copy or check is selected, the message <<Copy to and check of same folder not available.>> appears to disable copying or checking.

If the data protection is invalid

Copying of the program in the folder starts.

If a new folder is designated to a copy destination, create a folder and copy all the programs.

If a folder existing in a copy destination is designated, all the programs are copied to the copy destination folder. If a same number program exists, the Overwrite check screen is called.

For the Overwrite check screen, refer to the aforementioned "Copying/checking between folders."

If the data protection is valid

Crosscheck of the programs in the folder starts.

For checking, if the folder to be checked has no program to be checked, the operator message <<Required data not found>> appears to disable checking.

If nothing matches, the operator message <Check error> appears and the crosschecking is cancelled halfway. (Refer to the aforementioned "Copying/checking between folders.")

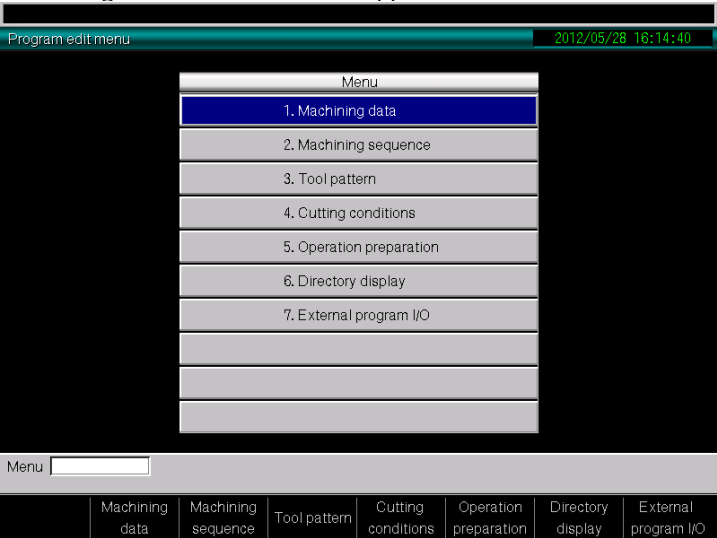
3. When the copying/checking finishes, the <Operate folder> screen is called back.
If an uneditable program is designated, the operator message <<Program is write-protected>> appears to disable copying or overwriting.

1.6 How to Convert G Codes

1.6.1 Operational Procedure

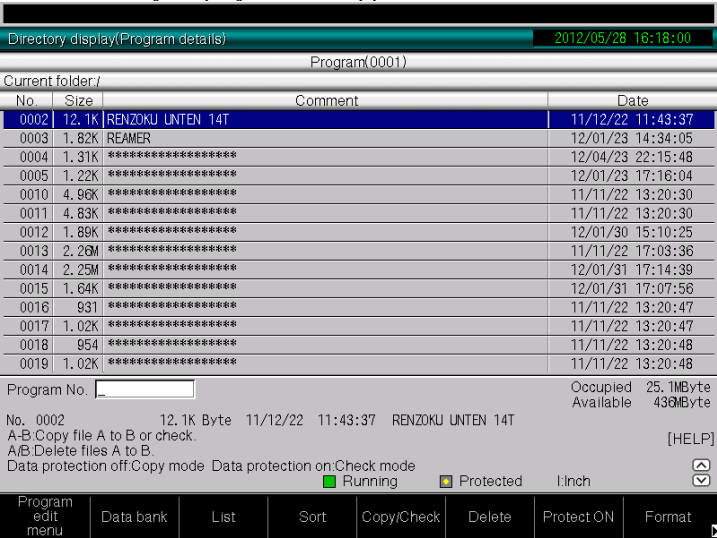
1. Set the [DATA PROTECTION] switch to the [OFF] position.
2. Push the [EDIT] key.

The <Program edit menu> screen appears.



3. Push the [6] key and then push the [ENT] key.

The <Directory display> screen appears.



If the <Directory display (Data bank)> screen is displayed, press the [F1] (Program) key.

- * The following methods can be used to select the desired menu on the <Program edit> screen.
- Use the cursor to select “6. Directory display” and press the [ENT] key.
- Press the [F6] (Directory display) key.

4. Press the [>>] key.
The function keys displayed change.

Directory display(Program details) 2012/05/28 16:20:09

Program(0001)

Current folder:/

No.	Size	Comment	Date
0002	12.1K	RENZOKU UNTEN 14T	11/12/22 11:43:37
0003	1.82K	REAMER	12/01/23 14:34:05
0004	1.31K	*****	12/04/23 22:15:48
0005	1.22K	*****	12/01/23 17:16:04
0010	4.96K	*****	11/11/22 13:20:30
0011	4.83K	*****	11/11/22 13:20:30
0012	1.89K	*****	12/01/30 15:10:25
0013	2.20M	*****	11/11/22 17:03:36
0014	2.25M	*****	12/01/31 17:14:39
0015	1.64K	*****	12/01/31 17:07:56
0016	931	*****	11/11/22 13:20:47
0017	1.02K	*****	11/11/22 13:20:47
0018	954	*****	11/11/22 13:20:48
0019	1.02K	*****	11/11/22 13:20:48

Program No. [] Occupied 25.1MByte
Available 430MByte

No. 0002 12.1K Byte 11/12/22 11:43:37 RENZOKU UNTEN 14T
A-B:Copy file A to B or check.
A/B:Delete files A to B. [HELP]
Data protection off:Copy mode Data protection on:Check mode
☒ Running ☐ Protected I.Inch

Program edit menu Convert G code Change program number Operate between folders

5. Press the [F1] (Convert G code) key.
The <Directory display (G code conversion)> screen appears.

Directory display(G code conversion details) 2012/05/28 16:27:23

Conversation program				NC program			
Current folder:/				Conversion destination:/			
No.	Size	Comment	Date	No.	Size	Date	
0001	2.47K	*****	12/05/24 14:04:00	0001	1.27K	12/05/24	
0002	12.1K	RENZOKU UNTEN 14T	11/12/22 11:43:37	0002	144	11/11/22	
0003	1.82K	REAMER	12/01/23 14:34:05	0004	74	11/11/22	
0004	1.31K	*****	12/04/23 22:15:48	0005	4	11/11/22	
0005	1.22K	*****	12/01/23 17:16:04	0010	64	11/11/22	
0010	4.96K	*****	11/11/22 13:20:30	0011	24	11/11/22	
0011	4.83K	*****	11/11/22 13:20:30	0012	14	11/11/22	
0012	1.89K	*****	12/01/30 15:10:25	0013	9	11/11/22	
0013	2.20M	*****	11/11/22 17:03:36	0014	14	11/11/22	
0014	2.25M	*****	12/01/31 17:14:39	0016	21	11/11/22	
0015	1.64K	*****	12/01/31 17:07:56	0017	8	11/11/22	
0016	931	*****	11/11/22 13:20:47	0018	8	11/11/22	
				0019	8	11/11/22	

Program No. [] Occupied 25.1MByte
Available 430MByte

No. 0001 2.47K Byte 12/05/24 14:04:00
Specify a conversation program number and press [F4].
A-B:Change conversation program A to NC program B and convert.
A/B:Convert conversation programs A to B. [HELP]
☒ Running ☐ Selected ☐ Can't convert ☐ Protected I.Inch

Back List Sort Select program Convert Move cursor right/left

The G-code convertible conversational language program is allocated with tools in the current unit system (metric system/inch system) for operation.

When the <Directory display (G code conversion)> screen appears, the content of the current folder on the <Directory display> screen appears both on the conversational program side and the NC program side.

As on the <Directory display> screen, if a folder name is selected by the cursor and the [ENT] key is pressed, the folder name can be moved to the folder.

Switch the cursor by using the after-mentioned <Switch cursor left and right> to move the folder on the NC program side.

(Note) Even if you input a folder name and push the [ENT] key, the folder cannot be moved.

→ To change between [List] and [Detail]

Press the [F1] (Details/list) key.

Each press of the [F1] key alternates the screen between [List] and [Detail].

<List>: Up to 28 conversational programs and 28 NC programs are displayed on one screen.

<Details>: Up to 14 conversational programs and 14 NC programs are displayed on one screen.
Comments are also displayed.

→ Sorting

Conversational programs or NC programs are sorted.

Programs in the current window are sorted.

1. Press the **[F2]** (Sort) key.

2. Select the sort criteria.

[F0] (Date (ascending)) :

Displays programs in the order from the oldest to the newest.

[F1] (Date (descending)) :

Displays programs in the order from the newest.

[F2] (Program No. (ascending)) :

Displays in the order from the smallest to the biggest.

Displays the folders in the alphabetical order of the folder names.

[F3] (Program No. (descending)) :

Displays in the order from the biggest to the smallest.

Displays the folders in the reverse alphabetical order of the folder names.

[F4] (Comment (ascending)) :

Displays in the alphabetical order of the character strings of the comment. The folder names are displayed in alphabetical order for the folders.

[F5] (Comment (descending)) :

Displays in the reverse alphabetical order of the character strings of the comment.

Displays the folders in the reverse alphabetical order of the folder names.

[F7] (Cancel) :

Cancels sorting.

Returns to the previous screen.

3. Programs are sorted and the screen returns to the previous screen.

* Refer to section “1.2.1 Program edit” further details on the sort function.

→ To select a program

Select the program to be converted.

1. Press the **[F3]** (Select program) key.

2. Select the program.

[F1] (Select/Cancel):

Selects the program at the cursor position.

When selecting an already selected program, the program is deselected.

[F2] (Start range selection):

Multiple programs can be selected from the cursor position by moving the cursor.

[F3] (Select all):

Selects all programs.

[F4] (Cancel all):

Deselects all programs.

[F7] (Back):

Returns to the previous screen.

(Note) Neither a folder nor a inconvertible program can be selected.

Directory display(G code conversion details)				2012/05/28 16:30:35		
Conversation program				NC program		
Current folder: /				Conversion destination: /		
No.	Size	Comment	Date	No.	Size	Date
SSSPPP/			12/05/25 11:41:31	VWV/		12/04/26
VWV/			12/05/25 11:41:31	0001	1.27K	12/05/24
0001	2.47K	*****	12/05/24 14:04:00	0002	144	11/11/22
0002	12.1K	RENZOKU UNTEN 14T	11/12/22 11:43:37	0004	74	11/11/22
0003	1.82K	REAMER	12/01/23 14:34:05	0005	4	11/11/22
0004	1.31K	*****	12/04/23 22:15:48	0010	64	11/11/22
0005	1.22K	*****	12/01/23 17:16:04	0011	24	11/11/22
0010	4.96K	*****	11/11/22 13:20:30	0012	14	11/11/22
0011	4.83K	*****	11/11/22 13:20:30	0013	9	11/11/22
0012	1.89K	*****	12/01/30 15:10:25	0014	14	11/11/22
0013	2.26M	*****	11/11/22 17:03:36	0016	21	11/11/22
0014	2.25M	*****	12/01/31 17:14:39	0017	8	11/11/22
0015	1.64K	*****	12/01/31 17:07:56	0018	8	11/11/22
0016	931	*****	11/11/22 13:20:47	0019	8	11/11/22
				Occupied 25.1MByte		
				Available 430MByte		
No. 0001 2.47K Byte 12/05/24 14:04:00				Occupied 17.1MByte		
Move the cursor to the program to be converted, and press[F1].				Available 430MByte		
				[HELP]		
				⏮		
				⏭		
Back	Select/ Deselect	Start selection	Select all	Deselect all		

→ To change cursor position to left/right

Change the position of the cursor between the conversational program window (left side) and the NC program window (right side).

1. Press the **[F7]** (Switch cursor left and right) key.

When the cursor is in the conversational program window, it moves to the NC program window.

When the cursor is in the NC program, a conversational program is called.

2. Select the program to be converted, and then press the **[F4]** (Convert) key.

→ How to select a program

1. Select multiple programs on the screen,
See “To select a program” on the previous page.
2. Enter the program number in the entry field at the lower part of the screen.

Entry example	Conversion operation
1234	Conversational program 1234 → NC program 1234
1234-5678 (Separate with a minus sign: “-”.)	Conversational program 1234 → NC program 5678
1000/1999 (Separate with a slash sign: “/”.)	Conversational programs that can be converted in the range of 1000 and 1999 → NC programs that can be converted in the range of 1000 and 1999

(Note) If a program is selected on the screen and the program number is input, an operator message is displayed.

If the same NC program number exists, the message <The same program number already exists> appears.

1. To convert
Press the **[F0]** (Yes) key.
2. To cancel conversion
Press the **[F1]** (No) key.

(Note) An alarm occurs when the measurement results are used although no measurement command is in the program. The link with the conversational program is disconnected once the program is converted.

1.6.1.1 Conversational Parameters

You can select which workpiece coordinates should be used when performing external conversion.
User parameter - Switch 1 <Workpiece zero conversion code>

You can select whether the G10 should be added to the program when performing external conversion.

User parameter - Switch 1 <Convert tool data>

You can select whether the tapping cycle (synchronous mode) (G77 or G78) should be used for tapping operation or the deep hole tapping cycle (synchronous mode) (G277 or G278) should be used for tapping operation.

User parameter - Switch 1 <Tap step return position>

1.6.2 Conversion Results

Basically, one operation in a conversational operation is converted to one block in an NC program.

1.6.2.1 Tool List Conversion

The tool list used in a conversational program can be reflected to the tool list in an NC program by G10 (programmable data input).

(Note) The tool numbers in the list data in the conversational mode are overwritten in the same tool numbers in the tool list data in the NC mode.

The tool length offset amount setting command is output to the head of a program.

The tool diameter offset amount setting command is output to the position where G41 or G42 code (tool diameter offset command) is commanded.

“0” is commanded for both the tool length wear offset and tool diameter wear offset.

No tool life setting command is output.

(No tool life is reflected, so change it if necessary.)

The extension information (tool wash/CTS) setting command of tool information is output to the head of a program.

If the user parameter <Convert tool data> is set <0: No>, the above tool list data setting command (G10) is output as a comment.

1.6.2.2 Conversion of Coordinate System Settings

Coordinate system setting commands are converted into G54-G59 (Workpiece coordinate system selection) according to the setting of the user parameter <Workpiece zero conversion code>.

The setting for each axis is output by G10 (programmable data input).

(Note) The workpiece coordinate zero point data set by the parameter is changed.

1.6.2.3 Comment Output

Comments set by a conversational program are output to the head of a program.

The number and name are output each time the job or tool is changed.

1.6.2.4 Notes

1. If a measurement number in a coordinate system setting job is set with other than “?” and no measurement motion is set in a motion call job before the coordinate system setting job, an error occurs and conversion becomes unavailable.
2. About a command position of additional axis (A- or B-axis) in the absolute mode:
(There is no problem in the incremental mode.)
Positioning (G0) command:
When strokes are controlled, the additional axis command position is regarded as the same point each time the axis rotates 360 degrees in the conversational mode. However, there is only one point in the NC language mode.
(This does not occur when the strokes are not controlled.)

Example)

If a stroke is between -360.000 and 720.000, 50.000 is at three positions (-310.000, 50.000, 410.000) in the conversational mode but is at only one position (50.000) in the NC language mode.

Cutting movement (G1) command:

Movement direction can be specified in the conversational mode, but it cannot be specified in the NC language mode, so the shortest path is always applied.

When specifying the movement direction, it is necessary to correct the program (e.g. using the incremental mode (G91)).

3. When the Z-axis is lowered in the incremental mode after the Z-axis has returned to the zero point, it lowers from the zero point by the specified value in the conversational mode (accordingly, the tool tip lowers by the tool length) . In the NC language mode, however, the tool length offset is enabled, so the tool tip moves to the specified position. Therefore, set the Z-axis to the position offset by the tool length for the Z-axis to operate the same way as in the conversational mode.

4. Cutter compensation

When cutter compensation is used in jobs that don't include motion calls (for example, contouring jobs), or when the current position is the same as the program command position after cutter compensation, for example:

```
G0 X0.000 Y0.000;
```

```
G41 D1 X0.000 Y0.000;
```

Cutter compensation is used according to the position of the next operation and the XY axes do travel in conversation language mode. However, they do not travel when in NC language mode. (Note that the axes do not travel for jobs that include motion calls, even when using conversation language mode.)

In NC language mode, G40 is used to cancel cutter compensation. However, an error is triggered when the next GM code is executed with an offset amount still remaining. Use G0 or G1 to add an XY command before the command that triggers the error in order to cancel the offset.

GM code that triggers an error

```
G28
```

```
G30
```

```
G121~G129
```

```
M410~M411
```

```
G133 • G134
```

When the tool change position is set for the XY axes while in offset mode, an error is triggered and the tool change is not possible.

Cancel the offset mode before the operation for the tool change positioning.

When using the conversation language, if there are 14 or more commands issued that do not include XY travel while in offset mode, the offset is applied at the travel end point in the vertical direction relative to the travel direction. However, in NC language mode, the offset is applied in the vertical direction when 3 or more motion commands (3 blocks) are issued.

5. Travel to tool change position during rotational transformation

When the tool change position is set for the XY axes while in rotational transformation mode, an error is triggered and the tool change is not possible. Therefore, cancel the rotational transformation before traveling to the tool change position.

6. Automatic workpiece measurement

An error is triggered when the automatic workpiece measurement is performed during rotational transformation in NC language mode.

Therefore, cancel the rotational transformation before performing the automatic workpiece measurement.

7. Spindle orientation to specified position

An error occurs when M19R* (* indicates an orientation angle) is used in arc modal state or during a canned cycle in the NC language mode. Cancel the arc modal state or the canned cycle.

8. To move 2nd/3rd zero return commands

When G-code conversion is performed, the 2nd and 3rd zero point return commands are converted to 2nd and 3rd reference point return commands. Set the same coordinate values as the 2nd and 3rd zero point return positions for the 2nd and 3rd reference point return position parameters after the mode has been changed to the NC language mode.

An error occurs when the 2nd or 3rd reference point return command is used during rotational transformation in the NC language mode. Cancel the rotational transformation before performing the 2nd or 3rd reference point return.

9. Amount of pallet shift

In NC language mode, there is no function to set the amount of shift for pallet 2. Therefore, set the workpiece coordinates for pallet 1 and 2 separately.

10. Z-axis stroke check

The Z-axis stroke is checked at the spindle end in the conversational mode but it is checked at the tool tip in the NC language mode.

1

1.6.2.5 Conversion Example

Conversational program example
“Conversational program”

Machining data		2012/05/28 16:36:54	
Program(0001)			
Workpiece zero	X	-200.000	Y -200.000 Z 200.000
	A	?	B ?
Return height(job change)	Height	80.000	
Workpieces		1	
Workpiece material		ADC12	
Job01	Signal output	Code	400
Job02	Tap	Torsion	Right thread
		Thread	M5.000x0.800
		Drilling depth	0.000 Chamfer 0.100
		Pattern	Circle
		Divisions	3.000 Holes 3
		Start angle	0.000 Radius 30.000
		Z-axis position	Bottom of hole Through Depth 8.000
		Height	50.000 Return height 52.000
		XY-axes position	X 50.000 Y 50.000
		Peripheral speed	Peripheral speed 100% Feed 100%
X			
Search:Job/line [HELP]			
Completion mode	Previous job	Next job	Search Add Set workpiece zero Copy job Paste job

Machining data		2012/05/28 16:37:45	
Program(0001) job 03			
Workpieces		1	
Workpiece material		ADC12	
Job01	Signal output	Code	400
Job02	Tap	Torsion	Right thread
		Thread	M5.000x0.800
		Drilling depth	0.000 Chamfer 0.100
		Pattern	Circle
		Divisions	3.000 Holes 3
		Start angle	0.000 Radius 30.000
		Z-axis position	Bottom of hole Through Depth 8.000
		Height	50.000 Return height 52.000
		XY-axes position	X 50.000 Y 50.000
		Peripheral speed	Peripheral speed 100% Feed 100%
Job03	Signal output	Code	401
Main program end			
Sub program end			
Job:			
0 Job end	6 Counter boring tap	12 Sub program	18 Facing
1 Center hole	7 Counter boring reamer	13 Program stop	19 Pocket milling
2 Hole	8 Milling	14 Coordinate system setting	20 Contouring
3 Tap	9 Coolant	15 Rolled tap	21 Chamfering
4 Ream	10 Axis travel	16 Counter boring rolled tap	
5 Counter boring hole	11 Signal output	17 Motion call	
[HELP]			
Completion mode	Previous job	Next job	Search Add Set workpiece zero Copy job Paste job

“Machining order”

Machining sequence		2012/05/28 16:40:05
Program(0001) ****		
1.Each job (1 pc.)	Workpieces are machined one at a time according to the preset job sequence. Machining proceeds according to the tool sequence in the job. When there are multiple workpieces, machining one workpiece is completed, and then the next workpiece is machined.	
2.Each tool (1 pc.)	Workpieces are basically machined one at a time. If common tools are used, priority is given to machining using these tools (previous machining must be completed). When there are multiple workpieces, machining one workpiece is completed, and then the next workpiece is machined.	
3.Each job	Jobs common to all workpieces on the table are executed first. Machining proceeds according to the tool sequence in the same job.	
4.Each tool (tool prioritized)	When common tools are used for other jobs that apply to all workpieces on the table (previous machining must be completed), priority is given to machining using these tools.	
5.Each tool (coord. sys. prioritized)	When there are coordinate system setting jobs in the program, all workpieces are machined according to the tool (priority given to tools) in the current coordinate system, and machining proceeds to the next coordinate system.	
Machining sequence		
Completion mode		

“Tool pattern”

Tool pattern		2012/05/28 16:41:28
Program(0001)		
	w/ chamfering	w/o chamfering
Center hole	1 Center drill	1 Center drill
Hole	2 Drill	2 Drill
	3 Chamfer tool	
Tap	1 Center drill	1 Center drill
	2 Drill	2 Drill
	3 Chamfer tool	3 Tapping
	4 Tapping	4 *
	5 *	
Rolled tap	1 Center drill	1 Center drill
	2 Drill	2 Drill
	3 Chamfer tool	3 Rolled tap
	4 Rolled tap	4 *
	5 *	
Tool		
1. Center drill	6. Chamfer tool	11. Reamer
2. Drill	7. CBR tool	12. Drill reamer
3. Carbide drill	8. Chamfer drill	13. End mill(R)
4. Tapping	9. Stepped drill	14. End mill(F)
5. Rolled tap	10. Drill tap	15. CB E-mill(R)
16. CB E-mill(F)		17. Chamf. cutter
Completion mode	Previous pattern	Next pattern

“Cutting conditions”

Cutting condition 1(boring tool)		2012/05/28 16:42:06
Program(0001) ****		
Material ADC12		m/min mm/rev.
Center drill	Center drill	Lengthwise 30.0 0.10
		Crosswise
Drill	Drill	- 2.999 50.0 0.10
		3.000 - 50.0 0.20
Carbide drill	Carbide drill	- 2.999 100.0 0.10
		3.000 - 100.0 0.20
Tap (Blind)	Spiral tap	- 2.999 30.0 -
		3.000 - 40.0 -
Tap (Through)	Point tap	- 2.999 30.0 -
		3.000 - 40.0 -
Rolled tap	Rolled tap	- 2.999 40.0 -
		3.000 - 50.0 -
Chamfering tool	Chamfer tool	- 35.0 0.30
Counter boring tool	CBR tool	- 35.0 0.15
Tool		
1. Center drill	6. Hand tap	11. Stepped drill
2. Drill	7. Rolled tap	12. Drill tap
3. Carbide drill	8. Chamfer tool	13. Reamer
4. Spiral tap	9. CBR tool	14. Drill reamer
5. Point tap	10. Chamfer drill	
Completion mode	Cutting condition 1 (boring tool)	Cutting condition 2 (Milling tool)
	Cutting condition 3 (two tools)	Previous tool
		Next tool

“Tool assignment”

Tool assignment				2012/05/28 16:55:08
Program(0001)				**** ????
Job02	1. Center hole	Tool No.	01	Center drill
Tap		Tool length offset		Cutter compensation **
Chamfering		Misc.func.before cutting		Misc.func.after cutting
		Dwell time		
	2. Hole	Tool No.	02	Drill 4.300
		Tool length offset		Cutter compensation **
		Misc.func.before cutting		Misc.func.after cutting
		Dwell time		
	3. Chamfering	Tool No.	04	Chamfer tool 10.000
		Tool length offset		Cutter compensation **
		Misc.func.before cutting		Misc.func.after cutting
		Dwell time		
	4. Tapping	Tool No.	03	Point tap M5.000
		Tool length offset		Cutter compensation **
		Misc.func.before cutting		Misc.func.after cutting
		Dwell time		
Tool No.				
-9999 Assign tools:Assign new -9998 Assign tools:Assign current tool first Assign tools: Only assign tools not yet assigned.				Tool assignment is completed. ? calculation was performed. [HELP]
Completion mode	Previous tool	Next tool	Change function	Calculation result
				Search
				Tool check
				Assign tools

“Tool list”

Tool list						2012/05/28 16:56:48
Tool No.	01(P:01)	02(P:02)	03(P:03)	04(P:04)	05	
Tool	Center drill	Drill	Point tap	Chamfer tool	Carbide drill	
Point angle	0.000	118.000	*	*	180.000	
Minor diameter	0.000	*	*	0.000	*	
Minor dia. length	0.000	*	*	*	*	
Center angle	90.000	*	*	90.000	*	
Invalid section	*	*	1.000	1.000	*	
Major/Nominal D	5.000	4.300	M5.000	10.000	9.999	
Pitch/Threads	*	*	0.800	*	*	
Length of teeth	*	30.000	30.000	*	99.999	
Torsion	*	*	Right thread	*	*	
Number of teeth	*	*	*	*	*	
Tool length	120.000	130.000	140.000	150.000	100.000	
Initial life	5000 (H)	10000 (H)	5000 (H)	**	99999pcs.	
Life warning	100 (H)	100 (H)	100 (H)	**	999pcs.	
Tool life	5000 (H)	10000 (H)	5000 (H)	**	99999pcs.	
Tool						
1.Center drill		4.Spiral tap		7.Rolled tap		10.Chamfer drill
2.Drill		5.Point tap		8.Chamfer tool		11.Stepped drill
3.Carbide drill		6.Hand tap		9.CBR tool		12.Drill tap
						[HELP]
Completion mode	Search	Tool length setting	Add	Copy	Paste	Display extension information
						Next menu

“User parameter settings”

Switch 1			2012/05/28 16:58:01
0113 Check tools to be used when assigned automatically [0:No 1:Yes]			1
0114 Workpc. zero conversion code [0:G54 1:G55 2:G56 3:G57 4:G58 5:G59]			0
0115 Convert tool data [0:No 1:Yes]			1
0116 Tap step return position [0:Return pos. 1:Start pos.]			0
0117 Magazine turn speed ratio 1(M436)	80	%	
0118 Magazine turn speed ratio 2(M437)	50	%	
0119 Turret type magazine speed [0:Normal 1:High]			
0120 Turret type magazine low-speed turn mode [0:Invalid 1:Valid]			
0121 Speed turret type magazine turns at low speed	20.0	min ⁻¹	
0122 Pallet select switch change [0:Invalid 1:Valid]			
0123 Lock shift key [0:No 1:Yes]			
0124 Max. actual cutting travel speed	20000	mm/min	
0125 Table load weight	250	kg	
0126 Arc radius error limit	0.000	mm	
0136 Back washing at end of program [0:No 1:Yes]			
0137 CTS pressurize check delay time	1.0	s	
0138 CTS depressurize check delay time	1.0	s	
Magazine turn speed ratio 1(M436) %			
			[HELP]
Completion mode	Switch 1	Switch 2	Switch 3
			Ref. position/Conversion zero point
			User registered misc. function
			Tool breakage detection
			Automatic workpiece meas.

NC program output results “NC program”

1

```

Edit 2012/05/28 17:00:43
Program(0001)
000001 (*****);
000002 ( -- TOOL LENG OFFSET DATA -- );
000003 G90 G10 L10 P1 R120.;
000004 G90 G10 L11 P1 R0.;
000005 G90 G10 L13 P1 R0.;
000006 G90 G10 L10 P2 R130.;
000007 G90 G10 L11 P2 R0.;
000008 G90 G10 L13 P2 R0.;
000009 G90 G10 L10 P4 R150.;
000010 G90 G10 L11 P4 R0.;
000011 G90 G10 L13 P4 R0.;
000012 G90 G10 L10 P3 R140.;
000013 G90 G10 L11 P3 R0.;
000014 G90 G10 L13 P3 R0.;
000015 ( ----- NC DATA ----- );
000016 G00 G40 G49 G64 G80 G69 G90 G99 G54;
000017 (#JOB01 SIGNL.);
000018 G90 G10 L2 P1 X-200. Y-200. Z200.;
000019 G91 G28 Z0;
000020 G90 M400;
000021 (#JOB02 TAP T1 C.DRILL D0.*90.);
000022 G100 T01 M03 S2387 X80. Y50. G43 H1 (120.) Z52.;
000023 G09 G01 Z48. F238.;
000024 G00 Z52.;

Total blocks 000063
[HELP]
Completion Search Replace Jump Copy Paste Cut Start auto
mode sequence
insertion

```

```

Edit 2012/05/28 17:01:16
Program(0001)
000024 G00 Z52.;
000025 G00 X35. Y75.981;
000026 G09 G01 Z48.;
000027 G00 Z52.;
000028 G00 X35. Y24.019;
000029 G09 G01 Z48.;
000030 G00 Z80.;
000031 (#JOB02 TAP T2 DRILL D4.3);
000032 G100 T02 M03 S3701 X80. Y50. G43 H2 (130.) Z52.;
000033 G83 Z38.708 R52. W51.6 V3.87 F740.;
000034 G00 X35. Y75.981;
000035 G83 Z38.708 R52. W51.6 V3.87 F740.;
000036 G00 X35. Y24.019;
000037 G83 Z38.708 R52. W51.6 V3.87 F740.;
000038 G00 Z80.;
000039 (#JOB02 TAP T4 CHFTOOL 90.*D10.);
000040 G100 T04 M03 S2475 X80. Y50. G43 H4 (150.) Z49.85;
000041 G09 G01 Z47.75 F742.;
000042 G00 Z52.;
000043 G00 X35. Y75.981;
000044 G00 Z49.85;
000045 G09 G01 Z47.75;
000046 G00 Z52.;
000047 G00 X35. Y24.019;

Total blocks 000063
[HELP]
Completion Search Replace Jump Copy Paste Cut Start auto
mode sequence
insertion

```

```

Edit 2012/05/28 17:01:34
Program(0001)
000047 G00 X35. Y24.019;
000048 G00 Z49.85;
000049 G09 G01 Z47.75;
000050 G00 Z80.;
000051 (#JOB02 TAP T3 P.TAP M05.*0.8);
000052 G100 T03 X80. Y50. G43 H3 (140.) Z52.;
000053 G77 Z39.2 R52. I0.8 W30. V1.5 S2546;
000054 G00 X35. Y75.981;
000055 G77 Z39.2 R52. I0.8 W30. V1.5 S2546;
000056 G00 X35. Y24.019;
000057 G77 Z39.2 R52. I0.8 W30. V1.5 S2546;
000058 G00 Z80.;
000059 (#JOB03 SIGNL.);
000060 M401;
000061 M05;
000062 M30;
000063

Total blocks 000063
[HELP]
Completion Search Replace Jump Copy Paste Cut Start auto
mode sequence
insertion

```


CHAPTER 2

DATA BANK

- 2.1 Overview of Data Bank**
- 2.2 Workpiece Coordinate Zero Point (NC)**
- 2.3 Tool Data (NC)**
- 2.4 Macro Variables (NC)**
- 2.5 User Parameter**
- 2.6 External I/O Signal**
- 2.7 Communication Parameter**
- 2.8 G/M Code Macro Settings**
- 2.9 Tool List (Conversation)**
- 2.10 Tool Pattern (Conversation)**
- 2.11 Tapping Drill Diameter (Conversation)**
- 2.12 Cutting Condition (Conversation)**

2.1 Overview of Data Bank

A data bank is a gathering of data necessary for machine running and operation. Kinds of the data bank used for the NC language mode differ from those for the conversational mode.

2.1.1 Outline

2.1.1.1 Data Bank for NC Language Mode

The data necessary for running the NC language mode are as follows:

1. Workpiece coordinate zero position
To set a zero point of a workpiece coordinate system.
2. Tool data
To register information like offset amount, life, etc. of a tool.
3. Macro variable
To set a macro variable
4. User parameter
To set a parameter for machine operation. This is an important datum for machine operation, so please change it prudently. Keep the parameter list attached on a machine at a safe place.
5. External I/O signal
To assign an external input/output signal for an input/output terminal and set input/output of an external alarm.
6. Communication parameter
To set a parameter for external input/output communication. Set this according to connection equipment.
7. Machine parameter
To set a parameter for machine operation. Unless this is set correct, the machine may break. So do not change the value set before machine delivery.
8. G/M code macro
Set the parameters related to G code and M code macros.
9. Special settings
The parameters can be set to support special cases. Normally no changes are required. Be extremely carefully when changing the settings because machine operation will change.
10. Machining load monitor
This sets the parameter for the function to measure the machining load, monitor it and assess whether the machining is normal. Refer to “11.13 Machining load monitor function” for further details. Refer to “Chapter 11.13 Machining load monitor function” in the Operation Manual for further details.

2.1.1.2 Data Bank for Conversational Language Mode

Data necessary for operation of the conversational mode are as follows:

1. Tool list
This registers information about kinds, sizes, shapes, lives, etc. of tools.
When memory operation is carried out, tools necessary for machining are automatically selected from the tool list.
2. Tool pattern
To set kinds and order of tools used for each machining process
Example) For tapping, necessary tools and order are automatically selected by specifying a tapping process.

Tapping	1	Center drill
	2	Drill
	3	Tap

3. Setting primary hole drilling diameter for tapping
To set a pilot hole size per kind of thread for tapping
(Example) When you want to tap, a necessary pilot hole size is sought and a corresponding drilling tool is automatically selected only by specifying the kind and size of a thread.

Metric unified thread M3.0×0.5	→	Setting primary hole drilling diameter for tapping 2.500	→	Tool List 2.500-diameter drill
-----------------------------------	---	---	---	-----------------------------------

4. Cutting conditions
To set cutting conditions other than peripheral speed, feed, and tool conditions per material or tool used.
(Example) When you want to drill, the peripheral speed and feed corresponding to the material and kind of drill are automatically selected.

S45C φ2.5 drill	→	Cutting condition 1 (drill) Peripheral speed 10 m/min Feed rate 0.05 mm/rev
--------------------	---	---

5. User parameters
Refer to “2.1.1.1. Data bank for the NC language mode”.
 6. External I/O signals
Refer to “2.1.1.1. Data bank for the NC language mode”.
 7. Communication parameters
Refer to “2.1.1.1. Data bank for the NC language mode”.
 8. Machine parameters
Refer to “2.1.1.1. Data bank for the NC language mode”.
 9. Special settings
Refer to “2.1.1.1. Data bank for NC language mode”.
 10. Machining load monitor
Refer to “2.1.1.1 Data bank for NC language mode”.
- * The data is the same and used in both NC language mode and conversation mode for user parameters, external I/O signals, communication parameters, machine parameters, special settings and the machining load monitor.
 - * The tool patterns and cutting conditions are possible to set by the program edit mode per machining datum. Refer to “Programming Manual, Conversational”.

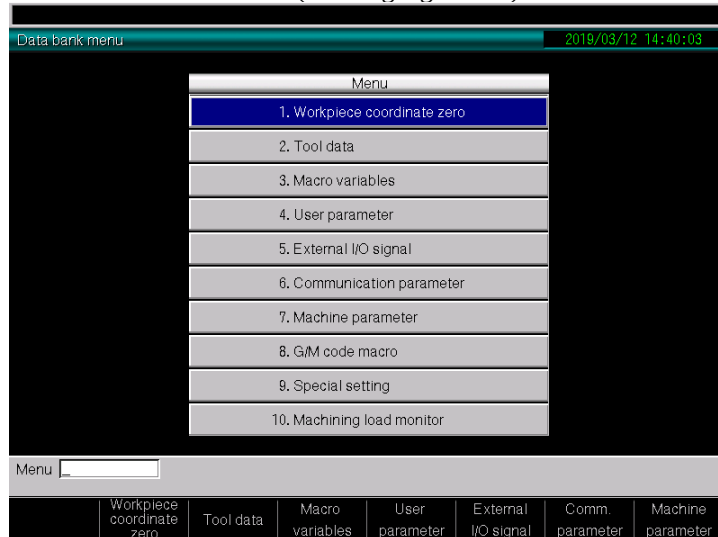
2.1.2 How to Edit

2.1.2.1 To Start Editing

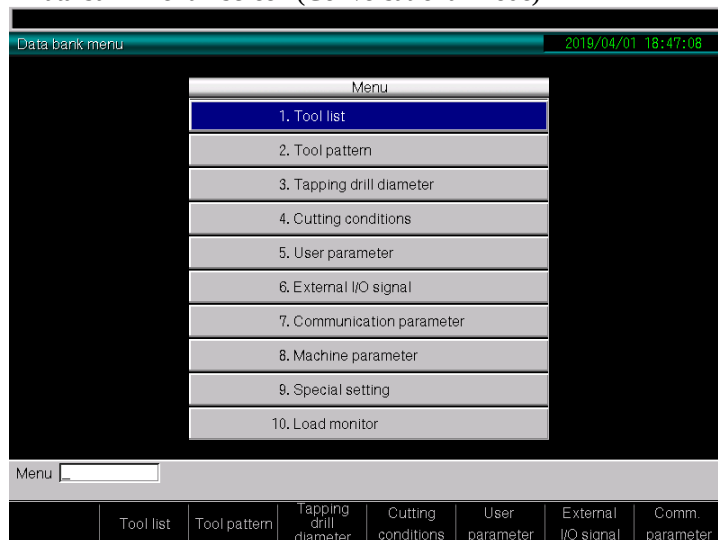
We are going to explain how to start editing the data bank.

1. Press the **[DATA BANK]** key to call the <Data bank menu> screen.

<Data bank menu> screen (NC language mode)



<Data bank menu> screen (Conversational mode)



2. Press the menu number and the **[ENT]** key or point the cursor and press the **[ENT]** key. Alternatively, press a function key. The selected data bank edit screen appears.

2.1.2.2 To Finish Editing

We are going to explain how to finish editing of the data bank.

1. On each data edit screen, press the **[F0]** (Completion mode) key.
The Edit completion mode screen appears.

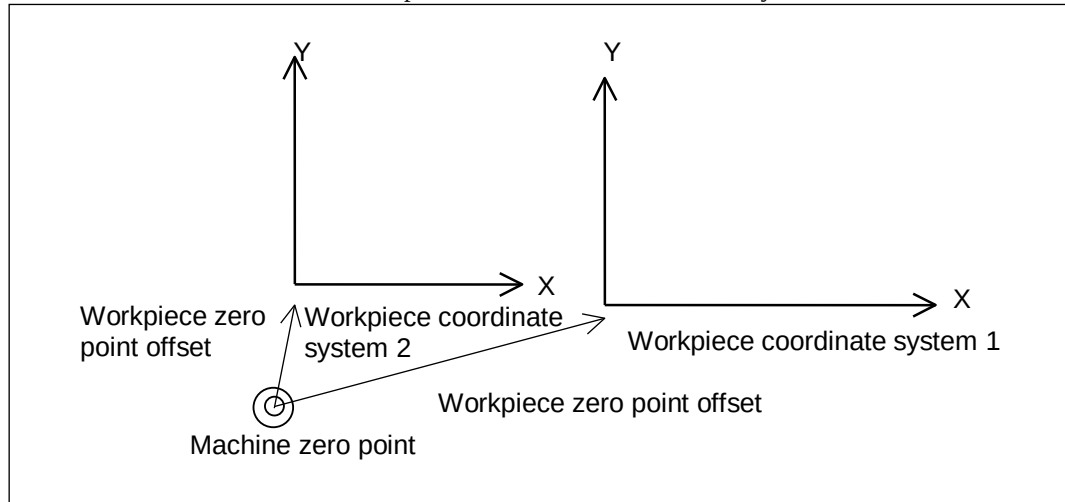
Switch 2		2012/12/07 14:19:12	
		^	
0501	4th-axis stroke control [0:No 1:Yes]	1	
0502	5th-axis stroke control [0:No 1:Yes]	1	
0503	6th-axis stroke control [0:No 1:Yes]	0	
0504	7th-axis stroke control [0:No 1:Yes]	0	
0505	8th-axis stroke control [0:No 1:Yes]	0	
0506	4th-axis stroke(-)	-180.000	deg.
0507	5th-axis stroke(-)	-360.000	deg.
0508	6th-axis stroke(-)	-360.000	deg.
0509	7th-axis stroke(-)	-360.000	deg.
0510	8th-axis stroke(-)	-360.000	deg.
0511	4th-axis stroke(+)	180.000	deg.
0512	5th-axis stroke(+)	360.000	deg.
0513	6th-axis stroke(+)	360.000	deg.
0514	7th-axis stroke(+)	360.000	deg.
0515	8th-axis stroke(+)	360.000	deg.
0516	Stroke limit 1 for X-axis(-)	-500.000	mm
0517	Stroke limit 1 for Y-axis(-)	-400.000	mm
		v	
Cancel w/ saving changes		Cancel w/o saving changes	
		Overwrite save	
		Edit mode	

2. Press the function key to save or cancel the data.
 - [F0]** (Cancel w/ saving changes) key Saves the edited content, finishes editing of the data bank, and returns to the menu screen.
 - [F3]** (Cancel w/o saving changes) key Invalidates the edited content and finishes editing of the data bank.
However, the overwritten content is not invalidated.
 - [F6]** Overwrite save) key Saves the edited content to continue editing of data bank and returns to the edit screen.
 - [F7]** (Edit mode) key Does not save the edited content but continues editing the data bank and returns to the edit screen.

2.2 Workpiece Coordinate Zero Point (NC)

2.2.1 Outline

Workpiece coordinate system zero points of six G codes (G54-G59), one external, and 48 extended workpiece coordinates 1-48 are set. The workpiece coordinate system zero points are set with the offset amount from the zero point of the machine coordinate system.



In addition, set the G54.2 reference rotary fixture offset.

2.2.1.1 Edit Screen

When the <Workpiece coordinate zero> is selected from the <Data bank menu> screen, the workpiece coordinate zero point edit screen appears.

Workpiece coordinate zero				2014/09/01 15:51:20
G54		G55	G56	G57
X 0.000	X 0.000	X 0.000	X 0.000	X 0.000
Y 0.000	Y 0.000	Y 0.000	Y 0.000	Y 0.000
Z 0.000	Z 0.000	Z 0.000	Z 0.000	Z 0.000
A 0.000	A 0.000	A 0.000	A 0.000	A 0.000
B 0.000	B 0.000	B 0.000	B 0.000	B 0.000
C 0.000	C 0.000	C 0.000	C 0.000	C 0.000
G58	G59	External		
X 0.000	X 0.000	X 0.000		
Y 0.000	Y 0.000	Y 0.000		
Z 0.000	Z 0.000	Z 0.000		
A 0.000	A 0.000	A 0.000		
B 0.000	B 0.000	B 0.000		
C 0.000	C 0.000	C 0.000		
G54 X mm				
[HELP]				
Completion mode	Extended coordinates	Automatic setting	Add	Search
		Parameter is taken in.	Rotary fixture offset	

If the **[F1]** (Extended coordinates) key is pressed, you can set the extended workpiece coordinates.

Workpiece coordinate zero(extended)		2014/09/01 15:51:45	
1	2	3	4
X 0.000	X 0.000	X 0.000	X 0.000
Y 0.000	Y 0.000	Y 0.000	Y 0.000
Z 0.000	Z 0.000	Z 0.000	Z 0.000
A 0.000	A 0.000	A 0.000	A 0.000
B 0.000	B 0.000	B 0.000	B 0.000
C 0.000	C 0.000	C 0.000	C 0.000
5	6	7	8
X 0.000	X 0.000	X 0.000	X 0.000
Y 0.000	Y 0.000	Y 0.000	Y 0.000
Z 0.000	Z 0.000	Z 0.000	Z 0.000
A 0.000	A 0.000	A 0.000	A 0.000
B 0.000	B 0.000	B 0.000	B 0.000
C 0.000	C 0.000	C 0.000	C 0.000
1 X mm			
[HELP]			
Completion mode	Standard coordinates	Automatic setting	Add
			Search
			Parameter is taken in.
			Rotary fixture offset

2

Press the **[F6]** (Rotary fixture offset) key to perform the reference rotary fixture offset.

Reference rotary fixture offset		2014/09/01 15:52:41	
	1	2	3
Reference angle A	0.000	0.000	0.000
Reference angle B	0.000	0.000	0.000
Reference angle C	0.000	0.000	0.000
Reference offset X	0.000	0.000	0.000
Reference offset Y	0.000	0.000	0.000
Reference offset Z	0.000	0.000	0.000
Axis for calculation	A	A	A
	5	6	7
Reference angle A	0.000	0.000	0.000
Reference angle B	0.000	0.000	0.000
Reference angle C	0.000	0.000	0.000
Reference offset X	0.000	0.000	0.000
Reference offset Y	0.000	0.000	0.000
Reference offset Z	0.000	0.000	0.000
Axis for calculation	A	A	A
1 Reference angle A deg.			
[HELP]			
Completion mode	Extended coordinates	Automatic setting	Add
			Search
			Parameter is taken in.
			Rotary fixture offset

Refer to “Chapter 12 (14) Rotary fixture offset function” in the Operation Manual for further details on each setting item.

2.2.2 How to Edit

2.2.2.1 To Move the Cursor

Move the cursor to where you want to change by the cursor keys.

If the extended workpiece coordinates are appearing, you can quickly move the cursor to the extended workpiece coordinates you want to change by searching. Input an extended workpiece coordinates number into the input area and press the **[F4]** (Search) key. The cursor moves to the extended workpiece coordinates you specified.

2.2.2.2 Direct Input of Data

1. Move the cursor to the item you want to change.
2. Input a value in the input area and press the **[ENT]** key.
The value is input at the cursor position and the cursor moves to the next item.

2.2.2.3 Automatic Setting of Data

Automatic setting for workpiece coordinate zero

A workpiece coordinate zero point offset amount which makes the current machine coordinate position to be a new absolute coordinate position is automatically calculated.

1. Move the cursor to the axis you want to change.
2. Press the **[F2]** (Automatic setting) key.
The <Automatic setting> screen appears according to the axis at the cursor position.
 - * If the **[F2]** is pressed when the cursor is at the axis which is not set with an option, the alarm <<No *-axis option>> appears and the <Automatic setting> screen is not displayed.
 - * When the <<Machine param. setting error (SYS 2)>> is appearing, the **[F2]** (Automatic setting) key is grayed out to disable operation.
- When the cursor is at the X axis (The Y and Z axes are the same.)

Automatic setting 2012/05/28 18:48:55

Machine coordinate pos.
X 0.000

Absolute coord. pos. after setting
X 10.000

Workpiece coord. zero after setting
X -10.000

G54 X mm

Set Not set

Input a value in the input area and press the **[ENT]** key. A new workpiece coordinate zero point offset amount (Workpiece coordinate zero point after setting) is automatically calculated and displayed so that the input value becomes a new absolute coordinate position (Absolute coordinate position after setting) for the current machine coordinate position.

If the **[DEL]** key is pressed, the value in the <Absolute coord. pos. after setting> is deleted to become a value currently set in the <Workpiece coord. zero after setting>

If the **[F0]** (Set) key is pressed, the value in the <Workpiece coord. zero after setting> is set in the X-axis to return to the previous screen.

If the **[F0]** (Set) key is pressed under the condition of the value in the <Absolute coord. pos. after setting> deleted, no value is set but the previous screen is called back.

If the **[F1]** (Not set) key is pressed, no value is set but the previous screen is called back.

If the **[F0]** (Set) key is pressed and the previous screen is called back, the cursor moves to the next item.

- If the cursor is at the A-axis (when one axis is used) (same as at the B or C axis),

Automatic setting2012/05/28 18:49:25

Machine coordinate pos.

A0.000

Absolute coord. pos. after setting

A10.000

Workpiece coord. zero after setting

A-10.000

G54 Adeg.

Set

Not set

Input a value in the input area and press the **[ENT]** key. A new workpiece coordinate zero point offset amount (Workpiece coordinate zero point after setting) is automatically calculated and displayed so that the input value becomes the value for the new absolute coordinate position (Absolute coordinate position after setting) for the current machine coordinate position.

If the **[DEL]** key is pressed, the value in the <Absolute coord. pos. after setting> is deleted and the value in the <Workpiece coord. zero after setting> become the currently set value.

If the **[F0]** (Set) key is pressed, the value in the <Workpiece coord. zero after setting> is set to the A-axis and the previous screen is called back.

If the **[F0]** (Set) key is pressed under the condition of the value in the <Absolute coord. pos. after setting> deleted, no value is set but the previous screen is called back.

If the **[F1]** (Not set) key is pressed, no value is set but the previous screen is called back.

If the **[F0]** (Set) key is pressed and the previous screen is called back, the cursor moves to the next item.

- If the cursor is at the A-axis (when two axes A1 and A2 are used) (Same as at the B-axis),

Automatic setting2012/05/28 18:53:02

Machine coordinate pos.

A10.000

A20.000

Absolute coord. pos. after setting

A10.000

Workpiece coord. zero after setting

A1-10.000

A2-10.000

G54 Adeg.

Set A1 axis

Set A2 axis

Not set

Input a value in the input area and press the **[ENT]** key. A new workpiece coordinate zero point offset amount (Workpiece coordinate zero point after setting) (A1/A2) is automatically calculated and displayed so that the input value becomes a value for the new absolute coordinate position (Absolute coordinate position after setting) for the current machine coordinate position (A1/A2).

If the **[DEL]** key is pressed, the value in the <Absolute coord. pos. after setting> is deleted and the value in the <Workpiece coord. zero after setting> becomes the currently set value.

If the **[F0]** (SetA1 axis) key is pressed, the value in the <Workpiece coord. zero after setting (A1)> is set in the A-axis and the previous screen is called back.

If the **[F1]** (Set A2 axis) key is pressed, the value in the <Workpiece coord. zero after setting (A2)> is set in the A-axis and the previous screen is called back. Press the **[F0]** (SetA1 axis) key or the **[F1]** (SetA2 axis) key, when the <Absolute coord. pos. after setting> value is deleted, to return to the previous screen without setting the value.

If the **[F2]** (Not set) key is pressed, no value is set but the previous screen is called back.

After pressing the **[F0]** (SetA1 axis) key, or the **[F1]** (SetA2 axis) key, and returning to the previous screen, the cursor will move to the next item.

Automatic setting for reference angle in reference rotary fixture offset

The current absolute coordinate position is integrated in the <Reference angle>.

1. Move the cursor to reference angle A, B or C.
2. Press the **[F2]** (Automatic setting) key.

The automatic setting screen, which corresponds to the axis where the cursor is, is displayed.

- * When the cursor is on an axis where the option is not set and the **[F2]** key is pressed, the alarm <<No *-axis option >> is triggered and the automatic setting screen does not display.
- When the cursor is on <Reference angle A> (when using 1 axis) (same applies for B and C)

Automatic setting		2014/09/01 15:54:03	
Machine coordinate pos.			
A	0.000		
Absolute coord.		Reference angle after setting	
A	0.000	A	0.000
Set	Not set		

Press **[F0]** (Set) to set the <Reference angle after setting> as the <Reference angle>, and then it will return to the previous screen.

Press the **[F1]** (Not set) key to cancel the setting operation and return to the previous screen.

- When the cursor is on <Reference angle A> (when using 2 axes for A1 and A2) (same applies for B- and C-axes)

Automatic setting		2014/09/01 16:45:44
Machine coordinate pos.		
A1	0.000	
A2	0.000	
Absolute coord.		
A1	0.000	
A2	0.000	
Reference angle after setting		
A1	0.000	
A2	0.000	
Set A1 axis	Set A2 axis	Not set

Press the **[F0]** (Set A1 axis) key to set <Reference angle A1 after setting> as <Reference angle A>, and then it will return to the previous screen.

Press the **[F1]** (Set A2 axis) key to set <Reference angle A2 after setting> as <Reference angle A>, and then it will return to the previous screen.

Press the **[F2]** (Not set) key to cancel the setting operation and return to the previous screen.

Automatic setting for reference offset in reference rotary fixture offset

The <Reference offset> is automatically calculated from the current machine coordinates, or from the input machine coordinates and the workpiece coordinate zero selection (G54 to G59 and G54.1) for the current selection.

(Note) This function can only be used when the rotation center coordinate for the index table is set to one of the workpiece coordinate zero point options. This function cannot be used when the rotation center coordinate is set in G92, G52 and the external workpiece coordinate system shift.

Set the reference offset from the current machine coordinates

- In memory or MDI operation, select the workpiece coordinate zero that is set as the rotation center coordinate on the index table.
- Move the cursor to reference offset X, Y or Z.
- Press the **[F2]** (Automatic setting) key.
- Press the **[F6]** (Current position setting) key.

The automatic setting screen, which corresponds to the axis where the cursor is, is displayed. The current machine coordinate position, the workpiece zero offset that is currently selected as the rotation center coordinate and the “machine coordinate position – rotation center coordinate” for the <Reference offset after setting> are automatically calculated and displayed.

- When the cursor is on <Reference offset X> (same applies for Y and Z)

Enter the value for the reference offset adjustment value and press the **[ENT]** key. Then, the value entered for “machine coordinate position – rotation center coordinate” are automatically calculated and displayed in the <Reference offset after setting>.

When the **[DEL]** key is pressed, the <Reference offset adjustment value> becomes 0. The value for “machine coordinate position – rotation center coordinate” is displayed in the <Reference offset after setting>.

Press **[F0]** (Set) to set the value in the <Reference offset after setting> as the <Reference offset>, and then it will return to the previous screen.

Press the **[F1]** (Not set) key to cancel the setting operation and return to the previous screen.

Set the reference offset from the machine coordinates entered

- In memory or MDI operation, select the workpiece coordinate zero that is set as the rotation center coordinate on the index table.
 - Move the cursor to reference offset X, Y or Z.
 - Press the **[F2]** (Automatic setting) key.
 - Press the **[F7]** (Machine coordinate setting) key.
- The automatic setting screen, which corresponds to the axis where the cursor is, is displayed.

- When the cursor is on <Reference offset X> (same applies for Y and Z)

Enter the machine coordinate you wish to set for the reference offset used in the setting machine coordinate position and press the **[ENT]** key. Then, the “setting machine coordinate position – rotation center coordinate” is automatically calculated and displayed in the <Reference offset after setting>.

Press **[F0]** (Set) to set the value in the <Reference offset after setting> as the <Reference offset>, and then it will return to the previous screen.

Press the **[F1]** (Not set) key to cancel the setting operation and return to the previous screen.

2.2.2.4 To Add Data

1. Move the cursor to the axis you want to change.
2. Input a value to be added in the input area and press the **[F3]** (Add) key.
The value input to the value set at the cursor position is added and the cursor moves the next item.

2.2.2.5 Data Set by Reading a Parameter

1. Move the cursor to the X-/Y-/Z-axis you want to change.
2. Press the **[F5]** (Parameter is taken in.) key.

Workpiece coordinate zero						2012/05/28 18:56:11	
G54		G55		G56		G57	
X	-9999.999	X	-9999.999	X	-9999.999	X	-9999.999
Y	-9999.999	Y	-9999.999	Y	-9999.999	Y	-9999.999
Z	-9999.999	Z	-9999.999	Z	-9999.999	Z	-9999.999
A	-9999.999	A	-9999.999	A	-9999.999	A	-9999.999
B	-9999.999	B	-9999.999	B	-9999.999	B	-9999.999
C	-9999.999	C	-9999.999	C	-9999.999	C	-9999.999
G58		G59		External			
X	-9999.999	X	-9999.999	X	-9999.999		
Y	-9999.999	Y	-9999.999	Y	-9999.999		
Z	-9999.999	Z	-9999.999	Z	-9999.999		
A	-9999.999	A	-9999.999	A	-9999.999		
B	-9999.999	B	-9999.999	B	-9999.999		
C	-9999.999	C	-9999.999	C	-9999.999		
The value for the user parameter "X-axis workpiece coordinate home (for taking in)" is set. OK?							
Yes	No						

The machine coordinates set in the user parameter <Workpiece coordinate zero point * axis (for capture)> is set.

2.2.2.6 To Finish Editing

When the setting is completed, press the **[F0]** (Completion mode) key and finish editing.
For how to finish editing, refer to “2.1.2.2 To finish editing”.

The workpiece coordinate system currently being used cannot be changed while in tilt plane machining mode.

The alarm << Feature coordinate manufacturing mode engaged >> is triggered when an attempt is made to exit editing.

2.2.3 Editing during Operation

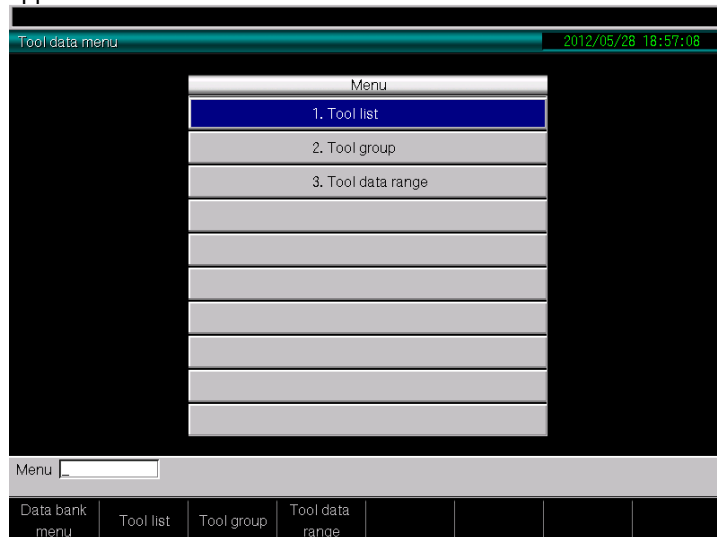
The workpiece coordinate zero point is used for calculation of a path commanded by a program in operation. The referenced timing is not by the execution pointer but is at the timing when the interpretation pointer is moved. So if a workpiece coordinate zero point is changed during operation, be cautious of the position of the interpretation pointer.

2.3 Tool Data (NC)

2.3.1 Outline

Information about the tools used for machining is set in the <Tool list>, <Tool group>, or <Tool data range>.

If the <Tool data> is selected from the <Data bank menu> screen, the <Tool data menu> screen appears.



2.3.2 To Set Tool List

Information about tool length offset, tool diameter offset, life and the like which vary depending on the kinds of tools is registered. Up to 99 tools can be registered.

If the <Tool list> is selected from the <Tool data menu> screen, the <Tool list> screen appears for editing.

Tool list					
2012/05/28 19:01:04					
Tool No.	01(P:01)	02(P:02)	03	04	05(P:03)
Tool name	CENTER DRILL1	BBB	CCC	DDD	EEE
Tool length offset	100.000	-999.999	-999.999	-999.999	-999.999
T length wear offset	0.000	-99.999	-99.999	-99.999	-99.999
Cutter comp.	100.000	-999.999	-999.999	-999.999	-999.999
Cutter wear offset	0.000	-99.999	-99.999	-99.999	-99.999
Initial life	100000(H)	999999(H)	999999(C)	*****	*****
Life warning	100(H)	100(H)	100(C)	*****	*****
Tool life	100000(H)	999999(H)	999999(C)	*****	*****
Tool No.	06	07(P:04)	08	09	10(P:05)
Tool name	FFF	GGG	HHH	III	JJJ
Tool length offset	50.000	0.000	0.000	0.000	0.000
T length wear offset	0.001	0.000	0.000	0.000	0.000
Cutter comp.	0.001	5.000	0.000	0.000	0.000
Cutter wear offset	0.001	0.000	0.000	0.000	0.000
Initial life	999999min	999999(H)	999999(C)	*****	*****
Life warning	100min	100(H)	100(C)	*****	*****
Tool life	999999min	999999(H)	999999(C)	*****	*****
Tool name [] [HELP]					
Completion mode	Search	Automatic setting	Add	Copy	Paste
				To switch displays	Tool data Delete

The tool whose number is in orange color indicates the tool registered in ATC.

If the [F6] (To switch displays) key is pressed, the following function keys appear:

- [F0] (Back) key : Displays the previous function.
- [F1] (List) key : Displays the tool data for 10 tools.
- [F2] (Expansion 1) key : Displays the screen that can set addable items like peripheral speed, rotary feed, or the like.
- [F3] (Expansion 2) key : Displays the screen that includes the setting items for the turning tool.

Back	List	Expansion 1	Expansion 2				
------	------	-------------	-------------	--	--	--	--

Tool list(Expansion 1)					
2012/05/28 19:07:29					
Tool No.	01(P:01)	02(P:02)	03	04	05(P:03)
Tool name	CENTER DRILL1	BBB	CCC	DDD	EEE
Tool length offset	100.000	-999.999	-999.999	-999.999	-999.999
T length wear offset	0.000	-99.999	-99.999	-99.999	-99.999
Cutter comp.	100.000	-999.999	-999.999	-999.999	-999.999
Cutter wear offset	0.000	-99.999	-99.999	-99.999	-99.999
Initial life	100000(H)	999999(H)	999999(C)	*****	*****
Life warning	100(H)	100(H)	100(C)	*****	*****
Tool life	100000(H)	999999(H)	999999(C)	*****	*****
Peripheral speed	999.9	999.9	999.9	999.9	999.9
Feed per rotation	9.99	9.99	9.99	9.99	9.99
S command value(G97)	99999	99999	99999	99999	99999
F command value(G94)	9999.99	9999.99	9999.99	9999.99	9999.99
Max. rotational frequency	99999	0	99999	99999	99999
Tool washing	Allowed	Not allowed	Allowed	Allowed	Allowed
CTS	Allowed	Not allowed	Allowed	Allowed	Allowed
Tool name					
[HELP]					
Completion mode	Search	Automatic setting	Add	Copy	Paste
				To switch displays	Tool data Delete

Tool list(Expansion 2)					
2016/06/21 18:24:29					
Tool No.	01(P:01)	02(P:02)	03	04	05(P:03)
Tool name	AAA	BBB	CCC	DDD	EEE
Tool length offset(Z)	-999.999	-999.999	-999.999	-999.999	-999.999
Tool len. wear offset(Z)	-99.999	-99.999	-99.999	-99.999	-99.999
Cutter/nose R comp.	-999.999	-999.999	-999.999	-999.999	-999.999
Cutter/nose R wear	-99.999	-99.999	-99.999	-99.999	-99.999
Tool position offset(X)	-999.999	-999.999	-999.999	-999.999	-999.999
Tool pos. wear offset(X)	-99.999	-99.999	-99.999	-99.999	-99.999
Tool position offset(Y)	-999.999	-999.999	-999.999	-999.999	-999.999
Tool pos. wear offset(Y)	-99.999	-99.999	-99.999	-99.999	-99.999
Imaginary tool nose dir.	0	9	9	9	9
Initial life	999999mIn	999999(H)	999999(C)	*****	*****
Life warning	999999mIn	999999(H)	999999(C)	*****	*****
Tool life	999999mIn	999999(H)	999999(C)	*****	*****
Max. speed	99999	0	99999	99999	99999
Tool name					
[HELP]					
Completion mode	Search	Automatic setting	Add	Copy	Paste
				To switch displays	Tool data Delete

2.3.2.1 List Screen

1. <Tool name>
Tool names are set. They are only displayed on the screen but are not used for operation.
If the [DEL] key is pressed when the cursor is at a <tool name>, the tool name is deleted. To delete the data of the tool where the cursor is, press [F7] (Tool data Delete).
2. <Tool length offset> and <T length wear offset>
The Z-axis offset amount when a tool length offset is commanded is set.
“Tool length offset amount” = <Tool length offset> + <Tlength wear offset>
3. <Cutter comp.> and <Cutter wear offset>
The offset amount when a tool diameter offset is commanded is set.
“Tool diameter offset amount” = <Cutter comp.> + <Cutter wear offset>
4. <Initial life>/<End-of-life>
When the cursor is moved to <Initial life>/<End-of-life>, you can set the life unit and the initial life/life end.
 - * By setting the user parameter <Tool life counting method>, the item name changes to <Initial life> or <End-of-life>.
 - When the value for the <Tool life counting method> is <0: Type 1>, the life control becomes the count-down type and the item name is displayed as <Initial life>.
 - When the value for the <Tool life counting method> is <1: Type 2>, the life control becomes the count-up type and the item name is displayed as <End-of-life>.

Life unit can be selected from the following 4:

Setting value	Unit	Explanation
<1. Not counted>	None	Tool life is not controlled. If <1: Not counted> is set, <*****> is indicated in <Initial life>/<Life end>, <Life warning>, and <Tool life>.
<2: Time>	min	Tool life is controlled by time (unit of minute). If a cutting motion is carried out for one minute, the life is subtracted or added.
<3: Drilling>	Hole	Tool life is controlled by number of machined holes. When performing drilling 1 using G81 in a drilling canned cycle, the tool life is decreased or increased. When using G392 or G376 in a canned cycle that is not for drilling, the tool life does not decrease or increase.
<4. Program>	Frequency	Tool life is controlled by program frequency. The life of a tool used for cutting by M30 is subtracted or added.

After the life unit is set, go to setting of <Initial life>/<End-of-life>.

Set an initial value or finish value of tool life for <Initial life>/<End-of-life>.

For the <Initial life>, if the <Initial life> is set, the same value is set to the <Tool life>.

For the <End-of-life>, if the <End-of-life> is set, 0 is set to the <Tool life>.

5. <Life warning>

To the <Life warning>, set a life value that predicts the end of a tool life.

When the <Tool life> reaches the value of the <Life warning>, the alarm <<Tool life ending (****)>> (The value in () is a tool number) appears.

Even if the alarm appears, start of memory operation is possible.

* Setting the <Life warning> to “0” in the counting-up type does not generate this alarm.

6. <Tool life>

Current value of the tool life is set. When the tool life is subtracted or added according to machining and the life is expired, the alarm <<Tool life end (****)>> (The value in () is a tool number) appears and start of memory operation becomes impossible.

* <Tool life> is not subtracted or added in MDI operation.

* Subtraction/addition and check of tool life are done only to the tool commanded in memory operation. Accordingly, if tools are changed by interruption of MDI in memory operation, subtraction/addition and check of tool life are not done to the tool even if the memory operation is restored thereafter.

* “Life is expired” in the countdown type means that the <Tool life> goes to 0. In the count-up type, it means that the <Tool life> goes to the same value as <End-of-life>.

* You can initialize the life of an expired tool from the <ATC tool> screen.

2.3.2.2 Extension 1 Screen

1. <Tool name>

This becomes the same data as those on the List screen.

2. <Tool length offset> and <Tool length wear offset>

This becomes the same data as those on the List screen.

3. <Cutter comp.> and <Cutter wear offset>

This becomes the same data as those on the List screen.

4. <Initial life>/<End-of-life>

This becomes the same data as that on the List screen.

5. <Life warning>

This becomes the same data as that on the List screen.

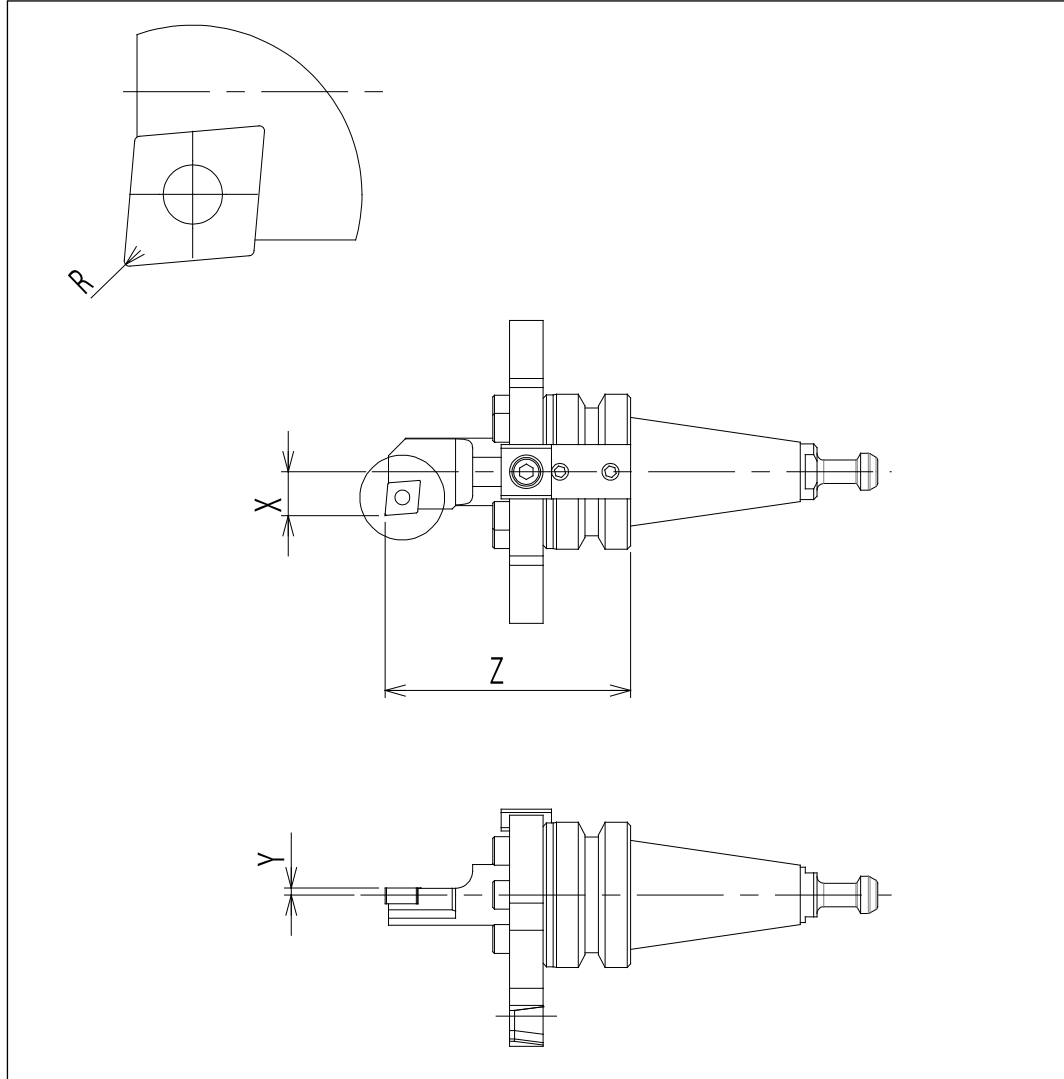
6. <Tool life>
This becomes the same data as that on the List screen.
7. <Peripheral speed>
Peripheral speed of a tool is set.
The unit in the metric system is m/min; the unit in the inch system is feet/min.
If the [DEL] key is pressed, no value is set.
8. <Feed per rotation>
Rotary feed of a tool is set.
The unit in the metric system is mm/rotation; the unit in the inch system is inch/rotation.
If the [DEL] key is pressed, no value is set.
9. <S command value (G97)>
If the setting <Automatically use S command when changing tool> in the user parameter is configured to <1:Yes>, the S command is issued automatically for the value that is set for a tool change used when canceling (G97) the constant peripheral speed control. Refer to section “5.7 Canned Cycle for Tool Change (Nonstop ATC) (G100)” in the Programming Manual NC for further details.
Press the [DEL] key to clear the value so that it is not set.
10. <F command value (G94)>
If the user parameter <Automatically use F command when changing tools> is set <1: Yes>, the feedrate per min feed set in tool change by (G94) is commanded.
For details, refer to “Programming Manual, NC”, “5.7 Canned cycle (Non-stop automatic tool change) (G100) for tool change”.
If the [DEL] key is pressed, no value is set.
11. <Max. rotational frequency>
The tool top rotational frequency when a tool is mounted in a spindle is set.
If the value is set to “0,” rotation is not possible. Configure the setting to “0” for tools that cannot rotate, such as measurement tools and turning tools.
For details, refer to “Programming Manual, NC,” “10.1 S Function”, “10.2 M Function (Spindle Control)”.
If the [DEL] key is pressed, no value is set.
12. <Tool wash>
If you do not want to splash coolant to a tool for tool wash, set with <1: Not allowed> so that no tool set with <Not allowed> is washed.
 - * The spindle air blow is not carried out even when the machine parameter <Spindle air blow/Tool wash> is set to <0: Spindle air blow type 1> or to <2: Spindle air blow type 2>.
 - * The discharge conditions can be changed in the user parameter (switch 1) <<Tool wash discharge conditions for tool data>>.

(Note) This function inhibits coolant from attaching on a tool in wash for tool change immediately before using a measuring tool or the like in the memory operation mode or MDI operation mode. However, it does not prevent other than coolant from attaching. In addition, regardless of this setting, a tool is washed with the [ATC] key in the manual mode. If you want to halt the tool wash in the manual mode, turn off the [CLT.P] key.
13. <CTS>
Set <1: Not allowed> for a tool not corresponding to center-through coolant. If a tool which is not set <Not allowed> is commanded for discharging CTS coolant, the alarm <<CTS unable tool>> appears and no CTS coolant is discharged.

2.3.2.3 Expansion 2 Screen

1. <Tool name>
The same data is used for the <tool name> on the list screen.
2. <Tool length offset (Z)> and <tool length wear offset (Z)>
These functions set the Z-axis offset when the commands for tool length offset (G43/G44) and tool position compensation (G143/G144) are issued.
 $\text{<Tool length offset>} = \text{<Tool length offset (Z)>} + \text{<T length wear offset (Z)>}$
The same data is used for the tool length offset and T length wear offset on the list screen.
3. <Cutter/Nose R compensation> and <Cutter/nose R wear>
These functions set the offset when the commands for cutter compensation (G41/G42) and nose R compensation (G141/G142) are issued.
“Cutter offset” = <Cutter/Nose R compensation> + <Cutter/nose R wear>
“Nose R offset” = <Cutter/Nose R compensation> + <Cutter/nose R wear>
The same data is used for the <Cutter compensation> and <Cutter wear offset> on the list screen.
4. <Tool position offset (X)> and <tool position wear offset (X)>
These functions set the X-axis offset when the commands for tool position compensation (G143/G144) are issued.
“X-axis tool position offset” = <Tool position offset (X)> + <T length wear offset (X)>
5. <Tool position offset (Y)> and <tool position wear offset (Y)>
These functions set the Y-axis offset when the commands for tool position compensation (G143/G144) are issued.
“Y-axis tool position offset” = <Tool position offset (Y)> + <T length wear offset (Y)>
6. <Imaginary tool nose direction>
This function uses a range of 0 to 9 to set the direction of the teeth, when looking from the center of nose R on the turning tool.
For further details, refer to “4.3.2 Virtual teeth” in the NC Programming Manual.
7. <Initial tool life> <End of tool life>
The same data is used for initial life / life end on the list screen.
8. <Life warning>
The same data is used for the life warning on the list screen.
9. <Tool life>
The same data is used for the tool life on the list screen.
10. <Maximum speed>
The same data is used for the <maximum speed> on the expansion 1 screen.

Turning tool setting items



Z: Tool length offset (Z) + Tool length wear offset (Z)

X: Tool position offset (X) + Tool position wear offset (X)

Y: Tool position offset (Y) + Tool position wear offset (Y)

R: Cutter / Nose R compensation + Cutter / Nose R wear offset

2.3.2.4 Movement of Cursor

Move the cursor to where you want to change by the cursor key.

You can quickly move the cursor to a desired tool number by searching. Input a tool number in the input area and press the **[F1]** (Search) key. The cursor moves to the specified number tool.

2.3.2.5 Direct Input of Data

1. Move the cursor to the item you want to change.
2. Input a value in the input area and press the **[ENT]** key.
The value is input at the cursor position and the cursor moves to the next item.

2.3.2.6 Automatic Setting

<Tool length offset>, <S command value(G97)>, and <F command value(G94)> can be automatically set. Point the cursor at the item you want to set and press the **[F2]** (Automatic setting) key.

- * In the list screen, the automatic setting is enabled for the <Tool length offset> regardless of the position of the cursor.

- <Automatic setting> of <Tool length offset>

Automatic setting 2012/05/28 19:16:11

Machine coordinate pos.

Z 480.000

Automatic tool length compensation

50.000

Tool length offset after setting (No. 01)

430.000

Set?

Yes No

Set the value (Current machine coordinate position value - User parameter <Automatic tool length compensation>) as a new tool length offset.

If the **[F0]** (Yes) key is pressed, the value of <Tool length offset after setting> is set in the tool length offset and the previous screen is called back.

If the **[F1]** (No) key is pressed, no value is set but the previous screen is called back.

- <Automatic setting> of <S command value(G97)> and <F command value(G94)>

Automatic setting 2012/05/28 19:17:19

Cutter comp.

100.000

Cutter wear offset

0.000

Peripheral speed (m/min)

999.9

S command value after setting

1591

Feed per rotation (mm/rotation)

9.99

F command value after setting

15894.09

Set?

Yes No

Set the value calculated from <Cutter comp.>, <Cutter wear offset>, <Peripheral speed (m/min)>, and <Feed per rotation (mm/rotation)> as a new S-command value or F-command value.

If the **[F0]** (Yes) key is pressed, the value of <S command value after setting> or <F command value after setting> is set as an S command value or F command value and the previous screen is called back.

If the **[F1]** (No) key is pressed, no value is set but the previous screen is called back.

- * Unless any of the <Cutter comp.>, <Peripheral speed (m/min)>, and <Feed per rotation (mm/rotation)> is set, the alarm <<Automatic setting items lack>> appears and the <Automatic setting> screen is not called back.
- * If <Cutter comp.> +<Cutter wear offset> becomes 0 or less, the alarm <<Automatic setting calculation is unable.>> appears and the <Automatic setting> screen is not called back.

2.3.2.7 To Add Data

1. Move the cursor to the item you want to change.
2. Input a value to be added in the input area and press the **[F3]** (Add) key.
The value set at the cursor position is added with an input value and the cursor moves to the next item.

2.3.2.8 To Copy Data

1. Move the cursor to the tool name you want to take in data.
2. Press the **[F4]** (Copy) key is.
Take in a set of tool data at the cursor position.
The taken data is taken into other data or held until power goes off.

2.3.2.9 To Delete Data

1. Move the cursor to the tool name of a tool from which you want to delete data.
2. If the **[F7]** (Tool data Delete) key is pressed, the deletion check screen appears.

Tool No. 12							
Are you sure you want to delete?							
Yes	No						

- If the **[F0]** (Yes) key is pressed, the data is taken in to delete the tool data at the cursor position.
- If the **[F1]** (No) key is pressed, nothing is deleted but the previous screen is called back.

The taken data is taken into other data or held until power goes off.

2.3.2.10 To Paste Data

1. Move the cursor to the tool name of the tool you want to paste the taken data.
2. Press the **[F5]** (Paste) key.
A set of tool data at the cursor position is changed into the taken data.

2.3.2.11 To Finish Editing

If setting completes, press the **[F0]** (Completion mode) key to finish editing.

2.3.3 To Set Tool Group

By registering multiple tools in a tool group, you can have the next tool selected from the group for machining when the tool life expires. You can register up to 30 groups.

2.3.3.1 Tool Group List

When the <Tool group> is selected from the <Tool data menu> screen, the Tool group screen appears.

Group list	
01	07
02	08
03	09
04	10
05	
06	

Group No. [HELP]

Completion mode

If no tool is registered in a group, the group is not displayed.

Point the cursor at the displayed tool group and press the **[DEL]** key. Then the tool group at the cursor position is deleted.

2.3.3.2 To Edit Tool Group

Register tools in the group selected from the Tool group screen.

One group can register up to 30 tools.

Input a desired group number in the input area on the Tool group screen and press the **[ENT]** key or point the cursor at a desired group and press the **[ENT]** key. Then the Tool group screen appears for editing.

Group01						
	Tool No.	Tool name	Tool length offset	Cutter comp.	Initial life	Current life
1	01	CENTER DRILL1	100.000	100.000	100000 (H)	100000 (H)
2	21	CENTER DRILL2	100.000	5.000	100000 (H)	100000 (H)
3	41	CENTER DRILL3	100.000	5.000	100000 (H)	100000 (H)
4	61	CENTER DRILL4	100.000	5.000	100000 (H)	100000 (H)
5	81	CENTER DRILL5	100.000	5.000	100000 (H)	100000 (H)
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						

Tool No.

Group list

Move the cursor to the position where you want to register a tool, input the tool number to be registered in the input area, and press the **[ENT]** key. When the tool is added, the order of the subsequent tools is shifted back by one.

In order to delete a registered tool, point the cursor at the position of the tool you want to delete and press the **[DEL]** key. When the tool is deleted, the order of the subsequent tools is shifted ahead by one.

* Life-remaining tools are searched and selected in the order set here.

When the tool registration finishes, press the **[F0]** (Group list) key to call back the Tool list screen.

2.3.3.3 To Finish Editing

When the setting completes, press the **[F0]** (Completion mode) key to finish editing.

2.3.4 To Set Tool Data Range

The input range is set for the tool list. On each tool, the <Range (+)> and <Range (-)> can be set for the <Tool length offset>, <T length wear offset>, <Cutter/Nose R compensation>, <Cutter/nose R wear>, <Tool position offset (X)>, <Tool pos. wear offset (X)>, <Tool position offset (Y)> and <Tool pos. wear offset (Y)>.

(Note) Some of the name on the screen may vary depending on the display format of the tool list.

Screen display format for tool list	Name for "tool diameter"
List	Tool dia.
Expansion 1	Tool dia.
Expansion 2	Tool dia./nose R

2.3.4.1 To Edit Tool Data Range

If the <Tool data range> is selected from the <Tool data menu> screen, the <Tool data range> screen appears for editing.

Tool data range (tool length)					
2012/05/28 19:19:40					
Tool No.	01 (P:01)	02 (P:02)	03	04	05 (P:03)
Tool name	CENTER DRILL	DR	DR	DR	DR
Tool length offset	100.000	999.999	999.999	999.999	999.999
Range (+)	110.000				
Range (-)	90.000				
T length wear offset	0.000	-99.999	-99.999	-99.999	-99.999
Range (+)	1.000				
Range (-)	-1.000				
Tool No.	06	07 (P:04)	08	09	10 (P:05)
Tool name	DR	DR	DR	DR	DR
Tool length offset	99.999	0.000	0.000	0.000	0.000
Range (+)					
Range (-)					
T length wear offset	0.000	0.000	0.000	0.000	0.000
Range (+)					
Range (-)					
Range (+) mm					
[HELP]					
Completion mode	Search	Copy	Paste	Tool length	Tool diameter
				Tool position X	Tool position Y

The following is displayed:

- [F4]** (Tool length) key : Displays the screen to set <Tool length offset> and <T length wear offset>
- [F5]** (Tool diameter) key : Displays the screen to set <Cutter comp.> and <Cutter wear offset>
- [F6]** (Tool position X) key : Screen for setting range for <Tool position offset (X)> and <Tool pos. wear offset (X)>
- [F7]** (Tool position Y) key : Screen is displayed for setting range for <Tool position offset (Y)> and <Tool pos. wear offset (Y)>.

If the current tool list screen display is for expansion 2, the **[F5]** key displays <Cutter/Nose R>.

Tool data range (Cutter/Nose R)						2016/06/21 16:13:26
Tool No.	01 (P:01)	02 (P:02)	03	04	05 (P:03)	
Tool name	CENTER DRILL1	DRILL	COG	COG	EE	
Cutter/nose R comp.	100.000	-99.999	-99.999	-99.999	-99.999	
Range (+)	110.000					
Range (-)	90.000					
Cutter/nose R wear	0.000	-99.999	-99.999	-99.999	-99.999	
Range (+)	1.000					
Range (-)	-1.000					
Tool No.	06	07 (P:04)	08	09	10 (P:05)	
Tool name	FR	FR	FR	FR	FR	
Cutter/nose R comp.	0.001	5.000	0.000	0.000	0.000	
Range (+)						
Range (-)						
Cutter/nose R wear	0.001	0.000	0.000	0.000	0.000	
Range (+)						
Range (-)						
Range (+) mm						[HELP]
Completion mode	Search	Copy	Paste	Tool length	Cutter/Nose R	Tool position X
						Tool position Y

Move the cursor to the item you want to change, input a desired value in the input area, and press the **[ENT]** key.

The **[F1]** (Search) key, **[F2]** (Copy) key, and **[F3]** (Paste) key are the same as those for the <Tool list> screen appears for editing.

- * Only the <Range (+) / (-)> can be set on this screen. The values cannot be changed for the tool list (<Tool length offset>, <T length wear offset>, <Cutter/Nose R compensation>, <Cutter/nose R wear>, <Tool position offset (X)>, <Tool pos. wear offset (X)>, <Tool position offset (Y)> and <Tool pos. wear offset (Y)>).
- * No value which becomes “Range (-) > Range (+)” cannot be set when setting <Range (+)/(-)>.

If you try to set a value other than in the range for each item of the tool list or a value which cannot be set in the <Range (+)/(-)> when the <Range (+)/(-)> is set, the warning screen appears.

Tool data setting range error		2012/05/28 19:22:01
Entered tool data is not within the setting range. Enter the correct value.		
Tool No.	01	
Type	Tool length offset (Z)	
Setting range	90.000 - 110.000	
Set value	120.000	
		Back

The input value is indicated in red color.
Press the **[F7]** (Back) key to return to the <Edit> screen and input a correct value again.

- * If any of the <Range (-)> and <Range (+)> is not set yet, it is considered that the range is not set. So the range is not checked.
- * The range is checked and the <Warning> screen is displayed even for setting by automatic setting or adding.
- * The range is also checked when tool data is set in memory operation by a macroprogram or G10.

- * A range check operation is performed when also executing Tool length offset / Cutter compensation (G41-44) and Tool position compensation / nose R compensation (G141 – G144) during memory operation. The alarm “Comm. issued to area other than (tool) data area” is triggered when the command area is outside of the range.

2.3.4.2 To Finish Editing

When setting completes, press the **[F0]** (completion mode) key to finish editing.

2.3.5 Edit during Operation

In operation, each data in the tool list, tool group list, and tool data range is referred in the timing when the interpretation pointer (not the execution pointer) moves. Change each data during operation. If you finish editing, be cautious of the interpretation pointer position.

2

2.4 Macro Variables (NC)

2.4.1 Outline

Macro variables in #100s, #500s, #600s, #700s, #800s, and #900s are set. The local variables (#001 to #33) are available only for reference.

- * The macro variables #100 to #199 are valid only while the power is ON. When the power is OFF, the value becomes vacant.

2.4.1.1 Edit Screen

When <Macro variable> is selected from the <Data bank menu> screen, the Macro variables screen appears for editing.

Macro variables									
2012/05/28 19:24:44									
Var.	Value	Var.	Value	Var.	Value	Var.	Value	Var.	Value
#100	10.000	#120		#140		#160		#180	
#101	-10.000	#121		#141		#161		#181	
#102	0.000	#122		#142		#162		#182	
#103	0.000	#123		#143		#163		#183	
#104	0.000	#124		#144		#164		#184	
#105	0.000	#125		#145		#165		#185	
#106	0.000	#126		#146		#166		#186	
#107	0.000	#127		#147		#167		#187	
#108	100.000	#128		#148		#168		#188	
#109	0.000	#129		#149		#169		#189	
#110	0.000	#130		#150		#170		#190	
#111	0.000	#131		#151		#171		#191	
#112	0.000	#132		#152		#172		#192	
#113	0.000	#133		#153		#173		#193	
#114	0.000	#134		#154		#174		#194	
#115	-300.000	#135		#155		#175		#195	
#116	0.000	#136		#156		#176		#196	
#117	0.000	#137		#157		#177		#197	
#118		#138		#158		#178		#198	
#119		#139		#159		#179		#199	
#100									
[HELP]									
Completion mode	Add	#100 - 199	#500 - 599	#600 - 699	#700 - 799	#800 - 899	#900 - 999		

By pressing the **[F2]** (#100 - 199) to **[F7]** (#900 - 999) keys, **[>>]** (Next function) key, and **[F1]** (Local variable) key, you can switch the displayed macro variables.

If a macro variable is set with a value out of the displayable range, it is indicated with <*****>.

If the number of digits after the decimal point exceeds the range that can be displayed, then it is rounded off to a value that can be displayed.

2.4.2 How to Edit

2.4.2.1 Direct Input of Data

1. Move the cursor to the macro variable you want to change.
2. Input a value in the input area and press the **[ENT]** key.
The value is input at the cursor position and the cursor moves to the next macro variable.

2.4.2.2 To Add Data

1. Move the cursor to the macro variable you want to change.
 2. Input a value to be added in the input area and press the **[F1]** (Add) key.
The value set at the cursor position is added with the input value and the cursor moves to the next macro variable.
- * You cannot add a value to the macro variable indicating <*****> (which shows that a value exceeding the displayable range is set).
- * When an addition operation is carried out for macro variables #100 to #199, it performs an addition operation for a value that can be displayed on the screen (value is rounded off to a value that can be displayed).

2.4.2.3 To Delete Data

1. Move the cursor to the macro variable you want to delete.
2. Press the **[DEL]** key.
The value for the macro variable is “blank” where the cursor is.

2.4.2.4 To Finish Editing

When the setting completes, press the **[F0]** (Completion mode) key to finish editing.

2.4.3 Editing during Operation

The macro variables that are referenced during operation occurs when interpretation pointer moves, not the execution pointer. When changing macro variables during operation, pay attention to the position of the interpretation pointer.

2.5 User Parameter

2.5.1 Outline

Parameters for machine operation are set. They are important data for operation of the machine. So change them prudently. Keep the parameter list attached on the machine in safe.

User parameters are categorized in the following categories:

Category name	Explanation
Switch 1	Sets the parameters for screen displays and motion types.
Switch 2	Sets the parameters for axial motions.
Switch 3	Sets the G code and M code used for checking if the positioning is finished, for displaying alarm numbers with lower priority and for displaying modals.
Reference position/Conversation zero point	Sets the parameters for reference points and conversational zero points.
User registered auxiliary function	Sets the user registered auxiliary function.
Tool breakage detection	Sets the parameters for tool breakage detectors.
Automatic workpiece measurement	Sets the parameters for automatic workpiece measurement.
Automatic thermal distortion compensation	Sets the parameters for automatic thermal distortion offset.
Quick table	Sets the parameters for quick tables. The machine which has no quick table is neglected.
High accuracy	Sets the parameters for high accuracy mode.
PLC	Sets the parameters for PLC.
PROFIBUS DP	Sets the parameters for PROFIBUS DP.
DeviceNet	Sets the parameters for DeviceNet.
CC-Link slave	Sets the parameters related to the CC-Link slave.
Rotary fixture offset	Set the parameters related to rotary fixture offset.
3D smoothing	Set the parameters related to 3D smoothing.
Feature coordinate setting	Set the parameters related to Feature coordinate setting.
CC-Link master	Sets the parameters related to the CC-Link master.

2

2.5.1.1 Edit Screen

When the <User parameter> is selected from the <Data bank menu> screen, the User parameter screen appears for editing.

Switch 1 2012/05/28 19:26:15

0001	Select mode [0: Manual mode 1: Memory operation mode]	0	
0002	Initial mode [0: High-speed travel 1: Low-speed travel 2: Step travel]	0	
0003	Select memory operation [0: Internal memory 1: Tape]	0	
0004	Program repeat [0: Stop 1: Repeat]	0	
0005	Workpiece counter when program is repeated [0: Off 1: On]	0	
0006	Tool life management when program is repeated [0: No 1: Yes]	0	
0007	Automatic tool length compensation	50,000	mm
0008	External signal reading time limit	0	s
0009	Dry run offset	50,000	mm
0010	Dry run offset lower limit	0,000	mm
0011	Program unit [0: Base 1: Least]	0	
0012	Program feed rate unit [0: basic 1: minimum]	0	
0013	G78 relief amount	5,000	mm
0014	G83 cutting start position	0,500	mm
0015	G77, G78 relief amount	5,000	Threads
0016	G76, G87 shift direction [0: +X dir. 1: -X dir. 2: +Y dir. 3: -Y dir.]	0	
0017	Tapping cycle return speed [0: S address command value, 1: Max. speed]	0	

Select mode

[HELP]

Completion mode Switch 1 Switch 2 Switch 3 Ref. position/Conversation zero point User registered misc. function Tool breakage detection Automatic workpiece meas.

Previously displayed categories are displayed. Pressing the function keys switches the categories.

Upper [F0] to [F7] and lower [F0] to [F7] of [>>] (Next function)

Completion mode	Switch 1	Switch 2	Switch 3	Ref. position, Conversation zero point	User registered misc.func.	Tool breakage detection	Automatic workpiece meas.
Completion mode	Auto thermal distortion comp.	Quick table	High accuracy	PLC	PROFIBUS DP	DeviceNet	CC-Link
Completion mode	Rotary fixture offset	3D smoothing	Feature coordinate setting				

(Note) In conversation language, <Rotary fixture offset> and <3D smoothing> and <Feature coordinate setting> are grayed out and cannot be selected.

<User registered miscellaneous functions> screen for editing

User registered miscellaneous functions						2012/05/28 19:28:34	
Auxiliary function	01	02	03	04	05		
Signal	999	999	999	999	999		
Name	12345678901234	12345678901234	12345678901234	12345678901234	12345678901234		
Auxiliary function	06	07	08	09	10		
Signal	999	999	999	999	999		
Name	12345678901234	12345678901234	12345678901234	12345678901234	12345678901234		
Auxiliary function	11	12	13	14	15		
Signal	999	999	999	999	999		
Name	12345678901234	12345678901234	12345678901234	12345678901234	12345678901234		
Auxiliary function	16	17	18	19	20		
Signal	999	999	999	999	999		
Name	12345678901234	12345678901234	12345678901234	12345678901234	12345678901234		
Name						Can be deleted	
[HELP]							
Completion mode	Switch 1	Switch 2	Switch 3	Ref. position/Conversation zero point	User registered misc. function	Tool breakage detection	Automatic workpiece meas.

2.5.2 How to Edit

2.5.2.1 Input of Data

1. Call the display of the item category you want to change.
2. Move the cursor to the item you want to change.
3. Input a value in the input area and press the [ENT] key.
The value at the cursor position is input and the cursor moves to the next item.

2.5.2.2 To Delete Data

When the cursor is at a deletable item, the screen displays <Can be deleted>.

1. Call the category of the item you want to delete.
2. Move the cursor to the item you want to delete.
3. Press the [DEL] key.

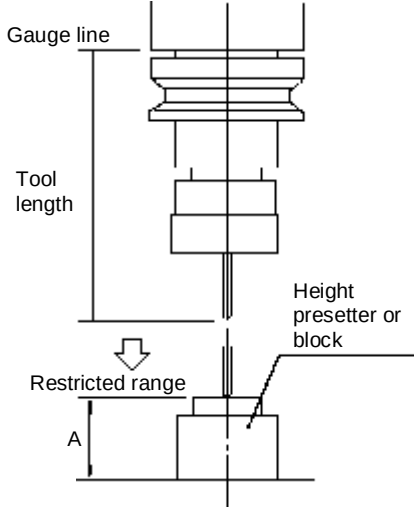
The value at the cursor position is deleted.

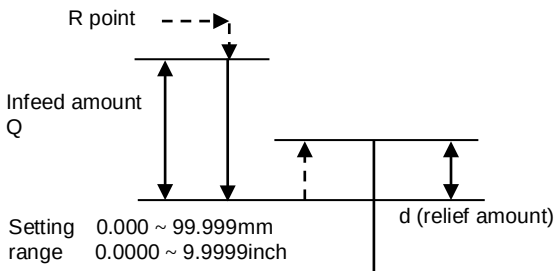
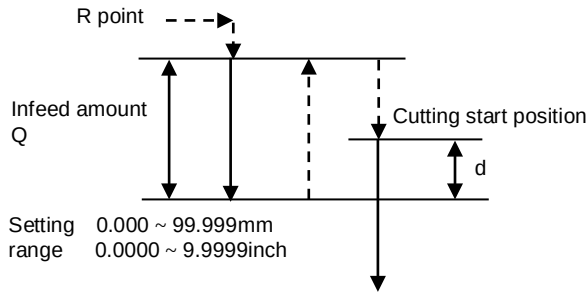
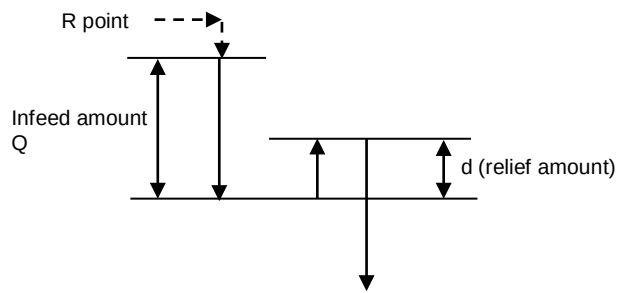
2.5.2.3 To Finish Editing

After setting completes, press the [F0] (Completion mode) key to finish editing.

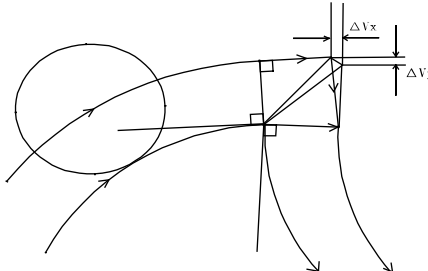
* Reset or power-off may be required depending on the changed item.

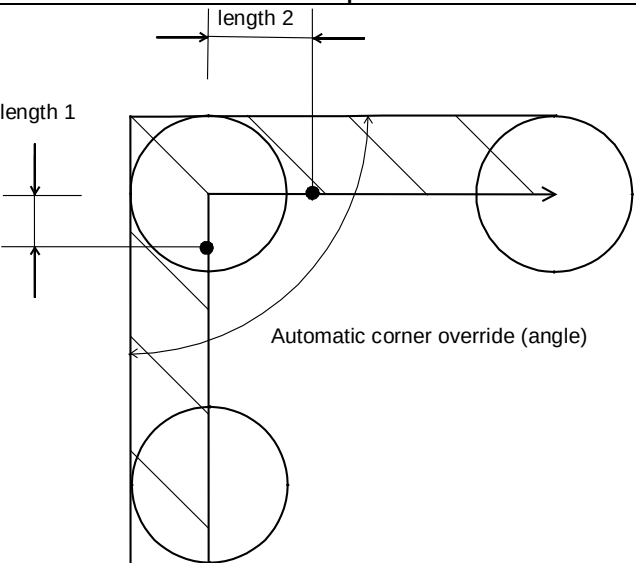
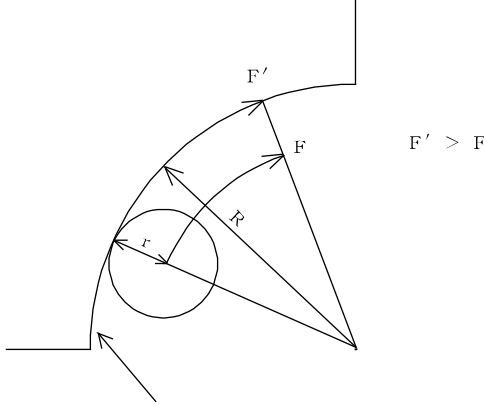
2.5.3 User Parameter 1 (Switch 1)

No.	Item name	Setting range	Description
0001	Select mode	0:Manual mode 1:Memory operation mode	Specify mode at the time of power input either by manual operation or memory operation.
0002	Initial mode	0: High-speed travel 1: Low-speed travel 2: Step travel	Specify movement mode by manual operation mode at the time of power input.
0003	Select memory operation (NC language only)	0: Internal memory operation 1: Tape operation	Set whether to perform memory operation or tape operation. When tape operation is selected, the function varies depending on the connection destination of the external I/O unit. When "0:General COMM device" is selected for "Connected to", a single program input from the communication port is run. When the connecting destination is set to a memory card, select and run the program from the recording device connected to the USB port.
0004	Program repeat	0:Stop 1:Repeat	Set whether to stop or continue operation after the program has ended in memory operation mode.
0005	Workpiece counter when program is repeated	0:Invalid 1:Valid	Set whether to reset the workpiece counter when the program is repeated.
0006	Tool life management when program is repeated	0:No 1:Yes	Set whether to decrement the remaining tool life counter when the program is repeated.
0007	Automatic tool length compensation	0~999.999mm	The tool length is measured by moving the Z-axis as shown in the diagram below, and can be registered as tool list data. In this situation, set the height (A in diagram) for the height presetter or block to be used as compensation. 
0008	External signal reading time limit	0~9999s	Set acceptable time by second from output of external signal to reception of end signal from outside. If the machine does not receive the completion signal within the time set here, a signal output timeout error will occur. When "0" is set, the machine does not proceed to the next operation until it receives the completion signal from the external device.
0009	Dry run offset	0~999.999mm	Offset amount for the Z-axis absolute command during dry run in the memory operation is set.
0010	Dry run offset lower limit	0~999.999mm	Generates an operator message <<Dry run offset amount:xxxx>> by turning the dry run on, when setting value of "Dry run offset amount" is lower than that of the parameter.
0011	Program unit (NC language only)	0:Base 1:Least	Set the position of the decimal point when it is not included in the program.
0012	Program feed rate unit (NC language only)	0:Base 1:Least	Set the position of a decimal point, when a decimal point is omitted in the program.

No.	Item name	Setting range	Description
0013	G73 relief amount (NC language only)	0~99.999mm	Set the relief amount (d) for G73 cycle.  Setting range 0.000 ~ 99.999mm 0.0000 ~ 9.9999inch
0014	G83 cutting start position (NC language only)	0~99.999mm	Set the cutting start position (d) for G83 cycle.  Setting range 0.000 ~ 99.999mm 0.0000 ~ 9.9999inch
0015	G77,G78 relief amount (NC language only)	0~9.999Threads	Set the relief amount (d) for G77 and G78 cycles.  Setting range 0.000~9.999 (No. of threads)
0016	G76,G87 shift direction (NC language only)	0:+X dir. 1:-X dir. 2:+Y dir. 3:-Y dir.	Set the shift direction after spindle orientation for G76 and G87.
0017	Tapping cycle return speed (NC language only)	0:S address command value 1:Max. speed	To select the spindle returning rotational frequency when an L address is omitted in G77, G78, G277, or G278 (tapping cycle) 0: S address command value (The spindle returns at the rotational frequency specified by an S address.) 1: Top rotational frequency (The spindle returns at the machine parameter "Top rotational frequency in tapping.")
0018	Program restart (NC language only)	0:Type1 1:Type2	Select a motion at the time of resuming automatic operation from a block in a middle of program. 0:Type1 Resumes a program in a state that blocks from the top to just before the specified operation start point have been executed. 1:Type2 Available to select either restart in which blocks from the top to just before the specified operation start point have been executed, or sequence search in which those blocks have been skipped. Performs sequence search when operation start point is specified by execution pointer.
0019	S.T.M. return (NC language only)	0:No 1:Yes	Designate whether or not to automatically restore final S, T command value and M code when resuming program.

No.	Item name	Setting range	Description
0020	Prevent restart from middle of program	0:No 1:Yes	Available to restrict program resumption(NC) in memory operation and start point setting(dialogue) function. Unavailable to set program resumption or start point by cursor key, page key and function key on the program screen, in case of "1:Restrict". Error message of <<Resuming operation from the middle of program is restricted>> is generated if the program is activated by changing the operation start point at the top using cursor key, etc.
0021 0022 0023 0024 0025 0026 0027 0028 0029	4th row display axis 5th row display axis 6th row display axis 7th row display axis 8th row display axis 9th row display axis 10th row display axis 11th row display axis 12th row display axis	0:Unset 1:4th axis 2:5th axis 3:6th axis 4:7th axis 5:8th axis 6:P1st axis 7:P2nd axis 8:P3rd axis 9:P4th axis	Specify respective additional axes to be displayed on additional axis display part(4th to 12th-row) on coordinate display. Axes with numbers corresponding to the row numbers are assigned if they haven't been set. *4th row:4th axis, 5th row:5th axis, 6th row:6th axis, 7th row:7th axis, 8th row:8th axis, 9th row:P1st axis, 10th row:P2nd axis, 11th row:P3rd axis, 12th row:P4th axis
0030	XYZ-axis absolute coord. display (NC language only)	0:Display 1 1:Display 2 2:Display 3	Concerning the display on the position screen for X-, Y- and Z-axes absolute coordinate positions Display 1: Length/Tool position compensation is not added. Display 2: Length/Tool position compensation is added when tool length offset/tool position offset is executing. Display 3: Length/Tool position compensation is added when the next operation to be executed is tool compensation mode.
0031	Display coordinates during rotational transformation	0:After 1:Before	Displays absolute coordinate position by coordinate value after rotation if "0:After conversion" is set while coordinate rotation is performed.
0032 0033 0034 0035 0036	Macro variable 1 for position screen display Macro variable 2 for position screen display Macro variable 3 for position screen display Macro variable 4 for position screen display Macro variable 5 for position screen display (NC language only)	1~33 100~199 500~999	Set macro variables to be displayed on "Modal 2" screen of position screen. Screen turns into blank display by deleting the setting.
0037	Block No. display color (NC language only)	0:W 1:BLU 2:PP 3:GRN 4:LBLU	Specify display color for block number on program screen in memory operation mode.
0038	Pointer display color (NC language only)	0:W 1:BLU 2:PP 3:GRN 4:LBLU	Specify display color for execution pointer on program screen in memory operation mode.

No.	Item name	Setting range	Description
0039	Travel of X, Y or Z axis when tool length/tool position offset is changed (NC language only)	0:Type1 1:Type2	Specify the operation when there is a tool length offset command without a Z-axis command, or when there is a tool position compensation command without a X-, Y- and Z-axes command. Type 1: Performs compensation operation. Type 2: Does not perform compensation operation, and carries out operation with compensation starting from the next axis travel command. (Note) When using type 2, do not perform the following commands before a compensation operation. - Scaling (G51) - Programmable mirror image (G51.1) - Rotational transformation (G68/G168) - Feature coordinate setting (G68.2) - Arm type tool change (G100/M06) using G91 modal - Shift amount setting (G92)
0040	Condition to offset tool length/tool position (NC language only)	0:Type1 1:Type2	Specify the timing to resume the length/tool position offset that was cancelled temporarily while travelling along the X-, Y- or Z-axes during a reference position return operation (G28/G30) or a machine coordinate selection operation (G53). Type1: When issuing an X-, Y- or Z- command, or when issuing an H command or G43/G44/G143/G144 command
0041	Start up/cancel	0:Type1 (shortcut) 1:Type2 (detour)	Set the tool travel type to type 1 or type 2 when starting up or cancelling the tool diameter/nose R compensation mode during an operation program.
0042	Corner travel limit	0~99.999mm	When cutting outside an acute angle that is close to 90° while in tool diameter/nose R compensation mode, cancel the travel operation if it stops short of the distance that is set here.  Travel is canceled when $\Delta V_x \leq \text{Setting}$ and $\Delta V_y \leq \text{Setting}$.
0043	Automatic corner override(angle)	0~179.999deg	Use the feed override for before and after a corner when the tool travels inside a corner so that the angle is smaller than the angle specified here while engaged in tool diameter/nose R compensation mode.
0044	Automatic corner override(length 1)	0~999.999mm	
0045	Automatic corner override(length 2)	0~999.999mm	

No.	Item name	Setting range	Description
0046	Automatic corner override(ratio)	1~100%	 length 1: Corner end point deceleration distance length 2: Corner start point deceleration distance Ratio: Deceleration ratio
0047	Inner arc override limit	1~100%	When performing circular cutting on the inner side of a workpiece while engaged in tool diameter/nose R compensation mode, the cutting surface speed on the inner side of the workpiece becomes faster than the speed of the tool center. To avoid this type of situation, set the automatic override value for the inner side of the arc relative to the feed that corresponds to the ratio between the command radius R and the tool radius/nose R radius r.  Cutting surface inside the workpiece
0048	Automatic centering speed 1	1~5000mm/min	Rapid traverse during automatic centering is set.
0049	Automatic centering speed 2	1~5000mm/min	Measuring feedrate during automatic centering is set.
0050	Automatic centering return	0.000~99.999mm	Return stroke after each detection during automatic centering cycle is set.
0051	Automatic centering compensation	0.000~9.999mm	Set the detection position compensation when performing automatic centering.
0052	Count end lamp status	0:No 1:Yellow 2:Red	Specify whether or not to light the pilot lamp when the workpiece counter ends is specified. (lamp lit: yellow or red) * The setting value is transmitted to built-in PLC and the function is implemented on a circuit of the built-in PLC. The function may not work according to the setting value, if the ladder before shipment was changed.
0053	Multiple M codes in one block (NC language only)	0:No 1:Yes	When "1:Yes" is set, commands of up to three M codes are simultaneously available for a block.

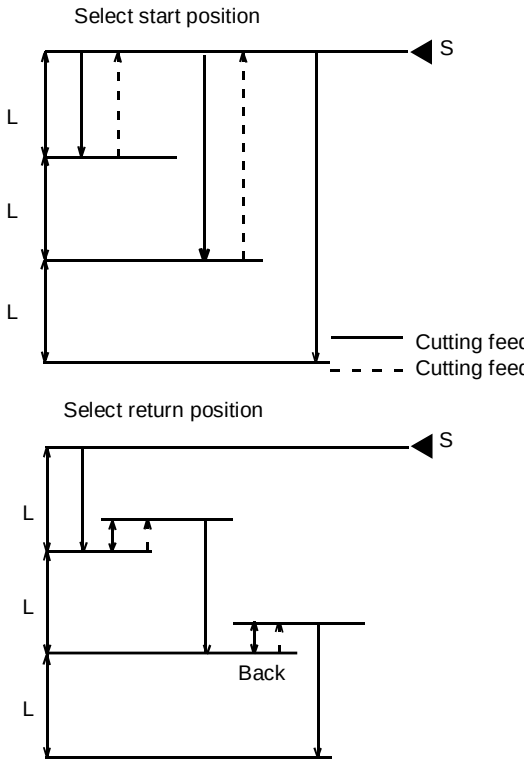
No.	Item name	Setting range	Description
0054	Unregistered M-code	0:Error 1:Invalid	Set whether to stop operation by generating an error or to continue the operation by ignoring M code, when a 3-digit M code without definition by the machine is commanded in memory operation.
0055	One shot signal M450 output time	0~99×0.1s	When signal 450 is output, it stays ON during the time set here. The next operation is performed after the set time has elapsed.
0056	One shot signal M451 output time	0~99×0.1s	When signal 451 is output, it stays ON during the time set here. The next operation is performed after the set time has elapsed.
0057	One shot signal M455 output time	0~99×0.1s	When signal 455 is output, it stays ON during the time set here. The next operation is performed before the set time has elapsed.
0058	One shot signal M456 output time	0~99×0.1s	When signal 456 is output, it stays ON during the time set here. The next operation is performed before the set time has elapsed.
0059	Automatic screen dimming time	0~999min	Automatically clears the screen when entry operation using switches are not performed within the time preset here. 0:Does not dim automatically
0060	Magazine panel automatic screen clear time	0~999min	(Reserved)
0061	Alarm display when screen is dimmed	0:Alarm screen 1:Dimming only	Set whether to change the screen when an alarm occurs while the screen is dimmed. 0:Alarm screen The alarm screen appears when an alarm occurs. 1:Dimming only The screen remains dimmed even if an alarm occurs.
0062	Machine No.	0~999999	Set numbers to machines to identify them when multiple machines are installed.
0063	Automatic door	0:No 1:Yes	Set whether or not to provide an optional automatic door.
0064	Area sensor pattern	0:Pattern 1 1:Pattern 2	Set the area sensor pattern to stop the automatic door immediately when the beam of area sensor is obstructed during automatic door operation. Pattern 1:The area sensor becomes active when the door is closing. Pattern 2:The area sensor becomes active when the door is opening or closing.
0065	Automatic door error level	0: Level 5 1: Level 4 2: Level 3 3: Level 2	Set alarm stop level for the alarm when the following errors are detected on the automatic door. - When it takes more than the time set (parameter value) for the automatic door open and close operations. - When the limit switch on the automatic door is turned ON at the end of the opening and closing operations. - When there is an error reported by the controller on the automatic door. Level 5: Servo is turned OFF Level 4: Operation stops temporarily Level 3: Operation stops at the block Level 2: Operation stops after 1 cycle The settings <2: Level 3> and <3: Level 2> are only available on machine models equipped with a QT. If these settings are used on any other machine model, the error <<User param. setting error (switch 1)>> is triggered.
0066	Automatic door operation time	0~99s	Set operating time for opening and closing of the automatic door. It falls into <<Door time-over>> error if opening or closing of door hasn't been completed within preset time. When "0" is set, the following operations aren't performed until opening or closing of door has been completed.
0067	Display alarm when door is open	0:No 1:Yes	Set whether or not to display the alarm on the screen while door is open.
0068	START lamp status when operating automatic door manually	0:Off 1:On	Set whether to turn the START lamp ON during door operation when the automatic door [MANUAL-AUTO] switch is set to "MANUAL".
0069	Override switch	0:Invalid 1:Valid	Set valid/invalid for override.

No.	Item name	Setting range	Description
0070	Alarm display when override is valid	0:No 1:Yes	Set whether or not to display the alarm when the override is not set to 100%.
0071	Prevent rapid traverse override	0:No 1:Yes	Set whether or not to prevent rapid traverse override. It is valid only when "1:Valid" is set for <Override switch>. When "1:Yes" is set, 100 % of the rapid traverse override is always applied.
0072	Prevent cutting feedrate override	0:No 1:Yes	Set whether or not to prevent cutting feedrate override. It is valid only when "1:Valid" is set for <Override switch>. When "1:Yes" is set, 100% of cutting feedrate override is always applied.
0073	Prevent spindle override	0:No 1:Yes	Set whether or not to prevent spindle override. It is valid only when "1:Valid" is set for <Override switch>. When "1:Yes" is set, 100 % of the spindle override is always applied.
0074	Change handle magnification	0:×1 1:×10	When "1:X10" is set, scaling magnification rate of handle is multiplied tenfold. This parameter is valid only when the least input increment is Type 2.
0075	Auto screen change at additional/PLC axis handle operation	0:No 1:Yes	When "1:Yes" is set, display of screen is automatically switched into a screen with manual condition if the switch of handle (axis selection) is turned to "4" in manual mode.
0076	Z-axis height to prohibit travel of additional axes	0~9999.999mm	Restricts rotation of additional axis (including QT axis, but additional axis on the side of pallet1 alone in case of QT machine) by jog operation in manual mode or handle operation when the following conditional equation is met. In this case, alarm of <<additional axis operation restriction area error>> is generated. Current Z axis machine coordinate value < "Z axis height for additional axis travel restriction"
0077	Z-axis height to prohibit travel of additional axes (pallet 2)	0~9999.999mm	Restricts rotation of additional axis (including QT axis) on the side of pallet 2 by jog operation in manual mode or handle operation when the following conditional equation is met. In this case, alarm of <<additional axis operation restriction area error>> is generated. Current Z axis machine coordinate value < "Z axis height for additional axis on the side of pallet 2 travel restriction"
0078	Prevent external operation for additional axes	0:No 1:Yes	When "1:Yes" is set, restricts rotation of additional axis of external pallet by jog operation in manual mode or handle operation. In this case, an operator message of <<external additional axis operation restriction is set>> is generated.
0079	Macro command single stop (NC language only)	0:No 1:Yes	Set whether or not to make a single stop in the block with only calculation commands or control commands (GOTO, DO, END, etc.). When "0" is set, macro command blocks are not regarded as movement blocks in cutter compensation mode.
0080	Interrupt macro interrupt type (NC language only)	0:Type1 1:Type2	Set how to interrupt a block currently being executed when interruption signal is detected. Type1:interrupt by suspending the execution Type2:interrupt without suspending the execution
0081	Interrupt macro call type (NC language only)	0:Type1 1:Type2	Set a way to call interruption program by interruption type macro. Type1:Subprogram type interruption Type2:Macro type interruption
0082	Interrupt macro acceptance type (NC language only)	0:Type1 1:Type2	Set how to receive interruption signal. Type1:Status trigger system Type2:Edge trigger system
0083	ATC simultaneous operation start position (NC language only)	0~9999.999mm	When 'R' command does not exist in G100, M6 block, the machine operates using the value preset here for the 'R' command. The value is set by machine coordinate value. This parameter is not used for ATC turret type machine.
0084	ATC reference tool length (NC language only)	-999.999 ~ 999.999mm	When tool length offset command does not exist in G100, M6 block, the machine operates using the value preset here for the tool length offset command. This parameter is not used for ATC turret type machine.

No.	Item name	Setting range	Description
0085	Automatically use S command when changing tool (NC language only)	0:No 1:Yes	Set whether or not to automatically give S command by a value registered in tool data "S command value" at tool replacement.
0086	Automatically use F command when changing tool (NC language only)	0:No 1:Yes	Set whether or not to automatically give F command by a value registered in tool data "F command value" at tool replacement.
0087	Tool data spindle rotation exceeded (NC language only)	0:Alarm 1:Clamp at max. speed	Set an operation to be performed when a rotational frequency of spindle beyond "Maximum rotational frequency" in tool data is commanded.
0088	Magazine index when changing tools of different size	0:Type1 1:Type2	Set the magazine index when Memory / MDI drive tool change to the difference radius size tool (Large tool to standard and the opposite tool change).(Medium tool handled as same as standard. Method1: Select the most nearest empty pot that present waiting pot to the next changing pot with short cut style. When present spindle cut time is shorter than magazine index time, tool change time is to be short. Method2: Select the most nearest empty pot from the next changing pot. When present spindle cut time is longer than magazine index time, tool change operation time is shortened. (Note) This parameter is not used when the machine is ATC turret type and no change to different radius tool is performed.
0089	Pot lowering start position	-999.999~999.999mm	Set the position where a pot starts to be tilted for tool replacing operation by a distance from return position at tool replacement. (It is available only for models that magazine operates in synchronization with the movement of Z axis.)
0090	XY axes movement when changing tool in manual mode	0:Invalid 1:Valid	X-/Y-axes movement is valid only on machine models where the magazine uses column movement in conjunction with X-/Y-axes. Set invalid/valid for the function to change tools at specified X-/Y-axes positions for a single tool change operation.
0091	X-axis position when changing tool in manual mode	-9999.999~9999.999mm	X-/Y-axes movement is valid only on machine models where the magazine uses column movement in conjunction with X-/Y-axes. Set the X-axis position to change tools in a single tool change operation.
0092	Y-axis position when changing tool in manual mode	-9999.999~9999.999mm	X-/Y-axes movement is valid only on machine models where the magazine uses column movement in conjunction with X-/Y-axes. Set the Y-axis position to change tools in a single tool change operation.
0093	Schedule program reset start position (Conversation only)	0:Program top 1:Schedule program top	Set a position to activate next time when reset key was pressed during schedule program operation.
0094	Program end stop in schedule program (Conversation only)	0:No 1:Yes	Set whether to stop the machine each time a machining program is completed or proceed to the next machining program when running a schedule program.
0095	Tool life counting method	0:Type1 1:Type2	Set the tool life counting method. The tool life is subtracted. When it becomes zero, tool life is expired. The tool life is added. When it reaches "Life end", tool life is expired.
0096	Change tool life display (Type2)	0:Type1 1:Type2	Valid when "1:Type2" is set for <Tool life counting method>, and sets the display method of tool life on ATC tool screen. When "0:Display1" is set, the tool life value is displayed directly When "1:Display2" is set, the remaining tool life value is displayed. (Life end minus life)

No.	Item name	Setting range	Description
0097	Spare tool change	0:At operation start 1:At tool change	Set the timing for replacing a tool to be used for program by spare tool when the tool life was expired. 0: Replaces with spare tool only at the first tool replacement after checking the life. (Don't be replaced with spare tool from the second time for replacement of the same tool.) 1: Replaces with spare tool every time when tool is replaced after the life was checked.
0098	Check tool life on first use (NC language only)	0:No 1:Yes	Set whether to check the life of the tool initially used after tape operation or extended memory operation has started. (Life is not checked when it is used for the second time and after.) This setting is valid during tool changes initiated by a T code co.
0099	External signal output time when MFIN is invalid (NC language only)	0~9999×0.1s	Set the time an M function (M00 - M99) is output when 'Does not wait for MFIN signal input' is set for macro #3003.
0100	Chip shower retention time	0~99s	Set the time that the chip shower remains ON after M401 is executed. The chip shower turns off when the time preset here has elapsed. * The setting value is transmitted to built-in PLC and the function is implemented on a circuit in the built-in PLC. The function may not work according to the setting value, if the ladder before shipment was changed.
0101	Automatic coolant off time	0~999min (0:Does not turn off automatically)	Set the time until coolant pump is turned off, when coolant pump operates in a state that the machine isn't operating (LEDs of both activation and stop are OFF). Auto-recovery is achieved by subsequent activation, external activation. Since it takes time from when pump is turned on until coolant is actually served in such a case, activation is delayed by the coolant delay time in case of auto-recovery. Coolant pump LED blinks when coolant pump was automatically turned OFF.
0102	Coolant delay time	0~99×0.1s	Set the time for waiting activation in case of coolant pump auto-recovery (activation, external activation). Activation LED is blinking even during waiting for preset time.
0103	Automatic machine light off time	0~999min (0:Does not turn off automatically)	Set the time after the machine light turns on until it turns off. When 1 or more (minute) is set, the machine light turns off when the preset time has elapsed after the machine light turned on.
0104	Standby mode time	0~999min (0:Does not turn off automatically)	Set the time after the servo motor is turned off until standby mode is entered when the machine is not operated. When no machine operation is performed even after the preset time has elapsed, the servo motors for all axes turn off and standby mode is entered, and the alarm message <<STANDBY MODE>> is displayed. (Note that this alarm is not registered in the alarm log.) Perform any of the following operations, and the servo controller turns on and operation is enabled. • All of operations on the operation panel • Operation activation from outside (activation, external activation, remote command for communication, etc.) operation
0105	Auto power off	0:No 1:Yes	Set active/inactive for automatic power off function. When "1:Yes" is set, power is automatically turned off when automatic power off time arrived. Power is also turned off during operation if the following conditions are satisfied: • Program end (Program is regarded to be ended even by restart, and external activation reservation) • Temporary suspension of operation or block stop
0106	Auto power off time	000000~235959 (00:00:00~23:59:59)	Set the time until power automatically turns off when "1:Yes" is set for "Auto power off". When "1:Yes" is set, "Auto off is on" is displayed on the screen.

No.	Item name	Setting range	Description
0107	Cancel editing when resetting	0:No 1:Yes	Set whether to cancel editing when the [RST] key is pressed.
0108	Save when mode of current program is changed (NC language only)	0:No 1:Overwrite save	Performs overwrite save of editing contents which have been made until then, if "1:Perform overwrite save" was set when mode was switched during editing a program in operation.
0109	Program edit start position during operation (NC language only)	0:Program top 1:Pointer	Displays cursor at the position of current operation pointer, if "1:Operation pointer" was set when editing of a program in operation started.
0110	Automatic sequence insertion increment (NC language only)	1~999	Set the sequence number increment when editing a program in automatic sequence number insertion mode.
0111	Set same number to edit/memory operation programs	0:No 1:Yes	Set whether or not to set the same number to edit operation program and memory operation program. When setting "0:No", specify each program number for edit/memory operation independently. When "1:Yes" is set, if either edit program number or memory program number is specified, the other program number becomes same number. A number is matched when edit/memory operation mode is switched. (Note 1)If Tape run is selected for memory operation in user parameter 1, this function is invalid. (Same as "0:No") (Note 2)This function is invalid (corresponding to "0:No") when "Operation program number", a detailed set term for current operation level, is inoperable.
0112	Assign tools when starting memory operation (Conversation only)	0:No 1:Yes	Performs tool reallocation automatically at the start of memory operation when "1:Yes" is set. It is available for a program in which tool relocation has been performed once.
0113	Check tools to be used when assigned automatically (Conversation only)	0:No 1:Yes	When tool reallocation was automatically performed at the start of memory operation by setting to "1:Yes", alarm is generated without starting operation, if a tool different from the tool which had been used until previous time is allocated.
0114	Workpc. zero conversion code (Conversation only)	0:G54,1:G55,2:G56,3:G57,4:G58,5:G59	Set the G code applied when externally converting a conversation program to a G-code program.
0115	Convert tool data (Conversation only)	0:No 1:Yes	Set whether to convert the tool list to the tool data using the G10 code when externally converting a conversation program to a G-code program. When "1:Yes" is set, the tool length compensation data is output at the top of the program and the cutter compensation data is output when performing cutter compensation.

No.	Item name	Setting range	Description
0116	Tap step return position (Conversation only)	0:Return pos. 1:Start pos.	Select the return position from the two selections below in tap step mode. 0:Return pos. The tool returns the distance set for “Tap step return” from the end position of each step. 1:Start pos. The tool always returns to the position that the fir  S: Tap Step starting position L: Cutting amount at a time * It is determined based on the method and proportion of tap step. Back: Returned amount of tap step. 2.0 fixed thread Converts like below when converting conversation program to G code program externally: 0: Return position - G77 or G78; 1: Recovery position - G277 or G278
0117	Magazine turn speed ratio 1(M436)	1~100%	Set the magazine turn speed changed by M436 as a ratio to the “Magazine rapid federate” This parameter is not used for ATC turret type machine.
0118	Magazine turn speed ratio 2(M437)	1~100%	Set the magazine turn speed changed by M437 as a ratio to the “Magazine rapid federate” This parameter is not used for ATC turret type machine.
0119	Turret type magazine speed	0: Standard 1: High-speed	Set the Turret type magazine rotating speed. “1:High speed” can be set if the condition is as follows. - The weight of per tool is not over 2Kg. - Total tool weight is not over 15Kg when the magazine is with the capacity of 14 tools. - Total tool weight is not over 21Kg when the magazine is with the capacity of 23 tools. Set it to "0:standard" when the condition isn't satisfied. Otherwise, error-stop may occur during magazine rotation. Disabled for S series.
0120	Turret type magazine low-speed turn mode	0:Invalid 1:Valid	0:Invalid, 1:Valid Set valid/invalid for turret type magazine low speed rotating mode. When "1:Valid" is set, magazine rotates by the speed specified in "turret type magazine low speed rotating".

No.	Item name	Setting range	Description
0121	Speed turret type magazine turns at low speed	0.1~500.0min ⁻¹	Set the speed of magazine rotation when "turret type magazine low speed rotation mode" is set to "1:Valid".
0122	Pallet key change	0:Invalid 1:Valid	Set whether to turn [PALLET] key to invalid or to valid. If it is turned to invalid, switching operation by [PALLET] key is unavailable.
0123	Lock shift key	0:No 1:Yes	When "1:Yes" is set, on-state and off-state is switched back and forth and the state is maintained each time when [SHIFT] key is pressed. When "0:No" is set, on-state is maintained only when [SHIFT] key is pressed.
0124	Max. actual cutting travel speed	1~999999 mm/min	Set the maximum cutting travel speed used for normal machining. As the setting decreases, the cutting time constant decreases, so the cycle time is also shorter. Note that alarm occurred the setting here applies even when a feed rate exceeding.
0125	Table load weight	0~999 kg	Set load weight on the table. Switching rapid traverse time constant by this setting value, speed-up is achieved by the shortened positioning time. Rapid traverse time constants are switched on the axes below: - Table movement model: XY axes; and Since input of false mass may cause some damage on the machine, <<load weight setting error>> is generated if any abnormality exists after checking whether the switched time constant is correct. However, if the user parameter <Positioning method> is set to <0: Non-linear interpolation type positioning>, when the rapid feed time constant changes, the rapid feed tool path changes. After this parameter is changed, check the tool path while idling, etc., and then resume the operation. (Note) Load weight estimate On machine models equipped with a moving table, the NC automatically estimates the load weight during rapid feed and displays it in the "Estimated results" on the I/O screen (Slave status display 2). When the estimate is calculated, the estimated results are displayed in [kg]. When there is no estimate, "*" is displayed. The conditions in order to estimate the load weight are as follows. - First rapid feed motion after resetting - Rapid traverse override is 100% - Travel distance is more than a certain value (When using non-linear interpolation type positioning) $\text{Travel amount} \geq (A/60) * B/1000$ (versions before W*0*.08.000) $\text{Travel amount} \geq (A/60) * (B+C) / 1000$ (versions after W*0*.08.000) (When using linear interpolation type positioning) $\text{Travel amount} \geq (A/60) * B/1000$ A: Rapid feedrate (mm/min) B: Rapid feed time constant (msec) C: Estimated speed check time (msec) There is no function to estimate the load weight on machine models equipped with column travel. The "Estimated results" on the I/O screen (Slave status display 2) always displays "*". (Machine models equipped with a quick table always display "*".)

No.	Item name	Setting range	Description
0126	Arc radius error limit	0~9999.999mm	Valid when an arc command is issued for I, J and K. Set an acceptable margin of error between the end position of a normal arc (obtained from the start position and center during a circular arc command) and the programmed end position. When the margin of error exceeds this set value, operation is stopped due to an error. Operation is stopped due to an error if the difference between the end point radius specified in the program and the end point radius that is calculated exceeds the set value. A check is performed for a spiral arc only when the radius incremental value (Q) is specified. When set to "0", it does not check.
0127	Scaling (NC language only)	0:Common scaling factor 1:Scaling factor for each axis	Set the scaling type applied. The machine will stop and an error will be output if the specified scaling type is different from the setting here.
0128	X-axis scaling (NC language only)	0:Invalid 1:Valid	Set whether X-axis scaling is valid. The machine will stop and an error will be output if X-axis scaling is validated in the program even when "Invalid" is set for the parameter.
0129	Y-axis scaling (NC language only)	0:Invalid 1:Valid	Set whether Y-axis scaling is valid. The machine will stop and an error will be output if X-axis scaling is validated in the program even when "Invalid" is set for the parameter.
0130	Z-axis scaling (NC language only)	0:Invalid 1:Valid	Set whether Z-axis scaling is valid. The machine will stop and an error will be output if X-axis scaling is validated in the program even when "Invalid" is set for the parameter.
0131	Scaling unit (NC language only)	0:0.001 1:0.00001	Set the minimum scaling unit. 0:0.001 Set 2000 for "Scaling" to double the scaling. 1:0.00001 Set 200000 for "Scaling" to double the scaling.
0132	Common scaling factor (NC language only)	-999999~ -1,1~999999	The scaling factor set here is applied when a program command does not include a scaling factor.
0133	X-axis scaling factor(I) (NC language only)	-999999~ -1,1~999999	The scaling factor set here is applied when a program command does not include a scaling factor.
0134	Y-axis scaling factor(J) (NC language only)	-999999~ -1,1~999999	The scaling factor set here is applied when a program command does not include a scaling factor.
0135	Z-axis scaling factor(K) (NC language only)	-999999~ -1,1~999999	The scaling factor set here is applied when a program command does not include a scaling factor.
0136	Back washing at end of program	0:No 1:Yes	Set whether the back washing cycle is performed when the program is completed. When "1:Yes" is selected with a CTS command included in the program, the back washing cycle is performed at the end of the program.
0137	CTS pressurize check delay time	0.0~9.9s	Set the time after the center-through-coolant is turned on by output signal M494 until the pump pressure check starts. When "0" is set, pressure check is not performed. * The setting value is transmitted to built-in PLC and the function is implemented on a circuit of the built-in PLC. The function may not work according to the setting value, if the ladder before shipment was changed.
0138	CTS depressurize check delay time	0.0~9.9s	Set the time until pump pressure check is started after turning center-through coolant off. When "0" is set, check is not performed. * The setting value is transmitted to built-in PLC and the function is implemented on a circuit of the built-in PLC. The function may not work according to the setting value, if the ladder before shipment was changed.

No.	Item name	Setting range	Description
0139	CTS pressure bleed time	0.0~9.9s	Set the time to bleed the pressure in the pump when the center-through-coolant is turned off. * The setting value is transmitted to built-in PLC and the function is implemented on a circuit of the built-in PLC. The function may not work according to the setting value, if the ladder before shipment was changed.
0140	CTS line filter alarm stop level	0:Level 2 1:Level 3 2:Level 4	Set the stop level of the alarm when clogging on CTS line filter is detected. Level 2: Operation stops after 1 cycle Level 3: Operation stops at the block Level 4: Operation stops temporarily
0141	ALM2 signal stop level	0:Level 2 1:Level 3 2:Level 4 3:Level 5	Set the stop level of alarm that turns ON external output signal <ALM2>. Stop levels are defined as follows. Level 2: Operation stops after 1 cycle Level 3: Operation stops at the block Level 4: Operation stops temporarily Level 5: Servo is turned OFF The ALM2 signal turns ON when an alarm that exceeds the preset stop level is triggered. However, the following alarms are excluded. <<*Door open>> <<*Standby mode>> <<*Enter your password.>> <<*[ENABLE] switch was turned OFF.>> <<*[MASTER ON] was turned OFF.>> <<*Handle operation switch is OFF>>
0142	Oiling/greasing when power is turned on	0:Type1 1:Type2	Set the method for oil feed or grease feed during power input, when automatic oil feed or grease feed function is valid. Method 1: Oil or grease feed is performed during power input when preset downtime has elapsed since the last oil feed or grease feed. Method 2: Oil feed or grease feed is performed during power input. If the power is turned OFF while the message <<Execute the automatic oiling / greasing cycle ON command>> is triggered, the oiling or greasing is carried out the next time the power is turned ON regardless of the setting noted above.
0143	Tool wash filter alarm stop level	0:Level 1 1:Level 2	Set the alarm stop level when an error is detected by confirmation of tool wash filter clogging. Level 1: Operation does not stop Level 2: Operation stops after 1 cycle (Note) The setting is valid only when all of the following conditions are met. <Spindle air blow/Tool wash> is set to <1:Tool wash type 1> or <3: Tool wash type 2>
0144	Tool wash filter check interval	1~999times	Set the confirmation interval of tool wash filter clogging. Confirmation of tool wash filter clogging is performed at tool replacement each time after M294 was performed by preset frequencies. (Note) The setting is valid only when all of the following conditions are met: Turret type ATC model <Spindle air blow/Tool wash> is set to <1: Tool wash type 1> or <3: Tool wash type 2> <Spindle air blow/Tool wash control method> is set to other than <3: Type 4> • If it has not been set, "1" is supposed to be selected; • Confirmation interval is cleared when parameter is changed.

No.	Item name	Setting range	Description
0145	Display tool wash filter check result	0:No 1:Yes	Set whether to display alarm when confirmation of tool wash filter clogging was normal or hasn't been performed. (Note) The setting is valid only when all of the following conditions are met: Turret type ATC model <Spindle air blow/Tool wash> is set to <1: Tool wash type 1> or <3: Tool wash type 2> <Spindle air blow/Tool wash control method> is set to other than <3: Type 4>.
0146	Faulty tool wash coolant level alarm stop(Type4)	0:Level 1 1:Level 2	Set alarm stop level for abnormal tool wash fluid surface. When "0" is set, tool wash is not turned off after alarm for abnormal tool wash fluid surface was generated. Operation is not stopped. When "1" is set, tool wash is turned off after alarm for abnormal tool wash fluid surface was generated. Operation is immediately stopped. (Note) The setting is valid only when all of the following conditions are met: <Spindle air blow/Tool wash> is set to <1: Tool wash type 1> or <3: Tool wash type 2> <Spindle air blow/Tool wash control method> is set to <3: Type 4>.
0147	Tool wash filter clogging warning alarm stop level(Type4)	0:Level 1 1:Level 2	Set the alarm stop level for the tool wash filter clogging warning. Level 1: Operation does not stop Level 2: Operation stops after 1 cycle (Note) The setting is valid only when all of the following conditions are met. <Spindle air blow/Tool wash> is set to <1: Tool wash type 1> or <3: Tool wash type 2>; <Spindle air blow/Tool wash control method> is set to <3: Type 4>
0148	No. of tool wash filter clogging warning checks(Type4)	0~999times	Set frequencies of confirmation for tool wash filter clogging notice. Notice alarm is generated if abnormality was concecutively confirmed by the preset frequencies. When "0" is set, clogging notice is not checked. (Note) The setting is valid only when all of the following conditions are met: <Spindle air blow/Tool wash> is set to <1: Tool wash type 1> or <3: Tool wash type 2> <Spindle air blow/Tool wash control method> is set to <3: Type 4>.
0149	Faulty tool wash coolant level sensor check interval(Type4)	0~999times	Set confirmation interval for tool wash fluid surface sensor abnormality. When "0" is set, abnormality confirmation for tool wash fluid surface sensor is not performed. (Note) The setting is valid only when all of the following conditions are met: <Spindle air blow/Tool wash> is set to <1: Tool wash type 1> or <3: Tool wash type 2> <Spindle air blow/Tool wash control method> is set to <3: Type 4>.
0150	Max. tapping speed (when returning tap)	1~999999 min ⁻¹	Set maximum rotational frequency for tap operation when tap return operation is performed.
0151	Motor insulation resistance measurement interval	0~999 Date	Set interval for generating operator message to urge motor insulation resistance measurement. Operator message of <<Perform motor insulation resistance measurement>> is generated during power input when preset number of days has passed since the previous measurement. When "0" is set, operator message is not generated.
0152	Motor insulation resistance monitoring acceptable value (all axes)	0.0~99.9 megohm	Set acceptable value of insulation resistance monitor for all axes measurement in motor insulation resistance measurement. If the monitor value for the insulation resistance is smaller than the set value, each axis is measured.

No.	Item name	Setting range	Description
0153	Motor insulation resistance monitoring acceptable value (abnormal)	0.0~99.9 megohm	Set acceptable value of insulation resistance monitor for single axis measurement in motor insulation resistance measurement. <<Motor insulation resistance monitor value error>> is generated when insulation resistance monitor value is lower than preset value.
0154	Motor insulation resistance monitoring acceptable value (alarm)	0.0~99.9 megohm	Set acceptable value of insulation resistance monitor for single axis measurement in motor insulation resistance measurement. <<Motor insulation resistance monitor value warning>> is generated when insulation resistance monitor value is lower than preset value.
0155	Spindle rotation return during restart	0: Do not return 1: Return	When pressing the [START] switch after returning to memory/MDI operation mode (after having changed the mode to manual mode during memory operation while the spindle (or lathe spindle) was rotating), this setting configures whether or not to return to the spindle (lathe spindle) speed/peripheral speed prior to manual operation.
0156	M function signal output delay time	1 to 9×0.01 sec.	The delay time is set for the M function signal output. - When the BCD signal turns ON, the MF turns ON after the delay time has elapsed. - When the MF turns OFF, the BCD signal turns OFF after the delay time has elapsed. (Note) This parameter is invalid for the M function signal on the built-in PLC.
0157	Complete axis travel during external signal output	0: Wait 1: Do not wait	This setting configures whether to wait or not for the axis travel to finish during the external signal output.
0158	Chip conveyor relay adhering check	0: OFF 1: ON	Set whether to check if the chip conveyor relay is sticking/adhering or not. If set to "1," it triggers the alarm <<Chip conveyor relay error> when the chip conveyor relay sticks or adheres. * The set value is transmitted to the built-in PLC and the function is implemented on a circuit in the built-in PLC. The function may not work according to the set value, if the factory-set ladder was changed.
0159	Error check when traveling during tool length/tool position offset cancel (NC language only)	0: ON 1: OFF	Set whether to check for errors or not when traveling occurs for a tool length offset cancel based on an H0 command, or a G49 command without a Z-axis command while in tool length offset mode. Set whether to check for errors or not when traveling occurs for a tool position offset cancel based on an H0 command, or a G49 command without an X-, Y- and Z-axes command while in tool position offset mode.
0160	Lathe machining infeed direction when power is turned ON (NC language only)	0 ~ 2	Sets the modal for the lathe machining infeed direction when the power is turned ON. 0: X-axis (G321) 1: Y-axis (G322) 2: Z-axis (G323)
0161	Cutting amount in thread cutting cycle (NC language only)	0 ~ 99.9	Sets the finishing amount in the thread cutting cycle. The finishing amount is the product when the lead of the thread (F command value) is multiplied by the set value. When 0 is set, the finishing is not carried out on the thread.
0162	Cutting angle in thread cutting cycle (NC language only)	0 ~ 89.999 degrees	Sets the finishing angle in the thread cutting cycle. When 0 is set, the finishing is not carried out on the thread.
0163	Thread angle in complex thread cutting cycle (NC language only)	0 ~ 120.000 degrees	Sets the thread angle in the complex thread cutting cycle.
0164	Minimum infeed amount in complex thread cutting cycle (NC language only)	0 ~ 999.999 mm	Sets the minimum infeed amount in the complex thread cutting cycle.
0165	Finishing count in complex thread cutting cycle (NC language only)	1 ~ 9999	Sets the finishing count in the complex thread cutting cycle.

No.	Item name	Setting range	Description
0166	Finishing allowance in complex thread cutting cycle (NC language only)	0 ~ 999.999 mm	Sets the finishing allowance in the complex thread cutting cycle.
0167	Signal batch reading/writing type (NC language only)	0: Type 1 1: Type 2	Set the type of data as signed or unsigned for the system variables that are read or written in batches for the interface input/output signal 32-bit. Type 1: Unsigned, Range: 0 to 4294967295 Type 2: Signed, Range: -2147483648 to 2147483647
0168	Z-axis travel method during tool change	0: Type 1 1: Type 2	Select the Z-axis travel method during a tool change. If this parameter is set to <1: Type 2>, the tool change time will take longer. However, the noise during the tool change can be decreased. (Air consumption during the tool change increases.) <1: Type 2> cannot be set depending on the model.
0169	The tapping synchronous error is too big - alarm stop level	0: Level 4 1: Level 5	Set the stop level for the alarm <<The tapping synchronous error is too big>>. Level 4: Operation stops temporarily Level 5: Servo is turned OFF
0170	Stop using feedrate override during rapid feed	0: Disabled 1: Enabled	Setting to <1: Enabled> stops operation when the feedrate override is at "0%" during rapid feed. This parameter stops operation during rapid feed only when the following conditions are met. Feedrate override is enabled Rapid traverse override is enabled [FEEDRATE OVERRIDE] switch is "0%"
0171	Arc radius R command error limit (NC language only)	0 ~ 9999.999 mm	Valid when an arc command is issued for radius R. When the radius R value is less than a value that is half of the distance between the arc start point position and the end point position, this difference represents a margin of error. When the margin of error is greater than the set value for this parameter, operation is stopped due to this error. When set to "0", this parameter uses the minimum value in accordance with the minimum unit setting.
0172	Positioning method (NC language only)	0: Non-linear interpolation type positioning 1: Linear interpolation type - Positioning method 1 2: Linear interpolation type - Positioning method 2	Select the positioning interpolation method. When set to <0: Non-linear interpolation type positioning>, the positioning operation for the X-, Y- and Z-axes and the additional axes is non-linear interpolation. When set to <1: Linear interpolation type - Positioning method 1>, the positioning operation for the X-, Y- and Z-axes and the additional axes is linear interpolation. However, there are some machines that have a restriction that only allows up to 1 additional axis to be used for simultaneous interpolation commands. In this situation, if a command is issued for two additional axes on the same block, the alarm <<Positioning command error for linear interpolation type>> is triggered. When set to <2: Linear interpolation type - Positioning method 2>, the positioning operation for the X-, Y- and Z-axes is linear interpolation, and the positioning operation for the additional axes is non-linear interpolation. Note that when the <Positioning method> is changed, the tool path during rapid feed changes. Check the path while idling, etc., and then start operation.
0173	Program load size (NC language only)	0: 2Mbyte 1: 8Mbyte	Set the possible total program size to be loaded in memory operation. If the total program size to be loaded is exceeded, the mode automatically changes to extended memory operation. If the setting <1: 8 Mbyte> is selected, it may take some time to load the program. The setting <1: 8 Mbyte> is not possible on an incompatible local PCB.

No.	Item name	Setting range	Description
0174	No tool detection function (when ATC rises)	0: Disable 1: Method 1 2: Method 2	Set the no tool detection function when ATC rises. When set to <1. Method 1>, if a tool is not fitted during the tool unclamp for the ATC operation, the alarm <<No spindle tool during ATC (when ATC rises)>> is triggered and operation stops at the Z-axis zero point after the ATC operation. When set to <2. Method 2>, the detection method is the same as <1. Method 1>, but operation does not stop after detection. In addition, the alarm <<No spindle tool during ATC (when ATC rises)>> does not trigger. The detection signal turns ON when detecting regardless of being set to <1. Method 1> or <2. Method 2>.
0175	No tool detection function (when ATC falls)	0: Disable 1: Method 1 2: Method 2	Set the no tool detection function when ATC falls. When set to <1. Method 1>, if a tool is not fitted during the tool clamp for the ATC operation, the alarm <<No spindle tool during ATC (when ATC falls)>> is triggered and operation stops at the Z-axis zero point after the ATC operation. When set to <2. Method 2>, the detection method is the same as <1. Method 1>, but operation does not stop after detection. In addition, the alarm <<No spindle tool during ATC (when ATC falls)>> does not trigger. The detection signal turns ON when detecting regardless of being set to <1. Method 1> or <2. Method 2>.
0176	Tool over key detection function	0: Disable 1: Method 1 2: Method 2	Set the tool over key detection function. When set to <1. Method 1>, if a tool goes over the spindle drive key during the tool clamp operation for the ATC operation, the alarm <<Spindle tool misaligned during ATC>> is triggered and operation stops at the Z-axis zero point after the ATC operation. When set to <2. Method 2>, the detection method is the same as <1. Method 1>, but operation does not stop after detection. In addition, the alarm <<Spindle tool misaligned during ATC>> does not trigger. The detection signal turns ON when detecting regardless of being set to <1. Method 1> or <2. Method 2>.
0177	Tool collision detection function during ATC	0: Disable 1: Enable	Enable or disable the tool collision detection function during ATC. When set to <1. Enable>, if a tool holder collides with the end of the spindle during the tool clamp operation for the ATC operation, the alarm <<Tool collision during ATC>> is triggered and operation immediately stops.
0178	Tool missing detection function	0: Disable 1: Method 1 2: Method 2	Set the tool missing detection function. When set to <1. Method 1>, if a tool is missing or is about to come off when the Z-axis raises, the alarm <<Spindle tool missing>> is triggered and operation stops. When set to <2. Method 2>, the detection method is the same as <1. Method 1>, but operation does not stop after detection. In addition, the alarm <<Spindle tool missing>> does not trigger. The detection signal turns ON when detecting regardless of being set to <1. Method 1> or <2. Method 2>.
0179	Spindle return method for program stop (NC language only)	0: Method 1 1: Method 2 2: Method 3	Set the spindle return method when pressing the [START] switch in M141 modal after the spindle stops when issuing a program stop command in memory operation. 0: Method 1 Automatically returns when operation resumes. 1: Method 2 Returns with M03/M04 command. 2: Method 3 Returns with S command and M03/M04 command..

No.	Item name	Setting range	Description
0180	Lathe spindle return method for program stop (NC language only)	0: Method 1 1: Method 2 2: Method 3	Set the lathe spindle return method when pressing the [START] switch in M142 modal after the lathe spindle stops when issuing a program stop command in memory operation. 0: Method 1 Automatically returns when operation resumes. 1: Method 2 Returns with M03/M04 command. 2: Method 3 Returns with S command and M03/M04 command..
0181	Coolant return method for program stop (NC language only)	0: Method 1 1: Method 2	Set the coolant return method when pressing the [START] switch after the coolant stops when issuing a program stop command in memory operation. 0: Method 1 Automatically returns when operation resumes. 1: Method 2 Returns with M08/M494 command. This parameter setting is for the return method of the coolant valve and the center-through-coolant.
0182	Tap accel setting	0 to 2	Set the modal for tap operation. 0: M252 (high speed) 1: M253 (medium speed) 2: M254 (low speed) This parameter initializes the set modal when the power is turned ON, when the [RST] key is pressed, when M02 or M30 is executed and when operation is reset. The settings <1:M253 (medium speed)> and <2:M254 (low speed)> are not possible depending on the machine model.
0183	Reset when [MASTER OM] is ON	0: No 1: Yes	Specify whether or not to execute a reset operation when the [MASTER ON] is set to ON.
0184	Screenshot	0: Disable 1: Enable	Enable or disable the screenshot function. When set to <1: Enable>, press the [SHIFT] + [RELSE] keys to take a screenshot of the current screen and save it to the memory card that is connected.
0185	Probe detection signal method	0: Method 1 1: Method 2	Set the accuracy of the detection signal that is obtained from the touch probe. Even if the <1: Method 2> increases the travel speed of the probe more than <0: Method 1>, there is less variation in the measurement values. As a result, the speed can be set higher than <Automatic centering speed 2> and <Measurement speed 2>. However, <1: Method 2> may not be possible depending on the machine structure, and an alarm may be triggered.
0186	PLC rapid traverse override function	0: Disable 1: Enable	Set whether to control the rapid traverse override from the PLC signal. The [RAPID TRAVERSE OVERRIDE] switch is used for control when set to <0: Disable>. Control is performed based on the values from the PLC BY378/BY379/BY37A when set to <1: Enable>. In the factory set ladder, the PLC BY378/BY379/BY37A is controlled in the same way as the [RAPID TRAVERSE OVERRIDE] switch for standard ladder program 1. (However, when the version of the factory set standard ladder program is earlier than IC00.09.B00, it is not installed.) (Note) When the user parameter (switch 1) <Prevent rapid traverse override> is set to <1: Yes>, or when the <Override switch> is set to <0: Invalid>, then the setting is fixed to 100% regardless of this parameter.

No.	Item name	Setting range	Description
0187	PLC feedrate override function	0: Disable 1: Enable	Set whether to control the cutting feedrate override from the PLC signal. The [FEEDRATE OVERRIDE] switch is used for control when set to <0: Disable>. Control is performed based on the values from the PLC BDY116 when set to <1: Enable>. In the factory set ladder, the PLC BDY116 is controlled in the same way as the [FEEDRATE OVERRIDE] switch for standard ladder program 1. (However, when the version of the factory set standard ladder program is earlier than IC00.09.B00, it is not installed.) (Note) When the user parameter (switch 1) <Prevent cutting feedrate override> is set to <1: Yes>, or when the <Override switch> is set to <0: Invalid>, then the setting is fixed to 100% regardless of this parameter.
0188	PLC spindle override function	0: Disable 1: Enable	Set whether to control the spindle override from the PLC signal. The [SPINDLE OVERRIDE] switch is used for control when set to <0: Disable>. Control is performed based on the values from the PLC BDY117 when set to <1: Enable>. In the factory set ladder, the PLC BDY117 is controlled in the same way as the [SPINDLE OVERRIDE] switch for standard ladder program 1. (However, when the version of the factory set standard ladder program is earlier than IC00.09.B00, it is not installed.) (Note) When the user parameter (switch 1) <Prevent spindle override> is set to <1: Yes>, or when the <Override switch> is set to <0: Invalid>, then the setting is fixed to 100% regardless of this parameter.
0189	ATC arm turn speed ratio 1 (M422)	1~100%	After an M422 command, set the ATC arm turn speed for the next tool change command using the standard tool speed ratio. When there is a turret type ATC mechanism, this parameter is not used. (Note) The setting cannot be used when the value is smaller than the machine parameter (system 3) <Lower limit for ATC arm turn speed ratio>.
0190	ATC arm turn speed ratio 1 (M423)	1~100%	After an M423 command, set the ATC arm turn speed for the next tool change command using the standard tool speed ratio. When there is a turret type ATC mechanism, this parameter is not used. (Note) The setting cannot be used when the value is smaller than the machine parameter (system 3) <Lower limit for ATC arm turn speed ratio>.

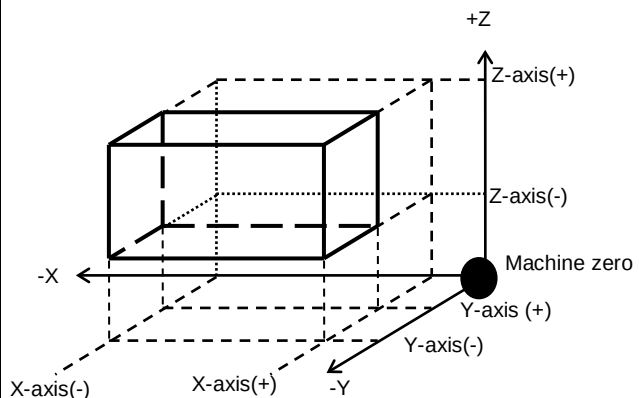
No.	Item name	Setting range	Description
0191	Tool wash discharge conditions for tool data	0: Type 1 1: Type 2	Set the discharge conditions for the <Tool wash> in the tool data. <0: Type 1> Discharge only when it is <Possible> on the current tool setting and the tool setting for the next tool. <1: Type 2> Discharge when the tool setting for the next tool is <Allowed>. (Note) In the setting <1: Type 2>, the tool wash may not discharge on a tool where it is <Not possible> due to the timing.
0192	Tool wash when changing a different size tool	0: Discharge when attaching tool to spindle 1: Discharge when returning tool to empty pot	Select whether to discharge the tool wash or not when returning the spindle's large or standard tool to an empty pot on a magazine after all the conditions below are met. - When changing a different size tool on an arm ATC mechanism - When the user parameter (switch 1) <Tool wash discharge conditions for tool data> is set to <0: Type 1> - When the <Tool wash> is set to <0: Possible> in the tool data for a spindle tool that is returned to an empty pot on a magazine <0: Discharge when attaching tool to spindle> Discharge is not carried out when returning the first large or standard spindle tool back to an empty pot on a magazine. Discharge is carried out when attaching the second standard or large tool from the magazine to the spindle. <1: Discharge when returning tool to empty pot> Discharge is carried out when returning the first large or standard spindle tool back to an empty pot on a magazine. Discharge is also carried out when attaching the standard or large tool in the second magazine to the spindle. (Note) - This parameter only applies to an arm type ATC mechanism. - When <1: Discharge when returning tool to empty pot> is selected, the discharge interval becomes shorter. Therefore, the coolant inside the chamber becomes empty and may discharge mist.
0193	Cutting speed exceeded for inverse time feed	0: Alarm 1: Clamp at maximum cutting speed	The cutting speed is calculated in the inverse time feed (G93). Set the operation when the speed in the following parameter is exceeded. - User parameter (switch 1) <Max. actual cutting travel speed> - Machine parameter (system 1) <X- to Z-axis max. cutting feedrate> - Machine parameter (system 2) <4th- to 8th-axis max. cutting rotation speed> 0: Alarm <<Max. speed exceeded>> or <<Feedrate error>> is triggered. 1: Clamp at maximum cutting speed Clamps at the minimum cutting speed set in the above parameter.
0194	Cutter compensation interference check (NC language only)	0: No 1: Type 1 2: Type 2	0: No interference check. 1: Checks for interference in the direction and arc angle 2: Checks for interference only in the arc angle
0195	Number of blocks to check cutter compensation interference (NC language only)	3 to 8	Specify the number of blocks for the interference check. (Note) A value cannot be set that is larger than the <Maximum blocks to check cutter compensation interference> in the machine parameter (system 1).

No.	Item name	Setting range	Description
0196	Involute interpolation override limit (NC language only)	0 to 100%	Set the minimum override value for the involute interpolation function when carrying out an involute interpolation while in cutter/nose R compensation mode. When setting this parameter to 0, the override value is fixed at 100% and operation is always carried out at the command speed.
0197	Involute interpolation error limit (NC language only)	0 to 9999.999 mm	When carrying out an involute interpolation command, if there is no end point on the involute curve that passes through the start point, set an allowable margin of error between the involute curve that passes through the start point and the end position of the involute curve that passes through the end point. If the margin of error exceeds the set value for this parameter, the alarm <<Involute interpolation error limit exceeded>> is triggered. There is no check carried out when this parameter is set to 0.
0198	Reset cycle time value	0: Program end 1: Memory operation start	Specify the timing when to reset the cycle time value. 0: Program end The value is reset to the <Current value> at the end of the program. However, the value does not reset when operation is interrupted or canceled in the middle of running. 1: Memory operation start The value is reset to the <Current value> at the start of memory operation.
0301	Conversation/NC language change	0:Conversation 1:NC	Set whether to use in conversation mode or in NC language mode.
0302	Operation time	0000:00:00 ~9999:59:59	Set initial value of automatic operation time (integration of hours of memory operation).
0303	Current date	2012/01/01 ~2037/12/31	Set the current date. Once it is set, the data is automatically updated.
0304	Current time	00:00:00 ~23:59:59	Set the current time. Once set, the data is automatically updated.
0305	Machine unit system	0:Metric 1:Inch	Set metric or imperial (inch) as the unit of length used for machining data.
0306	PLC ladder program protection	0:Invalid 1:Valid	0:Invalid The PLC ladder program can be written. If the communication parameter <Restrict Ethernet access> is set to <1: Yes>, communication with the PLC programming software is possible. 1:Valid Prohibit to write for PLC ladder program to protect the program. If the communication parameter <Restrict Ethernet access> is set to <1: Yes>, communication with the PLC programming software is not possible. Once the power is turned OFF, this parameter is automatically set to "1:Valid".

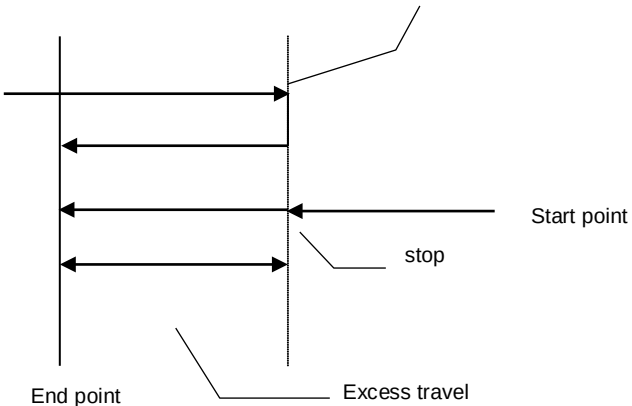
2.5.4 User Parameter 2 (Switch 2)

No.	Item name	Setting range	Description
0501	4th-axis stroke control	0:No 1:Yes	When the 4th-axis stroke is available, 1 is specified. When it is not available, 0 is specified. When <1: Yes> is specified, the values of 4th-axis stroke (-)/(+) and stroke limit for 4th-axis (-)/(+) are valid. In the following case, the error <<User param. setting error (switch 2)>> is occurs. <0: No> and <Additional axis stroke *> is set to <1: 4th axis>
0502	5th-axis stroke control	0:No 1:Yes	When the 5th-axis stroke is available, 1 is specified. When it is not available, 0 is specified. When <1: Yes> is specified, the values of 5th-axis stroke (-)/(+) and stroke limit for 5th-axis (-)/(+) are valid. If the settings are changed on the additional axis used for positioning (when the machine is not an M140X1), press the [RST] key and then turn the power OFF and ON. In the following case, the error <<User param. setting error (switch 2)>> is triggered. <1: Yes> and <Optional 5th-axis> is set to <Yes (lathe spindle)> - On M140X1 machine models, <0:No> and <Optional 5th-axis> is set to <Yes (positioning only)> <0: No> and <Additional axis stroke *> is set to <2: 5th axis>
0503	6th-axis stroke control	0:No 1:Yes	When the 6th-axis stroke is available, 1 is specified. When it is not available, 0 is specified. When <1: Yes> is specified, the values of 6th-axis stroke (-)/(+) and stroke limit for 6th-axis (-)/(+) are valid. If the settings are changed on the additional axis used for positioning (when the machine is not an M140X1), press the [RST] key and then turn the power OFF and ON. In the following case, the error <<User param. setting error (switch 2)>> is triggered. <1: Yes> and <Optional 6th-axis> is set to <Yes (lathe spindle)> - On M140X1 machine models, <0:No> and <Optional 6th-axis> is set to <Yes (positioning only)> <0: No> and <Additional axis stroke *> is set to <3: 6th axis>
0504	7th-axis stroke control	0:No 1:Yes	When the 7th-axis stroke is available, 1 is specified. When it is not available, 0 is specified. When <1: Yes> is specified, the values of 7th-axis stroke (-)/(+) and stroke limit for 7th-axis (-)/(+) are valid. If the settings are changed on the additional axis used for positioning (when the machine is not an M140X1), press the [RST] key and then turn the power OFF and ON. In the following case, the error <<User param. setting error (switch 2)>> is triggered. <1: Yes> and <Optional 7th-axis> is set to <Yes (lathe spindle)> - On M140X1 machine models, <0:No> and <Optional 7th-axis> is set to <Yes (positioning only)> <0: No> and <Additional axis stroke *> is set to <4: 7th axis>
0505	8th-axis stroke control	0:No 1:Yes	When the 8th-axis stroke is available, 1 is specified. When it is not available, 0 is specified. When <1: Yes> is specified, the values of 8th-axis stroke (-)/(+) and stroke limit for 8th-axis (-)/(+) are valid. If the settings are changed on the additional axis used for positioning (when the machine is not an M140X1), press the [RST] key and then turn the power OFF and ON. In the following case, the error <<User param. setting error (switch 2)>> is triggered. <1: Yes> and <Optional 8th-axis> is set to <Yes (lathe spindle)> - On M140X1 machine models, <0:No> and <Optional 8th-axis> is set to <Yes (positioning only)> <0: No> and <Additional axis stroke *> is set to <5: 8th axis>

No.	Item name	Setting range	Description
0506	4th-axis stroke(-)	-9999.999	When the 4-axis stroke limit is <1:Yes> the stroke limit value is input as an angle in the machine coordinates.
0511	4th-axis(+)	~9999.999deg	
0507	5th-axis stroke(-)	-9999.999	When the 5-axis stroke limit is <1:Yes> the stroke limit value is input as an angle in the machine coordinates.
0512	5th-axis(+)	~9999.999deg	
0508	6th-axis stroke(-)	-9999.999	When the 6-axis stroke limit is <1:Yes> the stroke limit value is input as an angle in the machine coordinates.
0513	6th-axis(+)	~9999.999deg	
0509	7th-axis stroke(-)	-9999.999	When the 7-axis stroke limit is <1:Yes> the stroke limit value is input as an angle in the machine coordinates.
0514	7th-axis(+)	~9999.999deg	
0510	8th-axis stroke(-)	-9999.999	When the 8-axis stroke limit is <1:Yes> the stroke limit value is input as an angle in the machine coordinates.
0515	8th-axis(+)	~9999.999deg	
0516	Stroke limit 1 for X-axis(-)	X, Y, Z Axis -9999.999 ~9999.999mm	Set the XY-axes travel range in machine coordinates. Z-axis setting differs between conversation program and NC language program. For conversation program, set the travel range of the tool top in machine coordinates. For NC language program, set the travel range of the spindle end face in machine coordinates. (-) and (+) indicate a negative and positive setting value respectively.
0517	Y-axis(-)		
0518	Z-axis(-)		
0524	X-axis(+)		
0525	Y-axis(+)		
0526	Z-axis(+)		
0519	Stroke limit 1 for 4th-axis(-)		
0527	4th-axis(+)		
0520	Stroke limit 1 for 5th-axis(-)	-9999.999 ~9999.999deg	When the 5-axis stroke limit is <1:Yes> the stroke limit value is input as an angle in the machine coordinates.
0528	5th-axis(+)		
0521	Stroke limit 1 for 6th-axis(-)	-9999.999 ~9999.999deg	When the 6-axis stroke limit is <1:Yes> the stroke limit value is input as an angle in the machine coordinates.
0529	6th-axis(+)		
0522	Stroke limit 1 for 7th-axis(-)	-9999.999 ~9999.999deg	When the 7-axis stroke limit is <1:Yes> the stroke limit value is input as an angle in the machine coordinates.
0530	7th-axis(+)		
0523	Stroke limit 1 for 8th-axis(-)	-9999.999 ~9999.999deg	When the 8-axis stroke limit is <1:Yes> the stroke limit value is input as an angle in the machine coordinates.
0531	8th-axis(+)		
0532	Stroke limit 2 for X-axis(-)	-9999.999 ~9999.999mm	Set the travel restricted range. Restricted area is not checked when the negative value is larger than, or equal to, the positive value. When XYZ-axes settings are all valid, the range defined by these settings is the restricted range. Conversation: The Z-axis travel restricted range is set by the tool tip position. NC language: The Z-axis travel restricted range is set by the spindle end face position.
0533	Y-axis(-)		
0534	Z-axis(-)		
0535	X-axis(+)		
0536	Y-axis(+)		
0537	Z-axis(+)		



No.	Item name	Setting range	Description
0538	Rapid traverse override speed 1	0~999999mm/min	Specify the feedrate of the Rapid traverse override switch on the operation panel. It becomes valid only during rapid traverse operation(XYZ axes).
0539	Rapid traverse override speed 2	0~999999mm/min	
0540	Rapid traverse override speed 3	0~999999mm/min	
0541	Rapid traverse override speed 4	0~999999mm/min	
0542	4th-axis override rotation speed 1	0.0~9999.9/min	Set the rotation speed when Speed 1 of Overdrive switch is selected.
0543	4th-axis override rotation speed 2	0.0~9999.9/min	Set the rotation speed when Speed 2 of Overdrive switch is selected.
0544	4th-axis override rotation speed 3	0.0~9999.9/min	Set the rotation speed when Speed 3 of Overdrive switch is selected.
0545	4th-axis override rotation speed 4	0.0~9999.9/min	Set the rotation speed when Speed 4 of Overdrive switch is selected.
0546	5th-axis override rotation speed 1	0.0~9999.9/min	Set the rotation speed when the override switch speed 1, 2, 3 or 4 is selected.
0547	5th-axis override rotation speed 2		If the machine parameter (system 2) <Optional 5th-axis> is set to <Yes (positioning only)> the following restrictions apply.
0548	5th-axis override rotation speed 3		· This parameter is invalid when using an M140X1.
0549	5th-axis override rotation speed 4		
0550	6th-axis override rotation speed 1	0.0~9999.9/min	Set the rotation speed when the override switch speed 1, 2, 3 or 4 is selected.
0551	6th-axis override rotation speed 2		If the machine parameter (system 2) <Optional 6th-axis> is set to <Yes (positioning only)> the following restrictions apply.
0552	6th-axis override rotation speed 3		· This parameter is invalid when using an M140X1.
0553	6th-axis override rotation speed 4		
0554	7th-axis override rotation speed 1	0.0~9999.9/min	Set the rotation speed when the override switch speed 1, 2, 3 or 4 is selected.
0555	7th-axis override rotation speed 2		If the machine parameter (system 2) <Optional 7th-axis> is set to <Yes (positioning only)> the following restrictions apply.
0556	7th-axis override rotation speed 3		· This parameter is invalid when using an M140X1.
0557	7th-axis override rotation speed 4		
0558	8th-axis override rotation speed 1	0.0~9999.9/min	Set the rotation speed when the override switch speed 1, 2, 3 or 4 is selected.
0559	8th-axis override rotation speed 2		If the machine parameter (system 2) <Optional 8th-axis> is set to <Yes (positioning only)> the following restrictions apply.
0560	8th-axis override rotation speed 3		· This parameter is invalid when using an M140X1
0561	8th-axis override rotation speed 4		
0562	X-axis position signal output range (-)	-9999.999 ~9999.999mm	External output signal XPOSSW turns on when the machine coordinate of the X-axis is within the range set for this parameter (including the boundary value) and turns off when it is not within the range. This signal does not turn on when dog zero return is completed.
0563	(+)		
0564	Y-axis position signal output range (-)	-9999.999 ~9999.999mm	External output signal YPOSSW turns on when the machine coordinate of the Y-axis is within the range set for this parameter (including the boundary value) and turns off when it is not within the range. This signal does not turn on when dog zero return is completed.
0565	(+)		
0566	Z-axis position signal output range (-)	-9999.999 ~9999.999mm	External output signal ZPOSSW turns on when the machine coordinate of the Z-axis is within the range set for this parameter (including the boundary value) and turns off when it is not within the range. This signal does not turn on when dog zero return is completed.
0567	(+)		

No.	Item name	Setting range	Description
0568 0569	4th-axis position signal output range (-) (+)	-9999.999 ~9999.999deg	External output signal 4POSSW turns on when the machine coordinate of the 4th-axis is within the range set for this parameter (including the boundary value) and turns off when it is not within the range. This signal does not turn on when dog zero return is completed.
0570 0571	5th-axis position signal output range (-) (+)	-9999.999 ~9999.999deg	External output signal 5POSSW turns on when the machine coordinate of the 5th-axis is within the range set for this parameter (including the boundary value) and turns off when it is not within the range. This signal does not turn on when dog zero return is completed.
0572 0573	6th-axis position signal output range (-) (+)	-9999.999 ~9999.999deg	External output signal 6POSSW turns on when the machine coordinate of the 6th-axis is within the range set for this parameter (including the boundary value) and turns off when it is not within the range. This signal does not turn on when dog zero return is completed.
0574 0575	7th-axis position signal output range (-) (+)	-9999.999 ~9999.999deg	External output signal 7POSSW turns on when the machine coordinate of the 7th-axis is within the range set for this parameter (including the boundary value) and turns off when it is not within the range. This signal does not turn on when dog zero return is completed.
0576 0577	8th-axis position signal output range (-) (+)	-9999.999 ~9999.999deg	External output signal 8POSSW turns on when the machine coordinate of the 8th-axis is within the range set for this parameter (including the boundary value) and turns off when it is not within the range. This signal does not turn on when dog zero return is completed.
0578 0579 0580 0581 0582 0583 0584 0585	single direction positioning excess travel amount X-axis Y-axis Z-axis 4th-axis 5th-axis 6th-axis 7th-axis 8th-axis (NC language only)	X,Y,ZAxis -99.999 ~99.999mm 4~8Axis -99.999 ~99.999deg	Set the single direction positioning travel amount for each axis. 
0586	OUTSIDE INDEX A switch address switching (NC language only)	0:No 1:A->C	Set it to "1:A->C" when "2:C" is set to machine parameter "5-axis address" - "8-axis address". Then it is possible to turn external C axis to rotate by using [OUTSIDE INDEX A] switch. Example "1:A->C" "5-axis address" "6-axis address": "2(C)" "7-axis address" "8-axis address": "1(B)" → [OUTSIDE INDEX A] : External C axis rotates [OUTSIDE INDEX B] : External B axis rotates
0587	External A-axis indexing(+)	0~999.999 deg	Set amount of travel on [OUTSIDE INDEX A] switch(+) side. * By setting <OUTSIDE INDEX A switch address switching>, the amount turns into indexing amount not of external A axis but of C axis.
0588	External A-axis indexing(-)	-999.999~0 deg	Set amount of travel on [OUTSIDE INDEX A] switch(-) side. * By setting <OUTSIDE INDEX A switch address switching>, the amount turns into indexing amount not of external A axis but of C axis.

No.	Item name	Setting range	Description
0589	OUTSIDE INDEX B switch address switching (NC language only)	0:No 1:B->C	Set it to "1:B->C" when "2:C" is set to machine parameter "5-axis address" - "8-axis address". Then it is possible to turn external C axis to rotate by using [OUTSIDE INDEX B] switch. Example "1:B->C" "5-axis address" "6-axis address": "2(C)" "7-axis address" "8-axis address": "0(A)" → [OUTSIDE INDEX A] : External A axis rotates [OUTSIDE INDEX B] : External C axis rotates
0590	External B-axis indexing(+)	0~999.999 deg	Set amount of travel on [OUTSIDE INDEX B] switch(+) side. * By setting <OUTSIDE INDEX B switch address switching>, the amount turns into indexing amount not of external B axis but of C axis.
0591	External B-axis indexing(-)	-999.999~0 deg	Set amount of travel on [OUTSIDE INDEX B] switch(-) side. * By setting <OUTSIDE INDEX B switch address switching>, the amount turns into indexing amount not of external B axis but of C axis.
0592	Z-axis lowering speed 1 when changing tool	1~999999 mm/min (0.1~99999.9 inch/min)	Set the positioning end check angle during M291 modal. From ATC zero-point to Z-axis zero-point) when changing tool in M293 modal. This parameter is used to extend the tool washing time. This is not used for the machine with Arm type ATC.
0593	Z-axis lowering speed 2 when changing tool	1~999999 mm/min (0.1~99999.9 inch/min)	Set the positioning end check angle during M292 modal. From ATC zero-point to Z-axis zero-point) when changing tool in M293 modal. This parameter is used to extend the tool washing time. This is not used for the machine with Arm type ATC.
0594	Z-axis lowering speed 3 when changing tool	1~999999 mm/min (0.1~99999.9 inch/min)	Set the positioning end check angle during M293 modal. From ATC zero-point to Z-axis zero-point) when changing tool in M293 modal. This parameter is used to extend the tool washing time. This is not used for the machine with Arm type ATC.
0595 0596	X-axis position signal output range 2 (-) (+)	-9999.999 ~9999.999mm	PLC signal XPOSSW2 turns OFF when the X-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0597 0598	Y-axis position signal output range 2 (-) (+)	-9999.999 ~9999.999mm	PLC signal YPOSSW2 turns OFF when the Y-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0599 0600	Z-axis position signal output range 2 (-) (+)	-9999.999 ~9999.999mm	PLC signal ZPOSSW2 turns OFF when the Z-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0601 0602	4th-axis position signal output range 2 (-) (+)	-9999.999 ~9999.999 deg	PLC signal 4POSSW2 turns OFF when the 4th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0603 0604	5th-axis position signal output range 2 (-) (+)	-9999.999 ~9999.999 deg	PLC signal 5POSSW2 turns OFF when the 5th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0605 0606	6th-axis position signal output range 2 (-) (+)	-9999.999 ~9999.999 deg	PLC signal 6POSSW2 turns OFF when the 6th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0607 0608	7th-axis position signal output range 2 (-) (+)	-9999.999 ~9999.999 deg	PLC signal 7POSSW2 turns OFF when the 7th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0609 0610	8th-axis position signal output range 2 (-) (+)	-9999.999 ~9999.999 deg	PLC signal 8POSSW2 turns OFF when the 8th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.

No.	Item name	Setting range	Description
0611 0612	X-axis position signal output range 3 (-) (+)	-9999.999 ~9999.999mm	PLC signal XPOSSW turns OFF when the X-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0613 0614	Y-axis position signal output range 3 (-) (+)	-9999.999 ~9999.999mm	PLC signal YPOSSW turns OFF when the Y-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0615 0616	Z-axis position signal output range 3 (-) (+)	-9999.999 ~9999.999mm	PLC signal ZPOSSW turns OFF when the Z-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0617 0618	4th-axis position signal output range 3 (-) (+)	-9999.999 ~9999.999deg	PLC signal 4POSSW turns OFF when the 4th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0619 0620	5th-axis position signal output range 3 (-) (+)	-9999.999 ~9999.999deg	PLC signal 5POSSW turns OFF when the 5th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0621 0622	6th-axis position signal output range 3 (-) (+)	-9999.999 ~9999.999deg	PLC signal 6POSSW turns OFF when the 6th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0623 0624	7th-axis position signal output range 3 (-) (+)	-9999.999 ~9999.999deg	PLC signal 7POSSW turns OFF when the 7th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0625 0626	8th-axis position signal output range 3 (-) (+)	-9999.999 ~9999.999deg	PLC signal 8POSSW turns OFF when the 8th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0627 0628	X-axis position signal output range 4 (-) (+)	-9999.999 ~9999.999mm	PLC signal XPOSSW2 turns OFF when the X-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0629 0630	Y-axis position signal output range 4 (-) (+)	-9999.999 ~9999.999mm	PLC signal YPOSSW2 turns OFF when the Y-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0631 0632	Z-axis position signal output range 4 (-) (+)	-9999.999 ~9999.999mm	PLC signal ZPOSSW2 turns OFF when the Z-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0633 0634	4th-axis position signal output range 4 (-) (+)	-9999.999 ~9999.999deg	PLC signal 4POSSW2 turns OFF when the 4th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0635 0636	5th-axis position signal output range 4 (-) (+)	-9999.999 ~9999.999deg	PLC signal 5POSSW2 turns OFF when the 5th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0637 0638	6th-axis position signal output range 4 (-) (+)	-9999.999 ~9999.999deg	PLC signal 6POSSW2 turns OFF when the 6th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0639 0640	7th-axis position signal output range 4 (-) (+)	-9999.999 ~9999.999deg	PLC signal 7POSSW2 turns OFF when the 7th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.
0641 0642	8th-axis position signal output range 4 (-) (+)	-9999.999 ~9999.999deg	PLC signal 8POSSW2 turns OFF when the 8th-axis machine coordinate is within the range set for this parameter (including the boundary value) and turns OFF when it is not within the range.

2.5.5 User Parameter 3 (Switch 3)

No.	Item name	Setting range	Description
1001	M271 positioning end check distance	0~99.999mm	Set the positioning end check distance during M271 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for XYZ-axes (linear axes) movement.
1002	M271 positioning end check angle	0~9.999deg	Set the positioning end check angle during M271 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for travel on additional axes (rotation axis), except for additional axes used for positioning.
1003	M272 positioning end check distance	0~99.999mm	Set the positioning end check distance during M272 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for XYZ-axes (linear axes) movement.
1004	M272 positioning end check angle	0~9.999deg	Set the positioning end check distance during M272 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for travel on additional axes (rotation axis), except for additional axes used for positioning.
1005	M273 positioning end check distance	0~99.999mm	Set the positioning end check distance during M273 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for XYZ-axes (linear axes) movement.
1006	M273 positioning end check angle	0~9.999deg	Set the positioning end check angle during M273 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for travel on additional axes (rotation axis), except for additional axes used for positioning.
1007	M274 positioning end check distance	0~99.999mm	Set the positioning end check distance during M274 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for XYZ-axes (linear axes) movement.
1008	M274 positioning end check angle	0~9.999deg	Set the positioning end check angle during M274 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for travel on additional axes (rotation axis), except for additional axes used for positioning.
1009	M275 positioning end check distance	0~99.999mm	Set the positioning end check distance during M275 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for XYZ-axes (linear axes) movement.
1010	M275 positioning end check angle	0~9.999deg	Set the positioning end check angle during M275 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for travel on additional axes (rotation axis), except for additional axes used for positioning.
1011	M276 positioning end check distance	0~99.999mm	Set the positioning end check angle during M276 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for XYZ-axes (linear axes) movement.
1012	M276 positioning end check angle	0~9.999deg	Set the positioning end check angle during M276 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for travel on additional axes (rotation axis), except for additional axes used for positioning.
1013	M277 positioning end check distance	0~99.999mm	Set the positioning end check angle during M277 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for XYZ-axes (linear axes) movement.

No.	Item name	Setting range	Description
1014	M277 positioning end check angle	0~9.999deg	Set the positioning end check angle during M277 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for travel on additional axes (rotation axis), except for additional axes used for positioning.
1015	M278 positioning end check distance	0~99.999mm	Set the positioning end check angle during M278 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for XYZ-axes (linear axes) movement.
1016	M278 positioning end check angle	0~9.999deg	Set the positioning end check angle during M278 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for travel on additional axes (rotation axis), except for additional axes used for positioning.
1017	M279 positioning end check distance	0~99.999mm	Set the positioning end check distance during M279 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for XYZ-axes (linear axes) movement.
1018	M279 positioning end check angle	0~9.999deg	Set the positioning end check angle during M279 modal. This setting is applied when the next operation is also positioning (when positioning is performed consecutively). This parameter is valid for travel on additional axes (rotation axis), except for additional axes used for positioning.
1019	Low priority indication alarm No. 1	0000 - 9999 : Operator message 010000 - 119999 : Alarm Top 2 digits:Category number (*1) 01:EX 02:EC 03:SV 04:NC 05:IO 06:SP 07:SM 08:SL 09:CM 10:ES 11:FC Last 4 digits: Alarm number	Set the alarm number not to flash when multiple alarms occur. The alarm message flashes in the instruction area of the screen. When multiple alarms are current, alarms with high priority flash. Although the priority is determined according to the importance of the alarm and time of occurrence, the lowest priority applies to the alarm number set for this parameter. However, this setting is invalid for alarms of release level 3. (*1) Preset value is displayed by category name. Input value: 041234 → Preset value display: NC1234
1020	Low priority indication alarm No. 2		
1021	Low priority indication alarm No. 3		
1022	Low priority indication alarm No. 4		
1023	Low priority indication alarm No. 5		
1024	Low priority indication alarm No. 6		
1025	Low priority indication alarm No. 7		
1026	Low priority indication alarm No. 8		
1027	Low priority indication alarm No. 9		
1028	Low priority indication alarm No. 10		

No.	Item name	Setting range	Description
1029	G code for modal display 1	0:No 1-18:G code for modal display	Sets the G code modal that is displayed in the <Modal info 1> screen and <MDI operation> screen.
1030	G code for modal display 2		The setting value and displayable modal are as follows:
1031	G code for modal display 3		0: Hide
1032	G code for modal display 4		1: (G00,G01,G02,G03,G02.2,G03.2,G102,G103,G202,G203,G33,G392)
1033	G code for modal display 5		2: (G17,G18,G19)
1034	G code for modal display 6		3: (G22,G23)
1035	G code for modal display 7		4: (G40,G41,G42,G141,G142)
1036	G code for modal display 8		5: (G43,G44,G143,G144,G49)
1037	G code for modal display 9		6: (G50,G51)
1038	G code for modal display 10		7: (G50.1,G51.1)
1039	G code for modal display 11		8: (G54,G55,G56,G57,G58,G59,G54.1)
1040	G code for modal display 12		9: (G54.2Pn) (Note)
1041	G code for modal display 13		10: (G61,G64)
1042	G code for modal display 14		11: (G66,G67)
1043	G code for modal display 15		12: (G68,G69,G168)
1044	G code for modal display 16 (NC language only)		13: (G73,G74,G76,G77,G78,G80,G81,G82, G83,G84,G85,G86,G87,G89,G177,G178, G181,G182,G185,G186,G189,G277,G278)
			14: (G90,G91)
			15: (G94,G95)
			16: (G96,G97)
			17: (G98,G99)
			18: (G321,G322,G323)
			(Note) The setting cannot be configured for <G code 4, 8, 12 and 16 for modal display>. In addition, the position following this setting value must be set to <0: Hide>

No.	Item name	Setting range	Description
1045	M code for modal display 1	0:No 1-36:M code for modal display	Sets the M code modal that is displayed in the <Modal info 1> screen and <MDI operation> screen.
1046	M code for modal display 2		The setting value and displayable modal are as follows:
1047	M code for modal display 3		0: Hide
1048	M code for modal display 4		1:(M03,M04,M05,M19,M111)
1049	M code for modal display 5		2:(M08,M09)
1050	M code for modal display 6		3:(M96,M97)
1051	M code for modal display 7		4:(M141,M142)
1052	M code for modal display 8		5:(M211,M221)
1053	M code for modal display 9		6:(M212,M222)
1054	M code for modal display 10		7:(M213,M223)
1055	M code for modal display 11		8:(M214,M224)
1056	M code for modal display 12		9:(M230,M231)
1057	M code for modal display 13		10:(M241,M242,M243,M244,M245,M246,M247,M248,M249,M250,M251,M252,M253,M254)
1058	M code for modal display 14		11:(M260,M261,M262,M263,M264,M265,M266,M267,M269,M280,M281,M282,M283,M284,M285,M286,M287,M289)
1059	M code for modal display 15		12:(M270,M271,M272,M273,M274,M275,M276,M277,M278,M279)
1060	M code for modal display 16 (NC language only)		13:(M290,M291,M292,M293)
			14:(M303,M304,M305)
			15:(M322,M323)
			16:(M400,M401)
			17:(M402,M403)
			18:(M404,M405)
			19:(M406,M407)
			20:(M408,M409)
			21:(M418,M419)
			22:(M430,M431)
			23:(M435,M436,M437)
			24:(M440,M441)
			25:(M442,M443)
			26:(M444,M445)
			27:(M480,M481)
			28:(M482,M483)
			29:(M484,M485)
			30:(M486,M487)
			31:(M494,M495)
			32:(M380,M381,M382,M383,M384,M385,M386,M387,M389)
			33:(M298,M299)
			35:(M420,M421,M422,M423)
			36:(M340,M341,M342,M343)
1061	Angle setting to check when additional axis for positioning is finished	0: Invalid 1: Error	When issuing an M code (M271 - M279) that sets the angle to check if the positioning operation is finished on the additional axis, use this setting to ignore the M code and continue operation or to trigger an alarm and stop operation.

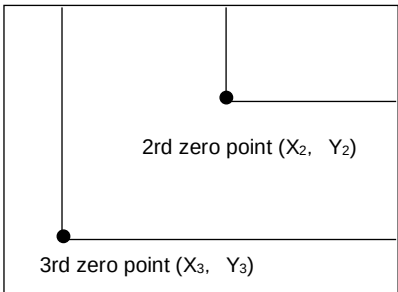
No.	Item name	Setting range	Description
1062	Specified alarm signal 2	0000 to 9999: Operator message 010000 to 119999: Alarm First two digits: Category number (Note) 01:EX 02:EC 03:SV 04:NC 05:IO 06:SP 07:SM 08:SL 09:CM 10:ES 11:FC Last four digits: Alarm number	PLC signal BX371 turns ON when the alarm/operator message specified in this parameter is triggered. (Note) The set value is shown in the category name. Input value: 041234 → Set value display: NC1234
1063	Specified alarm signal 3	0000 to 9999: Operator message 010000 to 119999: Alarm First two digits: Category number (Note) 01:EX 02:EC 03:SV 04:NC 05:IO 06:SP 07:SM 08:SL 09:CM 10:ES 11:FC Last four digits: Alarm number	PLC signal BX372 turns ON when the alarm/operator message specified in this parameter is triggered. (Note) The set value is shown in the category name. Input value: 041234 → Set value display: NC1234

No.	Item name	Setting range	Description
1064	Specified alarm signal 4	0000 to 9999: Operator message 010000 to 119999: Alarm First two digits: Category number (Note) 01:EX 02:EC 03:SV 04:NC 05:IO 06:SP 07:SM 08:SL 09:CM 10:ES 11:FC Last four digits: Alarm number	PLC signal BX373 turns ON when the alarm/operator message specified in this parameter is triggered. (Note) The set value is shown in the category name. Input value: 041234 → Set value display: NC1234
1065	Specified alarm signal 5	0000 to 9999: Operator message 010000 to 119999: Alarm First two digits: Category number (Note) 01:EX 02:EC 03:SV 04:NC 05:IO 06:SP 07:SM 08:SL 09:CM 10:ES 11:FC Last four digits: Alarm number	PLC signal BX374 turns ON when the alarm/operator message specified in this parameter is triggered. (Note) The set value is shown in the category name. Input value: 041234 → Set value display: NC1234

2.5.6 User Parameter 4 (Ref. Position/Conversation Zero Point)

No.	Item name	Setting range	Description
1501	Reference position X-axis (NC language only)	-9999.999~9999.999 mm	Set the position that the X axis should return after G28 command in the machine coordinates.
1502	Reference position Y-axis (NC language only)	-9999.999~9999.999 mm	Set the position that the Y axis should return after G28 command in the machine coordinates.
1503	Reference position Z-axis (NC language only)	-9999.999~9999.999 mm	Set the position that the Z axis should return after G28 command in the machine coordinates.
1504	Reference position 4th-axis (NC language only)	-9999.999~9999.999 deg	Set the position that the 4 axis should return after G28 command in the machine coordinates.
1505	Reference position 5th-axis (NC language only)	-9999.999~9999.999 deg	Set the position that the 5 axis should return after G28 command in the machine coordinates.
1506	Reference position 6th-axis (NC language only)	-9999.999~9999.999 deg	Set the position that the 6 axis should return after G28 command in the machine coordinates.
1507	Reference position 7th-axis (NC language only)	-9999.999~9999.999 deg	Set the position that the 7 axis should return after G28 command in the machine coordinates.
1508	Reference position 8th-axis (NC language only)	-9999.999~9999.999 deg	Set the position that the 8 axis should return after G28 command in the machine coordinates.
1509 1517 1525 1533 1541	2nd reference position X-axis 3rd reference position X-axis 4th reference position X-axis 5th reference position X-axis 6th reference position X-axis (NC language only)	-9999.999~9999.999 mm	Set the position that the X axis should return after G30 command in the machine coordinates.
1510 1518 1526 1534 1542	2nd reference position Y-axis 3rd reference position Y-axis 4th reference position Y-axis 5th reference position Y-axis 6th reference position Y-axis (NC language only)	-9999.999~9999.999 mm	Set the position that the Y axis should return after G30 command in the machine coordinates.
1511 1519 1527 1535 1543	2nd reference position Z-axis 3rd reference position Z-axis 4th reference position Z-axis 5th reference position Z-axis 6th reference position Z-axis (NC language only)	-9999.999~9999.999 mm	Set the position that the Z axis should return after G30 command in the machine coordinates.

No.	Item name	Setting range	Description
1512	2nd reference position 4th-axis	-9999.999~9999.999 deg	Set the position that the 4 axis should return after G30 command in the machine coordinates.
1520	3rd reference position 4th-axis		
1528	4th reference position 4th-axis		
1536	5th reference position 4th-axis		
1544	6th reference position 4th-axis (NC language only)		
1513	2nd reference position 5th-axis	-9999.999~9999.999 deg	Set the position that the 5 axis should return after G30 command in the machine coordinates.
1521	3rd reference position 5th-axis		
1529	4th reference position 5th-axis		
1537	5th reference position 5th-axis		
1545	6th reference position 5th-axis (NC language only)		
1514	2nd reference position 6th-axis	-9999.999~9999.999 deg	Set the position that the 6 axis should return after G30 command in the machine coordinates.
1522	3rd reference position 6th-axis		
1530	4th reference position 6th-axis		
1538	5th reference position 6th-axis		
1546	6th reference position 6th-axis (NC language only)		
1515	2nd reference position 7th-axis	-9999.999~9999.999 deg	Set the position that the 7 axis should return after G30 command in the machine coordinates.
1523	3rd reference position 7th-axis		
1531	4th reference position 7th-axis		
1539	5th reference position 7th-axis		
1547	6th reference position 7th-axis (NC language only)		
1516	2nd reference position 8th-axis	-9999.999~9999.999 deg	Set the position that the 8 axis should return after G30 command in the machine coordinates.
1524	3rd reference position 8th-axis		
1532	4th reference position 8th-axis		
1540	5th reference position 8th-axis		
1548	6th reference position 8th-axis (NC language only)		
1549	Reference point return complete signal output at servo off (NC language only)	0:No 1:Yes	When <Reference point return complete signal output at servo off> is set to "1:Yes" and distance from the reference point is within the range preset by "Reference point acceptable range at servo off", reference point return complete signal is continuously turned on even at servo off.
1550	Reference point allowable range at servo off (for XYZ axes) (NC language only)	0~9.999mm 0~0.9999inch	

No.	Item name	Setting range	Description
1551	Reference point allowable range at servo off (for additional axes) (NC language only)	0~9.999deg	
1552	X-axis zero return pos.(No.2) (Conversation only)	-9999.999~9999.999 mm (X2 in the righe figure)	Set the second and third zero point return positions in machine coordinates. 
1553	Y-axis zero return pos.(No.2) (Conversation only)	-9999.999~9999.999 mm (Y2 in the righe figure)	
1559	X-axis zero return pos.(No.3) (Conversation only)	-9999.999~9999.999 mm (X3 in the righe figure)	
1560	Y-axis zero return pos.(No.3) (Conversation only)	-9999.999~9999.999 mm (Y3 in the righe figure)	
1554	4th-axis zero return pos.(No.2) (Conversation only)	-9999.999~9999.999 deg or ? (No return)	
1555	5th-axis zero return pos.(No.2) (Conversation only)	-9999.999~9999.999 deg or ? (No return)	Set the second zero return position in machine coordinates for the 4th, 5th, 6th, 7th, and 8th axes. When "?" is set, the axis does not return to the zero point.
1556	6th-axis zero return pos.(No.2) (Conversation only)	-9999.999~9999.999 deg or ? (No return)	
1557	7th-axis zero return pos.(No.2) (Conversation only)	-9999.999~9999.999 deg or ? (No return)	
1558	8th-axis zero return pos.(No.2) (Conversation only)	-9999.999~9999.999 deg or ? (No return)	
1561	4th-axis zero return pos.(No.3) (Conversation only)	-9999.999~9999.999 deg or ? (No return)	
1562	5th-axis zero return pos.(No.3) (Conversation only)	-9999.999~9999.999 deg or ? (No return)	Set the third zero return position in machine coordinates for the 4th, 5th, 6th, 7th, and 8th axes. When "?" is set, the axis does not return to the zero point.
1563	6th-axis zero return pos.(No.3) (Conversation only)	-9999.999~9999.999 deg or ? (No return)	
1564	7th-axis zero return pos.(No.3) (Conversation only)	-9999.999~9999.999 deg or ? (No return)	
1565	8th-axis zero return pos.(No.3) (Conversation only)	-9999.999~9999.999 deg or ? (No return)	
1566	Zero point return complete signal output at servo off (Conversation only)	0:No 1:Yes	
1567	Zero point allowable range at servo off (for XYZ axes) (Conversation only)	0~9.999mm 0~0.9999inch	When <Zero point return complete signal output at servo off> is set to "1:Yes" and distance from zero point return position is within the range preset by "Zero point return acceptable range at servo off", zero point return complete signal is continuously turned on even at servo off.
1568	Zero point allowable range at servo off (for additional axes) (Conversation only)	0~9.999deg	

No.	Item name	Setting range	Description
1569	Workpiece coordinate zero point X axis (for capture) (NC language only)	-9999.999 ~9999.999mm	Set machine coordinate to be input when the function key "Parameter extraction" on the workpiece coordinate zero point screen is pressed.
1570	Workpiece coordinate zero point Y axis (for capture) (NC language only)	-9999.999 ~9999.999mm	
1571	Workpiece coordinate zero point Z axis (for capture) (NC language only)	-9999.999 ~9999.999mm	

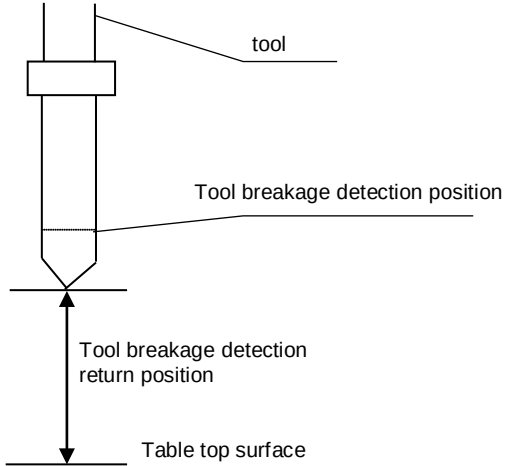
2.5.7 User Parameter 5 (User Registered Auxiliary Functions)

Setting items for option 01 to 20

No.	Item name	Setting range	Description
01~20	signal	10 - 95, 800 - 899, 900 - 999 (19 and 30 are excluded.)	Set a signal to be displayed on M code list (customer definition) screen in monitor screen.
01~20	Name	Characters that can be input from the panel: Max. 14	Register displaying name of the signal set in the above item. Display part includes operation displaying part for conversation operation in addition to monitor screen. Up to 14 characters to be input from the panel are available for registration.

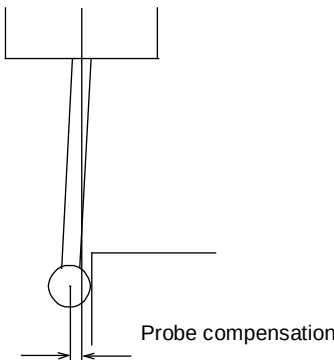
2.5.8 User Parameter 6 (Tool Breakage Detection)

No.	Item name	Setting range	Description
1801	Tool breakage detection OP	0:Not equipped 1:Type1 2:Type2 3:Type3 4:Type4	0:No 1:Type1 Tool breakage is detected by the Z-axis travel and the detection arm swing. 2:Type2 Set "Type2" when detecting tool breakage inside the magazine for arm-turn type ATC. 3:Type3 Set "Type3" when detecting tool breakage by touch signal with the table fixed. For QT installed machine, set pallet to be equipped with tool breakage detector for "Pallet 1 tool breakage detector" and "Pallet 2 tool breakage detector". "4:Type4" is set for the type of machine to perform tool breakage detection within the magazine in case of ATC with arm turning system. (Tool breakage detection is performed based on fixed tool length.) (Note) Setting is only possible for <0: Not equipped> and <3: Type 3>
1802	Pallet 1 tool breakage detector	0:No 1:Yes	For QT installed machine, when the tool breakage detector is equipped to pallet 1, set <1:Equipped>. This number is not used for the machine without QT.
1803	Pallet 2 tool breakage detector	0:No 1:Yes	For QT installed machine, when the tool breakage detector is equipped to pallet 2, set <1:Equipped>. This number is not used for the machine without QT.
1804	Tool breakage detection execution interval(No. of operations) (Conversation only)	1~999times	Set the program operation interval to execute tool breakage detection (number of times the program is run). For example, when "5" is set, tool breakage detection is executed each time the program has been run five times. When "1" is set, tool breakage detection is executed each time the program has been run once.

No.	Item name	Setting range	Description
1805	Reference tool length offset No. (NC language only)	0~99	When the program has been created using the reference tool, set the tool length data number containing the preset reference tool length.(When not using the reference tool, set "0") This parameter is valid only during the tool breakage detection. Tool detect operation is performed by adding tool length that set the tool number here and tool length that commanded tool number by length offset (H).
1806	Tool breakage detection return position (Type1)	-9999.999 ~9999.999mm	Set tool end return position in machine coordinate before tool breakage detection motion. 
1807	Tool breakage detection position X-axis (Type1)	-9999.999~0mm	Set the tool breakage detection position on the machine coordinate.
1808	Tool breakage detection position Y-axis (Type1)	-9999.999~0mm	Set the tool breakage detection position on the machine coordinate.
1809	Tool breakage detection offset (Type2)	0~99.999mm	Set the distance from the tool tip to the detection position when "2:Type2" is set for "Tool breakage detection option".
1810	Tool breakage detection position X-axis (Type3)	-9999.999 ~0.000mm	Set the position that tool breakage detection signal is turned ON on the machine coordinate. For QT installed machine, the set value is applied on pallet 1.
1811	Tool breakage detection position Y-axis (Type3)	-9999.999 ~0.000mm	Set the position that tool breakage detection signal is turned ON on the machine coordinate. For QT installed machine, the set value is applied on pallet 1.
1812	Tool breakage detection position Z-axis (Type3)	-9999.999 ~9999.999mm	Set the position that tool breakage detection signal is turned ON on the machine coordinate. For QT installed machine, the set value is applied on pallet 1.
1813	Tool breakage detection start distance (Type3) (Conversation only)	0~99.999mm	Set the cutting feed start position of tool breakage detection movement by the distance from <Tool breakage detection position Z-axis (Type3)>. For QT installed machine, the set value is applied on pallet 1.
1814	Tool breakage detection stop distance (Type3) (Conversation only)	0~99.999mm	Set the cutting feed stop position of tool breakage detection movement by the distance from <Tool breakage detection position Z-axis (Type3)>. For QT installed machine, the set value is applied on pallet 1.
1815	Tool breakage detection speed (Type3) (Conversation only)	1~5000mm/min	Set the cutting feed speed of tool breakage detection movement. For QT installed machine, the set value is applied on pallet 1.
1816	Tool breakage detection coolant OFF (Type3) (Conversation only)	0:No 1:Yes	Set whether the coolant and CT coolant is automatically performed during tool breakage detection movement. When "1:Yes" is set, coolant operation stops at the start of tool breakage detection movement and then it starts again after the detection movement is finished. However, CT coolant operation is not performed after the detection movement For QT installed machine, the set value is applied on pallet 1.

No.	Item name	Setting range	Description
1817	Pallet 2 tool breakage detection pos. X-axis (Type3)	-9999.999~0mm (-999.9999~0inch)	Set the position that tool breakage detection signal is turned ON on the machine coordinate. For QT installed machine, the set value is applied on pallet 2. This number is not used for the machine without QT.
1818	Pallet 2 tool breakage detection pos. Y-axis (Type3)	-9999.999~0mm (-999.9999~0inch)	Set the position that tool breakage detection signal is turned ON on the machine coordinate. For QT installed machine, the set value is applied on pallet 2. This number is not used for the machine without QT.
1819	Pallet 2 tool breakage detection pos. Z-axis (Type3)	-9999.999 ~9999.999mm (-999.9999 ~999.9999inch)	Set the position that tool breakage detection signal is turned ON on the machine coordinate. For QT installed machine, the set value is applied on pallet 2. This number is not used for the machine without QT.
1820	Pallet 2 tool breakage detection start distance (Type3) (Conversation only)	0~99.999mm (0~9.9999inch)	Set the cutting feed start position of tool breakage detection movement by the distance from <Tool breakage detection position Z-axis (Type3)>. For QT installed machine, the set value is applied on pallet 2. This number is not used for the machine without QT.
1821	Pallet 2 tool breakage detection stop distance (Type3) (Conversation only)	0~99.999mm (0~9.9999inch)	Set the cutting feed stop position of tool breakage detection movement by the distance from <Tool breakage detection position Z-axis (Type3)>. For QT installed machine, the set value is applied on pallet 2. This number is not used for the machine without QT.
1822	Pallet 2 tool breakage detection speed (Type3) (Conversation only)	1~5000mm/min (0.1~196.8inch/min)	Set the cutting feed speed of tool breakage detection movement. For QT installed machine, the set value is applied on pallet 2. This number is not used for the machine without QT.
1823	Pallet 2 tool breakage detection coolant OFF (Type3) (Conversation only)	0:No 1:Yes	Set whether the coolant and CT coolant is automatically performed during tool breakage detection movement. When <1:Yes> is set, coolant operation stops at the start of tool breakage detection movement and then it starts again after the detection movement is finished. However, CT coolant operation is not performed after the detection movement. For QT installed machine, the set value is applied on pallet 2. This number is not used for the machine without QT.

2.5.9 User Parameter 7 (Automatic Workpiece Measurement)

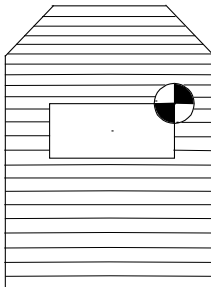
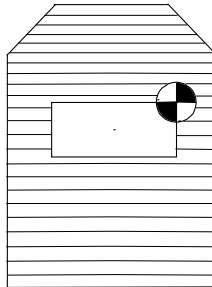
No.	Item name	Setting range	Description
2001 2002	Probe compensation 1 Probe compensation 2	-9.999~9.999mm	Set the difference between the stylus tip ball center and the spindle when the detection signal is turned on while the touch probe is attached to the spindle. X component and Y component of the difference are probe offset 1 and 2 respectively. To set the difference, carry out "Probe compensation (Corner)" of automatic centering. 

No.	Item name	Setting range	Description
2003 2004	Probe compensation 3 Probe compensation 4	-9.999~9.999mm	Set the difference between center of circle obtained by three point measurement and that of actual circle. X component and Y component of the difference correspond to probe correction amount 3 and probe correction amount 4 respectively. Perform automatic centering "Probe correction amount(3 points on circle)" at the time of automatic setting.
2005	Measurement motion	0:Type1 1:Type2	0:Type1 Performs measurement using Measurement speed 1 and Measurement speed 2. 1:Type2 Performs measurement using Measurement speed 2.
2006	Measurement speed 1	1~5000mm/min	Set the first measurement speed for Type1 measurement motion.
2007	Measurement speed 2	1~5000mm/min	Set the second measurement speed for Type1 measurement motion and the measurement speed for Type2 measurement motion.
2008	Stop point before measurement	0.000~99.999mm	Set the distance between the measurement probe end face and the workpiece surface before starting measurement skip.
2009	Measurement travel limit distance	0.000~99.999mm	Set the overtravel when the measurement skip operation exceeds the predicted value (programmed value).
2010	Measurement tolerance 1	0.000~99.999mm	When the difference between the measurement result and the predicted value (programmed value) exceeds this setting, the <<Large error in measured value (1)>> alarm occurs. When "0" is set, error checking is not performed.
2011	Measurement tolerance 2	0.000~99.999mm	When the difference between the current measurement result and the previous measurement result exceeds this setting, the <<Large error in measured value (2)>> alarm occurs. When "0" is set, error checking is not performed.
2012	Return distance after measurement	0.000~99.999mm	Set the return distance from the position the measurement probe contacted the workpiece during automatic measurement.

2.5.10 User Parameter 8 (Automatic Thermal Distortion Compensation)

No.	Item name	Setting range	Description
2501	Z-axis compensation	-999~999(%)	Adjust the Z-axis compensation within the range of -400% to 400%.
2502	Spindle compensation (Z-axis)	-999~999(%)	Adjust the spindle compensation for Z-axis within the range of -400% to 400%.
2503	X-axis compensation	-999~999(%)	Adjust the X-axis compensation within the range of -400% to 400%.
2504	Spindle compensation (X-axis)	-999~999(%)	Adjust the spindle compensation for X-axis within the range of -400% to 400%.
2505	Y-axis compensation	-999~999(%)	Adjust the Y-axis compensation within the range of -400% to 400%.
2506	Spindle compensation (Y-axis)	-999~999(%)	Adjust the spindle compensation for Y-axis within the range of -400% to 400%.
2507	Auto thermal distortion compen. at memory operation start	0:No 1:Yes	Set whether or not to perform thermal distortion compensation automatically at the start of memory operation. When "1:Yes" is set, thermal distortion compensation for XYZ axes is performed automatically on start-up of memory operation. This parameter is valid only for the machine installed with thermal distortion compensation software. Even if "1:Yes" is set to machine without the software, no operation is performed.
2508	Turning spindle compensation (For Z-axis)	-999~999(%)	Adjust the compensation value amount for the Z-axis spindle. This parameter is only valid for M type models.
2509	Turning spindle compensation (For X-axis)	-999~999(%)	Adjust the compensation value amount for the X-axis spindle. This parameter is only valid for M type models.
2510	Turning spindle compensation (For Y-axis)	-999~999(%)	Adjust the compensation value amount for the Y-axis spindle. This parameter is only valid for M type models.

2.5.11 User Parameter 9 (Quick Table)

No.	Item name	Setting range	Description
3001	Pallet program folder (NC)	Up to 8 characters among alphameric characters and period are available to input from a panel.	Set the program number processed with each pallet. Only such folders of one-hierarchy under root folder are available for setting. * When setting has not been performed,it is designated to be root folder. * When the external program selection input signal is turned on, external selection is effective.
3002	Pallet 1 program No.(NC)	1~8999	Set program number for machining by using pallet 1 and pallet 2.
3003	Pallet 2 program No.(NC)		* External program selection is valid when external program selection input signal is turned on.
3004	Pallet program folder (Conversation.)	Up to 8 characters among alphameric characters and period are available to input from a panel.	Set the program number processed with each pallet. Only such folders of one-hierarchy under root folder are available for setting. * When setting has not been performed,it is designated to be root folder. * When the external program selection input signal is turned on, external selection is effective.
3005	Pallet 1 program No. (Conversation.)	1~8999 9900~9999	Set program number for machining by using pallet 1 and pallet 2.
3006	Pallet 2 program No. (Conversation.)		* External program selection is valid when external program selection input signal is turned on.
3007	Pallet 2 shift(X-axis) (Conversation only)	-99.999~99.999	Set this parameter when machining the same workpiece on pallets 1 and 2 using the same program. The shift amount set here is applied to the machining zero point of pallet 1 when machining using pallet 2.  Pallet 1 machining zero point of pallet 1 X-100.000 Y-60.000 Z 70.000  Pallet 2 machining zero point of pallet 2 X-110.000 Y-55.000 Z 71.000 Program creation procedure is as below. 1. Create a program with the workpiece placed on pallet 1. 2. Set a workpiece on pallet 2 and measure the workpiece zero point to calculate the pallet 2 zero point. 3. Set the program numbers for pallet 1 and pallet 2. (Generally, the same program number is set.)
3008	Pallet 2 shift(Y-axis) (Conversation only)		
3009	Pallet 2 shift(Z-axis) (Conversation only)		

No.	Item name	Setting range	Description
3010 3011	Pallet 1 loading Pallet 2 loading	0:At operation start 1:At 1st tool change	Set whether to load the pallet at the start of memory operation when the first tool change is performed. This can be set for pallet 1 and pallet 2 respectively. 0:At operation start The pallet is loaded immediately after memory operation starts. After the pallet is loaded, machining starts from the top process. 1: At first tool change After memory operation starts, the pallet is loaded when the first tool change command is specified, leading to reduction of non-machining time. However, commands specified before the first tool change command are executed without the pallet loaded. The pallet is loaded at the start of operation in the following situations when using the NC language, regardless of the settings above. • When "Select memory operation" is set to "1: Tape". • When operating in the extended memory operation mode and the program size that is loaded exceeds 2MB. R650X2 When equipped with the 40 MG option: jig turning diameter ϕ enlarged, and <Move to safe position when turning pallet> (machine parameter (system 1)) is set to <1: Type 2>, the parameter cannot be set to <1:At 1st tool change>. If set to that parameter, the alarm <<User parameter setting error (Quick table)>> is triggered.

2.5.12 User Parameter 10 (High Accuracy)

(High accuracy (Adjustment))

There are machine models where the parameter does not display and it cannot be used.

No.	Item name	Setting range	Description
	Machining mode	0 ~ 9	Set the machining mode for high accuracy mode. When cutting and travelling, it operates in the mode set here. Set the desired value to support the machining content. 0: High accuracy mode off 1: Standard (standard setting that balances accuracy and smoothness) 2: Rough (setting with shorter machining time) 3: Medium rough (setting between rough and finishing) 4: Medium rough S (setting focused on machining surface quality) 5: Finishing (setting focused on true accuracy for program command) 6: Finishing S (setting focused on machining surface quality and true accuracy of program commands, machining time is longer) 7: Adjustment A (machining mode A set by customer) 8: Adjustment B (machining mode B set by customer) 9: Adjustment C (machining mode C set by customer)
	Standard mode	1 ~ 6	Adjust according to the mode set here when adjusting the machining mode. Set from <1> to <6> when using the machining mode adjustment. 0: OFF 1: Standard 2: Rough 3: Medium rough 4: Medium rough S 5: Finishing 6: Finishing S
	Accuracy	-5 ~ 5	Set the accuracy level when adjusting the machining mode. The larger the value is, the higher the machining accuracy becomes. The smaller the value, the faster the machining becomes.
	Smoothness	-5 ~ 5	Set the smoothness level on the machining surface when adjusting the machining mode. The larger the value is, the smoother the machining becomes.

(High accuracy (Accuracy specification))

There are machine models where the parameter does not display and it cannot be used.

No.	Item name	Setting range	Description
	Accuracy specification mode	0 ~ 3	The machining automatically decelerates so that if a margin of error, between the actual path and the programmed path, occurs when performing linear or circular interpolation, the difference becomes less than the value specified in the <Accuracy level>. 0: OFF 1: Accuracy spec. A 2: Accuracy spec. B 3: Accuracy spec. C
	Accuracy level	0.000 ~ 5.000 mm	Set the accuracy level for accuracy specification mode. When a value other than <0> is set, automatic deceleration is carried out so that the margin of error from the programmed path is less than the set value. When set to <0>, it does not decelerate.
	Smoothness	-5 ~ 5	Set the smoothness level on the machining surface for the accuracy specification mode. The larger the value is, the smoother the machining becomes.

(High accuracy mode A)

No.	Item name	Setting range	Description
	Corner deceleration override	0 ,10 ~ 9999 %	Used for automatic corner deceleration function. Automatic corner deceleration is performed at 100 based on the machine's unique deceleration rate. When a value larger than "100" is set, the deceleration rate is smaller, making machining time shorter. When a value smaller than "100" is set, the deceleration rate is larger, resulting in more accurate machining. When "0" is set, automatic corner deceleration is not performed. When the <Accuracy level> is set to another value besides 0, this parameter is invalid.
	Arc deceleration override	0 ,10 ~ 9999 %	Used for automatic arc deceleration. When "100" is set, automatic arc deceleration is performed based on the machine's unique deceleration rate. When a value larger than "100" is set, the deceleration rate is smaller, making machining time shorter. When a value smaller than "100" is set, the deceleration rate is larger, resulting in more accurate machining. When "0" is set, automatic arc deceleration is not performed. When the <Accuracy level> is set to another value besides 0, this parameter is invalid.
	Curve approximation deceleration override	0 ,10 ~ 9999 %	Used for automatic curve approximation deceleration. When "100" is set, automatic curve approximation deceleration is performed based on the machine's unique deceleration rate. When a value larger than "100" is set, the deceleration rate is smaller, making machining time shorter. When a value smaller than "100" is set, the deceleration rate is larger, resulting in more accurate machining. When "0" is set, automatic curve approximation deceleration is not performed. When the <Accuracy level> is set to another value besides 0, this parameter is invalid.
	Smooth path offset level	1 ~ 5	Used when <Smooth path offset function> (described later) is set to <Valid>. The larger the value is, the smoother the path becomes. Making the set value larger may cause a significant deviation from the program path. Be extra careful when changing the value.
	Smooth override	10 ~ 9999 %	When 100, it uses the cutting feed time constant that is selected in the <Cutting feed time constant selection>. Setting to a value larger than 100 results in smoother machining, and one that is smaller than 100 leads to highly accurate machining.

No.	Item name	Setting range	Description
	Cutting time constant selection	0 ~ 2	Selects the cutting feed time constant for high accuracy mode A. 0: Group 1 Uses <High accuracy A cutting feed time const. 1-3 (Group 1)> in the machine parameter (high accuracy). 1: Group 2 Uses <High accuracy A cutting feed time const. 1-3 (Group 2)> in the machine parameter (high accuracy). 2: Group 3 Uses <High accuracy A cutting feed time const. 1-3 (Group 3)> in the machine parameter (high accuracy).
	Minute block deletion distance	0.000 ~ 2.000 mm	If the travel distance of one block is less than the parameter value, the block is deleted. If <Smooth path offset function> is set to <Valid>, it is executed at a programmed point after path approximation.
	Accuracy level	0.000 ~ 5.000 mm	Used with the fully automatic deceleration function. When a value other than 0 is set, automatic deceleration is carried out so that the margin of error from the command path is less than the set value. In this situation, the following are disabled: <Corner deceleration override>, <Arc deceleration override> and <Curve approximation deceleration override>. When set to 0, the fully automatic deceleration function is disabled and the following are enabled: <Corner deceleration override>, <Arc deceleration override> and <Curve approximation deceleration override>.

(High accuracy mode B)

No.	Item name	Setting range	Description
	Corner deceleration override	0,10~9999 %	The operation is the same as high accuracy mode A.
	Arc deceleration override	0,10~9999 %	The operation is the same as high accuracy mode A.
	Curve approximation deceleration override	0,10~9999 %	The operation is the same as high accuracy mode A.
	Smooth path offset level	1~5	The operation is the same as high accuracy mode A.
	Smooth override	100~9999 %	Use machine parameter(high accuracy) "High accuracy B cutting feed time constant" at a value of 100. Setting of value larger than 100 results in smoother machining.
	Minute block deletion distance	0~2.000 mm	The operation is the same as high accuracy mode A.
	Smooth override type	0: Type 1 1: Type 2	Change the smooth override type. 0: Type 1 Even if the smooth override setting is changed, the speed difference that is permitted between blocks does not change. 1: Type 2 The speed difference that is permitted between blocks is equal to: machine parameter (high accuracy) "speed difference" × "smooth override value"/100. When type 2 is selected, increasing the set value for the smooth override makes it more difficult to decelerate at corners. Therefore, the machining time can be reduced.

(High accuracy common)

No.	Item name	Setting range	Description
3201	Smooth path offset function	0: Invalid 1: Valid	Set Smooth path offset function to valid/invalid.
3202	Smooth path offset cancel distance	0.000~99.999 mm	Use when <Smooth path offset function> is set to <Valid>. If the travel distance of one block is larger than the set value, smooth path offset is cancelled. Making the set value larger may cause a significant deviation from the program path. Be extra careful when changing the value.
3203	Smooth path offset cancel angle	0.000~120.000 deg	Use when <Smooth path offset function> is set to <Valid>. If the angle difference between blocks is larger than the set value, smooth path offset is cancelled. Making the set value larger may cause a significant deviation from the program path. Be extra careful when changing the value.
3204	Smooth path offset angle determination invalid distance	0.000~9.999 mm	Use when <Smooth path offset function> is set to <Valid>. Even if a block is cancelled smooth path offset by the determination of <Smooth path offset cancel angle>, if the travel distance of the block is equal to the set value or less, smooth path offset is not cancelled.
3205	Smooth path offset reference length	0.000~9.999 mm	Use when <Smooth path offset function> is set to <Valid>. If the travel distance of one block is longer than the set value, the path is offset using the set value.
3206	Minute block deletion when smooth path offset is cancelled	0: No 1: Yes	Used when the <Smooth path offset function> is set to <Valid>. When this parameter is set to <1: Yes>, depending on the settings for <Smooth path offset cancel distance> and the <Smooth path offset cancel angle>, the <Minute block deletion distance> applies the deletion even if the smooth path offset is cancelled for the block. (Note) Update the software version to W*10.** or later in order to change the parameter setting to <1: Yes>.
3207	Accuracy level method	0:Type1 1:Type2	Set the deceleration method when the accuracy level is specified in high accuracy mode A.

2.5.13 User Parameter 11 (PLC)

No.	Item name	Setting range	Description
3501	P1-axis stroke control	0:No 1:Yes	When stroke is set, designate "1", and when it is not set, designate "0", if PLC axis type is set to rotation axis. When <1:Yes> is selected, values of stroke P1-P4 axes (-), (+) are valid.
3502	P2-axis stroke control		
3503	P3-axis stroke control		
3504	P4-axis stroke control		
3505	P1-axis stroke(-)	-9999999~999999 99	Inputs value of stroke by machine coordinate, when PLC axis type is a linear axis or a rotation axis with a setting of "1: Yes".
3506	P2-axis stroke(-)		
3507	P3-axis stroke(-)		
3508	P4-axis stroke(-)		
3509	P1-axis stroke(+)		
3510	P2-axis stroke(+)		
3511	P3-axis stroke(+)		
3512	P4-axis stroke(+)		
3513	Reference point P1 axis	-9999999 ~ 9999999	Set a position to travel for based on reference point return command by using machine coordinate position.
3514	Reference point P2 axis		
3515	Reference point P3 axis		
3516	Reference point P4 axis		

No.	Item name	Setting range	Description
3517	P1-axis override speed 1	0 ~ 9999999	Specify override speed.
3518	P1-axis override speed 2		
3519	P1-axis override speed 3		
3520	P1-axis override speed 4		
3521	P2-axis override speed 1		
3522	P2-axis override speed 2		
3523	P2-axis override speed 3		
3524	P2-axis override speed 4		
3525	P3-axis override speed 1		
3526	P3-axis override speed 2		
3527	P3-axis override speed 3		
3528	P3-axis override speed 4		
3529	P4-axis override speed 1		
3530	P4-axis override speed 2		
3531	P4-axis override speed 3		
3532	P4-axis override speed 4		
3533	Collis. detct.-P1ax stop meth.	0: Method 1 1: Method 2	Select the stop method for the PLC axis when a collision detection alarm is triggered.
3534	Collis. detct.-P2ax stop meth.		If <0: Method 1> is selected, it stops at the same time as the rapid feed when a collision detection alarm is triggered.
3535	Collis. detct.-P3ax stop meth.		It takes a longer time to stop, but no vibrations occur caused by sudden stopping.
3536	Collis. detct.-P4ax stop meth.		If <1: Method 2> is selected, it stops in the shortest time when a collision detection alarm is triggered. Vibrations can occur due to a sudden stop.

2.5.14 User Parameter 12 (PROFIBUS DP)

Refer to “Chapter 12 (18) 1. PROFIBUS DP (Slave)” in the Operation Manual.

2.5.15 User Parameter 13 (DeviceNet)

Refer to “Chapter 12 (18) 2. DeviceNet (Slave)” in the Operation Manual.

2.5.16 User Parameter 14 (CC-Link Slave)

Refer to “Chapter 12 (18) 3. CC-Link (Remote device station)” in the Operation Manual.

2.5.17 User Parameter 15 (Rotary Fixture Offset)

Refer to “Chapter 12 (14) Rotary fixture offset function” in the Operation Manual.

2.5.18 User Parameter 17 (Feature Coordinate Setting)

Refer to "Chapter 12 (19) Feature Coordinate Setting Function” in the Operation Manual for further details.

2.5.19 User Parameter 18 (CC-Link Master)

Refer to “Chapter 12 (18) 4. CC-Link (Master station)” in the Operation Manual.

2.5.20 List of Default Values for User Parameter

2.5.20.1 Metric(S500X1,S700X1,S300X1,S1000X1, M140X1,R450X1)

User Parameter 1 (Switch 1)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0001	Select mode		0		
0002	Initial mode		0		
0003	Select memory operation		0		
0004	Program repeat		0		
0005	Workpiece counter when program is repeated		0		
0006	Tool life management when program is repeated		0		
0007	Automatic tool length compensation		0.0000		
0008	External signal reading time limit		0		
0009	Dry run offset amount		0.0000		
0010	Dry run offset amount lower limit		0.0000		
0011	Program unit		0		
0012	Program feed rate unit		0		
0013	G73 relief amount		0.5000		
0014	G83 cutting start position		0.5000		
0015	G77,G78 relief amount		1.000		
0016	G76,G87 shift direction		0		
0017	Tapping cycle return speed		0		
0018	Program restart		0		
0019	S.T.M. return		1		
0020	Prevent restart from middle of program		0		
0021	4th row display axis		1		
0022	5th row display axis		2		
0023	6th row display axis		3		
0024	7th row display axis		4		
0025	8th row display axis		5		
0026	9th row display axis		6		
0027	10th row display axis		7		
0028	11th row display axis		8		
0029	12th row display axis		9		
0030	XYZ-axis absolute coord. Display		2		
0031	Display coordinates during rotational transformation		0		
0032	Macro variable 1 for position screen display				
0033	Macro variable 2 for position screen display				
0034	Macro variable 3 for position screen display				
0035	Macro variable 4 for position screen display				
0036	Macro variable 5 for position screen display				
0037	Block No. display color		4		
0038	Pointer display color		3		
0039	X/Y/Z-axis travel at tool length/tool pos. offset change		0		
0040	Condition to offset tool length/tool position		0		
0041	Start up/Cancel		0		
0042	Corner travel limit		0.0000		
0043	Automatic corner override(angle)		0.0000		
0044	Automatic corner override(length 1)		0.0000		
0045	Automatic corner override(length 2)		0.0000		
0046	Automatic corner override(ratio)		100		
0047	Inner arc override limit		100		
0048	Automatic centering speed 1		500.0		
0049	Automatic centering speed 2		10.0		
0050	Automatic centering return		0.5000		

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0051	Automatic centering compensation	0.0000			
0052	Lamp status when count ends	0			
0053	Multiple M codes in one block	0	0	1	0
0054	Unregistered M-code	0			
0055	One shot signal M450 output time	5			
0056	One shot signal M451 output time	5			
0057	One shot signal M455 output time	5			
0058	One shot signal M456 output time	5			
0059	Automatic screen dimming time	0			
0060	Magazine panel automatic screen clear time	0			
0061	Alarm display when screen is dimmed	0			
0062	Machine No.	0			
0063	Automatic door	0			
0064	Area sensor pattern	1			
0065	Automatic door error level	1			
0066	Automatic door operation time	5			
0067	Display alarm when door is open	0			
0068	START lamp status when operating automatic door manually	0			
0069	Override switch	1			
0070	Alarm display when override is valid	1			
0071	Prevent rapid traverse override	0			
0072	Prevent cutting feedrate override	0			
0073	Prevent spindle override	0			
0074	Change handle magnification	1			
0075	Auto screen change at additional/PLC axis handle operation	0			
0076	Z-axis height to prohibit travel of additional axes	0.0000	0.0000	200.0000	0.0000
0077	Z-axis height to prohibit travel of additional axes(pallet 2)	0.0000			
0078	Prevent external operation for additional axes	0			
0079	Macro command single stop	0			
0080	Interrupt macro interrupt type	0			
0081	Interrupt macro call type	0			
0082	Interrupt macro acceptance type	0			
0083	ATC simultaneous operation start position	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250) 830.0000 (High column350)	405.0000	505.0000 (Standard) 555.0000 (Low)
0084	ATC reference tool length	200.0000			
0085	Automatically use S command when changing tool	0			
0086	Automatically use F command when changing tool	0			
0087	Tool data spindle rotation exceeded	0			
0088	Magazine index when changing tools of different sizes	0			
0089	Pot lowering start position	0.0000			
0090	XY axes movement when changing tool in manual mode	0	0	0	1

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0091	X-axis position when changing tool in manual mode	0.0000	0.0000	0.0000	-450.0000 (14tools) 0.0000 (22tools)
0092	Y-axis position when changing tool in manual mode	0.0000	0.0000	0.0000	-320.0000 (14tools) 0.0000 (22tools)
0093	Schedule program reset start position	0			
0094	Program end stop in schedule program	0			
0095	Tool life counting method	0			
0096	Change tool life display(Type2)	0			
0097	Spare tool change	0			
0098	Check tool life on first use	1			
0099	External signal output time when MFIN is invalid	5			
0100	Chip shower retention time	0			
0101	Automatic coolant off time	0			
0102	Coolant delay time	0			
0103	Automatic machine light off time	0			
0104	Standby mode time	0			
0105	Auto power off	0			
0106	Auto power off time				
0107	Cancel editing when resetting	1			
0108	Save when mode of current program is changed	0			
0109	Program edit start position during operation	0			
0110	Automatic sequence insertion increment	10			
0111	Set same number to edit/memory operation programs	0			
0112	Assign tools when starting memory operation	1			
0113	Check tools to be used when assigned automatically	1			
0114	Workpc. zero conversion code	0			
0115	Convert tool data	1			
0116	Tap step return position	0			
0117	Magazine turn speed ratio 1(M436)	80			
0118	Magazine turn speed ratio 2(M437)	50			
0119	Turret type magazine speed	0			
0120	Turret type magazine low-speed turn mode	0			
0121	Speed turret type magazine turns at low speed	20.00			
0122	Pallet select key change	1			
0123	Lock shift key	0			
0124	Max. actual cutting travel speed	20000.0			
0125	Table load weight	250	300	40	240
0126	Arc radius error limit	0.0000			
0127	Scaling	0			
0128	X-axis scaling	0			
0129	Y-axis scaling	0			
0130	Z-axis scaling	0			
0131	Scaling unit	0			
0132	Common scaling factor	1000			
0133	X-axis scaling factor(I)	1000			
0134	Y-axis scaling factor(J)	1000			
0135	Z-axis scaling factor(K)	1000			
0136	Back washing at end of program	1			
0137	CTS pressurize check delay time	1.0			
0138	CTS depressurize check delay time	1.0			
0139	CTS pressure bleed time	0.0			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0140	CTS line filter alarm stop level	1			
0141	ALM2 signal stop level	0			
0142	Oiling/greasing when power is turned on	0			
0143	Tool wash filter alarm stop level	0			
0144	Tool wash filter check interval	1			
0145	Display tool wash filter check result	0			
0146	Faulty tool wash coolant level alarm stop(Type4)	1			
0147	Tool wash filter clogging warning alarm stop level(Type4)	0			
0148	No. of tool wash filter clogging warning checks(Type4)	1			
0149	Faulty tool wash coolant level sensor check interval(Type4)	200			
0150	Max. tapping speed (when returning tap)	100			
0151	Motor insulation resistance measurement interval	0			
0152	Motor insulation resistance monitoring acceptable value (all axes)	5.0			
0153	Motor insulation resistance monitoring acceptable value (abnormal)	1.1			
0154	Motor insulation resistance monitoring acceptable value (alarm)	5.0			
0155	Spindle rotation return during restart	0	0	1	0
0156	M function signal output delay time	1			
0157	Finish axis travel during external signal output	0			
0158	Chip conveyor relay adhering check	0			
0159	Error check when traveling during tool length/tool position offset cancel	0			
0160	Lathe machining infeed direction when power is turned ON	0			
0161	Cutting amount in thread cutting cycle	1.5			
0162	Cutting angle in thread cutting cycle	45.0000			
0163	Thread angle in complex thread cutting cycle	60.0000			
0164	Minimum infeed amount in complex thread cutting cycle	0.0500			
0165	Finishing count in complex thread cutting cycle	1			
0166	Finishing allowance in complex thread cutting cycle	0.0250			
0167	Signal batch reading/writing type	0			
0168	Z-axis travel method during tool change	0			
0169	Tapping sync. error too big - alarm stop level	0			
0170	Stop using feedrate override during rapid feed	0			
0171	Arc radius R command error limit	0.0010			
0172	Positioning method	0			
0173	Program load size	0			
0174	No tool detection function (when ATC rises)	0			
0175	No tool detection function (when ATC falls)	0			
0176	Tool over key detection function	0			
0177	Tool collision detection function during ATC	0			
0178	Tool missing detection function	0			
0179	Spindle return method for program stop	1			
0180	Lathe spindle return method for program stop	2			
0181	Coolant return method for program stop	1			
0182	Tap accel. Setting	0			
0183	Reset when operation preparation is ON	0			
0184	Screenshot	0			
0185	Probe detection signal method	0			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0186	PLC rapid traverse override function	0			
0187	PLC feedrate override function	0			
0188	PLC spindle override function	0			
0189	ATC arm turn speed ratio 1 (M422)	100			
0190	ATC arm turn speed ratio 2 (M423)	100			
0191	Tool wash discharge conditions for tool data	0			
0192	Tool wash when changing a different size tool	0			
0193	Cutting speed exceeded for inverse time feed	0			
0194	Cutter compensation interference check	0			
0195	Number of blocks to check cutter compensation interference	8			
0196	Involute interpolation override limit	0			
0197	Involute interpolation error limit	0.1000			
0198	Reset cycle time value	0			

User Parameter 2 (Switch 2)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0501	4th-axis stroke control	1			
0502	5th-axis stroke control	0	0	1	0
0503	6th-axis stroke control	0			
0504	7th-axis stroke control	0			
0505	8th-axis stroke control	0			
0506	4th-axis stroke(-)	-180.0000			
0507	5th-axis stroke(-)	0.0000	0.0000	-95.0000	0.0000
0508	6th-axis stroke(-)	0.0000			
0509	7th-axis stroke(-)	0.0000			
0510	8th-axis stroke(-)	0.0000			
0511	4th-axis stroke(+)	0.0000			
0512	5th-axis stroke(+)	0.0000	0.0000	5.0000	0.0000
0513	6th-axis stroke(+)	0.0000			
0514	7th-axis stroke(+)	0.0000			
0515	8th-axis stroke(+)	0.0000			
0516	Stroke limit 1 for X-axis(-)	-500.0000 (Standard) -700.0000 (Long) -300.0000 (Short)	-1000.0000	-200.0000	-450.0000
0517	Stroke limit 1 for Y-axis(-)	-400.0000	-500.0000	-440.0000	-320.0000
0518	Stroke limit 1 for Z-axis(-)	0.0000			
0519	Stroke limit 1 for 4th-axis(-)	-180.0000			
0520	Stroke limit 1 for 5th-axis(-)	0.0000	0.0000	-95.0000	0.0000
0521	Stroke limit 1 for 6th-axis(-)	0.0000			
0522	Stroke limit 1 for 7th-axis(-)	0.0000			
0523	Stroke limit 1 for 8th-axis(-)	0.0000			
0524	Stroke limit 1 for X-axis(+)	0.0000			
0525	Stroke limit 1 for Y-axis(+)	0.0000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0526	Stroke limit 1 for Z-axis(+)	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250) 830.0000 (High column350)	405.0000	505.0000 (Standard) 555.0000 (Low)
0527	Stroke limit 1 for 4th-axis(+)	0.0000			
0528	Stroke limit 1 for 5th-axis(+)	0.0000	0.0000	5.0000	0.0000
0529	Stroke limit 1 for 6th-axis(+)	0.0000			
0530	Stroke limit 1 for 7th-axis(+)	0.0000			
0531	Stroke limit 1 for 8th-axis(+)	0.0000			
0532	Stroke limit 2 for X-axis(-)	0.0000			
0533	Stroke limit 2 for Y-axis(-)	0.0000			
0534	Stroke limit 2 for Z-axis(-)	0.0000			
0535	Stroke limit 2 for X-axis(+)	0.0000			
0536	Stroke limit 2 for Y-axis(+)	0.0000			
0537	Stroke limit 2 for Z-axis(+)	0.0000			
0538	Rapid traverse override speed 1	0.0			
0539	Rapid traverse override speed 2	500.0			
0540	Rapid traverse override speed 3	5000.0			
0541	Rapid traverse override speed 4	10000.0			
0542	4th-axis override rotation speed 1	0.00			
0543	4th-axis override rotation speed 2	0.30			
0544	4th-axis override rotation speed 3	2.50			
0545	4th-axis override rotation speed 4	5.00			
0546	5th-axis override rotation speed 1	0.00			
0547	5th-axis override rotation speed 2	0.50			
0548	5th-axis override rotation speed 3	5.00			
0549	5th-axis override rotation speed 4	10.00			
0550	6th-axis override rotation speed 1	0.00			
0551	6th-axis override rotation speed 2	0.50			
0552	6th-axis override rotation speed 3	5.00			
0553	6th-axis override rotation speed 4	10.00			
0554	7th-axis override rotation speed 1	0.00			
0555	7th-axis override rotation speed 2	0.50	0.50	1.00	0.50
0556	7th-axis override rotation speed 3	5.00	5.00	10.00	5.00
0557	7th-axis override rotation speed 4	10.00	10.00	30.00	10.00
0558	8th-axis override rotation speed 1	0.00			
0559	8th-axis override rotation speed 2	0.50			
0560	8th-axis override rotation speed 3	5.00			
0561	8th-axis override rotation speed 4	10.00			
0562	X-axis position signal output range(-)	0.0000			
0563	X-axis position signal output range(+)	0.0000			
0564	Y-axis position signal output range(-)	0.0000			
0565	Y-axis position signal output range(+)	0.0000			
0566	Z-axis position signal output range(-)	0.0000			
0567	Z-axis position signal output range(+)	0.0000			
0568	4th-axis position signal output range(-)	0.0000			
0569	4th-axis position signal output range(+)	0.0000			
0570	5th-axis position signal output range(-)	0.0000			
0571	5th-axis position signal output range(+)	0.0000			
0572	6th-axis position signal output range(-)	0.0000			
0573	6th-axis position signal output range(+)	0.0000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0574	7th-axis position signal output range(-)		0.0000		
0575	7th-axis position signal output range(+)		0.0000		
0576	8th-axis position signal output range(-)		0.0000		
0577	8th-axis position signal output range(+)		0.0000		
0578	X-axis single direction positioning excess travel amount		0.0000		
0579	Y-axis single direction positioning excess travel amount		0.0000		
0580	Z-axis single direction positioning excess travel amount		0.0000		
0581	4th-axis single direction positioning excess travel amount		0.0000		
0582	5th-axis single direction positioning excess travel amount		0.0000		
0583	6th-axis single direction positioning excess travel amount		0.0000		
0584	7th-axis single direction positioning excess travel amount		0.0000		
0585	8th-axis single direction positioning excess travel amount		0.0000		
0586	External index A switch address switching		0		
0587	External A-axis indexing(+)		180.0000		
0588	External A-axis indexing(-)		-180.0000		
0589	External index B switch address switching		0		
0590	External B-axis indexing(+)		180.0000		
0591	External B-axis indexing(-)		-180.0000		
0592	Z-axis lowering speed 1 when changing tool		35000.0		
0593	Z-axis lowering speed 2 when changing tool		35000.0		
0594	Z-axis lowering speed 3 when changing tool		35000.0		
0595	X-axis position signal output range 2 (-)		0.0000		
0596	X-axis position signal output range 2 (+)		0.0000		
0597	Y-axis position signal output range 2 (-)		0.0000		
0598	Y-axis position signal output range 2 (+)		0.0000		
0599	Z-axis position signal output range 2 (-)		0.0000		
0600	Z-axis position signal output range 2 (+)		0.0000		
0601	4-axis position signal output range 2 (-)		0.0000		
0602	4-axis position signal output range 2 (+)		0.0000		
0603	5-axis position signal output range 2 (-)		0.0000		
0604	5-axis position signal output range 2 (+)		0.0000		
0605	6-axis position signal output range 2 (-)		0.0000		
0606	6-axis position signal output range 2 (+)		0.0000		
0607	7-axis position signal output range 2 (-)		0.0000		
0608	7-axis position signal output range 2 (+)		0.0000		
0609	8-axis position signal output range 2 (-)		0.0000		
0610	8-axis position signal output range 2 (+)		0.0000		
0611	X-axis position signal output range 3 (-)		0.0000		
0612	X-axis position signal output range 3 (+)		0.0000		
0613	Y-axis position signal output range 3 (-)		0.0000		
0614	Y-axis position signal output range 3 (+)		0.0000		
0615	Z-axis position signal output range 3 (-)		0.0000		
0616	Z-axis position signal output range 3 (+)		0.0000		
0617	4-axis position signal output range 3 (-)		0.0000		
0618	4-axis position signal output range 3 (+)		0.0000		
0619	5-axis position signal output range 3 (-)		0.0000		
0620	5-axis position signal output range 3 (+)		0.0000		
0621	6-axis position signal output range 3 (-)		0.0000		
0622	6-axis position signal output range 3 (+)		0.0000		
0623	7-axis position signal output range 3 (-)		0.0000		

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0624	7-axis position signal output range 3 (+)		0.0000		
0625	8-axis position signal output range 3 (-)		0.0000		
0626	8-axis position signal output range 3 (+)		0.0000		
0627	X-axis position signal output range 4 (-)		0.0000		
0628	X-axis position signal output range 4 (+)		0.0000		
0629	Y-axis position signal output range 4 (-)		0.0000		
0630	Y-axis position signal output range 4 (+)		0.0000		
0631	Z-axis position signal output range 4 (-)		0.0000		
0632	Z-axis position signal output range 4 (+)		0.0000		
0633	4-axis position signal output range 4 (-)		0.0000		
0634	4-axis position signal output range 4 (+)		0.0000		
0635	5-axis position signal output range 4 (-)		0.0000		
0636	5-axis position signal output range 4 (+)		0.0000		
0637	6-axis position signal output range 4 (-)		0.0000		
0638	6-axis position signal output range 4 (+)		0.0000		
0639	7-axis position signal output range 4 (-)		0.0000		
0640	7-axis position signal output range 4 (+)		0.0000		
0641	8-axis position signal output range 4 (-)		0.0000		
0642	8-axis position signal output range 4 (+)		0.0000		

User Parameter 3 (Switch 3)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1001	M271 positioning end check distance		0.3000		
1002	M271 positioning end check angle		0.3000		
1003	M272 positioning end check distance		0.3000		
1004	M272 positioning end check angle		0.3000		
1005	M273 positioning end check distance		0.3000		
1006	M273 positioning end check angle		0.3000		
1007	M274 positioning end check distance		0.3000		
1008	M274 positioning end check angle		0.3000		
1009	M275 positioning end check distance		0.3000		
1010	M275 positioning end check angle		0.3000		
1011	M276 positioning end check distance		0.3000		
1012	M276 positioning end check angle		0.3000		
1013	M277 positioning end check distance		0.3000		
1014	M277 positioning end check angle		0.3000		
1015	M278 positioning end check distance		0.3000		
1016	M278 positioning end check angle		0.3000		
1017	M279 positioning end check distance		0.3000		
1018	M279 positioning end check angle		0.3000		
1019	Low priority indication alarm No. 1				
1020	Low priority indication alarm No. 2				
1021	Low priority indication alarm No. 3				
1022	Low priority indication alarm No. 4				
1023	Low priority indication alarm No. 5				
1024	Low priority indication alarm No. 6				
1025	Low priority indication alarm No. 7				
1026	Low priority indication alarm No. 8				
1027	Low priority indication alarm No. 9				
1028	Low priority indication alarm No. 10				
1029	G code for modal display 1		1		
1030	G code for modal display 2		2		
1031	G code for modal display 3		3		
1032	G code for modal display 4		4		
1033	G code for modal display 5		5		
1034	G code for modal display 6		6		

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1035	G code for modal display 7	7			
1036	G code for modal display 8	10			
1037	G code for modal display 9	8			
1038	G code for modal display 10	11			
1039	G code for modal display 11	12			
1040	G code for modal display 12	13			
1041	G code for modal display 13	14			
1042	G code for modal display 14	15			
1043	G code for modal display 15	16			
1044	G code for modal display 16	17			
1045	M code for modal display 1	1			
1046	M code for modal display 2	2			
1047	M code for modal display 3	3			
1048	M code for modal display 4	4			
1049	M code for modal display 5	5			
1050	M code for modal display 6	6			
1051	M code for modal display 7	7			
1052	M code for modal display 8	8			
1053	M code for modal display 9	9			
1054	M code for modal display 10	11			
1055	M code for modal display 11	12			
1056	M code for modal display 12	14			
1057	M code for modal display 13	16			
1058	M code for modal display 14	31			
1059	M code for modal display 15	0			
1060	M code for modal display 16	0			
1061	Angle setting to check when additional axis for positioning is finished	0			
1062	Specified alarm signal 2				
1063	Specified alarm signal 3				
1064	Specified alarm signal 4				
1065	Specified alarm signal 5				

User Parameter 4 (Reference position/Conversation zero point)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1501	Reference position X-axis	0.0000			
1502	Reference position Y-axis	0.0000			
1503	Reference position Z-axis	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250) 830.0000 (High column350)	405.0000	505.0000 (Standard) 555.0000 (Low)
1504	Reference position 4th-axis	0.0000			
1505	Reference position 5th-axis	0.0000			
1506	Reference position 6th-axis	0.0000			
1507	Reference position 7th-axis	0.0000			
1508	Reference position 8th-axis	0.0000			
1509	2nd reference position X-axis	0.0000			
1510	2nd reference position Y-axis	0.0000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1511	2nd reference position Z-axis	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250) 830.0000 (High column350)	405.0000	505.0000 (Standard) 555.0000 (Low)
1512	2nd reference position 4th-axis	0.0000			
1513	2nd reference position 5th-axis	0.0000			
1514	2nd reference position 6th-axis	0.0000			
1515	2nd reference position 7th-axis	0.0000			
1516	2nd reference position 8th-axis	0.0000			
1517	3rd reference position X-axis	0.0000			
1518	3rd reference position Y-axis	0.0000			
1519	3rd reference position Z-axis	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250) 830.0000 (High column350)	405.0000	505.0000 (Standard) 555.0000 (Low)
1520	3rd reference position 4th-axis	0.0000			
1521	3rd reference position 5th-axis	0.0000			
1522	3rd reference position 6th-axis	0.0000			
1523	3rd reference position 7th-axis	0.0000			
1524	3rd reference position 8th-axis	0.0000			
1525	4th reference position X-axis	0.0000			
1526	4th reference position Y-axis	0.0000			
1527	4th reference position Z-axis	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250) 830.0000 (High column350)	405.0000	505.0000 (Standard) 555.0000 (Low)
1528	4th reference position 4th-axis	0.0000			
1529	4th reference position 5th-axis	0.0000			
1530	4th reference position 6th-axis	0.0000			
1531	4th reference position 7th-axis	0.0000			
1532	4th reference position 8th-axis	0.0000			
1533	5th reference position X-axis	0.0000			
1534	5th reference position Y-axis	0.0000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1535	5th reference position Z-axis	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250) 830.0000 (High column350)	405.0000	505.0000 (Standard) 555.0000 (Low)
1536	5th reference position 4th-axis	0.0000			
1537	5th reference position 5th-axis	0.0000			
1538	5th reference position 6th-axis	0.0000			
1539	5th reference position 7th-axis	0.0000			
1540	5th reference position 8th-axis	0.0000			
1541	6th reference position X-axis	0.0000			
1542	6th reference position Y-axis	0.0000			
1543	6th reference position Z-axis	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250) 830.0000 (High column350)	405.0000	505.0000 (Standard) 555.0000 (Low)
1544	6th reference position 4th-axis	0.0000			
1545	6th reference position 5th-axis	0.0000			
1546	6th reference position 6th-axis	0.0000			
1547	6th reference position 7th-axis	0.0000			
1548	6th reference position 8th-axis	0.0000			
1549	Reference point return complete signal output at servo off	0			
1550	Reference point allowable range at servo off (for XYZ axes)	0.0000			
1551	Reference point allowable range at servo off (for additional axes)	0.0000			
1552	X-axis zero return pos.(No.2)	0.0000			
1553	Y-axis zero return pos.(No.2)	0.0000			
1554	4th-axis zero return pos.(No.2)	0.0000			
1555	5th-axis zero return pos.(No.2)	0.0000			
1556	6th-axis zero return pos.(No.2)	0.0000			
1557	7th-axis zero return pos.(No.2)	0.0000			
1558	8th-axis zero return pos.(No.2)	0.0000			
1559	X-axis zero return pos.(No.3)	0.0000			
1560	Y-axis zero return pos.(No.3)	0.0000			
1561	4th-axis zero return pos.(No.3)	0.0000			
1562	5th-axis zero return pos.(No.3)	0.0000			
1563	6th-axis zero return pos.(No.3)	0.0000			
1564	7th-axis zero return pos.(No.3)	0.0000			
1565	8th-axis zero return pos.(No.3)	0.0000			
1566	Zero point return complete signal output at servo off	0			
1567	Zero point allowable range at servo off (for XYZ axes)	0.0000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1568	Zero point allowable range at servo off (for additional axes)	0.0000			
1569	Workpiece coordinate zero point X axis (for capture)	0.0000			
1570	Workpiece coordinate zero point Y axis (for capture)	0.0000			
1571	Workpiece coordinate zero point Z axis (for capture)	0.0000			

user parameter 5(User registered miscellaneous functions)

There is not default value.

User Parameter 6 (Tool breakage detection)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1801	Tool breakage detection option	0			
1802	Pallet 1 tool breakage detector	0			
1803	Pallet 2 tool breakage detector	0			
1804	Tool breakage detection execution interval(No. of operations)	1			
1805	Reference tool length offset No.	0			
1806	Tool breakage detection return position(Type1)	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250) 830.0000 (High column350)	405.0000	505.0000 (Standard) 555.0000 (Low)
1807	Tool breakage detection position X-axis(Type1)	0.0000			
1808	Tool breakage detection position Y-axis(Type1)	0.0000			
1809	Tool breakage detection offset(Type2)	0.0000			
1810	Tool breakage detection position X-axis(Type3)	0.0000			
1811	Tool breakage detection position Y-axis(Type3)	0.0000			
1812	Tool breakage detection position Z-axis(Type3)	0.0000			
1813	Tool breakage detection start distance(Type3)	2.0000			
1814	Tool breakage detection stop distance(Type3)	1.0000			
1815	Tool breakage detection speed(Type3)	500.0			
1816	Tool breakage detection coolant OFF(Type3)	1			
1817	Pallet 2 tool breakage detection pos. X-axis(Type3)	0.0000			
1818	Pallet 2 tool breakage detection pos. Y-axis(Type3)	0.0000			
1819	Pallet 2 tool breakage detection pos. Z-axis(Type3)	0.0000			
1820	Pallet 2 tool breakage detection start distance(Type3)	2.0000			
1821	Pallet 2 tool breakage detection stop distance(Type3)	1.0000			
1822	Pallet 2 tool breakage detection speed(Type3)	500.0			
1823	Pallet 2 tool breakage detection coolant OFF(Type3)	1			

User Parameter 7 (Automatic workpiece measurement)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
2001	Probe compensation 1	0.0000			
2002	Probe compensation 2	0.0000			
2003	Probe compensation 3	0.0000			
2004	Probe compensation 4	0.0000			
2005	Measurement motion	0			
2006	Measurement speed 1	500.0			
2007	Measurement speed 2	100.0			
2008	Stop point before measurement	5.0000			
2009	Measurement travel limit distance	2.0000			
2010	Measurement tolerance 1	0.0000			
2011	Measurement tolerance 2	0.0000			
2012	Return distance after measurement	1.0000			

User Parameter 8 (Automatic thermal distortion compensation)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
2501	Z-axis compensation	100			
2502	Spindle compensation(Z-axis)	100			
2503	X-axis compensation	100			
2504	Spindle compensation(X-axis)	100			
2505	Y-axis compensation	100			
2506	Spindle compensation(Y-axis)	100			
2507	Auto. thermal distortion compen. at memory operation start	0			
2508	Lathe spindle compensation(Z-axis)	0			
2509	Lathe spindle compensation(X-axis)	0			
2510	Lathe spindle compensation(Y-axis)	0			

User Parameter 9 (Quick table)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
3001	Pallet program folder				
3002	Pallet 1 program No.				
3003	Pallet 2 program No.				
3004	Pallet program folder				
3005	Pallet 1 program No.				
3006	Pallet 2 program No.				
3007	Pallet 2 shift(X-axis)	0.0000			
3008	Pallet 2 shift(Y-axis)	0.0000			
3009	Pallet 2 shift(Z-axis)	0.0000			
3010	Pallet 1 loading	0			
3011	Pallet 2 loading	0			

**User Parameter 10(High accuracy)
High accuracy A (M260)**

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	120			
	Arc deceleration override	200			
	Curve approximation deceleration override	200			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M261)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	60			
	Arc deceleration override	140			
	Curve approximation deceleration override	140			
	Smooth path offset level	1			
	Smooth override	50			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M262)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	70			
	Arc deceleration override	70			
	Curve approximation deceleration override	70			
	Smooth path offset level	1			
	Smooth override	600			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M263)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	120			
	Arc deceleration override	200			
	Curve approximation deceleration override	200			
	Smooth path offset level	1			
	Smooth override	900			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M264)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	120			
	Arc deceleration override	200			
	Curve approximation deceleration override	200			
	Smooth path offset level	1			
	Smooth override	2500			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M265)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	600			
	Arc deceleration override	1000			
	Curve approximation deceleration override	1000			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Accuracy level	0.0000			

High accuracy A (M266)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	100			
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M267)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	100			
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy B (M280)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	0			
	Curve approximation deceleration override	0			
	Smooth path offset level	3			
	Smooth override	100			
	Minute block deletion distance	0.0100			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M281)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M282)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M283)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M284)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M285)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M286)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M287)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	100			

High accuracy common

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
3201	Smooth path offset function	0			
3202	Smooth path offset cancel distance	10.0000			
3203	Smooth path offset cancel angle	10.0000			
3204	Smooth path offset angle determination invalid distance	0.0100			
3205	Smooth path offset reference length	1.0000			
3206	Minute block deletion when smooth path offset is cancelled	0			
3207	Accuracy level method	0			

High accuracy(Adjust) common

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Machining mode	0 (Standard) (Long) (Short)			

High accuracy(Adjust) Adjust.A

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Standard mode	0 (Standard) (Long) (Short)			
	Accuracy	0 (Standard) (Long) (Short)			
	Smoothness	0 (Standard) (Long) (Short)			

High accuracy(Adjust) Adjust.B

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Standard mode	0 (Standard) (Long) (Short)			
	Accuracy	0 (Standard) (Long) (Short)			
	Smoothness	0 (Standard) (Long) (Short)			

High accuracy(Adjust) Adjust.C

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Standard mode	0 (Standard) (Long) (Short)			
	Accuracy	0 (Standard) (Long) (Short)			
	Smoothness	0 (Standard) (Long) (Short)			

High accuracy(Accuracy spec.) common

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Accuracy spec. mode	0 (Standard) (Long) (Short)			

High accuracy(Accuracy spec.) Accuracy spec. A

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Accuracy level	0.1000 (Standard) (Long) (Short)			
	Smoothness	0 (Standard) (Long) (Short)			

High accuracy(Accuracy spec.) Accuracy spec. B

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Accuracy level	0.0300 (Standard) (Long) (Short)			
	Smoothness	0 (Standard) (Long) (Short)			

High accuracy(Accuracy spec.) Accuracy spec. C

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Accuracy level	0.0100 (Standard) (Long) (Short)			
	Smoothness	0 (Standard) (Long) (Short)			

User Parameter 11 (PLC)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
3501	P1-axis stroke control	0			
3502	P2-axis stroke control	0			
3503	P3-axis stroke control	0			
3504	P4-axis stroke control	0			
3505	P1-axis stroke(-)	0			
3506	P2-axis stroke(-)	0			
3507	P3-axis stroke(-)	0			
3508	P4-axis stroke(-)	0			
3509	P1-axis stroke(+)	0			
3510	P2-axis stroke(+)	0			
3511	P3-axis stroke(+)	0			
3512	P4-axis stroke(+)	0			
3513	Reference point P1-axis	0			
3514	Reference point P2-axis	0			
3515	Reference point P3-axis	0			
3516	Reference point P4-axis	0			
3517	P1-axis override speed 1	0			
3518	P1-axis override speed 2	500			
3519	P1-axis override speed 3	5000			
3520	P1-axis override speed 4	10000			
3521	P2-axis override speed 1	0			
3522	P2-axis override speed 2	500			
3523	P2-axis override speed 3	5000			
3524	P2-axis override speed 4	10000			
3525	P3-axis override speed 1	0			
3526	P3-axis override speed 2	500			
3527	P3-axis override speed 3	5000			
3528	P3-axis override speed 4	10000			
3529	P4-axis override speed 1	0			
3530	P4-axis override speed 2	500			
3531	P4-axis override speed 3	5000			
3532	P4-axis override speed 4	10000			
3533	Collis. detct.-P1ax stop meth.	0			
3534	Collis. detct.-P2ax stop meth.	0			
3535	Collis. detct.-P3ax stop meth.	0			
3536	Collis. detct.-P4ax stop meth.	0			

User Parameter 12 (Profibus)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
4001	Node address	0			
4002	I/O setting 1				
4003	I/O setting 2				
4004	I/O setting 3				
4005	I/O setting 4				
4006	I/O setting 5				
4007	I/O setting 6				
4008	I/O setting 7				
4009	I/O setting 8				
4010	I/O setting 9				
4011	I/O setting 10				
4012	I/O setting 11				
4013	I/O setting 12				
4014	I/O setting 13				
4015	I/O setting 14				
4016	I/O setting 15				

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
4017	I/O setting 16				
4018	I/O setting 17				
4019	I/O setting 18				
4020	I/O setting 19				
4021	I/O setting 20				

User Parameter 13 (DeviceNet)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
4201	MAC ID			0	
4202	Transfer rate			0	
4203	Input bytes			0	
4204	Output bytes			0	

User Parameter 14 (CC-Link Slave)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
4401	Station number			1	
4402	Slave station category			0	
4403	Occupied number of stations			1	
4404	Transfer rate			0	
4405	Specify data clear in case of communication error			0	

User Parameter 15 (Rotary fixture offset)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
4601	XYZ-axes travel during current rotary fixture offset change			0	
4602	Rotary fixture offset reset method			0	
4603	Tilt axis 1	0	0	2	0
4604	Rotation axis 1	0	0	4	0
4605	Tilt axis 2			0	
4606	Rotation axis 2			0	
4607	Tilt axis 3			0	
4608	Rotation axis 3			0	
4609	Tilt axis 4			0	
4610	Rotation axis 4			0	
4611	Forward direction for the coordinate system on tilt axis 1	0	0	1	0
4612	Forward direction for the coordinate system on rotation axis 1	0	0	5	0
4613	Forward direction for the coordinate system on tilt axis 2			0	
4614	Forward direction for the coordinate system on rotation axis 2			0	
4615	Forward direction for the coordinate system on tilt axis 3			0	
4616	Forward direction for the coordinate system on rotation axis 3			0	
4617	Forward direction for the coordinate system on tilt axis 4			0	
4618	Forward direction for the coordinate system on rotation axis 4			0	

User Parameter 16 (3D smoothing function)
3D smoothing function(M380)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M381)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M382)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M383)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M384)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M385)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M386)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M387)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function common

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
4801	3D smoothing circular interpolation command				
4802	3D smoothing sequence number command				

User Parameter 17 (feature coordinate setting)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
10001	Tilt axis 1	0	0	2	0
10002	Rotation axis 1	0	0	4	0
10003	Tilt axis 2	0			
10004	Rotation axis 2	0			
10005	Forward direction for the coordinate system on tilt axis 1	0	0	1	0
10006	Forward direction for the coordinate system on rotation axis 1	0	0	5	0
10007	Forward direction for the coordinate system on tilt axis 2	0			
10008	Forward direction for the coordinate system on rotation axis 2	0			
10009	Rotation center X coordinate offset for tilt axis 1	0.0000			
10010	Rotation center Y coordinate offset for tilt axis 1	0.0000			
10011	Rotation center Z coordinate offset for tilt axis 1	0.0000			
10012	Rotation center X coordinate offset for rotation axis 1	0.0000			
10013	Rotation center Y coordinate offset for rotation axis 1	0.0000			
10014	Rotation center Z coordinate offset for rotation axis 1	0.0000			
10015	Rotation center X coordinate offset for tilt axis 2	0.0000			
10016	Rotation center Y coordinate offset for tilt axis 2	0.0000			
10017	Rotation center Z coordinate offset for tilt axis 2	0.0000			
10018	Rotation center X coordinate offset for rotation axis 2	0.0000			
10019	Rotation center Y coordinate offset for rotation axis 2	0.0000			
10020	Rotation center Z coordinate offset for rotation axis 2	0.0000			
10021	Absolute coordinate display format	0			

User Parameter 18 (CC-Link master)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
10201	Transfer rate	0			
10202	Self-station number	0			
10203	Interface board setting	0			
10204	Occupied number of stations (for intelligent device stations)	0			
10205	Standby master station number	0			
10206	CC-Link version	0			
10207	Retry attempts	3			
10208	Number of automatic return nodes	1			
10209	Specify action when there is a driver error	0			
10210	Specify station data clear in case of any error	0			

2.5.20.2 Inch(S500X1,S700X1,S300X1,S1000X1, M140X1,R450X1)

User Parameter 1 (Switch 1)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0001	Select mode		0		
0002	Initial mode		0		
0003	Select memory operation		0		
0004	Program repeat		0		
0005	Workpiece counter when program is repeated		0		
0006	Tool life management when program is repeated		0		
0007	Automatic tool length compensation		0.00000		
0008	External signal reading time limit		0		
0009	Dry run offset amount		0.00000		
0010	Dry run offset amount lower limit		0.00000		
0011	Program unit		0		
0012	Program feed rate unit		0		
0013	G73 relief amount		0.01960		
0014	G83 cutting start position		0.01960		
0015	G77,G78 relief amount		1.000		
0016	G76,G87 shift direction		0		
0017	Tapping cycle return speed		0		
0018	Program restart		0		
0019	S.T.M. return		1		
0020	Prevent restart from middle of program		0		
0021	4th row display axis		1		
0022	5th row display axis		2		
0023	6th row display axis		3		
0024	7th row display axis		4		
0025	8th row display axis		5		
0026	9th row display axis		6		
0027	10th row display axis		7		
0028	11th row display axis		8		
0029	12th row display axis		9		
0030	XYZ-axis absolute coord. Display		2		
0031	Display coordinates during rotational transformation		0		
0032	Macro variable 1 for position screen display				
0033	Macro variable 2 for position screen display				
0034	Macro variable 3 for position screen display				
0035	Macro variable 4 for position screen display				
0036	Macro variable 5 for position screen display				
0037	Block No. display color		4		
0038	Pointer display color		3		
0039	X/Y/Z-axis travel at tool length/tool pos. offset change		0		
0040	Condition to offset tool length/tool position		0		
0041	Start up/Cancel		0		
0042	Corner travel limit		0.00000		
0043	Automatic corner override(angle)		0.0000		
0044	Automatic corner override(length 1)		0.00000		
0045	Automatic corner override(length 2)		0.00000		
0046	Automatic corner override(ratio)		100		
0047	Inner arc override limit		100		
0048	Automatic centering speed 1		19.60		
0049	Automatic centering speed 2		0.30		
0050	Automatic centering return		0.01960		
0051	Automatic centering compensation		0.00000		
0052	Lamp status when count ends		0		

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0053	Multiple M codes in one block	0	0	1	0
0054	Unregistered M-code	0			
0055	One shot signal M450 output time	5			
0056	One shot signal M451 output time	5			
0057	One shot signal M455 output time	5			
0058	One shot signal M456 output time	5			
0059	Automatic screen dimming time	0			
0060	Magazine panel automatic screen clear time	0			
0061	Alarm display when screen is dimmed	0			
0062	Machine No.	0			
0063	Automatic door	0			
0064	Area sensor pattern	1			
0065	Automatic door error level	1			
0066	Automatic door operation time	5			
0067	Display alarm when door is open	0			
0068	START lamp status when operating automatic door manually	0			
0069	Override switch	1			
0070	Alarm display when override is valid	1			
0071	Prevent rapid traverse override	0			
0072	Prevent cutting feedrate override	0			
0073	Prevent spindle override	0			
0074	Change handle magnification	1			
0075	Auto screen change at additional/PLC axis handle operation	0			
0076	Z-axis height to prohibit travel of additional axes	0.00000	0.00000	7.87400	0.00000
0077	Z-axis height to prohibit travel of additional axes(pallet 2)	0.00000			
0078	Prevent external operation for additional axes	0			
0079	Macro command single stop	0			
0080	Interrupt macro interrupt type	0			
0081	Interrupt macro call type	0			
0082	Interrupt macro acceptance type	0			
0083	ATC simultaneous operation start position	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250) 32.67720 (High column350)	15.94480	19.88180 (Standard) 21.85030 (Low)
0084	ATC reference tool length	7.87400			
0085	Automatically use S command when changing tool	0			
0086	Automatically use F command when changing tool	0			
0087	Tool data spindle rotation exceeded	0			
0088	Magazine index when changing tools of different sizes	0			
0089	Pot lowering start position	0.00000			
0090	XY axes movement when changing tool in manual mode	0	0	0	1

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0091	X-axis position when changing tool in manual mode	0.00000	0.00000	0.00000	-17.71650 (14tools) 0.00000 (22tools)
0092	Y-axis position when changing tool in manual mode	0.00000	0.00000	0.00000	-12.59840 (14tools) 0.00000 (22tools)
0093	Schedule program reset start position	0			
0094	Program end stop in schedule program	0			
0095	Tool life counting method	0			
0096	Change tool life display(Type2)	0			
0097	Spare tool change	0			
0098	Check tool life on first use	1			
0099	External signal output time when MFIN is invalid	5			
0100	Chip shower retention time	0			
0101	Automatic coolant off time	0			
0102	Coolant delay time	0			
0103	Automatic machine light off time	0			
0104	Standby mode time	0			
0105	Auto power off	0			
0106	Auto power off time				
0107	Cancel editing when resetting	1			
0108	Save when mode of current program is changed	0			
0109	Program edit start position during operation	0			
0110	Automatic sequence insertion increment	10			
0111	Set same number to edit/memory operation programs	0			
0112	Assign tools when starting memory operation	1			
0113	Check tools to be used when assigned automatically	1			
0114	Workpc. zero conversion code	0			
0115	Convert tool data	1			
0116	Tap step return position	0			
0117	Magazine turn speed ratio 1(M436)	80			
0118	Magazine turn speed ratio 2(M437)	50			
0119	Turret type magazine speed	0			
0120	Turret type magazine low-speed turn mode	0			
0121	Speed turret type magazine turns at low speed	20.00			
0122	Pallet select key change	1			
0123	Lock shift key	0			
0124	Max. actual cutting travel speed	787.40			
0125	Table load weight	250	300	40	240
0126	Arc radius error limit	0.00000			
0127	Scaling	0			
0128	X-axis scaling	0			
0129	Y-axis scaling	0			
0130	Z-axis scaling	0			
0131	Scaling unit	0			
0132	Common scaling factor	1000			
0133	X-axis scaling factor(I)	1000			
0134	Y-axis scaling factor(J)	1000			
0135	Z-axis scaling factor(K)	1000			
0136	Back washing at end of program	1			
0137	CTS pressurize check delay time	1.0			
0138	CTS depressurize check delay time	1.0			
0139	CTS pressure bleed time	0.0			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0140	CTS line filter alarm stop level	1			
0141	ALM2 signal stop level	0			
0142	Oiling/greasing when power is turned on	0			
0143	Tool wash filter alarm stop level	0			
0144	Tool wash filter check interval	1			
0145	Display tool wash filter check result	0			
0146	Faulty tool wash coolant level alarm stop(Type4)	1			
0147	Tool wash filter clogging warning alarm stop level(Type4)	0			
0148	No. of tool wash filter clogging warning checks(Type4)	1			
0149	Faulty tool wash coolant level sensor check interval(Type4)	200			
0150	Max. tapping speed (when returning tap)	100			
0151	Motor insulation resistance measurement interval	0			
0152	Motor insulation resistance monitoring acceptable value (all axes)	5.0			
0153	Motor insulation resistance monitoring acceptable value (abnormal)	1.1			
0154	Motor insulation resistance monitoring acceptable value (alarm)	5.0			
0155	Spindle rotation return during restart	0	0	1	0
0156	M function signal output delay time	1			
0157	Finish axis travel during external signal output	0			
0158	Chip conveyor relay adhering check	0			
0159	Error check when traveling during tool length/tool position offset cancel	0			
0160	Lathe machining infeed direction when power is turned ON	0			
0161	Cutting amount in thread cutting cycle	1.5			
0162	Cutting angle in thread cutting cycle	45.0000			
0163	Thread angle in complex thread cutting cycle	60.0000			
0164	Minimum infeed amount in complex thread cutting cycle	0.00190			
0165	Finishing count in complex thread cutting cycle	1			
0166	Finishing allowance in complex thread cutting cycle	0.00090			
0167	Signal batch reading/writing type	0			
0168	Z-axis travel method during tool change	0			
0169	Tapping sync. error too big - alarm stop level	0			
0170	Stop using feedrate override during rapid feed	0			
0171	Arc radius R command error limit	0.00010			
0172	Positioning method	0			
0173	Program load size	0			
0174	No tool detection function (when ATC rises)	0			
0175	No tool detection function (when ATC falls)	0			
0176	Tool over key detection function	0			
0177	Tool collision detection function during ATC	0			
0178	Tool missing detection function	0			
0179	Spindle return method for program stop	1			
0180	Lathe spindle return method for program stop	2			
0181	Coolant return method for program stop	1			
0182	Tap accel. Setting	0			
0183	Reset when operation preparation is ON	0			
0184	Screenshot	0			
0185	Probe detection signal method	0			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0186	PLC rapid traverse override function	0			
0187	PLC feedrate override function	0			
0188	PLC spindle override function	0			
0189	ATC arm turn speed ratio 1 (M422)	100			
0190	ATC arm turn speed ratio 2 (M423)	100			
0191	Tool wash discharge conditions for tool data	0			
0192	Tool wash when changing a different size tool	0			
0193	Cutting speed exceeded for inverse time feed	0			
0194	Cutter compensation interference check	0			
0195	Number of blocks to check cutter compensation interference	8			
0196	Involute interpolation override limit	0			
0197	Involute interpolation error limit	0.00390			
0198	Reset cycle time value	0			

User Parameter 2 (Switch 2)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0501	4th-axis stroke control	1			
0502	5th-axis stroke control	0	0	1	0
0503	6th-axis stroke control	0			
0504	7th-axis stroke control	0			
0505	8th-axis stroke control	0			
0506	4th-axis stroke(-)	-180.0000			
0507	5th-axis stroke(-)	0.0000	0.0000	-95.0000	0.0000
0508	6th-axis stroke(-)	0.0000			
0509	7th-axis stroke(-)	0.0000			
0510	8th-axis stroke(-)	0.0000			
0511	4th-axis stroke(+)	0.0000			
0512	5th-axis stroke(+)	0.0000	0.0000	5.0000	0.0000
0513	6th-axis stroke(+)	0.0000			
0514	7th-axis stroke(+)	0.0000			
0515	8th-axis stroke(+)	0.0000			
0516	Stroke limit 1 for X-axis(-)	-19.68500 (Standard) -27.55900 (Long) -11.81110 (Short)	-39.37000	-7.87400	-17.71650
0517	Stroke limit 1 for Y-axis(-)	-15.74800	-19.09450	-17.32280	-12.59840
0518	Stroke limit 1 for Z-axis(-)	0.00000			
0519	Stroke limit 1 for 4th-axis(-)	-180.0000			
0520	Stroke limit 1 for 5th-axis(-)	0.0000	0.0000	-95.0000	0.0000
0521	Stroke limit 1 for 6th-axis(-)	0.0000			
0522	Stroke limit 1 for 7th-axis(-)	0.0000			
0523	Stroke limit 1 for 8th-axis(-)	0.0000			
0524	Stroke limit 1 for X-axis(+)	0.00000			
0525	Stroke limit 1 for Y-axis(+)	0.00000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0526	Stroke limit 1 for Z-axis(+)	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250) 32.67720 (High column350)	15.94480	19.88180 (Standard) 21.85030 (Low)
0527	Stroke limit 1 for 4th-axis(+)	0.0000			
0528	Stroke limit 1 for 5th-axis(+)	0.0000	0.0000	5.0000	0.0000
0529	Stroke limit 1 for 6th-axis(+)	0.0000			
0530	Stroke limit 1 for 7th-axis(+)	0.0000			
0531	Stroke limit 1 for 8th-axis(+)	0.0000			
0532	Stroke limit 2 for X-axis(-)	0.00000			
0533	Stroke limit 2 for Y-axis(-)	0.00000			
0534	Stroke limit 2 for Z-axis(-)	0.00000			
0535	Stroke limit 2 for X-axis(+)	0.00000			
0536	Stroke limit 2 for Y-axis(+)	0.00000			
0537	Stroke limit 2 for Z-axis(+)	0.00000			
0538	Rapid traverse override speed 1	0.00			
0539	Rapid traverse override speed 2	19.60			
0540	Rapid traverse override speed 3	196.80			
0541	Rapid traverse override speed 4	393.70			
0542	4th-axis override rotation speed 1	0.00			
0543	4th-axis override rotation speed 2	0.30			
0544	4th-axis override rotation speed 3	2.50			
0545	4th-axis override rotation speed 4	5.00			
0546	5th-axis override rotation speed 1	0.00			
0547	5th-axis override rotation speed 2	0.50			
0548	5th-axis override rotation speed 3	5.00			
0549	5th-axis override rotation speed 4	10.00			
0550	6th-axis override rotation speed 1	0.00			
0551	6th-axis override rotation speed 2	0.50			
0552	6th-axis override rotation speed 3	5.00			
0553	6th-axis override rotation speed 4	10.00			
0554	7th-axis override rotation speed 1	0.00			
0555	7th-axis override rotation speed 2	0.50	0.50	1.00	0.50
0556	7th-axis override rotation speed 3	5.00	5.00	10.00	5.00
0557	7th-axis override rotation speed 4	10.00	10.00	30.00	10.00
0558	8th-axis override rotation speed 1	0.00			
0559	8th-axis override rotation speed 2	0.50			
0560	8th-axis override rotation speed 3	5.00			
0561	8th-axis override rotation speed 4	10.00			
0562	X-axis position signal output range(-)	0.00000			
0563	X-axis position signal output range(+)	0.00000			
0564	Y-axis position signal output range(-)	0.00000			
0565	Y-axis position signal output range(+)	0.00000			
0566	Z-axis position signal output range(-)	0.00000			
0567	Z-axis position signal output range(+)	0.00000			
0568	4th-axis position signal output range(-)	0.0000			
0569	4th-axis position signal output range(+)	0.0000			
0570	5th-axis position signal output range(-)	0.0000			
0571	5th-axis position signal output range(+)	0.0000			
0572	6th-axis position signal output range(-)	0.0000			
0573	6th-axis position signal output range(+)	0.0000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0574	7th-axis position signal output range(-)		0.0000		
0575	7th-axis position signal output range(+)		0.0000		
0576	8th-axis position signal output range(-)		0.0000		
0577	8th-axis position signal output range(+)		0.0000		
0578	X-axis single direction positioning excess travel amount		0.00000		
0579	Y-axis single direction positioning excess travel amount		0.00000		
0580	Z-axis single direction positioning excess travel amount		0.00000		
0581	4th-axis single direction positioning excess travel amount		0.0000		
0582	5th-axis single direction positioning excess travel amount		0.0000		
0583	6th-axis single direction positioning excess travel amount		0.0000		
0584	7th-axis single direction positioning excess travel amount		0.0000		
0585	8th-axis single direction positioning excess travel amount		0.0000		
0586	External index A switch address switching		0		
0587	External A-axis indexing(+)		180.0000		
0588	External A-axis indexing(-)		-180.0000		
0589	External index B switch address switching		0		
0590	External B-axis indexing(+)		180.0000		
0591	External B-axis indexing(-)		-180.0000		
0592	Z-axis lowering speed 1 when changing tool		1377.90		
0593	Z-axis lowering speed 2 when changing tool		1377.90		
0594	Z-axis lowering speed 3 when changing tool		1377.90		
0595	X-axis position signal output range 2 (-)		0.00000		
0596	X-axis position signal output range 2 (+)		0.00000		
0597	Y-axis position signal output range 2 (-)		0.00000		
0598	Y-axis position signal output range 2 (+)		0.00000		
0599	Z-axis position signal output range 2 (-)		0.00000		
0600	Z-axis position signal output range 2 (+)		0.00000		
0601	4-axis position signal output range 2 (-)		0.0000		
0602	4-axis position signal output range 2 (+)		0.0000		
0603	5-axis position signal output range 2 (-)		0.0000		
0604	5-axis position signal output range 2 (+)		0.0000		
0605	6-axis position signal output range 2 (-)		0.0000		
0606	6-axis position signal output range 2 (+)		0.0000		
0607	7-axis position signal output range 2 (-)		0.0000		
0608	7-axis position signal output range 2 (+)		0.0000		
0609	8-axis position signal output range 2 (-)		0.0000		
0610	8-axis position signal output range 2 (+)		0.0000		
0611	X-axis position signal output range 3 (-)		0.00000		
0612	X-axis position signal output range 3 (+)		0.00000		
0613	Y-axis position signal output range 3 (-)		0.00000		
0614	Y-axis position signal output range 3 (+)		0.00000		
0615	Z-axis position signal output range 3 (-)		0.00000		
0616	Z-axis position signal output range 3 (+)		0.00000		
0617	4-axis position signal output range 3 (-)		0.0000		
0618	4-axis position signal output range 3 (+)		0.0000		
0619	5-axis position signal output range 3 (-)		0.0000		
0620	5-axis position signal output range 3 (+)		0.0000		
0621	6-axis position signal output range 3 (-)		0.0000		
0622	6-axis position signal output range 3 (+)		0.0000		
0623	7-axis position signal output range 3 (-)		0.0000		

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
0624	7-axis position signal output range 3 (+)		0.0000		
0625	8-axis position signal output range 3 (-)		0.0000		
0626	8-axis position signal output range 3 (+)		0.0000		
0627	X-axis position signal output range 4 (-)		0.00000		
0628	X-axis position signal output range 4 (+)		0.00000		
0629	Y-axis position signal output range 4 (-)		0.00000		
0630	Y-axis position signal output range 4 (+)		0.00000		
0631	Z-axis position signal output range 4 (-)		0.00000		
0632	Z-axis position signal output range 4 (+)		0.00000		
0633	4-axis position signal output range 4 (-)		0.0000		
0634	4-axis position signal output range 4 (+)		0.0000		
0635	5-axis position signal output range 4 (-)		0.0000		
0636	5-axis position signal output range 4 (+)		0.0000		
0637	6-axis position signal output range 4 (-)		0.0000		
0638	6-axis position signal output range 4 (+)		0.0000		
0639	7-axis position signal output range 4 (-)		0.0000		
0640	7-axis position signal output range 4 (+)		0.0000		
0641	8-axis position signal output range 4 (-)		0.0000		
0642	8-axis position signal output range 4 (+)		0.0000		

User Parameter 3 (Switch 3)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1001	M271 positioning end check distance		0.01180		
1002	M271 positioning end check angle		0.3000		
1003	M272 positioning end check distance		0.01180		
1004	M272 positioning end check angle		0.3000		
1005	M273 positioning end check distance		0.01180		
1006	M273 positioning end check angle		0.3000		
1007	M274 positioning end check distance		0.01180		
1008	M274 positioning end check angle		0.3000		
1009	M275 positioning end check distance		0.01180		
1010	M275 positioning end check angle		0.3000		
1011	M276 positioning end check distance		0.01180		
1012	M276 positioning end check angle		0.3000		
1013	M277 positioning end check distance		0.01180		
1014	M277 positioning end check angle		0.3000		
1015	M278 positioning end check distance		0.01180		
1016	M278 positioning end check angle		0.3000		
1017	M279 positioning end check distance		0.01180		
1018	M279 positioning end check angle		0.3000		
1019	Low priority indication alarm No. 1				
1020	Low priority indication alarm No. 2				
1021	Low priority indication alarm No. 3				
1022	Low priority indication alarm No. 4				
1023	Low priority indication alarm No. 5				
1024	Low priority indication alarm No. 6				
1025	Low priority indication alarm No. 7				
1026	Low priority indication alarm No. 8				
1027	Low priority indication alarm No. 9				
1028	Low priority indication alarm No. 10				
1029	G code for modal display 1			1	
1030	G code for modal display 2			2	
1031	G code for modal display 3			3	
1032	G code for modal display 4			4	
1033	G code for modal display 5			5	
1034	G code for modal display 6			6	

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1035	G code for modal display 7	7			
1036	G code for modal display 8	10			
1037	G code for modal display 9	8			
1038	G code for modal display 10	11			
1039	G code for modal display 11	12			
1040	G code for modal display 12	13			
1041	G code for modal display 13	14			
1042	G code for modal display 14	15			
1043	G code for modal display 15	16			
1044	G code for modal display 16	17			
1045	M code for modal display 1	1			
1046	M code for modal display 2	2			
1047	M code for modal display 3	3			
1048	M code for modal display 4	4			
1049	M code for modal display 5	5			
1050	M code for modal display 6	6			
1051	M code for modal display 7	7			
1052	M code for modal display 8	8			
1053	M code for modal display 9	9			
1054	M code for modal display 10	11			
1055	M code for modal display 11	12			
1056	M code for modal display 12	14			
1057	M code for modal display 13	16			
1058	M code for modal display 14	31			
1059	M code for modal display 15	0			
1060	M code for modal display 16	0			
1061	Angle setting to check when additional axis for positioning is finished	0			
1062	Specified alarm signal 2				
1063	Specified alarm signal 3				
1064	Specified alarm signal 4				
1065	Specified alarm signal 5				

User Parameter 4 (Reference position/Conversation zero point)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1501	Reference position X-axis	0.00000			
1502	Reference position Y-axis	0.00000			
1503	Reference position Z-axis	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250) 32.67720 (High column350)	15.94480	19.88180 (Standard) 21.85030 (Low)
1504	Reference position 4th-axis	0.0000			
1505	Reference position 5th-axis	0.0000			
1506	Reference position 6th-axis	0.0000			
1507	Reference position 7th-axis	0.0000			
1508	Reference position 8th-axis	0.0000			
1509	2nd reference position X-axis	0.00000			
1510	2nd reference position Y-axis	0.00000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1511	2nd reference position Z-axis	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250) 32.67720 (High column350)	15.94480	19.88180 (Standard) 21.85030 (Low)
1512	2nd reference position 4th-axis	0.0000			
1513	2nd reference position 5th-axis	0.0000			
1514	2nd reference position 6th-axis	0.0000			
1515	2nd reference position 7th-axis	0.0000			
1516	2nd reference position 8th-axis	0.0000			
1517	3rd reference position X-axis	0.00000			
1518	3rd reference position Y-axis	0.00000			
1519	3rd reference position Z-axis	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250) 32.67720 (High column350)	15.94480	19.88180 (Standard) 21.85030 (Low)
1520	3rd reference position 4th-axis	0.0000			
1521	3rd reference position 5th-axis	0.0000			
1522	3rd reference position 6th-axis	0.0000			
1523	3rd reference position 7th-axis	0.0000			
1524	3rd reference position 8th-axis	0.0000			
1525	4th reference position X-axis	0.00000			
1526	4th reference position Y-axis	0.00000			
1527	4th reference position Z-axis	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250) 32.67720 (High column350)	15.94480	19.88180 (Standard) 21.85030 (Low)
1528	4th reference position 4th-axis	0.0000			
1529	4th reference position 5th-axis	0.0000			
1530	4th reference position 6th-axis	0.0000			
1531	4th reference position 7th-axis	0.0000			
1532	4th reference position 8th-axis	0.0000			
1533	5th reference position X-axis	0.00000			
1534	5th reference position Y-axis	0.00000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1535	5th reference position Z-axis	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250) 32.67720 (High column350)	15.94480	19.88180 (Standard) 21.85030 (Low)
1536	5th reference position 4th-axis	0.0000			
1537	5th reference position 5th-axis	0.0000			
1538	5th reference position 6th-axis	0.0000			
1539	5th reference position 7th-axis	0.0000			
1540	5th reference position 8th-axis	0.0000			
1541	6th reference position X-axis	0.00000			
1542	6th reference position Y-axis	0.00000			
1543	6th reference position Z-axis	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250) 32.67720 (High column350)	15.94480	19.88180 (Standard) 21.85030 (Low)
1544	6th reference position 4th-axis	0.0000			
1545	6th reference position 5th-axis	0.0000			
1546	6th reference position 6th-axis	0.0000			
1547	6th reference position 7th-axis	0.0000			
1548	6th reference position 8th-axis	0.0000			
1549	Reference point return complete signal output at servo off	0			
1550	Reference point allowable range at servo off (for XYZ axes)	0.00000			
1551	Reference point allowable range at servo off (for additional axes)	0.0000			
1552	X-axis zero return pos.(No.2)	0.00000			
1553	Y-axis zero return pos.(No.2)	0.00000			
1554	4th-axis zero return pos.(No.2)	0.0000			
1555	5th-axis zero return pos.(No.2)	0.0000			
1556	6th-axis zero return pos.(No.2)	0.0000			
1557	7th-axis zero return pos.(No.2)	0.0000			
1558	8th-axis zero return pos.(No.2)	0.0000			
1559	X-axis zero return pos.(No.3)	0.00000			
1560	Y-axis zero return pos.(No.3)	0.00000			
1561	4th-axis zero return pos.(No.3)	0.0000			
1562	5th-axis zero return pos.(No.3)	0.0000			
1563	6th-axis zero return pos.(No.3)	0.0000			
1564	7th-axis zero return pos.(No.3)	0.0000			
1565	8th-axis zero return pos.(No.3)	0.0000			
1566	Zero point return complete signal output at servo off	0			
1567	Zero point allowable range at servo off (for XYZ axes)	0.00000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1568	Zero point allowable range at servo off (for additional axes)	0.0000			
1569	Workpiece coordinate zero point X axis (for capture)	0.00000			
1570	Workpiece coordinate zero point Y axis (for capture)	0.00000			
1571	Workpiece coordinate zero point Z axis (for capture)	0.00000			

user parameter 5(User registered miscellaneous functions)

There is not default value.

User Parameter 6 (Tool breakage detection)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
1801	Tool breakage detection option	0			
1802	Pallet 1 tool breakage detector	0			
1803	Pallet 2 tool breakage detector	0			
1804	Tool breakage detection execution interval(No. of operations)	1			
1805	Reference tool length offset No.	0			
1806	Tool breakage detection return position(Type1)	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250) 32.67720 (High column350)	15.94480	19.88180 (Standard) 21.85030 (Low)
1807	Tool breakage detection position X-axis(Type1)	0.00000			
1808	Tool breakage detection position Y-axis(Type1)	0.00000			
1809	Tool breakage detection offset(Type2)	0.00000			
1810	Tool breakage detection position X-axis(Type3)	0.00000			
1811	Tool breakage detection position Y-axis(Type3)	0.00000			
1812	Tool breakage detection position Z-axis(Type3)	0.00000			
1813	Tool breakage detection start distance(Type3)	0.07870			
1814	Tool breakage detection stop distance(Type3)	0.03930			
1815	Tool breakage detection speed(Type3)	19.60			
1816	Tool breakage detection coolant OFF(Type3)	1			
1817	Pallet 2 tool breakage detection pos. X-axis(Type3)	0.00000			
1818	Pallet 2 tool breakage detection pos. Y-axis(Type3)	0.00000			
1819	Pallet 2 tool breakage detection pos. Z-axis(Type3)	0.00000			
1820	Pallet 2 tool breakage detection start distance(Type3)	0.07870			
1821	Pallet 2 tool breakage detection stop distance(Type3)	0.03930			
1822	Pallet 2 tool breakage detection speed(Type3)	19.60			
1823	Pallet 2 tool breakage detection coolant OFF(Type3)	1			

User Parameter 7 (Automatic workpiece measurement)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
2001	Probe compensation 1	0.00000			
2002	Probe compensation 2	0.00000			
2003	Probe compensation 3	0.00000			
2004	Probe compensation 4	0.00000			
2005	Measurement motion	0			
2006	Measurement speed 1	19.60			
2007	Measurement speed 2	3.90			
2008	Stop point before measurement	0.19680			
2009	Measurement travel limit distance	0.07870			
2010	Measurement tolerance 1	0.00000			
2011	Measurement tolerance 2	0.00000			
2012	Return distance after measurement	0.03930			

User Parameter 8 (Automatic thermal distortion compensation)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
2501	Z-axis compensation	100			
2502	Spindle compensation(Z-axis)	100			
2503	X-axis compensation	100			
2504	Spindle compensation(X-axis)	100			
2505	Y-axis compensation	100			
2506	Spindle compensation(Y-axis)	100			
2507	Auto. thermal distortion compen. at memory operation start	0			
2508	Lathe spindle compensation(Z-axis)	0			
2509	Lathe spindle compensation(X-axis)	0			
2510	Lathe spindle compensation(Y-axis)	0			

User Parameter 9 (Quick table)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
3001	Pallet program folder				
3002	Pallet 1 program No.				
3003	Pallet 2 program No.				
3004	Pallet program folder				
3005	Pallet 1 program No.				
3006	Pallet 2 program No.				
3007	Pallet 2 shift(X-axis)	0.00000			
3008	Pallet 2 shift(Y-axis)	0.00000			
3009	Pallet 2 shift(Z-axis)	0.00000			
3010	Pallet 1 loading	0			
3011	Pallet 2 loading	0			

**User Parameter 10(High accuracy)
High accuracy A (M260)**

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	120			
	Arc deceleration override	200			
	Curve approximation deceleration override	200			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M261)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	60			
	Arc deceleration override	140			
	Curve approximation deceleration override	140			
	Smooth path offset level	1			
	Smooth override	50			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M262)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	70			
	Arc deceleration override	70			
	Curve approximation deceleration override	70			
	Smooth path offset level	1			
	Smooth override	600			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M263)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	120			
	Arc deceleration override	200			
	Curve approximation deceleration override	200			
	Smooth path offset level	1			
	Smooth override	900			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M264)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	120			
	Arc deceleration override	200			
	Curve approximation deceleration override	200			
	Smooth path offset level	1			
	Smooth override	2500			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M265)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	600			
	Arc deceleration override	1000			
	Curve approximation deceleration override	1000			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Accuracy level	0.00000			

High accuracy A (M266)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	100			
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M267)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	100			
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy B (M280)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	0			
	Curve approximation deceleration override	0			
	Smooth path offset level	3			
	Smooth override	100			
	Minute block deletion distance	0.00030			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M281)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M282)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M283)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M284)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M285)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M286)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M287)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Corner deceleration override	100			

High accuracy common

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
3201	Smooth path offset function	0			
3202	Smooth path offset cancel distance	0.39300			
3203	Smooth path offset cancel angle	10.0000			
3204	Smooth path offset angle determination invalid distance	0.00030			
3205	Smooth path offset reference length	0.03930			
3206	Minute block deletion when smooth path offset is cancelled	0			
3207	Accuracy level method	0			

High accuracy(Adjust) common

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Machining mode	0 (Standard) (Long) (Short)			

High accuracy(Adjust) Adjust.A

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Standard mode	0 (Standard) (Long) (Short)			
	Accuracy	0 (Standard) (Long) (Short)			
	Smoothness	0 (Standard) (Long) (Short)			

High accuracy(Adjust) Adjust.B

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Standard mode	0 (Standard) (Long) (Short)			
	Accuracy	0 (Standard) (Long) (Short)			
	Smoothness	0 (Standard) (Long) (Short)			

High accuracy(Adjust) Adjust.C

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Standard mode	0 (Standard) (Long) (Short)			
	Accuracy	0 (Standard) (Long) (Short)			
	Smoothness	0 (Standard) (Long) (Short)			

High accuracy(Accuracy spec.) common

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Accuracy spec. mode	0 (Standard) (Long) (Short)			

High accuracy(Accuracy spec.) Accuracy spec. A

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Accuracy level	0.00390 (Standard) (Long) (Short)			
	Smoothness	0 (Standard) (Long) (Short)			

High accuracy(Accuracy spec.) Accuracy spec. B

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Accuracy level	0.00120 (Standard) (Long) (Short)			
	Smoothness	0 (Standard) (Long) (Short)			

High accuracy(Accuracy spec.) Accuracy spec. C

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Accuracy level	0.00040 (Standard) (Long) (Short)			
	Smoothness	0 (Standard) (Long) (Short)			

User Parameter 11 (PLC)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
3501	P1-axis stroke control	0			
3502	P2-axis stroke control	0			
3503	P3-axis stroke control	0			
3504	P4-axis stroke control	0			
3505	P1-axis stroke(-)	0			
3506	P2-axis stroke(-)	0			
3507	P3-axis stroke(-)	0			
3508	P4-axis stroke(-)	0			
3509	P1-axis stroke(+)	0			
3510	P2-axis stroke(+)	0			
3511	P3-axis stroke(+)	0			
3512	P4-axis stroke(+)	0			
3513	Reference point P1-axis	0			
3514	Reference point P2-axis	0			
3515	Reference point P3-axis	0			
3516	Reference point P4-axis	0			
3517	P1-axis override speed 1	0			
3518	P1-axis override speed 2	20			
3519	P1-axis override speed 3	200			
3520	P1-axis override speed 4	400			
3521	P2-axis override speed 1	0			
3522	P2-axis override speed 2	20			
3523	P2-axis override speed 3	200			
3524	P2-axis override speed 4	400			
3525	P3-axis override speed 1	0			
3526	P3-axis override speed 2	20			
3527	P3-axis override speed 3	200			
3528	P3-axis override speed 4	400			
3529	P4-axis override speed 1	0			
3530	P4-axis override speed 2	20			
3531	P4-axis override speed 3	200			
3532	P4-axis override speed 4	400			
3533	Collis. detct.-P1ax stop meth.	0			
3534	Collis. detct.-P2ax stop meth.	0			
3535	Collis. detct.-P3ax stop meth.	0			
3536	Collis. detct.-P4ax stop meth.	0			

User Parameter 12 (Profibus)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
4001	Node address	0			
4002	I/O setting 1				
4003	I/O setting 2				
4004	I/O setting 3				
4005	I/O setting 4				
4006	I/O setting 5				
4007	I/O setting 6				
4008	I/O setting 7				
4009	I/O setting 8				
4010	I/O setting 9				
4011	I/O setting 10				
4012	I/O setting 11				
4013	I/O setting 12				
4014	I/O setting 13				
4015	I/O setting 14				
4016	I/O setting 15				

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
4017	I/O setting 16				
4018	I/O setting 17				
4019	I/O setting 18				
4020	I/O setting 19				
4021	I/O setting 20				

User Parameter 13 (DeviceNet)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
4201	MAC ID		0		
4202	Transfer rate		0		
4203	Input bytes		0		
4204	Output bytes		0		

User Parameter 14 (CC-Link Slave)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
4401	Station number		1		
4402	Slave station category		0		
4403	Occupied number of stations		1		
4404	Transfer rate		0		
4405	Specify data clear in case of communication error		0		

User Parameter 15 (Rotary fixture offset)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
4601	XYZ-axes travel during current rotary fixture offset change		0		
4602	Rotary fixture offset reset method		0		
4603	Tilt axis 1	0	0	2	0
4604	Rotation axis 1	0	0	4	0
4605	Tilt axis 2		0		
4606	Rotation axis 2		0		
4607	Tilt axis 3		0		
4608	Rotation axis 3		0		
4609	Tilt axis 4		0		
4610	Rotation axis 4		0		
4611	Forward direction for the coordinate system on tilt axis 1	0	0	1	0
4612	Forward direction for the coordinate system on rotation axis 1	0	0	5	0
4613	Forward direction for the coordinate system on tilt axis 2		0		
4614	Forward direction for the coordinate system on rotation axis 2		0		
4615	Forward direction for the coordinate system on tilt axis 3		0		
4616	Forward direction for the coordinate system on rotation axis 3		0		
4617	Forward direction for the coordinate system on tilt axis 4		0		
4618	Forward direction for the coordinate system on rotation axis 4		0		

**User Parameter 16 (3D smoothing function)
3D smoothing function(M380)**

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M381)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M382)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M383)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M384)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M385)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M386)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M387)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function common

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
4801	3D smoothing circular interpolation command				
4802	3D smoothing sequence number command				

User Parameter 17 (feature coordinate setting)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
10001	Tilt axis 1	0	0	2	0
10002	Rotation axis 1	0	0	4	0
10003	Tilt axis 2	0			
10004	Rotation axis 2	0			
10005	Forward direction for the coordinate system on tilt axis 1	0	0	1	0
10006	Forward direction for the coordinate system on rotation axis 1	0	0	5	0
10007	Forward direction for the coordinate system on tilt axis 2	0			
10008	Forward direction for the coordinate system on rotation axis 2	0			
10009	Rotation center X coordinate offset for tilt axis 1	0.00000			
10010	Rotation center Y coordinate offset for tilt axis 1	0.00000			
10011	Rotation center Z coordinate offset for tilt axis 1	0.00000			
10012	Rotation center X coordinate offset for rotation axis 1	0.00000			
10013	Rotation center Y coordinate offset for rotation axis 1	0.00000			
10014	Rotation center Z coordinate offset for rotation axis 1	0.00000			
10015	Rotation center X coordinate offset for tilt axis 2	0.00000			
10016	Rotation center Y coordinate offset for tilt axis 2	0.00000			
10017	Rotation center Z coordinate offset for tilt axis 2	0.00000			
10018	Rotation center X coordinate offset for rotation axis 2	0.00000			
10019	Rotation center Y coordinate offset for rotation axis 2	0.00000			
10020	Rotation center Z coordinate offset for rotation axis 2	0.00000			
10021	Absolute coordinate display format	0			

User Parameter 18 (CC-Link master)

No.	Item	S500X1 S700X1 S300X1	S1000X1	M140X1	R450X1
10201	Transfer rate	0			
10202	Self-station number	0			
10203	Interface board setting	0			
10204	Occupied number of stations (for intelligent device stations)	0			
10205	Standby master station number	0			
10206	CC-Link version	0			
10207	Retry attempts	3			
10208	Number of automatic return nodes	1			
10209	Specify action when there is a driver error	0			
10210	Specify station data clear in case of any error	0			

2.5.20.3 Metric(R650X1,M140X1,S500X2,S700X2, S300X2,F600X1)

User Parameter 1 (Switch 1)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0001	Select mode		0		
0002	Initial mode		0		
0003	Select memory operation		0		
0004	Program repeat		0		
0005	Workpiece counter when program is repeated		0		
0006	Tool life management when program is repeated		0		
0007	Automatic tool length compensation		0.0000		
0008	External signal reading time limit		0		
0009	Dry run offset amount		0.0000		
0010	Dry run offset amount lower limit		0.0000		
0011	Program unit		0		
0012	Program feed rate unit		0		
0013	G73 relief amount		0.5000		
0014	G83 cutting start position		0.5000		
0015	G77,G78 relief amount		1.000		
0016	G76,G87 shift direction		0		
0017	Tapping cycle return speed		0		
0018	Program restart		0		
0019	S.T.M. return		1		
0020	Prevent restart from middle of program		0		
0021	4th row display axis		1		
0022	5th row display axis		2		
0023	6th row display axis		3		
0024	7th row display axis		4		
0025	8th row display axis		5		
0026	9th row display axis		6		
0027	10th row display axis		7		
0028	11th row display axis		8		
0029	12th row display axis		9		
0030	XYZ-axis absolute coord. Display		2		
0031	Display coordinates during rotational transformation		0		
0032	Macro variable 1 for position screen display				
0033	Macro variable 2 for position screen display				
0034	Macro variable 3 for position screen display				
0035	Macro variable 4 for position screen display				
0036	Macro variable 5 for position screen display				
0037	Block No. display color		4		
0038	Pointer display color		3		
0039	X/Y/Z-axis travel at tool length/tool pos. offset change		0		
0040	Condition to offset tool length/tool position		0		
0041	Start up/Cancel		0		
0042	Corner travel limit		0.0000		
0043	Automatic corner override(angle)		0.0000		
0044	Automatic corner override(length 1)		0.0000		
0045	Automatic corner override(length 2)		0.0000		
0046	Automatic corner override(ratio)		100		
0047	Inner arc override limit		100		
0048	Automatic centering speed 1		500.0		
0049	Automatic centering speed 2		10.0		
0050	Automatic centering return		0.5000		
0051	Automatic centering compensation		0.0000		
0052	Lamp status when count ends		0		

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0053	Multiple M codes in one block	0	1	0	0
0054	Unregistered M-code	0			
0055	One shot signal M450 output time	5			
0056	One shot signal M451 output time	5			
0057	One shot signal M455 output time	5			
0058	One shot signal M456 output time	5			
0059	Automatic screen dimming time	0			
0060	Magazine panel automatic screen clear time	0			
0061	Alarm display when screen is dimmed	0			
0062	Machine No.	0			
0063	Automatic door	0			
0064	Area sensor pattern	1			
0065	Automatic door error level	1			
0066	Automatic door operation time	5			
0067	Display alarm when door is open	0			
0068	START lamp status when operating automatic door manually	0			
0069	Override switch	1			
0070	Alarm display when override is valid	1			
0071	Prevent rapid traverse override	0			
0072	Prevent cutting feedrate override	0			
0073	Prevent spindle override	0			
0074	Change handle magnification	1			
0075	Auto screen change at additional/PLC axis handle operation	0			
0076	Z-axis height to prohibit travel of additional axes	0.0000	200.0000	0.0000	0.0000
0077	Z-axis height to prohibit travel of additional axes(pallet 2)	0.0000			
0078	Prevent external operation for additional axes	0			
0079	Macro command single stop	0			
0080	Interrupt macro interrupt type	0			
0081	Interrupt macro call type	0			
0082	Interrupt macro acceptance type	0			
0083	ATC simultaneous operation start position	555.0000 (Standard) 625.0000 (Low)	455.0000	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	550.0000
0084	ATC reference tool length	200.0000			
0085	Automatically use S command when changing tool	0			
0086	Automatically use F command when changing tool	0			
0087	Tool data spindle rotation exceeded	0			
0088	Magazine index when changing tools of different sizes	0			
0089	Pot lowering start position	0.0000			
0090	XY axes movement when changing tool in manual mode	1	0	0	0
0091	X-axis position when changing tool in manual mode	-650.0000	0.0000	0.0000	0.0000
0092	Y-axis position when changing tool in manual mode	0.0000			
0093	Schedule program reset start position	0			
0094	Program end stop in schedule program	0			
0095	Tool life counting method	0			
0096	Change tool life display(Type2)	0			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0097	Spare tool change	0			
0098	Check tool life on first use	1			
0099	External signal output time when MFIN is invalid	5			
0100	Chip shower retention time	0			
0101	Automatic coolant off time	0			
0102	Coolant delay time	0			
0103	Automatic machine light off time	0			
0104	Standby mode time	0			
0105	Auto power off	0			
0106	Auto power off time				
0107	Cancel editing when resetting	1			
0108	Save when mode of current program is changed	0			
0109	Program edit start position during operation	0			
0110	Automatic sequence insertion increment	10			
0111	Set same number to edit/memory operation programs	0			
0112	Assign tools when starting memory operation	1			
0113	Check tools to be used when assigned automatically	1			
0114	Workpc. zero conversion code	0			
0115	Convert tool data	1			
0116	Tap step return position	0			
0117	Magazine turn speed ratio 1(M436)	80			
0118	Magazine turn speed ratio 2(M437)	50			
0119	Turret type magazine speed	0			
0120	Turret type magazine low-speed turn mode	0			
0121	Speed turret type magazine turns at low speed	20.00			
0122	Pallet select key change	1			
0123	Lock shift key	0			
0124	Max. actual cutting travel speed	20000.0			
0125	Table load weight	400	40	250	400
0126	Arc radius error limit	0.0000			
0127	Scaling	0			
0128	X-axis scaling	0			
0129	Y-axis scaling	0			
0130	Z-axis scaling	0			
0131	Scaling unit	0			
0132	Common scaling factor	1000			
0133	X-axis scaling factor(I)	1000			
0134	Y-axis scaling factor(J)	1000			
0135	Z-axis scaling factor(K)	1000			
0136	Back washing at end of program	1			
0137	CTS pressurize check delay time	1.0			
0138	CTS depressurize check delay time	1.0			
0139	CTS pressure bleed time	0.0			
0140	CTS line filter alarm stop level	1			
0141	ALM2 signal stop level	0			
0142	Oiling/greasing when power is turned on	0			
0143	Tool wash filter alarm stop level	0			
0144	Tool wash filter check interval	1			
0145	Display tool wash filter check result	0			
0146	Faulty tool wash coolant level alarm stop(Type4)	1			
0147	Tool wash filter clogging warning alarm stop level(Type4)	0			
0148	No. of tool wash filter clogging warning checks(Type4)	1			
0149	Faulty tool wash coolant level sensor check interval(Type4)	200			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0150	Max. tapping speed (when returning tap)	100			
0151	Motor insulation resistance measurement interval	0			
0152	Motor insulation resistance monitoring acceptable value (all axes)	5.0			
0153	Motor insulation resistance monitoring acceptable value (abnormal)	1.1			
0154	Motor insulation resistance monitoring acceptable value (alarm)	5.0			
0155	Spindle rotation return during restart	0	1	0	0
0156	M function signal output delay time	1			
0157	Finish axis travel during external signal output	0			
0158	Chip conveyor relay adhering check	0			
0159	Error check when traveling during tool length/tool position offset cancel	0			
0160	Lathe machining infeed direction when power is turned ON	0			
0161	Cutting amount in thread cutting cycle	1.5			
0162	Cutting angle in thread cutting cycle	45.0000			
0163	Thread angle in complex thread cutting cycle	60.0000			
0164	Minimum infeed amount in complex thread cutting cycle	0.0500			
0165	Finishing count in complex thread cutting cycle	1			
0166	Finishing allowance in complex thread cutting cycle	0.0250			
0167	Signal batch reading/writing type	0			
0168	Z-axis travel method during tool change	0			
0169	Tapping sync. error too big - alarm stop level	0			
0170	Stop using feedrate override during rapid feed	0			
0171	Arc radius R command error limit	0.0010			
0172	Positioning method	0			
0173	Program load size	0	0	1	1
0174	No tool detection function (when ATC rises)	0			
0175	No tool detection function (when ATC falls)	0			
0176	Tool over key detection function	0			
0177	Tool collision detection function during ATC	0			
0178	Tool missing detection function	0			
0179	Spindle return method for program stop	1	1	0	0
0180	Lathe spindle return method for program stop	2	2	0	0
0181	Coolant return method for program stop	1	1	0	0
0182	Tap accel. Setting	0	0	1 (10K) 1 (10K(High torque)) 1 (16K) 0 (27K)	1
0183	Reset when operation preparation is ON	0			
0184	Screenshot	0			
0185	Probe detection signal method	0	0	1	1
0186	PLC rapid traverse override function	0			
0187	PLC feedrate override function	0			
0188	PLC spindle override function	0			
0189	ATC arm turn speed ratio 1 (M422)	100			
0190	ATC arm turn speed ratio 2 (M423)	100			
0191	Tool wash discharge conditions for tool data	0			
0192	Tool wash when changing a different size tool	0			
0193	Cutting speed exceeded for inverse time feed	0			
0194	Cutter compensation interference check	0			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0195	Number of blocks to check cutter compensation interference	8			
0196	Involute interpolation override limit	0			
0197	Involute interpolation error limit	0.1000			
0198	Reset cycle time value	0			

User Parameter 2 (Switch 2)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0501	4th-axis stroke control	1			
0502	5th-axis stroke control	0	1	0	0
0503	6th-axis stroke control	0			
0504	7th-axis stroke control	0			
0505	8th-axis stroke control	0			
0506	4th-axis stroke(-)	-180.0000			
0507	5th-axis stroke(-)	0.0000	-30.0000	0.0000	0.0000
0508	6th-axis stroke(-)	0.0000			
0509	7th-axis stroke(-)	0.0000			
0510	8th-axis stroke(-)	0.0000			
0511	4th-axis stroke(+)	0.0000			
0512	5th-axis stroke(+)	0.0000	120.0000	0.0000	0.0000
0513	6th-axis stroke(+)	0.0000			
0514	7th-axis stroke(+)	0.0000			
0515	8th-axis stroke(+)	0.0000			
0516	Stroke limit 1 for X-axis(-)	-650.0000	-200.0000	-500.0000 (Standard) -700.0000 (Long) -300.0000 (Short)	-600.0000
0517	Stroke limit 1 for Y-axis(-)	-400.0000	-440.0000	-400.0000	-400.0000
0518	Stroke limit 1 for Z-axis(-)	0.0000			
0519	Stroke limit 1 for 4th-axis(-)	-180.0000			
0520	Stroke limit 1 for 5th-axis(-)	0.0000	-30.0000	0.0000	0.0000
0521	Stroke limit 1 for 6th-axis(-)	0.0000			
0522	Stroke limit 1 for 7th-axis(-)	0.0000			
0523	Stroke limit 1 for 8th-axis(-)	0.0000			
0524	Stroke limit 1 for X-axis(+)	0.0000			
0525	Stroke limit 1 for Y-axis(+)	0.0000			
0526	Stroke limit 1 for Z-axis(+)	555.0000 (Standard) 625.0000 (Low)	455.0000	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	550.0000
0527	Stroke limit 1 for 4th-axis(+)	0.0000			
0528	Stroke limit 1 for 5th-axis(+)	0.0000	120.0000	0.0000	0.0000
0529	Stroke limit 1 for 6th-axis(+)	0.0000			
0530	Stroke limit 1 for 7th-axis(+)	0.0000			
0531	Stroke limit 1 for 8th-axis(+)	0.0000			
0532	Stroke limit 2 for X-axis(-)	0.0000			
0533	Stroke limit 2 for Y-axis(-)	0.0000			
0534	Stroke limit 2 for Z-axis(-)	0.0000			
0535	Stroke limit 2 for X-axis(+)	0.0000			
0536	Stroke limit 2 for Y-axis(+)	0.0000			
0537	Stroke limit 2 for Z-axis(+)	0.0000			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0538	Rapid traverse override speed 1	0.0			
0539	Rapid traverse override speed 2	500.0			
0540	Rapid traverse override speed 3	5000.0			
0541	Rapid traverse override speed 4	10000.0			
0542	4th-axis override rotation speed 1	0.00			
0543	4th-axis override rotation speed 2	0.30			
0544	4th-axis override rotation speed 3	2.50			
0545	4th-axis override rotation speed 4	5.00			
0546	5th-axis override rotation speed 1	0.00			
0547	5th-axis override rotation speed 2	0.50			
0548	5th-axis override rotation speed 3	5.00			
0549	5th-axis override rotation speed 4	10.00			
0550	6th-axis override rotation speed 1	0.00			
0551	6th-axis override rotation speed 2	0.50			
0552	6th-axis override rotation speed 3	5.00			
0553	6th-axis override rotation speed 4	10.00			
0554	7th-axis override rotation speed 1	0.00			
0555	7th-axis override rotation speed 2	0.50	1.00	0.50	0.50
0556	7th-axis override rotation speed 3	5.00	10.00	5.00	5.00
0557	7th-axis override rotation speed 4	10.00	30.00	10.00	10.00
0558	8th-axis override rotation speed 1	0.00			
0559	8th-axis override rotation speed 2	0.50			
0560	8th-axis override rotation speed 3	5.00			
0561	8th-axis override rotation speed 4	10.00			
0562	X-axis position signal output range(-)	0.0000			
0563	X-axis position signal output range(+)	0.0000			
0564	Y-axis position signal output range(-)	0.0000			
0565	Y-axis position signal output range(+)	0.0000			
0566	Z-axis position signal output range(-)	0.0000			
0567	Z-axis position signal output range(+)	0.0000			
0568	4th-axis position signal output range(-)	0.0000			
0569	4th-axis position signal output range(+)	0.0000			
0570	5th-axis position signal output range(-)	0.0000			
0571	5th-axis position signal output range(+)	0.0000			
0572	6th-axis position signal output range(-)	0.0000			
0573	6th-axis position signal output range(+)	0.0000			
0574	7th-axis position signal output range(-)	0.0000			
0575	7th-axis position signal output range(+)	0.0000			
0576	8th-axis position signal output range(-)	0.0000			
0577	8th-axis position signal output range(+)	0.0000			
0578	X-axis single direction positioning excess travel amount	0.0000			
0579	Y-axis single direction positioning excess travel amount	0.0000			
0580	Z-axis single direction positioning excess travel amount	0.0000			
0581	4th-axis single direction positioning excess travel amount	0.0000			
0582	5th-axis single direction positioning excess travel amount	0.0000			
0583	6th-axis single direction positioning excess travel amount	0.0000			
0584	7th-axis single direction positioning excess travel amount	0.0000			
0585	8th-axis single direction positioning excess travel amount	0.0000			
0586	External index A switch address switching	0			
0587	External A-axis indexing(+)	180.0000			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0588	External A-axis indexing(-)	-180.0000			
0589	External index B switch address switching	0			
0590	External B-axis indexing(+)	180.0000			
0591	External B-axis indexing(-)	-180.0000			
0592	Z-axis lowering speed 1 when changing tool	35000.0			
0593	Z-axis lowering speed 2 when changing tool	35000.0			
0594	Z-axis lowering speed 3 when changing tool	35000.0			
0595	X-axis position signal output range 2 (-)	0.0000			
0596	X-axis position signal output range 2 (+)	0.0000			
0597	Y-axis position signal output range 2 (-)	0.0000			
0598	Y-axis position signal output range 2 (+)	0.0000			
0599	Z-axis position signal output range 2 (-)	0.0000			
0600	Z-axis position signal output range 2 (+)	0.0000			
0601	4-axis position signal output range 2 (-)	0.0000			
0602	4-axis position signal output range 2 (+)	0.0000			
0603	5-axis position signal output range 2 (-)	0.0000			
0604	5-axis position signal output range 2 (+)	0.0000			
0605	6-axis position signal output range 2 (-)	0.0000			
0606	6-axis position signal output range 2 (+)	0.0000			
0607	7-axis position signal output range 2 (-)	0.0000			
0608	7-axis position signal output range 2 (+)	0.0000			
0609	8-axis position signal output range 2 (-)	0.0000			
0610	8-axis position signal output range 2 (+)	0.0000			
0611	X-axis position signal output range 3 (-)	0.0000			
0612	X-axis position signal output range 3 (+)	0.0000			
0613	Y-axis position signal output range 3 (-)	0.0000			
0614	Y-axis position signal output range 3 (+)	0.0000			
0615	Z-axis position signal output range 3 (-)	0.0000			
0616	Z-axis position signal output range 3 (+)	0.0000			
0617	4-axis position signal output range 3 (-)	0.0000			
0618	4-axis position signal output range 3 (+)	0.0000			
0619	5-axis position signal output range 3 (-)	0.0000			
0620	5-axis position signal output range 3 (+)	0.0000			
0621	6-axis position signal output range 3 (-)	0.0000			
0622	6-axis position signal output range 3 (+)	0.0000			
0623	7-axis position signal output range 3 (-)	0.0000			
0624	7-axis position signal output range 3 (+)	0.0000			
0625	8-axis position signal output range 3 (-)	0.0000			
0626	8-axis position signal output range 3 (+)	0.0000			
0627	X-axis position signal output range 4 (-)	0.0000			
0628	X-axis position signal output range 4 (+)	0.0000			
0629	Y-axis position signal output range 4 (-)	0.0000			
0630	Y-axis position signal output range 4 (+)	0.0000			
0631	Z-axis position signal output range 4 (-)	0.0000			
0632	Z-axis position signal output range 4 (+)	0.0000			
0633	4-axis position signal output range 4 (-)	0.0000			
0634	4-axis position signal output range 4 (+)	0.0000			
0635	5-axis position signal output range 4 (-)	0.0000			
0636	5-axis position signal output range 4 (+)	0.0000			
0637	6-axis position signal output range 4 (-)	0.0000			
0638	6-axis position signal output range 4 (+)	0.0000			
0639	7-axis position signal output range 4 (-)	0.0000			
0640	7-axis position signal output range 4 (+)	0.0000			
0641	8-axis position signal output range 4 (-)	0.0000			
0642	8-axis position signal output range 4 (+)	0.0000			

User Parameter 3 (Switch 3)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1001	M271 positioning end check distance			0.3000	
1002	M271 positioning end check angle			0.3000	
1003	M272 positioning end check distance			0.3000	
1004	M272 positioning end check angle			0.3000	
1005	M273 positioning end check distance			0.3000	
1006	M273 positioning end check angle			0.3000	
1007	M274 positioning end check distance			0.3000	
1008	M274 positioning end check angle			0.3000	
1009	M275 positioning end check distance			0.3000	
1010	M275 positioning end check angle			0.3000	
1011	M276 positioning end check distance			0.3000	
1012	M276 positioning end check angle			0.3000	
1013	M277 positioning end check distance			0.3000	
1014	M277 positioning end check angle			0.3000	
1015	M278 positioning end check distance			0.3000	
1016	M278 positioning end check angle			0.3000	
1017	M279 positioning end check distance			0.3000	
1018	M279 positioning end check angle			0.3000	
1019	Low priority indication alarm No. 1				
1020	Low priority indication alarm No. 2				
1021	Low priority indication alarm No. 3				
1022	Low priority indication alarm No. 4				
1023	Low priority indication alarm No. 5				
1024	Low priority indication alarm No. 6				
1025	Low priority indication alarm No. 7				
1026	Low priority indication alarm No. 8				
1027	Low priority indication alarm No. 9				
1028	Low priority indication alarm No. 10				
1029	G code for modal display 1			1	
1030	G code for modal display 2			2	
1031	G code for modal display 3			3	
1032	G code for modal display 4			4	
1033	G code for modal display 5			5	
1034	G code for modal display 6			6	
1035	G code for modal display 7			7	
1036	G code for modal display 8			10	
1037	G code for modal display 9			8	
1038	G code for modal display 10			11	
1039	G code for modal display 11			12	
1040	G code for modal display 12			13	
1041	G code for modal display 13			14	
1042	G code for modal display 14			15	
1043	G code for modal display 15			16	
1044	G code for modal display 16			17	
1045	M code for modal display 1			1	
1046	M code for modal display 2			2	
1047	M code for modal display 3			3	
1048	M code for modal display 4			4	
1049	M code for modal display 5			5	
1050	M code for modal display 6			6	
1051	M code for modal display 7			7	
1052	M code for modal display 8			8	
1053	M code for modal display 9			9	
1054	M code for modal display 10			11	
1055	M code for modal display 11			12	
1056	M code for modal display 12			14	

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1057	M code for modal display 13	16			
1058	M code for modal display 14	31			
1059	M code for modal display 15	0			
1060	M code for modal display 16	0			
1061	Angle setting to check when additional axis for positioning is finished	0			
1062	Specified alarm signal 2				
1063	Specified alarm signal 3				
1064	Specified alarm signal 4				
1065	Specified alarm signal 5				

User Parameter 4 (Reference position/Conversation zero point)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1501	Reference position X-axis	0.0000			
1502	Reference position Y-axis	0.0000			
1503	Reference position Z-axis	555.0000 (Standard) 625.0000 (Low)	455.0000	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	550.0000
1504	Reference position 4th-axis	0.0000			
1505	Reference position 5th-axis	0.0000			
1506	Reference position 6th-axis	0.0000			
1507	Reference position 7th-axis	0.0000			
1508	Reference position 8th-axis	0.0000			
1509	2nd reference position X-axis	0.0000			
1510	2nd reference position Y-axis	0.0000			
1511	2nd reference position Z-axis	555.0000 (Standard) 625.0000 (Low)	455.0000	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	550.0000
1512	2nd reference position 4th-axis	0.0000			
1513	2nd reference position 5th-axis	0.0000			
1514	2nd reference position 6th-axis	0.0000			
1515	2nd reference position 7th-axis	0.0000			
1516	2nd reference position 8th-axis	0.0000			
1517	3rd reference position X-axis	0.0000			
1518	3rd reference position Y-axis	0.0000			
1519	3rd reference position Z-axis	555.0000 (Standard) 625.0000 (Low)	455.0000	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	550.0000
1520	3rd reference position 4th-axis	0.0000			
1521	3rd reference position 5th-axis	0.0000			
1522	3rd reference position 6th-axis	0.0000			
1523	3rd reference position 7th-axis	0.0000			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1524	3rd reference position 8th-axis	0.0000			
1525	4th reference position X-axis	0.0000			
1526	4th reference position Y-axis	0.0000			
1527	4th reference position Z-axis	555.0000 (Standard) 625.0000 (Low)	455.0000	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	550.0000
1528	4th reference position 4th-axis	0.0000			
1529	4th reference position 5th-axis	0.0000			
1530	4th reference position 6th-axis	0.0000			
1531	4th reference position 7th-axis	0.0000			
1532	4th reference position 8th-axis	0.0000			
1533	5th reference position X-axis	0.0000			
1534	5th reference position Y-axis	0.0000			
1535	5th reference position Z-axis	555.0000 (Standard) 625.0000 (Low)	455.0000	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	550.0000
1536	5th reference position 4th-axis	0.0000			
1537	5th reference position 5th-axis	0.0000			
1538	5th reference position 6th-axis	0.0000			
1539	5th reference position 7th-axis	0.0000			
1540	5th reference position 8th-axis	0.0000			
1541	6th reference position X-axis	0.0000			
1542	6th reference position Y-axis	0.0000			
1543	6th reference position Z-axis	555.0000 (Standard) 625.0000 (Low)	455.0000	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	550.0000
1544	6th reference position 4th-axis	0.0000			
1545	6th reference position 5th-axis	0.0000			
1546	6th reference position 6th-axis	0.0000			
1547	6th reference position 7th-axis	0.0000			
1548	6th reference position 8th-axis	0.0000			
1549	Reference point return complete signal output at servo off	0			
1550	Reference point allowable range at servo off (for XYZ axes)	0.0000			
1551	Reference point allowable range at servo off (for additional axes)	0.0000			
1552	X-axis zero return pos.(No.2)	0.0000			
1553	Y-axis zero return pos.(No.2)	0.0000			
1554	4th-axis zero return pos.(No.2)	0.0000			
1555	5th-axis zero return pos.(No.2)	0.0000			
1556	6th-axis zero return pos.(No.2)	0.0000			
1557	7th-axis zero return pos.(No.2)	0.0000			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1558	8th-axis zero return pos.(No.2)	0.0000			
1559	X-axis zero return pos.(No.3)	0.0000			
1560	Y-axis zero return pos.(No.3)	0.0000			
1561	4th-axis zero return pos.(No.3)	0.0000			
1562	5th-axis zero return pos.(No.3)	0.0000			
1563	6th-axis zero return pos.(No.3)	0.0000			
1564	7th-axis zero return pos.(No.3)	0.0000			
1565	8th-axis zero return pos.(No.3)	0.0000			
1566	Zero point return complete signal output at servo off	0			
1567	Zero point allowable range at servo off (for XYZ axes)	0.0000			
1568	Zero point allowable range at servo off (for additional axes)	0.0000			
1569	Workpiece coordinate zero point X axis (for capture)	0.0000			
1570	Workpiece coordinate zero point Y axis (for capture)	0.0000			
1571	Workpiece coordinate zero point Z axis (for capture)	0.0000			

user parameter 5(User registered miscellaneous functions)

There is not default value.

User Parameter 6 (Tool breakage detection)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1801	Tool breakage detection option	0			
1802	Pallet 1 tool breakage detector	0			
1803	Pallet 2 tool breakage detector	0			
1804	Tool breakage detection execution interval(No. of operations)	1			
1805	Reference tool length offset No.	0			
1806	Tool breakage detection return position(Type1)	555.0000 (Standard) 625.0000 (Low)	455.0000	480.0000 (Standard) 630.0000 (High column150) 730.0000 (High column250)	550.0000
1807	Tool breakage detection position X-axis(Type1)	0.0000			
1808	Tool breakage detection position Y-axis(Type1)	0.0000			
1809	Tool breakage detection offset(Type2)	0.0000			
1810	Tool breakage detection position X-axis(Type3)	0.0000			
1811	Tool breakage detection position Y-axis(Type3)	0.0000			
1812	Tool breakage detection position Z-axis(Type3)	0.0000			
1813	Tool breakage detection start distance(Type3)	2.0000			
1814	Tool breakage detection stop distance(Type3)	1.0000			
1815	Tool breakage detection speed(Type3)	500.0			
1816	Tool breakage detection coolant OFF(Type3)	1			
1817	Pallet 2 tool breakage detection pos. X-axis(Type3)	0.0000			
1818	Pallet 2 tool breakage detection pos. Y-axis(Type3)	0.0000			
1819	Pallet 2 tool breakage detection pos. Z-axis(Type3)	0.0000			
1820	Pallet 2 tool breakage detection start distance(Type3)	2.0000			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1821	Pallet 2 tool breakage detection stop distance(Type3)	1.0000			
1822	Pallet 2 tool breakage detection speed(Type3)	500.0			
1823	Pallet 2 tool breakage detection coolant OFF(Type3)	1			

User Parameter 7 (Automatic workpiece measurement)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
2001	Probe compensation 1	0.0000			
2002	Probe compensation 2	0.0000			
2003	Probe compensation 3	0.0000			
2004	Probe compensation 4	0.0000			
2005	Measurement motion	0			
2006	Measurement speed 1	500.0			
2007	Measurement speed 2	100.0			
2008	Stop point before measurement	5.0000			
2009	Measurement travel limit distance	2.0000			
2010	Measurement tolerance 1	0.0000			
2011	Measurement tolerance 2	0.0000			
2012	Return distance after measurement	1.0000			

User Parameter 8 (Automatic thermal distortion compensation)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
2501	Z-axis compensation	100			
2502	Spindle compensation(Z-axis)	100			
2503	X-axis compensation	100			
2504	Spindle compensation(X-axis)	100			
2505	Y-axis compensation	100			
2506	Spindle compensation(Y-axis)	100			
2507	Auto. thermal distortion compen. at memory operation start	0			
2508	Lathe spindle compensation(Z-axis)	0			
2509	Lathe spindle compensation(X-axis)	0			
2510	Lathe spindle compensation(Y-axis)	0			

User Parameter 9 (Quick table)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
3001	Pallet program folder				
3002	Pallet 1 program No.				
3003	Pallet 2 program No.				
3004	Pallet program folder				
3005	Pallet 1 program No.				
3006	Pallet 2 program No.				
3007	Pallet 2 shift(X-axis)	0.0000			
3008	Pallet 2 shift(Y-axis)	0.0000			
3009	Pallet 2 shift(Z-axis)	0.0000			
3010	Pallet 1 loading	0			
3011	Pallet 2 loading	0			

User Parameter 10(High accuracy)
High accuracy A (M260)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override			120	
	Arc deceleration override			200	
	Curve approximation deceleration override			200	
	Smooth path offset level			1	
	Smooth override			100	
	Cutting feed time constant selection			0	
	Minute block deletion distance			0.0000	
	Accuracy level			0.0000	

High accuracy A (M261)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override			60	
	Arc deceleration override			140	
	Curve approximation deceleration override			140	
	Smooth path offset level			1	
	Smooth override			50	
	Cutting feed time constant selection			0	
	Minute block deletion distance			0.0000	
	Accuracy level			0.0000	

High accuracy A (M262)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override			70	
	Arc deceleration override			70	
	Curve approximation deceleration override			70	
	Smooth path offset level			1	
	Smooth override			600	
	Cutting feed time constant selection			0	
	Minute block deletion distance			0.0000	
	Accuracy level			0.0000	

High accuracy A (M263)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override			120	
	Arc deceleration override			200	
	Curve approximation deceleration override			200	
	Smooth path offset level			1	
	Smooth override			900	
	Cutting feed time constant selection			0	
	Minute block deletion distance			0.0000	
	Accuracy level			0.0000	

High accuracy A (M264)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override			120	
	Arc deceleration override			200	
	Curve approximation deceleration override			200	
	Smooth path offset level			1	
	Smooth override			2500	
	Cutting feed time constant selection			0	

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M265)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override	600			
	Arc deceleration override	1000			
	Curve approximation deceleration override	1000			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M266)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override	100			
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M267)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override	100			
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy B (M280)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	0			
	Curve approximation deceleration override	0			
	Smooth path offset level	3			
	Smooth override	100			
	Minute block deletion distance	0.0100			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M281)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M282)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M283)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M284)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M285)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M286)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M287)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy common

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
3201	Smooth path offset function	0			
3202	Smooth path offset cancel distance	10.0000			
3203	Smooth path offset cancel angle	10.0000			
3204	Smooth path offset angle determination invalid distance	0.0100			
3205	Smooth path offset reference length	1.0000			
3206	Minute block deletion when smooth path offset is cancelled	0			
3207	Accuracy level method	0	0	1	1

High accuracy(Adjust) common

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Machining mode			0	

High accuracy(Adjust) Adjust.A

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Standard mode			0	
	Accuracy			0	
	Smoothness			0	

High accuracy(Adjust) Adjust.B

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Standard mode			0	
	Accuracy			0	
	Smoothness			0	

High accuracy(Adjust) Adjust.C

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Standard mode			0	
	Accuracy			0	
	Smoothness			0	

High accuracy(Accuracy spec.) common

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Accuracy spec. mode			0	

High accuracy(Accuracy spec.) Accuracy spec. A

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Accuracy level			0.1000	
	Smoothness			0	

High accuracy(Accuracy spec.) Accuracy spec. B

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Accuracy level			0.0300	
	Smoothness			0	

High accuracy(Accuracy spec.) Accuracy spec. C

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Accuracy level			0.0100	
	Smoothness			0	

User Parameter 11 (PLC)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
3501	P1-axis stroke control		0		
3502	P2-axis stroke control		0		
3503	P3-axis stroke control		0		
3504	P4-axis stroke control		0		
3505	P1-axis stroke(-)		0		
3506	P2-axis stroke(-)		0		
3507	P3-axis stroke(-)		0		
3508	P4-axis stroke(-)		0		
3509	P1-axis stroke(+)		0		
3510	P2-axis stroke(+)		0		
3511	P3-axis stroke(+)		0		
3512	P4-axis stroke(+)		0		
3513	Reference point P1-axis		0		
3514	Reference point P2-axis		0		
3515	Reference point P3-axis		0		
3516	Reference point P4-axis		0		
3517	P1-axis override speed 1		0		
3518	P1-axis override speed 2		500		
3519	P1-axis override speed 3		5000		
3520	P1-axis override speed 4		10000		
3521	P2-axis override speed 1		0		
3522	P2-axis override speed 2		500		
3523	P2-axis override speed 3		5000		
3524	P2-axis override speed 4		10000		
3525	P3-axis override speed 1		0		
3526	P3-axis override speed 2		500		
3527	P3-axis override speed 3		5000		
3528	P3-axis override speed 4		10000		
3529	P4-axis override speed 1		0		
3530	P4-axis override speed 2		500		
3531	P4-axis override speed 3		5000		
3532	P4-axis override speed 4		10000		

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
3533	Collis. detct.-P1ax stop meth.	0			
3534	Collis. detct.-P2ax stop meth.	0			
3535	Collis. detct.-P3ax stop meth.	0			
3536	Collis. detct.-P4ax stop meth.	0			

User Parameter 12 (Profibus)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
4001	Node address	0			
4002	I/O setting 1				
4003	I/O setting 2				
4004	I/O setting 3				
4005	I/O setting 4				
4006	I/O setting 5				
4007	I/O setting 6				
4008	I/O setting 7				
4009	I/O setting 8				
4010	I/O setting 9				
4011	I/O setting 10				
4012	I/O setting 11				
4013	I/O setting 12				
4014	I/O setting 13				
4015	I/O setting 14				
4016	I/O setting 15				
4017	I/O setting 16				
4018	I/O setting 17				
4019	I/O setting 18				
4020	I/O setting 19				
4021	I/O setting 20				

User Parameter 13 (DeviceNet)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
4201	MAC ID	0			
4202	Transfer rate	0			
4203	Input bytes	0			
4204	Output bytes	0			

User Parameter 14 (CC-Link Slave)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
4401	Station number	1			
4402	Slave station category	0			
4403	Occupied number of stations	1			
4404	Transfer rate	0			
4405	Specify data clear in case of communication error	0			

User Parameter 15 (Rotary fixture offset)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
4601	XYZ-axes travel during current rotary fixture offset change	0			
4602	Rotary fixture offset reset method	0			
4603	Tilt axis 1	0	2	0	0
4604	Rotation axis 1	0	4	0	0
4605	Tilt axis 2	0			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
4606	Rotation axis 2	0			
4607	Tilt axis 3	0			
4608	Rotation axis 3	0			
4609	Tilt axis 4	0			
4610	Rotation axis 4	0			
4611	Forward direction for the coordinate system on tilt axis 1	0	1	0	0
4612	Forward direction for the coordinate system on rotation axis 1	0	5	0	0
4613	Forward direction for the coordinate system on tilt axis 2	0			
4614	Forward direction for the coordinate system on rotation axis 2	0			
4615	Forward direction for the coordinate system on tilt axis 3	0			
4616	Forward direction for the coordinate system on rotation axis 3	0			
4617	Forward direction for the coordinate system on tilt axis 4	0			
4618	Forward direction for the coordinate system on rotation axis 4	0			

User Parameter 16 (3D smoothing function)
3D smoothing function(M380)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M381)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M382)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M383)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M384)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M385)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M386)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M387)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function common

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
4801	3D smoothing circular interpolation command				
4802	3D smoothing sequence number command				

User Parameter 17 (feature coordinate setting)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
10001	Tilt axis 1	0	2	0	0
10002	Rotation axis 1	0	4	0	0
10003	Tilt axis 2		0		
10004	Rotation axis 2		0		
10005	Forward direction for the coordinate system on tilt axis 1	0	1	0	0
10006	Forward direction for the coordinate system on rotation axis 1	0	5	0	0
10007	Forward direction for the coordinate system on tilt axis 2		0		
10008	Forward direction for the coordinate system on rotation axis 2		0		
10009	Rotation center X coordinate offset for tilt axis 1		0.0000		
10010	Rotation center Y coordinate offset for tilt axis 1		0.0000		
10011	Rotation center Z coordinate offset for tilt axis 1		0.0000		
10012	Rotation center X coordinate offset for rotation axis 1		0.0000		
10013	Rotation center Y coordinate offset for rotation axis 1		0.0000		
10014	Rotation center Z coordinate offset for rotation axis 1		0.0000		
10015	Rotation center X coordinate offset for tilt axis 2		0.0000		
10016	Rotation center Y coordinate offset for tilt axis 2		0.0000		
10017	Rotation center Z coordinate offset for tilt axis 2		0.0000		
10018	Rotation center X coordinate offset for rotation axis 2		0.0000		
10019	Rotation center Y coordinate offset for rotation axis 2		0.0000		
10020	Rotation center Z coordinate offset for rotation axis 2		0.0000		
10021	Absolute coordinate display format		0		

User Parameter 18 (CC-Link master)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
10201	Transfer rate	0			
10202	Self-station number	0			
10203	Interface board setting	0			
10204	Occupied number of stations (for intelligent device stations)	0			
10205	Standby master station number	0			
10206	CC-Link version	0			
10207	Retry attempts	3			
10208	Number of automatic return nodes	1			
10209	Specify action when there is a driver error	0			
10210	Specify station data clear in case of any error	0			

2.5.20.4 Inch(R650X1,M140X1,S500X2,S700X2, S300X2,F600X1)

User Parameter 1 (Switch 1)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0001	Select mode		0		
0002	Initial mode		0		
0003	Select memory operation		0		
0004	Program repeat		0		
0005	Workpiece counter when program is repeated		0		
0006	Tool life management when program is repeated		0		
0007	Automatic tool length compensation		0.00000		
0008	External signal reading time limit		0		
0009	Dry run offset amount		0.00000		
0010	Dry run offset amount lower limit		0.00000		
0011	Program unit		0		
0012	Program feed rate unit		0		
0013	G73 relief amount		0.01960		
0014	G83 cutting start position		0.01960		
0015	G77,G78 relief amount		1.000		
0016	G76,G87 shift direction		0		
0017	Tapping cycle return speed		0		
0018	Program restart		0		
0019	S.T.M. return		1		
0020	Prevent restart from middle of program		0		
0021	4th row display axis		1		
0022	5th row display axis		2		
0023	6th row display axis		3		
0024	7th row display axis		4		
0025	8th row display axis		5		
0026	9th row display axis		6		
0027	10th row display axis		7		
0028	11th row display axis		8		
0029	12th row display axis		9		
0030	XYZ-axis absolute coord. Display		2		
0031	Display coordinates during rotational transformation		0		
0032	Macro variable 1 for position screen display				
0033	Macro variable 2 for position screen display				
0034	Macro variable 3 for position screen display				
0035	Macro variable 4 for position screen display				
0036	Macro variable 5 for position screen display				
0037	Block No. display color		4		
0038	Pointer display color		3		
0039	X/Y/Z-axis travel at tool length/tool pos. offset change		0		
0040	Condition to offset tool length/tool position		0		
0041	Start up/Cancel		0		
0042	Corner travel limit		0.00000		
0043	Automatic corner override(angle)		0.0000		
0044	Automatic corner override(length 1)		0.00000		
0045	Automatic corner override(length 2)		0.00000		
0046	Automatic corner override(ratio)		100		
0047	Inner arc override limit		100		
0048	Automatic centering speed 1		19.60		
0049	Automatic centering speed 2		0.30		
0050	Automatic centering return		0.01960		
0051	Automatic centering compensation		0.00000		
0052	Lamp status when count ends		0		

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0053	Multiple M codes in one block	0	1	0	0
0054	Unregistered M-code	0			
0055	One shot signal M450 output time	5			
0056	One shot signal M451 output time	5			
0057	One shot signal M455 output time	5			
0058	One shot signal M456 output time	5			
0059	Automatic screen dimming time	0			
0060	Magazine panel automatic screen clear time	0			
0061	Alarm display when screen is dimmed	0			
0062	Machine No.	0			
0063	Automatic door	0			
0064	Area sensor pattern	1			
0065	Automatic door error level	1			
0066	Automatic door operation time	5			
0067	Display alarm when door is open	0			
0068	START lamp status when operating automatic door manually	0			
0069	Override switch	1			
0070	Alarm display when override is valid	1			
0071	Prevent rapid traverse override	0			
0072	Prevent cutting feedrate override	0			
0073	Prevent spindle override	0			
0074	Change handle magnification	1			
0075	Auto screen change at additional/PLC axis handle operation	0			
0076	Z-axis height to prohibit travel of additional axes	0.00000	7.87400	0.00000	0.00000
0077	Z-axis height to prohibit travel of additional axes(pallet 2)	0.00000			
0078	Prevent external operation for additional axes	0			
0079	Macro command single stop	0			
0080	Interrupt macro interrupt type	0			
0081	Interrupt macro call type	0			
0082	Interrupt macro acceptance type	0			
0083	ATC simultaneous operation start position	21.85030 (Standard) 24.60620 (Low)	17.91330	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	21.65350
0084	ATC reference tool length	7.87400			
0085	Automatically use S command when changing tool	0			
0086	Automatically use F command when changing tool	0			
0087	Tool data spindle rotation exceeded	0			
0088	Magazine index when changing tools of different sizes	0			
0089	Pot lowering start position	0.00000			
0090	XY axes movement when changing tool in manual mode	1	0	0	0
0091	X-axis position when changing tool in manual mode	-25.59050	0.00000	0.00000	0.00000
0092	Y-axis position when changing tool in manual mode	0.00000			
0093	Schedule program reset start position	0			
0094	Program end stop in schedule program	0			
0095	Tool life counting method	0			
0096	Change tool life display(Type2)	0			

Chapter2 Data bank

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0097	Spare tool change	0			
0098	Check tool life on first use	1			
0099	External signal output time when MFIN is invalid	5			
0100	Chip shower retention time	0			
0101	Automatic coolant off time	0			
0102	Coolant delay time	0			
0103	Automatic machine light off time	0			
0104	Standby mode time	0			
0105	Auto power off	0			
0106	Auto power off time				
0107	Cancel editing when resetting	1			
0108	Save when mode of current program is changed	0			
0109	Program edit start position during operation	0			
0110	Automatic sequence insertion increment	10			
0111	Set same number to edit/memory operation programs	0			
0112	Assign tools when starting memory operation	1			
0113	Check tools to be used when assigned automatically	1			
0114	Workpc. zero conversion code	0			
0115	Convert tool data	1			
0116	Tap step return position	0			
0117	Magazine turn speed ratio 1(M436)	80			
0118	Magazine turn speed ratio 2(M437)	50			
0119	Turret type magazine speed	0			
0120	Turret type magazine low-speed turn mode	0			
0121	Speed turret type magazine turns at low speed	20.00			
0122	Pallet select key change	1			
0123	Lock shift key	0			
0124	Max. actual cutting travel speed	787.40			
0125	Table load weight	400	40	250	400
0126	Arc radius error limit	0.00000			
0127	Scaling	0			
0128	X-axis scaling	0			
0129	Y-axis scaling	0			
0130	Z-axis scaling	0			
0131	Scaling unit	0			
0132	Common scaling factor	1000			
0133	X-axis scaling factor(I)	1000			
0134	Y-axis scaling factor(J)	1000			
0135	Z-axis scaling factor(K)	1000			
0136	Back washing at end of program	1			
0137	CTS pressurize check delay time	1.0			
0138	CTS depressurize check delay time	1.0			
0139	CTS pressure bleed time	0.0			
0140	CTS line filter alarm stop level	1			
0141	ALM2 signal stop level	0			
0142	Oiling/greasing when power is turned on	0			
0143	Tool wash filter alarm stop level	0			
0144	Tool wash filter check interval	1			
0145	Display tool wash filter check result	0			
0146	Faulty tool wash coolant level alarm stop(Type4)	1			
0147	Tool wash filter clogging warning alarm stop level(Type4)	0			
0148	No. of tool wash filter clogging warning checks(Type4)	1			
0149	Faulty tool wash coolant level sensor check interval(Type4)	200			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0150	Max. tapping speed (when returning tap)	100			
0151	Motor insulation resistance measurement interval	0			
0152	Motor insulation resistance monitoring acceptable value (all axes)	5.0			
0153	Motor insulation resistance monitoring acceptable value (abnormal)	1.1			
0154	Motor insulation resistance monitoring acceptable value (alarm)	5.0			
0155	Spindle rotation return during restart	0	1	0	0
0156	M function signal output delay time	1			
0157	Finish axis travel during external signal output	0			
0158	Chip conveyor relay adhering check	0			
0159	Error check when traveling during tool length/tool position offset cancel	0			
0160	Lathe machining infeed direction when power is turned ON	0			
0161	Cutting amount in thread cutting cycle	1.5			
0162	Cutting angle in thread cutting cycle	45.0000			
0163	Thread angle in complex thread cutting cycle	60.0000			
0164	Minimum infeed amount in complex thread cutting cycle	0.00190			
0165	Finishing count in complex thread cutting cycle	1			
0166	Finishing allowance in complex thread cutting cycle	0.00090			
0167	Signal batch reading/writing type	0			
0168	Z-axis travel method during tool change	0			
0169	Tapping sync. error too big - alarm stop level	0			
0170	Stop using feedrate override during rapid feed	0			
0171	Arc radius R command error limit	0.00010			
0172	Positioning method	0			
0173	Program load size	0	0	1	1
0174	No tool detection function (when ATC rises)	0			
0175	No tool detection function (when ATC falls)	0			
0176	Tool over key detection function	0			
0177	Tool collision detection function during ATC	0			
0178	Tool missing detection function	0			
0179	Spindle return method for program stop	1	1	0	0
0180	Lathe spindle return method for program stop	2	2	0	0
0181	Coolant return method for program stop	1	1	0	0
0182	Tap accel. Setting	0	0	1 (10K) 1 (10K(High torque)) 1 (16K) 0 (27K)	1
0183	Reset when operation preparation is ON	0			
0184	Screenshot	0			
0185	Probe detection signal method	0	0	1	1
0186	PLC rapid traverse override function	0			
0187	PLC feedrate override function	0			
0188	PLC spindle override function	0			
0189	ATC arm turn speed ratio 1 (M422)	100			
0190	ATC arm turn speed ratio 2 (M423)	100			
0191	Tool wash discharge conditions for tool data	0			
0192	Tool wash when changing a different size tool	0			
0193	Cutting speed exceeded for inverse time feed	0			
0194	Cutter compensation interference check	0			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0195	Number of blocks to check cutter compensation interference	8			
0196	Involute interpolation override limit	0			
0197	Involute interpolation error limit	0.00390			
0198	Reset cycle time value	0			

User Parameter 2 (Switch 2)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0501	4th-axis stroke control	1			
0502	5th-axis stroke control	0	1	0	0
0503	6th-axis stroke control	0			
0504	7th-axis stroke control	0			
0505	8th-axis stroke control	0			
0506	4th-axis stroke(-)	-180.0000			
0507	5th-axis stroke(-)	0.0000	-30.0000	0.0000	0.0000
0508	6th-axis stroke(-)	0.0000			
0509	7th-axis stroke(-)	0.0000			
0510	8th-axis stroke(-)	0.0000			
0511	4th-axis stroke(+)	0.0000			
0512	5th-axis stroke(+)	0.0000	120.0000	0.0000	0.0000
0513	6th-axis stroke(+)	0.0000			
0514	7th-axis stroke(+)	0.0000			
0515	8th-axis stroke(+)	0.0000			
0516	Stroke limit 1 for X-axis(-)	-25.59050	-7.87400	-19.68500 (Standard) -27.55900 (Long) -11.81110 (Short)	-23.62200
0517	Stroke limit 1 for Y-axis(-)	-15.74800	-17.32280	-15.74800	-15.74800
0518	Stroke limit 1 for Z-axis(-)	0.00000			
0519	Stroke limit 1 for 4th-axis(-)	-180.0000			
0520	Stroke limit 1 for 5th-axis(-)	0.0000	-30.0000	0.0000	0.0000
0521	Stroke limit 1 for 6th-axis(-)	0.0000			
0522	Stroke limit 1 for 7th-axis(-)	0.0000			
0523	Stroke limit 1 for 8th-axis(-)	0.0000			
0524	Stroke limit 1 for X-axis(+)	0.00000			
0525	Stroke limit 1 for Y-axis(+)	0.00000			
0526	Stroke limit 1 for Z-axis(+)	21.85030 (Standard) 24.60620 (Low)	17.91330	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	21.65350
0527	Stroke limit 1 for 4th-axis(+)	0.0000			
0528	Stroke limit 1 for 5th-axis(+)	0.0000	120.0000	0.0000	0.0000
0529	Stroke limit 1 for 6th-axis(+)	0.0000			
0530	Stroke limit 1 for 7th-axis(+)	0.0000			
0531	Stroke limit 1 for 8th-axis(+)	0.0000			
0532	Stroke limit 2 for X-axis(-)	0.00000			
0533	Stroke limit 2 for Y-axis(-)	0.00000			
0534	Stroke limit 2 for Z-axis(-)	0.00000			
0535	Stroke limit 2 for X-axis(+)	0.00000			
0536	Stroke limit 2 for Y-axis(+)	0.00000			
0537	Stroke limit 2 for Z-axis(+)	0.00000			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0538	Rapid traverse override speed 1	0.00			
0539	Rapid traverse override speed 2	19.60			
0540	Rapid traverse override speed 3	196.80			
0541	Rapid traverse override speed 4	393.70			
0542	4th-axis override rotation speed 1	0.00			
0543	4th-axis override rotation speed 2	0.30			
0544	4th-axis override rotation speed 3	2.50			
0545	4th-axis override rotation speed 4	5.00			
0546	5th-axis override rotation speed 1	0.00			
0547	5th-axis override rotation speed 2	0.50			
0548	5th-axis override rotation speed 3	5.00			
0549	5th-axis override rotation speed 4	10.00			
0550	6th-axis override rotation speed 1	0.00			
0551	6th-axis override rotation speed 2	0.50			
0552	6th-axis override rotation speed 3	5.00			
0553	6th-axis override rotation speed 4	10.00			
0554	7th-axis override rotation speed 1	0.00			
0555	7th-axis override rotation speed 2	0.50	1.00	0.50	0.50
0556	7th-axis override rotation speed 3	5.00	10.00	5.00	5.00
0557	7th-axis override rotation speed 4	10.00	30.00	10.00	10.00
0558	8th-axis override rotation speed 1	0.00			
0559	8th-axis override rotation speed 2	0.50			
0560	8th-axis override rotation speed 3	5.00			
0561	8th-axis override rotation speed 4	10.00			
0562	X-axis position signal output range(-)	0.00000			
0563	X-axis position signal output range(+)	0.00000			
0564	Y-axis position signal output range(-)	0.00000			
0565	Y-axis position signal output range(+)	0.00000			
0566	Z-axis position signal output range(-)	0.00000			
0567	Z-axis position signal output range(+)	0.00000			
0568	4th-axis position signal output range(-)	0.0000			
0569	4th-axis position signal output range(+)	0.0000			
0570	5th-axis position signal output range(-)	0.0000			
0571	5th-axis position signal output range(+)	0.0000			
0572	6th-axis position signal output range(-)	0.0000			
0573	6th-axis position signal output range(+)	0.0000			
0574	7th-axis position signal output range(-)	0.0000			
0575	7th-axis position signal output range(+)	0.0000			
0576	8th-axis position signal output range(-)	0.0000			
0577	8th-axis position signal output range(+)	0.0000			
0578	X-axis single direction positioning excess travel amount	0.00000			
0579	Y-axis single direction positioning excess travel amount	0.00000			
0580	Z-axis single direction positioning excess travel amount	0.00000			
0581	4th-axis single direction positioning excess travel amount	0.0000			
0582	5th-axis single direction positioning excess travel amount	0.0000			
0583	6th-axis single direction positioning excess travel amount	0.0000			
0584	7th-axis single direction positioning excess travel amount	0.0000			
0585	8th-axis single direction positioning excess travel amount	0.0000			
0586	External index A switch address switching	0			
0587	External A-axis indexing(+)	180.0000			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
0588	External A-axis indexing(-)	-180.0000			
0589	External index B switch address switching	0			
0590	External B-axis indexing(+)	180.0000			
0591	External B-axis indexing(-)	-180.0000			
0592	Z-axis lowering speed 1 when changing tool	1377.90			
0593	Z-axis lowering speed 2 when changing tool	1377.90			
0594	Z-axis lowering speed 3 when changing tool	1377.90			
0595	X-axis position signal output range 2 (-)	0.00000			
0596	X-axis position signal output range 2 (+)	0.00000			
0597	Y-axis position signal output range 2 (-)	0.00000			
0598	Y-axis position signal output range 2 (+)	0.00000			
0599	Z-axis position signal output range 2 (-)	0.00000			
0600	Z-axis position signal output range 2 (+)	0.00000			
0601	4-axis position signal output range 2 (-)	0.0000			
0602	4-axis position signal output range 2 (+)	0.0000			
0603	5-axis position signal output range 2 (-)	0.0000			
0604	5-axis position signal output range 2 (+)	0.0000			
0605	6-axis position signal output range 2 (-)	0.0000			
0606	6-axis position signal output range 2 (+)	0.0000			
0607	7-axis position signal output range 2 (-)	0.0000			
0608	7-axis position signal output range 2 (+)	0.0000			
0609	8-axis position signal output range 2 (-)	0.0000			
0610	8-axis position signal output range 2 (+)	0.0000			
0611	X-axis position signal output range 3 (-)	0.00000			
0612	X-axis position signal output range 3 (+)	0.00000			
0613	Y-axis position signal output range 3 (-)	0.00000			
0614	Y-axis position signal output range 3 (+)	0.00000			
0615	Z-axis position signal output range 3 (-)	0.00000			
0616	Z-axis position signal output range 3 (+)	0.00000			
0617	4-axis position signal output range 3 (-)	0.0000			
0618	4-axis position signal output range 3 (+)	0.0000			
0619	5-axis position signal output range 3 (-)	0.0000			
0620	5-axis position signal output range 3 (+)	0.0000			
0621	6-axis position signal output range 3 (-)	0.0000			
0622	6-axis position signal output range 3 (+)	0.0000			
0623	7-axis position signal output range 3 (-)	0.0000			
0624	7-axis position signal output range 3 (+)	0.0000			
0625	8-axis position signal output range 3 (-)	0.0000			
0626	8-axis position signal output range 3 (+)	0.0000			
0627	X-axis position signal output range 4 (-)	0.00000			
0628	X-axis position signal output range 4 (+)	0.00000			
0629	Y-axis position signal output range 4 (-)	0.00000			
0630	Y-axis position signal output range 4 (+)	0.00000			
0631	Z-axis position signal output range 4 (-)	0.00000			
0632	Z-axis position signal output range 4 (+)	0.00000			
0633	4-axis position signal output range 4 (-)	0.0000			
0634	4-axis position signal output range 4 (+)	0.0000			
0635	5-axis position signal output range 4 (-)	0.0000			
0636	5-axis position signal output range 4 (+)	0.0000			
0637	6-axis position signal output range 4 (-)	0.0000			
0638	6-axis position signal output range 4 (+)	0.0000			
0639	7-axis position signal output range 4 (-)	0.0000			
0640	7-axis position signal output range 4 (+)	0.0000			
0641	8-axis position signal output range 4 (-)	0.0000			
0642	8-axis position signal output range 4 (+)	0.0000			

User Parameter 3 (Switch 3)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1001	M271 positioning end check distance			0.01180	
1002	M271 positioning end check angle			0.3000	
1003	M272 positioning end check distance			0.01180	
1004	M272 positioning end check angle			0.3000	
1005	M273 positioning end check distance			0.01180	
1006	M273 positioning end check angle			0.3000	
1007	M274 positioning end check distance			0.01180	
1008	M274 positioning end check angle			0.3000	
1009	M275 positioning end check distance			0.01180	
1010	M275 positioning end check angle			0.3000	
1011	M276 positioning end check distance			0.01180	
1012	M276 positioning end check angle			0.3000	
1013	M277 positioning end check distance			0.01180	
1014	M277 positioning end check angle			0.3000	
1015	M278 positioning end check distance			0.01180	
1016	M278 positioning end check angle			0.3000	
1017	M279 positioning end check distance			0.01180	
1018	M279 positioning end check angle			0.3000	
1019	Low priority indication alarm No. 1				
1020	Low priority indication alarm No. 2				
1021	Low priority indication alarm No. 3				
1022	Low priority indication alarm No. 4				
1023	Low priority indication alarm No. 5				
1024	Low priority indication alarm No. 6				
1025	Low priority indication alarm No. 7				
1026	Low priority indication alarm No. 8				
1027	Low priority indication alarm No. 9				
1028	Low priority indication alarm No. 10				
1029	G code for modal display 1			1	
1030	G code for modal display 2			2	
1031	G code for modal display 3			3	
1032	G code for modal display 4			4	
1033	G code for modal display 5			5	
1034	G code for modal display 6			6	
1035	G code for modal display 7			7	
1036	G code for modal display 8			10	
1037	G code for modal display 9			8	
1038	G code for modal display 10			11	
1039	G code for modal display 11			12	
1040	G code for modal display 12			13	
1041	G code for modal display 13			14	
1042	G code for modal display 14			15	
1043	G code for modal display 15			16	
1044	G code for modal display 16			17	
1045	M code for modal display 1			1	
1046	M code for modal display 2			2	
1047	M code for modal display 3			3	
1048	M code for modal display 4			4	
1049	M code for modal display 5			5	
1050	M code for modal display 6			6	
1051	M code for modal display 7			7	
1052	M code for modal display 8			8	
1053	M code for modal display 9			9	
1054	M code for modal display 10			11	
1055	M code for modal display 11			12	
1056	M code for modal display 12			14	

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1057	M code for modal display 13	16			
1058	M code for modal display 14	31			
1059	M code for modal display 15	0			
1060	M code for modal display 16	0			
1061	Angle setting to check when additional axis for positioning is finished	0			
1062	Specified alarm signal 2				
1063	Specified alarm signal 3				
1064	Specified alarm signal 4				
1065	Specified alarm signal 5				

User Parameter 4 (Reference position/Conversation zero point)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1501	Reference position X-axis	0.00000			
1502	Reference position Y-axis	0.00000			
1503	Reference position Z-axis	21.85030 (Standard) 24.60620 (Low)	17.91330	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	21.65350
1504	Reference position 4th-axis	0.0000			
1505	Reference position 5th-axis	0.0000			
1506	Reference position 6th-axis	0.0000			
1507	Reference position 7th-axis	0.0000			
1508	Reference position 8th-axis	0.0000			
1509	2nd reference position X-axis	0.00000			
1510	2nd reference position Y-axis	0.00000			
1511	2nd reference position Z-axis	21.85030 (Standard) 24.60620 (Low)	17.91330	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	21.65350
1512	2nd reference position 4th-axis	0.0000			
1513	2nd reference position 5th-axis	0.0000			
1514	2nd reference position 6th-axis	0.0000			
1515	2nd reference position 7th-axis	0.0000			
1516	2nd reference position 8th-axis	0.0000			
1517	3rd reference position X-axis	0.00000			
1518	3rd reference position Y-axis	0.00000			
1519	3rd reference position Z-axis	21.85030 (Standard) 24.60620 (Low)	17.91330	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	21.65350
1520	3rd reference position 4th-axis	0.0000			
1521	3rd reference position 5th-axis	0.0000			
1522	3rd reference position 6th-axis	0.0000			
1523	3rd reference position 7th-axis	0.0000			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1524	3rd reference position 8th-axis	0.0000			
1525	4th reference position X-axis	0.00000			
1526	4th reference position Y-axis	0.00000			
1527	4th reference position Z-axis	21.85030 (Standard) 24.60620 (Low)	17.91330	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	21.65350
1528	4th reference position 4th-axis	0.0000			
1529	4th reference position 5th-axis	0.0000			
1530	4th reference position 6th-axis	0.0000			
1531	4th reference position 7th-axis	0.0000			
1532	4th reference position 8th-axis	0.0000			
1533	5th reference position X-axis	0.00000			
1534	5th reference position Y-axis	0.00000			
1535	5th reference position Z-axis	21.85030 (Standard) 24.60620 (Low)	17.91330	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	21.65350
1536	5th reference position 4th-axis	0.0000			
1537	5th reference position 5th-axis	0.0000			
1538	5th reference position 6th-axis	0.0000			
1539	5th reference position 7th-axis	0.0000			
1540	5th reference position 8th-axis	0.0000			
1541	6th reference position X-axis	0.00000			
1542	6th reference position Y-axis	0.00000			
1543	6th reference position Z-axis	21.85030 (Standard) 24.60620 (Low)	17.91330	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	21.65350
1544	6th reference position 4th-axis	0.0000			
1545	6th reference position 5th-axis	0.0000			
1546	6th reference position 6th-axis	0.0000			
1547	6th reference position 7th-axis	0.0000			
1548	6th reference position 8th-axis	0.0000			
1549	Reference point return complete signal output at servo off	0			
1550	Reference point allowable range at servo off (for XYZ axes)	0.00000			
1551	Reference point allowable range at servo off (for additional axes)	0.0000			
1552	X-axis zero return pos.(No.2)	0.00000			
1553	Y-axis zero return pos.(No.2)	0.00000			
1554	4th-axis zero return pos.(No.2)	0.0000			
1555	5th-axis zero return pos.(No.2)	0.0000			
1556	6th-axis zero return pos.(No.2)	0.0000			
1557	7th-axis zero return pos.(No.2)	0.0000			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1558	8th-axis zero return pos.(No.2)	0.0000			
1559	X-axis zero return pos.(No.3)	0.00000			
1560	Y-axis zero return pos.(No.3)	0.00000			
1561	4th-axis zero return pos.(No.3)	0.0000			
1562	5th-axis zero return pos.(No.3)	0.0000			
1563	6th-axis zero return pos.(No.3)	0.0000			
1564	7th-axis zero return pos.(No.3)	0.0000			
1565	8th-axis zero return pos.(No.3)	0.0000			
1566	Zero point return complete signal output at servo off	0			
1567	Zero point allowable range at servo off (for XYZ axes)	0.00000			
1568	Zero point allowable range at servo off (for additional axes)	0.0000			
1569	Workpiece coordinate zero point X axis (for capture)	0.00000			
1570	Workpiece coordinate zero point Y axis (for capture)	0.00000			
1571	Workpiece coordinate zero point Z axis (for capture)	0.00000			

user parameter 5(User registered miscellaneous functions)

There is not default value.

User Parameter 6 (Tool breakage detection)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1801	Tool breakage detection option	0			
1802	Pallet 1 tool breakage detector	0			
1803	Pallet 2 tool breakage detector	0			
1804	Tool breakage detection execution interval(No. of operations)	1			
1805	Reference tool length offset No.	0			
1806	Tool breakage detection return position(Type1)	21.85030 (Standard) 24.60620 (Low)	17.91330	18.89760 (Standard) 24.80310 (High column150) 28.74010 (High column250)	21.65350
1807	Tool breakage detection position X-axis(Type1)	0.00000			
1808	Tool breakage detection position Y-axis(Type1)	0.00000			
1809	Tool breakage detection offset(Type2)	0.00000			
1810	Tool breakage detection position X-axis(Type3)	0.00000			
1811	Tool breakage detection position Y-axis(Type3)	0.00000			
1812	Tool breakage detection position Z-axis(Type3)	0.00000			
1813	Tool breakage detection start distance(Type3)	0.07870			
1814	Tool breakage detection stop distance(Type3)	0.03930			
1815	Tool breakage detection speed(Type3)	19.60			
1816	Tool breakage detection coolant OFF(Type3)	1			
1817	Pallet 2 tool breakage detection pos. X-axis(Type3)	0.00000			
1818	Pallet 2 tool breakage detection pos. Y-axis(Type3)	0.00000			
1819	Pallet 2 tool breakage detection pos. Z-axis(Type3)	0.00000			
1820	Pallet 2 tool breakage detection start distance(Type3)	0.07870			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
1821	Pallet 2 tool breakage detection stop distance(Type3)	0.03930			
1822	Pallet 2 tool breakage detection speed(Type3)	19.60			
1823	Pallet 2 tool breakage detection coolant OFF(Type3)	1			

User Parameter 7 (Automatic workpiece measurement)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
2001	Probe compensation 1	0.00000			
2002	Probe compensation 2	0.00000			
2003	Probe compensation 3	0.00000			
2004	Probe compensation 4	0.00000			
2005	Measurement motion	0			
2006	Measurement speed 1	19.60			
2007	Measurement speed 2	3.90			
2008	Stop point before measurement	0.19680			
2009	Measurement travel limit distance	0.07870			
2010	Measurement tolerance 1	0.00000			
2011	Measurement tolerance 2	0.00000			
2012	Return distance after measurement	0.03930			

User Parameter 8 (Automatic thermal distortion compensation)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
2501	Z-axis compensation	100			
2502	Spindle compensation(Z-axis)	100			
2503	X-axis compensation	100			
2504	Spindle compensation(X-axis)	100			
2505	Y-axis compensation	100			
2506	Spindle compensation(Y-axis)	100			
2507	Auto. thermal distortion compen. at memory operation start	0			
2508	Lathe spindle compensation(Z-axis)	0			
2509	Lathe spindle compensation(X-axis)	0			
2510	Lathe spindle compensation(Y-axis)	0			

User Parameter 9 (Quick table)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
3001	Pallet program folder				
3002	Pallet 1 program No.				
3003	Pallet 2 program No.				
3004	Pallet program folder				
3005	Pallet 1 program No.				
3006	Pallet 2 program No.				
3007	Pallet 2 shift(X-axis)	0.00000			
3008	Pallet 2 shift(Y-axis)	0.00000			
3009	Pallet 2 shift(Z-axis)	0.00000			
3010	Pallet 1 loading	0			
3011	Pallet 2 loading	0			

User Parameter 10(High accuracy)
High accuracy A (M260)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override			120	
	Arc deceleration override			200	
	Curve approximation deceleration override			200	
	Smooth path offset level			1	
	Smooth override			100	
	Cutting feed time constant selection			0	
	Minute block deletion distance			0.00000	
	Accuracy level			0.00000	

High accuracy A (M261)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override			60	
	Arc deceleration override			140	
	Curve approximation deceleration override			140	
	Smooth path offset level			1	
	Smooth override			50	
	Cutting feed time constant selection			0	
	Minute block deletion distance			0.00000	
	Accuracy level			0.00000	

High accuracy A (M262)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override			70	
	Arc deceleration override			70	
	Curve approximation deceleration override			70	
	Smooth path offset level			1	
	Smooth override			600	
	Cutting feed time constant selection			0	
	Minute block deletion distance			0.00000	
	Accuracy level			0.00000	

High accuracy A (M263)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override			120	
	Arc deceleration override			200	
	Curve approximation deceleration override			200	
	Smooth path offset level			1	
	Smooth override			900	
	Cutting feed time constant selection			0	
	Minute block deletion distance			0.00000	
	Accuracy level			0.00000	

High accuracy A (M264)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override			120	
	Arc deceleration override			200	
	Curve approximation deceleration override			200	
	Smooth path offset level			1	
	Smooth override			2500	
	Cutting feed time constant selection			0	

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M265)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override	600			
	Arc deceleration override	1000			
	Curve approximation deceleration override	1000			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M266)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override	100			
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M267)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Corner deceleration override	100			
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy B (M280)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	0			
	Curve approximation deceleration override	0			
	Smooth path offset level	3			
	Smooth override	100			
	Minute block deletion distance	0.00030			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M281)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M282)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M283)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M284)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M285)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M286)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M287)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy common

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
3201	Smooth path offset function	0			
3202	Smooth path offset cancel distance	0.39300			
3203	Smooth path offset cancel angle	10.0000			
3204	Smooth path offset angle determination invalid distance	0.00030			
3205	Smooth path offset reference length	0.03930			
3206	Minute block deletion when smooth path offset is cancelled	0			
3207	Accuracy level method	0	0	1	1

High accuracy(Adjust) common

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Machining mode			0	

High accuracy(Adjust) Adjust.A

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Standard mode			0	
	Accuracy			0	
	Smoothness			0	

High accuracy(Adjust) Adjust.B

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Standard mode			0	
	Accuracy			0	
	Smoothness			0	

High accuracy(Adjust) Adjust.C

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Standard mode			0	
	Accuracy			0	
	Smoothness			0	

High accuracy(Accuracy spec.) common

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Accuracy spec. mode			0	

High accuracy(Accuracy spec.) Accuracy spec. A

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Accuracy level			0.00390	
	Smoothness			0	

High accuracy(Accuracy spec.) Accuracy spec. B

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Accuracy level			0.00120	
	Smoothness			0	

High accuracy(Accuracy spec.) Accuracy spec. C

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Accuracy level			0.00040	
	Smoothness			0	

User Parameter 11 (PLC)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
3501	P1-axis stroke control		0		
3502	P2-axis stroke control		0		
3503	P3-axis stroke control		0		
3504	P4-axis stroke control		0		
3505	P1-axis stroke(-)		0		
3506	P2-axis stroke(-)		0		
3507	P3-axis stroke(-)		0		
3508	P4-axis stroke(-)		0		
3509	P1-axis stroke(+)		0		
3510	P2-axis stroke(+)		0		
3511	P3-axis stroke(+)		0		
3512	P4-axis stroke(+)		0		
3513	Reference point P1-axis		0		
3514	Reference point P2-axis		0		
3515	Reference point P3-axis		0		
3516	Reference point P4-axis		0		
3517	P1-axis override speed 1		0		
3518	P1-axis override speed 2		20		
3519	P1-axis override speed 3		200		
3520	P1-axis override speed 4		400		
3521	P2-axis override speed 1		0		
3522	P2-axis override speed 2		20		
3523	P2-axis override speed 3		200		
3524	P2-axis override speed 4		400		
3525	P3-axis override speed 1		0		
3526	P3-axis override speed 2		20		
3527	P3-axis override speed 3		200		
3528	P3-axis override speed 4		400		
3529	P4-axis override speed 1		0		
3530	P4-axis override speed 2		20		
3531	P4-axis override speed 3		200		
3532	P4-axis override speed 4		400		

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
3533	Collis. detct.-P1ax stop meth.	0			
3534	Collis. detct.-P2ax stop meth.	0			
3535	Collis. detct.-P3ax stop meth.	0			
3536	Collis. detct.-P4ax stop meth.	0			

User Parameter 12 (Profibus)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
4001	Node address	0			
4002	I/O setting 1				
4003	I/O setting 2				
4004	I/O setting 3				
4005	I/O setting 4				
4006	I/O setting 5				
4007	I/O setting 6				
4008	I/O setting 7				
4009	I/O setting 8				
4010	I/O setting 9				
4011	I/O setting 10				
4012	I/O setting 11				
4013	I/O setting 12				
4014	I/O setting 13				
4015	I/O setting 14				
4016	I/O setting 15				
4017	I/O setting 16				
4018	I/O setting 17				
4019	I/O setting 18				
4020	I/O setting 19				
4021	I/O setting 20				

User Parameter 13 (DeviceNet)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
4201	MAC ID	0			
4202	Transfer rate	0			
4203	Input bytes	0			
4204	Output bytes	0			

User Parameter 14 (CC-Link Slave)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
4401	Station number	1			
4402	Slave station category	0			
4403	Occupied number of stations	1			
4404	Transfer rate	0			
4405	Specify data clear in case of communication error	0			

User Parameter 15 (Rotary fixture offset)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
4601	XYZ-axes travel during current rotary fixture offset change	0			
4602	Rotary fixture offset reset method	0			
4603	Tilt axis 1	0	2	0	0
4604	Rotation axis 1	0	4	0	0
4605	Tilt axis 2	0			

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
4606	Rotation axis 2			0	
4607	Tilt axis 3			0	
4608	Rotation axis 3			0	
4609	Tilt axis 4			0	
4610	Rotation axis 4			0	
4611	Forward direction for the coordinate system on tilt axis 1	0	1	0	0
4612	Forward direction for the coordinate system on rotation axis 1	0	5	0	0
4613	Forward direction for the coordinate system on tilt axis 2			0	
4614	Forward direction for the coordinate system on rotation axis 2			0	
4615	Forward direction for the coordinate system on tilt axis 3			0	
4616	Forward direction for the coordinate system on rotation axis 3			0	
4617	Forward direction for the coordinate system on tilt axis 4			0	
4618	Forward direction for the coordinate system on rotation axis 4			0	

User Parameter 16 (3D smoothing function)
3D smoothing function(M380)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M381)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M382)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M383)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M384)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M385)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M386)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M387)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function common

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
4801	3D smoothing circular interpolation command				
4802	3D smoothing sequence number command				

User Parameter 17 (feature coordinate setting)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
10001	Tilt axis 1	0	2	0	0
10002	Rotation axis 1	0	4	0	0
10003	Tilt axis 2		0		
10004	Rotation axis 2		0		
10005	Forward direction for the coordinate system on tilt axis 1	0	1	0	0
10006	Forward direction for the coordinate system on rotation axis 1	0	5	0	0
10007	Forward direction for the coordinate system on tilt axis 2		0		
10008	Forward direction for the coordinate system on rotation axis 2		0		
10009	Rotation center X coordinate offset for tilt axis 1		0.00000		
10010	Rotation center Y coordinate offset for tilt axis 1		0.00000		
10011	Rotation center Z coordinate offset for tilt axis 1		0.00000		
10012	Rotation center X coordinate offset for rotation axis 1		0.00000		
10013	Rotation center Y coordinate offset for rotation axis 1		0.00000		
10014	Rotation center Z coordinate offset for rotation axis 1		0.00000		
10015	Rotation center X coordinate offset for tilt axis 2		0.00000		
10016	Rotation center Y coordinate offset for tilt axis 2		0.00000		
10017	Rotation center Z coordinate offset for tilt axis 2		0.00000		
10018	Rotation center X coordinate offset for rotation axis 2		0.00000		
10019	Rotation center Y coordinate offset for rotation axis 2		0.00000		
10020	Rotation center Z coordinate offset for rotation axis 2		0.00000		
10021	Absolute coordinate display format		0		

User Parameter 18 (CC-Link master)

No.	Item	R650X1	M140X2	S500X2 S700X2 S300X2	F600X1
10201	Transfer rate	0			
10202	Self-station number	0			
10203	Interface board setting	0			
10204	Occupied number of stations (for intelligent device stations)	0			
10205	Standby master station number	0			
10206	CC-Link version	0			
10207	Retry attempts	3			
10208	Number of automatic return nodes	1			
10209	Specify action when there is a driver error	0			
10210	Specify station data clear in case of any error	0			

2.5.20.5 Metric(R650X2_40MG,R450X2,R650X2,M200X3,M300X3)

User Parameter 1 (Switch 1)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0001	Select mode		0		
0002	Initial mode		0		
0003	Select memory operation		0		
0004	Program repeat		0		
0005	Workpiece counter when program is repeated		0		
0006	Tool life management when program is repeated		0		
0007	Automatic tool length compensation		0.0000		
0008	External signal reading time limit		0		
0009	Dry run offset amount		0.0000		
0010	Dry run offset amount lower limit		0.0000		
0011	Program unit		0		
0012	Program feed rate unit		0		
0013	G73 relief amount		0.5000		
0014	G83 cutting start position		0.5000		
0015	G77,G78 relief amount		1.000		
0016	G76,G87 shift direction		0		
0017	Tapping cycle return speed		0		
0018	Program restart		0		
0019	S.T.M. return		1		
0020	Prevent restart from middle of program		0		
0021	4th row display axis		1		
0022	5th row display axis		2		
0023	6th row display axis		3		
0024	7th row display axis		4		
0025	8th row display axis		5		
0026	9th row display axis		6		
0027	10th row display axis		7		
0028	11th row display axis		8		
0029	12th row display axis		9		
0030	XYZ-axis absolute coord. Display		2		
0031	Display coordinates during rotational transformation		0		
0032	Macro variable 1 for position screen display				
0033	Macro variable 2 for position screen display				
0034	Macro variable 3 for position screen display				
0035	Macro variable 4 for position screen display				
0036	Macro variable 5 for position screen display				
0037	Block No. display color		4		
0038	Pointer display color		3		
0039	X/Y/Z-axis travel at tool length/tool pos. offset change		0		
0040	Condition to offset tool length/tool position		0		
0041	Start up/Cancel		0		
0042	Corner travel limit		0.0000		
0043	Automatic corner override(angle)		0.0000		
0044	Automatic corner override(length 1)		0.0000		
0045	Automatic corner override(length 2)		0.0000		
0046	Automatic corner override(ratio)		100		
0047	Inner arc override limit		100		
0048	Automatic centering speed 1		500.0		
0049	Automatic centering speed 2		10.0		
0050	Automatic centering return		0.5000		
0051	Automatic centering compensation		0.0000		
0052	Lamp status when count ends		0		

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0053	Multiple M codes in one block	0	0	0	1
0054	Unregistered M-code	0			
0055	One shot signal M450 output time	5			
0056	One shot signal M451 output time	5			
0057	One shot signal M455 output time	5			
0058	One shot signal M456 output time	5			
0059	Automatic screen dimming time	0			
0060	Magazine panel automatic screen clear time	0			
0061	Alarm display when screen is dimmed	0			
0062	Machine No.	0			
0063	Automatic door	0			
0064	Area sensor pattern	1			
0065	Automatic door error level	1			
0066	Automatic door operation time	5			
0067	Display alarm when door is open	0			
0068	START lamp status when operating automatic door manually	0			
0069	Override switch	1			
0070	Alarm display when override is valid	1			
0071	Prevent rapid traverse override	0			
0072	Prevent cutting feedrate override	0			
0073	Prevent spindle override	0			
0074	Change handle magnification	1			
0075	Auto screen change at additional/PLC axis handle operation	0			
0076	Z-axis height to prohibit travel of additional axes	0.0000	0.0000	0.0000	200.0000
0077	Z-axis height to prohibit travel of additional axes(pallet 2)	0.0000			
0078	Prevent external operation for additional axes	0			
0079	Macro command single stop	0			
0080	Interrupt macro interrupt type	0			
0081	Interrupt macro call type	0			
0082	Interrupt macro acceptance type	0			
0083	ATC simultaneous operation start position	435.0000 (Standard) 505.0000 (Low)	505.0000 (Standard) 555.0000 (Low)	555.0000 (Standard) 625.0000 (Low)	455.0000 (Standard) 505.0000 (Long)
0084	ATC reference tool length	250.0000	200.0000	200.0000	200.0000
0085	Automatically use S command when changing tool	0			
0086	Automatically use F command when changing tool	0			
0087	Tool data spindle rotation exceeded	0			
0088	Magazine index when changing tools of different sizes	0			
0089	Pot lowering start position	0.0000			
0090	XY axes movement when changing tool in manual mode	0	1	1	0
0091	X-axis position when changing tool in manual mode	0.0000	-450.0000 (14tools) 0.0000 (22tools)	-650.0000	0.0000
0092	Y-axis position when changing tool in manual mode	0.0000	-320.0000 (14tools) 0.0000 (22tools)	0.0000	0.0000
0093	Schedule program reset start position	0			
0094	Program end stop in schedule program	0			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0095	Tool life counting method	0			
0096	Change tool life display(Type2)	0			
0097	Spare tool change	0			
0098	Check tool life on first use	1			
0099	External signal output time when MFIN is invalid	5			
0100	Chip shower retention time	0			
0101	Automatic coolant off time	0			
0102	Coolant delay time	0			
0103	Automatic machine light off time	0			
0104	Standby mode time	0			
0105	Auto power off	0			
0106	Auto power off time				
0107	Cancel editing when resetting	1			
0108	Save when mode of current program is changed	0			
0109	Program edit start position during operation	0			
0110	Automatic sequence insertion increment	10			
0111	Set same number to edit/memory operation programs	0			
0112	Assign tools when starting memory operation	1			
0113	Check tools to be used when assigned automatically	1			
0114	Workpc. zero conversion code	0			
0115	Convert tool data	1			
0116	Tap step return position	0			
0117	Magazine turn speed ratio 1(M436)	80			
0118	Magazine turn speed ratio 2(M437)	50			
0119	Turret type magazine speed	0			
0120	Turret type magazine low-speed turn mode	0			
0121	Speed turret type magazine turns at low speed	20.00			
0122	Pallet select key change	1			
0123	Lock shift key	0			
0124	Max. actual cutting travel speed	20000.0			
0125	Table load weight	400	240	400	40 (Standard) 75 (Long)
0126	Arc radius error limit	0.0000			
0127	Scaling	0			
0128	X-axis scaling	0			
0129	Y-axis scaling	0			
0130	Z-axis scaling	0			
0131	Scaling unit	0			
0132	Common scaling factor	1000			
0133	X-axis scaling factor(I)	1000			
0134	Y-axis scaling factor(J)	1000			
0135	Z-axis scaling factor(K)	1000			
0136	Back washing at end of program	1			
0137	CTS pressurize check delay time	1.0			
0138	CTS depressurize check delay time	1.0			
0139	CTS pressure bleed time	0.0			
0140	CTS line filter alarm stop level	1			
0141	ALM2 signal stop level	0			
0142	Oiling/greasing when power is turned on	0			
0143	Tool wash filter alarm stop level	0			
0144	Tool wash filter check interval	1			
0145	Display tool wash filter check result	0			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0146	Faulty tool wash coolant level alarm stop(Type4)	1			
0147	Tool wash filter clogging warning alarm stop level(Type4)	0			
0148	No. of tool wash filter clogging warning checks(Type4)	1			
0149	Faulty tool wash coolant level sensor check interval(Type4)	200			
0150	Max. tapping speed (when returning tap)	100			
0151	Motor insulation resistance measurement interval	0			
0152	Motor insulation resistance monitoring acceptable value (all axes)	5.0			
0153	Motor insulation resistance monitoring acceptable value (abnormal)	1.1			
0154	Motor insulation resistance monitoring acceptable value (alarm)	5.0			
0155	Spindle rotation return during restart	0			
0156	M function signal output delay time	1			
0157	Finish axis travel during external signal output	0			
0158	Chip conveyor relay adhering check	0			
0159	Error check when traveling during tool length/tool position offset cancel	0			
0160	Lathe machining infeed direction when power is turned ON	0			
0161	Cutting amount in thread cutting cycle	1.5			
0162	Cutting angle in thread cutting cycle	45.0000			
0163	Thread angle in complex thread cutting cycle	60.0000			
0164	Minimum infeed amount in complex thread cutting cycle	0.0500			
0165	Finishing count in complex thread cutting cycle	1			
0166	Finishing allowance in complex thread cutting cycle	0.0250			
0167	Signal batch reading/writing type	0			
0168	Z-axis travel method during tool change	0			
0169	Tapping sync. error too big - alarm stop level	0			
0170	Stop using feedrate override during rapid feed	0			
0171	Arc radius R command error limit	0.0010			
0172	Positioning method	0			
0173	Program load size	1			
0174	No tool detection function (when ATC rises)	0			
0175	No tool detection function (when ATC falls)	0			
0176	Tool over key detection function	0			
0177	Tool collision detection function during ATC	0			
0178	Tool missing detection function	0			
0179	Spindle return method for program stop	0			
0180	Lathe spindle return method for program stop	0			
0181	Coolant return method for program stop	0			
0182	Tap accel. Setting	1			
0183	Reset when operation preparation is ON	0			
0184	Screenshot	0			
0185	Probe detection signal method	1			
0186	PLC rapid traverse override function	0			
0187	PLC feedrate override function	0			
0188	PLC spindle override function	0			
0189	ATC arm turn speed ratio 1 (M422)	80	100	100	100
0190	ATC arm turn speed ratio 2 (M423)	50	100	100	100
0191	Tool wash discharge conditions for tool data	1	0	0	0

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0192	Tool wash when changing a different size tool	0			
0193	Cutting speed exceeded for inverse time feed	0			
0194	Cutter compensation interference check	0			
0195	Number of blocks to check cutter compensation interference	8			
0196	Involute interpolation override limit	0			
0197	Involute interpolation error limit	0.1000			
0198	Reset cycle time value	0			

User Parameter 2 (Switch 2)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0501	4th-axis stroke control	1			
0502	5th-axis stroke control	0	0	0	1
0503	6th-axis stroke control	0			
0504	7th-axis stroke control	0			
0505	8th-axis stroke control	0			
0506	4th-axis stroke(-)	-180.0000			
0507	5th-axis stroke(-)	0.0000	0.0000	0.0000	-30.0000
0508	6th-axis stroke(-)	0.0000			
0509	7th-axis stroke(-)	0.0000			
0510	8th-axis stroke(-)	0.0000			
0511	4th-axis stroke(+)	0.0000			
0512	5th-axis stroke(+)	0.0000	0.0000	0.0000	120.0000
0513	6th-axis stroke(+)	0.0000			
0514	7th-axis stroke(+)	0.0000			
0515	8th-axis stroke(+)	0.0000			
0516	Stroke limit 1 for X-axis(-)	-650.0000	-450.0000	-650.0000	-200.0000 (Standard) -300.0000 (Long)
0517	Stroke limit 1 for Y-axis(-)	-400.0000	-320.0000	-400.0000	-440.0000
0518	Stroke limit 1 for Z-axis(-)	0.0000			
0519	Stroke limit 1 for 4th-axis(-)	-180.0000			
0520	Stroke limit 1 for 5th-axis(-)	0.0000	0.0000	0.0000	-30.0000
0521	Stroke limit 1 for 6th-axis(-)	0.0000			
0522	Stroke limit 1 for 7th-axis(-)	0.0000			
0523	Stroke limit 1 for 8th-axis(-)	0.0000			
0524	Stroke limit 1 for X-axis(+)	0.0000			
0525	Stroke limit 1 for Y-axis(+)	0.0000			
0526	Stroke limit 1 for Z-axis(+)	685.0000 (Standard) 755.0000 (Low)	505.0000 (Standard) 555.0000 (Low)	555.0000 (Standard) 625.0000 (Low)	455.0000 (Standard) 505.0000 (Long)
0527	Stroke limit 1 for 4th-axis(+)	0.0000			
0528	Stroke limit 1 for 5th-axis(+)	0.0000	0.0000	0.0000	120.0000
0529	Stroke limit 1 for 6th-axis(+)	0.0000			
0530	Stroke limit 1 for 7th-axis(+)	0.0000			
0531	Stroke limit 1 for 8th-axis(+)	0.0000			
0532	Stroke limit 2 for X-axis(-)	0.0000			
0533	Stroke limit 2 for Y-axis(-)	0.0000			
0534	Stroke limit 2 for Z-axis(-)	0.0000			
0535	Stroke limit 2 for X-axis(+)	0.0000			
0536	Stroke limit 2 for Y-axis(+)	0.0000			
0537	Stroke limit 2 for Z-axis(+)	0.0000			
0538	Rapid traverse override speed 1	0.0			
0539	Rapid traverse override speed 2	500.0			
0540	Rapid traverse override speed 3	5000.0			
0541	Rapid traverse override speed 4	10000.0			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0542	4th-axis override rotation speed 1	0.00			
0543	4th-axis override rotation speed 2	0.30			
0544	4th-axis override rotation speed 3	2.50			
0545	4th-axis override rotation speed 4	5.00			
0546	5th-axis override rotation speed 1	0.00			
0547	5th-axis override rotation speed 2	0.50			
0548	5th-axis override rotation speed 3	5.00			
0549	5th-axis override rotation speed 4	10.00			
0550	6th-axis override rotation speed 1	0.00			
0551	6th-axis override rotation speed 2	0.50			
0552	6th-axis override rotation speed 3	5.00			
0553	6th-axis override rotation speed 4	10.00			
0554	7th-axis override rotation speed 1	0.00			
0555	7th-axis override rotation speed 2	0.50	0.50	0.50	1.00
0556	7th-axis override rotation speed 3	5.00	5.00	5.00	10.00
0557	7th-axis override rotation speed 4	10.00	10.00	10.00	30.00
0558	8th-axis override rotation speed 1	0.00			
0559	8th-axis override rotation speed 2	0.50			
0560	8th-axis override rotation speed 3	5.00			
0561	8th-axis override rotation speed 4	10.00			
0562	X-axis position signal output range(-)	0.0000			
0563	X-axis position signal output range(+)	0.0000			
0564	Y-axis position signal output range(-)	0.0000			
0565	Y-axis position signal output range(+)	0.0000			
0566	Z-axis position signal output range(-)	0.0000			
0567	Z-axis position signal output range(+)	0.0000			
0568	4th-axis position signal output range(-)	0.0000			
0569	4th-axis position signal output range(+)	0.0000			
0570	5th-axis position signal output range(-)	0.0000			
0571	5th-axis position signal output range(+)	0.0000			
0572	6th-axis position signal output range(-)	0.0000			
0573	6th-axis position signal output range(+)	0.0000			
0574	7th-axis position signal output range(-)	0.0000			
0575	7th-axis position signal output range(+)	0.0000			
0576	8th-axis position signal output range(-)	0.0000			
0577	8th-axis position signal output range(+)	0.0000			
0578	X-axis single direction positioning excess travel amount	0.0000			
0579	Y-axis single direction positioning excess travel amount	0.0000			
0580	Z-axis single direction positioning excess travel amount	0.0000			
0581	4th-axis single direction positioning excess travel amount	0.0000			
0582	5th-axis single direction positioning excess travel amount	0.0000			
0583	6th-axis single direction positioning excess travel amount	0.0000			
0584	7th-axis single direction positioning excess travel amount	0.0000			
0585	8th-axis single direction positioning excess travel amount	0.0000			
0586	External index A switch address switching	0			
0587	External A-axis indexing(+)	180.0000			
0588	External A-axis indexing(-)	-180.0000			
0589	External index B switch address switching	0			
0590	External B-axis indexing(+)	180.0000			
0591	External B-axis indexing(-)	-180.0000			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0592	Z-axis lowering speed 1 when changing tool	50000.0	35000.0	35000.0	35000.0
0593	Z-axis lowering speed 2 when changing tool	50000.0	35000.0	35000.0	35000.0
0594	Z-axis lowering speed 3 when changing tool	50000.0	35000.0	35000.0	35000.0
0595	X-axis position signal output range 2 (-)	0.0000			
0596	X-axis position signal output range 2 (+)	0.0000			
0597	Y-axis position signal output range 2 (-)	0.0000			
0598	Y-axis position signal output range 2 (+)	0.0000			
0599	Z-axis position signal output range 2 (-)	0.0000			
0600	Z-axis position signal output range 2 (+)	0.0000			
0601	4-axis position signal output range 2 (-)	0.0000			
0602	4-axis position signal output range 2 (+)	0.0000			
0603	5-axis position signal output range 2 (-)	0.0000			
0604	5-axis position signal output range 2 (+)	0.0000			
0605	6-axis position signal output range 2 (-)	0.0000			
0606	6-axis position signal output range 2 (+)	0.0000			
0607	7-axis position signal output range 2 (-)	0.0000			
0608	7-axis position signal output range 2 (+)	0.0000			
0609	8-axis position signal output range 2 (-)	0.0000			
0610	8-axis position signal output range 2 (+)	0.0000			
0611	X-axis position signal output range 3 (-)	0.0000			
0612	X-axis position signal output range 3 (+)	0.0000			
0613	Y-axis position signal output range 3 (-)	0.0000			
0614	Y-axis position signal output range 3 (+)	0.0000			
0615	Z-axis position signal output range 3 (-)	0.0000			
0616	Z-axis position signal output range 3 (+)	0.0000			
0617	4-axis position signal output range 3 (-)	0.0000			
0618	4-axis position signal output range 3 (+)	0.0000			
0619	5-axis position signal output range 3 (-)	0.0000			
0620	5-axis position signal output range 3 (+)	0.0000			
0621	6-axis position signal output range 3 (-)	0.0000			
0622	6-axis position signal output range 3 (+)	0.0000			
0623	7-axis position signal output range 3 (-)	0.0000			
0624	7-axis position signal output range 3 (+)	0.0000			
0625	8-axis position signal output range 3 (-)	0.0000			
0626	8-axis position signal output range 3 (+)	0.0000			
0627	X-axis position signal output range 4 (-)	0.0000			
0628	X-axis position signal output range 4 (+)	0.0000			
0629	Y-axis position signal output range 4 (-)	0.0000			
0630	Y-axis position signal output range 4 (+)	0.0000			
0631	Z-axis position signal output range 4 (-)	0.0000			
0632	Z-axis position signal output range 4 (+)	0.0000			
0633	4-axis position signal output range 4 (-)	0.0000			
0634	4-axis position signal output range 4 (+)	0.0000			
0635	5-axis position signal output range 4 (-)	0.0000			
0636	5-axis position signal output range 4 (+)	0.0000			
0637	6-axis position signal output range 4 (-)	0.0000			
0638	6-axis position signal output range 4 (+)	0.0000			
0639	7-axis position signal output range 4 (-)	0.0000			
0640	7-axis position signal output range 4 (+)	0.0000			
0641	8-axis position signal output range 4 (-)	0.0000			
0642	8-axis position signal output range 4 (+)	0.0000			

User Parameter 3 (Switch 3)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1001	M271 positioning end check distance	0.3000			
1002	M271 positioning end check angle	0.3000			
1003	M272 positioning end check distance	0.3000			
1004	M272 positioning end check angle	0.3000			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1005	M273 positioning end check distance		0.3000		
1006	M273 positioning end check angle		0.3000		
1007	M274 positioning end check distance		0.3000		
1008	M274 positioning end check angle		0.3000		
1009	M275 positioning end check distance		0.3000		
1010	M275 positioning end check angle		0.3000		
1011	M276 positioning end check distance		0.3000		
1012	M276 positioning end check angle		0.3000		
1013	M277 positioning end check distance		0.3000		
1014	M277 positioning end check angle		0.3000		
1015	M278 positioning end check distance		0.3000		
1016	M278 positioning end check angle		0.3000		
1017	M279 positioning end check distance		0.3000		
1018	M279 positioning end check angle		0.3000		
1019	Low priority indication alarm No. 1				
1020	Low priority indication alarm No. 2				
1021	Low priority indication alarm No. 3				
1022	Low priority indication alarm No. 4				
1023	Low priority indication alarm No. 5				
1024	Low priority indication alarm No. 6				
1025	Low priority indication alarm No. 7				
1026	Low priority indication alarm No. 8				
1027	Low priority indication alarm No. 9				
1028	Low priority indication alarm No. 10				
1029	G code for modal display 1		1		
1030	G code for modal display 2		2		
1031	G code for modal display 3		3		
1032	G code for modal display 4		4		
1033	G code for modal display 5		5		
1034	G code for modal display 6		6		
1035	G code for modal display 7		7		
1036	G code for modal display 8		10		
1037	G code for modal display 9		8		
1038	G code for modal display 10		11		
1039	G code for modal display 11		12		
1040	G code for modal display 12		13		
1041	G code for modal display 13		14		
1042	G code for modal display 14		15		
1043	G code for modal display 15		16		
1044	G code for modal display 16		17		
1045	M code for modal display 1		1		
1046	M code for modal display 2		2		
1047	M code for modal display 3		3		
1048	M code for modal display 4		4		
1049	M code for modal display 5		5		
1050	M code for modal display 6		6		
1051	M code for modal display 7		7		
1052	M code for modal display 8		8		
1053	M code for modal display 9		9		
1054	M code for modal display 10		11		
1055	M code for modal display 11		12		
1056	M code for modal display 12		14		
1057	M code for modal display 13		16		
1058	M code for modal display 14		31		
1059	M code for modal display 15		0		
1060	M code for modal display 16		0		
1061	Angle setting to check when additional axis for positioning is finished		0		

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1062	Specified alarm signal 2				
1063	Specified alarm signal 3				
1064	Specified alarm signal 4				
1065	Specified alarm signal 5				

User Parameter 4 (Reference position/Conversation zero point)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1501	Reference position X-axis	0.0000			
1502	Reference position Y-axis	0.0000			
1503	Reference position Z-axis	685.0000 (Standard) 755.0000 (Low)	505.0000 (Standard) 555.0000 (Low)	555.0000 (Standard) 625.0000 (Low)	455.0000 (Standard) 505.0000 (Long)
1504	Reference position 4th-axis	0.0000			
1505	Reference position 5th-axis	0.0000			
1506	Reference position 6th-axis	0.0000			
1507	Reference position 7th-axis	0.0000			
1508	Reference position 8th-axis	0.0000			
1509	2nd reference position X-axis	0.0000			
1510	2nd reference position Y-axis	0.0000			
1511	2nd reference position Z-axis	685.0000 (Standard) 755.0000 (Low)	505.0000 (Standard) 555.0000 (Low)	555.0000 (Standard) 625.0000 (Low)	455.0000 (Standard) 505.0000 (Long)
1512	2nd reference position 4th-axis	0.0000			
1513	2nd reference position 5th-axis	0.0000			
1514	2nd reference position 6th-axis	0.0000			
1515	2nd reference position 7th-axis	0.0000			
1516	2nd reference position 8th-axis	0.0000			
1517	3rd reference position X-axis	0.0000			
1518	3rd reference position Y-axis	0.0000			
1519	3rd reference position Z-axis	685.0000 (Standard) 755.0000 (Low)	505.0000 (Standard) 555.0000 (Low)	555.0000 (Standard) 625.0000 (Low)	455.0000 (Standard) 505.0000 (Long)
1520	3rd reference position 4th-axis	0.0000			
1521	3rd reference position 5th-axis	0.0000			
1522	3rd reference position 6th-axis	0.0000			
1523	3rd reference position 7th-axis	0.0000			
1524	3rd reference position 8th-axis	0.0000			
1525	4th reference position X-axis	0.0000			
1526	4th reference position Y-axis	0.0000			
1527	4th reference position Z-axis	685.0000 (Standard) 755.0000 (Low)	505.0000 (Standard) 555.0000 (Low)	555.0000 (Standard) 625.0000 (Low)	455.0000 (Standard) 505.0000 (Long)
1528	4th reference position 4th-axis	0.0000			
1529	4th reference position 5th-axis	0.0000			
1530	4th reference position 6th-axis	0.0000			
1531	4th reference position 7th-axis	0.0000			
1532	4th reference position 8th-axis	0.0000			
1533	5th reference position X-axis	0.0000			
1534	5th reference position Y-axis	0.0000			
1535	5th reference position Z-axis	685.0000 (Standard) 755.0000 (Low)	505.0000 (Standard) 555.0000 (Low)	555.0000 (Standard) 625.0000 (Low)	455.0000 (Standard) 505.0000 (Long)
1536	5th reference position 4th-axis	0.0000			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1537	5th reference position 5th-axis	0.0000			
1538	5th reference position 6th-axis	0.0000			
1539	5th reference position 7th-axis	0.0000			
1540	5th reference position 8th-axis	0.0000			
1541	6th reference position X-axis	0.0000			
1542	6th reference position Y-axis	0.0000			
1543	6th reference position Z-axis	685.0000 (Standard) 755.0000 (Low)	505.0000 (Standard) 555.0000 (Low)	555.0000 (Standard) 625.0000 (Low)	455.0000 (Standard) 505.0000 (Long)
1544	6th reference position 4th-axis	0.0000			
1545	6th reference position 5th-axis	0.0000			
1546	6th reference position 6th-axis	0.0000			
1547	6th reference position 7th-axis	0.0000			
1548	6th reference position 8th-axis	0.0000			
1549	Reference point return complete signal output at servo off	0			
1550	Reference point allowable range at servo off (for XYZ axes)	0.0000			
1551	Reference point allowable range at servo off (for additional axes)	0.0000			
1552	X-axis zero return pos.(No.2)	0.0000			
1553	Y-axis zero return pos.(No.2)	0.0000			
1554	4th-axis zero return pos.(No.2)	0.0000			
1555	5th-axis zero return pos.(No.2)	0.0000			
1556	6th-axis zero return pos.(No.2)	0.0000			
1557	7th-axis zero return pos.(No.2)	0.0000			
1558	8th-axis zero return pos.(No.2)	0.0000			
1559	X-axis zero return pos.(No.3)	0.0000			
1560	Y-axis zero return pos.(No.3)	0.0000			
1561	4th-axis zero return pos.(No.3)	0.0000			
1562	5th-axis zero return pos.(No.3)	0.0000			
1563	6th-axis zero return pos.(No.3)	0.0000			
1564	7th-axis zero return pos.(No.3)	0.0000			
1565	8th-axis zero return pos.(No.3)	0.0000			
1566	Zero point return complete signal output at servo off	0			
1567	Zero point allowable range at servo off (for XYZ axes)	0.0000			
1568	Zero point allowable range at servo off (for additional axes)	0.0000			
1569	Workpiece coordinate zero point X axis (for capture)	0.0000			
1570	Workpiece coordinate zero point Y axis (for capture)	0.0000			
1571	Workpiece coordinate zero point Z axis (for capture)	0.0000			

user parameter 5(User registered miscellaneous functions)

There is not default value.

User Parameter 6 (Tool breakage detection)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1801	Tool breakage detection option	0			
1802	Pallet 1 tool breakage detector	0			
1803	Pallet 2 tool breakage detector	0			
1804	Tool breakage detection execution interval(No. of operations)	1			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1805	Reference tool length offset No.	0			
1806	Tool breakage detection return position(Type1)	685.0000 (Standard) 755.0000 (Low)	505.0000 (Standard) 555.0000 (Low)	555.0000 (Standard) 625.0000 (Low)	455.0000 (Standard) 505.0000 (Long)
1807	Tool breakage detection position X-axis(Type1)	0.0000			
1808	Tool breakage detection position Y-axis(Type1)	0.0000			
1809	Tool breakage detection offset(Type2)	0.0000			
1810	Tool breakage detection position X-axis(Type3)	0.0000			
1811	Tool breakage detection position Y-axis(Type3)	0.0000			
1812	Tool breakage detection position Z-axis(Type3)	0.0000			
1813	Tool breakage detection start distance(Type3)	2.0000			
1814	Tool breakage detection stop distance(Type3)	1.0000			
1815	Tool breakage detection speed(Type3)	500.0			
1816	Tool breakage detection coolant OFF(Type3)	1			
1817	Pallet 2 tool breakage detection pos. X-axis(Type3)	0.0000			
1818	Pallet 2 tool breakage detection pos. Y-axis(Type3)	0.0000			
1819	Pallet 2 tool breakage detection pos. Z-axis(Type3)	0.0000			
1820	Pallet 2 tool breakage detection start distance(Type3)	2.0000			
1821	Pallet 2 tool breakage detection stop distance(Type3)	1.0000			
1822	Pallet 2 tool breakage detection speed(Type3)	500.0			
1823	Pallet 2 tool breakage detection coolant OFF(Type3)	1			

User Parameter 7 (Automatic workpiece measurement)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
2001	Probe compensation 1	0.0000			
2002	Probe compensation 2	0.0000			
2003	Probe compensation 3	0.0000			
2004	Probe compensation 4	0.0000			
2005	Measurement motion	0			
2006	Measurement speed 1	500.0			
2007	Measurement speed 2	100.0			
2008	Stop point before measurement	5.0000			
2009	Measurement travel limit distance	2.0000			
2010	Measurement tolerance 1	0.0000			
2011	Measurement tolerance 2	0.0000			
2012	Return distance after measurement	1.0000			

User Parameter 8 (Automatic thermal distortion compensation)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
2501	Z-axis compensation	100			
2502	Spindle compensation(Z-axis)	100			
2503	X-axis compensation	100			
2504	Spindle compensation(X-axis)	100			
2505	Y-axis compensation	100			
2506	Spindle compensation(Y-axis)	100			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
2507	Auto. thermal distortion compen. at memory operation start	0			
2508	Lathe spindle compensation(Z-axis)	0			
2509	Lathe spindle compensation(X-axis)	0			
2510	Lathe spindle compensation(Y-axis)	0			

User Parameter 9 (Quick table)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
3001	Pallet program folder				
3002	Pallet 1 program No.				
3003	Pallet 2 program No.				
3004	Pallet program folder				
3005	Pallet 1 program No.				
3006	Pallet 2 program No.				
3007	Pallet 2 shift(X-axis)	0.0000			
3008	Pallet 2 shift(Y-axis)	0.0000			
3009	Pallet 2 shift(Z-axis)	0.0000			
3010	Pallet 1 loading	0			
3011	Pallet 2 loading	0			

**User Parameter 10(High accuracy)
High accuracy A (M260)**

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	120			
	Arc deceleration override	200			
	Curve approximation deceleration override	200			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M261)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	60			
	Arc deceleration override	140			
	Curve approximation deceleration override	140			
	Smooth path offset level	1			
	Smooth override	50			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M262)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	70			
	Arc deceleration override	70			
	Curve approximation deceleration override	70			
	Smooth path offset level	1			
	Smooth override	600			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M263)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	120			
	Arc deceleration override	200			
	Curve approximation deceleration override	200			
	Smooth path offset level	1			
	Smooth override	900			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M264)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	120			
	Arc deceleration override	200			
	Curve approximation deceleration override	200			
	Smooth path offset level	1			
	Smooth override	2500			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M265)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	600			
	Arc deceleration override	1000			
	Curve approximation deceleration override	1000			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M266)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	100			
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy A (M267)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	100			
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.0000			
	Accuracy level	0.0000			

High accuracy B (M280)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	0			
	Curve approximation deceleration override	0			
	Smooth path offset level	3			
	Smooth override	100			
	Minute block deletion distance	0.0100			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M281)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M282)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M283)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M284)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M285)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M286)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M287)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.0000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy common

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
3201	Smooth path offset function	0			
3202	Smooth path offset cancel distance	10.0000			
3203	Smooth path offset cancel angle	10.0000			
3204	Smooth path offset angle determination invalid distance	0.0100			
3205	Smooth path offset reference length	1.0000			
3206	Minute block deletion when smooth path offset is cancelled	0			
3207	Accuracy level method	1			

High accuracy(Adjust) common

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Machining mode	0			

High accuracy(Adjust) Adjust.A

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Standard mode	0			
	Accuracy	0			
	Smoothness	0			

High accuracy(Adjust) Adjust.B

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Standard mode	0			
	Accuracy	0			
	Smoothness	0			

High accuracy(Adjust) Adjust.C

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Standard mode	0			
	Accuracy	0			
	Smoothness	0			

High accuracy(Accuracy spec.) common

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Accuracy spec. mode	0			

High accuracy(Accuracy spec.) Accuracy spec. A

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Accuracy level	0.1000			
	Smoothness	0			

High accuracy(Accuracy spec.) Accuracy spec. B

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Accuracy level	0.0300			
	Smoothness	0			

High accuracy(Accuracy spec.) Accuracy spec. C

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Accuracy level	0.0100			
	Smoothness	0			

User Parameter 11 (PLC)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
3501	P1-axis stroke control	0			
3502	P2-axis stroke control	0			
3503	P3-axis stroke control	0			
3504	P4-axis stroke control	0			
3505	P1-axis stroke(-)	0			
3506	P2-axis stroke(-)	0			
3507	P3-axis stroke(-)	0			
3508	P4-axis stroke(-)	0			
3509	P1-axis stroke(+)	0			
3510	P2-axis stroke(+)	0			
3511	P3-axis stroke(+)	0			
3512	P4-axis stroke(+)	0			
3513	Reference point P1-axis	0			
3514	Reference point P2-axis	0			
3515	Reference point P3-axis	0			
3516	Reference point P4-axis	0			
3517	P1-axis override speed 1	0			
3518	P1-axis override speed 2	500			
3519	P1-axis override speed 3	5000			
3520	P1-axis override speed 4	10000			
3521	P2-axis override speed 1	0			
3522	P2-axis override speed 2	500			
3523	P2-axis override speed 3	5000			
3524	P2-axis override speed 4	10000			
3525	P3-axis override speed 1	0			
3526	P3-axis override speed 2	500			
3527	P3-axis override speed 3	5000			
3528	P3-axis override speed 4	10000			
3529	P4-axis override speed 1	0			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
3530	P4-axis override speed 2	500			
3531	P4-axis override speed 3	5000			
3532	P4-axis override speed 4	10000			
3533	Collis. detct.-P1ax stop meth.	0			
3534	Collis. detct.-P2ax stop meth.	0			
3535	Collis. detct.-P3ax stop meth.	0			
3536	Collis. detct.-P4ax stop meth.	0			

User Parameter 12 (Profibus)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
4001	Node address	0			
4002	I/O setting 1				
4003	I/O setting 2				
4004	I/O setting 3				
4005	I/O setting 4				
4006	I/O setting 5				
4007	I/O setting 6				
4008	I/O setting 7				
4009	I/O setting 8				
4010	I/O setting 9				
4011	I/O setting 10				
4012	I/O setting 11				
4013	I/O setting 12				
4014	I/O setting 13				
4015	I/O setting 14				
4016	I/O setting 15				
4017	I/O setting 16				
4018	I/O setting 17				
4019	I/O setting 18				
4020	I/O setting 19				
4021	I/O setting 20				

User Parameter 13 (DeviceNet)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
4201	MAC ID	0			
4202	Transfer rate	0			
4203	Input bytes	0			
4204	Output bytes	0			

User Parameter 14 (CC-Link Slave)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
4401	Station number	1			
4402	Slave station category	0			
4403	Occupied number of stations	1			
4404	Transfer rate	0			
4405	Specify data clear in case of communication error	0			

User Parameter 15 (Rotary fixture offset)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
4601	XYZ-axes travel during current rotary fixture offset change	0			
4602	Rotary fixture offset reset method	0			
4603	Tilt axis 1	0	0	0	2
4604	Rotation axis 1	0	0	0	4
4605	Tilt axis 2	0			
4606	Rotation axis 2	0			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
4607	Tilt axis 3	0			
4608	Rotation axis 3	0			
4609	Tilt axis 4	0			
4610	Rotation axis 4	0			
4611	Forward direction for the coordinate system on tilt axis 1	0	0	0	1
4612	Forward direction for the coordinate system on rotation axis 1	0	0	0	5
4613	Forward direction for the coordinate system on tilt axis 2	0			
4614	Forward direction for the coordinate system on rotation axis 2	0			
4615	Forward direction for the coordinate system on tilt axis 3	0			
4616	Forward direction for the coordinate system on rotation axis 3	0			
4617	Forward direction for the coordinate system on tilt axis 4	0			
4618	Forward direction for the coordinate system on rotation axis 4	0			

User Parameter 16 (3D smoothing function)
3D smoothing function(M380)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M381)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M382)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M383)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M384)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M385)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M386)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M387)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function common

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
4801	3D smoothing circular interpolation command				
4802	3D smoothing sequence number command				

User Parameter 17 (feature coordinate setting)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
10001	Tilt axis 1	0	0	0	2
10002	Rotation axis 1	0	0	0	4
10003	Tilt axis 2	0			
10004	Rotation axis 2	0			
10005	Forward direction for the coordinate system on tilt axis 1	0	0	0	1
10006	Forward direction for the coordinate system on rotation axis 1	0	0	0	5
10007	Forward direction for the coordinate system on tilt axis 2	0			
10008	Forward direction for the coordinate system on rotation axis 2	0			
10009	Rotation center X coordinate offset for tilt axis 1	0.0000			
10010	Rotation center Y coordinate offset for tilt axis 1	0.0000			
10011	Rotation center Z coordinate offset for tilt axis 1	0.0000			
10012	Rotation center X coordinate offset for rotation axis 1	0.0000			
10013	Rotation center Y coordinate offset for rotation axis 1	0.0000			
10014	Rotation center Z coordinate offset for rotation axis 1	0.0000			
10015	Rotation center X coordinate offset for tilt axis 2	0.0000			
10016	Rotation center Y coordinate offset for tilt axis 2	0.0000			
10017	Rotation center Z coordinate offset for tilt axis 2	0.0000			
10018	Rotation center X coordinate offset for rotation axis 2	0.0000			
10019	Rotation center Y coordinate offset for rotation axis 2	0.0000			
10020	Rotation center Z coordinate offset for rotation axis 2	0.0000			
10021	Absolute coordinate display format	0			

User Parameter 18 (CC-Link master)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
10201	Transfer rate	0			
10202	Self-station number	0			
10203	Interface board setting	0			
10204	Occupied number of stations (for intelligent device stations)	0			
10205	Standby master station number	0			
10206	CC-Link version	0			
10207	Retry attempts	3			
10208	Number of automatic return nodes	1			
10209	Specify action when there is a driver error	0			
10210	Specify station data clear in case of any error	0			

2.5.20.6 Inch(R650X2_40MG,R450X2,R650X2,M200X3,M300X3)

User Parameter 1 (Switch 1)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0001	Select mode		0		
0002	Initial mode		0		
0003	Select memory operation		0		
0004	Program repeat		0		
0005	Workpiece counter when program is repeated		0		
0006	Tool life management when program is repeated		0		
0007	Automatic tool length compensation		0.00000		
0008	External signal reading time limit		0		
0009	Dry run offset amount		0.00000		
0010	Dry run offset amount lower limit		0.00000		
0011	Program unit		0		
0012	Program feed rate unit		0		
0013	G73 relief amount		0.01960		
0014	G83 cutting start position		0.01960		
0015	G77,G78 relief amount		1.000		
0016	G76,G87 shift direction		0		
0017	Tapping cycle return speed		0		
0018	Program restart		0		
0019	S.T.M. return		1		
0020	Prevent restart from middle of program		0		
0021	4th row display axis		1		
0022	5th row display axis		2		
0023	6th row display axis		3		
0024	7th row display axis		4		
0025	8th row display axis		5		
0026	9th row display axis		6		
0027	10th row display axis		7		
0028	11th row display axis		8		
0029	12th row display axis		9		
0030	XYZ-axis absolute coord. Display		2		
0031	Display coordinates during rotational transformation		0		
0032	Macro variable 1 for position screen display				
0033	Macro variable 2 for position screen display				
0034	Macro variable 3 for position screen display				
0035	Macro variable 4 for position screen display				
0036	Macro variable 5 for position screen display				
0037	Block No. display color		4		
0038	Pointer display color		3		
0039	X/Y/Z-axis travel at tool length/tool pos. offset change		0		
0040	Condition to offset tool length/tool position		0		
0041	Start up/Cancel		0		
0042	Corner travel limit		0.00000		
0043	Automatic corner override(angle)		0.0000		
0044	Automatic corner override(length 1)		0.00000		
0045	Automatic corner override(length 2)		0.00000		
0046	Automatic corner override(ratio)		100		
0047	Inner arc override limit		100		
0048	Automatic centering speed 1		19.60		
0049	Automatic centering speed 2		0.30		
0050	Automatic centering return		0.01960		
0051	Automatic centering compensation		0.00000		
0052	Lamp status when count ends		0		

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0053	Multiple M codes in one block	0	0	0	1
0054	Unregistered M-code	0			
0055	One shot signal M450 output time	5			
0056	One shot signal M451 output time	5			
0057	One shot signal M455 output time	5			
0058	One shot signal M456 output time	5			
0059	Automatic screen dimming time	0			
0060	Magazine panel automatic screen clear time	0			
0061	Alarm display when screen is dimmed	0			
0062	Machine No.	0			
0063	Automatic door	0			
0064	Area sensor pattern	1			
0065	Automatic door error level	1			
0066	Automatic door operation time	5			
0067	Display alarm when door is open	0			
0068	START lamp status when operating automatic door manually	0			
0069	Override switch	1			
0070	Alarm display when override is valid	1			
0071	Prevent rapid traverse override	0			
0072	Prevent cutting feedrate override	0			
0073	Prevent spindle override	0			
0074	Change handle magnification	1			
0075	Auto screen change at additional/PLC axis handle operation	0			
0076	Z-axis height to prohibit travel of additional axes	0.00000	0.00000	0.00000	7.87400
0077	Z-axis height to prohibit travel of additional axes(pallet 2)	0.00000			
0078	Prevent external operation for additional axes	0			
0079	Macro command single stop	0			
0080	Interrupt macro interrupt type	0			
0081	Interrupt macro call type	0			
0082	Interrupt macro acceptance type	0			
0083	ATC simultaneous operation start position	17.12590 (Standard) 19.88180 (Low)	19.88180 (Standard) 21.85030 (Low)	21.85030 (Standard) 24.60620 (Low)	17.91330 (Standard) 19.88180 (Long)
0084	ATC reference tool length	9.84250	7.87400	7.87400	7.87400
0085	Automatically use S command when changing tool	0			
0086	Automatically use F command when changing tool	0			
0087	Tool data spindle rotation exceeded	0			
0088	Magazine index when changing tools of different sizes	0			
0089	Pot lowering start position	0.00000			
0090	XY axes movement when changing tool in manual mode	0	1	1	0
0091	X-axis position when changing tool in manual mode	0.00000	-17.71650 (14tools) 0.00000 (22tools)	-25.59050	0.00000
0092	Y-axis position when changing tool in manual mode	0.00000	-12.59840 (14tools) 0.00000 (22tools)	0.00000	0.00000
0093	Schedule program reset start position	0			
0094	Program end stop in schedule program	0			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0095	Tool life counting method	0			
0096	Change tool life display(Type2)	0			
0097	Spare tool change	0			
0098	Check tool life on first use	1			
0099	External signal output time when MFIN is invalid	5			
0100	Chip shower retention time	0			
0101	Automatic coolant off time	0			
0102	Coolant delay time	0			
0103	Automatic machine light off time	0			
0104	Standby mode time	0			
0105	Auto power off	0			
0106	Auto power off time				
0107	Cancel editing when resetting	1			
0108	Save when mode of current program is changed	0			
0109	Program edit start position during operation	0			
0110	Automatic sequence insertion increment	10			
0111	Set same number to edit/memory operation programs	0			
0112	Assign tools when starting memory operation	1			
0113	Check tools to be used when assigned automatically	1			
0114	Workpc. zero conversion code	0			
0115	Convert tool data	1			
0116	Tap step return position	0			
0117	Magazine turn speed ratio 1(M436)	80			
0118	Magazine turn speed ratio 2(M437)	50			
0119	Turret type magazine speed	0			
0120	Turret type magazine low-speed turn mode	0			
0121	Speed turret type magazine turns at low speed	20.00			
0122	Pallet select key change	1			
0123	Lock shift key	0			
0124	Max. actual cutting travel speed	787.40			
0125	Table load weight	400	240	400	40 (Standard) 75 (Long)
0126	Arc radius error limit	0.00000			
0127	Scaling	0			
0128	X-axis scaling	0			
0129	Y-axis scaling	0			
0130	Z-axis scaling	0			
0131	Scaling unit	0			
0132	Common scaling factor	1000			
0133	X-axis scaling factor(I)	1000			
0134	Y-axis scaling factor(J)	1000			
0135	Z-axis scaling factor(K)	1000			
0136	Back washing at end of program	1			
0137	CTS pressurize check delay time	1.0			
0138	CTS depressurize check delay time	1.0			
0139	CTS pressure bleed time	0.0			
0140	CTS line filter alarm stop level	1			
0141	ALM2 signal stop level	0			
0142	Oiling/greasing when power is turned on	0			
0143	Tool wash filter alarm stop level	0			
0144	Tool wash filter check interval	1			
0145	Display tool wash filter check result	0			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0146	Faulty tool wash coolant level alarm stop(Type4)	1			
0147	Tool wash filter clogging warning alarm stop level(Type4)	0			
0148	No. of tool wash filter clogging warning checks(Type4)	1			
0149	Faulty tool wash coolant level sensor check interval(Type4)	200			
0150	Max. tapping speed (when returning tap)	100			
0151	Motor insulation resistance measurement interval	0			
0152	Motor insulation resistance monitoring acceptable value (all axes)	5.0			
0153	Motor insulation resistance monitoring acceptable value (abnormal)	1.1			
0154	Motor insulation resistance monitoring acceptable value (alarm)	5.0			
0155	Spindle rotation return during restart	0			
0156	M function signal output delay time	1			
0157	Finish axis travel during external signal output	0			
0158	Chip conveyor relay adhering check	0			
0159	Error check when traveling during tool length/tool position offset cancel	0			
0160	Lathe machining infeed direction when power is turned ON	0			
0161	Cutting amount in thread cutting cycle	1.5			
0162	Cutting angle in thread cutting cycle	45.0000			
0163	Thread angle in complex thread cutting cycle	60.0000			
0164	Minimum infeed amount in complex thread cutting cycle	0.00190			
0165	Finishing count in complex thread cutting cycle	1			
0166	Finishing allowance in complex thread cutting cycle	0.00090			
0167	Signal batch reading/writing type	0			
0168	Z-axis travel method during tool change	0			
0169	Tapping sync. error too big - alarm stop level	0			
0170	Stop using feedrate override during rapid feed	0			
0171	Arc radius R command error limit	0.00010			
0172	Positioning method	0			
0173	Program load size	1			
0174	No tool detection function (when ATC rises)	0			
0175	No tool detection function (when ATC falls)	0			
0176	Tool over key detection function	0			
0177	Tool collision detection function during ATC	0			
0178	Tool missing detection function	0			
0179	Spindle return method for program stop	0			
0180	Lathe spindle return method for program stop	0			
0181	Coolant return method for program stop	0			
0182	Tap accel. Setting	1			
0183	Reset when operation preparation is ON	0			
0184	Screenshot	0			
0185	Probe detection signal method	1			
0186	PLC rapid traverse override function	0			
0187	PLC feedrate override function	0			
0188	PLC spindle override function	0			
0189	ATC arm turn speed ratio 1 (M422)	80	100	100	100
0190	ATC arm turn speed ratio 2 (M423)	50	100	100	100
0191	Tool wash discharge conditions for tool data	1	0	0	0

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0192	Tool wash when changing a different size tool	0			
0193	Cutting speed exceeded for inverse time feed	0			
0194	Cutter compensation interference check	0			
0195	Number of blocks to check cutter compensation interference	8			
0196	Involute interpolation override limit	0			
0197	Involute interpolation error limit	0.00390			
0198	Reset cycle time value	0			

User Parameter 2 (Switch 2)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0501	4th-axis stroke control	1			
0502	5th-axis stroke control	0	0	0	1
0503	6th-axis stroke control	0			
0504	7th-axis stroke control	0			
0505	8th-axis stroke control	0			
0506	4th-axis stroke(-)	-180.0000			
0507	5th-axis stroke(-)	0.0000	0.0000	0.0000	-30.0000
0508	6th-axis stroke(-)	0.0000			
0509	7th-axis stroke(-)	0.0000			
0510	8th-axis stroke(-)	0.0000			
0511	4th-axis stroke(+)	0.0000			
0512	5th-axis stroke(+)	0.0000	0.0000	0.0000	120.0000
0513	6th-axis stroke(+)	0.0000			
0514	7th-axis stroke(+)	0.0000			
0515	8th-axis stroke(+)	0.0000			
0516	Stroke limit 1 for X-axis(-)	-25.59050	-17.71650	-25.59050	-7.87400 (Standard) -11.81100 (Long)
0517	Stroke limit 1 for Y-axis(-)	-15.74800	-12.59840	-15.74800	-17.32280
0518	Stroke limit 1 for Z-axis(-)	0.00000			
0519	Stroke limit 1 for 4th-axis(-)	-180.0000			
0520	Stroke limit 1 for 5th-axis(-)	0.0000	0.0000	0.0000	-30.0000
0521	Stroke limit 1 for 6th-axis(-)	0.0000			
0522	Stroke limit 1 for 7th-axis(-)	0.0000			
0523	Stroke limit 1 for 8th-axis(-)	0.0000			
0524	Stroke limit 1 for X-axis(+)	0.00000			
0525	Stroke limit 1 for Y-axis(+)	0.00000			
0526	Stroke limit 1 for Z-axis(+)	26.96850 (Standard) 29.72440 (Low)	19.88180 (Standard) 21.85030 (Low)	21.85030 (Standard) 24.60620 (Low)	17.91330 (Standard) 19.88180 (Long)
0527	Stroke limit 1 for 4th-axis(+)	0.0000			
0528	Stroke limit 1 for 5th-axis(+)	0.0000	0.0000	0.0000	120.0000
0529	Stroke limit 1 for 6th-axis(+)	0.0000			
0530	Stroke limit 1 for 7th-axis(+)	0.0000			
0531	Stroke limit 1 for 8th-axis(+)	0.0000			
0532	Stroke limit 2 for X-axis(-)	0.00000			
0533	Stroke limit 2 for Y-axis(-)	0.00000			
0534	Stroke limit 2 for Z-axis(-)	0.00000			
0535	Stroke limit 2 for X-axis(+)	0.00000			
0536	Stroke limit 2 for Y-axis(+)	0.00000			
0537	Stroke limit 2 for Z-axis(+)	0.00000			
0538	Rapid traverse override speed 1	0.00			
0539	Rapid traverse override speed 2	19.60			
0540	Rapid traverse override speed 3	196.80			
0541	Rapid traverse override speed 4	393.70			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0542	4th-axis override rotation speed 1	0.00			
0543	4th-axis override rotation speed 2	0.30			
0544	4th-axis override rotation speed 3	2.50			
0545	4th-axis override rotation speed 4	5.00			
0546	5th-axis override rotation speed 1	0.00			
0547	5th-axis override rotation speed 2	0.50			
0548	5th-axis override rotation speed 3	5.00			
0549	5th-axis override rotation speed 4	10.00			
0550	6th-axis override rotation speed 1	0.00			
0551	6th-axis override rotation speed 2	0.50			
0552	6th-axis override rotation speed 3	5.00			
0553	6th-axis override rotation speed 4	10.00			
0554	7th-axis override rotation speed 1	0.00			
0555	7th-axis override rotation speed 2	0.50	0.50	0.50	1.00
0556	7th-axis override rotation speed 3	5.00	5.00	5.00	10.00
0557	7th-axis override rotation speed 4	10.00	10.00	10.00	30.00
0558	8th-axis override rotation speed 1	0.00			
0559	8th-axis override rotation speed 2	0.50			
0560	8th-axis override rotation speed 3	5.00			
0561	8th-axis override rotation speed 4	10.00			
0562	X-axis position signal output range(-)	0.00000			
0563	X-axis position signal output range(+)	0.00000			
0564	Y-axis position signal output range(-)	0.00000			
0565	Y-axis position signal output range(+)	0.00000			
0566	Z-axis position signal output range(-)	0.00000			
0567	Z-axis position signal output range(+)	0.00000			
0568	4th-axis position signal output range(-)	0.0000			
0569	4th-axis position signal output range(+)	0.0000			
0570	5th-axis position signal output range(-)	0.0000			
0571	5th-axis position signal output range(+)	0.0000			
0572	6th-axis position signal output range(-)	0.0000			
0573	6th-axis position signal output range(+)	0.0000			
0574	7th-axis position signal output range(-)	0.0000			
0575	7th-axis position signal output range(+)	0.0000			
0576	8th-axis position signal output range(-)	0.0000			
0577	8th-axis position signal output range(+)	0.0000			
0578	X-axis single direction positioning excess travel amount	0.00000			
0579	Y-axis single direction positioning excess travel amount	0.00000			
0580	Z-axis single direction positioning excess travel amount	0.00000			
0581	4th-axis single direction positioning excess travel amount	0.0000			
0582	5th-axis single direction positioning excess travel amount	0.0000			
0583	6th-axis single direction positioning excess travel amount	0.0000			
0584	7th-axis single direction positioning excess travel amount	0.0000			
0585	8th-axis single direction positioning excess travel amount	0.0000			
0586	External index A switch address switching	0			
0587	External A-axis indexing(+)	180.0000			
0588	External A-axis indexing(-)	-180.0000			
0589	External index B switch address switching	0			
0590	External B-axis indexing(+)	180.0000			
0591	External B-axis indexing(-)	-180.0000			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
0592	Z-axis lowering speed 1 when changing tool	1968.50	1377.90	1377.90	1377.90
0593	Z-axis lowering speed 2 when changing tool	1968.50	1377.90	1377.90	1377.90
0594	Z-axis lowering speed 3 when changing tool	1968.50	1377.90	1377.90	1377.90
0595	X-axis position signal output range 2 (-)	0.00000			
0596	X-axis position signal output range 2 (+)	0.00000			
0597	Y-axis position signal output range 2 (-)	0.00000			
0598	Y-axis position signal output range 2 (+)	0.00000			
0599	Z-axis position signal output range 2 (-)	0.00000			
0600	Z-axis position signal output range 2 (+)	0.00000			
0601	4-axis position signal output range 2 (-)	0.0000			
0602	4-axis position signal output range 2 (+)	0.0000			
0603	5-axis position signal output range 2 (-)	0.0000			
0604	5-axis position signal output range 2 (+)	0.0000			
0605	6-axis position signal output range 2 (-)	0.0000			
0606	6-axis position signal output range 2 (+)	0.0000			
0607	7-axis position signal output range 2 (-)	0.0000			
0608	7-axis position signal output range 2 (+)	0.0000			
0609	8-axis position signal output range 2 (-)	0.0000			
0610	8-axis position signal output range 2 (+)	0.0000			
0611	X-axis position signal output range 3 (-)	0.00000			
0612	X-axis position signal output range 3 (+)	0.00000			
0613	Y-axis position signal output range 3 (-)	0.00000			
0614	Y-axis position signal output range 3 (+)	0.00000			
0615	Z-axis position signal output range 3 (-)	0.00000			
0616	Z-axis position signal output range 3 (+)	0.00000			
0617	4-axis position signal output range 3 (-)	0.0000			
0618	4-axis position signal output range 3 (+)	0.0000			
0619	5-axis position signal output range 3 (-)	0.0000			
0620	5-axis position signal output range 3 (+)	0.0000			
0621	6-axis position signal output range 3 (-)	0.0000			
0622	6-axis position signal output range 3 (+)	0.0000			
0623	7-axis position signal output range 3 (-)	0.0000			
0624	7-axis position signal output range 3 (+)	0.0000			
0625	8-axis position signal output range 3 (-)	0.0000			
0626	8-axis position signal output range 3 (+)	0.0000			
0627	X-axis position signal output range 4 (-)	0.00000			
0628	X-axis position signal output range 4 (+)	0.00000			
0629	Y-axis position signal output range 4 (-)	0.00000			
0630	Y-axis position signal output range 4 (+)	0.00000			
0631	Z-axis position signal output range 4 (-)	0.00000			
0632	Z-axis position signal output range 4 (+)	0.00000			
0633	4-axis position signal output range 4 (-)	0.0000			
0634	4-axis position signal output range 4 (+)	0.0000			
0635	5-axis position signal output range 4 (-)	0.0000			
0636	5-axis position signal output range 4 (+)	0.0000			
0637	6-axis position signal output range 4 (-)	0.0000			
0638	6-axis position signal output range 4 (+)	0.0000			
0639	7-axis position signal output range 4 (-)	0.0000			
0640	7-axis position signal output range 4 (+)	0.0000			
0641	8-axis position signal output range 4 (-)	0.0000			
0642	8-axis position signal output range 4 (+)	0.0000			

User Parameter 3 (Switch 3)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1001	M271 positioning end check distance	0.01180			
1002	M271 positioning end check angle	0.3000			
1003	M272 positioning end check distance	0.01180			
1004	M272 positioning end check angle	0.3000			

Chapter2 Data bank

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1005	M273 positioning end check distance		0.01180		
1006	M273 positioning end check angle		0.3000		
1007	M274 positioning end check distance		0.01180		
1008	M274 positioning end check angle		0.3000		
1009	M275 positioning end check distance		0.01180		
1010	M275 positioning end check angle		0.3000		
1011	M276 positioning end check distance		0.01180		
1012	M276 positioning end check angle		0.3000		
1013	M277 positioning end check distance		0.01180		
1014	M277 positioning end check angle		0.3000		
1015	M278 positioning end check distance		0.01180		
1016	M278 positioning end check angle		0.3000		
1017	M279 positioning end check distance		0.01180		
1018	M279 positioning end check angle		0.3000		
1019	Low priority indication alarm No. 1				
1020	Low priority indication alarm No. 2				
1021	Low priority indication alarm No. 3				
1022	Low priority indication alarm No. 4				
1023	Low priority indication alarm No. 5				
1024	Low priority indication alarm No. 6				
1025	Low priority indication alarm No. 7				
1026	Low priority indication alarm No. 8				
1027	Low priority indication alarm No. 9				
1028	Low priority indication alarm No. 10				
1029	G code for modal display 1		1		
1030	G code for modal display 2		2		
1031	G code for modal display 3		3		
1032	G code for modal display 4		4		
1033	G code for modal display 5		5		
1034	G code for modal display 6		6		
1035	G code for modal display 7		7		
1036	G code for modal display 8		10		
1037	G code for modal display 9		8		
1038	G code for modal display 10		11		
1039	G code for modal display 11		12		
1040	G code for modal display 12		13		
1041	G code for modal display 13		14		
1042	G code for modal display 14		15		
1043	G code for modal display 15		16		
1044	G code for modal display 16		17		
1045	M code for modal display 1		1		
1046	M code for modal display 2		2		
1047	M code for modal display 3		3		
1048	M code for modal display 4		4		
1049	M code for modal display 5		5		
1050	M code for modal display 6		6		
1051	M code for modal display 7		7		
1052	M code for modal display 8		8		
1053	M code for modal display 9		9		
1054	M code for modal display 10		11		
1055	M code for modal display 11		12		
1056	M code for modal display 12		14		
1057	M code for modal display 13		16		
1058	M code for modal display 14		31		
1059	M code for modal display 15		0		
1060	M code for modal display 16		0		
1061	Angle setting to check when additional axis for positioning is finished		0		

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1062	Specified alarm signal 2				
1063	Specified alarm signal 3				
1064	Specified alarm signal 4				
1065	Specified alarm signal 5				

User Parameter 4 (Reference position/Conversation zero point)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1501	Reference position X-axis	0.00000			
1502	Reference position Y-axis	0.00000			
1503	Reference position Z-axis	26.96850 (Standard) 29.72440 (Low)	19.88180 (Standard) 21.85030 (Low)	21.85030 (Standard) 24.60620 (Low)	17.91330 (Standard) 19.88180 (Long)
1504	Reference position 4th-axis	0.0000			
1505	Reference position 5th-axis	0.0000			
1506	Reference position 6th-axis	0.0000			
1507	Reference position 7th-axis	0.0000			
1508	Reference position 8th-axis	0.0000			
1509	2nd reference position X-axis	0.00000			
1510	2nd reference position Y-axis	0.00000			
1511	2nd reference position Z-axis	26.96850 (Standard) 29.72440 (Low)	19.88180 (Standard) 21.85030 (Low)	21.85030 (Standard) 24.60620 (Low)	17.91330 (Standard) 19.88180 (Long)
1512	2nd reference position 4th-axis	0.0000			
1513	2nd reference position 5th-axis	0.0000			
1514	2nd reference position 6th-axis	0.0000			
1515	2nd reference position 7th-axis	0.0000			
1516	2nd reference position 8th-axis	0.0000			
1517	3rd reference position X-axis	0.00000			
1518	3rd reference position Y-axis	0.00000			
1519	3rd reference position Z-axis	26.96850 (Standard) 29.72440 (Low)	19.88180 (Standard) 21.85030 (Low)	21.85030 (Standard) 24.60620 (Low)	17.91330 (Standard) 19.88180 (Long)
1520	3rd reference position 4th-axis	0.0000			
1521	3rd reference position 5th-axis	0.0000			
1522	3rd reference position 6th-axis	0.0000			
1523	3rd reference position 7th-axis	0.0000			
1524	3rd reference position 8th-axis	0.0000			
1525	4th reference position X-axis	0.00000			
1526	4th reference position Y-axis	0.00000			
1527	4th reference position Z-axis	26.96850 (Standard) 29.72440 (Low)	19.88180 (Standard) 21.85030 (Low)	21.85030 (Standard) 24.60620 (Low)	17.91330 (Standard) 19.88180 (Long)
1528	4th reference position 4th-axis	0.0000			
1529	4th reference position 5th-axis	0.0000			
1530	4th reference position 6th-axis	0.0000			
1531	4th reference position 7th-axis	0.0000			
1532	4th reference position 8th-axis	0.0000			
1533	5th reference position X-axis	0.00000			
1534	5th reference position Y-axis	0.00000			
1535	5th reference position Z-axis	26.96850 (Standard) 29.72440 (Low)	19.88180 (Standard) 21.85030 (Low)	21.85030 (Standard) 24.60620 (Low)	17.91330 (Standard) 19.88180 (Long)
1536	5th reference position 4th-axis	0.0000			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1537	5th reference position 5th-axis	0.0000			
1538	5th reference position 6th-axis	0.0000			
1539	5th reference position 7th-axis	0.0000			
1540	5th reference position 8th-axis	0.0000			
1541	6th reference position X-axis	0.00000			
1542	6th reference position Y-axis	0.00000			
1543	6th reference position Z-axis	26.96850 (Standard) 29.72440 (Low)	19.88180 (Standard) 21.85030 (Low)	21.85030 (Standard) 24.60620 (Low)	17.91330 (Standard) 19.88180 (Long)
1544	6th reference position 4th-axis	0.0000			
1545	6th reference position 5th-axis	0.0000			
1546	6th reference position 6th-axis	0.0000			
1547	6th reference position 7th-axis	0.0000			
1548	6th reference position 8th-axis	0.0000			
1549	Reference point return complete signal output at servo off	0			
1550	Reference point allowable range at servo off (for XYZ axes)	0.00000			
1551	Reference point allowable range at servo off (for additional axes)	0.0000			
1552	X-axis zero return pos.(No.2)	0.00000			
1553	Y-axis zero return pos.(No.2)	0.00000			
1554	4th-axis zero return pos.(No.2)	0.0000			
1555	5th-axis zero return pos.(No.2)	0.0000			
1556	6th-axis zero return pos.(No.2)	0.0000			
1557	7th-axis zero return pos.(No.2)	0.0000			
1558	8th-axis zero return pos.(No.2)	0.0000			
1559	X-axis zero return pos.(No.3)	0.00000			
1560	Y-axis zero return pos.(No.3)	0.00000			
1561	4th-axis zero return pos.(No.3)	0.0000			
1562	5th-axis zero return pos.(No.3)	0.0000			
1563	6th-axis zero return pos.(No.3)	0.0000			
1564	7th-axis zero return pos.(No.3)	0.0000			
1565	8th-axis zero return pos.(No.3)	0.0000			
1566	Zero point return complete signal output at servo off	0			
1567	Zero point allowable range at servo off (for XYZ axes)	0.00000			
1568	Zero point allowable range at servo off (for additional axes)	0.0000			
1569	Workpiece coordinate zero point X axis (for capture)	0.00000			
1570	Workpiece coordinate zero point Y axis (for capture)	0.00000			
1571	Workpiece coordinate zero point Z axis (for capture)	0.00000			

user parameter 5(User registered miscellaneous functions)

There is not default value.

User Parameter 6 (Tool breakage detection)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1801	Tool breakage detection option	0			
1802	Pallet 1 tool breakage detector	0			
1803	Pallet 2 tool breakage detector	0			
1804	Tool breakage detection execution interval(No. of operations)	1			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
1805	Reference tool length offset No.	0			
1806	Tool breakage detection return position(Type1)	26.96850 (Standard) 29.72440 (Low)	19.88180 (Standard) 21.85030 (Low)	21.85030 (Standard) 24.60620 (Low)	17.91330 (Standard) 19.88180 (Long)
1807	Tool breakage detection position X-axis(Type1)	0.00000			
1808	Tool breakage detection position Y-axis(Type1)	0.00000			
1809	Tool breakage detection offset(Type2)	0.00000			
1810	Tool breakage detection position X-axis(Type3)	0.00000			
1811	Tool breakage detection position Y-axis(Type3)	0.00000			
1812	Tool breakage detection position Z-axis(Type3)	0.00000			
1813	Tool breakage detection start distance(Type3)	0.07870			
1814	Tool breakage detection stop distance(Type3)	0.03930			
1815	Tool breakage detection speed(Type3)	19.60			
1816	Tool breakage detection coolant OFF(Type3)	1			
1817	Pallet 2 tool breakage detection pos. X-axis(Type3)	0.00000			
1818	Pallet 2 tool breakage detection pos. Y-axis(Type3)	0.00000			
1819	Pallet 2 tool breakage detection pos. Z-axis(Type3)	0.00000			
1820	Pallet 2 tool breakage detection start distance(Type3)	0.07870			
1821	Pallet 2 tool breakage detection stop distance(Type3)	0.03930			
1822	Pallet 2 tool breakage detection speed(Type3)	19.60			
1823	Pallet 2 tool breakage detection coolant OFF(Type3)	1			

User Parameter 7 (Automatic workpiece measurement)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
2001	Probe compensation 1	0.00000			
2002	Probe compensation 2	0.00000			
2003	Probe compensation 3	0.00000			
2004	Probe compensation 4	0.00000			
2005	Measurement motion	0			
2006	Measurement speed 1	19.60			
2007	Measurement speed 2	3.90			
2008	Stop point before measurement	0.19680			
2009	Measurement travel limit distance	0.07870			
2010	Measurement tolerance 1	0.00000			
2011	Measurement tolerance 2	0.00000			
2012	Return distance after measurement	0.03930			

User Parameter 8 (Automatic thermal distortion compensation)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
2501	Z-axis compensation	100			
2502	Spindle compensation(Z-axis)	100			
2503	X-axis compensation	100			
2504	Spindle compensation(X-axis)	100			
2505	Y-axis compensation	100			
2506	Spindle compensation(Y-axis)	100			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
2507	Auto. thermal distortion compen. at memory operation start	0			
2508	Lathe spindle compensation(Z-axis)	0			
2509	Lathe spindle compensation(X-axis)	0			
2510	Lathe spindle compensation(Y-axis)	0			

User Parameter 9 (Quick table)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
3001	Pallet program folder				
3002	Pallet 1 program No.				
3003	Pallet 2 program No.				
3004	Pallet program folder				
3005	Pallet 1 program No.				
3006	Pallet 2 program No.				
3007	Pallet 2 shift(X-axis)	0.00000			
3008	Pallet 2 shift(Y-axis)	0.00000			
3009	Pallet 2 shift(Z-axis)	0.00000			
3010	Pallet 1 loading	0			
3011	Pallet 2 loading	0			

User Parameter 10(High accuracy)
High accuracy A (M260)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	120			
	Arc deceleration override	200			
	Curve approximation deceleration override	200			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M261)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	60			
	Arc deceleration override	140			
	Curve approximation deceleration override	140			
	Smooth path offset level	1			
	Smooth override	50			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M262)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	70			
	Arc deceleration override	70			
	Curve approximation deceleration override	70			
	Smooth path offset level	1			
	Smooth override	600			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M263)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	120			
	Arc deceleration override	200			
	Curve approximation deceleration override	200			
	Smooth path offset level	1			
	Smooth override	900			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M264)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	120			
	Arc deceleration override	200			
	Curve approximation deceleration override	200			
	Smooth path offset level	1			
	Smooth override	2500			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M265)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	600			
	Arc deceleration override	1000			
	Curve approximation deceleration override	1000			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M266)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	100			
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy A (M267)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Corner deceleration override	100			
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Cutting feed time constant selection	0			
	Minute block deletion distance	0.00000			
	Accuracy level	0.00000			

High accuracy B (M280)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	0			
	Curve approximation deceleration override	0			
	Smooth path offset level	3			
	Smooth override	100			
	Minute block deletion distance	0.00030			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M281)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M282)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M283)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M284)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M285)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M286)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy B (M287)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Arc deceleration override	100			
	Curve approximation deceleration override	100			
	Smooth path offset level	1			
	Smooth override	100			
	Minute block deletion distance	0.00000			
	Smooth override type	0			
	Corner deceleration override	100			

High accuracy common

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
3201	Smooth path offset function	0			
3202	Smooth path offset cancel distance	0.39300			
3203	Smooth path offset cancel angle	10.0000			
3204	Smooth path offset angle determination invalid distance	0.00030			
3205	Smooth path offset reference length	0.03930			
3206	Minute block deletion when smooth path offset is cancelled	0			
3207	Accuracy level method	1			

High accuracy(Adjust) common

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Machining mode	0			

High accuracy(Adjust) Adjust.A

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Standard mode	0			
	Accuracy	0			
	Smoothness	0			

High accuracy(Adjust) Adjust.B

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Standard mode	0			
	Accuracy	0			
	Smoothness	0			

High accuracy(Adjust) Adjust.C

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Standard mode	0			
	Accuracy	0			
	Smoothness	0			

High accuracy(Accuracy spec.) common

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Accuracy spec. mode	0			

High accuracy(Accuracy spec.) Accuracy spec. A

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Accuracy level	0.00390			
	Smoothness	0			

High accuracy(Accuracy spec.) Accuracy spec. B

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Accuracy level	0.00120			
	Smoothness	0			

High accuracy(Accuracy spec.) Accuracy spec. C

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Accuracy level	0.00040			
	Smoothness	0			

User Parameter 11 (PLC)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
3501	P1-axis stroke control	0			
3502	P2-axis stroke control	0			
3503	P3-axis stroke control	0			
3504	P4-axis stroke control	0			
3505	P1-axis stroke(-)	0			
3506	P2-axis stroke(-)	0			
3507	P3-axis stroke(-)	0			
3508	P4-axis stroke(-)	0			
3509	P1-axis stroke(+)	0			
3510	P2-axis stroke(+)	0			
3511	P3-axis stroke(+)	0			
3512	P4-axis stroke(+)	0			
3513	Reference point P1-axis	0			
3514	Reference point P2-axis	0			
3515	Reference point P3-axis	0			
3516	Reference point P4-axis	0			
3517	P1-axis override speed 1	0			
3518	P1-axis override speed 2	20			
3519	P1-axis override speed 3	200			
3520	P1-axis override speed 4	400			
3521	P2-axis override speed 1	0			
3522	P2-axis override speed 2	20			
3523	P2-axis override speed 3	200			
3524	P2-axis override speed 4	400			
3525	P3-axis override speed 1	0			
3526	P3-axis override speed 2	20			
3527	P3-axis override speed 3	200			
3528	P3-axis override speed 4	400			
3529	P4-axis override speed 1	0			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
3530	P4-axis override speed 2	20			
3531	P4-axis override speed 3	200			
3532	P4-axis override speed 4	400			
3533	Collis. detct.-P1ax stop meth.	0			
3534	Collis. detct.-P2ax stop meth.	0			
3535	Collis. detct.-P3ax stop meth.	0			
3536	Collis. detct.-P4ax stop meth.	0			

User Parameter 12 (Profibus)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
4001	Node address	0			
4002	I/O setting 1				
4003	I/O setting 2				
4004	I/O setting 3				
4005	I/O setting 4				
4006	I/O setting 5				
4007	I/O setting 6				
4008	I/O setting 7				
4009	I/O setting 8				
4010	I/O setting 9				
4011	I/O setting 10				
4012	I/O setting 11				
4013	I/O setting 12				
4014	I/O setting 13				
4015	I/O setting 14				
4016	I/O setting 15				
4017	I/O setting 16				
4018	I/O setting 17				
4019	I/O setting 18				
4020	I/O setting 19				
4021	I/O setting 20				

User Parameter 13 (DeviceNet)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
4201	MAC ID	0			
4202	Transfer rate	0			
4203	Input bytes	0			
4204	Output bytes	0			

User Parameter 14 (CC-Link Slave)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
4401	Station number	1			
4402	Slave station category	0			
4403	Occupied number of stations	1			
4404	Transfer rate	0			
4405	Specify data clear in case of communication error	0			

User Parameter 15 (Rotary fixture offset)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
4601	XYZ-axes travel during current rotary fixture offset change	0			
4602	Rotary fixture offset reset method	0			
4603	Tilt axis 1	0	0	0	2
4604	Rotation axis 1	0	0	0	4
4605	Tilt axis 2	0			
4606	Rotation axis 2	0			

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
4607	Tilt axis 3	0			
4608	Rotation axis 3	0			
4609	Tilt axis 4	0			
4610	Rotation axis 4	0			
4611	Forward direction for the coordinate system on tilt axis 1	0	0	0	1
4612	Forward direction for the coordinate system on rotation axis 1	0	0	0	5
4613	Forward direction for the coordinate system on tilt axis 2	0			
4614	Forward direction for the coordinate system on rotation axis 2	0			
4615	Forward direction for the coordinate system on tilt axis 3	0			
4616	Forward direction for the coordinate system on rotation axis 3	0			
4617	Forward direction for the coordinate system on tilt axis 4	0			
4618	Forward direction for the coordinate system on rotation axis 4	0			

User Parameter 16 (3D smoothing function)
3D smoothing function(M380)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M381)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M382)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M383)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M384)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M385)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M386)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function(M387)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
	Command path shape				
	Cross-section setting method				
	Cross-section center X				
	Cross-section center Y				
	Angle between sections				
	Distance between sections				
	Smoothing level				
	Commnd. path division length				
	Max commd pts btw sections				
	Dist to clear path divisions				

3D smoothing function common

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
4801	3D smoothing circular interpolation command				
4802	3D smoothing sequence number command				

User Parameter 17 (feature coordinate setting)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
10001	Tilt axis 1	0	0	0	2
10002	Rotation axis 1	0	0	0	4
10003	Tilt axis 2	0			
10004	Rotation axis 2	0			
10005	Forward direction for the coordinate system on tilt axis 1	0	0	0	1
10006	Forward direction for the coordinate system on rotation axis 1	0	0	0	5
10007	Forward direction for the coordinate system on tilt axis 2	0			
10008	Forward direction for the coordinate system on rotation axis 2	0			
10009	Rotation center X coordinate offset for tilt axis 1	0.00000			
10010	Rotation center Y coordinate offset for tilt axis 1	0.00000			
10011	Rotation center Z coordinate offset for tilt axis 1	0.00000			
10012	Rotation center X coordinate offset for rotation axis 1	0.00000			
10013	Rotation center Y coordinate offset for rotation axis 1	0.00000			
10014	Rotation center Z coordinate offset for rotation axis 1	0.00000			
10015	Rotation center X coordinate offset for tilt axis 2	0.00000			
10016	Rotation center Y coordinate offset for tilt axis 2	0.00000			
10017	Rotation center Z coordinate offset for tilt axis 2	0.00000			
10018	Rotation center X coordinate offset for rotation axis 2	0.00000			
10019	Rotation center Y coordinate offset for rotation axis 2	0.00000			
10020	Rotation center Z coordinate offset for rotation axis 2	0.00000			
10021	Absolute coordinate display format	0			

User Parameter 18 (CC-Link master)

No.	Item	R650X2_40MG	R450X2	R650X2	M200X3 M300X3
10201	Transfer rate	0			
10202	Self-station number	0			
10203	Interface board setting	0			
10204	Occupied number of stations (for intelligent device stations)	0			
10205	Standby master station number	0			
10206	CC-Link version	0			
10207	Retry attempts	3			
10208	Number of automatic return nodes	1			
10209	Specify action when there is a driver error	0			
10210	Specify station data clear in case of any error	0			

2.6 External I/O Signal

2.6.1 Outline

External I/O signals are allocated to I/O terminals and external alarms are set.

External I/O signals are divided into the following categories.

Menu page	Category name	Description
1	External input signals	External input signals are allocated to the input terminals (X) to the machine or the built-in PLC functions.
	External output signals	External output signals (Y) are allocated to the output terminals from the machine or the built-in PLC functions.
	External alarms	External alarms are set.
2	PROFIBUS DP -> NC	External input signals (X or D) are allocated to the input terminals to the machine or the built-in PLC functions from the PROFIBUS DP (slave).
	NC -> PROFIBUS DP	External output signals (Y or D) are allocated to the input terminals to the PROFIBUS DP (slave) from the machine or the built-in PLC functions.
	PROFIBUS DP status	Inputting destinations of the PROFIBUS DP (slave) status information to the machine or the built-in PLC functions (D) are allocated.
3	DeviceNet -> NC	External input signals (X or D) are allocated to the input terminals to the machine or the built-in PLC functions from the DeviceNet (slave).
	NC -> DeviceNet	External input signals (Y or D) are allocated to the input terminals to the DeviceNet (slave) from the machine or the built-in PLC functions.
	DeviceNet status	The inputting destinations (D) of DeviceNet (slave) status information to the machine or the built-in PLC functions are allocated.
4	RY/RWw (CC-Link -> NC)	External input signals are allocated to the input terminals (X or D) to the machine or the built-in PLC functions from the CC-Link (slave).
	RX/RWr (NC -> CC-Link)	External input signals (Y or D) are allocated to the input terminals to the CC-Link (slave) from the machine or the built-in PLC functions.
	CC-Link status	Inputting destinations (D) of the CC-Link (slave) status information are allocated to the machine or the built-in PLC functions.
5	External panel 1	External panel 1 are set. The range of data that are input/output between the built-in PLC functions and the external panel 1 is designated.
	External panel 2	Settings for external panel 2 are made. The range of data that are input/output between the built-in PLC functions and the external panel 2 is designated.

2.6.1.1 Edit Screen

Selecting <External I/O signal> on the <Data bank menu> screen displays the Edit screen for external I/O signals.

External input signal							
2012/05/29 10:33:13							
Standard terminal block							
Terminal	Signal	Terminal	Signal	Terminal	Signal	Terminal	Signal
No.000	EXER10	No.008		No.016		No.024	
No.001	EXER11	No.009	OPSP1P	No.017		No.025	
No.002	EXER12	No.010	EXRST	No.018		No.026	
No.003	EXER13	No.011	PR01	No.019		No.027	
No.004	EXER14	No.012		No.020		No.028	
No.005	PRLOCK	No.013		No.021		No.029	
No.006	EDLOCK	No.014		No.022		No.030	
No.007	D1R0K	No.015		No.023		No.031	

No.000						Check OK	Can be deleted
1. PR01	11. EXRST	21. EXER15	31. EXER25	41. EXER35	51. EXER45		
2. PR02	12. EXORG	22. EXER16	32. EXER26	42. EXER36	52. EXER46		
3. PR04	13. EXZORG	23. EXER17	33. EXER27	43. EXER37	53. EXER47		
4. PR08	14. CTURN	24. EXER18	34. EXER28	44. EXER38	54. EXER48		
5. PR016	15. MF IN	25. EXER19	35. EXER29	45. EXER39	55. EXER49		
6. PR032	16. EXER10	26. EXER20	36. EXER30	46. EXER40	56. EXER50		
7. PR064	17. EXER11	27. EXER21	37. EXER31	47. EXER41	57. EXER51		
8. EXSTR1	18. EXER12	28. EXER22	38. EXER32	48. EXER42	58. EXER52		
9. OUTST	19. EXER13	29. EXER23	39. EXER33	49. EXER43	59. EXER53		
10. EXSTOP	20. EXER14	30. EXER24	40. EXER34	50. EXER44	60. EXER54		

Completion mode	External input signal	External output signal	External alarm	Overlapping terminal check	Search	Next menu

The previously displayed category is displayed. By pressing the function key, the displayed category can be switched to another one.

The list of external signals to be displayed in the lower section of the screen for the category <External input signal> or <External output signal> can be switched to the next page by pressing the [F7] (Next menu) key.

- * The signals shown in red signify those which cannot be supported in the current language mode (Conversation/NC).
- * The terminals whose terminal number is grayed out signify those which are not connected (cannot be recognized). Although signals can be set to such terminals, however, please note that signals are not input/output to/from these terminals.

External alarm edit screen

External alarm							
2012/05/29 10:34:35							
External alarm 10		External alarm 11		External alarm 12		External alarm 13	
AAA		BBB		CCC		DDD	
Stop level	1	Stop level	2	Stop level	3	Stop level	4
Release level	2	Release level	2	Release level	2	Release level	2
Alarm log registrati	Yes	Alarm log registrati	Yes	Alarm log registrati	Yes	Alarm log registrati	Yes
External alarm 14		External alarm 15		External alarm 16		External alarm 17	
EEE		FFF		GGG		HHH	
Stop level	5	Stop level	1	Stop level	1	Stop level	1
Release level	2	Release level	4	Release level	4	Release level	4
Alarm log registrati	No	Alarm log registrati	No	Alarm log registrati	No	Alarm log registrati	No
External alarm 18		External alarm 19		External alarm 20		External alarm 21	
III		JJJ		KKK		LLL	
Stop level	1	Stop level	1	Stop level	1	Stop level	1
Release level	4	Release level	4	Release level	4	Release level	4
Alarm log registrati	No	Alarm log registrati	No	Alarm log registrati	No	Alarm log registrati	No

Message						Can be deleted

Completion mode	External input signal	External output signal	External alarm	Overlapping terminal check	Search	Next menu

2.6.2 How to Edit

2.6.2.1 Locating the Cursor

Locate the cursor at the item to be changed using the cursor key.

Search can be used to immediately locate the cursor at a desired “terminal block” or “External alarm.” Enter a terminal block number (0: standard terminal block, 1 to 64: extended terminal blocks 1 to 64) in the input area, enter an external alarm number, and press the **[F6]** (Search) key. Then the cursor is located at the designated number.

2.6.2.2 Data Input

1. Display the category to be changed.
2. Locate the cursor at the item to be changed.
3. Enter a value in the input area and press **[ENT]**.
For the <External input signal> and <External output signal>, enter the number of the external signal to be allocated to the terminal.

This inputs the value at the cursor position and moves the cursor to the next item.

- * As to <900:PLCX000-00F> through <961:PLCX3D0-3DF> in <External input signal> and <900:PLCY000-00F> through <961:PLCY3D0-3DF> in <External output signal>, set them for every 16 terminals collectively at a time.
When setting them, they are set to 16 terminals including the one with the terminal number at the cursor position (NO.**00 to 15 or NO.**16 to 31) collectively at a time. Please note that the preset signals are overwritten.

When entering the number of other signal in the input area with the cursor located at any of the 16 terminals (NO.*00 to 15 or NO.*16 to 31) to which signals <900:PLCX000-00F> through <961:PLCX3D0-3DF> or <900:PLCY000-00F> through <961:PLCY3D0-3DF> are set and pressing **[ENT]**, only the signal at the cursor position is set and the settings for the other 15 terminals are deleted.

Terminal		Signal		Terminal		Signal		Terminal		Signal	
No.000	PLCX000-00F	No.008	PLCX000-00F	No.016	OPSK IP	No.024		No.032			
No.001	PLCX000-00F	No.009	PLCX000-00F	No.017	OPSTOP	No.025		No.033			
No.002	PLCX000-00F	No.010	PLCX000-00F	No.018	EXER21	No.026		No.034			
No.003	PLCX000-00F	No.011	PLCX000-00F	No.019	EXER22	No.027		No.035			
No.004	PLCX000-00F	No.012	PLCX000-00F	No.020	CTURN	No.028		No.036			
No.005	PLCX000-00F	No.013	PLCX000-00F	No.021	4LOCK	No.029		No.037			
No.006	PLCX000-00F	No.014	PLCX000-00F	No.022	5LOCK	No.030		No.038			
No.007	PLCX000-00F	No.015	PLCX000-00F	No.023	6LOCK	No.031		No.039			

Terminal		Signal		Terminal		Signal		Terminal		Signal	
No.000		No.008		No.016		No.024		No.032		No.040	
61. EXER55	71. EXER65	81. EXER75	91. EXER85	101. EXER95	111. M468						
62. EXER56	72. EXER66	82. EXER76	92. EXER86	102. EXER96	112. SPLOCK						
63. EXER57	73. EXER67	83. EXER77	93. EXER87	103. EXER97	113. ATCLCK						
64. EXER58	74. EXER68	84. EXER78	94. EXER88	104. EXER98	114. XYLOCK						
65. EXER59	75. EXER69	85. EXER79	95. EXER89	105. EXER99	115. ZLOCK						
66. EXER60	76. EXER70	86. EXER80	96. EXER90	106. ALMRLS	116. 4LOCK						
67. EXER61	77. EXER71	87. EXER81	97. EXER91	107. M460	117. 5LOCK						
68. EXER62	78. EXER72	88. EXER82	98. EXER92	108. M462	118. 6LOCK						
69. EXER63	79. EXER73	89. EXER83	99. EXER93	109. M464	119. 7LOCK						
70. EXER64	80. EXER74	90. EXER84	100. EXER94	110. M466	120. 8LOCK						

Completion mode	External input signal	External output signal	External alarm	Overlapping terminal check	Search	Next menu

External input signals		
Item name	Settingrange	Description
Terminal number	1~176 (Common)	According to the terminal numbers, external input signals can be set as desired. They can be set to “no setting” using [DEL] key.
Standard terminal blocks	501~640 (NC language only)	
NO.000~031	701~706 (Conversation only)	
Extended terinal blocks 1	900~993 (PLC input)	
NO.000~031	999 (RESERV)	For details of external input signals, refer to “2.6.3 External input signals”.
Extended terminal blocks 2	or no setting	
NO.000~031	For PLC input, set every 16 terminals collectively, excluding PLCX3E0-3EF.	
Extended terminal blocks 3		
NO.000~031		
Extended terminal blocks 4		
NO.000~031	900:PLCX000-00F	
:	901:PLCX010-01F	
:	:	
Extended terminal blocks 63	960:PLCX3C0-3CF	
NO.000~031	961:PLCX3D0-3DF	
Extended terminal blocks 64	962~977:PLCX3E0-3EF	
NO.000~031	978~993:PLCX3F0-3FF	

List of external input signals

Setting No.	Signal	Setting No.	Signal	Setting No.	Signal	Setting No.	Signal
1	PRO1	129	WCEDOK	170	OVTR5M	900	PLCX000-00F
2	PRO2	130	MGEDOK	171	OVTR6P	:	:
3	PRO4	131	MTEDOK	172	OVTR6M	961	PLCX3D0-3DF
4	PRO8	132	PREDOK	173	OVTR7P		
5	PRO16	133	IOEDOK	174	OVTR7M	962	PLCX3E0
6	PRO32	134	CPEDOK	175	OVTR8P	963	PLCX3E1
7	PRO64	135	DIROK	176	OVTR8M	964	PLCX3E2
8	EXSTRT	136	MODEOK			965	PLCX3E3
9	OUTST	137	LEVEL1	501	EXREF	966	PLCX3E4
10	EXSTOP	138	LEVEL2	502	EXREF2	967	PLCX3E5
11	EXRST	139	LEVEL3	503	EXREF3	968	PLCX3E6
12	EXORG	140	LEVEL4	504	EXREF4	969	PLCX3E7
13	EXZORG	141	WKCLR1	505	EXREF5	970	PLCX3E8
14	CTURN	142	WKCLR2	506	EXREF6	971	PLCX3E9
15	MFIN	143	WKCLR3	507	OPSKIP	972	PLCX3EA
16	EXER10	144	WKCLR4	508	OPSTOP	973	PLCX3EB
:	:	145	DRLOCK	509	EXWORK	974	PLCX3EC
105	EXER99	146	ATDROP	510	WPEDOK	975	PLCX3ED
106	ALMRLS	147	ATDRCL	511	MCEDOK	976	PLCX3EE
107	M460	148	OUTAP	512	UINT	977	PLCX3EF
108	M462	149	OUTAM	513	#1000	978	PLCX3F0
109	M464	150	OUTBP	:	:	979	PLCX3F1
110	M466	151	OUTBM	544	#1031	980	PLCX3F2
111	M468	152	OUTLCK	545	#1200	981	PLCX3F3
112	SPLOCK	153	UNCLS5	:	:	982	PLCX3F4
113	ATCLCK	154	UNCLS6	640	#1295	983	PLCX3F5
114	XYLOCK	155	UNCLS7			984	PLCX3F6
115	ZLOCK	156	UNCLS8	701	PR9900	985	PLCX3F7
116	4LOCK	157	UNCSP1	702	EXORG2	986	PLCX3F8
117	5LOCK	158	UNCSP2	703	EXORG3	987	PLCX3F9
118	6LOCK	159	UNCSP3	704	TPEDOK	988	PLCX3FA
119	7LOCK	160	UNCSP4	705	THEDOK	989	PLCX3FB
120	8LOCK	161	CLPLS5	706	CTEDOK	990	PLCX3FC
121	MDLOCK	162	CLPLS6			991	PLCX3FD
122	KYLOCK	163	CLPLS7			992	PLCX3FE
123	PRLOCK	164	CLPLS8			993	PLCX3FF
124	EDLOCK	165	CLPSP1				
125	MGPLCK	166	CLPSP2			999	RESERV
126	TLEDOK	167	CLPSP3				

Setting No.	Signal	Setting No.	Signal	Setting No.	Signal	Setting No.	Signal
127	UPEDOK	168	CLPSP4				
128	MPEDOK	169	OVTR5P				

External output signals

Item name	Setting range	Description
Terminal number	1~138 (Common)	According to the terminal number, external output signals can be set as desired.
Standard terminal blocks	501~636 (NC language only)	
NO.100~131	701~702 (Conversation only)	They can be set to “no setting” using [DEL] key.
Extended terminal blocks 1	900~993 (PLC output)	
NO.100~131	999 (RESERV)	For details of external output signals, refer to “2.6.4 External input signals”.
Extended terminal blocks 2	or no setting	
NO.100~131	For PLC input, set every 16 terminals collectively, excluding	
Extended terminal blocks 3	PLCY3E-3EF.	
NO.100~131	900:PLCY000-00F	
Extended terminal blocks 4	901:PLCY010-01F	
NO.100~131	:	
:	:	
Extended terminal blocks 63	960:PLCY3C0-3CF	
NO.100~131	961:PLCY3D0-3DF	
Extended terminal blocks 64	962~977:PLCY3E0-3EF	
NO.100~131	978~993:PLCY3F0-3FF	

List of external output signals

Setting No.	Signal	Setting No.	Signal	Setting No.	Signal	Setting No.	Signal
1	M00	51	ZP4	99	ZPOSSW	508	LSSLCT
2	M30/1	52	ZP5	100	4POSSW	509	#1100
3	M30/2	53	ZP6	101	5POSSW	:	:
4	M08	54	ZP7	102	6POSSW	540	#1131
5	M11	55	ZP8	103	7POSSW	541	#1300
6	M12	56	PFIN1	104	8POSSW	:	:
7	M14	57	PFIN2	105	OUTAON	636	#1395
8	M18	58	RSTOUT	106	OUTBON		
9	M21	59	TLBRKE	107	UNCLP5	701	ORGFN2
10	M22	60	RUNSTP	108	UNCLP6	702	ORGFN3
11	M24	61	BATALM	109	UNCLP7		
12	M28	62	MNTALM	110	UNCLP8	900	PLCY000-00F
13	MF	63	GRN	111	UCLPP1	:	:
14	M900	64	YEL	112	UCLPP2	961	PLCY3D0-3DF
15	M400	65	RED	113	UCLPP3	962	PLCY3E0
16	M402	66	CNTUP	114	UCLPP4	963	PLCY3E1
17	M404	67	CNTPRE	115	MGZ1	964	PLCY3E2
18	M406	68	STPOUT	116	MGZ2	965	PLCY3E3
19	M408	69	STPLED	117	MGZ4	966	PLCY3E4
20	M480	70	DROPEN	118	MGZ8	967	PLCY3E5
21	M482	71	DRCLS	119	MGZ16	968	PLCY3E6
22	M484	72	INDRCL	120	MGZ32	969	PLCY3E7
23	M486	73	SDDRCL	121	UNLOCK	970	PLCY3E8
24	M450	74	XDR1CL	122	SPIALM	971	PLCY3E9
25	M451	75	XDR2CL	123	DRHOLD	972	PLCY3EA
26	M455	76	ATDRSL	124	DRORFN	973	PLCY3EB
27	M456	77	DRRDY	125	RDBTAL	974	PLCY3EC
28	AUTO	78	DROFIN	126	HWMO	975	PLCY3ED
29	STL	79	DRCFIN	127	RYCHK1	976	PLCY3EE
30	MEMOK	80	DRALM1	128	XPSSW2	977	PLCY3EF
31	MEMMOD	81	DRALM2	129	YPSSW2	978	PLCY3F0
32	EXPRUN	82	RDYLED	130	ZPSSW2	979	PLCY3F1
33	NCOK	83	DRYRUN	131	4PSSW2	980	PLCY3F2
34	ALM	84	SINGL	132	5PSSW2	981	PLCY3F3
35	ALM2	85	RESTAT	133	6PSSW2	982	PLCY3F4
36	ALMLV1	86	RPD100	134	7PSSW2	983	PLCY3F5
37	ALMLV2	87	SPN100	135	8PSSW2	984	PLCY3F6
38	ASTLV1	88	FED100	136	TLDHI	985	PLCY3F7

Setting No.	Signal	Setting No.	Signal	Setting No.	Signal	Setting No.	Signal
39	ASTLV2	89	RELESE	137	TLDLO	986	PLCY3F8
40	ASTLV3			138	TLDON	987	PLCY3F9
41	ASTLV4					988	PLCY3FA
42	ASTLV5	90	SAFETY			989	PLCY3FB
43	TOOL	91	COOLSW			990	PLCY3FA
44	TOLPRE	92	CHPSW	501	REFIN	991	PLCY3FD
45	ORGFIN	93	PULOFF	502	REFIN2	992	PLCY3FE
46	SPTURN	94	QTSEL1	503	REFIN3	993	PLCY3FF
47	ZPX	95	QTSEL2	504	REFIN4		
48	ZPY	96	WAVDSP	505	REFIN5	999	RESERV
49	ZPZ	97	XPOSSW	506	REFIN6		
50	ZPSP	98	YPOSSW	507	LSTURN		

External alarms

Item name	Setting range	Description
External alarms 10~99	Message	Up to 20 characters Register alarm messages. Up to 20 characters that can be input through the panel can be registered. They can be set to “no setting” using [DEL] key. When no alarm message is set, “External alarms 10 to 99” are displayed. When the communication parameter <Extended comment input> is set to <1: Yes>, the external I/O signal file (EXIO*.NC) (* refers to the data bank number) is transferred and input. This enables input for characters that could not be entered using the machine keys. Refer to section “5.5 Communication Protocol” in the Operation Manual (Data) for further details.
	Stop levels	1: Level 1 2: Level 2 3: Level 3 4: Level 4 5: Level 5 Select the level at which the operation is affected by an alarm (external alarms 10 to 99) that occurs upon input of any of EXER10 to 99 signals, from the five levels below. 1: Operation does not stop. 2: Operation stops upon completion of the current cycle. 3: Operation is stopped upon execution of the current block. 4: Operation is paused. 5: Servo control is turned off. When the field is empty or blank, the default setting is <1: Level 1>.
	Release levels	1: Level 1 2: Level 2 3: Level 3 4: ALMRLS signal Select the means for releasing an alarm (external alarms 10 to 99) that occurs upon input of any of EXER signals 10 to 99, from the four levels below. 1: Any key 2: [RST] key 3: Power off 4: ALMRLS signal or [RST] key When the field is empty or blank, the default setting is <1: Level 1>.
	Alarm log registration	0: No 1: Yes Set whether or not an alarm is registered in the alarm log when such an alarm occurs.

- * When stop level has been set to any level from 3 to 5, the release level is treated as set to level 2 even if it is set to level 1 or 4.
In this case, an <<External alarm setting error>> (stop level 1, release level 1) occurs when resetting a changed parameter or displaying the External alarm setting screen.
- * When set to stop level 2 through 5, the alarm does not cancel or reset during memory operation (excluding stoppage) even for cancel level 1 alarms.

2.6.2.3 Data Deletion

When the cursor is located at a deletable item, <Can be deleted> is displayed.

1. Display a category which contains the item to be deleted.
2. Locate the cursor at the item to be deleted.
3. Press the **[DEL]** key.

This deletes the value at the cursor position.

- * As to <900:PLCX000-00F> through <961:PLCX3D0-3DF> in <External input signal> and <900:PLCY000-00F> through <961:PLCY3D0-3DF> in <External output signal>, delete them for every 16 terminals collectively at a time.

When pressing the **[DEL]** key with the cursor located at any of the 16 terminals (NO.*00 to 15 or NO.*16 to 31) to which signals <900:PLCX000-00F> through <961:PLCX3D0-3DF> or <900:PLCY000-00F> through <961:PLCY3D0-3DF> are set, the settings for the 16 terminals including the one with the terminal number at the cursor position are deleted collectively.

2.6.2.4 Overlapping Input Terminal Check

The values set on each screen for one setting item are only valid for 1 terminal on the following screens: external input signal (standard terminal block, extension terminal block 1 to 64), the PROFIBUS DP slave (PROFIBUS DP → NC)/status, , the DeviceNet slave (DeviceNet → NC)/status and the CC-Link slave RY/RWw (CC-Link → NC)/status screens. If the same value is overlappingly set to multiple setting items, only the first set terminal is made valid in the order shown below and the subsequent setting items are treated as “no setting”.

- Smaller terminal number of standard terminal blocks
- Smaller terminal number of extended terminal blocks
- Terminal that is displayed on top in the PROFIBUS DP slave status
- Small terminal numbers for the PROFIBUS DP slave (PROFIBUS DP → NC)
- Terminal that is displayed on top in the DeviceNet slave status
- Small terminal numbers for the DeviceNet slave (DeviceNet → NC)
- Terminal that is displayed on top in the CC-Link slave status
- Small terminal numbers for the CC-Link slave (RY/RWw (CC-Link → NC))

When pressing the **[F5]** (Overlapping terminal check) on any of the input signal screens, check for overlappingly set terminals is performed. (Search the files being edited.)

When there are no overlapping terminals, <Check OK> is displayed in the teaching area.

When there is any overlapping terminal, the names of the signals to which the terminal is overlappingly set, the terminal block in which the overlapping terminal is placed and the terminal number of the overlapping terminal are displayed as shown below.

- When there are multiple overlapping terminals, they are displayed in serial order of their set numbers to the extent that they can be displayed.
- When there are overlapping terminals too many to be displayed in the display area, they are not displayed.
- Overlapping terminal check is performed also for the terminal blocks not connected to the machine.
- When the CM PCB is an intelligent device station, the terminal name that is duplicated appears as follows. (Refer to the screen below for a display example of RWw.)

When Rx is duplicated: CM Rx+*** (Note 1)

When RWr is duplicated: CM RWr+*** (Note 2)

(Note 1) * refers to a code between 000 and 3FF.**

(Note 2) * refers to a code between 000 and 07F.**

7577 The external input terminal is duplicated.

External input signal 2019/06/15 10:00:23

Standard terminal block

Terminal	Signal	Terminal	Signal	Terminal	Signal	Terminal	Signal
No.000	EXER22	No.008		No.016		No.024	
No.001	EXER23	No.009		No.017		No.025	
No.002	EXER24	No.010		No.018		No.026	
No.003		No.011		No.019		No.027	
No.004		No.012		No.020		No.028	
No.005		No.013		No.021		No.029	
No.006		No.014		No.022		No.030	
No.007		No.015		No.023		No.031	

[HELP]

No.000 Overlap. terminals exist ☒ Can be deleted

Overlap. terminals exist (28. EXER22)
 Standard No.000
 Ext.01 No.000
 CM RX000

Overlap. terminals exist (29. EXER23)
 Standard No.001
 Ext.01 No.001
 CM RX001

Overlap. terminals exist (30. EXER24)
 Standard No.002

Completion mode	External input signal	External output signal	External alarm		Overlapping terminal check	Search	Overlapping terminal display end

However, when overlapping terminal check has been performed during editing even just once, the previously searched values are displayed. Pressing again the **[F5]** (Overlapping terminal check) performs the check again.

7587 or 24

External input signal 2012/05/29 10:47:32

Extended terminal block54

Terminal	Signal	Terminal	Signal	Terminal	Signal	Terminal	Signal
No.000	PLCX000-00F	No.008	PLCX000-00F	No.016	OPSK IP	No.024	
No.001	PLCX000-00F	No.009	PLCX000-00F	No.017	OPSTOP	No.025	
No.002	PLCX000-00F	No.010	PLCX000-00F	No.018	EXER21	No.026	
No.003	PLCX000-00F	No.011	PLCX000-00F	No.019	EXER22	No.027	
No.004	PLCX000-00F	No.012	PLCX000-00F	No.020	CTURN	No.028	
No.005	PLCX000-00F	No.013	PLCX000-00F	No.021	4LOCK	No.029	
No.006	PLCX000-00F	No.014	PLCX000-00F	No.022	5LOCK	No.030	
No.007	PLCX000-00F	No.015	PLCX000-00F	No.023	6LOCK	No.031	

No.000 Overlap. terminals exist ☒ Can be deleted

Overlap. terminals exist (27. EXER21)
 Extensionf No.008
 Extensionf No.018

Overlap. terminals exist (28. EXER22)
 Extensionf No.009
 Extensionf No.019

Overlap. terminals exist (507. OPSK IP)
 Standard No.009
 Extensionf No.016

Completion mode	External input signal	External output signal	External alarm		Overlapping terminal check	Search	Overlapping terminal display end

At this time, switching to the output terminal screens, external alarms, PROFIBUS DP status, DeviceNet status and CC-Link status functions are grayed out.

Pressing the **[F7]** (Overlapping terminal display end) returns to the teaching of the original set value.

Even if modification to overlapping terminals is attempted through setting, the display in the teaching area remains unchanged.

If the editing operation closes due to a duplication terminal error and the **[RST]** key is pressed, the operator message: <<The external input terminal is duplicated>>, and the error message: <<Duplicated errors of expansion input board (*)>> are displayed.

2.6.2.5 Ending Editing

Upon completion of the setting, press the **[F0]** (Completion mode) key to end the editing.

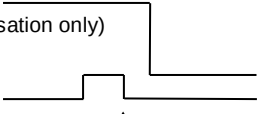
* When any change has been made, resetting is prompted.

2.6.3 External Input Signals

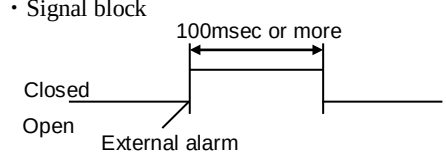
Described in this section is how to use external program selection signals and cautions on it.

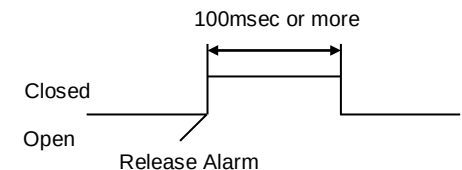
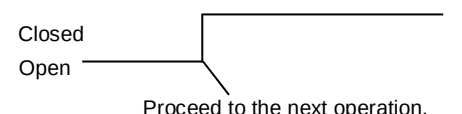
2.6.3.1 List of Input Signals

*1 signifies program editing.

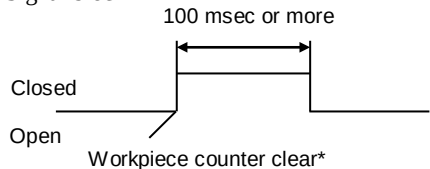
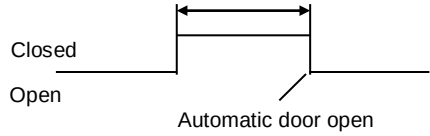
Signal name	Symbol	Logic		Enabled mode				External input signal key registration	Function
		Normally open	Normally closed	Manual	MDI	Memory	Editing *1		
External program selection	PRO1 PRO2 PRO4 PRO8 PRO16 PRO32 PRO64 PR9900 (Conversation only)	○				○		○	To be used when preliminarily created program is specified externally. Refer to: For details of how to use, refer to “External program selection.” PRO1,2,4,8,16, 32,64 PR9900 (Conversation only) Start  Read program selection signal
External start	EXSTRT	○			○	○		×	Equivalent to [START] switch on the operation panel
External pallet start	OUTST	○				○		×	Equivalent to [PALLET START] switch on the operation panel
External stop	EXSTOP	○		○	○	○		×	Equivalent to [FEED HOLD] switch on the operation panel
External reset	EXRST	○		○	○	○	○	×	Equivalent to [RST] key on the operation panel

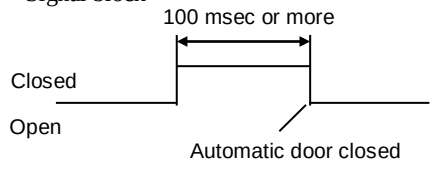
Signal name	Symbol	Logic		Enabled mode				External input signal key registration	Function
		Normally open	Normally closed	Manual	MDI	Memory	Editing *1		
External zero return	EXORG	○				○		×	The axis moves to the zero point when the contact of signal EXORG has closed and then opened again. (Note 1)Axis movement is done at positioning feed. (Note 2)Axes move in the following order: Z→XY (simultaneously) (Note 3)The travel order on the additional axis follows the order set in the machine parameters <4th- to 8th-axis zero return order>. (Note 4)Axes do not move to the middle point. (Note 5)Input the signal for 100 msec or more. (Note 6)During external Z-axis zero return, Automatic operation starting (STL) signal is not turned on. (Note 7)When the machine is equipped with turning spindle and the turning spindle is currently selected, zero return is not made for the additional axis that serves as the turning spindle. <u>Effective conditions</u> (1) Automatic operation ongoing signal (STL) is off. (2) The [FEED HOLD] switch is not pressed. (3) The external stop signal (EXSTOP) is off. (4) An alarm higher than stop level 2 has not occurred. (5) The door is closed. (6) The screen <Move start position> is not currently displaying.
External Z-axis zero return	EXZORG	○				○		×	The axis moves to the zero point when the contact of signal EXZORG has closed and then opened again. (Note 1)Axis movement is done at positioning feed. (Note 2)Axes do not move to the middle point. (Note 3)Input the signal for 100 msec or more. (Note 4)During external Z-axis zero return, Automatic operation starting (STL) signal is not turned on. <u>Enabling conditions</u> (1) Automatic operation ongoing signal (STL) is off. (2) The [FEED HOLD] switch is not pressed. (3) The external stop signal (EXSTOP) is off. (4) An alarm higher than stop level 2 has not occurred. (5) The door is closed. (6) The screen <Move start position> is not currently displaying.

Signal name	Symbol	Logic		Enabled mode				External input signal key registration	Function
		Normally open	Normally closed	Manual	MDI	Memory	Editing *1		
External pallet indexing	CTURN	○				○		×	Indexing pallet is performed when the contact of signal CTRUN is closed and then opened again. Operation is the same as when the [P. INDEX] key is pressed. <u>Effective conditions</u> (1) Automatic operation ongoing signal (STL) is off. (2) The [FEED HOLD] switch is not pressed. (3) The external stop signal (EXSTOP) is off. (4) An alarm higher than stop level 2 has not occurred. (5) The door is closed. (6) The screen <Move start position> is not currently displaying.
M function completion signal	MFIN	○			○	○		×	To the M function output M00 to M99, signal MFIN is input as peripheral equipment operation completion signal. (Refer to:) For details including signal timing, refer to “2.6.4 External output signals.”
External alarm 10 : External alarm 99	EXER10 : EXER99	○		○	○	○	○	×	When the any of contact of signal from EXER10 to EXER 99 is closed, one of <<External error 10>> to <<External error 99>> is displayed on the screen. Alarm level can be set by the <Stop level> of <External alarm 10> through <External alarm 99> in the <External I/O signal>, and the following levels are available. Stop level 1: Display only Stop level 2: Stops upon completion of the current cycle. Stop level 3: Block stop Stop level 4: Pause Stop level 5: Servo controller off Release level can be set by the <Cancel levels> of <External alarm 10> through <External alarm 99> in the <External I/O signal>, and the following levels are available. Release level 1: Any key operation Release level 2: [RST] key. Release level 3: Power-off required Release ALMRLS signal: ALMRLS signal * When <Stop level> has been set to any level from 3 to 5, the <Cancel level> is treated as set to level 2 even if it is set to level 1 or 4. • Signal block 

Signal name	Symbol	Logic		Enabled mode				External input signal key registration	Function
		Normally open	Normally closed	Manual	MDI	Memory	Editing *1		
Release alarms	ALMRLS	○		○	○	○	○	×	When the machine is stopped by the generation of the following alarm, the alarm at stop level 1 or 2 can be released by inputting ALMRLS signal in the external input signals. In the case of an <<External error***>> (where **:10 to 99) with the <Cancel level> in the external I/O signal setting set to <4: ALMRLS signal> 
Wait signals	M460 M462 M464 M466 M468	○		○	○	○	○	×	These are input signals used for signal output functions 460 and are used to halt the next motion during operation. When the contacts of signal M460 are closed, signal output functions 460 proceed to the next motion. Signal output 461 waits for the opening of the contacts of signal M460 and proceeds to the next operation. (The signals M462 through M468 operate in the same manner as that of signal M460.)  (Note) Maximum standby time represents the time set in <External signal reading time limit> in the user parameters. Setting it to "0" results in infinite wait.
Prevent spindle operation	SPLOCK	○		○	○	○		×	Prohibit spindle rotation. Spindle stop and orientation are possible. When this signal is turned ON during spindle rotation, the spindle does not stop.
Prevent ATC operation	ATCLCK	○		○	○	○		×	Prohibits ATC operations: magazine swivel, use of [ATC] key, tool replacement in MDI or memory mode, and next tool preparation. The automatic setting operation for the ATC monitoring parameter is prohibited. The [RELSE] key + [M] key are prohibited on the screen used to configure the <Automatic setting for ATC monitoring parameter>.
Prevent XY-axes operation	XYLOCK	○		○	○	○		×	Prohibits XY-axis operation. When the program is created so that other axes operate simultaneously, these axes do not operate either.
Prevent Z-axis operation	ZLOCK	○		○	○	○		×	Prohibits Z-axis operation.

Signal name	Symbol	Logic		Enabled mode				External input signal key registration	Function
		Normally open	Normally closed	Manual	MDI	Memory	Editing *1		
Prevent 4th axis operation Prevent 5th axis operation Prevent 6th axis operation Prevent 7th axis operation Prevent 8th axis operation	4LOCK 5LOCK 6LOCK 7LOCK 8LOCK	○		○	○	○		×	Prohibits 4 ~ 8th axis operation. Also prohibits clamping and unclamping operations. (Note) Turning on this signal during rotation of the axis that serves as the turning spindle does not stop the turning spindle.
Prevent mode change	MDLOCK	○		○	○	○	○	×	Prohibits mode changeover.
Prevent key operation	KYLOCK	○		○	○	○	○	×	Prevents key operations other than those using [RESET], [RELEASE], [+X], [-X], [+Y], [-Y], [+Z], [-Z], [+4] and [-4] keys. As long as the [EMERGENCY] switch is ON, however, the prevent signal is released. Switch operations using [OVERRIDE] (spindle, cutting, positioning), [EMERGENCY], [START], [FEED HOLD] and [PALLET START] switches are not prevented.
Prevent program edit	PRLOCK	○					○	×	Prevents data change operations (insert, delete, set) in edit mode (Program screen). Deleting and copying files in the directory display is also prohibited. In addition, operations that change the data during preparation are prohibited. As long as the [EMERGENCY] switch is ON, however, the prevent signal is released.
Prevent edit	EDLOCK	○		○	○	○	○	○	Prevents program editing. Prevents data bank editing. Also prevents editing of workpiece counter, ATC tools and maintenance notice. As long as the [EMERGENCY] switch is ON, however, the prevent signal is released.
Prevent magazine panel operation	MGPLCK								Reserved
Permit editing Tool data (NC) and Tool list (Conversation)	TLEDOK	○		○	○	○	○	○	As long as this signal is on and even when "EDLOCK" is on, the tool data can be edited.
Permit editing user parameters	UPEDOK	○		○	○	○	○	○	As long as this signal is on and even when "EDLOCK" is on, the user parameter can be edited.
Permit editing machine parameters	MPEDOK	○		○	○	○	○	○	As long as this signal is on and even when "EDLOCK" is on, the machine parameter can be edited.
Permit editing workpiece counter	WCEDOK	○		○	○	○	○	○	As long as this signal is on and even when "EDLOCK" is on, the workpiece counter can be edited.
Permit editing ATC tools	MGEDOK	○		○	○	○	○	○	As long as this signal is on and even when "EDLOCK" is on, the ATC tool can be edited.

Signal name	Symbol	Logic		Enabled mode				External input signal key registration	Function
		Normally open	Normally closed	Manual	MDI	Memory	Editing *1		
Permit editing maintenance notice	MTEDOK	○		○	○	○	○	○	Edition of maintenance notice is enabled as long as this signal is ON, even if Prevent editing (EDLOCK) is ON.
Permit program editing	PREDOK	○		○	○	○	○	○	Edition of program is enabled as long as this signal is ON, even if Prevent editing (EDLOCK) is ON or Prevent program editing (PRLOCK) is ON.
Permit editing external I/O signals	IOEDOK	○		○	○	○	○	○	Edition of external I/O signal screen is enabled as long as this signal is ON, even if Prevent editing (EDLOCK) is ON.
Permit editing communication parameters	CPEDOK	○		○	○	○	○	○	Edition of the Communication parameters screen is enabled as long as this signal is ON, even if Prevent editing (EDLOCK) is ON.
Permit editing directory display	DIROK	○		○	○	○	○	○	Operation of directory display is enabled as long as this signal is ON, even if Prevent editing (EDLOCK) is ON or Prevent program editing (PRLOCK) is ON.
Permit mode change	MODEOK	○		○	○	○	○	○	Mode change is enabled as long as this signal is ON, even if Prevent mode change (MDLOCK) is ON.
Operation level 1	LEVEL1	○		○	○	○	○	○	Set the operation level. When multiple operation level signals are ON, the operation level is set to the highest level enabled. When all the signals are OFF, the operation level is set to level "0."
Operation level 2	LEVEL2								
Operation level 3	LEVEL3								
Operation level 4	LEVEL4								
Workpiece counter clear 1	WKCLR1	○		○	○	○	○	×	Initializes the current value for each workpiece counter. • Signal block  100 msec or more Closed Open Workpiece counter clear*
Workpiece counter clear 2	WKCLR2								
Workpiece counter clear 3	WKCLR3								
Workpiece counter clear 4	WKCLR4								
Workpiece counter clear 4	WKCLR4								
Door lock	DRLOCK	○		○	○	○	○	×	Setting the signal DRLOCK to "Contact closed" locks the door closed.
Automatic door opening	ATDROP	○		○	○	○	○	×	Similar to the opening of [AUTOMATIC DOOR OPEN/CLOSE] switch. Opens the door. The operation is available regardless of the status of the [AUTOMATIC DOOR MANUAL/AUTO] switch. • Signal block  Closed Open Automatic door open

Signal name	Symbol	Logic		Enabled mode				External input signal key registration	Function
		Normally open	Normally closed	Manual	MDI	Memory	Editing *1		
Automatic door closing	ATDRCL	○		○	○	○	○	×	Similar to the opening of [AUTOMATIC DOOR OPEN/CLOSE] switch. Closes the automatic door when opened. The operation is available regardless of the status of the [AUTOMATIC DOOR MANUAL/AUTO] switch. Signal block 
Outer A-axis(+) indexing	OUTAP	○				○		×	If the ladder program is not changed from the default status at shipment, the A-axis of the outer pallet is rotated by the amount of the set vale for <External A-axis indexing (+)> in the User parameters. Starts the operation at the falling edge of the signal. Does not operate when the outer door (side door) is open. * By means of the setting for <External index A switch address swiching> in the User parameters, the C-axis instead of the A-axis of the external pallet can be rotated.
Outer A-axis(-) indexing	OUTAM	○				○		×	If the ladder program is not changed from the default status at shipment, the A-axis of the outer pallet is rotated by the amount of the set vale for <External A-axis indexing (-)> in the User parameters. Starts the operation at the falling edge of the signal. Does not operate when the outer door (side door) is open. * By means of the setting for <External index A switch address swiching> in the User parameters, the C-axis instead of the A-axis of the outer pallet can be rotated.
Outer B-axis(+) indexing	OUTBP	○				○		×	If the ladder program is not changed from the default status at shipment, the B-axis of the outer pallet is rotated by the amount of the set vale for <External B-axis indexing (+)> in the User parameters. Starts the operation at the falling edge of the signal. Does not operate when the outer door (side door) is open. * By means of the setting for <External index B switch address swiching> in the User parameters, the C-axis instead of the B-axis of the outer pallet can be rotated.

Signal name	Symbol	Logic		Enabled mode				External input signal key registration	Function
		Normally open	Normally closed	Manual	MDI	Memory	Editing *1		
Outer B-axis(-) indexing	OUTBM	○				○		×	If the ladder program is not changed from the default status at shipment, the B-axis of the outer pallet is rotated by the amount of the set vale for <External B-axis indexing (-)> in the User parameters. Starts the operation at the falling edge of the signal. Does not operate when the outer door (side door) is open. * By means of the setting for <External index B switch address swiching> in the User parameters, the C-axis instead of the B-axis of the outer pallet can be rotated.
Prevent outer additional axis indexing	OUTLCK	○		○	○	○	○	×	Prevents the indexing of outer A-,B- and C-axes as long as this signal is ON. Prevents the panel switch as well as the external input signals described above.
5th axis unclamp	UNCLS5	○		○	○	○	○	×	Inputs the unclamping signal status of the 5th to 8th axes respectively. Whether the unclamping signal status is checked by ON or OFF is set by <5th (to 8th) axis unclamp check input signal> in the machine parameters.
6th axis unclamp	UNCLS6								
7th axis unclamp	UNCLS7								
8th axis unclamp	UNCLS8								
P1-axis unclamp	UNCSP1	○		○	○	○	○	×	Inputs the unclamping signal status of the P1 to P4 axes respectively. Whether the unclamping signal status is checked by ON or OFF is set by <P1 (to P4) axis unclamp check input signal> in the machine parameters.
P2-axis unclamp	UNCSP2								
P3-axis unclamp	UNCSP3								
P4-axis unclamp	UNCSP4								
5th axis clamp	CLPLS5	○		○	○	○	○	×	Inputs the clamping signal status of the 5th to 8th axes respectively. Whether the clamping signal status is checked by ON or OFF is set by <5th (to 8th) axis clamp check input signal> in the machine parameters.
6th axis clamp	CLPLS6								
7th axis clamp	CLPLS7								
8th axis clamp	CLPLS8								
P1-axis clamp	CLPSP1	○		○	○	○	○	×	Inputs the clamping signal status of the P1 to P4 axes respectively. Whether the unclamping signal status is checked by ON or OFF is set by <P1 (to P4) axis clamp check input signal> in the machine parameters.
P2-axis clamp	CLPSP2								
P3-axis clamp	CLPSP3								
P4-axis clamp	CLPSP4								
5th axis Overtravel (+)	OVTR5P	○		○	○	○	○	×	Inputs the 5th axis overtravel (+). When this signal turns on, the machine will output the <<*Overtravel (+5)>> error.
5th-axis overtravel (-)	OVTR5M	○		○	○	○	○	×	Inputs the 5th axis overtravel (-). When this signal turns on, the machine will output the <<*Overtravel (-5)>> error.
6th-axis overtravel (+)	OVTR6P	○		○	○	○	○	×	Inputs the 6th axis overtravel (+). When this signal turns on, the machine will output the <<*Overtravel (+6)>> error.
6th-axis overtravel (-)	OVTR6M	○		○	○	○	○	×	Inputs the 6th axis overtravel (-). When this signal turns on, the machine will output the <<*Overtravel (-6)>> error.

Signal name	Symbol	Logic		Enabled mode				External input signal key registration	Function
		Normally open	Normally closed	Manual	MDI	Memory	Editing *1		
7th-axis overtravel (+)	OVTR7P	○		○	○	○	○	×	Inputs the 7th axis overtravel (+). When this signal turns on, the machine will output the <<*Overtravel (+7)>> error.
7th-axis overtravel (-)	OVTR7M	○		○	○	○	○	×	Inputs the 7th axis overtravel (-). When this signal turns on, the machine will output the <<*Overtravel (-7)>> error.
8th-axis overtravel (+)	OVTR8P	○		○	○	○	○	×	Inputs the 8th axis overtravel (+). When this signal turns on, the machine will output the <<*Overtravel (+8)>> error.
8th-axis overtravel (-)	OVTR8M	○		○	○	○	○	×	Inputs the 8th axis overtravel (-). When this signal turns on, the machine will output the <<*Overtravel (-8)>> error.
PLC input X000 : PLC input X3FF	PLCX000 : PLCX3FF	○		○	○	○	○	○	Turn this signal on to turn on the PLC input X00~X3FF signal.
Reservation	RESERV								Already set the poart by the machine parameter when left the factory to avoid setting of duplication. (This is not signal, as same as unset.))

1. NC language only

Signal name	Symbol	Logic		Enabled mode				External input signal key registration	Function
		Normally open	Normally closed	MANU	MDI	Memory	Edit*1		
External 1st reference point return	EXREF	○				○		×	The machine can be returned to the 1st~6th reference point set by the parameter, by opening and then closing the contact of the EXREF, EXREF2~EXREF6. (Note 1) Axis movement is done at positioning feed. (Note 2) The movement order of axes is Z and then X and Y (simultaneous). (Note 3) The travel order on the additional axis follows the order set in the machine parameters <4th- to 8th-axis zero return order>. (Note 4) Axes do not move to the middle point. (Note 5) After normal completion of 1st ~ 6th reference point return, REFIN/REFIN2~REFIN6 signal is output. (Note 6) Input signal for 100 msec or more. (Note 7) During execution of external 1st to 6th reference point return, Automatic operation starting (STL) signal is not turned on. (Note 8) When the machine is equipped with turning spindle and the turning spindle is currently selected, reference point return is not made for the additional axis that serves as the turning spindle. <u>Effective conditions</u> (1) Automatic operation ongoing signal (STL) is off. (2) The [FEED HOLD] switch is not pressed. (3) The external stop signal (EXSTOP) is off. (4) An alarm higher than stop level 2 has not occurred. (5) The door is closed. (6) The screen <Move start position> is not currently displaying.
External 2nd reference point return	EXREF2								
External 3rd reference point return	EXREF3								
External 4th reference point return	EXREF4								
External 5th reference point return	EXREF5								
External 6th reference point return	EXREF6								
External optional skip	OPSKIP	○			○	○		×	When this signal is ON, memory operation is skipped at the block which contains slash (/) code.
External optional stop	OPSTOP	○				○		×	When this signal is ON, memory operation is temporary stopped at the execution of M01.
Shifting external workpiece coordinates	EXWORK	○			○	○		×	As long as this signal is on, the workpiece coordinates zero point (external) setting is valid.
Permit editing workpiece coord. zero	WPEDOK	○		○	○	○	○	○	Edition of workpiece coordinate zero is enabled as long as this signal is ON, even if Prevent editing (EDLOCK) is ON.
Permit editing macro data	MCEDOK	○		○	○	○	○	○	Edition of macro data is enabled as long as this signal is ON, even if Prevent editing (EDLOCK) is ON.
Interrupt macro	UINT	○				○		×	When the interrupt macro (UINT) signal is turned on during program execution, the execution of the current program is suspended and the program of the commanded number is executed.

Signal name	Symbol	Logic		Enabled mode				External input signal key registration	Function
		Normally open	Normally closed	MANU	MDI	Memory	Edit*1		
Macro variables #1000 : #1031	#1000 : #1031	○			○	○		○	Retains values for macro variables #1000 to #1031 and #1200 to #1295 internally. The ON status is treated as “1” while the OFF status as “0.”
Macro variables #1200 : #1295	#1200 : #1295								

2. Conversation only

Signal name	Symbol	Logic		Enabled mode				External input signal registration	Function
		Normally open	Normally closed	Manual	MDI	Memory	Editing*1		
External 2nd zero return External 3rd zero return	EXORG2 EXORG3	○				○		×	The machine can be returned to its 2nd or 3rd zero point set with the parameter by opening and then closing the contact of the signal EXORG2 or EXORG3. Completion of zero return can be checked by signal ORGFN2 or ORGFN3. (Reference) For details, refer to the section of completion of 2nd zero return. (Note 1) Travel occurs at the positioning speed. (Note 2) The travel order is Z → XY (simultaneous). (Note 3) The travel order for the additional axis follows the order set in the machine parameters <4th- to 8th-axis zero return order>. (Note 4) The signals EXORG2 and EXORG3 are enabled when the signal STL is OFF. (Note 5) During external 2nd or 3rd zero return, the signal STL is not output. (Note 6) Under the edit-during- operation mode, input the signal for 500 msec or more. (Note 7) The door is closed.
Prevent editing tool pattern	TPEDOK	○		○	○	○	○	○	Edition of tool pattern is enabled as long as this signal is ON, even if Prevent editing (EDLOCK) is ON.
Prevent editing tap prepared hole drill diameter	THEDOK	○		○	○	○	○	○	Edition of tap prepared hole drill diameter is enabled as long as this signal is ON, even if Prevent editing (EDLOCK) is ON.
Prevent editing cutting conditions	CTEDOK	○		○	○	○	○	○	Edition of cutting conditions is enabled as long as this signal is ON, even if Prevent editing (EDLOCK) is ON.

2.6.3.2 Detailed Description of External Input Signals

External program selection

Described in this section is how to use external program selection signals and cautions on it.

1. How to Use

Under the memory operation mode, input signals to PRO1, PRO2, PRO4, PRO8, PRO16, PRO32, PRO64 and PR9900.

Then inputting the start signal causes the NC to read the signals and execute the program.

Program number	PR9900	PR064	PR032	PR016	PR008	PR004	PR002	PR001
Program number entered last	×	×	×	×	×	×	×	×
0001	×	×	×	×	×	×	×	○
0002	×	×	×	×	×	×	○	×
0003	×	×	×	×	×	×	○	○
0004	×	×	×	×	×	○	×	×
0005	×	×	×	×	×	○	×	○
0006	×	×	×	×	×	○	○	×
0007	×	×	×	×	×	○	○	○
0008	×	×	×	×	○	×	×	×
0009	×	×	×	×	○	×	×	○
0010	×	×	×	×	○	×	○	×
0011	×	×	×	×	○	×	○	○
0012	×	×	×	×	○	○	×	×
0013	×	×	×	×	○	○	×	○
0014	×	×	×	×	○	○	○	×
0015	×	×	×	×	○	○	○	○
0016	×	×	×	○	×	×	×	×
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
0032	×	×	○	×	×	×	×	×
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
0064	×	○	×	×	×	×	×	×
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
0099	×	○	○	×	×	×	○	○
0100	×	○	○	×	×	○	×	×
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
0127	×	○	○	○	○	○	○	○

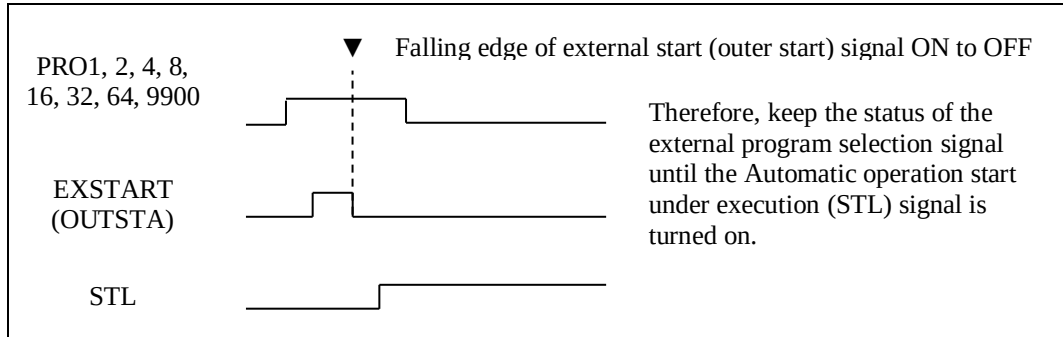
○: Contact closed ×: Contact open

Program number	PR9900	PR064	PR032	PR016	PR008	PR004	PR002	PR001
Program number entered last	○	×	×	×	×	×	×	×
9901	○	×	×	×	×	×	×	○
9902	○	×	×	×	×	×	○	×
9903	○	×	×	×	×	×	○	○
9904	○	×	×	×	×	○	×	×
9905	○	×	×	×	×	○	×	○
9906	○	×	×	×	×	○	○	×
9907	○	×	×	×	×	○	○	○
9908	○	×	×	×	○	×	×	×
9909	○	×	×	×	○	×	×	○
9910	○	×	×	×	○	×	○	×
9911	○	×	×	×	○	×	○	○
9912	○	×	×	×	○	○	×	×
9913	○	×	×	×	○	○	×	○
9914	○	×	×	×	○	○	○	×
9915	○	×	×	×	○	○	○	○
9916	○	×	×	○	×	×	×	×
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
9932	○	×	○	×	×	×	×	×
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
9964	○	○	×	×	×	×	×	×
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•
9999	○	○	○	×	×	×	○	○
External program selection number error	○	○	○	×	×	○	×	×
	•	•	•	•	•	•	•	•
	•	•	•	•	•	•	•	•
	•	•	•	•	•	•	•	•
External program selection number error	○	○	○	○	○	○	○	○
	○	○	○	○	○	○	○	○
	○	○	○	○	○	○	○	○

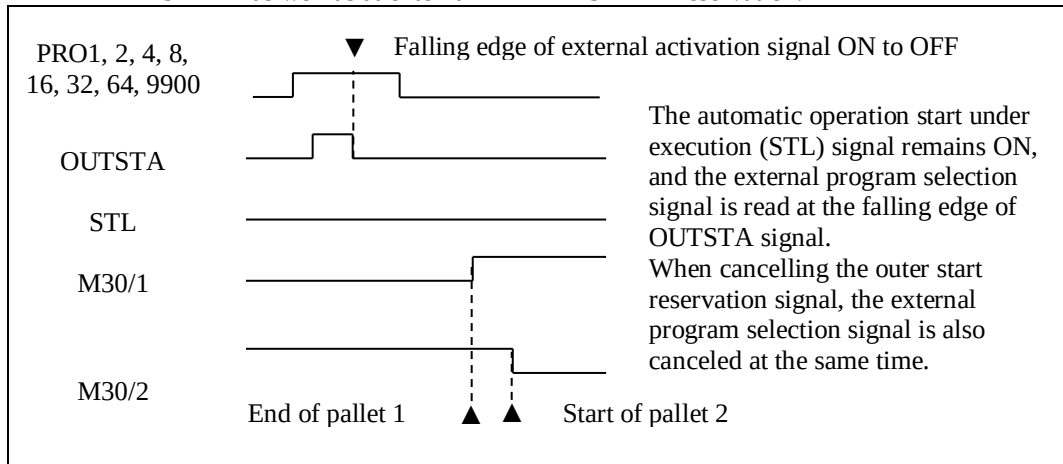
○: Contact closed ×: Contact open

2. Cautions

- (1) Reading of signals is started by the start signal in the operation reset status (with neither start lamp nor stop lamp lit) or by the PALLET START signal with the PALLET START reservation lamp not lit.
- (2) External program selection signals (PRO1, PRO2, PRO4, PRO8, PRO16, PRO32, PRO64) are input to the NC according to the following timing chart:
The same goes at start as well as at external start, PALLET START, and external PALLET START.



3. The external program selection signal is read at the falling edge of OUTSTA signal at PALLET START as well as at external PALLET START reservation.



When the start reservation signal is canceled, the external program selection signal having read is also canceled.

4. Input the external start (EXSTART) signal for 100 msec or more.
Effective conditions
Memory operation is in reset status. (Except block stop and halt)
5. When an operating program number is issued multiple times, the program number is listed as follows: (1)>(2)-1>(2)-2>(2)-3>(3)
 - (1) The program number specified through communication
 - (2)-1 Program number entered in the external input signal <External program selection>
 - 2 Program number entered in the internal PLC output data <External program selection No.>
 - 3 Program number entered in the internal PLC output signal <External program selection>
 - (3) The program number specified during memory operation or the <Pallet 1(2) program No.> in the User parameters.
6. PR9900 signal is valid only in conversation mode.
(This is for conversation schedule programs.)
7. External program selection numbers, or programs that can be selected using <External program selection No.> are only those programs that are located in the root directory for the NC (those contained in the root folder in the memory card in the case of tape operation).
When a program number in a sub folder has been commanded, the program of the same program number in the root folder is selected, or an alarm <<No program>> is generated if there is no program of the same program number in the root folder.

Input signals accompanying axis movement

For the following input signals accompanying axis movement, input is accepted only during operation stop or single stop.

When the following input signal is input during single stop in memory operation, the memory operation is reset.

EXORG

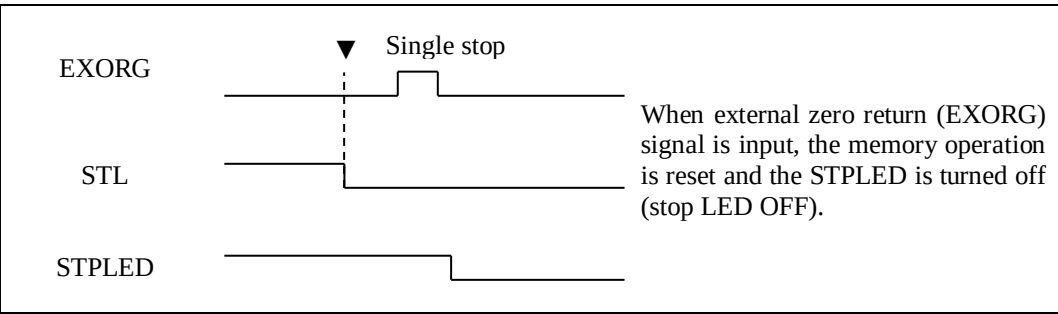
EXZORG

EXREF

CTURN

EXREF2~6 (NC language only)

EXORG2,3 (Conversation only)



External input signal key registration

Inserting the memory card that has been created as external input signal keys in the USB port, the designated external input signal is turned on.

2.6.4 External Output Signals

2.6.4.1 List of Output Signals

*1 signifies program editing.

Signal name	Symbol	Output controlled modes				Function
		Manual	MDI	Memory	Edit*1	
Program stop	M00			○		This signal is activated when a 'Program stop' is executed during memory operation and is deactivated in response to a start or reset signal. This signal is deactivated in response to the a start or reset signal. (Note) The signal undergoes no change in case an alarm occurs.
Program end	M30/1 M30/2			○		This signal is activated when a 'Program end' is executed during memory operation. • Turns on upon execution of program end for pallet 1. • Turns on upon execution of program end for pallet 2. Turns off by Start signal or resetting. (Note) For the machine not equipped with QT, only M30/1 is output. For the machine equipped with QT, M30/1 is output regardless of the pallet when the [PALLET] key is "OFF."
Coolant operation running	M08		○	○		Indicates that the coolant is being circulated.
M function output	M11 M12 M14 M18 M21 M22 M24 M28 MF M900		○	○		Sends the 2-digit decimal number BCD code to a peripheral by specifying any of signal outputs M01 to M99 (during MDI or memory operation. For M901 to M999, the M900 signal turns on at the same time. The last two digits are the same as the above. (Note) Even if the <Multiple M codes in one block> is set to <1: Yes> in the User parameters, only one M function signal output can be specified in one block. It cannot be specified together with PLC output.

Signal name	Symbol	Output controlled modes				Function																														
		Manual	MDI	Memory	Edit*1																															
External signal output	M400		○	○		Can be turned on and off according to the program during MDI or memory operation. (Example) Turns on when the output signal 402 is executed in the program and turns off when the signal output 403 is executed. Unlike M11 to M28, this signal does not check the M function complete signal MFIN.																														
	M402																																			
	M404																																			
	M406																																			
	M408																																			
	M480																																			
	M482																																			
	M484																																			
	M486																																			
						<table><tr><th>Signal name</th><th>ON</th><th>OFF</th></tr><tr><td>M400</td><td>M400 is commanded in program.</td><td>“Chip shower retention time” has passed since M401 was commanded in program.</td></tr><tr><td>M402</td><td>M402 is commanded in program.</td><td>M403 is commanded in program.</td></tr><tr><td>M404</td><td>M404 is commanded in program.</td><td>M405 is commanded in program.</td></tr><tr><td>M406</td><td>M406 is commanded in program.</td><td>M407 is commanded in program.</td></tr><tr><td>M408</td><td>M408 is commanded in program.</td><td>M409 is commanded in program.</td></tr><tr><td>M480</td><td>M480 is commanded in program.</td><td>M481 is commanded in program.</td></tr><tr><td>M482</td><td>M482 is commanded in program.</td><td>M483 is commanded in program.</td></tr><tr><td>M484</td><td>M484 is commanded in program.</td><td>M485 is commanded in program.</td></tr><tr><td>M486</td><td>M486 is commanded in program.</td><td>M487 is commanded in program.</td></tr></table>	Signal name	ON	OFF	M400	M400 is commanded in program.	“Chip shower retention time” has passed since M401 was commanded in program.	M402	M402 is commanded in program.	M403 is commanded in program.	M404	M404 is commanded in program.	M405 is commanded in program.	M406	M406 is commanded in program.	M407 is commanded in program.	M408	M408 is commanded in program.	M409 is commanded in program.	M480	M480 is commanded in program.	M481 is commanded in program.	M482	M482 is commanded in program.	M483 is commanded in program.	M484	M484 is commanded in program.	M485 is commanded in program.	M486	M486 is commanded in program.	M487 is commanded in program.
Signal name	ON	OFF																																		
M400	M400 is commanded in program.	“Chip shower retention time” has passed since M401 was commanded in program.																																		
M402	M402 is commanded in program.	M403 is commanded in program.																																		
M404	M404 is commanded in program.	M405 is commanded in program.																																		
M406	M406 is commanded in program.	M407 is commanded in program.																																		
M408	M408 is commanded in program.	M409 is commanded in program.																																		
M480	M480 is commanded in program.	M481 is commanded in program.																																		
M482	M482 is commanded in program.	M483 is commanded in program.																																		
M484	M484 is commanded in program.	M485 is commanded in program.																																		
M486	M486 is commanded in program.	M487 is commanded in program.																																		
						(Note 1) Turns OFF at input of reset signal. (Note 2) Does not turn OFF at the generation of an alarm.																														
One shot output M450 M451	M450		○	○		This signal is output by one shot. Set the output time by the <One shot signal M450 (M451) output time> in the User parameters. Ends the signal output and proceeds to the next operation.																														
	M451																																			
						<table><tr><td>M450</td><td>Next operation</td></tr></table> One shot signal M450 output time 1. Turns off upon input of reset signal. 2. Does not turn off on the generation of an alarm.	M450	Next operation																												
M450	Next operation																																			
One shot output M455 M456	M455		○	○		This signal is output by one shot. Set the output time by the <One shot signal M455 (M456) Output time > in the User parameters. As soon as the signal output is started, proceeds to the next operation.																														
	M456																																			
						<table><tr><td>M455</td><td>Next operation</td></tr></table> One shot signal M455 output time 1. Turns off upon input of reset signal. 2. Does not turn off on the generation of an alarm.	M455	Next operation																												
M455	Next operation																																			

Signal name	Symbol	Output controlled modes				Function
		Manual	MDI	Memory	Edit*1	
Automatic operation running	AUTO		○	○		In MDI operation or memory operation, Condition for ON: Initial start input Conditions for OFF: 1. Resetting 2. An alarm at stop level 5 (servo OFF). (Note) Does not turn on upon external zero return.
Automatic operation starting	STL		○	○		During MDI or memory operation: On conditions: During an operation Off conditions: 1 End of a motion 2 Single block stop 3 Stop 4 Reset 5 Alarm (Note) Does not turn ON upon external zero return.
Memory operation enabled	MEMOK			○		This signal turns ON when memory operation mode is enabled and is ready to start memory operation. Turns OFF in the following situations. • When [SINGL], [DRY] and [M.LCK] keys are ON • While reset is processing • When a door is open • When one of the following alarms or operator messages are triggered Alarms with a stop level 2 or higher <<Relocation was detected>> <<Reset the relocation detection status>> <<Execute automatic oiling / greasing cycle ON command>> • Travel screen is displayed for start position at program restart
Memory operation mode	MEMMOD			○		This signal turns on when you switch to memory operation mode or edit mode during operation. This signal is turns off in any other mode.
Extension memory running	EXPRUN		○	○		Turns ON when the program that is displayed in memory operation mode is for an extended memory operation.
NC ready	NCOK	○	○	○	○	This signal is turned ON when the NC unit is ready for any operation such as axis movement. The signal turns OFF on the occurrence of an alarm that causes the servo control to be turned OFF. Turns ON when the NC preparations are complete and all operations such as axis travel are OK and ready to be received. It turns OFF when an alarm is triggered that turns OFF the servo. (Example of an alarm when the servo is turned OFF) • Stop level 5 alarm (such as <<*axis servo error>>) • <<*Emergency button activated>> (When the [EMERGENCY] switch is pressed) • <<*Door open>> (When the front or side door is open and the [DOOR INTERLOCK] switch is enabled) • <<*Standby mode>> (When in standby mode)

Signal name	Symbol	Output controlled modes				Function															
		Manual	MDI	Memory	Edit*1																
Abnormal output	ALM	○	○	○	○	Outputs when a stop level 5 (servo OFF) or a cancel level 3 (power OFF required) alarm is triggered. However, the following alarms are excluded. • <<*Door open>> • <<*Standby mode>> • <<*Enter your password.>> • <<*[ENABLE] switch was turned OFF.>> (Triggered during a teaching pendant (option) operation) • <<*[MASTER ON] was turned OFF.>> Remove the cause and reset to restore operation. Note that the ALM signal does not output when the power is OFF because the transistor outputs.															
Abnormal output 2	ALM2	○	○	○	○	This signal turns ON on the generation of an alarm at a stop level beyond the preset level set in <ALM2 signal stop level> in the User parameters, excluding the following alarms: • <<*Door open>> • <<*Standby mode>> • <<*Enter your password>> • <<*[ENABLE] switch was turned OFF.>> (Triggered during a teaching pendant (option) operation) • <<*[MASTER ON] was turned OFF.>> • <<*[Handle operation switch OFF]>>															
Alarm level	ALMLV1 ALMLV2	○	○	○	○	<table><tr><td>ALMLV1</td><td>ALMLV2</td><td>Alarm status</td></tr><tr><td>OFF</td><td>OFF</td><td>No error</td></tr><tr><td>ON</td><td>OFF</td><td>Level causing stop at the end of the cycle</td></tr><tr><td>OFF</td><td>ON</td><td>Level causing stop at the end of the block</td></tr><tr><td>ON</td><td>ON</td><td>Level causing pause or higher</td></tr></table>	ALMLV1	ALMLV2	Alarm status	OFF	OFF	No error	ON	OFF	Level causing stop at the end of the cycle	OFF	ON	Level causing stop at the end of the block	ON	ON	Level causing pause or higher
ALMLV1	ALMLV2	Alarm status																			
OFF	OFF	No error																			
ON	OFF	Level causing stop at the end of the cycle																			
OFF	ON	Level causing stop at the end of the block																			
ON	ON	Level causing pause or higher																			
Alarm stop level 1 Alarm stop level 2 Alarm stop level 3 Alarm stop level 4 Alarm stop level 5	ASTLV1 ASTLV2 ASTLV3 ASTLV4 ASTLV5	○	○	○	○	The signal at the relevant level turns ON on the generation of an alarm at any level. When alarms are generated at multiple levels, the signals at the relevant levels turn ON.															

Signal name	Symbol	Output controlled modes				Function
		Manual	MDI	Memory	Edit*1	
Tool abnormality output	TOOL			○		This signal turns ON on the occurrence of a <<Tool breakage error>> in tool breakage detection or when the end of the life of the tool used in the current program has been reached and the current program is ended. (It does not turn ON when a spare tool is registered.) To release the signal and restart, perform one of the followings. (1) Register a spare tool then press the [RST] key. (2) Reset the tool life then press the [RST] key. (Note 1)The signal is momentarily deactivated if the [RST] key is pressed without registering a spare tool or resetting the tool life. M120(M121) When the measurement signal (TOUCH) is OFF (ON) in the execution of M120 (M121), a <<Tool breakage error>> is generated and the TOOL signal is turned ON. (Note 2)The behavior upon occurrence of a <<Tool breakage error>> takes place as described above, regardless of the value for the <Tool breakage detection option> in the “User parameters”.
Tool life end warning	TOLPRE			○		Turns on if the service life of the tool used for operation has reached the estimated service life.ON timing is the same as for TOOL signal.
Zero point return completion	ORGFIN	○	○	○	○	This signal is always activated.
Spindle rotation	SPTURN	○	○	○	○	This signal is active while the spindle rotates and is inactive while it is in stop.The signal, however, is not active during an oriented spindle stop and a tapping.
Zero position output	ZPX ZPY ZPZ ZPSP ZP4 ZP5 ZP6 ZP7 ZP8	○	○	○	○	Upon completion of zero return, the ZPX, ZPY, ZPZ, ZP4, ZP5, ZP6, ZP7, and ZP8 signals turn ON. Independent signals are provided for each axis. Then as the machine moves from the machine zero, these signals turn OFF, and they turn ON again when the machine returns to the machine origin again. The ZPSP signal turns ON upon completion of spindle orientation and turns OFF by spindle rotation, the reset signal, or [SP.STOP] key. (Note 1)They turn OFF on occurrence of an alarm at stop level 5 (servo off level). (Note 2)For the machine equipped with QT, the ZP4 signal turns ON when the 4th axis machine coordinate is “0.000.” It does not turn ON at any indexed position.

Signal name	Symbol	Output controlled modes				Function
		Manual	MDI	Memory	Edit*1	
Pallet indexing complete	PFIN1 PFIN2	○	○	○	○	This signal turns ON as follows: PFIN1: When the pallet 1 is indexed as outer pallet PFIN2: When the pallet 2 is indexed as outer pallet If you move the pallet from the index point, the corresponding signal will go off. Returning the pallet to the index point will turn the signal on. (Note) The signal will be turned off, if the servo control will be off such as an emergency stop. Always output "OFF" for QT uninstalled mashine.
Output during resetting	RSTOUT	○	○	○		Outputs as one shot (300 msec. or while reset lamp is lit, whichever is longer) when the reset LED is lit.
Tool breakage alarm	TLBRKE		○	○		This signal turns ON on the occurrence of a <<Tool breakage error>> and OFF when the error is reset.
Operation stop alarm	RUNSTP	○	○	○	○	Turns ON when an alarm is triggered that stops operation. It turns OFF when the alarm is cleared and reset.
Battery alarm	BATALM	○	○	○	○	The signal turns ON when the battery for maintaining the absolute encoder data is low.
Maintenance request	MNTALM	○	○	○	○	This signal turns on when the status of maintenance notice is alarm.
Green lamp	GRN		○	○		Outputs the signal with the same condition as the green lamp. Condition for ON: STL = ON Condition for OFF: Other than the conditions for ON above
Yellow lamp	YEL	○	○	○	○	Outputs the signal with the same condition as yellow lamp. Condition for ON: Any of the following conditions is satisfied: - M30/1 or M30/2 is ON. - The <Count end lamp status> is set to <Yellow> in the User parameters and the workpiece counter has ended. Condition for OFF: Other than the condition for ON above
Red lamp	RED	○	○	○	○	The signal in the same status as that of the red lamp is output. Condition for ON: Any of the following conditions is satisfied: - ALM signal is ON. <Count end lamp status> is set to <Red> in the User parameters and the workpiece counter has ended. Condition for OFF: Other than the condition for ON above
Workpiece counter end	CNTUP	○	○	○	○	Condition for ON: Workpiece counter end status Condition for OFF: Other than the condition for ON above
Workpiece counter end warning	CNTPRE	○	○	○	○	Condition for ON: Workpiece counter end warning status Condition for OFF: Other than the condition for ON above
Stop	STPOUT	○	○	○	○	Equivalent to the [FEED HOLD] switch input
Stopping	STPLED	○	○	○	○	Indicates that the operation is currently stopped. Equivalent to the stop LED.

Signal name	Symbol	Output controlled modes				Function
		Manual	MDI	Memory	Edit*1	
Door (Outside door/front door) opening end	DROPEN	○	○	○	○	Indicates that the outside door (front door) is open.
Door (Outer door/front door) closing end	DRCLS	○	○	○	○	Indicates that the outer door (front door) is closed.
Inner door closing end	INDRCL	○	○	○	○	Indicates that the inner door is closed.
Side door closing end	SDDRCL	○	○	○	○	Indicates that the side door is closed.
Outside door 1 closing end	XDR1CL	○	○	○	○	Indicates that the outside door 1 is closed.
Outside door 2 closing end	XDR2CL	○	○	○	○	Indicates that the outside door 2 is closed.
Automatic door manual/automatic selector switch	ATDRSL	○	○	○	○	Outputs whether the automatic door is in the automatic status or the manual status is output. This signal turns ON in the automatic status and OFF in the manual status.
Automatic door READY	DRRDY	○	○	○	○	Indicates that the automatic door is READY. (Electric cylinder)
Automatic door opening operation complete	DROFIN	○	○	○	○	Indicates that the opening of the automatic door has completed. (Electric cylinder)
Automatic door closing operation complete	DRCFIN	○	○	○	○	Indicates that the closing of the automatic door has completed. (Electric cylinder)
Automatic door alarm 1	DRALM1	○	○	○	○	Shows the automatic door alarm (cancel level 3) status. (Electric cylinder type) When the <Automatic door type> in the machine parameter (system 1) is set to <1: Electric cylinder type 1> → OFF: Alarm not triggered ON: Alarm triggered When the <Automatic door type> in the machine parameter (system 1) is set to <2: Electric cylinder type 2> → OFF: Alarm triggered ON: Alarm not triggered
Automatic door alarm 2	DRALM2	○	○	○	○	Shows the automatic door alarm (cancel level 2) status. (Electric cylinder type) When the <Automatic door type> in the machine parameter (system 1) is set to <1: Electric cylinder type 1> → OFF: Alarm not triggered ON: Alarm triggered When the <Automatic door type> in the machine parameter (system 1) is set to <2: Electric cylinder type 2> → OFF: Alarm triggered ON: Alarm not triggered
PALLET START LED	RDYLED			○		Indicates that start is reserved.
Dry run	DRYRUN	○	○	○	○	Indicates that the machine is in dry run mode.
Single operation	SINGL	○	○	○	○	Indicates that the machine is in single operation.
Restart	RESTAT	○	○	○	○	Indicates that the machine is in restart.
Rapid feed override	RPD100	○	○	○	○	Indicates that the machine is in rapid feed override state. ON: 100% OFF: Other than 100%

Signal name	Symbol	Output controlled modes				Function		
		Manual	MDI	Memory	Edit*1			
Spindle override	SPN100	○	○	○	○	Indicates that the machine is in spindle override state. ON: 100% OFF: Other than 100%		
Cutting feed override	FED100	○	○	○	○	Indicates that the machine is in cutting feed override state. ON: 100% OFF: Other than 100%		
Release key kept pressed	RELESE	○	○	○	○	Indicates that the [RELSE] key is kept pressed.		
Door lock enabled	SAFETY	○	○	○	○	Indicates that the [DOOR INTERLOCK] switch is enabled.		
Coolant key	COOLSW	○	○	○	○	Indicates that [CLT.P] key is ON.		
Chip shower key	CHIPSW	○	○	○	○	Indicates that [CHP.F] key is ON.		
Handle OFF	PULOFF	○	○	○	○	Turns ON when axis operation is not possible using either the handle operation or the teaching pendant.		
						- The signal turns ON when the handle is not connected and when the teaching pendant is not connected. - When the handle or the teaching pendant is connected and the axis selection is OFF, the signal turns ON if the mode is not set to <Manual operation>. If the mode is set to <Manual operation>, the signal turns ON following the table below.		
						When an alarm is triggered that is stop level 2 or higher.	ON	
						While the [FEED HOLD] switch is being pressed down.	ON	
						External stop (EXSTOP) is ON.	ON	
PALLET key	QTSEL1 QTSEL2	○	○	○	○	Indicates: [PALLET] key.		
						[PALLET] key	QTSEL1	QTSEL2
						OFF	OFF	OFF
						1	ON	OFF
						2	OFF	ON
1←→2	ON	ON						
Waveform display notice signal	WAVDSP	○	○	○	○	This signal turns OFF as soon as the waveform display function is started. The signal turns ON when the condition for triggering the waveform display notice signal is satisfied.		
X-axis position signal	XPOSSW	○	○	○	○	Signal turns ON when the machine coordinates fall within the range (including the boundary value) indicated in user parameter <*-axis position signal output range (-)/(+)> or <*-axis position signal output range 3 (-)/(+)>. The signal turns OFF when the machine coordinate is outside the range. BY375 determines which parameter will be used.		
Y-axis position signal	YPOSSW	○	○	○	○			
Z-axis position signal	ZPOSSW	○	○	○	○			
4th axis position signal	4POSSW	○	○	○	○			
5th axis position signal	5POSSW	○	○	○	○			
6th axis position signal	6POSSW	○	○	○	○			
7th axis position signal	7POSSW	○	○	○	○			
8h axis position signal	8POSSW	○	○	○	○			

Signal name	Symbol	Output controlled modes				Function
		Manual	MDI	Memory	Edit*1	
Outer A-axis indexing under execution	OUTAON	○	○	○	○	This signal stays on while the outer pallet A-axis is being indexed as specified by a panel switch or input signal. (Note) By means of the setting for <External index A switch address switching> in the User parameters, this signal can be turned ON while the C-axis instead of A-axis is being indexed.
Outer B-axis indexing under execution	OUTBON	○	○	○	○	This signal stays on while the outer pallet A-axis is being indexed as specified by a panel switch or input signal. (Note) By means of the setting for <External index B switch address switching> in the User parameters, this signal can be turned ON while the C-axis instead of B-axis is being indexed.
5th axis unclamp 6th axis unclamp 7th axis unclamp 8th axis unclamp	UNCLP5 UNCLP6 UNCLP7 UNCLP8	○	○	○	○	These are unclamp output signals for the 5th to 8th axes. “ON” is output when unclamping and “OFF” is output when clamping.
P1-axis unclamp P2-axis unclamp P3-axis unclamp P4-axis unclamp	UCLPP1 UCLPP2 UCLPP3 UCLPP4	○	○	○	○	These are unclamp output signals for the P1 to P4 axes. “ON” is output when unclamping and “OFF” is output when clamping.
Indexing magazine number 1 Indexing magazine number 2 Indexing magazine number 4 Indexing magazine number 8 Indexing magazine number 16 Indexing magazine number 32	MGZ1 MGZ2 MGZ4 MGZ8 MGZ16 MGZ32	○	○	○	○	These signals output respective indexing magazine numbers (pot numbers). Example 1: When the indexing magazine number is “1” MGZ1 = ON MGZ2 = OFF MGZ4 = OFF MGZ8 = OFF MGZ16 = OFF MGZ32 = OFF Example 2: When the indexing magazine number is “21” MGZ1 = ON MGZ2 = OFF MGZ4 = ON MGZ8 = OFF MGZ16 = ON MGZ32 = OFF Example 3: When no magazine is indexed (=0) MGZ1 = OFF MGZ2 = OFF MGZ4 = OFF MGZ8 = OFF MGZ16 = OFF MGZ32 = OFF
Data protection switch	UNLOCK	○	○	○	○	Turns ON when the [DATA PROTECTION] switch is set to “Disabled”.
Spindle run-in operation required to be executed	SPIALM	○	○	○	○	Turns ON when spindle run-in operation is required to be executed.
Automatic door stop detection	DRHOLD	○	○	○	○	The signal turns ON when the automatic door stop is detected and the <Automatic door type> in the machine parameter (system 1) is set to <2: Electric cylinder type 2>. It is OFF for all other situations.
Automatic door machine zero return finished	DRORFN	○	○	○	○	The signal turns ON when the automatic door machine zero return is completed and the <Automatic door type> in the machine parameter (system 1) is set to <2: Electric cylinder type 2>. It is OFF for all other situations.
Battery alarm on relocation detection device	RDBTAL	○	○	○	○	The signal turns ON when the battery for the relocation detection device is low.

Signal name	Symbol	Output controlled modes				Function
		Manual	MDI	Memory	Edit*1	
[MASTER ON]	HWMO	○	○	○	○	Indicates the status of operation preparation. OFF: Operation preparation OFF ON: Operation preparation ON Indicates the same status as the LED for the operation preparation switch.
X-axis position signal 2	XPSSW2	○	○	○	○	Signal turns ON when the machine coordinates fall within the range (including the boundary value) indicated in user parameter <*-axis position signal output range 2 (-)/(+)> or <*-axis position signal output range 4 (-)/(+)>.
Y-axis position signal 2	YPSSW2	○	○	○	○	
Z-axis position signal 2	ZPSSW2	○	○	○	○	
4th axis position signal 2	4PSSW2	○	○	○	○	The signal turns OFF when the machine coordinate is outside the range. BY376 determines which parameter will be used.
5th axis position signal 2	5PSSW2	○	○	○	○	
6th axis position signal 2	6PSSW2	○	○	○	○	
7th axis position signal 2	7PSSW2	○	○	○	○	
8th axis position signal 2	8PSSW2	○	○	○	○	
Servo relay 1 contact ON	RYCHK1	○	○	○	○	Indicates the status of the servo relay 1 contact. OFF: Relay contact OFF ON: Relay contact ON
Machining load maximum	TLDHI	○	○	○	○	Turns ON when the machining load monitor function detects the alarm <<Machining load maximum>>. It turns OFF when the alarm is not detected or when the alarm is reset.
Machining load minimum	TLDLO	○	○	○	○	Turns ON when the machining load monitor function detects the alarm <<Machining load minimum>>. It turns OFF when the alarm is not detected or when the alarm is reset.
Machining load monitor ON	TLDON	○		○		Turns ON when the machining load monitor function is currently monitoring the machining load. It turns OFF when the function is not monitoring or when the monitoring has ended.
PLC output Y000 : PLC output Y3EF	PLCY000 : PLCY3FF	○	○	○	○	The values for PLC output Y000 to Y3EF signals are output. (Note) Even if the <Multiple M codes in one block> is set to <1: Yes> in the User parameters, only one PLC output signal can be specified in one block. It cannot be specified together with M function signal output.
Reservation	RESERV					Preset at shipment in order to avoid overlapping setting. (This signal is not treated as a signal but treated as no setting. When this signal is set, avoid using the relevant terminal of the terminal block set.)

1. NC only

Signal name	Symbol	Output controlled modes				Function
		Manual	MDI	Memory	Edit ^{*1}	
1st reference point return complete 2nd reference point return complete 3rd reference point return complete 4th reference point return complete 5th reference point return complete 6th reference point return complete	REFIN REFIN2 REFIN3 REFIN4 REFIN5 REFIN6		○	○		These signals turn ON when all the X, Y, Z, 4th, 5th, 6th, 7th, and 8th axes have returned to the 1st to 6th reference points by the execution of G30 command in MDI or memory operation or by external signals (EXREF, EXREF2 to EXREF6). Conditions for OFF: 1. When any one of X, Y, Z, 4th, 5th, 6th, 7th, and 8th axes is moved from the 1st to 6th reference points, or 2. When an alarm at stop level 5 is generated. (However, the signal remain ON even if the servo control is turned OFF, provided that the <Reference point return complete signal output at servo off> is set to <1: Yes> in the User parameters and the distance of the relevant axis from the relevant reference point is within the range set in the <Reference point allowable range at servo off>. (Note 1) The signal does not turn on when all axes return to their reference position in manual mode. (Note 2) When the turning spindle is currently selected, these signals turn ON when all the axes excluding the turning spindle have returned to the 1st to 6th reference points.
Turning spindle is rotating	LSTURN	○	○	○	○	There is a turning spindle and turns ON when turning spindle is rotating. It turns OFF when the turning spindle stops.
Select turning spindle	LSSLCT	○	○	○	○	Turn ON when there is a turning spindle and the turning spindle is selected.
Macro variables #1100 : #1131 Macro variables #1300 : #1395	#1100 : #1131 #1300 : #1395		○	○		These signals output the values for macro variables #1100 to #1131 and #1300 to #1395. “OFF” is output when set to “0” while “ON” is output when set to “1.”

2. Conversation only

Signal name	Symbol	Output controlled modes				Function
		Manual	MDI	Memory	Edit ^{*1}	
2nd zero return complete 3rd zero return complete	ORGFN2 ORGFN3		○	○		This signal is output when 2nd or 3rd zero point return is completed as specified by a program command or external signal. OFF 1. Any one of the axes is not at the 2nd or 3rd zero point. 2. The servo controller turns off due to an emergency stop or similar. (Note) The signal is not output when the axis moves after it has been positioned to the 2nd or 3rd zero point in manual mode or after the axis has been positioned to the 2nd or 3rd zero point by a positioning command. The signal is not output when the relevant axis has been positioned at or passed through the 2nd or 3rd zero point by a positioning command. The signal remains ON even if the servo control is turned OFF, provided that the <Zero point return complete signal output at servo off> is set to <1: Yes> in the User parameters and the distance of the relevant axis from the relevant reference point is within the range set in the <Zero point allowable range at servo off>.

2.6.4.2 Detailed Description of External Output Signals

M function signal output

Usage of the M function output signals and precautions are described below.

1. How to Use

2-digit number in BCD code is sent to a peripheral by specifying any of signal output 01 to 99 in MDI or memory operation mode.

This signal is output, and the signal MF is output after the <M function signal output delay time> set in the user parameter has elapsed. Read the M function signal output indicated by this signal MF.

Once the peripheral equipment operation finishes (based on the M function signal output), wait at least 20 ms before inputting the M function end signal MFIN.

Signal input in program	High-order digits					Low-order digits			
	Signal name					Signal name			
	M900	M28	M24	M22	M21	M18	M14	M12	M11
Signal output 01	×	×	×	×	×	×	×	×	○
Signal output 02	×	×	×	×	×	×	×	○	×
Signal output 03	×	×	×	×	×	×	×	○	○
Signal output 04	×	×	×	×	×	×	○	×	×
Signal output 05	×	×	×	×	×	×	○	×	○
Signal output 06	×	×	×	×	×	×	○	○	×
Signal output 07	×	×	×	×	×	×	○	○	○
Signal output 08	×	×	×	×	×	○	×	×	×
Signal output 09	×	×	×	×	×	○	×	×	○
Signal output 10	×	×	×	×	○	×	×	×	×
Signal output 11	×	×	×	×	○	×	×	×	○
•	•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•	•
Signal output 99	×	○	×	×	○	○	×	×	○
Signal output 901	○	×	×	×	×	×	×	×	○
•	•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•	•
Signal output 909	○	×	×	×	×	○	×	×	○
Signal output 910	○	×	×	×	○	×	×	×	×
Signal output 911	○	×	×	×	○	×	×	×	○
•	•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•	•
Signal output 999	○	○	×	×	○	○	×	×	○

○: ON

×: OFF

M function signals M00 to M99 and M901 to M999 can be created by the combination of high-order digits and low-order digits.

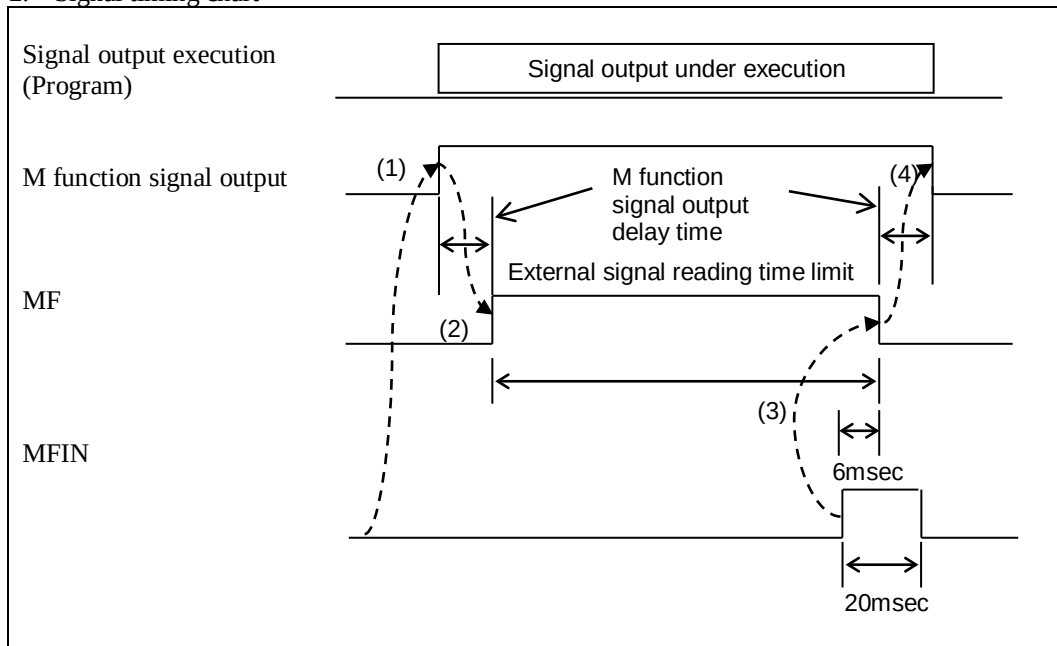
However, the following codes are for internal use and are not output externally.

Conversation: M00, M01, M02, M03, M04, M05, M06, M08, M09, M19, M30

NC language: M00, M01, M02, M03, M04, M05, M06, M08, M09, M19, M30, M96, M97, M98, M99

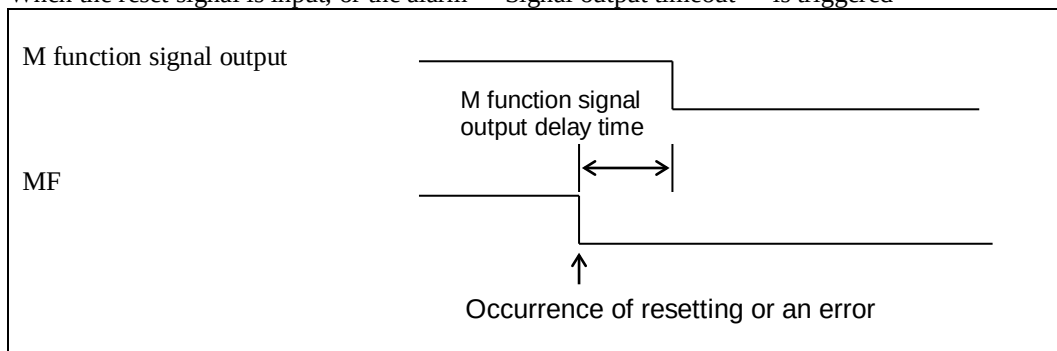
(Note) Even if the <Multiple M codes in one block> is set to <1: Yes> in the User parameters, only one M function signal output can be specified in one block. It cannot be specified together with PLC output.

2. Signal timing chart



- (1) The M function signal is output.
During this time, if the MFIN is input, the M function signal is output after the MFIN input stops.
- (2) The MF is output after the delay time has elapsed for the M function signal output.
- (3) If the MFIN signal is input within the setting specified per the <External signal reading time limit> in the user parameter, then the MF output stops.
- (4) The M function signal output stops after the delay time has elapsed for the M function signal output.

When the reset signal is input, or the alarm <<Signal output timeout>> is triggered



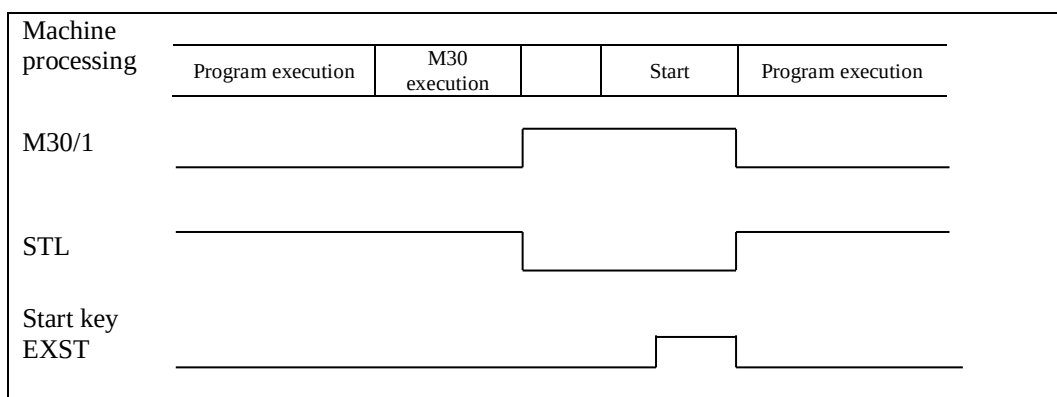
The MF output stops if the reset signal is input, or if the signal output timeout is triggered. The M function signal output stops after the delay time has elapsed for the M function signal output.

Program end, M30/1,M30/2

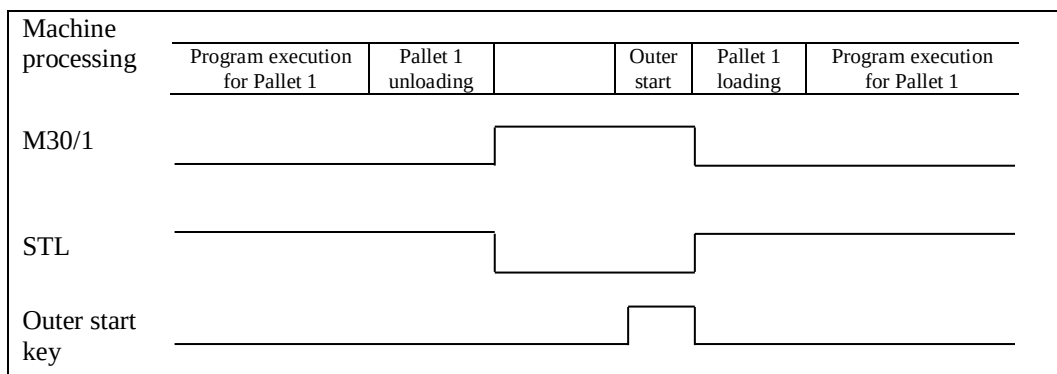
For the machine not equipped with QT, the program end is treated as the same as the **[PALLET]** key set to “OFF.”

1. When <Program repeat> is set to <0: Stop>

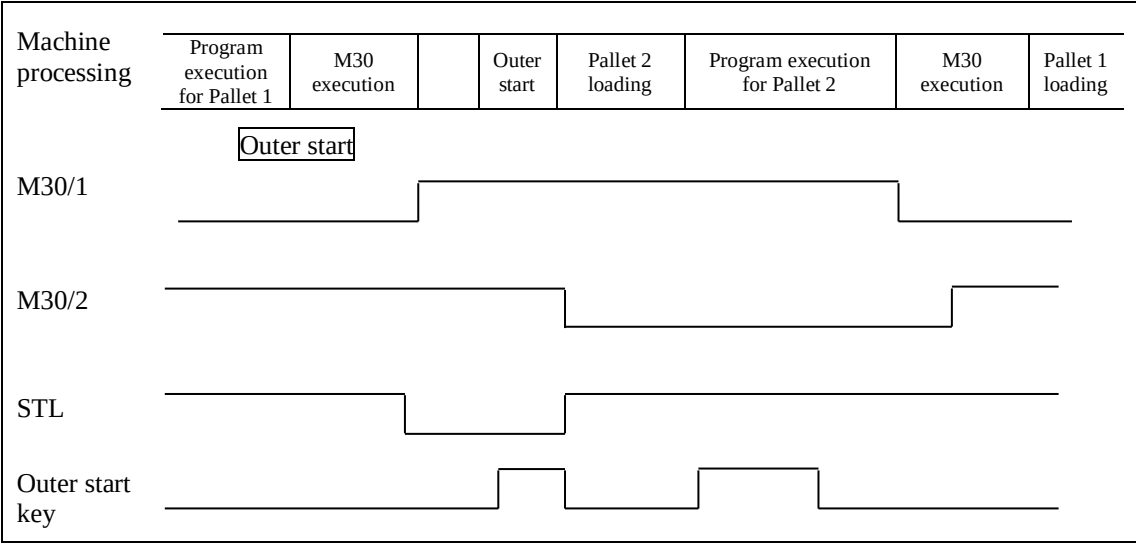
PALLET key	Off	1 or 2	1⇔2
Condition for ON	M30/1 goes on irrespective of the used pallet when you execute the program end.	M30/1 or M30/2 goes on when you execute the program end and take out the pallet.	M30/1 or M30/2 goes on when you execute the program end.
OFF	1. When you input the reset signal. 2. When a start input is accepted.	1. When you input the reset signal. 2. When an outside start input is accepted.	

(1) When **[PALLET]** key is set to “OFF.”

(Note 1) When EXST signal is input from an external unit, check that M30 signal is ON and STL signal OFF.

(2) When **[PALLET]** key is set to “1” or “2”.

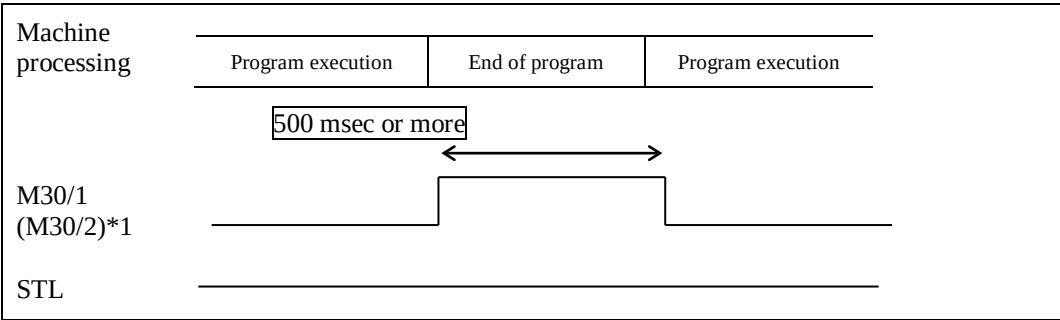
(3) When [PALLET] key is set to “1<->2”



2

2. When the machine is started by Start signal with the <Program repeat> set to <1: Repeat> When the [PALLET] key is set to other than “OFF,” the program for the pallet at the inner position at the start of the machine is repeated.
When the [PALLET] key is set to “OFF,” the program displayed on the screen is repeated.

Condition for ON When the end of program is executed during memory operation
Condition for OFF 1. When the machine is restarted upon completion of the program end processing
2. When the reset signal is input

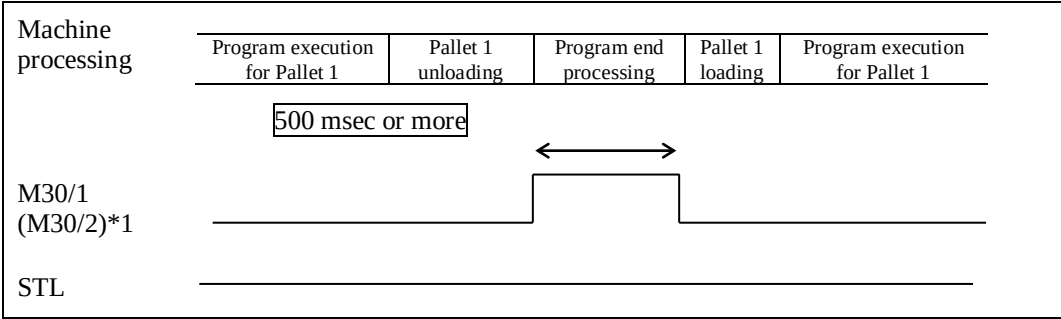


*1 M30/2 turns ON when the [PALLET] key is set to “1<->2” or “2” and the machine is started with the Pallet 2 at the inner position.

- (Note 1) The signal undergoes no change in case an alarm occurs.
(Note 2) The automatic door (optional) works only one time at start. Thereafter it will not work.
(Note 3) STL does not turn off even during M30/1 or M30/2 output.

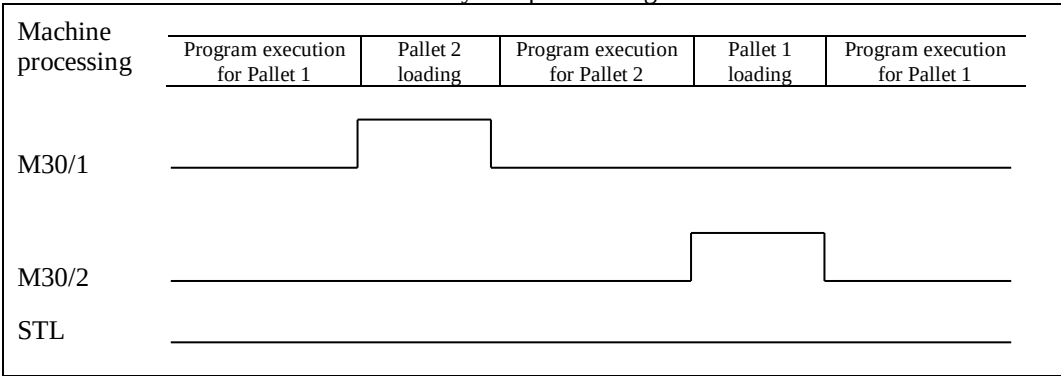
3. When the machine is started by PALLET STARTing with the <Program repeat> set to <1: Repeat>
(1) When [PALLET] key is set to “OFF.”
When [PALLET] key h is set to “OFF.”
Turning ON the PALLET START generates an alarm <<The pallet selection key is OFF>>.

- (2) When **[PALLET]** key is set to “1” or “2”.
- | | |
|-------------------|--|
| Condition for ON | When the program end is executed and the pallet unloading has completed during memory operation |
| Condition for OFF | 1. When the program end processing has completed and the operation is restarted
2. When you input the reset signal. |



- (Note 1) These signals do not change at the generation of an alarm.
- (Note 2) The optional automatic door operates just once at start-up and does not operate after that.
- (Note 3) The STL signal does not turn OFF during output of M30/1 or M30/2.

- (3) When the **[PALLET]** key h is set to “1<->2”
- | | |
|-------------------|---|
| Condition for ON | When the program end is executed during memory operation |
| Condition for OFF | 1. When the pallet on the other side is loaded and the program execution is started
2. When you input reset signal |



- (Note 1) These signals do not change at the generation of an alarm.
- (Note 2) The optional automatic door operates just once at start-up and does not operate after that.
- (Note 3) The STL signal does not turn OFF during output of M30/1 or M30/2.

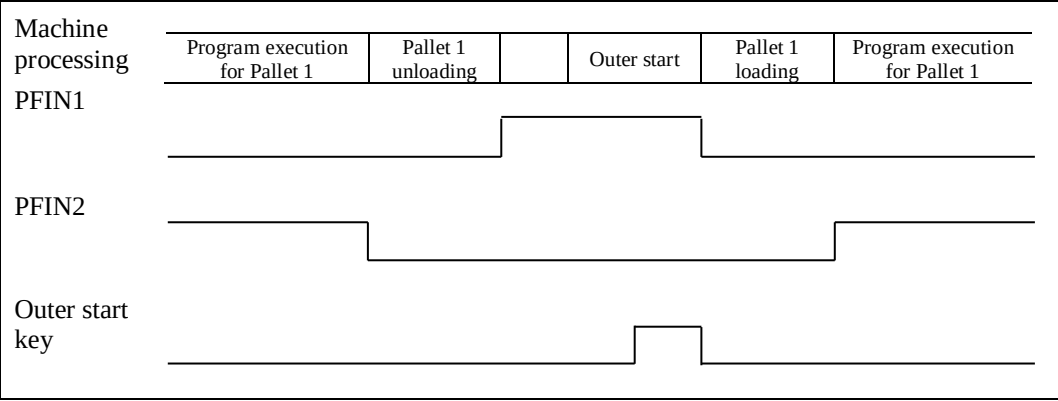
2.6.4.3 Pallet Indexing Complete PFIN1, PFIN2

These signals turn OFF when the pallet has moved from the indexed position and turn OFF when it has returned to the indexed position.

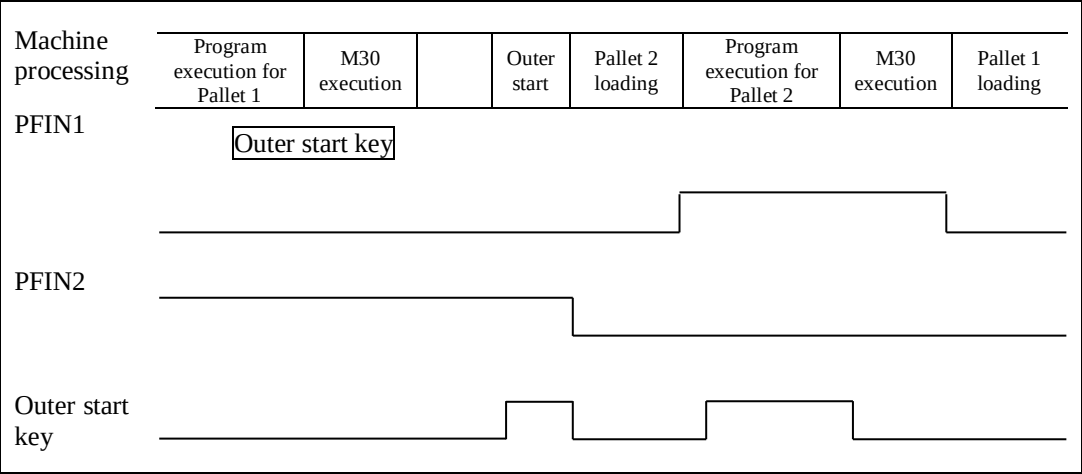
At program end or reset, or when an alarm is generated , they remain ON.

1. When <Program repeat> is set to <0: Stop>
(1) When the **[PALLET]** key is “OFF,” the PFIN1/PFIN2 signal for the pallet at the outer position always turns ON.

- (2) When the **[PALLET]** key is set to “1” (or “2”)

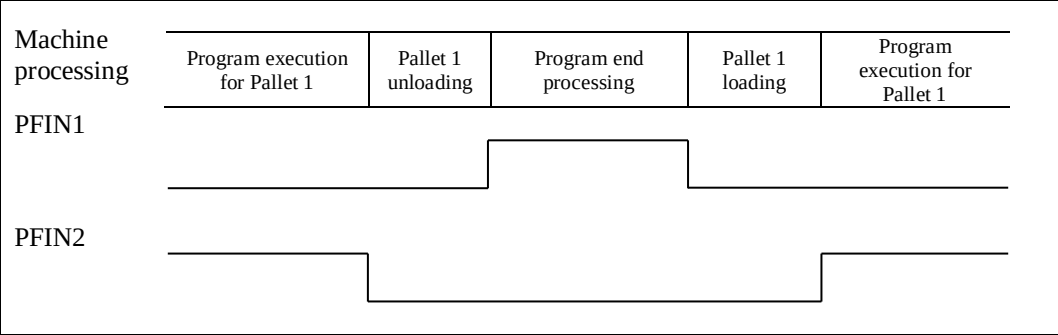


- (3) When the **[PALLET]** key is set to “1<->2”

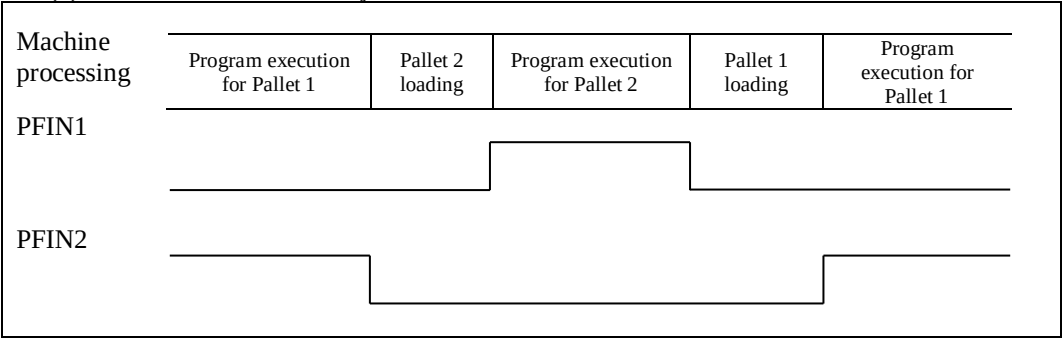


2. When the machine is started by Start signal with the <Program repeat> set to <1: Repeat>
The PFIN1/PFIN2 signal for the pallet at the outer position at start-up always turns ON.
3. When the machine has been started by PALLET START with the <Program repeat> set to <1: Repeat>
(1) When the **[PALLET]** key is “OFF,” the machine does not operate by PALLET START.
When the PALLET START is turned ON, an alarm <<The pallet selection key is OFF>> is generated.

- (2) When1” the **[PALLET]** key is set to “1” (or “2”)



(3) When the [PALLET] key is set to “1<->2”



2.7 Communication Parameter

2.7.1 Outline

Set parameters for external input/output communication. Make setting in accordance with the connected device.

Communication parameters are divided into the following categories.

Category name	Description
<Serial port>	Sets the communication parameters for serial port.
<Ethernet/FTP>	Sets the communication parameters for Ethernet and FTP.
<Memory card>	Sets the communication parameters for memory card.
<Auto notification function>	Set the parameter for the auto notification function.

2

2.7.1.1 Edit Screen

Selecting <Communication parameter> on the <Data bank menu> screen displays the communication parameter edit screen.

Displays the previously displayed category. Category can be switched over by pressing a function key.

2.7.2 How to Edit

2.7.2.1 Data Input/Output

1. Display the category that includes the item to be changed.
2. Locate the cursor at the item to be changed.
3. Enter a value in the input area and press the **[ENT]**.
This inputs the value at the cursor position and moves the cursor to the next item.

* For details of the items, see para. 2.7.3 or later.

2.7.2.2 Data Deletion

When the cursor is located at an item that is deletable, <Can be deleted> is displayed.

1. Display the category that includes the item to be deleted.
2. Locate the cursor at the item to be deleted.
3. Press the **[Delete]** key.
This deletes the value at the cursor position.

2.7.2.3 Ending Editing

Upon completion of the settings, press the **[F0]** (End mode) key to end the editing.

For how to end editing, refer to “2.1 General Outline of Data Bank”.

(Note) When communication is in progress, a reset may be requested when exiting editing depending on the item(s) that was changed. In this situation, press the **[RST]** key after the communication has finished in order to reflect the parameter change.

2.7.3 Common

No.	Items	Setting range	Description
	Reset (slave station)	0: Invalid 1: Valid	Set whether to terminate communication by the [RST] key when the machine is in slave communication mode. When 0:Invalid is set, communication is not terminated even if the [RST] key is pressed.
	Data overwriting (slave station)	0: No 1: Yes	Set whether to overwrite the same program or data existing in the machine when the machine is in slave communication mode (input). When 0:No is set, the data is not overwritten.
	Remote operation	0: Invalid 1: Valid	Set whether to enable or disable the remote operation via communication, such as zero return, memory operation start and stop.
	Slave station command alarm display	0: No 1: Yes	When “1: Yes” is set, notification alarm is generated when slave station communication has completed.
	Method for batch program input	0: Type1 1: Type 2	Designate the method for program division when performing batch input of NC program. type 1: When a program end code (M30, M02, M99) is found out, the part from the top to the block is input as a program. type2: When a program number (O****) is found out, the part from the top to the previous block is input as a program.
	File name added as comment	0: No 1: Yes	When “1: Yes” is set, original file name is added to the first block as a comment line if NC program other than that output from the machine (O****) is input.
	Comment input extension	0: No 1: Yes	When “1: Yes” is set, characters which are unavailable to the key operation of the machine can be used when inputting comments of program or character string of data bank. For details, refer to “5.5 Communication protocol.”
	External output - Output destination (NC language only)	0: Master stat. output destination 1: Memory card 2: FTP server	Specify the output destination for the macro (external output function). When outputting to a general communications device, set to <0: Master stat. output destination>, and set the master station connection to <General COMM device>.

2.7.4 Serial Port

No.	Items	Setting range	Description
	Connected device	10 of one-byte characters	Input name of the communication device connected to the serial port.
	NC-exclusive communication device communication protocol (master station)	0: Type 1 1: Type 2	Select protocol used for communication. Choose which protocol do you use from the one compatible with the conventional machine or expanded one available for remote control.
	Baud rate	0:300 1:600 2:1200 3:2400 4:4800 5:9600 6:19200 7:38400 8:57600	Set the baud rate for data transmission.
	Parity	0: None 1: Odd 2: Even	Set the type of parity bit to be added to one-character data.

No.	Items	Setting range	Description
	Stop bit	0 : 1 bits 1 : 2 bits 0: 1 bits 1: 2 bits	Set the distance, which discriminates the end of one-character data.
	Character length	0: 7 bits 1: 8 bits	Set the length of one-character data.
	Communication mode	0: Line 1: Code 1 2: Code 2	Select communication type among the followings. 0: Line (Control line system) Communication is controlled by using RS and CS. 1: Code 1 (Control code system 1) Communication is performed by control code system on the master station and control line system on the slave station. 2: Code 2 (Control code system 2) Communication is performed by control code system on both master and slave stations. Communication is controlled by using DC1 and DC3 codes. The head and end of data are suffixed with DC2 and DC4, respectively.
	Response monitoring time	0~999 sec 0 to 999 sec	Allowable time interval between the receipt of the one character and the next one. Also allowable time between the sending of one character and the next one. Setting to "0" results in infinite standby state.
	Invalid data (For NC language only)	0: Conv 1: Error 2: Ignor	Designate how to process data not supported by NC if it has been read. When "0: Conversion" is set, such data is converted into "?" and the program is displayed as "?" on the edit screen.
	Sending data code (For NC language only)	0:ISO 1:EIA 2:ASCII	Set the transfer code when transferring the data to the receiving station.
	End of block (For NC language only)	0:CR、LF 1:LF	Set the code for the end of one block (EOB). (When "Send data code" is set to "0: ISO code")
	TV check	0: None 1: Odd 2: Even	Set whether to perform a one block parity check.
	Output dot character	0: No 1: Yes	Designate whether or not to type out ornate initials during output when connection destination is a general communication device.
	Reset recovery time	0 to 99 sec	Turns ER and RS signals of machine OFF for the period of time set here, when any of the following conditions occurs: - When [RST] key is pressed during main station communication; - When [RST] key is pressed during machine slave station communication, when "Reset (slave station)" is set to "1: Valid"; - When an error occurs when communication has not been normally ended; or - When the operation has shifted to maintenance menu during machine slave station communication.
	Checksum	0: Invalid 1: Valid	Set whether or not to detect errors using the checksum in the communication of NC communication device protocol type 2.
	EIA character code (#) (For NC language only)	0~255	Set the EIA character code for character '#.' Set a code that is not identical to the existing EIA code.
	EIA character code (*) (For NC language only)	0~255	Set the EIA character code for character '*.' Set a code that is not identical to the existing EIA code.

No.	Items	Setting range	Description
	EIA character code (=) (For NC language only)	0~255	Set the EIA character code for character '='. Set a code that is not identical to the existing EIA code.
	EIA character code ([) (For NC language only)	0~255	Set the EIA character code for '[' character. Set a code that is not identical to the existing EIA code.
	EIA character code (]) (For NC language only)	0~255	Set the EIA character code for ']' character. Set a code that is not identical to the existing EIA code.
	Check DR signal	0: No 1: Yes	When "1: Yes" is set, DR signal is checked during communication. Communication is unavailable when DR signal is OFF. When "0: No" is set, check is not performed.
	Communication end DC3 code	0: None 1: Yes	When "1: Yes" is set, waits for transmission or reception of DC3 code at the end of communication in communication methods of codes 1 and 2. (When receiving data) On receiving all the external data, DC3 code is transmitted to external device. (When transmitting data) On transmitting all the data to external device, the system is ready to receive DC3 code. The communication terminates upon receiving DC3 code. When 0: None is set, this communication end DC3 code is not available.
	DC1 code DC2 code DC3 code DC4 code	0~255	Set control codes in communication methods of codes 1 and 2. Make setting by using decimal numbers. DC1: 17=11H (hexadecimal) DC2: 18=12H DC3: 147=93H DC4: 20=14H (Meaning of the DC code) DC1: Start of communication or restart of the communication interrupted by DC3. DC2: Indication of the head of the data. DC3: Temporary interruption of the communication in case of buffer overflow for example. DC4: Indication of the end of the data.
	Leading zero suppression (DPRNT) (For NC language only)	0: Type 1 1: Type 2	Set the output method for high-order zero at the time of macro variable output under DPRNT command. Type 1: Outputs space Type 2: Nothing is output
	External output - FTP output destination	255 one-byte characters	Specify the output directory on the FTP server for the macro (external output function).
	External output - FTP output name	32 one-byte characters	Specify the character string for the output file name on the FTP server for the macro (external output function).

2.7.5 Ethernet/FTP

When the network function is used, please consult with your network manager. In addition, be careful when configuring the IP address and other settings and be sure to test and check the communication settings. If the IP address or other settings are configured incorrectly, the communication failure or error can impact the entire network and cause a malfunction in other equipment. Therefore, take extra care when configuring the settings.

No.	Items	Setting range	Description
	Host name	15 of one-byte characters	Set a name for identifying the machine on the network. Usable characters are 'A' to 'Z' and '0' to '9'. Character string consisting only of numeric characters is unavailable. Character string headed or ended with '-' is also unavailable.
	Use DHCP	0: No 1: Yes	Set whether or not to use DHCP. When "1: Yes" is set, IP address is automatically acquired from DHCP server.
	IP address	0.0.0.0~ 255.255.255.255	Set IP address to be assigned to the machine. When "Use DHCP" is set to "1: Yes", it becomes invalid (set automatically). Enter the address in 12 figures without using "." (period). Example: To set "10.172.83.1", enter "010172083001". (Note) "0.0.0.0", "127.*.*" and "255.*.*" cannot be set.
	Mask bit	top 2 bit (192.0.0.0)~ top 30 bit (255.255.255.252)	Set the subnet mask of the network the machine belongs to. When "Use DHCP" is set to "1: Yes", it becomes invalid (set automatically). Enter the address in 12 figures without using "." (period). Example: To set "255.255.0.0", enter "255255000000".
	Default gateway	0.0.0.0~ 255.255.255.255	Set the IP address of the default gateway of network to which the machine belongs. When 0 is set, there is no default gate way. When "Use DHCP" is set to "1: Yes", it becomes invalid (set automatically). Enter the address in 12 figures without using "." (period). Example: To enter "10.172.83.1", enter "010172083001". (Note) "0.0.0.0", "127.*.*" and "255.*.*" cannot be set.
	Automatic acquisition of DNS server address	0: No 1: Yes	Automatically acquires an address of DNS server. When "1: Yes" is set, address of DNS server is acquired from DHCP server. When "Use DHCP" is set to "0: No", it becomes invalid (uses "DNS server address 1/2").
	DNS server address 1	0.0.0.0~ 255.255.255.255	Set the address of first DNS server. When "Automatic acquisition of DNS server address" is set to "1: Yes", it becomes invalid. Enter the address in 12 figures without using "." (period). Example: To set "10.172.83.1", enter "010172083001". (Note) "0.0.0.0", "127.*.*" and "255.*.*" cannot be set.

No.	Items	Setting range	Description
	DNS server address 2	0.0.0.0~ 255.255.255.255	Set the address of second DNS server. When “Automatic acquisition of DNS server address” is set to “1: Yes”, it becomes invalid. Enter the address in 12 figures without using “.” (period). Example: To set “10.172.83.1”, enter “010172083001”. (Note) “0.0.0.0”, “127.*.*.” and “255.*.*” cannot be set.
	Port number	10000~65535	Set a port number to be used for slave station communication by communication protocol type 2 of NC-dedicated communication device.
	PLC port No.	10000~65535	Set a port number to be used when communicating PLC data in slave station communication based on PLC PROGRAMMING SOFTWARE.
	FTP server name (IP address)	255 of one-byte characters	Set the host name or IP address of FTP server to be connected with the machine (FTP client) when connection destination is the FTP server. To designate IP address, enter the address in 12 figures without using “.” (period). Example: To set “10.172.83.1”, enter “010172083001”. (Note) “0.0.0.0”, “127.*.*.” and “255.*.*” cannot be set. When specifying using the host name, it takes time for the name to be set.
	FTP server port number	0~65535	Set a port number of an FTP server to be connected with the machine (FTP client) when the connection destination is the FTP server.
	User name as client	20 of one-byte characters	Set a user name for connecting the machine (FTP client) with an FTP server when connection destination is the FTP server.
	Password as client	20 of one-byte characters	Set a password for connecting the machine (FTP client) with an FTP server when connection destination is the FTP server. The password having been set is displayed as “*”.*. This parameter is saved using plain text in a file. Therefore, please note this when managing the files.
	Use passive mode	0: No 1: Yes	Set whether or not to use passive mode for data transfer when connection destination is an FTP server. When “0: No” is set, data transfer is performed in active mode.
	Server user name	20 of one-byte characters	Use when <Restrict Ethernet access> is set to <0: No>. Set the user name to permit connection when the machine serves as the FTP server. Permission to connect from the client side is not given when not set. The user for this parameter is not valid when <Restrict Ethernet access> is set to <1: Yes>.

No.	Items	Setting range	Description
	Password as server	20 of one-byte characters	Use when <Restrict Ethernet access> is set to <0: No>. Set a password to permit connection when the machine serves as the FTP server. The password having been set is displayed as “*”.* If the password is not set yet, the password for the <Server user name> will not set. It is possible to login by only using the user name. This parameter is saved using plain text in a file. Therefore, please note this when managing the files.
	FTP time zone	-12:00~12:00	If the world standard time (GMT) is used for time information on FTP, enter the difference between the current date and time of the machine and GMT.
	Response monitoring time	0~999 sec.	Set allowable time interval between the receipt of one character and that of the next one or between the sending of one character and that of the next one. Further, connection is cut off when no command has been sent from a client during the period of time set here when the machine serves as the FTP server. When “0” is set, it becomes infinite standby state.
	Invalid data (For NC language only)	0: Conv 1: Error 2: Ignor	Designate how to process data not supported by NC if such data has been read. When “0: Conversion” is set, such data is converted into “?” and the program is displayed as “?” on the edit screen.
	End of block (For NC language only)	0:CR、 LF 1:LF	Set the code for the end of one block (EOB).
	TV check (For NC language only)	0: None 1: Odd 2: Even	Set whether to perform a one block parity check. It does not apply to the FTP.
	Checksum	0: Invalid 1: Valid	Set whether or not to detect errors using the checksum in the communication by communication protocol type 2 of NC-dedicated communication device.
	Header/footer (For NC language only)	0: None 1: Yes	Designate how to treat header and footer when connection destination is FTP server. None: At the time of output, header and footer are not added. At the time of input, data are inputted as they are without checking header and footer. Yes: At the time of output, header and footer are added. At the time of input, header and footer are checked and deleted.
	FTP response file size	0: Type 1 1: Type 2	Specified the file size that responds to the LIST and SIZE commands when this machine is used as an FTP server. Type 1: Outputs the larger size, taking into account the conversion of the “End of block” code during output, etc. Type 2: Outputs the NC internal file size. This type is appropriate for communication within the NC.
	Leading zero suppression (DPRNT) (NC language only)	0: Type 1 1: Type 2	Set the output method for high-order zeros when macro variables are output via DPRNT commands. Type 1: Outputs spaces. Type 2: Nothing is output.

No.	Items	Setting range	Description
	External output - FTP output destination (NC language only)	255 one-byte characters	Specify the output directory on the FTP server for the macro (external output function). The following characters can be used as a folder delimiter: "A" to "Z", "0" to "9", parentheses "(" and ")", a hyphen "-" as well as the forward slash "/".
	External output - FTP output name (NC language only)	32 one-byte characters	Specify the character string for the output file name on the FTP server for the macro (external output function). The following characters can be used: "A" to "Z", "0" to "9", parentheses "(" and ")" and a hyphen "-".
	Restrict Ethernet access	0: No 1: Yes	Set whether to restrict access to use the Ethernet function or not.
	Common user name	20 one-byte (half-width) characters	Use when <Restrict Ethernet access> is set to <1: Yes>. Set the user name for authorizing the connection when the machine serves as the FTP server. Set the user name to give access to this machine using communication protocol type 2 for the NC communication device. The only characters that can be used are those that can be entered on the operation panel (Upper case letters, numbers and symbol characters). However, a space character cannot be used. Permission to connect from the FTP client side is not given when this is not set. The access restriction, for using communication protocol type 2 for an NC communication device, cannot be lifted temporarily. This parameter is converted to a character string that cannot be read as a file. The user for this parameter is not valid when <Restrict Ethernet access> is set to <0: No>.
	Common password	8 to 20 one-byte (half-width) characters	Use when <Restrict Ethernet access> is set to <1: Yes>. Set the password for authorizing the connection when the machine serves as the FTP server. Set the password to give access to this machine using communication protocol type 2 for an NC communication device. The only characters that can be used are those that can be entered on the operation panel (Upper case letters, numbers and symbol characters). However, a space character cannot be used. At least two of the following character types must be used: Upper case letters, numbers or symbols. Asterisks (*) are displayed for the password that is set. (When creating a PC software that displays this parameter, be sure to include a way to conceal what is displayed, such as using asterisks.) Permission to connect from the FTP client side is not given when this is not set. The access restriction, for using communication protocol type 2 for an NC communication device, cannot be lifted temporarily. This parameter is converted to a character string that cannot be read as a file.

No.	Items	Setting range	Description
	Access timeout period	0 ~ 999(min.)	Restrict access to this machine using communication protocol type 2 for an NC communication device by setting <Restrict Ethernet access> to <1: Yes>. Set the time period to lift the access restriction temporarily. When this is set to <0>, the access restriction is lifted temporarily until the connection is released. When this is not set, it functions in the same way as when it is set to <0>.

2.7.6 Memory Card

No.	Items	Setting range	Description
	Invalid data (For NC language only)	0: Conv 1: Error 2: Ignor	Designate how to process data not supported by NC if it has been read. When "0: Conversion" is set, such data is converted into "?" and the program is displayed as "?" on the edit screen.
	End of block (For C language only)	0:CR, LF 1:LF	Set the code for the end of one block (EOB).
	TV check (For NC language only)	0: None 1: Odd 2: Even	Set whether to perform a one block parity check.
	Header/footer (For NC language only)	0: None 1: Yes	Designate how to treat header and footer when connection destination is FTP server. None: At the time of output, header and footer are not added. At the time of input, data are inputted as they are without checking header and footer. Yes: At the time of output, header and footer are added. At the time of input, header and footer are checked and deleted.
	Leading zero suppression (DPRNT) (For NC language only)	0: Type 1 1: Type 2	Set the output method for high-order zero at the time of macro variable output under DPRNT command. Type 1: Outputs space Type 2: Nothing is output

2.7.7 Auto notification function

No.	Item name	Setting range	Description
	Auto notification function	0: Disable 1: Enable	Enable or disable the auto notification function.
	Start operation at power ON	0: No 1: Yes	When set to <0: No>, the auto notification function is activated by issuing a slave command. When set to <1: Yes>, the function is activated when the power is turned ON.
	Notification host name (IP address)	255 one-byte characters	Set the host name or IP address for the notification destination. When setting using an IP address, enter the address using a 12 digit number without inserting any periods (.). Ex: When setting the address "10.172.83.1", enter the address number as follows: "010172083001". (Note) "0.0.0.0", "127.*.*.*" and "255.*.*.*" cannot be set. When specifying using the host name, it takes time for the name to be set.
	Notification port number	10000 to 65535	Set the port number for the notification host.
	Retry attempts	0 to 99 times	Set the number retry attempts when a connection with the host has failed.
	Retry wait time	0 to 999 sec.	Set the wait time before a retry attempt is made.
	Control method	0: Standard 1: Simple	Select the control method for the auto notification function.
	Response monitoring time	0 to 999 sec.	After sending an auto notification, the timing that is set here is used to sever the connection when a response from the host is not sent. When set to <0>, it waits forever.
	Data register address for storing error code	0 to 2047	Set the address for data register (D) in the PLC where the auto notification function stores the error code. This value is invalid when the <Control method> is set to <1: Simple>.
	Alarm display for send error	0: No display 1: Display	Select whether to display or not display an alarm when a send error is triggered for the auto notification.
	Method to specify sending data	0: Direct 1: Indirect	Select the method to specify the data, and either specify indirectly using the internal output data (BDY) in the PLC, or specify directly in the parameters for <Type of sending data>, <Leading address for sending data> and <Sending data length> in the auto notification.
	Stop level for error when sending data is specified indirectly	1 to 5	Specify the stop level for an alarm that is triggered if specifying a ladder program is invalid when the <Method to specify sending data> is set to <1: Indirect>. 1: Operation does not stop 2: Operation stops after 1 cycle 3: Operation stops at the block 4: Operation stops temporarily 5: Servo is turned OFF This value is invalid when set to <0: Direct>.
	Cancel level for error when sending data is specified indirectly	1 to 3	Specify the cancel level for an alarm that is triggered if specifying a ladder program is invalid when the <Method to specify sending data> is set to <1: Indirect>. 1: Any key 2: [RST] key 3: Power OFF This value is invalid when set to <0: Direct>.
	Type of sending data	0: No data 1: Internal relay (M) 2: Data register (D) 16 bit 3: Data register (D) 32 bit	Set the type of data sent in the auto notification when the <Method to specify sending data> is set to <0: Indirect>. No data, PLC internal relay (M) and PLC data register (D) 16 bit/32 bit can be set. This value is invalid when set to <1: Indirect>.

No.	Item name	Setting range	Description
	Leading address for sending data	0 to 8191	Set the leading address for the data sent in the auto notification when the <Method to specify sending data> is set to <0: Indirect>. This value is invalid when set to <1: Indirect>. The value that can be set changes depending on the setting for <Type of sending data>. <1: Internal relay (M)> 0 to 8191 <2: Data register (D) 16 bit> 0 to 2047 <3: Data register (D) 32 bit> Even numbers from 0 to 2046
	Sending data length	1 to 600 packets	Set the data length sent in the auto notification when the <Method to specify sending data> is set to <0: Indirect>. This value is invalid when set to <1: Indirect>. The value that can be set changes depending on the setting for <Type of sending data>. <1: Internal relay (M)> 1 to 600 packets <2: Data register (D) 16 bit> 1 to 100 packets <3: Data register (D) 32 bit> 1 to 50 packets

- * When setting a value from 3 to 5 for the <Stop level for error when sending data is specified indirectly>, even if the <Cancel level for error when sending data is specified indirectly> is set to "1", it is processed as a cancel level 2.
In this situation, the <<Communication parameter setting error (auto notification function)>> (Stop level 1, cancel level1) is triggered when there is a reset after a parameter change, or when the communication parameter setting screen is displayed.
- * When setting a value from 2 to 5 for the <Stop level for error when sending data is specified indirectly>, even if the <Cancel level for error when sending data is specified indirectly> is set to "1", the alarm is not cancelled during memory operation (except during a stoppage).

2.7.8 List of Communication Parameter Default Values

Serial port

Item	Same for all models
Reset(slave)	0
Data overwrite(slave)	1
Remote operation	1
Display slave command alarm	1
Program batch input system	0
Add comment of file name	0
Extended comment input	1
External output - Output destination	0
Connected device	
NC communication device protocol(master)	0
Baud rate	5
Parity	0
Stop bit	0
Character length	0
Communication mode	0
Response monitoring time	60
Invalid data	1
Sending data code	0
End of block	0
TV check	0
Output dot character	0
Time for recovery after resetting	1
Checksum	0
EIA character code(#)	173
EIA character code(*)	186
EIA character code(=)	194
EIA character code(l)	161
EIA character code(j)	162
Check DR signal	1
Communication end DC3 code	0
DC1 code	17
DC2 code	18
DC3 code	147
DC4 code	20
Leading zero suppression (DPRNT)	0

Ethernet/FTP

Item	Same for all models
Reset(slave)	0
Data overwrite(slave)	1
Remote operation	1
Display slave command alarm	1
Program batch input system	0
Add comment of file name	0
Extended comment input	1
External output - Output destination	0
Host name	
Use DHCP	0
IP address	1.1.1.1
Mask bit	255.255.255.0
Default gateway	
Auto acquisition of DNS server address	0
DNS server address 1	
DNS server address 2	
Port No.	10000
PLC port number	20000
FTP server name(IP address)	

Item	Same for all models
FTP server port number	21
User name as client	
Password as client	
Use passive mode	1
Server user name	
Password as server	
FTP time zone	0
Response monitoring time	60
Invalid data	1
End of block	0
TV check	0
Checksum	0
Header/footer	0
FTP response file size	0
Leading zero suppression (DPRNT)	0
External output - FTP output destination	
External output - FTP output name	
Restrict Ethernet access	
Common user name	
Common password	
Access timeout period	

Memory card

Item	Same for all models
Reset(slave)	0
Data overwrite(slave)	1
Remote operation	1
Display slave command alarm	1
Program batch input system	0
Add comment of file name	0
Extended comment input	1
External output - Output destination	0
Invalid data	1
End of block	0
TV check	0
Header/footer	0
Leading zero suppression (DPRNT)	0

Auto notification function

Item	Same for all models
Auto notification function	0
Start operation at power ON	0
Notification host name (IP address)	
Notification port number	30000
Retry attempts	3
Retry wait time	3
Control method	0
Response monitoring time	60
Data register address for storing error code	300
Alarm display for send error	1
Method to specify sending data	0
Stop level for error when sending data is specified indirectly	4
Cancel level for error when sending data is specified indirectly	2
Type of sending data	0
Leading address for sending data	100
Sending data length	10

2.8 G/M Code Macro Settings

2.8.1 Overview

This section is for setting the parameters related to the G code macro and M code macro functions.

The G/M code macros are divided into the following categories.

Category name	Description
Common parameter	Sets the parameters that are the same for G and M code macros.
G code macro	Displays the parameters related to G code macros.
M code macro	Displays the parameters related to M code macros.

2.8.1.1 Edit Screen

Select “G/M code macro” on the data bank menu screen to display the G/M code macro edit screen.

The category that was previously displayed is shown. Press the function key to switch the category.

F0	F1	F2	F3	F4	F5	F6	F7
Completion mode	Common parameter	G code macro	M code macro				

G code macro edit screen

M code macro edit screen

2

2.8.2 Edit Method

2.8.2.1 Enter Data Directly

1. Display the category with the item you wish to change.
 2. Move the cursor to the item you wish to change.
 3. Enter the value into the data input field, and press the **[ENT]** key.
- The value is input at the cursor's position, and the cursor moves to the next item.

* Refer to the section “2.8.3 List of G/M code macro parameters” for further details on the items.

2.8.2.2 Delete Data

If the cursor is positioned at an item that can be deleted, then the message “Can be deleted” will appear.

1. Display the category with the item you wish to delete.
2. Move the cursor to the item you wish to delete.
3. Press the **[DEL]** key.

The value at the cursor position is deleted.

2.8.2.3 Exit Editing

After the setting is completed, press the **[F0]** (Completion mode) key to complete and exit editing. Refer to “2.1 Overview of data bank” for further details on the ways to exit editing mode.

2.8.3 List of G/M Code Macro Parameters

2.8.3.1 Common Parameter

No.	Item name	Setting range	Description
	Macro program call control type	0: Method 1 1: Method 2	Set the control method for the M code in the macro program that is called using G code or for G code in the macro program that is called using M code. Method 1: The G/M code in the macro program runs as a G/M code in the system. Method 2: The macro program can be called using the G/M code in the macro program.
	G code macro program folder	Up to 8 alphanumeric characters including periods can be entered from the panel.	Set the folder name for the program number that is called using the G code macro command. * When not set, the root folder is used.

No.	Item name	Setting range	Description
	M code macro program folder	Up to 8 alphanumeric characters including periods can be entered from the panel.	Set the folder name for the program number that is called using the M code macro command. * When not set, the root folder is used.
	System G/M code calling during MDI operation	0: No call 1: Call	If the setting is configured to <1: Call> and a G/M code is registered as a <G code number> or <M code number> for MDI operation mode, when that code is executed, the machine operates using the system G/M code. When executing the call with the parameter set to <0: No call>, the alarm <<Macro program call not possible>> is triggered.

2.8.3.2 G Code Macro

Setting items for registration numbers 1 to 20

No.	Item name	Setting range	Description
	G code number	1 to 999	Sets the G code when calling the macro program using G code. (However, 65, 66 and 67 cannot be set.) The function is disabled when the field is left blank.
	Program No.	1 to 8999	Sets the program number that is called by the macro.
	Call method	0: Method 1 1: Method 2	Specifies the call type in the macro program. Method 1: A call equivalent to G65 (Simple call) is carried out. Method 2: A call equivalent to G66 (Modal call) is carried out.

- * When the [RST] key is pressed after exiting edit mode, if a G code number has been duplicated or is redundant, the alarm <<G code macro duplicate error>> is triggered.

2.8.3.3 M Code Macro

Setting items for registration numbers 1 to 20

No.	Item name	Setting range	Description
	M code number	1 to 999	Sets the M code when calling the macro program using M code. (However, 1, 2, 30, 96 - 99 and 203 cannot be set.) The function is disabled when the field is left blank.
	Program No.	1 to 8999	Sets the program number that is called by the macro.

- * When the [RST] key is pressed after exiting edit mode, if a G code number has been duplicated or is redundant, the alarm <<M code macro duplicate error>> is triggered.

2.9 Tool list (Conversation)

2.9.1 Outline

Information about type, dimension, shape and life of the tools to be used for machining are registered. 20 types of tools are available and up to 99 tools can be registered.

If tool dimensions or shape are modified, be sure to correct the data. And perform tool assignment check to see if the tool is usable.

- * If the registered dimensions and shape of the tools are different from those of the tools actually mounted on the magazine, collision of tools, workpiece and jigs or an accident may occur. Be sure to register data correctly.
- * When storing tools, apply an identification label on them so that the actual tools is identical with registered tool data.

2

2.9.1.1 Edit Screen

Selecting the “Tool list” on the Data bank menu screen displays the edit screen for tool list.

Tool list					
2012/05/29 11:38:51					
Tool No.	01(P:01)	02(P:02)	03(P:04)	04(P:03)	05
Tool	Center drill	Drill	Point tap	Chamfer tool	Carbide drill
Point angle	0.000	118.000	*	*	180.000
Minor diameter	0.000	*	*	0.000	*
Minor dia. length	0.000	*	*	*	*
Center angle	90.000	*	*	90.000	*
Invalid section	*	*	1.000	1.000	*
Major/Nominal D	5.000	4.300	M5.000	10.000	9.999
Pitch/Threads	*	*	0.800	*	*
Length of teeth	*	30.000	30.000	*	99.999
Torsion	*	*	Right thread	*	*
Number of teeth	*	*	*	*	*
Tool length	120.000	130.000	140.000	150.000	100.000
Initial life	5000(H)	10000(H)	5000(H)	**	99999pcs.
Life warning	100(H)	100(H)	100(H)	**	999pcs.
Tool life	5000(H)	10000(H)	5000(H)	**	99999pcs.
Tool					
1.Center drill 4.Spiral tap 7.Rolled tap 10.Chamfer drill 2.Drill 5.Point tap 8.Chamfer tool 11.Stepped drill 3.Carbide drill 6.Hand tap 9.CBR tool 12.Drill tap					
[HELP]					
Completion mode	Search	Tool length setting	Add	Copy	Paste
		Display extension information	Next menu		

The tools whose tool number is shown in orange are those which are registered in the ATC tools. Tool numbers are displayed with pot numbers.

Pressing the [F6] (Display extension information) key displays the edit screen for tool list that includes extended information.

Tool list (extension information)					
2012/05/29 11:40:57					
Tool No.	01(P:01)	02(P:02)	03(P:04)	04(P:03)	05
Tool	Center drill	Drill	Point tap	Chamfer tool	Carbide drill
Point angle	0.000	118.000	*	*	180.000
Minor diameter	0.000	*	*	0.000	*
Minor dia. length	0.000	*	*	*	*
Center angle	90.000	*	*	90.000	*
Invalid section	*	*	1.000	1.000	*
Major/Nominal D	5.000	4.300	M5.000	10.000	9.999
Pitch/Threads	*	*	0.800	*	*
Length of teeth	*	30.000	30.000	*	99.999
Torsion	*	*	Right thread	*	*
Number of teeth	*	*	*	*	*
Tool length	120.000	130.000	140.000	150.000	100.000
Initial life	5000(H)	10000(H)	5000(H)	**	99999pcs.
Life warning	100(H)	100(H)	100(H)	**	999pcs.
Tool life	5000(H)	10000(H)	5000(H)	**	99999pcs.
Tool washing	Allowed	Not allowed	Allowed	Allowed	Allowed
C/TS	Allowed	Not allowed	Allowed	Allowed	Allowed
Tool					
1.Center drill 4.Spiral tap 7.Rolled tap 10.Chamfer drill 2.Drill 5.Point tap 8.Chamfer tool 11.Stepped drill 3.Carbide drill 6.Hand tap 9.CBR tool 12.Drill tap					
[HELP]					
Completion mode	Search	Tool length setting	Add	Copy	Paste
		Hide extension information	Next menu		

When the [$\rightarrow$] (Next function) key is pressed

Completion mode	Tool data								
	Delete								

2.9.1.2 Setting Items

1. Tools

Designate the tool type.

Select from the 20 types of tools displayed below the input area.

- * 12 types of tools are initially displayed below the input area. Pressing the [F7] (Next menu) key displays the remaining 8 types of tools.

When designating a tool type, asterisk "*" is displayed for the following items that do not require input. The other items for which "*" is not displayed require input. Be sure to input values for them. Leaving any item field blank causes a failure to recognize the normal tool shape.

2. Point angle: α

Set the angle of the tool tip.

3. Minor diameter: d

Set the minor diameter of the tool tip part.

4. Minor dia. length: L

Set the minor diameter length of the tool tip part.

5. Center angle: θ

Set the large diameter angle of the center drill or the point angle of the chamfering tool or corner round cutter.

6. Invalid section: Lo

Set the length of the bevel lead.

- * For the tap, specify the pitch or the threads.

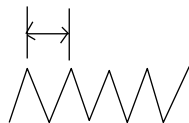
7. Major/Nominal: D

Set the Outside diameter of the tool.

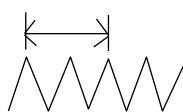
- * For the tap, specify the screw type and size (outside diameter).

8. Set the pitch or the threads of the tap.

For the metric thread,
set the pitch.



For the inch thread,
set the threads per inch.



9. Length of teeth: L

Set the length of the cutting tooth.

10. Torsion

Set the direction of the torsion of the tap (right thread/left thread).

11. Number of teeth

Set the number of teeth of the end mill or corner round cutter.

12. Tool length: LL

Set the length from the gauge line of the tool holder to the tip of the tool

13. Initial life/life end

Locating the cursor at the “Initial life/life end” enables the life unit and the initial life/life end to be set.

- * Dependin on the setting for the <Tool life counting method> in the User parameters, the item name changes to either “Initial life” or “Life end.”
 - When the value of the <Tool life counting method> is set to “0: Type 1,” the tool life management is performed by counting down, and the item name is displayed as “Initial life”.
 - When the value of the <Tool life counting method> is set to “1: Type 2,” the tool life management is performed by counting up, and the item name is displayed as “Life end”.

Life unit is selectable from the following four units.

Set value	Unit	Description
1: Not counted	None	Tool life management is not performed “1: Setting to “1: Not counted,” the “Tool life/life end” is displayed as “Life warning” and “***” is displayed in the “Life” item.
2: Time	Min	Tool life management is performed in time (in the unit of minute). As the cutting travel is performed for one minute, the tool life is subtracted/added.
3: Drilled holes	Holes	Manages the tool life based on the number of holes drilled. When 1 hole is drilled in a drilling job, or when a motion call is executed for 1 job, the count for the tool life is added or subtracted accordingly.
4: Workpieces	Number of workpieces	Tool life management is performed in workpieces. At the end of operation, the tool life of the tool with which cutting travel has been done is subtracted/added.

Setting the tool life unit goes to the setting of the initial life/life end.

For the initial life/life end, st the initial value/end value of the tool life.

For the initial life, setting a initial life sets the same value for the tool life.

For the life end, setting a life end sets “0” for the tool life.

14. Life warning

For the life warning, set the tool life value at which the end of the tool life is warned.

When the tool life reaches the value for the life warning, an alarm <<Tool life ending (****)>> (where the parenthesized value is a tool number) is generated.

Even after the generation of such an alarm, starting of memory operation remain enabled.

- * Setting the life warning to “0” in the counting-up type does not generate this alarm.

15. Life

Set the current value for the tool life. When the end of life is reached by subtraction/addition as a result of machining, an alarm <<Tool life end (****)>> (where the parenthesized value is a tool number) is generated. Upon generation of this alarm, starting of memory operation is prevented.

- * In the MDI operation, the tool life is not subtracted/added.
- * Only for the tools commanded in memory operation, tool life subtraction/addition and tool life check are performed. Thus when a tool change has been made through MDI operation during memory operation, tool life subtraction/addition and tool life check are not performed even after returning to the memory operation.
- * “End of life reached” means that the tool life reaches “0” in the counting-down method or that the tool life reached the same as the value set for the end of life in the counting-up method.
- * The tool life for the tool whose life end has reached can be initialized on the ATC tool screen.

16. Tool wash

For a tool that you would like to prevent from being washed with coolant, set “1: Disabled.” This disables the tool wash for the tool for which “Disabled” is set.

- * The discharge conditions can be changed in the user parameter (switch 1) <Tool wash discharge conditions for tool data>.

17. CTS

For a tool which does not support center through coolant, set “1: Disabled.” When the CTS coolant discharge” is commanded for a tool that is set to “Disabled” for CTS coolant, CTS coolant discharge is prevented with an alarm “CTS disabled tool” generated.

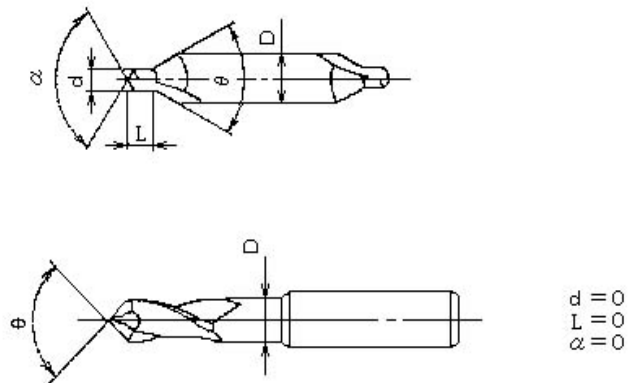
2

List of items permitting tool type input

Tool \ Item	Center drill	Drill	Carbide drill	Spiral tap	Point tap	Hand tap	Rolled tap	Chamfering tool	Counter boring tool	Drill with chamfering tool	Drill with counter boring tool	Drill tap	Reamer	Drill reamer	Rough finishing end mill (rough)	Finishing end mill (finish)	Rough finishing carbide end mill	Finishing carbide end mill (finish)	Corner round cutter	Centering tool
Point angle: α	○	○	○	*	*	*	*	*	*	○	○	○	*	○	*	*	*	*	*	*
Minor dia.: d	○	*	*	*	*	*	*	○	○	○	○	○	*	○	*	*	*	*	○	○
Minor dia.length: L	○	*	*	*	*	*	*	*	○	○	○	○	*	○	*	*	*	*	*	*
Center angle: θ	○	*	*	*	*	*	*	○	*	○	*	*	*	*	*	*	*	*	○	*
Invalid section: Lo	*	*	*	○	○	○	○	○	*	*	*	○	○	○	*	*	*	*	*	*
Major/Nominal: D	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	*
Pitch/Treads: P	*	*	*	○	○	○	○	*	*	*	*	○	*	*	*	*	*	*	*	*
Length of teeth: L	*	○	○	○	○	○	○	*	○	○	○	○	○	○	○	○	○	○		*
Torsion	*	*	*	○	○	○	○	*	*	*	*	○	*	*	*	*	*	*	*	*
Number of teeth	*	*	*	*	*	*	*	*	*	*	*	*	*	*	○	○	○	○	○	*
Tool length: LL	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
Initial life/Life end	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	*
Life warning	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	*
Tool life	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	*
Tool wash	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
CTS	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○

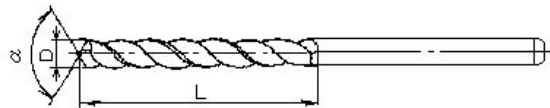
Tool shape and positions of dimensions of the tools registered in the tool list

1. Center drill

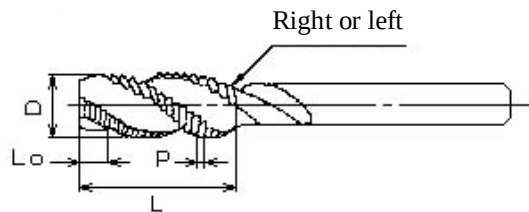


2. Drill

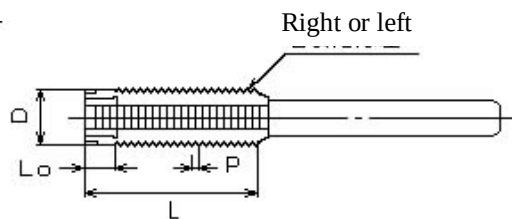
3. Carbide drill



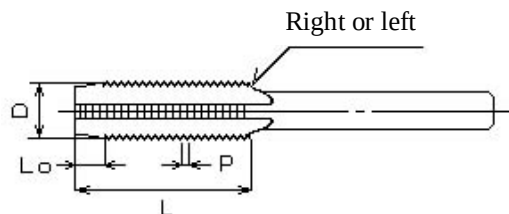
4. Spiral tap



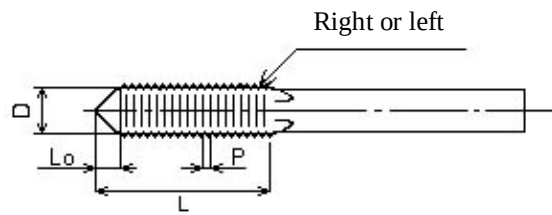
5. Point tap



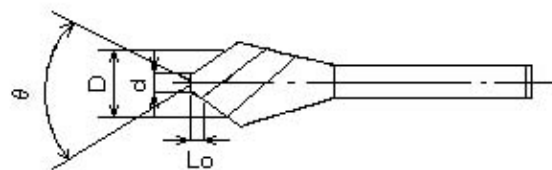
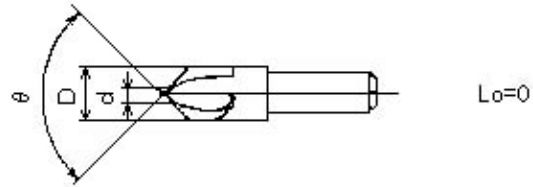
6. Hand tap



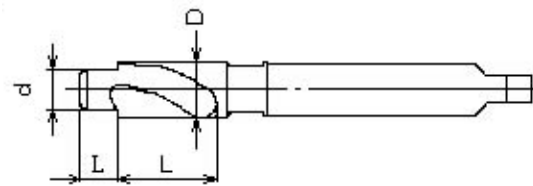
7. Rolled tap



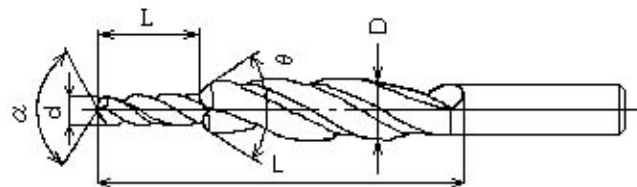
8. Chamfering tool



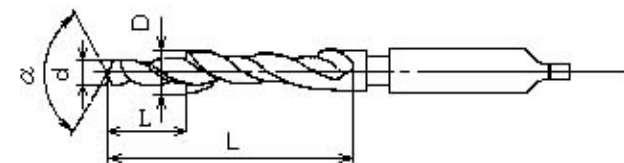
9. Counter boring tool



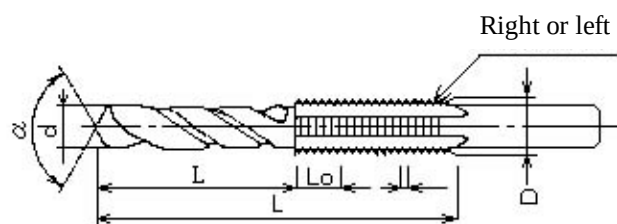
10. Drill with chamfering tool



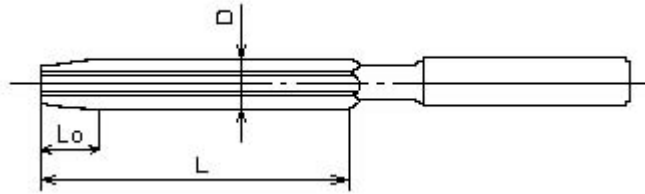
11. Drill with counter boring tool



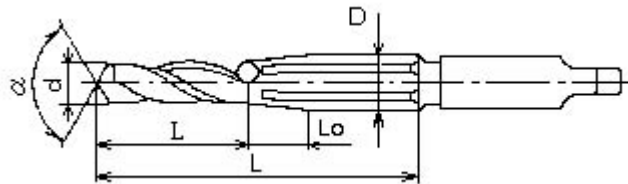
12. drill tap



13. Reamer



14. Drill reamer

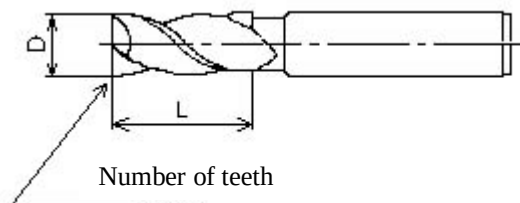


15. End mill (rough) (Rough finishing tool)

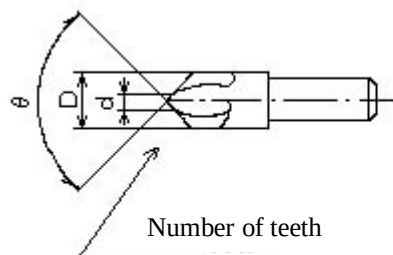
16. End mill (finish) (Finishing tool)

17. Carbide end mill (Rough finishing tool)

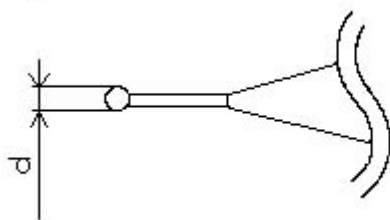
18. Carbide end mill (finish) (Finishing tool)



19. Corner round cutter

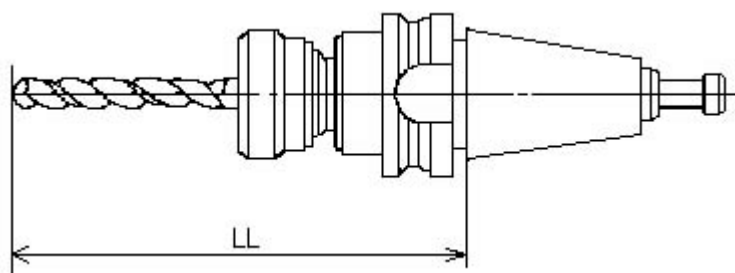


20. Centering tool



2

Tool length



2.9.2 How to Edit

2.9.2.1 Locating Cursor

Locate the cursor at the position to be changed using the cursor key.

Search can be used for immediately search the tool number to be changed. Enter a tool number in the input area and press the **[F1]** (Search) key. This locates the cursor at the tool of the designated tool number.

* Depending on the tool type, some tool does not require input. For such items not requiring input, "*" is displayed. Such item are skipped in locating the cursor.

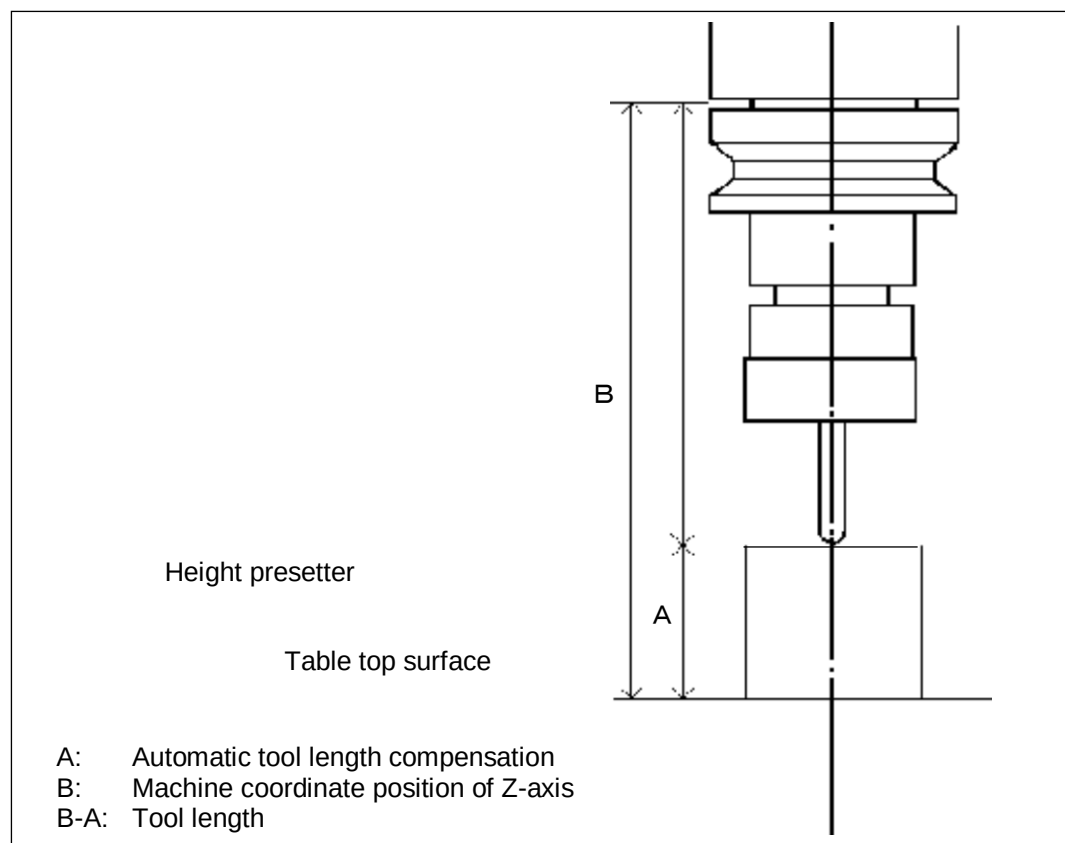
2.9.2.2 Direct Data Input

1. Locate the cursor at the item to be changed.
2. Enter a value in the input area and press the **[ENT]** key.
This inputs the data at the cursor position and locates the cursor at the next item.

2.9.2.3 Setting of Tool Length

Calculation is made using the current position and the "Automatic tool length compensation" in the User parameters with the tool whose tool length is to be set attached to the spindle.

Set the height (A) of the height presetter in the user parameter <Automatic tool length compensation> beforehand.



1. Attache the tool whose tool length is to be set to the spindle.
For details, refer to "4.4 ATC tools" "8.2.3.1 Attaching the tool" in the operation manual.
2. Operate the machine in manual operation mode and lower the Z-axis to the position where the tool tip slightly touches the surface of the height presetter.
3. Display the Edit screen for tool list and press the **[F2]** (Tool length setting) key.

Tool length setting		2012/05/29 11:42:12	
Machine coordinate pos	Tool No	01 (P:01)	
Z	480.000	Center drill	
		Point angle	0.000
		Minor diameter	0.000
		Minor dia. length	0.000
		Center angle	90.000
		Invalid section	*
Automatic tool length compensation	Major/Nominal D	5.000	
	50.000	Pitch/Threads	*
		Length of teeth	*
		Torsion	*
		Number of teeth	*
		Tool length	120.000
		Initial life	5000 (H)
		Life warning	100 (H)
		Tool life	5000 (H)
		Tool washing	Allowed
		CTS	Allowed
Set?			
Yes	No		

Subtract the <Automatic tool length compensation> value in the user parameter (switch 1) from the current machine coordinate position, and then use the difference as the value to set the new tool length.

Pressing the **[F0]** (Yes) key sets the value for “Tool length after setting” and returns to the previous screen.

Pressing the **[F1]** (No) key returns to the previous screen without making the setting.

*Tool length setting is disabled when no tool is set to the spindle.

2.9.2.4 Data Addition

1. Locate the cursor at the item to be changed.
2. Enter a value to be added in the input area and press the **[F3]** (Addition) key.
This adds the input value to the set value at the cursor position and locates the cursor at the next item.

* Addition is disabled when the cursor is located at any of the following items: Tool, Screw, Nominal, Number of teeth, Life related items, Tool wash and CTS.

2.9.2.5 Data Copy

1. Locate the cursor at a specific “Tool” for which data is to be captured.
2. Press the **[F4]** (Copy) key.
This captures a set of data for the tool at which the cursor is located.
The captured data remain retained until another data is captured or the power is turned OFF.

2.9.2.6 Data Deletion

1. Locate the cursor at a specific “Tool” for which data is to be deleted.
2. Pressing the **[Delete]** key or the **[F1]** (Tool data deletion) key displays the Deletion confirmation screen.

Tool number 12							
Are you sure you want to delete?							
Yes	No						

- Pressing the **[F0]** (Yes) key captures data and deletes the tool data at the cursor position.
- Pressing **[F1]** (No) key returns to the previous screen without deleting the data.

The captured data remains retained until another data is captured or the power is turned OFF.

2.9.2.7 Data Paste

1. Locate the cursor at a specific “Tool” for which the captured data is to be pasted.
2. Press the [F5] (Paste) key.
This changes the set of tool data at the cursor position to the data captured.

2.9.2.8 Registering Tools with Auxiliary Code

Tools can be registered with any of auxiliary codes 1 to 9. By giving auxiliary codes to tools beforehand, it is possible to designate the tool to be used where there are identical tools.

When specifying a tool type, giving a hyphen [-] and any of auxiliary codes [1] to [9] following the tool type number and pressing the [ENT] key adds the auxiliary code to the tool type number of the particular tool.

In the example below, setting for a drill is performed with auxiliary code “1” given to the tool No.11.

Locate the cursor at the “Tool” No.11, enter “2-1” in the input area and press the [ENT] key.

Tool list		2012/12/07 14:20:25			
Tool No.	10	11	12	13	14
Tool	Center drill	Drill			
Point angle	0.000				
Minor diameter	0.000	*			
Minor dia. leng	0.000	*			
Center angle	90.000	*			
Invalid section	*	*			
Major/Nominal D	10.000				
Pitch/Threads	*	*			
Length of teeth	*				
Torsion	*	*			
Number of teeth	*	*			
Tool length	150.000				
Initial life	**				
Life warning	**				
Tool life	**				
Point angle deg.					
[HELP]					
Completion mode	Search	Tool length setting	Add	Copy	Paste
					Display extension information
					Next menu

By setting an auxiliary code in the tool list and setting a tool registered in the cutting condition 1/2 using the same auxiliary code beforehand, the tool with the auxiliary code is selected in tool allocation.

2.9.2.9 End of Editing

Upon completion of the setting, press the [F0] (End mode) key to end the editing.

2.9.3 Editing during Operation

An operator message <<Operating>> is generated and editing is prevented from being ended.

2.10 Tool pattern (Conversation)

2.10.1 Outline

Set the tool type to be used in the machining processes and their order of use. Then, in tool allocation, the tools required for the respective machining processes are automatically selected according to the contents set here.

Two types of tool patterns with or without chamfering can be set for each machining process.

Normally tools are automatically selected according to the default tool pattern set at shipment. However, some machining process may require pattern to be changed.

2.10.1.1 Edit Screen

Selecting the “Tool pattern” on the Data bank menu screen displays the edit screen for tool patterns.

Tool pattern				2012/05/29 11:43:51	
	w/ chamfering	w/o chamfering			
Center hole		1 Center drill			
Hole	1 Center drill 2 Drill 3 Chamfer tool	1 Center drill 2 Drill			
Tap	1 Center drill 2 Drill 3 Chamfer tool 4 Tapping 5 *	1 Center drill 2 Drill 3 Tapping 4 *			
Rolled tap	1 Center drill 2 Drill 3 Chamfer tool 4 Rolled tap 5 *	1 Center drill 2 Drill 3 Rolled tap 4 *			
Tool					
1. Center drill 6. Chamfer tool 11. Reamer 16. CB E-mill(F) 2. Drill 7. CBR tool 12. Drill reamer 17. Chamf. cutter 3. Carbide drill 8. Chamfer drill 13. End mill(R) 4. Tapping 9. Stepped drill 14. End mill(F) 5. Rolled tap 10. Drill tap 15. CB E-mill(R)					
Completion mode	Previous pattern	Next pattern			

2.10.2 How to Edit

2.10.2.1 Locating cursor

Locate the cursor at the item to be changed using the cursor key.

Pressing the **[F1]** (Previous pattern) or the **[F2]** (Next pattern) key locates the cursor at the top of the previous or next machining process.

2.10.2.2 Data Change

Change the tool at the cursor position to another type of tool.

1. Locate the cursor at the tool to be changed.
2. Select a tool number from the tool types displayed below the input area, enter a value in the input area and press the **[ENT]** key.

This overwrites the tool at the cursor position and loates the cursor at the next tool.

2.10.2.3 Data Insertion

Add a new data before the tool at the cursor position.

1. Locate the cursor at the position to which the data is to be added.
2. Press **[INS]** key.

This displays “*” and brings down the subsequent data by one.

The tool which was set as the last tool in that machining process is deleted.

3. Set the tool at the cursor position according to the procedure of “2.9.2.2 Data change.”

2.10.2.4 Data Deletion

Delete the tool at the cursor position.

1. Locate the cursor at the tool to be deleted.
2. Press **[DEL]** key.

This deletes the tool at the cursor position and brings up the subsequent data by one.

An asterisk “*” is displayed at the end of that machining process.

2.10.2.5 Ending Editing

Upon completion of the setting, press the **[F0]** (End mode) key to end the editing.

2.10.3 List of Default Values

The tool pattern as shown below is initially set for each machining process before shipment. Modify the tool pattern as required according to the tools, workpiece materials to be used.

Machining process	With chamfer	Without chamfer
Center hole drilling	-	■1 Center drill
Drilling	■1 Center drill ■2 Drill ■3 Chamfering tool	■1 Center drill ■2 Drill
Tapping	■1 Center drill ■2 Drill ■3 Chamfering tool ■4 Tap ■5	■1 Center drill ■2 Drill ■3 Tap ■4
Rolled tap	■1 Center drill ■2 Drill ■3 Chamfering tool ■4 Rolled tap ■5	■1 Center drill ■2 Drill ■3 Rolled tap ■4
Reamer	■1 Center drill ■2 Drill ■3 Chamfering tool ■4 End mill R ■5 Reamer	■1 Center drill ■2 Drill ■3 End mill R ■4 Reamer
Counterboring	■1 Center drill ■2 Drill ■3 Counterboring tool ■4 Chamfering tool	■1 Center drill ■2 Drill ■3 Counterboring tool
Counterbore tapping	■1 Center drill ■2 Drill ■3 Counterboring tool ■4 Chamfering tool ■5 Tap ■6	■1 Center drill ■2 Drill ■3 Counterboring tool ■4 Tap ■5
Counterbore tapping by rolled tap	■1 Center drill ■2 Drill ■3 Counterboring tool ■4 Chamfering tool ■5 Rolled tap ■6	■1 Center drill ■2 Drill ■3 Counterboring tool ■4 Rolled tap ■5
Counterbore reamer	■1 Center drill ■2 Drill ■3 Counterboring tool ■4 Chamfering tool ■5 End mill R ■6 Reamer	■1 Center drill ■2 Drill ■3 Counterboring tool ■4 End mill R ■5 Reamer
Face milling	-	■1 End mill R
Facing	-	■1 End mill R ■2 End mill F
Pocket	■1 End mill R ■2 End mill F ■3 Chamfering mill	■1 End mill R ■2 End mill F
Counterboring	-	■1 End mill R ■2 End mill F
Chamfering	■1 Chamfering mill	-

2.11 Tapping Drill Diameter (Conversation)

2.11.1 Outline

Set the tap prepare hole drill diameters. Set them for each screw type, nominal diameter and pitch/threads. Further set taps separately for cutting and rolling.

- * Actually there is allowance for hole extension according to the manner the tool holder is attached, and tool runout and shape. Pefrom thorough management of the tool diameters and prepared holes.

2.11.1.1 Edit Screen

Selecting the “Tap prepared hole drill diameter” on the Data bank menu screen displays the edit screen for tap prepared hole drill diameters.

Metric coarse thread		2012/05/29 11:44:37			
	Cutting	Rolled tapping		Cutting	Rolled tapping
M 1.0x0.25	0.750	0.880	M 7.0x1.0	6.100	6.510
M 1.1x0.25	0.850	0.980	M 8.0x1.25	6.800	7.380
M 1.2x0.25	0.950	1.080	M 9.0x1.25	7.800	8.380
M 1.4x0.3	1.100	1.280	M10.0x1.5	8.600	9.220
M 1.6x0.35	1.300	1.440	M12.0x1.75	10.400	11.080
M 1.7x0.35	1.400	1.540	M14.0x2.0	12.200	12.950
M 1.8x0.35	1.500	1.640	M16.0x2.0	14.200	14.950
M 2.0x0.4	1.650	1.820	M18.0x2.5	15.700	16.670
M 2.2x0.45	1.800	1.990	M20.0x2.5	17.700	18.670
M 2.3x0.4	2.000	2.120	M22.0x2.5	19.700	20.670
M 2.5x0.45	2.100	2.300	M24.0x3.0	21.200	22.410
M 2.6x0.45	2.200	2.400	M27.0x3.0	24.200	
M 3.0x0.5	2.500	2.770	M30.0x3.5	26.700	
M 3.5x0.6	3.000	3.210			
M 4.0x0.7	3.300	3.640			
M 4.5x0.75	3.800	4.140			
M 5.0x0.8	4.300	4.620			
M 6.0x1.0	5.100	5.520			

By pressing the [F1] (Metric coarse thread) to [F7] (BSW) key or [F1] (BSF) to [F2] (Miniature screw thread) key for [>>] (Next function), the thread type can be changed.

2.11.2 How to Edit

2.11.2.1 Data Change

1. Locate the cursor at the item to be changed.
2. Enter a value in the input area and press the [ENT] key.
This overwrites the prepared hole drill diameter at the cursor position and locates the cursor at the next item.

2.11.2.2 Ending an Edit

Upon completion of the setting, press the [F0] (End mode) key to end the editing.

2.11.3.1 Drill Diameters for Tapping Primary Hole

2.11.3.1 Drill Diameters for Tapping Primary Hole

1. Metric coarse screw

2. Metric fine screw

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2. Metric fine screw (cont.)

Nominal pitch	Drill hole dia.
M24.0 × 1.0	23.100(0.9106)
M25.0 × 2.0	23.200(0.9134)
M25.0 × 1.5	23.600(0.9299)
M25.0 × 1.0	24.100(0.9500)
M26.0 × 1.5	24.600(0.9693)
M27.0 × 2.0	25.200(0.9921)
M27.0 × 1.5	25.600(1.0087)
M27.0 × 1.0	26.100(1.0287)
M28.0 × 2.0	26.200(1.0315)
M28.0 × 1.5	26.600(1.0480)
M28.0 × 1.0	27.100(1.0681)
M30.0 × 3.0	27.200(1.0709)
M30.0 × 2.0	28.200(1.1102)
M30.0 × 1.5	28.600(1.1268)
M30.0 × 1.0	29.100(1.1469)

3. Unified coarse screw

Number of threads	Primary hole drill diameter
UNC 1-64	1.550(0.0610)
UNC 2-56	1.850(0.0728)
UNC 3-48	2.100(0.0827)
UNC 4-40	2.300(0.0906)
UNC 5-40	2.600(0.1024)
UNC 6-32	2.800(0.1102)
UNC 8-32	3.500(0.1378)
UNC 10-24	3.900(0.1535)
UNC 12-24	4.500(0.1772)
UNC 1/4-20	5.200(0.2047)
UNC 5/16-18	6.700(0.2638)
UNC 3/8-16	8.100(0.3189)
UNC 7/16-14	9.500(0.3740)
UNC 1/2-13	11.000(0.4331)
UNC 9/16-12	12.300(0.4844)
UNC 5/8-11	13.800(0.5469)
UNC 3/4-10	16.800(0.6614)
UNC 7/8-9	19.600(0.7717)

4. Unified fine screw

Number of threads	primary hole drill diameter
UNF 0-80	1.300(0.0512)
UNF 1-72	1.600(0.0630)
UNF 2-64	1.900(0.0748)
UNF 3-56	2.100(0.0827)
UNF 4-48	2.400(0.0945)
UNF 5-44	2.700(0.1063)
UNF 6-40	3.000(0.1181)
UNF 8-36	3.500(0.1378)
UNF 10-32	4.100(0.1614)
UNF 12-28	4.700(0.1850)
UNF 1/4-28	5.500(0.2165)
UNF 5/16-24	7.000(0.2756)
UNF 3/8-24	8.600(0.3386)
UNF 7/16-20	10.100(0.3970)
UNF 1/2-20	11.600(0.4688)
UNF 9/16-18	13.000(0.5118)
UNF 5/8-18	14.600(0.5781)
UNF 3/4-16	17.600(0.6929)
UNF 7/8-14	20.600(0.8110)

- Where the machine parameter “Change screw type” is set to “0: Language set” and the “Display language” is set to “Japanese” or where the “Change screw type” is set to “1: For Japan”.

5. Sewing machine thread

Number of threads	primary hole drill diameter
SM 1/16-80	1.250
SM 5/64-64	1.500
SM 3/32-100	2.100
SM 3/32-56	1.900
SM 1/8-48	2.600
SM 1/8-44	2.500
SM 1/8-40	2.500
SM 9/64-40	2.900
SM 11/64-40	3.700
SM 11/64-32	3.500
SM 3/16-40	4.100
SM 3/16-32	3.900
SM 3/16-28	3.800
SM 3/16-24	3.600
SM 13/64-32	4.300
SM 7/32-32	4.700
SM 15/64-28	5.000
SM 1/4-40	5.600
SM 1/4-28	5.400
SM 1/4-24	5.200
SM 9/32-28	6.200
SM 9/32-20	5.800
SM 5/16-28	7.000
SM 5/16-24	6.800
SM 5/16-18	6.300
SM 11/32-28	7.800
SM 3/8-28	8.500
SM 3/8-18	7.900
SM 7/16-28	10.100
SM 7/16-16	9.300
SM 1/2-28	11.700
SM 1/2-20	11.300
SM 1/2-12	10.300
SM 9/16-20	13.000

6. Standard straight pipe thread PS

Number of threads	primary hole drill diameter
PS 1/16-28	6.500
PS 1/8-28	8.500
PS 1/4-19	11.400
PS 3/8-19	15.000
PS 1/2-14	18.500

7. Standard taper pipe thread PT

Number of threads	primary hole drill diameter
PT 1/16-28	6.400
PT 1/8-28	8.400
○PT 1/4-19	11.200
○PT 3/8-19	14.700
○PT 1/2-14	18.000

The taper pipe thread with circle(○) applies the machining with deeper cuts.

Change the cutting conditions (cutting speed, number of steps etc.) as needed.

8. Standard parallel fastening thread PF

Number of threads	primary hole drill diameter
PF 1/16-28	6.800
PF 1/8-28	8.800
PF 1/4-19	11.900
PF 3/8-19	15.400
PF 1/2-14	19.000

9. Miniature screw thread

Number of threads	primary hole drill diameter
S0.4×0.1	0.310
S0.5×0.125	0.390
S0.6×0.15	0.470
S0.7×0.175	0.550
S0.8×0.2	0.630
S0.9×0.225	0.710

- Where the machine parameter 1 “Change screw type” is set to “0: Language set” and the “Display language” is set to other than “Japanese” or where the “Change screw type” is set to “2: For other than Japan”.

5. NPS

Number of threads	primary hole drill diameter
NPS 1/16-27	6.630(0.2610)
NPS 1/8-27	8.840(0.3480)
NPS 1/4-18	11.510(0.4531)
NPS 3/8-18	15.080(0.5937)

6. BSP

Number of threads	primary hole drill diameter
BSP 1/16-28	6.700(0.2638)
BSP 1/8-28	8.800(0.3465)
BSP 1/4-19	11.900(0.4688)
BSP 3/8-19	15.400(0.6094)

7. BSW

Number of threads	primary hole drill diameter
BSW 1/8-40	2.600(0.1024)
BSW 3/16-24	3.700(0.1457)
BSW 1/4-20	5.100(0.2008)
BSW 5/16-18	6.600(0.2598)
BSW 3/8-16	8.000(0.3150)
BSW 7/16-14	9.400(0.3701)
BSW 1/2-12	10.600(0.4219)
BSW 9/16-12	12.200(0.4844)
BSW 5/8-11	13.600(0.5469)
BSW 11/16-11	15.100(0.5945)

8. BSF

Number of threads	primary hole drill diameter
BSF 3/16-32	4.000(0.1575)
BSF 7/32-28	4.700(0.1850)
BSF 1/4-26	5.400(0.2126)
BSF 9/32-26	6.200(0.2441)
BSF 5/16-22	6.800(0.2677)
BSF 3/8-20	8.300(0.3268)
BSF 7/16-18	9.800(0.3858)
BSF 1/2-16	11.200(0.4375)
BSF 9/16-16	12.800(0.5118)
BSF 5/8-14	14.100(0.5625)
BSF 11/16-14	15.700(0.6250)

9. BA

Number of threads	primary hole drill diameter
BA 0 × 1.00	5.100(0.2008)
BA 1 × 0.90	4.500(0.1772)
BA 2 × 0.81	4.000(0.1575)
BA 3 × 0.73	3.500(0.1378)
BA 4 × 0.66	3.100(0.1220)
BA 5 × 0.59	2.700(0.1063)
BA 6 × 0.53	2.400(0.0945)
BA 7 × 0.48	2.100(0.0827)
BA 8 × 0.43	1.800(0.0709)
BA 9 × 0.39	1.600(0.0630)
BA 10 × 0.35	1.400(0.0551)
BA 11 × 0.31	1.200(0.0472)
BA 12 × 0.28	1.100(0.0433)
BA 13 × 0.25	1.000(0.0394)
BA 14 × 0.23	0.800(0.0315)

10. NPT

Number of threads	primary hole drill diameter
NPT1 1/16-27	6.400(0.2500)
NPT1 1/8-27	8.600(0.3390)
NPT1 1/4-18	11.400(0.4531)
NPT1 3/8-18	14.700(0.5781)

11. Sewing machine thread

Number of threads	primary hole drill diameter
SM 1/16-80	1.250(0.0492)
SM 5/64-64	1.500(0.0591)
SM 3/32-100	2.100(0.0827)
SM 3/32-56	1.900(0.0748)
SM 1/8-48	2.600(0.1024)
SM 1/8-44	2.500(0.0984)
SM 1/8-40	2.500(0.0984)
SM 9/64-40	2.900(0.1142)
SM 11/64-40	3.700(0.1457)
SM 11/64-32	3.500(0.1378)
SM 3/16-40	4.100(0.1614)
SM 3/16-32	3.900(0.1535)
SM 3/16-28	3.800(0.1496)
SM 3/16-24	3.600(0.1417)
SM 13/64-32	4.300(0.1693)
SM 7/32-32	4.700(0.1850)
SM 15/64-28	5.000(0.1969)
SM 1/4-40	5.600(0.2205)
SM 1/4-28	5.400(0.2126)
SM 1/4-24	5.200(0.2047)
SM 9/32-28	6.200(0.2441)
SM 9/32-20	5.800(0.2283)
SM 5/16-28	7.000(0.2756)
SM 5/16-24	6.800(0.2677)
SM 5/16-18	6.300(0.2480)
SM 11/32-28	7.800(0.3071)
SM 3/8-28	8.500(0.3346)
SM 3/8-18	7.900(0.3110)
SM 7/16-28	10.100(0.3976)
SM 7/16-16	9.300(0.3661)
SM 1/2-28	11.700(0.4606)
SM 1/2-20	11.300(0.4449)
SM 1/2-12	10.300(0.4055)
SM 9/16-20	13.000(0.5118)

12. Miniature screw thread

Number of threads	primary hole drill diameter
S0.4 × 0.1	0.310
S0.5 × 0.125	0.390
S0.6 × 0.1	0.470
S0.7 × 0.1	0.550
S0.8 × 0.1	0.630
S0.9 × 0.1	0.710

2.11.3.2 Drill Diameters for Tapping Primary Hole by Rolled Tap

The values as shown below are initially set before shipment. Modify the values as required according to the tools and workpiece materials to be used.

1. Metric coarse screw

Nominal pitch	primary hole drill diameter
M1.0 × 0.25	0.880
M1.1 × 0.25	0.980
M1.2 × 0.25	1.080
M1.4 × 0.3	1.260
M1.6 × 0.35	1.440
M1.7 × 0.35	1.540
M1.8 × 0.35	1.640
M2.0 × 0.4	1.820
M2.2 × 0.45	1.990
M2.3 × 0.4	2.120
M2.5 × 0.45	2.300
M2.6 × 0.45	2.400
M3.0 × 0.5	2.770
M3.5 × 0.6	3.210
M4.0 × 0.7	3.640
M4.5 × 0.75	4.140
M5.0 × 0.8	4.620
M6.0 × 1.0	5.520
M7.0 × 1.0	6.510
M8.0 × 1.25	7.380
M9.0 × 1.25	8.380
M10.0 × 1.5	9.220
M12.0 × 1.75	11.080
M14.0 × 2.0	12.950
M16.0 × 2.0	14.950
M18.0 × 2.5	16.670
M20.0 × 2.5	18.670
M22.0 × 2.5	20.670
M24.0 × 3.0	22.410

2. Metric fine screw

Nominal pitch	primary hole drill diameter
M1.0 × 0.2	0.900
M1.1 × 0.2	1.000
M1.2 × 0.2	1.100
M1.4 × 0.2	1.300
M1.6 × 0.2	1.500
M1.8 × 0.2	1.700
M2.0 × 0.25	1.890
M2.2 × 0.25	2.090
M2.5 × 0.35	2.340
M3.0 × 0.35	2.840
M3.5 × 0.35	3.340
M4.0 × 0.5	3.770
M4.5 × 0.5	4.270
M5.0 × 0.5	4.770
M5.5 × 0.5	5.270
M6.0 × 0.75	5.640
M7.0 × 0.75	6.640
M8.0 × 1.0	7.510
M8.0 × 0.75	7.640
M9.0 × 1.0	8.510
M9.0 × 0.75	8.640
M10.0 × 1.25	9.380
M10.0 × 1.0	9.520
M10.0 × 0.75	9.660
M12.0 × 1.5	11.220
M12.0 × 1.25	11.380
M12.0 × 1.0	11.520
M14.0 × 1.5	13.220
M14.0 × 1.0	13.520
M15.0 × 1.5	14.220
M15.0 × 1.0	14.520
M16.0 × 1.5	15.220
M16.0 × 1.0	15.520
M17.0 × 1.5	16.270
M17.0 × 1.0	16.520
M18.0 × 2.0	16.950
M18.0 × 1.5	17.230
M18.0 × 1.0	17.520
M20.0 × 2.0	18.950
M20.0 × 1.5	19.230
M20.0 × 1.0	19.520
M22.0 × 2.0	20.950

2. (Metric fine screw) (Continued)

Nominal pitch	Primary hole drill diameter
M22.0 × 1.5	21.230
M22.0 × 1.0	21.520
M24.0 × 2.0	22.950
M24.0 × 1.5	23.230
M24.0 × 1.0	23.520
M25.0 × 2.0	23.950
M25.0 × 1.5	24.230
M25.0 × 1.0	24.520
M26.0 × 1.5	25.230
M27.0 × 2.0	25.950
M27.0 × 1.5	26.770
M27.0 × 1.0	26.520
M28.0 × 2.0	26.950
M28.0 × 1.5	27.230
M28.0 × 1.0	27.520
M30.0 × 3.0	28.390
M30.0 × 2.0	28.950
M30.0 × 1.5	29.230
M30.0 × 1.0	29.520

4. Unified fine screw

Nominal pitch	Primary hole drill diameter
UNF 0-80	1.410
UNF 1-72	1.720
UNF 2-64	2.040
UNF 3-56	2.330
UNF 4-48	2.630
UNF 5-44	2.950
UNF 6-40	3.230
UNF 8-36	3.870
UNF 10-32	4.470
UNF 12-28	5.070
UNF 1/4-28	5.940
UNF 5/16-24	7.430
UNF 3/8-24	9.020
UNF 7/16-20	10.490
UNF 1/2-20	12.050
UNF 9/16-18	13.580
UNF 5/8-18	15.170
UNF 3/4-16	18.250
UNF 7/8-14	21.270

3. Unified coarse screw

Nominal pitch	Primary hole drill diameter
UNC 1-64	1.710
UNC 2-56	2.020
UNC 3-48	2.320
UNC 4-40	2.590
UNC 5-40	2.920
UNC 6-32	3.170
UNC 8-32	3.830
UNC 10-24	4.350
UNC 12-24	4.980
UNC 1/4-20	5.770
UNC 5/16-18	7.240
UNC 3/8-16	8.730
UNC 7/16-14	10.190
UNC 1/2-13	11.680
UNC 9/16-12	13.240
UNC 5/8-11	14.670
UNC 3/4-10	17.740
UNC 7/8-9	20.750

9(12). Miniature screw threads (*)

Nominal pitch	Primary hole drill diameter
S0.4 × 0.1	0.350
S0.5 × 0.125	0.440
S0.6 × 0.15	0.530
S0.7 × 0.175	0.620
S0.8 × 0.2	0.710
S0.9 × 0.225	0.800

(*)

The display number changes depending on the settings for the machine parameters “Display language” and “Change screw type”.

2.12 Cutting Condition (Conversation)

2.12.1 Outline

Set the cutting conditions for the tool according to the material of workpiece. Up to 10 types of workpiece materials can be registered. Make setting for 19 types of tools for each workpiece material.

Workpiece materials Nos. 1 to 5 are preset at shipment, and for workpiece materials Nos. 6 to 10, desired cutting conditions can be set.

2.12.1.1 Edit Screen

Selecting the “Cutting conditions” on the Data bank menu screen displays the workpiece material menu screen.

Material menu 2012/05/29 11:45:28

Menu

- 1. S45C
- 2. SCM435
- 3. FC20
- 4. ADC12
- 5. SUS304
- 6. Special material 1
- 7. Special material 2
- 8. Special material 3
- 9. Special material 4
- 10. Special material 5

Menu

Data bank menu S45C SCM435 FC20 ADC12 SUS304 Special material 1 Special material 2

Selecting the workpiece material for which the cutting conditions are to be set on the workpiece material menu screen displays the edit screen for cutting conditions for the material.

Cutting condition 1(boring tool) 2012/05/29 11:45:45

Material S45C m/min mm/rev.

Tool	Lengthwise	Crosswise
Center drill	12.0	0.05
Drill	- 2.999	10.0
Carbide drill	- 2.999	20.0
Tap (Blind)	- 2.999	12.0
Tap (Through)	- 2.999	15.0
Rolled tap	- 2.999	10.0
Chamfering tool	-	15.0
Counter boring tool	-	15.0

Tool

- 1. Center drill
- 2. Drill
- 3. Carbide drill
- 4. Spiral tap
- 5. Point tap
- 6. Hand tap
- 7. Rolled tap
- 8. Chamfer tool
- 9. CBR tool
- 10. Chamfer drill
- 11. Stepped drill
- 12. Drill tap
- 13. Reamer
- 14. Drill reamer

Completion mode Cutting condition 1 boring tool Cutting condition 2 Milling tool Cutting condition 3 (w/o tools) Previous tool Next tool

Pressing any of the [F1] to [F3] keys changes to the edit screen for the relevant cutting conditions.

- Cutting conditions 1 drilling tools
Set the tool types and cutting conditions for drilling tools.
- Cutting conditions 2 milling tools
Set the tool type and cutting conditions for milling tools.
- Cutting conditions 3 other than tool conditions
Enter the conditions other than tool conditions.

2.12.2 How to Edit

2.12.2.1 Locating Cursor

Locate the cursor at the item to be changed.

For the edit screen for “Cutting condition 1 drilling tools” or the “Cutting conditions 2 milling tools,” pressing the [F4] (Previous tool) or [F5] (Next tool) key locates the cursor at the top of the previous or next tool.

2.12.2.2 Data Change

1. Locate the cursor at the item to be changed.

2. Enter a value in the input area and press the [ENT] key.

This inputs the value at the cursor position and locates the cursor at the next item.

2.12.2.3 Ending an Edit

Upon completion of the setting, press the [F0] (End mode) key to end the editing.

2.12.3 List of Default Values

2.12.3.1 Cutting Conditions 1 (Drill)

1. S45C

Processing method		Registered tool name	Peripheral speed (m/min)	Feed (mm/revolution)	Remarks
Center drill	Lengthwise	Center drill Center drill 1~9	12	0.05	Lengthwise :For center hole cutting Crosswise :For chamfer milling by forced assignment of center drill
	Crosswise		*	*	
Drill	~2.999	Drill Drill 1-9	10	0.05	~2.999: Drill OD less than 2.999mm 3~: Drill OD 3mm or over
	3~		18	0.10	
Carbide drill	~2.999	Carbide drill Carbide drill 1-9	20	0.05	~2.999: Drill OD less than 2.999mm 3~: Drill OD 3mm or over
	3~		30	0.10	
Tapping (blind)	~2.999	Spiral tap Spiral tap 1-9 Point tap Point tap 1-9 Hand tap Hand tap 1-9	12	*	-2.999 :Tap dia. less than 2.999mm 3- :Tap dia. 3mm or over
	3~		12	*	
Tapping (through)	~2.999	Spiral tap Spiral tap 1-9 Point tap Point tap 1-9 Hand tap Hand tap 1-9	15	*	-2.999 :Tap dia. less than 2.999mm 3- :Tap dia. 3mm or over
	3~		15	*	
Rolled tap	~2.999	Rolled tap Rolled tap 1-9	10	*	-2.999 :Tap dia. less than 2.999mm 3- :Tap dia. 3mm or over
	3~		10	*	
Drill with chamfering tool	*	Chamfering tool Chamfering tool 1-9	15	0.10	
Drill with counter boring tool	*	Counterboring tool Counterboring tool 1-9	15	0.07	
Chamfered drill	hole	Chamfered drill Chamfered drill 1~9	18	0.10	For drilling, specify the cutting conditions for drilling with a drill using chamfering tool; for chamfering, specify those for chamfering using a drill with chamfering tool.
	Chamfering		15	0.10	

Processing method		Registered tool name	Peripheral speed (m/min)	Feed (mm/revolution)	Remarks
Stepped drill	hole	Stepped drill Stepped drill 1~9	18	0.10	For drilling, specify the cutting conditions for drilling using a drill with counter boring tool; for counter boring, specify those for counter boring using a drill with counter boring tool.
	Chamfering		15	0.07	
Drill tap	hole	Drill tap Drill tap 1~9	18	0.10	For drilling, specify the cutting conditions for drilling using a drill tap; for tapping, specify those for thread cutting using a drill tap.
	TAP		15	*	
Reamer	~2.999	Reamer Reamer 1-9	6	0.10	~2.999:Reamer OD less than 2.999mm 3~ :Reamer OD 3mm or over
	3~		12	0.20	
Drill reamer	hole	Drill reamer Drill reamer 1~9	18	0.10	For drilling, specify the cutting conditions for drilling using a drill reamer; for reaming, specify those for reaming using a drill reamer.
	Reamer		12	0.20	

2. SCM435

Processing method		Registered tool name	Peripheral speed (mm/min)	Feed (mm/rev)	Remarks
Center drill	Lengthwise	Center drill Center drill 1~9	10	0.05	Lengthwise: For center hole cutting Crosswise: For chamfer milling by forced assignment of center drill
	Crosswise		*	*	
Drill	~2.999	Drill Drill 1-9	7	0.03	~2.999: Drill OD less than 2.999mm 3~: Drill OD 3mm or over
	3~		14	0.08	
Carbide drill	~2.999	Carbide drill Carbide drill 1-9	15	0.03	~2.999: Drill OD less than 2.999mm 3~: Drill OD 3mm or over
	3~		25	0.08	
Tapping (blind)	~2.999	Spiral tap Spiral tap 1-9 Point tap Point tap 1-9 Hand tap Hand tap 1-9	12	*	-2.999 :Tap dia. less than 2.999mm 3- :Tap dia. 3mm or over
	3~		12	*	
Tapping (through)	~2.999	Spiral tap Spiral tap 1-9 Point tap Point tap 1-9 Hand tap Hand tap 1-9	12	*	-2.999 :Tap dia. less than 2.999mm 3- :Tap dia. 3mm or over
	3~		12	*	
Rolled tap	~2.999	Rolled tap Rolled tap 1-9	10	*	-2.999 :Tap dia. less than 2.999mm 3- :Tap dia. 3mm or over
	3~		10	*	
Chamfering tool	*	Chamfering tool Chamfering tool 1-9	12	0.08	
Counterboring tool	*	Counterboring tool Counterboring tool 1-9	12	0.05	

Processing method		Registered tool name	Peripheral speed (mm/min)	Feed (mm/rev)	Remarks
Drill with chamfering tool	hole	Chamfered drill Chamfered drill 1~9	14	0.08	For drilling, specify the cutting conditions for drilling using a drill with chamfering tool; for chamfering, specify those for chamfering using a drill with chamfering tool.
	Chamfering		12	0.08	
Drill with counter boring tool	hole	Stepped drill Stepped drill 1~9	14	0.08	For drilling, specify the cutting conditions for drilling using a drill with counter boring tool; for counter boring, specify those for counter boring using a drill with counter boring tool.
	Counter boring		12	0.05	
Drill tap	hole	Drill tap Drill tap 1~9	14	0.08	For drilling, specify the cutting conditions for drilling using a drill tap; for tapping, specify those for thread cutting using a drill tap.
	Tap		12	*	
Reamer	~2.999	Reamer Reamer 1-9	5	0.10	-2.999 :Reamer OD less than 2.999mm 3- :Reamer OD 3mm or over
	3~		10	0.20	
Drill reamer	hole	Drill reamer Drill reamer 1~9	14	0.08	For drilling, specify the cutting conditions for drilling using a drill reamer; for reaming, specify those for reaming using a drill reamer.
	Reamer		10	0.20	

3. FC20

Processing method		Registered tool name	Peripheral speed (mm/min)	Feed (mm/rev)	Remarks
Center drill	Lengthwise	Center drill Center drill 1~9	15	0.08	Lengthwise is the cutting conditions for center hole cutting, and Crosswise is those for chamfer milling by forced assignment of center drill.
	Crosswise		*	*	
Drill	~2.999	Drill Drill 1-9	13	0.06	~2.999: Drill OD less than 2.999mm 3~: Drill OD 3mm or over
	3~		25	0.15	
Carbide drill	~2.999	Carbide drill Carbide drill 1-9	30	0.06	~2.999: Drill OD less than 2.999mm 3~: Drill OD 3mm or over
	3~		50	0.15	
Tapping (blind)	~2.999	Spiral tap Spiral tap 1-9 Point tap Point tap 1-9 Hand tap Hand tap 1-9	15	*	-2.999 :Tap dia. less than 2.999mm 3- :Tap dia. 3mm or over
	3~		15	*	
Tapping (through)	~2.999	Spiral tap Spiral tap 1-9 Point tap Point tap 1-9 Hand tap Hand tap 1-9	15	*	-2.999 :Tap dia. less than 2.999mm 3- :Tap dia. 3mm or over
	3~		15	*	
Rolled tap	~2.999	Rolled tap Rolled tap 1-9	*	*	-2.999 :Tap dia. less than 2.999mm 3- :Tap dia. 3mm or over
	3~		*	*	
Chamfering tool	*	Chamfering tool Chamfering tool 1-9	15	0.15	

Processing method		Registered tool name	Peripheral speed (mm/min)	Feed (mm/rev)	Remarks
Counterboring tool	*	Counterboring tool Counterboring tool 1-9	15	0.10	
Drill with chamfering tool	hole	Chamfered drill Chamfered drill 1~9	25	0.15	For drilling, specify the cutting conditions for drilling using a drill with chamfering tool; for chamfering, specify those for chamfering using a drill with chamfering tool.
	Chamfering		15	0.15	
Drill with counter boring tool	Drilling	Stepped drill Stepped drill 1~9	25	0.15	For drilling, specify the cutting conditions for drilling using a drill with counter boring tool; for counter boring, specify those for counter boring using a drill with counter boring tool.
	Counter boring		15	0.10	
Drill tap	hole	Drill tap Drill tap 1~9	25	0.15	For drilling, specify the cutting conditions for drilling using a drill tap; for tapping, specify those for thread cutting using a drill tap.
	Tapping		15	*	
Reamer	~2.999	Reamer Reamer 1-9	8	0.10	-2.999 :Reamer OD less than 2.999mm 3- :Reamer OD 3mm or over
	3~		15	0.20	
Drill reamer	hole	Drill reamer Drill reamer1~9	25	0.15	For drilling, specify the cutting conditions for drilling using a drill reamer; for reaming, specify those for reaming using a drill reamer.
	Reamer		15	0.20	

4. ADC12

Processing method		Registered tool name	Peripheral speed (mm/min)	Feed (mm/rev)	Remarks
Center drill	Lengthwise	Center drill Center drill 1~9	30	0.10	Lengthwise: For center hole cutting Crosswise: For chamfer milling by forced assignment of center drill
	Crosswise		*	*	
Drill	~2.999	Drill Drill 1-9	50	0.10	~2.999: Drill OD less than 2.999mm 3~: Drill OD 3mm or over
	3~		50	0.20	
Carbide drill	~2.999	Carbide drill Carbide drill 1-9	100	0.10	~2.999: Drill OD less than 2.999mm 3~: Drill OD 3mm or over
	3~		100	0.20	
Tapping (blind)	~2.999	Spiral tap Spiral tap 1-9 Point tap Point tap 1-9 Hand tap Hand tap 1-9	30	*	-2.999 :Tap dia. less than 2.999mm 3- :Tap dia. 3mm or over
	3~		40	*	
Tapping (through)	~2.999	Spiral tap Spiral tap 1-9 Point tap Point tap 1-9 Hand tap Hand tap 1-9	30	*	-2.999 :Tap dia. less than 2.999mm 3- :Tap dia. 3mm or over
	3~		40	*	

Processing method		Registered tool name	Peripheral speed (mm/min)	Feed (mm/rev)	Remarks
Rolled tap	~2.999	Rolled tap Rolled tap 1-9	40	*	-2.999 : Tap dia. less than 2.999mm 3- : Tap dia. 3mm or over
	3~		50	*	
Chamfering tool	*	Chamfering tool Chamfering tool 1-9	35	0.30	
Counterboring tool	*	Counterboring tool Counterboring tool 1-9	35	0.15	
Drill with chamfering tool	hole	Chamfered drill Chamfered drill 1~9	50	0.20	For drilling, specify the cutting conditions for drilling using a drill with chamfering tool; for chamfering, specify those for chamfering using a drill with chamfering tool.
	Chamfering		35	0.30	
Drill with counter boring tool	hole	Stepped drill Stepped drill 1~9	50	0.20	For drilling, specify the cutting conditions for drilling using a drill with counter boring tool; for counter boring, specify those for counter boring using a drill with counter boring tool.
	Chamfering		35	0.15	
Drill tap	hole	Drill tap Drill tap 1~9	50	0.20	For drilling, specify the cutting conditions for drilling using a drill tap; for tapping, specify those for thread cutting using a drill tap.
	Tapping		30	*	
Reamer	~2.999	Reamer Reamer 1-9	15	0.10	“~2.999” is the cutting conditions for a reamer with a diameter of not more than 2.999 mm; “3~” is those for a reamer with a diameter of not less than 3 mm.
	3~		30	0.20	
Drill reamer	hole	Drill reamer Drill reamer 1~9	50	0.20	For drilling, specify the cutting conditions for drilling using a drill reamer; for reaming, specify those for reaming using a drill reamer.
	Reamer		30	0.20	

5. SUS304

Processing method		Registered tool name	Peripheral speed (m/min)	Feed (mm/R ev.)	Remarks
Center drill	Lengthwise	Center drill Center drill 1~9	8	0.05	Lengthwise is the cutting conditions for center hole cutting, and Crosswise is those for chamfer milling by forced assignment of center drill.
	Crosswise		*	*	
Drill	~2.999	Drill Drill 1-9	6	0.03	“~2.999” is the cutting conditions for a drill of not more than 2.999 mm in OD; “3~” is those for a drill of not less than 3 mm in OD.
	3~		10	0.07	
Carbide drill	~2.999	Carbide drill Carbide drills 1 to 9	10	0.03	“~2.999” is the cutting conditions for a drill of not more than 2.999 mm in OD; “3~” is those for a drill with of not less than 3 mm in OD.
	3~		15	0.07	
Tap (blind)	~2.999	S tap S taps 1~9 P tap P taps 1~9 H tap H taps 1~9	6	*	“~2.999” is the cutting conditions for a tap with a of not more than 2.999 mm in nominal dia.; “3~” is those for a tap with of not less than 3 mm in nominal dia.
	3~		6	*	

Processing method		Registered tool name	Peripheral speed (m/min)	Feed (mm/R ev.)	Remarks
Tap (through)	~2.999	S tap	6	*	“~2.999” is the cutting conditions for a tap of not more than 2.999 mm in nominal dia.; “3~” is those for a tap of not less than 3 mm in nominal dia.
	3~	S taps 1~9 P tap P taps 1~9 H tap H tap 1~9	6	*	
Rolled tap	~2.999	Rolled tap Rolled taps 1~9	6	*	“~2.999” is the cutting conditions for a tap of not more than 2.999 mm in nominal dia.; “3~” is those for a tap with of not less than 3 mm in nominal dia.
	3~		6	*	
Chamfering tool	*	Chamfering tool Chamfering tools 1~9	10	0.07	
Counter boring tool	*	Counter boring tool Counter boring tools 1~9	10	0.05	
Drill with chamfering tool	Drilling	Chamfer drill Chamfer drill 1~9	10	0.07	For drilling, specify the cutting conditions for drilling using a drill with chamfering tool; for chamfering, specify those for chamfering using a drill with chamfering tool.
	Chamfering		10	0.07	
Drill with counter boring tool	Drilling	Stepped drill Stepped drill 1~9	10	0.07	For drilling, specify the cutting conditions for drilling using a drill with counter boring tool; for counter boring, specify those for counter boring using a drill with counter boring tool.
	Counter boring		10	0.05	
Drill tap	Drilling	Drill tap Drill taps 1~9	10	0.07	For drilling, specify the cutting conditions for drilling using a drill tap; for tapping, specify those for thread cutting using a drill tap.
	Tapping		6	*	
Reamer	~2.999	Reamer Reamers 1~9	4	0.10	“~2.999” is the cutting conditions for a reamer of not more than 2.999 mm in OD; “3~” is those for a reamer of not less than 3 mm in OD.
	3~		8	0.20	
Drill reamer	Drilling	Drill reamer Drill reamers 1~9	10	0.07	For drilling, specify the cutting conditions for drilling using a drill reamer; for reaming, specify those for reaming using a drill reamer.
	Reaming		8	0.20	

2.12.3.2 Cutting Conditions 2 (Milling Tools)

1. S45C

Processing method			Registered tool name	Peripheral speed (m/min)	Feed (mm/Rev.)	Remarks
End mill (rough) (Rough finishing end mill)	■ Rough finishing (End mill for drilling/counter boring)	Vertical	End mill (rough) End mills (rough) 1 to 9	15	0.03	End mills (rough) are used for rough finishing; end mills (finish) are used for finishing. For each tool, set the cutting conditions to “rough finishing” when performing rough finishing in facing, pocketing or contouring, or set to “finishing” when performing finishing.
		Horizontal		15	0.05	
	Finishing	Vertical		20	0.03	
		Horizontal		20	0.03	
End mill (finish) (Finishing end mill)	Rough finishing	Vertical	End mill (finish) End mills (finish) 1 to 9	15	0.03	• Vertical and horizontal For vertical, set the cutting conditions for processing with end mill cutter in vertical feed; for horizontal, set those for processing with end mill cutter in horizontal feed. • Face milling For face milling, the cutting conditions marked with ■ on the left hand are effective.
		Horizontal		15	0.05	
	■ Finishing (End mill for drilling/counter boring)	Vertical		20	0.03	
		Horizontal		20	0.03	
Carbide end mill (rough) (Rough finishing end mill)	■ Rough finishing (End mill for drilling/counter boring)	Vertical	Carbide end mill (rough) Carbide end mills (rough) 1 to 9	30	0.03	
		Horizontal		30	0.05	
	Finishing	Vertical		40	0.03	
		Horizontal		40	0.03	
Carbide end mill (finishing) (Carbide end mill for finishing)	Rough finishing	Vertical	Carbide end mill (finishing) Carbide end mills (finishing) 1 to 9	30	0.03	• Face milling For face milling, the cutting conditions marked with ■ on the left hand are effective.
		Horizontal		30	0.05	
	■ Finishing (End mill for drilling/counter boring)	Vertical		40	0.03	
		Horizontal		40	0.03	
Corner rounding cutter		Vertical	Corner rounding cutter Corner rounding cutters 1 to 9	15	0.03	For vertical, set the cutting conditions for processing with corner rounding cutter in vertical feed; for horizontal, set those for processing with corner rounding cutter in horizontal feed.
		Horizontal		15	0.05	

- When performing drilling, finishing or counter boring using an end mill, the cutting conditions marked with ■ are effective as in face milling.
- The unit of feed for milling tool is feed per tooth.

2. SCM435

Processing method			Registered tool name	Peripheral speed (m/min)	Feed (mm/Rev.)	Remarks
End mill (rough) (Rough finishing end mill)	■ Rough finishing (End mill for drilling/counter boring)	Vertical	End mill (rough) End mills (rough) 1 to 9	12	0.03	- For rough finishing and finishing - End mills (rough) are used for rough finishing; end mills (finish) are used for finishing.
		Crosswise		12	0.05	
	Finishing	Vertical		15	0.03	
		Horizontal		15	0.03	
End mill (finish) (Finishing end mill)	Rough finishing	Vertical	End mill (finish) End mills (finish) 1 to 9	12	0.03	Cutting conditions are set for rough finishing when performing a facing, pocketing or contouring and are set for finishing when performing a finishing.
		Horizontal		12	0.05	
	■ Finishing (End mill for drilling/counter boring)	Vertical		15	0.03	
		Horizontal		15	0.03	
Carbide end mill (rough) (Rough finishing end mill)	■ Rough finishing (End mill for drilling/counter boring)	Vertical	Carbide end mill (rough) Carbide end mills (rough) 1 to 9	20	0.03	Lengthwise/Crosswise: For vertical, set the cutting conditions for processing with end mill cutter in vertical feed; for horizontal, set those for processing with end mill cutter in horizontal feed.
		Horizontal		20	0.05	
	Finishing	Vertical		25	0.03	
		Horizontal		25	0.03	
Carbide end mill (finishing) (Carbide end mill for finishing)	Rough finishing	Vertical	1~9 Carbide end mill (finish) Carbide end mills (finish) 1 to 9	20	0.03	FACE MILL Effective on the cutting condition marked with ■.
		Horizontal		20	0.05	
	■ Finishing (End mill for drilling/counter boring)	Vertical		25	0.03	
		Horizontal		25	0.03	
Corner rounding cutter		Vertical	Corner rounding cutter Corner rounding cutters 1 to 9	12	0.03	For vertical, set the cutting conditions for processing with corner rounding cutter in vertical feed; for horizontal, set those for processing with corner rounding cutter in horizontal feed.
		Horizontal		12	0.05	

- In hole-machining, hole-facing or spot-facing by end mill, cutting condition marked with ■ is effective as in milling.
- Feed of a mill is per tooth.

3. FC20

Cutting method			Registered tool	Peripheral speed (mm/min)	Feed (mm/rev)	Remarks
End mill (rough) (Rough finishing end mill)	■ Rough finishing (End mill for drilling/counter boring)	Vertical	End mill (rough) End mills (rough) 1 to 9			• For rough finish ing and finishing End mills (rough) are used for rough finishing; end mills (finish) are used for finishing.
		Horizontal				
	Finishing	Vertical				
		Horizontal				
End mill (finish) (Finishing end mill)	Rough finishing	Vertical	End mill (finish) End mills (finish) 1 to 9			Cutting conditions are set for rough finishing when performing a facing, pocketing or contouring and are set for finishing when performing a finishing.
		Horizontal				
	■ Finishing (End mill for drilling/counter boring)	Vertical				
		Horizontal				
Carbide end mill (rougnt) (Rough finishing end mill)	■ Rough finishing (End mill for drilling/counter boring)	Vertical	Carbide end mill (rough) Carbide end mills (rough) 1 to 9	25	0.03	
		Horizontal		25	0.05	
	Finishing	Vertical		30	0.03	
		Horizontal		30	0.03	
Carbide end mill (finishing) (Carbide end mill for finishing)	Rough finishing	Vertical	1~9 Carbide end mill (finishing) Carbide end mills (finishing) 1 to 9	25	0.03	• Vertical and horizontal For vertical, set the cutting conditions for processing with end mill cutter in vertical feed; for horizontal, set those for processing with end mill cutter in hozirontal feed.
		Horizontal		25	0.05	
		Vertical		30	0.03	
	■ Finishing (End mill for drilling/counter boring)	Horizontal		30	0.03	
Corner rounding cutter		Vertial	Corner rounding cutter Corner rounding cutters 1 to 9	15	0.03	For vertical, set the cutting conditions for processing with corner rounding cutter in vertical feed; for horizontal, set those for processing with corner rounding cutter in hozirontal feed.
		Horizontal		15	0.05	

- In hole-machining, hole-facing or spot-facing by end mill, cutting condition marked with ■ is effective as in milling.
- The unit of feed for milling tool is feed per tooth.

4. ADC12

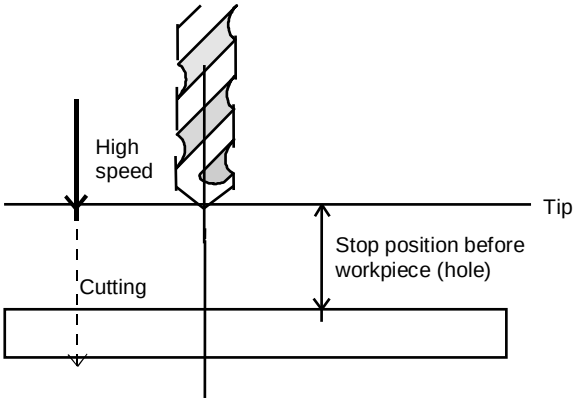
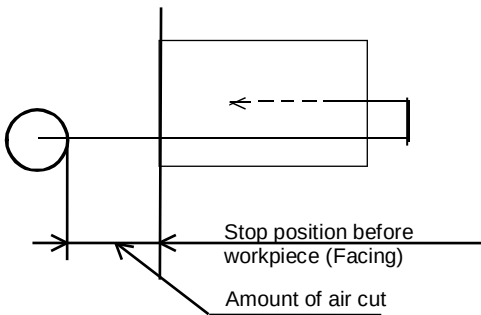
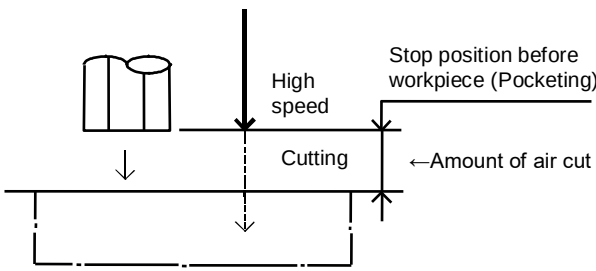
Processing method			Registered tool name	Peripheral speed (m/min)	Feed (mm/Rev.)	Remarks
End mill (rough) (Rough finishing end mill)	■ Rough finishing (End mill for drilling/counter boring)	Vertical	End mill (rough) End mills (rough) 1 to 9	50	0.05	• For rough finishing and finishing End mills (rough) are used for rough finishing; end mills (finish) are used for finishing.
		Horizontal		50	0.10	
	Finishing	Vertical		75	0.05	
		Horizontal		75	0.05	
		Horizontal		75	0.05	
End mill (finish) (Finishing end mill)	Rough finishing	Vertical	End mill (finish) End mills (finish) 1 to 9	50	0.05	For each tool, set the cutting conditions to “rough finishing” when performing rough finishing in facing, pocketing or contouring, or set to “finishing” when performing finishing.
		Horizontal		50	0.10	
	■ Finishing (End mill for drilling/counter boring)	Vertical		75	0.05	
		Horizontal		75	0.05	
		Horizontal		75	0.05	
Carbide end mill (rough) (Rough finishing end mill)	■ Rough finishing (End mill for drilling/counter boring)	Vertical	Carbide end mill (rough) Carbide end mills (rough) 1 to 9	100	0.05	
		Horizontal		100	0.10	
	Finishing	Vertical		150	0.03	
		Horizontal		150	0.03	
		Horizontal		150	0.03	
Carbide end mill (finishing) (Carbide end mill for finishing)	Rough finishing	Vertical	Carbide end mill (finish) Carbide end mills (finish) 1 to 9	100	0.05	• Vertical and horizontal For vertical, set the cutting conditions for processing with end mill cutter in vertical feed; for horizontal, set those for processing with end mill cutter in horizontal feed. • Face milling For face milling, the cutting conditions marked with ■ on the left hand are effective.
		Horizontal		100	0.10	
	■ Finishing (End mill for drilling/counter boring)	Vertical		150	0.03	
		Horizontal		150	0.03	
		Horizontal		150	0.03	
Corner rounding cutter		Vertical	Corner rounding cutter Corner rounding cutters 1 to 9	50	0.05	For vertical, set the cutting conditions for processing with corner rounding cutter in vertical feed; for horizontal, set those for processing with corner rounding cutter in horizontal feed.
		Horizontal		50	0.10	

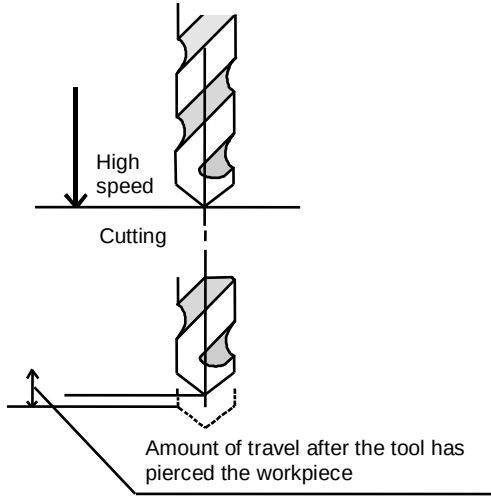
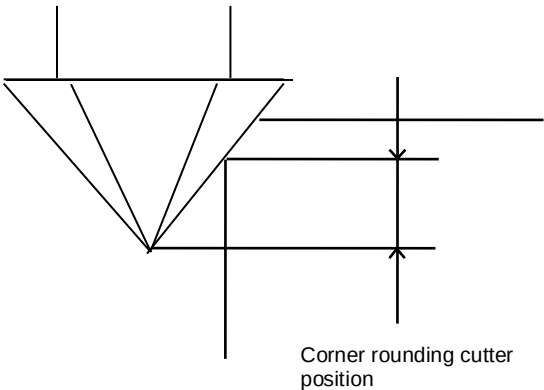
- When performing drilling, finishing or counter boring using an end mill, the cutting conditions marked with ■ are effective as in face milling
- The unit of feed for milling tool is feed per tooth.

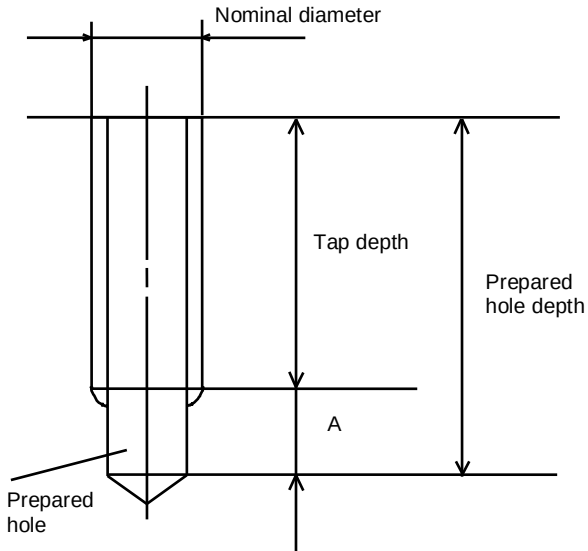
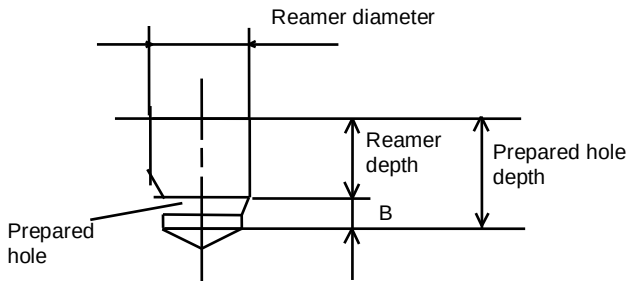
5. SUS304

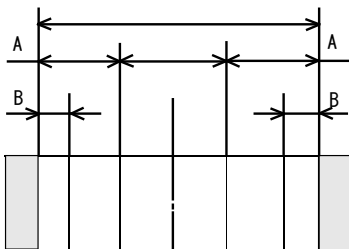
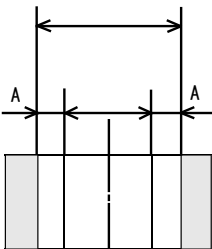
Processing method			Registered tool name	Peripheral speed (m/min)	Feed (mm/Rev.)	Remarks
End mill (rough) (Rough finishing end mill)	■ Rough finishing (End mill for drilling/counter boring)	Vertical	End mill (rough) End mills (rough) 1 to 9	10	0.03	• For rough finishing and finishing End mills (rough) are used for rough finishing; end mills (finish) are used for finishing.
		Horizontal		10	0.05	
	Finishing	Vertical		12	0.03	
		Horizontal		12	0.03	
End mill (finish) (Finishing end mill)	Rough finishing	Vertical	End mill (finish) End mills (finish) 1 to 9	10	0.03	For each tool, set the cutting conditions to “rough finishing” when performing rough finishing in facing, pocketing or contouring, or set to “finishing” when performing finishing.
		Horizontal		10	0.05	
	■ Finishing (End mill for drilling/counter boring)	Vertical		12	0.03	
		Horizontal		12	0.03	
Carbide end mill (rough) (Rough finishing end mill)	■ Rough finishing (End mill for drilling/counter boring)	Vertical	Carbide end mill (rough) Carbide end mills (rough) 1 to 9	15	0.03	
		Horizontal		15	0.05	
	Finishing	Vertical		20	0.03	
		Horizontal		20	0.03	
Carbide end mill (finishing) (Carbide end mill for finishing)	Rough finishing	Vertical	Carbide end mill (finish) Carbide end mills (finish) 1 to 9	15	0.03	• Vertical and horizontal For vertical, set the cutting conditions for processing with end mill cutter in vertical feed; for horizontal, set those for processing with end mill cutter in horizontal feed.
		Horizontal		15	0.05	
	■ Finishing (End mill for drilling/counter boring)	Vertical		20	0.03	
		Horizontal		20	0.03	
		Horizontal		20	0.03	
		Horizontal		20	0.03	
Corner rounding cutter		Vertical	Corner rounding cutter Corner rounding cutters 1 to 9	10	0.03	For vertical, set the cutting conditions for processing with corner rounding cutter in vertical feed; for horizontal, set those for processing with corner rounding cutter in horizontal feed.
		Horizontal		10	0.05	

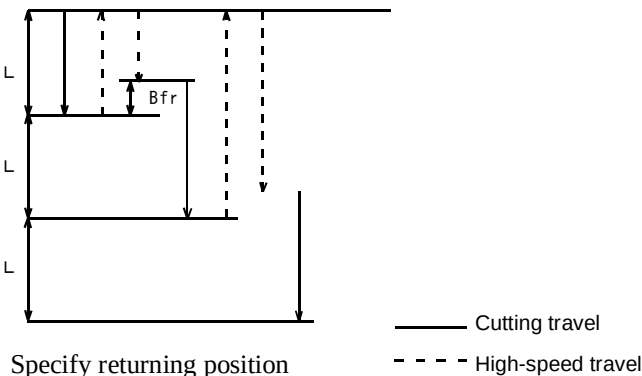
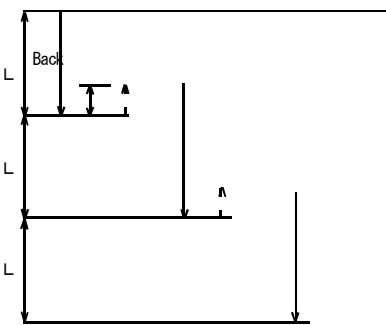
2.12.3.3 Cutting Conditions (Other than Tool Conditions)

Item name	Setting range	Description
Stop pos. before workpiece (hole)	0~99.999 mm	* Hole machining Distance (air cut stroke) between the switchover point (from rapid traverse to cutting feed rate) and a machining surface is set.  If workpiece thickness is excessively uneven due to forged surface, set value requires some allowance. Take top surface cutting into account.
Stop position before workpiece (facing)	0~99.999 mm	• Facing Distance (air cut stroke) between a workpiece and a cutting start position is set. 
Stop position before workpiece (pocketing)	0~99.999 mm	• Distance (air cut stroke) between the switchover point (from rapid traverse to cutting feed rate) and a machining surface is set. 

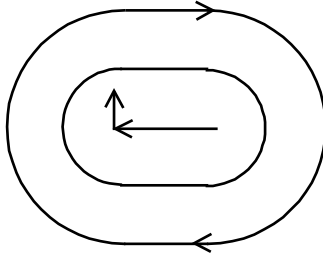
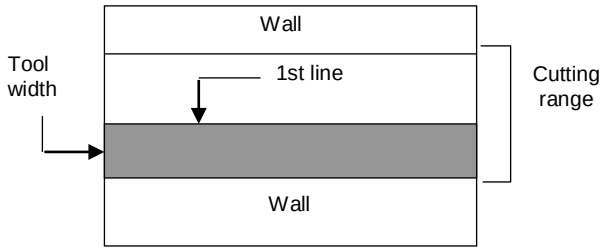
Item name	Setting range	Description
Cutting length after drilling	0~99.999 mm	Set an amount of travel after the tool has pierced the workpiece when drilling a through hole. 
Chamfering tool position	0~99.999 mm	Relative positions of chamfering tool and workpiece in pocketing or chamfering are set. 

Item name	Setting range	Description
Tap prepared hole depth margin	0~99.999 thread	Difference in depth between primary hole and tapped hole is set in terms of number of threads If drilled hole is less deep than tapped hole in blind hole cutting, use the parameter. $\text{Tap prepared hole depth margin} = \frac{\text{Prepared hole depth} - \text{Tap}}{\text{Pitch}} = \frac{A}{\text{Pitch}}$  • When the value of the prepare hole depth of processing data during tap-related processing is equal to that of tap depth or smaller than that (0 included), processing is executed according to the formula: (Prepare hole depth) = (Tap depth) + (Tap prepared hole depth margin) × (Pitch).
Reamer prepared hole margin depth	0~99.999 times	Difference in depth between primary hole and reamed hole is set by its ratio to reamer diameter If primary hole is less deep than reamed hole in blind hole cutting, use the parameter. $\text{Reamer prepared hole depth margin} = \frac{\text{Prepared hole depth} - \text{Reamer}}{\text{Reamer diameter}} = \frac{B}{\text{Reamer diameter}}$  • When the value of the prepare hole depth of processing data during reamer-related processing is equal to that of reamer depth or smaller than that (0 included), processing is executed according to the formula: (Prepare hole depth) = (Reamer depth) + (Reamer prepared hole margin depth) × (Pitch).

Item name	Setting range	Description																																
Reamer finishing margin 1 (~2.999) (3.000~)	0~99.999 mm	Finishing allowance for 1-step finish or for the 1st step finish in 2-step finish reaming. Set A+A.  1-step finish 2-step finish * 0~2.999(0~0.1180):For reamer diameter less than 2.999mm (3mm.) * 3~(0.1181~) :For reamer diameter 3mm(0.1181in.) or over																																
Reamer finish margin 2 (~2.999) (3.000~)	0 ~ 99.999 mm	Finishing allowance for the 2nd step finish in 2- step finish reaming. Set B+B, referring to the figure above. When using a drill and reamer in this order to prepare a hole in a tap job, use a drill with a diameter of (Prepare hole drill diameter – Reamer finishing margin 2).																																
Drill step method	0~2	Select the stepping mode in drilling from among the following 3:																																
Drill step ratio	0~999%	0: No step Does not step. 1 : Constant Depth of cut is constant in all steps. Depth of cut can be calculated in the following formula. $L(\text{mm}) = \frac{\text{Drill outer dia. (mm)} \times \text{Drill step ratio (\%)}}{100}$ 2 : REDUCING Cut of depth becomes smaller as drilled depth becomes larger.Depth of cut can be calculated in the following formula. $L(\text{mm}) = \frac{\alpha \times \text{Drill Outer dia. (mm)} \times \text{Drill step ratio (\%)}}{100}$ L_n : Nth time infeed α : Cutting coefficient <table><tr><td>N</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>α</td><td>4.0</td><td>3.3</td><td>2.7</td><td>2.1</td><td>1.7</td><td>1.4</td><td>1.2</td></tr></table> <table><tr><td>N</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td></tr><tr><td>α</td><td>0.9</td><td>0.7</td><td>0.6</td><td>0.4</td><td>0.4</td><td>0.3</td><td>0.3</td></tr></table> *For the 15th and after, use the step coefficient of the 14th. *The infeed amount is calculated by dividing the pitch by 10, when the infeed amount is less than the quotient of pitch/10 in step method 1 (constant) or step method 2 (variable).	N	1	2	3	4	5	6	7	α	4.0	3.3	2.7	2.1	1.7	1.4	1.2	N	8	9	10	11	12	13	14	α	0.9	0.7	0.6	0.4	0.4	0.3	0.3
N	1	2	3	4	5	6	7																											
α	4.0	3.3	2.7	2.1	1.7	1.4	1.2																											
N	8	9	10	11	12	13	14																											
α	0.9	0.7	0.6	0.4	0.4	0.3	0.3																											

Item name	Setting range	Description
Drill step return position	0~1	Select a required drill POINT step return position from the following 2 types:
Drill step return stroke	0~99.999 mm	0: Start position The tool returns to the start position of the first drill step. 1: Return position The tools returns to a position obtained by combining the start position of the last drill step and the drill step return amount. Drill step return amount is effective only when the drill step return position is set to "1: Return pos". Designation of start point  Specify returning position  Drill step start position L: Depth of cut for 1 step *This stroke is determined by the selected drill step and ratio. Bfr: The stop position before cutting in drill step mode when start position is set is fixed to 0.5 mm. Back: Drill step return

Item name	Setting range	Description																																
Tap step mode	0~2	Select the step method for tapping from the following three methods. 0: No step Does not step. 1: Constant Depth of cut is constant in all steps. Depth of cut can be calculated in the following way: $L(\text{mm}) = \frac{\text{Tap outer dia. (mm)} \times \text{Tap step ratio (\%)}}{100}$ 2: REDUCING Cut of depth becomes smaller as tapped depth comes larger. Depth of cut can be calculated in the following way: $L(\text{mm}) = \frac{\alpha \times \text{Tap outer dia. (mm)} \times \text{Tap step ratio (\%)}}{100}$ L_n : Nth time infeed α: Cutting coefficient <table><tr><td>N</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>α</td><td>2.0</td><td>1.7</td><td>1.3</td><td>1.1</td><td>0.8</td><td>0.7</td><td>0.6</td></tr></table> <table><tr><td>N</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td></tr><tr><td>α</td><td>0.4</td><td>0.4</td><td>0.3</td><td>0.2</td><td>0.2</td><td>0.1</td><td>0.1</td></tr></table> *For the 15th and after, use the step coefficient of the 14th. *The infeed amount is calculated by dividing the pitch by 10, when the infeed amount is less than the quotient of pitch/10 in step method 1 (constant) or step method 2 (variable).	N	1	2	3	4	5	6	7	α	2.0	1.7	1.3	1.1	0.8	0.7	0.6	N	8	9	10	11	12	13	14	α	0.4	0.4	0.3	0.2	0.2	0.1	0.1
N	1	2	3	4	5	6	7																											
α	2.0	1.7	1.3	1.1	0.8	0.7	0.6																											
N	8	9	10	11	12	13	14																											
α	0.4	0.4	0.3	0.2	0.2	0.1	0.1																											
Tap return speed (ratio)	100~999 %	With respect to the one in downward motion. * The feed rate never exceeds the max. tapping revolving speed.																																
Dwell (chamfering)	0 ~ 99×0.1sec	Specify Z-axis dwell time for a post-chamfering stop for better accuracy. This setting is effective also for the drill, drill tap, drill reamer and stepped drill forcibly assigned for chamfering and for machining with end mill R and F and carbide end mill R and F.																																
Dwell (counter boring)	0 ~ 99×0.1sec	Set the stop time for temporarily stopping the Z-axis to increase accuracy after counter boring. This setting is also valid when machining using an end mill (rough, finish) or carbide end mill (rough, finish).																																

Item name	Setting range	Description
Pocket Override	1~100%	During a pocketing, a tool under-goes cutting load. Therefore, apply the override to a feed rate when the tool undergoes the load. Feed rate can be calculated by the following formula. $\text{Actual feed (mm/revolution)} = \text{F (Feed obtained based on cutting conditions) (mm/revolution)} \times \frac{\text{Pocket override}}{100}$  Pocketing override is effective on the part indicated by the bold arrowed line on the figure on the left side.
Facing Override	1~100%	This setting is valid when the cutting direction in a facing job is in the X-axis or Y-axis direction (wall on both sides). Because the first line cutting is heavy cutting, override is applied on the feed calculated from the machining data. 
ATC simultaneous operation start position	-9999.99 9~ 9999.999 mm	Indicates the start position of the Z axis in the machine coordinates when the X, Y, and Z axes have moved upon tool replacement. This value is indicated in the machine coordinates. (Note) Considering that the X and Y axes move upon tool replacement, set the start position so that these axes do not make contact with the workpiece.
Bottom hole depth tolerance	0.00~ 0.999mm	Compound tools for drilling blind holes can be assigned in the setting range according to the depth of drill hole bottom and drilling length.
Reamer return speed 0:Rapid feed 1:Cutting feed	0~1	Reamer return feed rate is set: rapid feed rate or jog feed rate
Z-axis zero return at memory operation start 0: No 1: Yes	0~1	When starting a memory operation, whether to return Z axis to zero point or not is set.

List of default values for cutting condition 3

Items	metric	inch
Stop pos. before workpiece (hole)	2.000	0.1000
Stop pos. before workpiece (facing)	2.000	0.1000
Stop pos. before workpiece (pocket)	2.000	0.1000
Cutting length after drilling	2.000	0.1000
Corner rounding cutter position	2.000	0.1000
Tap hole	6.000	6.000
Reamer hole	1.000	1.000
Reamer finish margin 1 (- 2.999)	0.000	0.0000
Reamer finish margin 2 (- 2.999)	0.000	0.0000
Reamer finish margin 1 (- 3.000)	0.000	0.0000
Reamer finish margin 2 (- 3.000)	0.000	0.0000
Drill step mode	2	2
Step ratio		
[S45C]	100	100
[SCM435]	80	80
[FC20]	200	200
[ADC]	300	300
[SUS304]	70	70
Drill step return	0	0
Drill step return	0.500	0.0200
Tap step mode	2	2
Tap step ratio		
[S45C]	100	100
[SCM435]	80	80
[FC20]	200	200
[ADC]	300	300
[SUS304]	70	70
Tap return speed (ratio)	100	100
Dwell (chamfering)	0	0
Dwell (counter boring)	0	0
Pocket override	70	70
Facing override	70	70
ATC simultaneous operation start position		
S300X1/S500X1/S700X1/S1000X1 (Standard) S300X2/S500X2/S700X2 (Standard)	480.000	18.8976
S300X1/S500X1/S700X1/S1000X1 (High column 150) S300X2/S500X2/S700X2 (High column 150)	630.000	24.8031
S300X1/S500X1/S700X1/S1000X1 (High column 250) S300X2/S500X2/S700X2 (High column 250)	730.000	28.7410
S300X1/S500X1/S700X1 (High column 300) S300X2/S500X2/S700X2 (High column 300)	780.000	30.70870
S1000X1 (High column 350)	830.000	32.67720
F600X1	550.000	21.6535
R450X1 (Standard) R450X2 (Standard)	505.000	19.8818
R450X1 (Low base) R450X2 (Low base)	555.000	21.8503
R650X1 (Standard) R650X2 14/22MG (Standard)	555.000	21.8503
R650X1 (Low base) R650X2 14/22MG (Low base)	625.000	24.6062
R650X2 40MG (Standard)	350.000	13.7796
R650X2 40MG (Low base)	420.000	16.5355
Bottom hole depth tolerance	0.100	0.100
Reamer return speed	0	0
Z-axis zero return at memory operation start	1	1

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CHAPTER 3

3

GRAPHIC

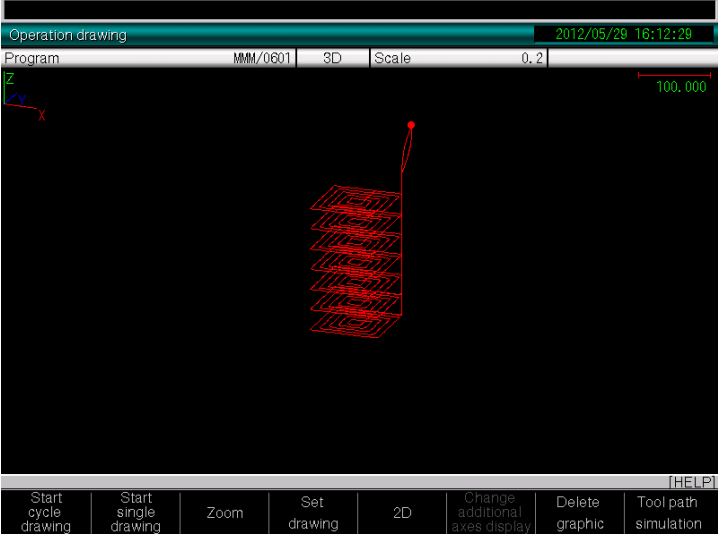
- 3.1** **Outline**
- 3.2** **Operation Drawing**
- 3.3** **Tool Path Simulation**

3.1 Outline

3.1.1 Outline

The program path can be confirmed by drawing it on the <Operation drawing> screen.
The drawn path indicates the tool path.

1. Press the **[GRAPH]** key.
The <Operation drawing> screen appears.



- Scale Displays the graphic scale.
- Scale mark A red line (one tenth of the width of the screen) with green digits is displayed at the top right of the screen. The green digits indicate the dimensional relationship of the display to the actual size of the object.

There are two graphic modes: operation drawing mode and tool path simulation mode.

	Outline
Operation drawing	The operation path of programs stored for memory operation in the memory operation mode is drawn. This is available only in the memory operation mode. Drawing during operation: The movement path during memory operation, manual operation, or MDI operation is drawn.
Tool path simulation	The operation path of the program is drawn in any mode. Since any program number can be specified and executed, the graphic of the next memory operation program can be checked during the current memory operation.

There are two graphic display modes: 2D (two-dimensional) and 3D (three-dimensional).

	Outline
2D	Displays the graphic when viewed from the plane set for [2D drawing plane]. When [3-plane drawing] is selected, the screen is divided into four segments, and a view from one of three directions is displayed in three of these segments.
3D	Displays the graphic with the coordinate system set for [3D drawing plane] rotated by the angle set for [Horizontal rotation angle] or [Vertical rotation angle].

Graphic drawing is subject to the following display restrictions:

	Restrictions
Operation drawing	When the path of the program run at high speed is drawn, the displayed path may become rough. When drawing before operation or drawing during operation is performed for a long time, “Too many drawing points (operation before drawing/operation during drawing)” is displayed at the bottom of the section, and old points may not be displayed when the viewpoint is changed.
Tool path simulation	When tool path simulation is performed for a long time, “Too many drawing points (tool path simulation)” is displayed at the bottom of the section, and old points may not be displayed when the viewpoint is changed.

3.1.2 Changing between Operation Drawing and Tool Path Simulation

- Changing from operation drawing to tool path simulation
 - Press the **[F7]** (Tool path simulation) key.
- Changing from tool path simulation to operation drawing
 - Press the **[F7]** (Operation drawing) key.

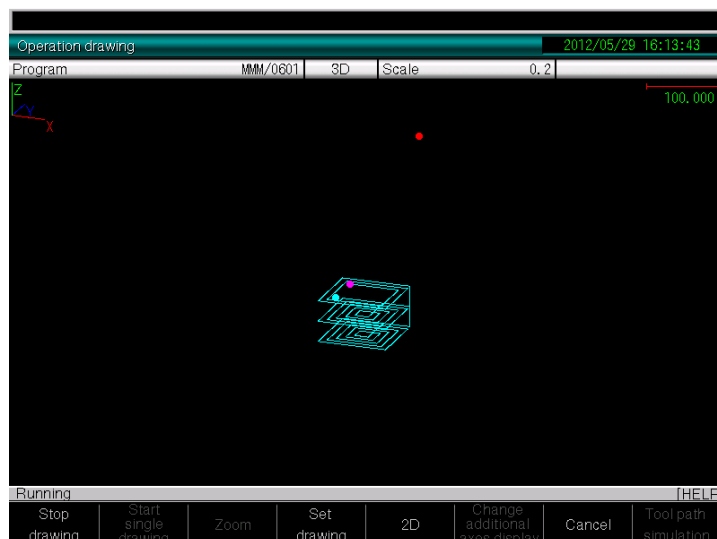
(Note) Changing the mode between operation drawing and tool path simulation is not possible during cycle drawing or single drawing. (It is possible while block stop is applied.)

3.1.3 Changing between 2D and 3D

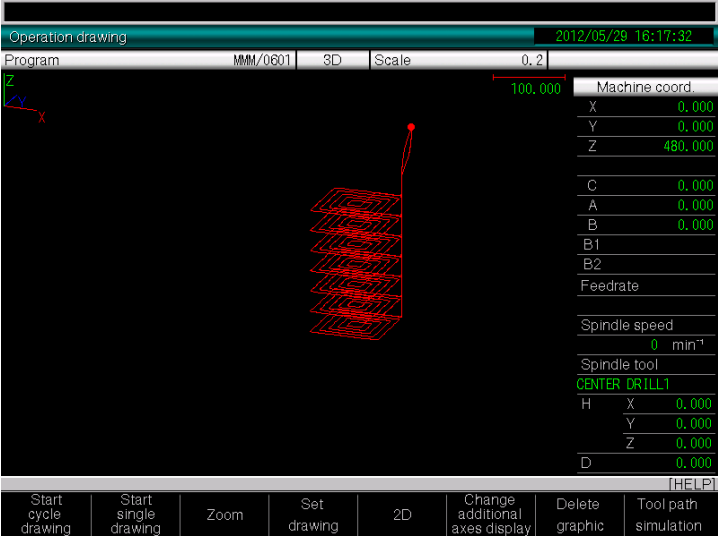
- Changing from 2D to 3D
 - Press the **[F4]** (3D) key.
- Changing from 3D to 2D
 - Press the **[F4]** (2D) key.

3.1.4 Drawing Size and Coordinate Display

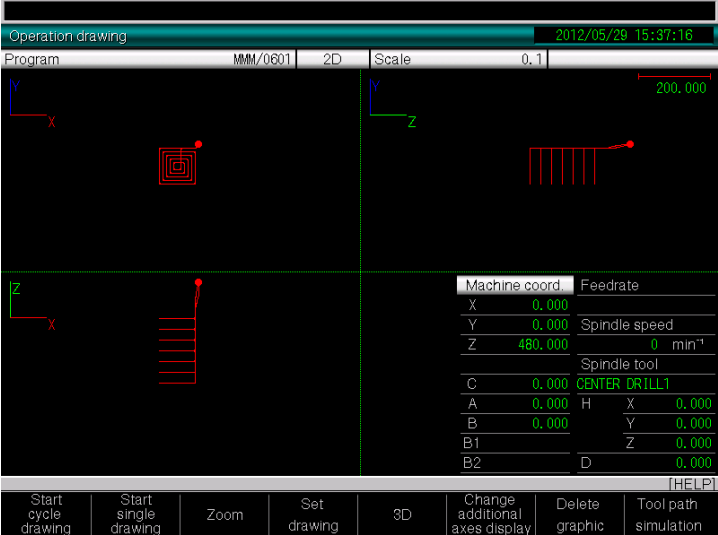
The current drawing data can be displayed on the screen by changing the settings for <Coord. display> on the drawing setting screen during both operation drawing and tool path simulation. When <0: No> is set for <Coord. display>, the entire graphic is displayed on the screen.



When <1: Machine coord.>, <2: Absolute coord.>, or <3: Relative coord.> is set for <Coord. display>, the drawing data is displayed on the right of the graphic.



For 3-plane drawings, the drawing data is displayed at the bottom right of the screen.



The display for the absolute coordinate position when <2: Absolute coordinates> is selected is the same as the screen.

Every time when the [F5] (To change additional axis displays) key is pressed, the axis where the display of an additional is set in the user parameter <4-8th row display axis> and the axis set in <9-12th row display axis> are alternately displayed.

	Actual operation information display (*1)	Simulation information display (*1)
Coordinates (X-, Y- and Z-axes) (*2)	Displays the current coordinates in the specified coordinate system. If using the NC language, when adding the tool length/tool position offset to the absolute coordinate position, the (H) display is added to the X-, Y- and Z-axes coordinates and the tool number is displayed under the Z-axis. (*3)	Displays the current simulation coordinates in the specified coordinate system. If using the NC language, when adding the tool length/tool position compensation while the absolute coordinates are displayed, the (H) display is added to the X-, Y- and Z-axes coordinates and the tool number is displayed under the Z-axis. (*3)
Coordinates (4-8 axes, PLC axis)	For an additional axis (4-8 axes, PLC axis), the axis set in the user parameters <4th row display axis> to <8th row display axis> and the axis set in the <9th row display axis> to <12th row display axis> are alternately displayed every time when the [F5] (Change additional axis display) key is pressed.	For an additional axis (4-8 axes, PLC axis), the axis set in the user parameters <4th row display axis> to <8th row display axis> and the axis set in the <9th row display axis> to <12th row display axis> are alternately displayed every time when the [F5] (Change additional axis display) key is pressed. However, the value of the PLC axis is not displayed.
Feedrate	Displays the machine's actual feed rate.	Displays the feed rate of a simulation program.
Spindle speed	Displays the machine's actual spindle speed.	Displays the spindle speed of a simulation program.
Spindle tool	Displays the name of the tool currently attached to the spindle.	Displays the name of the tool regarded as the spindle tool in simulation.
H (Tool length)	Displays the tool length/tool position compensation during actual operation.	Displays the tool length/tool position compensation in the simulation.
D (Tool diameter/nose R)	Displays the cutter/nose R compensation during actual operation.	Displays the cutter/nose R compensation in the simulation.

*1 The <Operation drawing> screen displays the simulation data when drawing before operation is on and the actual operation data when it is off.
The <Tool path simulation> screen always displays the simulation data.

*2 If absolute coordinates are displayed and the <Display coordinates during rotational transformation> user parameter has been set to <1: Before>, a "(R)" will be displayed together with the coordinate values before turning while coordinate turning is in progress.
(Example: 123.456(R)) If <0: After> has been set, the values after coordinate turning has been applied will be displayed. (The "R" will not be displayed.)

*3 If the absolute coordinates are displayed in the NC language, when adding the tool length/tool position offset to the position of the absolute coordinates and when the <XYZ-axis absolute coord. Display> setting in the user parameter is configured to "1: Display 2" or "2: Display 3," that tool number is displayed under the Z-axis.
The tool number is displayed similar to "(H99)".

3.1.5 Display of Drawing Alarm

Alarm messages and information about alarm occurrence are displayed when an alarm occurs in a drawing program during drawing. When <Coord. display> is set to other than<0: No> on the <Tool path simulation> screen, the coordinates are displayed. Information when the alarm occurs is also displayed in the tool information display. Such information is displayed until new drawing starts or drawing programs are changed (during tool path drawing).

- For NC language
An alarm message is displayed at the bottom of the screen. The block number of alarm occurrence and the block are displayed.



- For conversational language
An alarm message is displayed at the bottom of the screen. If the alarm occurs in a program, information about motion and machining when the alarm occurs are displayed at the top of the screen.



3.2 Operation Drawing

3.2.1 Outline

You can draw a rough graphic of a program reserved for memory operation in the memory operation mode.

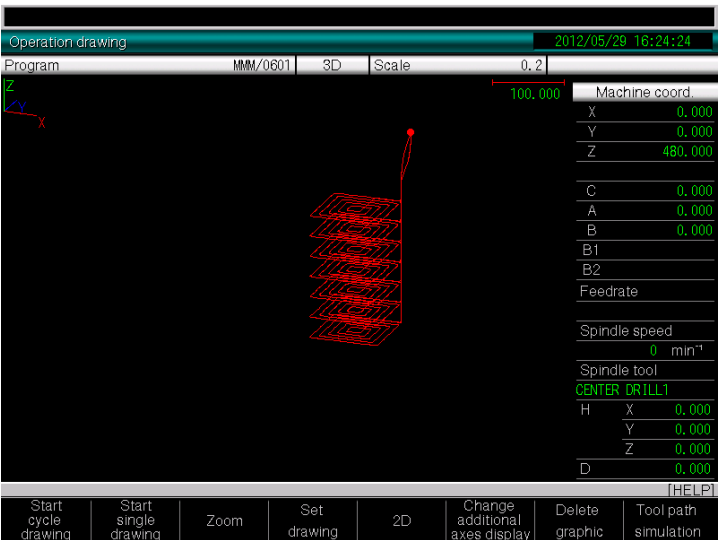
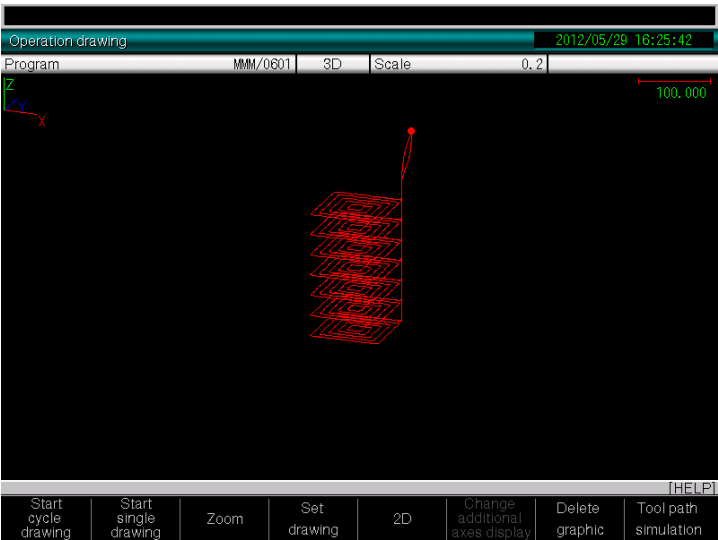
If an operation program is in a root folder, only program number is displayed. In other case, the “Folder name/Program number” is displayed.

Drawing is available only in memory operation. Drawing before operation automatically performs simulation and displays the path.

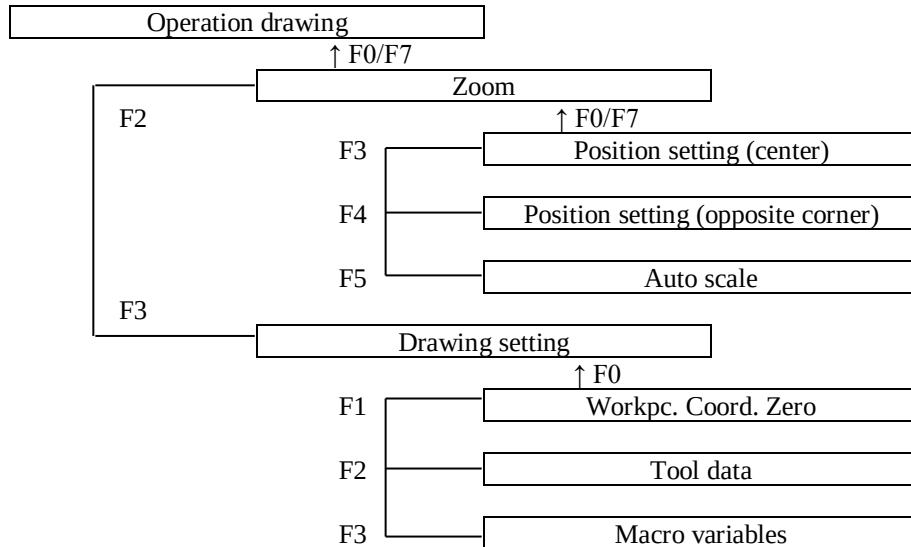
In addition, you can draw a rough movement path in memory operation, manual operation, and MDI operation and check the tool path. (Drawing during operation)

OPERATION

1. Press the **[MEM]** key to call the memory operation mode.
2. Press the **[GRAPH]** key to display the <Operation drawing> screen.
3. Check that <Operation drawing> is displayed at the top left of the screen.
If <Tool path simulation> is displayed, press the **[F7]** (Operation drawing) key.



Screen flow



3

3.2.2 Drawing during Operation

The path of machine movement is drawn.

The tool path is drawn in a specified color.

The current tool position is indicated with a red “●” symbol.

The tool's next position is indicated with a blue “●” symbol in memory operation.

(Note 1) The path is not drawn in the following cases:

When <0: No> is set for <Drawing during operation>

When manual or MDI operation mode is entered while <1: Memory mode only> is set for <Drawing during operation>

During memory operation block not within the range set for “Start / End sequence No.” (NC language)

When performing a job not within the range set for “Start / End job No.” (conversational language)

(Note 2) The drawn path is plotted per specified time based on an NC program. So the actual path at a corner slightly differs from the drawn path. In addition, a smooth override path in the high accuracy mode is not reflected to the drawn path.

3.2.3 Cycle Drawing

The program saved and reversed for memory operation is drawn.

The current position of a simulation tool is indicated with a purple “●” symbol in drawing.

The next position of the simulation tool is indicated with a blue “●” symbol.

Operation

1. Press the [F0] (Start cycle drawing) key.
2. Drawing starts.

(Note 1) Cycle drawing is possible only in the memory operation mode.

The drawing start position is set in <Start sequence NO.> on the <drawing setting> screen.

Drawing stops at a block containing M00. Press the [F0] (Cycle drawing start) key again to continue.

Cycle drawing is not possible in tape operation.

(Note 2) A drawing path is created according to NC programs. So the actual path at a corner slightly differs from the drawn path. In addition, a smooth override path in the high accuracy mode is not reflected to the drawn path.

3.2.4 Single Drawing

The path of the program stored in memory operation is drawn one block at a time.

Operation

1. Press the **[F1]** (Start single drawing) key.
2. Press the **[F1]** (Start single drawing) key again.
The next block is drawn.

(Note 1) Cycle drawing is possible only in the memory operation mode.
The drawing start position is set in <Start sequence NO.> on the <drawing setting> screen.

Cycle drawing is not possible in tape operation.

(Note 2) If the user stops a single operation in the pre-operation graph, after changing to another program in the memory operation screen and returning to the operation graph screen, the number of the program, to which the user changed, is displayed in the memory operation screen. In this situation, press the **[F1]** key (start single graph) to open the graph again. The program number is displayed on the screen for the program that was stopped. However, if the user stops the newly opened program, it reverts back to the previous program number on the memory operation screen.

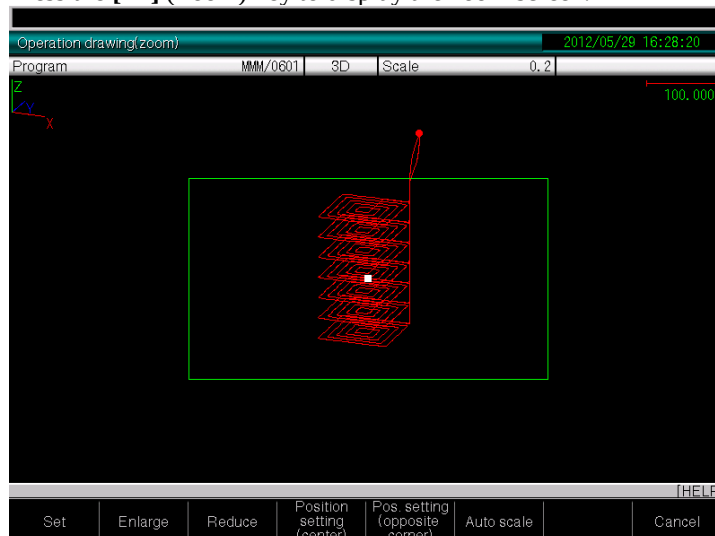
3.2.5 Zoom

The scale of graphic and the center of display can be changed.

Operation

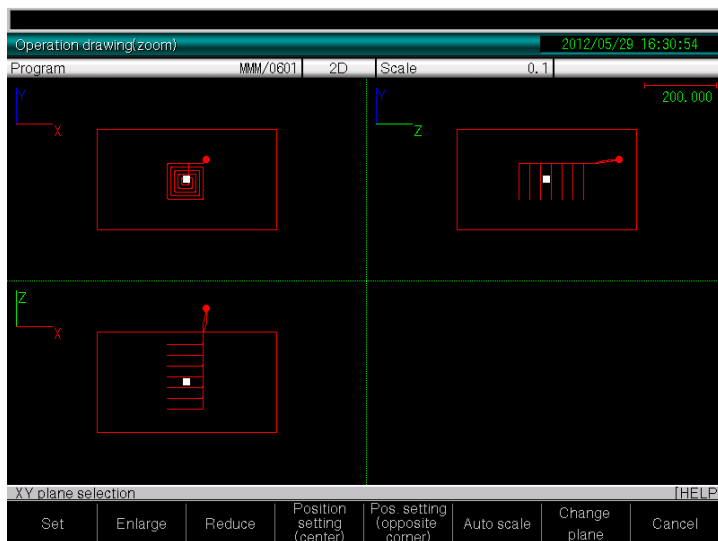
1. Call the zoom screen

Press the **[F2]** (Zoom) key to display the zoom screen.



2. □ is displayed at the center of the screen. (*)
3. Use the cursor keys to move the □ to the position you want to set as a screen center.
4. Set the screen display range.
[F1] (Enlarge) : Enlarges the display.
[F2] (Reduce) : Reduces the display.
Red square frame : Indicates enlargement of the graphic. (The area within the red square is displayed over the entire screen.)
Green square frame : Indicates reduction of the graphic. (The graphic displayed on the screen is reduced to the size of the green square.)
5. Press the **[F0]** (Set) key.
The center of the graphic and the size are changed, and the graphic is drawn again.

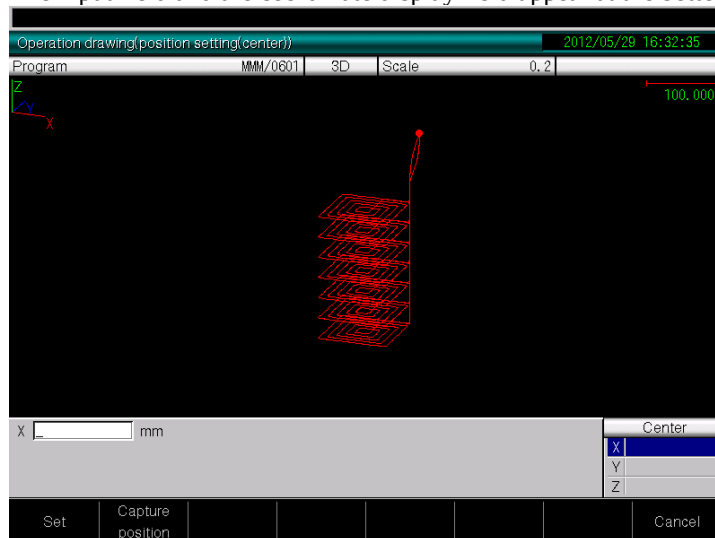
- * Note: When using 3-plane drawing, □ is displayed at each of the three planes. <plane selection> is displayed at the bottom left of the screen for the plane currently selected. Up and down movements using the cursor keys and enlargement and reduction of the graphic size are applied to the selected plane. Each press of the **[F6]** (Change plane) key cycles the plane in the following order; XY plane, XZ plane, ZY plane, and XY plane.



→Position setting (center)

The center of the screen is manually input.

1. Call the zoom screen.
2. Press the **[F3]** (Position setting (center)) key.
3. The input field and the coordinate display field appear at the bottom of the screen.

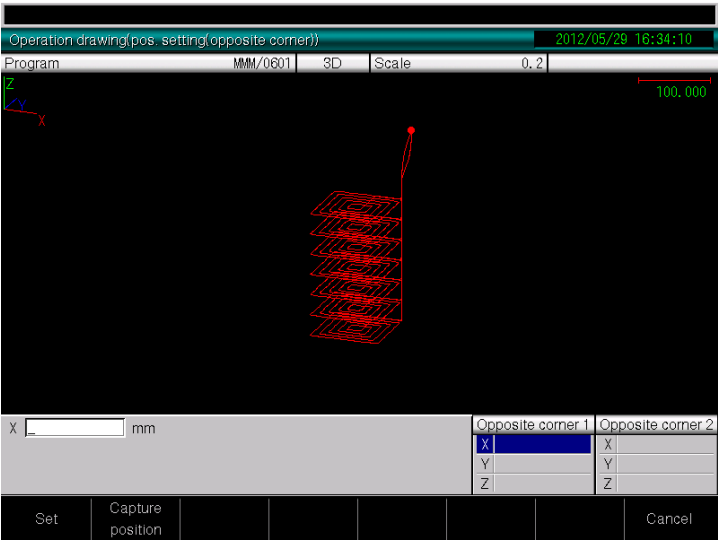


4. Set a new center.
Pressing the **[F1]** (Capture position) key captures the current machine coordinates and sets this point as the center.
5. Press the **[F0]** (Set) key.
The center of the graphic is changed and the graphic is drawn again.

→Position setting (opposite corner)

The scale is changed so that the two points specified are displayed on the screen. The screen center will be the mid-point of the two points entered.

1. Call the zoom screen.
2. Press the **[F4]** (Pos. setting (opposite center)) key.
3. The input field and the coordinate display field appear at the bottom of the screen.



4. Input two points to be displayed on the screen.
Pressing the **[F1]** (Capture position) key captures the current machine coordinates and sets this point as the center.
5. Press the **[F0]** (Set) key.
The center of the graphic and the scale are changed, and the graphic is drawn again.

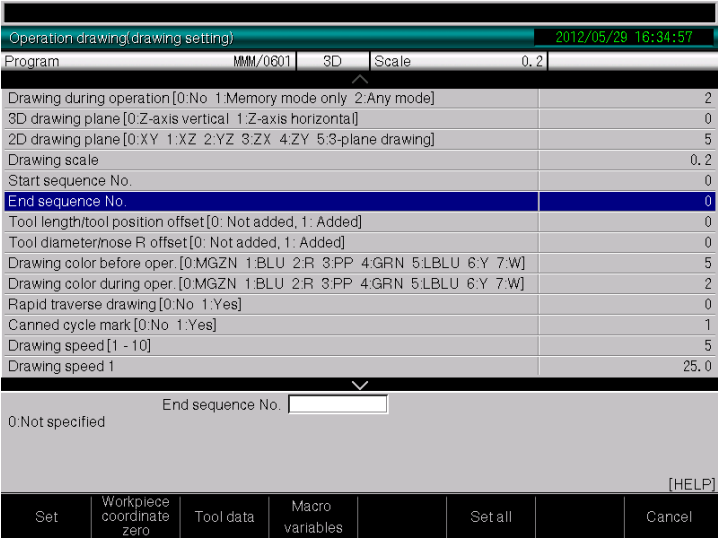
→ Auto scale
The center of the drawing and the scale setting are automatically calculated.

1. Press the **[F5]** (Auto scale) key on the zoom screen.
2. The center and scaling setting are automatically calculated.
A new scale and the center are automatically set, and the drawn graphic is cleared.

(Note) Processing is not completed for a program containing an infinitely recursive statement.
Press the **[F7]** (Cancel) key.

3.2.6 Drawing Setting

Drawing conditions must be set before drawing.
Press the **[F3]** (Set drawing) key.



Setting	Outline												
Drawing during operation	Set whether to draw the operation path when the machine is operating. <0: No>: Does not display the machine operation path. <1: Memory mode only>: Displays the operation path only in the memory operation mode. <2: Any mode>: Displays the operation path in all modes, including operations in manual and MDI modes.												
3D drawing plane	Set the reference direction when displaying the graphic in the 3D mode.												
2D drawing plane	Set the reference direction when displaying the graphic in the 2D mode.												
Drawing scale	Set the graphic scale. When this value is changed, the green digits under the scale mark change accordingly.												
Start sequence No. (Start job No.)	NC language: When the value is not “0,” drawing starts from the block that contains the sequence number set here. When “0” is set, drawing starts from the top of the program. Conversational language: When the value is not “0,” drawing starts from the job that contains the job number set here. When “0” is set, drawing starts from the top of the program.												
End sequence No. (End job No.)	NC language: When the value is not “0,” drawing continues to the next block that contains the sequence number set here. When “0” is set, drawing continues to the end of the program. Conversational language: When the value is not “0,” drawing continues to the next job that contains the job number set here. When “0” is set, drawing continues to the end of the program.												
Tool length/tool position offset (Tool length offset)	(When using the NC language) Specify the X-, Y- and Z-axes graph position when using the tool length/tool position offset. This function is valid only for pre-operation graphs. To not add the offset, a “0” is used for the tool length/tool position compensation. To add the offset, a value for the tool length/tool position compensation in the tool list is used to offset the positions of the X-, Y- and Z-axes. <table><tr><td></td><td>During tool length offset (G43/G44)</td><td>During tool position compensation (G143/G144)</td></tr><tr><td>0: Not added</td><td>Ignores the <Tool length offset> and plots the graph using the spindle tip position.</td><td>Ignores the <Tool length offset> and <Tool position offset> and plots the graph using the spindle tip position.</td></tr><tr><td>1: Added (Type 1)</td><td>Uses the <Tool length offset> to offset the Z-axis position and plots the graph using the spindle tip position.</td><td>Uses the <Tool length offset> and <Tool position offset> to offset the positions for the X-, Y- and Z-axes and plots the graph using the spindle tip position.</td></tr><tr><td>2: Added (Type 2)</td><td>Uses the <Tool length offset> to offset the Z-axis position and plots the graph using the tool tip position.</td><td>Uses the <Tool length offset> and <Tool position offset> to offset the positions for the X-, Y- and Z-axes and plots the graph using the tool tip position.</td></tr></table> (When using the conversation language) Specify the Z-axis graph position when using the tool length offset. This function is valid only for pre-operation graphs. To not add the offset, a “0” is used for the tool length offset. To add the offset, a value for the tool length position in the tool list is used to offset the Z-axis position. <0: Not added>: Ignores the <Tool length offset> and plots the graph using the spindle tip position. <1: Added (Type 1)>: Uses the <Tool length offset> to offset the Z-axis position and plots the graph using the spindle tip position. <2: Added (Type 2)>: Uses the <Tool length offset> to offset the Z-axis position and plots the graph using the tool tip position.		During tool length offset (G43/G44)	During tool position compensation (G143/G144)	0: Not added	Ignores the <Tool length offset> and plots the graph using the spindle tip position.	Ignores the <Tool length offset> and <Tool position offset> and plots the graph using the spindle tip position.	1: Added (Type 1)	Uses the <Tool length offset> to offset the Z-axis position and plots the graph using the spindle tip position.	Uses the <Tool length offset> and <Tool position offset> to offset the positions for the X-, Y- and Z-axes and plots the graph using the spindle tip position.	2: Added (Type 2)	Uses the <Tool length offset> to offset the Z-axis position and plots the graph using the tool tip position.	Uses the <Tool length offset> and <Tool position offset> to offset the positions for the X-, Y- and Z-axes and plots the graph using the tool tip position.
	During tool length offset (G43/G44)	During tool position compensation (G143/G144)											
0: Not added	Ignores the <Tool length offset> and plots the graph using the spindle tip position.	Ignores the <Tool length offset> and <Tool position offset> and plots the graph using the spindle tip position.											
1: Added (Type 1)	Uses the <Tool length offset> to offset the Z-axis position and plots the graph using the spindle tip position.	Uses the <Tool length offset> and <Tool position offset> to offset the positions for the X-, Y- and Z-axes and plots the graph using the spindle tip position.											
2: Added (Type 2)	Uses the <Tool length offset> to offset the Z-axis position and plots the graph using the tool tip position.	Uses the <Tool length offset> and <Tool position offset> to offset the positions for the X-, Y- and Z-axes and plots the graph using the tool tip position.											

Setting	Outline		
Cutter/Nose R compensation (Cutter compensation)	(When using the NC language) Specify the graph position for the axis that uses the cutter/nose R compensation. This function is valid only for pre-operation graphs.		
		During cutter compensation (G41/G42)	During nose R compensation (G141/G142)
	0: Not added	Displays the program path.	Displays the program path.
	1: Added	Displays the tool center path.	Displays the path that is offset with nose R compensation from the program path.
	(When using the conversation language) Specifies the X- and Y-axes graph position when using the cutter compensation. This function is valid only for pre-operation graphs. <0: Not added>: Displays the program path. <1: added>: Displays the tool center path.		
Drawing color before operation	Set the color for the paths drawn during drawing before operation. When “0” is set, each tool is displayed in a different color using the colors registered for the ATC tools.		
Drawing color during operation	Set the color for the paths drawn during drawing in operation. When “0” is set, each tool is displayed in a different color using the colors registered for the ATC tools.		
Rapid traverse drawing	Set whether to draw G0 blocks during drawing before operation. <0: No>: Does not draw any G0 blocks. <1: Yes>: Draws the G0 blocks in a dashed line.		
Canned cycle mark	Sets whether to add the symbol “●” to the section where canned cycle drilling is executed. (This function is valid when the XY plane is displayed on the screen.)		
Drawing speed	Set the drawing speed when a graphic is drawn before operation. The drawing speed is divided into ten levels from <Drawing speed 1> to <Drawing speed 2>. Set the desired level number. If the setting for <Drawing speed 1> is smaller than that for <Drawing speed 2>, the former is automatically applied, regardless of the setting here.		
Drawing speed 1	Set the longest time taken for drawing diagonal lines on the screen.		
Drawing speed 2	Set the shortest time taken for drawing diagonal lines on the screen.		
Coordinate display	Set whether to display the spindle position and spindle tool data on the screen. <0: No>: Coordinates and tool information are not displayed. <1: Machine coord. / 2: Absolute coord. / 3: Relative coord.>: Specified coordinates value and tool information are displayed.		
Clear screen when changing tool	Set whether to clear the drawn graphic when tools are changed.		
Clear screen when rotating additional axis	Set whether to clear the drawn graphic when the additional axis is rotated.		
Horizontal rotation angle of view point	Set the angle that the workpiece rotates relative to the horizontal plane in 3D display mode.		
Vertical rotation angle of view point	Set the angle at which the workpiece rotates relative to the vertical plane in 3D display mode.		
Drawing start position X	Set an X-axis position (in machine coordinates) when drawing starts before operation.		
Drawing start position Y	Set a Y-axis position (in machine coordinates) when drawing starts before operation.		
Drawing start position Z	Set a Z-axis position (in machine coordinates) when drawing starts before operation.		
Machining pallet	Specify a machining pallet. This setting is valid only for the machine equipping a quick table.		

- Workpiece coordinates (NC language)
The workpiece coordinate zero, the extended coordinates and the reference rotary fixture offset used in the pre-operation graphs can be referenced.
Press the **[F1]** (Workpiece coordinate) key to display the following functions.
[F0] (Back) key: Returns to the graph settings screen.
[F1] (Standard coordinates/Extended coordinates) key: Displays the workpiece coordinate zero or extended coordinates.
Every time the **[F1]** key is pressed, the function changes.
[F6] (Rotary fixture offset) key: Displays the reference rotary fixture offset.
- Tool data (NC language)
You can check the tool data used when a graphic is drawn before operation.
If the **[F2]** (Tool data) key is pressed, the following function appears:
[F0] (Return) key: Returns to the <Drawing setting> screen.
[F1] (List) key: Displays tool data for 10 tools.
[F2] (Expansion 1) key: Displays the available setting items screen such as peripheral speed and rotation feed.
[F3] (Expansion 2) key: Displays the screen that includes the setting items for the turning tool.
- Macro variable (NC language)
You can check the macro variable used when a graphic is drawn before operation.
Press the **[F3]** (Macro variable) key.
- Batch setting (NC language)
You can set all the drawing parameters in the same conditions as those in the current memory operation.
The same settings are applied to workpiece coordinates, tool data, and macro variables.
 1. Press the **[F5]** (Batch setting) key.
 2. Press the **[F0]** (Yes) key.

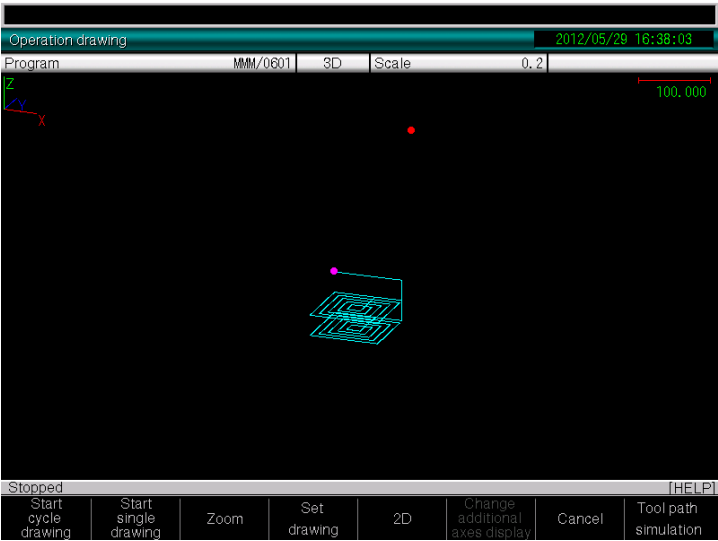
3.2.7 To Stop Drawing

A block stop is applied to cycle drawing.

1. Press the **[F0]** (Stop drawing) key during drawing.
2. Drawing continues to the end of the current block, and stops.

(Note 1) If mode change is attempted during drawing before operation, the same operation as when drawing is stopped applies.

(Note 2) If the user stops operation in the pre-operation graph, after changing to another program in the memory operation screen and returning to the operation graph screen, the number of the program, to which the user changed, is displayed in the memory operation screen.
In this situation, launch the graph to open it again. The program number is displayed on the screen for the program that was stopped. However, if the user stops the newly opened program, it reverts back to the previous program number on the memory operation screen.



3.2.8 To Cancel Drawing

Cycle drawing or single drawing is cancelled.

Press the **[F6]** (Cancel) key during drawing.



3.2.9 To Delete Graphic

The drawn graphic is deleted.

1. Press the **[F6]** (Delete graphic) key when cycle drawing or single drawing is not performed. (Stop drawing first if cycle drawing or single drawing is being carried out.)
2. The graphic displayed on the screen is completely deleted.

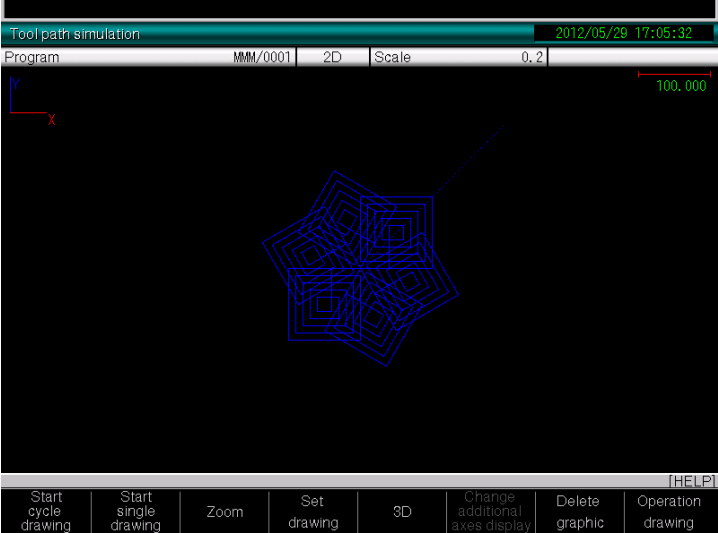
3.3 Tool Path Simulation

3.3.1 Outline

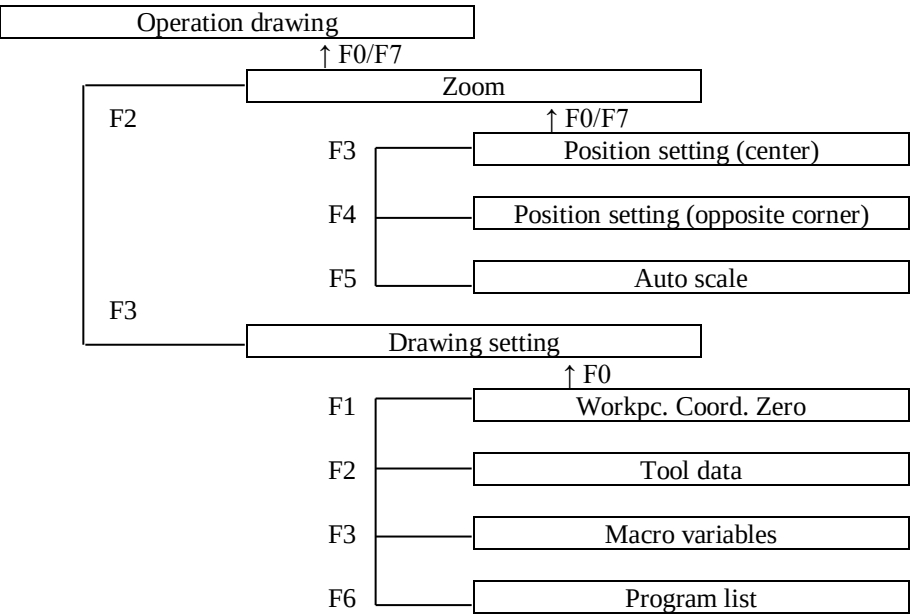
You can check the tool path of a rough graphic of a program specified by you.
If the specified program is in a root folder, only the program number is displayed. For other than that, the “Folder name/program number” is displayed.
Drawing is possible in all modes. When the tool path simulation mode is activated, simulation is automatically performed and the path is drawn and displayed.

Operation

- 1. Press the **[GRAPH]** key to display the <Operation drawing> screen.
- 2. Check that “Tool path simulation” is displayed at the top left of the screen.
If “Operation drawing” is displayed, press the **[F7]** (Tool path simulation) key.



Screen flow



3.3.2 Cycle Drawing

The program specified by the user is drawn.

The current position of the simulation tool is indicated with a purple “●” symbol during drawing operation.

The next position of the simulation tool is indicated with a blue “●” symbol.

Operation

1. Press the [F0] (Start cycle drawing) key.
2. Drawing starts.

(Note 1) Set the program number of the drawing with the “Drawing program folder” and “Drawing program number” on the <Drawing setting> screen. The drawing start position is set in “Start sequence NO.” on the <Drawing setting> screen.

Drawing stops at the block containing M00. Press the [F0] (Cycle drawing start) key again to continue.

Cycle drawing is not possible in tape operation.

(Note 2) Drawing path is created according to NC programs. So the actual path at a corner slightly differs from the drawn path. In addition, a smooth override path in the high accuracy mode is not reflected to the drawn path.

3.3.3 Single Drawing

The path of the program specified by the user is drawn one block at a time.

Operation

1. Press the [F1] (Start single drawing) key.
2. Press the [F1] (Start single drawing) key again.
The next block is drawn.

(Note) Set the program number of the drawing with the “Drawing program folder” and “Drawing program number” on the <Drawing setting> screen. The drawing start position is set in “Start sequence NO.” on the <Drawing setting> screen.

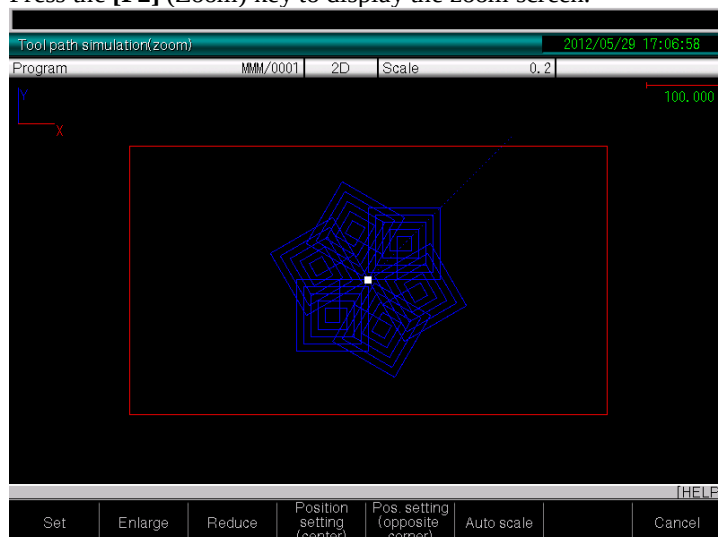
Cycle drawing is not possible in tape operation.

3.3.4 Zoom

The scale of the graphic and the center of display can be changed.

Operation

1. Call the zoom screen.
Press the [F2] (Zoom) key to display the zoom screen.

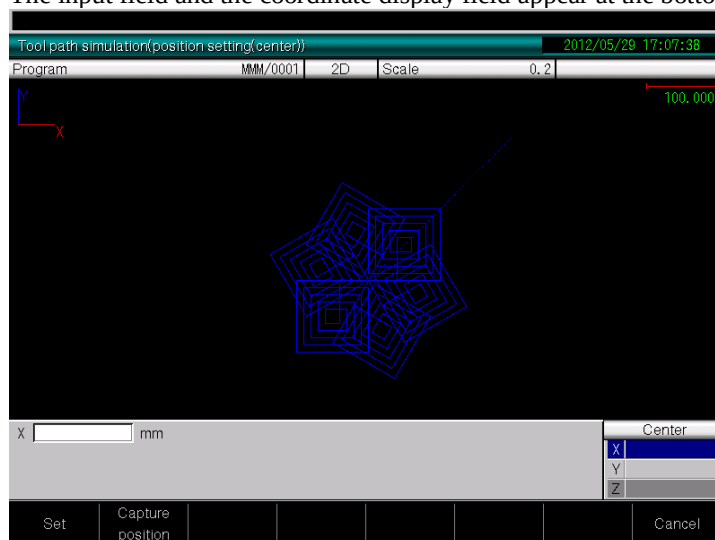


2. □ is displayed at the center of the screen. (*)
 3. Use the cursor keys to move the □ to the position you want to set as a screen center.
 4. Set the screen display range.
[F1] (Enlarge) : Enlarges the display.
[F2] (Reduce) : Reduces the display.
 Red square frame : Indicates enlargement of the graphic. (The area within the red square is displayed over the entire screen.)
 Green square frame : Indicates reduction of the graphic. (The graphic displayed on the screen is reduced to the size of the green square.)
 5. Press the **[F0]** (Set) key.
 The center of the graphic and the size are changed, and the graphic is drawn again.
- * Note: When using 3-plane drawing, □ is displayed in each of the three planes. <** plane selection> is displayed at the bottom left of the screen for the plane currently selected. Up and down movements using the cursor keys and enlargement and reduction of the graphic size are applied to the selected plane. Each press of the **[F6]** (Change plane) key cycles the plane in the following order; XY plane, XZ plane, ZY plane, and XY plane.

→ Position setting (center)

The center of the screen is manually input.

1. Call the zoom screen
2. Press the **[F3]** (Position setting (center)) key.
3. The input field and the coordinate display field appear at the bottom of the screen.



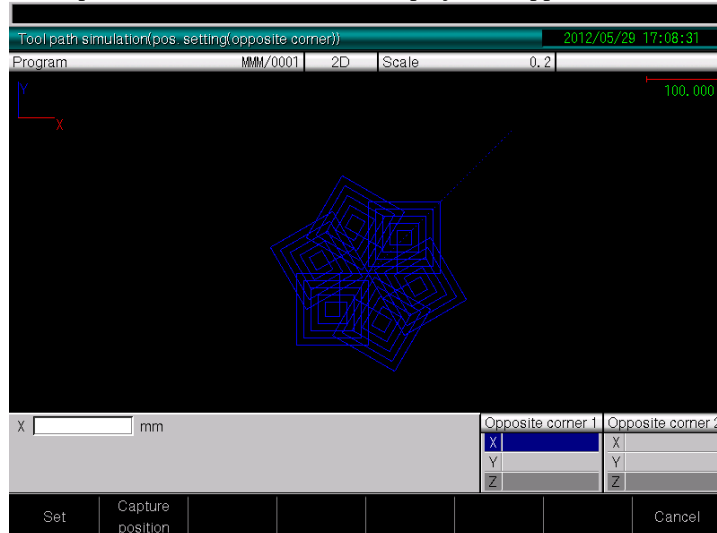
4. Set a new center.
 Pressing the **[F1]** (Capture position) key captures the current machine coordinates and sets this point as the center.
5. Press the **[F0]** (Set) key.
 The center of the graphic is changed and the graphic is drawn again.

→ Position setting (opposite corner)

The scale is changed so that the two points specified are displayed on the screen.

The screen center will be the mid-point of the two points entered.

1. Call the zoom screen.
2. Press the **[F4]** (Pos. setting (opposite center)) key.
3. The input field and the coordinate display field appear at the bottom of the screen.



4. Input two points to be displayed on the screen.
Pressing the **[F1]** (Capture position) key captures the current machine coordinates and sets this point as the center.
5. Press the **[F0]** (Set) key.
The center of the graphic and the scale are changed, and the graphic is drawn again.

→ Auto scale

The center of the drawing and the scale setting are automatically calculated.

1. Press the **[F5]** (Auto scale) key on the zoom screen.
2. The center and scaling setting are automatically calculated.
A new scale and the center are automatically set, and the drawn graphic is cleared.

(Note) Processing is not completed for programs containing an infinitely recursive statement.
Press the **[F7]** (Cancel) key.

3.3.5 Drawing Setting

Drawing conditions must be set before drawing.
Press the **[F3]** (Set drawing) key.

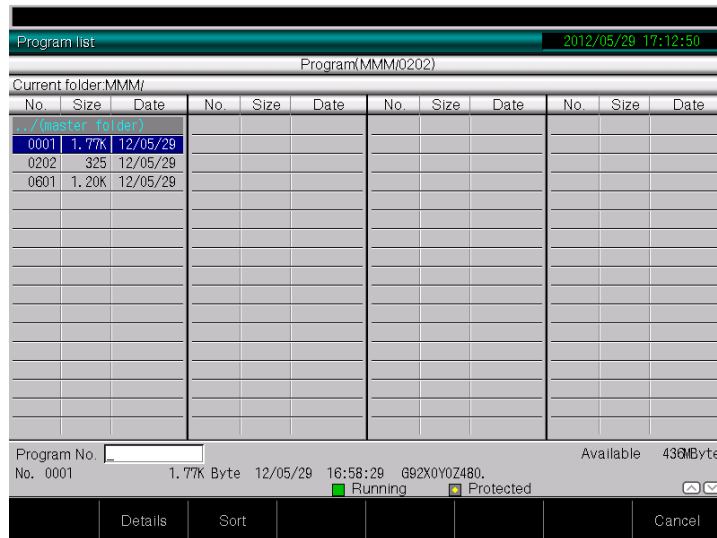
Tool path simulation(drawing setting)		2012/05/29 17:09:30
Program	MMM/0001	2D Scale 0.2
^		
Drawing program No.	1	
3D drawing plane [0:Z-axis vertical 1:Z-axis horizontal]	0	
2D drawing plane [0:XY 1:XZ 2:YZ 3:ZX 4:ZY 5:3-plane drawing]	0	
Drawing scale	0.2	
Start sequence No.	0	
End sequence No.	0	
Tool length/tool position offset [0: Not added, 1: Added]	0	
Tool diameter/nose R offset [0: Not added, 1: Added]	0	
Drawing color [0:Each tool change 1:BLU 2:R 3:PP 4:GRN 5:LBLU 6:Y 7:W]	1	
Rapid traverse drawing [0:No 1:Yes]	1	
Canned cycle mark [0:No 1:Yes]	0	
Drawing speed [1 - 10]	1	
Drawing speed 1	10.0	
Drawing speed 2	1.0	
v		
Drawing speed		
1: Slowest ----- 10: Fastest		
[HELP]		
Set	Workpiece coordinate zero	Tool data
		Macro variables
Set all	Program list	Cancel

Setting	Outline		
Drawing program folder	Specify the folder of a program drawn in the tool path simulation. If a program in the root folder is specified, do not set it.		
Drawing program No.	Set the program number to be drawn in the tool path simulation mode.		
3D drawing plane	Set the reference direction when displaying the graphic in the 3D mode.		
2D drawing plane	Set the reference direction when displaying the graphic in the 2D mode.		
Drawing scale	Set the graphic scale. When this value is changed, the green digits under the scale mark change accordingly.		
Start sequence No. (Start job No.)	NC language: When the value is not “0,” drawing starts from the block that contains the sequence number set here. When “0” is set, drawing starts from the top of the program. Conversational language: When the value is not “0,” drawing starts from the job that contains the job number set here. When “0” is set, drawing starts from the top of the program.		
End sequence No. (End job No.)	NC language: When the value is not “0,” drawing continues to the next block that contains the sequence number set here. When “0” is set, drawing continues to the end of the program. Conversational language: When the value is not “0”, drawing continues to the next job that contains the job number set here. When “0” is set, drawing continues to the end of the program.		
Tool length/tool position offset (Tool length offset)	Specify the X-, Y- and Z-axes graph position when using the tool length/tool position offset. To not add the offset, a “0” is used for the tool length/tool position compensation. To add the offset, a value for the tool length/tool position compensation in the tool list is used to offset the positions of the X-, Y- and Z-axes.		
		During tool length offset (G43/G44)	During tool position compensation (G143/G144)
	0: Not added	Ignores the <Tool length offset> and plots the graph using the spindle tip position.	Ignores the <Tool length offset> and <Tool position offset> and plots the graph using the spindle tip position.
	1: Added (Type 1)	Uses the <Tool length offset> to offset the Z-axis position and plots the graph using the spindle tip position.	Uses the <Tool length offset> and <Tool position offset> to offset the positions for the X-, Y- and Z-axes and plots the graph using the spindle tip position.
	2: Added (Type 2)	Uses the <Tool length offset> to offset the Z-axis position and plots the graph using the tool tip position.	Uses the <Tool length offset> and <Tool position offset> to offset the positions for the X-, Y- and Z-axes and plots the graph using the tool tip position.
Tool diameter offset	Set the X/Y axis drawing position with tool offset applied. <0: Not added>: Displays the program path. <1: Added>: Displays the program path and the tool center position.		
Drawing color	Set the color for the path drawn during the tool path simulation. When “0” is set, tools are displayed in a different color each time the tool is changed.		
Rapid traverse drawing	Set whether to draw G0 blocks during the tool path simulation. <0: No>: Does not draw any G0 blocks. <1: Yes>: Draws the G0 blocks in a dashed line.		
Canned cycle mark	Set whether to add the “●” symbol to the section where canned cycle drilling is executed. (This function is enabled when the XY plane is displayed on the screen.)		

Setting	Outline
Drawing speed	Set the drawing speed for the tool path simulation. The drawing speed is divided into ten levels from <Drawing speed 1> to <Drawing speed 2>. Set the desired level number. If the setting for <Drawing speed 1> is smaller than that for <Drawing speed 2>, the former is automatically applied, regardless of the setting here.
Drawing speed 1	Set the longest time taken for drawing diagonal lines on the screen.
Drawing speed 2	Set the shortest time taken for drawing diagonal lines on the screen.
Coordinate display	Set whether to display the spindle position and spindle tool data on the screen. <0: No>: Coordinates and tool information are not displayed. <1: Machine coord. / 2: Absolute coord. / 3: Relative coord.>: Specified coordinates value and tool information are displayed.
Clear screen when changing tool	Set whether to clear the drawn graphic when tools are changed.
Clear screen when rotating additional axis	Set whether to clear the drawn graphic when the additional axis is rotated.
Horizontal rotation angle of view point	Set the angle that the workpiece rotates relative to the horizontal plane in the 3D display mode.
Vertical rotation angle of view point	Set the angle that the workpiece rotates relative to the vertical plane in the 3D display mode.
Drawing start position X	Set an X-axis position (in machine coordinates) when the tool path simulation starts.
Drawing start position Y	Set a Y-axis position (in machine coordinates) when the tool path simulation starts.
Drawing start position Z	Set a Z-axis position (in machine coordinates) when the tool path simulation starts.
Machining pallet	Specify the machining pallet. This setting is valid only for the machine equipping a quick table.

- Workpiece coordinates (NC language)
The workpiece coordinate zero, the extended coordinates and the reference rotary fixture offset used in the tool path simulation can be referenced.
Press the **[F1]** (Workpiece coordinate) key to display the following functions.
[F0] (Back) key: Returns to the graph settings screen.
[F1] (Standard coordinates/Extended coordinates) key: Displays the workpiece coordinate zero or extended coordinates.
Every time the **[F1]** key is pressed, the function changes.
[F6] (Rotary fixture offset) key: Displays the reference rotary fixture offset.
- Tool data (NC language)
You can check the tool data used for tool path simulation.
When the **[F2]** (Tool data) key is pressed, the following function appears:
[F0] (Return) key: Return to the drawing set screen.
[F1] (List) key: Displays tool data for 10 tools.
[F2] (Expansion 1) key: Displays the available setting items screen such as peripheral speed and rotation feed.
[F3] (Expansion 2) key: Displays the screen that includes the setting items for the turning tool.
- Macro variable (NC language)
You can check the macro variables used for tool path simulation.
Press the **[F3]** (Macro variables) key.
- Batch setting (NC language)
You can set all the drawing parameters to the same conditions as those for the current memory operation.
The same settings are applied to workpiece coordinates, tool data, and macro variables.
1. Press the **[F5]** (Set all) key.
2. Press the **[F0]** (Yes) key.

- **Program list**
A list of the programs currently registered is displayed. Sorting priority can be selected from among date, program number, and comment. Programs selected from this screen can be designated for tool path simulation.
Move the cursor to [Drawing program], and press the **[F6]** (Program list) key.

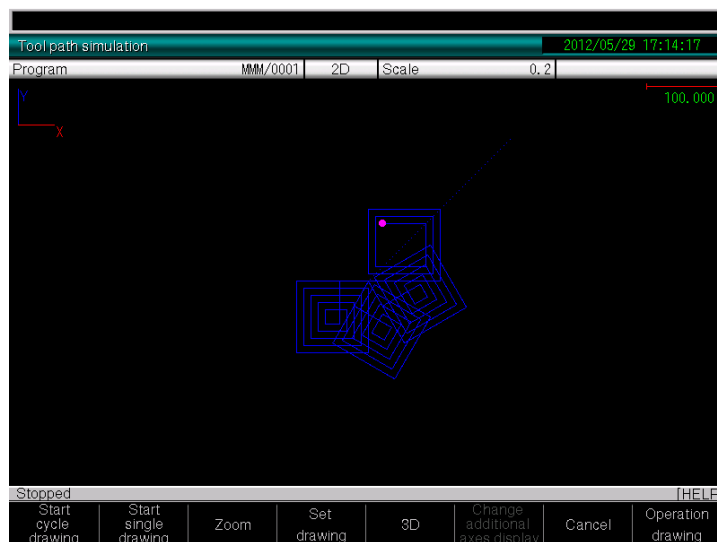


If a program is selected, the drawing program folder is also set. (If a program is selected from a root folder, the drawing program folder is not set.)

3.3.6 To Stop Drawing

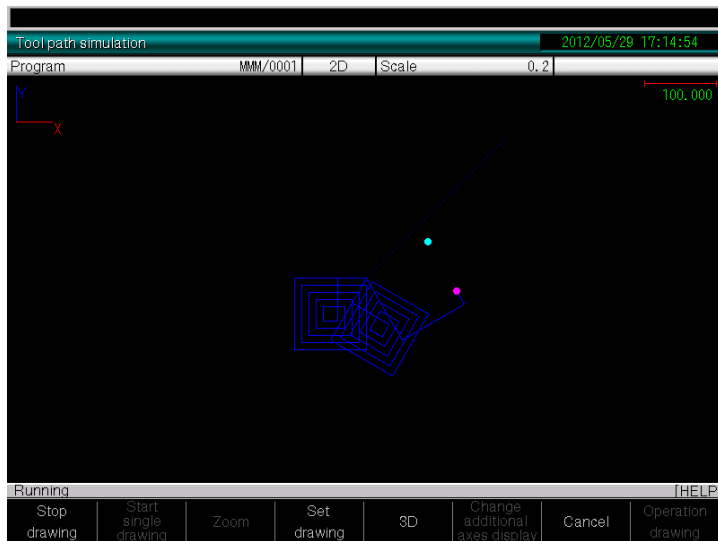
A block stop is applied to cycle drawing.

1. Press the **[F0]** (Stop drawing) key during drawing.
2. Drawing continues to the end of the current block, and stops.



3.3.7 To Cancel Drawing

Cycle drawing or single drawing is cancelled.
Press the **[F6]** (Cancel) key during drawing.



3.3.8 To Delete Graphic

The drawn graphic is deleted.

1. Press the **[F6]** (Delete graphic) key when cycle drawing or single drawing is not performed.
(Stop drawing first if cycle drawing or single drawing is being carried out.)
2. The graphic displayed on the screen is deleted completely.

CHAPTER 4

TAPE OPERATION MODE

4

- 4.1 How to Perform Tape Run
- 4.2 Configuration of Tape Run Program
- 4.3 Notices

4.1 How to Perform Tape Run

The tape run function operates the machine while reading the program from an external Input device.

This function is used when executing a large program that cannot be stored in the NC.

Two types of tape run can be operated.

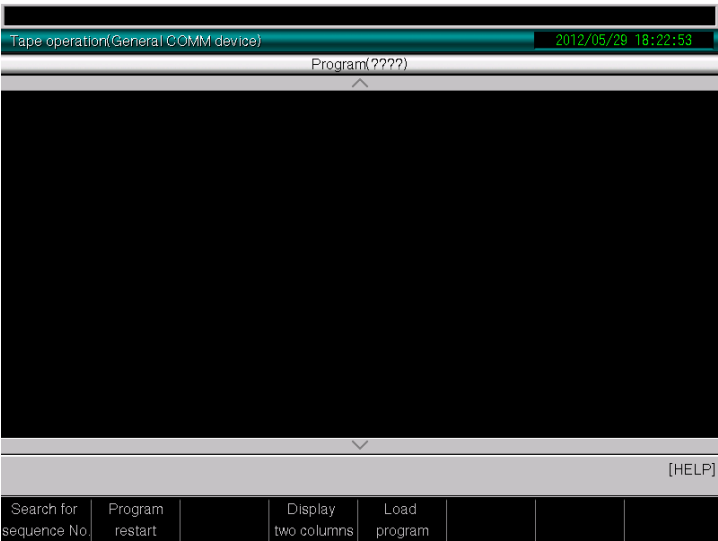
1. One method is to read a program through serial port with the connected device set to “General communication device”.
(Hereinafter referred to as “Tape run (General COMM device)”)
2. Another method is to read a program through USB port with the connected device set to “Memory card”.
(Hereinafter referred to as “Tape run (Memory card)”)

4.1.1 Tape Run (General COMM Device)

Set the followings before performing tape run (General COMM device).

- Set <Select memory operation> in the user parameters (switch 1) to <1: Tape>
- Set “General COMM device” for “connected dedevice“ by <Change connected device> of external Input device.
- For how to change the connected device, refer to “5.2.2 Changing connected device.”Check the settings for the communication parameter (general COMM device) and for the external input device.

When memory run mode is selected, the <Tape run> screen appears.



Press the [START] key and load the program from the external input device. Tape run starts.

- Program loading
In order to check the contents of the program in advance before starting the operation, press [F4] (program load) key.
The top section of the loaded program is displayed on the screen so you can check the contents.

```

Tape operation(General COMM device) 2012/05/29 18:24:11
Program(3210)
>> 000001 (03210 TAPE);
000002 G90G00X0. Y0. Z300. ;
000003 G68P3211;
000004 G00X-20. ;
000005 G67;
000006 ;
000007 G68P3211L2;
000008 G00X-30. ;
000009 G67;
000010 ;
000011 G68P3211A1. ;
000012 G00X-40. ;
000013 G67;
000014 ;
000015 M98P3212;
000016 ;
000017 M30;
[HELP]
Search for Program Display Load
sequence No. restart two columns program
  
```

If the program is OK, press the [START] switch. Then the operation starts.
 If the program is not the desired one, reset the external input device, press the [F4] (program load) key again, and then send the desired program from the external input device.

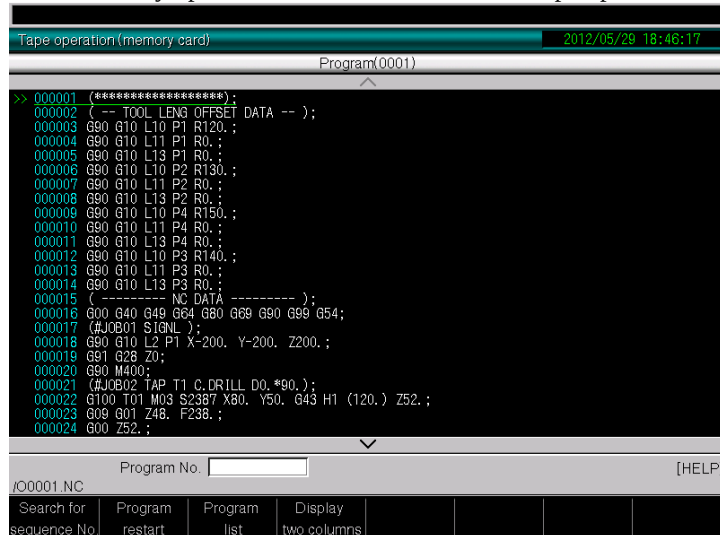
- Program restart
 Pressing the [F1] (program restart) key displays the program restart screen.
 For details of program restart, refer to “7.15 Program restart function” in the operation manual.
- Checking the input program number
 For the machine equipped with quick table, since a certain value is preset for the <Pallet 1 program NO.> and <Pallet 2 program NO.>, respectively, in the user parameters (quick table), an alarm occurs if the program numbers read from the external input device do not coincide with those preset for the pallets, when the [PALLET] key is set at other than “OFF”. Program number check is not performed when no values are preset in the parameters.

4.1.2 Tape Run (Memory Card)

Set the followings before performing tape run (Memory card).

- Set the <Select memory operation> in the user parameters (switch 1) to “1: Tape”.
- Set “Memory card” for “connected device” by <Change connected device> of the external input device.
For how to change the connected device, refer to “5.2.2 Changing connected device.”
- Check the settings for the communication parameters (memory card).

When memory operation mode is selected, the <Tape operation> screen appears.



- Select program 1
Enter a program number as in the normal memory run, and press the **[ENT]** key.
For selecting a program other than that output by the machine (O****NC), use Select program 2.
 - Select program 2
Press the **[F2]** (program list) key to display the program list screen.
As in the normal memory run, select a program number with the cursor or enter a program number, and then press the **[ENT]** key.
When the memory card has been replaced, press the **[F3]** (latest directory) key to update the display.
For displaying a program in a sub folder, select a sub folder with the cursor and press the **[ENT]** key.
For displaying a program other than that output by the machine (O****NC), press the **[F5]** (display of all files) key to display the entire files.
- * When a program other than that output by the machine (O****NC) has been selected, the program number is displayed as <????>.
 - * Programs stored in the machine cannot be called up.
 - * To display the display contents, access the USB port. During the operation, be sure to never remove the memory card.
 - * Details (comments) cannot be displayed. Sorting by the comment contents is also unavailable.

Program list										2012/05/29 18:48:33	
Program(0001)											
Current folder: /											
No	Size	Date	No	Size	Date	No	Size	Date	No	Size	Date
0001	1.33K	12/05/29	0024	10	12/05/29	0041	19	12/05/29	0058	11	12/05/29
0002	165	12/05/29	0025	10	12/05/29	0042	33	12/05/29	0059	46	12/05/29
0004	85	12/05/29	0026	11	12/05/29	0043	19	12/05/29	0060	48	12/05/29
0005	5	12/05/29	0027	14	12/05/29	0044	14	12/05/29	0061	78	12/05/29
0010	77	12/05/29	0028	10	12/05/29	0045	22	12/05/29	0068	64	12/05/29
0011	29	12/05/29	0029	17	12/05/29	0046	1.99M	12/05/29	0069	5	12/05/29
0012	17	12/05/29	0030	17	12/05/29	0047	17	12/05/29	0070	269	12/05/29
0013	11	12/05/29	0031	17	12/05/29	0048	17	12/05/29	0071	45	12/05/29
0014	17	12/05/29	0032	17	12/05/29	0049	15	12/05/29	0072	17	12/05/29
0016	25	12/05/29	0033	53	12/05/29	0050	15	12/05/29	0073	17	12/05/29
0017	10	12/05/29	0034	29	12/05/29	0051	172	12/05/29	0074	17	12/05/29
0018	10	12/05/29	0035	73	12/05/29	0052	48	12/05/29	0075	17	12/05/29
0019	10	12/05/29	0036	11	12/05/29	0053	11	12/05/29	0076	17	12/05/29
0020	10	12/05/29	0037	11	12/05/29	0054	167	12/05/29	0077	27	12/05/29
0021	10	12/05/29	0038	11	12/05/29	0055	2.25M	12/05/29	0078	44	12/05/29
0022	10	12/05/29	0039	11	12/05/29	0056	23	12/05/29	0079	52	12/05/29
0023	10	12/05/29	0040	27	12/05/29	0057	64	12/05/29	0080	61	12/05/29
Program No.											
No. 0001 1.33K Byte 12/05/29 18:40:52											
Sort Latest directory Display all files Cancel											

If the displayed program is OK, press the **[START]** switch. Then the operation starts.

- * During the operation (including the pause state (with the stop LED lit)), never remove the memory card.

- Program restart

Pressing the **[F1]** (program restart) key displays the program restart screen.

For details of program restart, refer to “7.15 Program restart function” in the operation manual.

- About operation program

For the machine equipped with quick table, since a certain value is preset from the <Pallet program folder>, <Pallet 1 program NO.> and <Pallet 2 program NO.>, respectively, in the user parameters (quick table), operation program is switched to that having the program number preset in the set folder, when the **[PALLET]** key is set at other than “OFF.”

When the external program selection is valid, it is switched to the selected program (for all machines).

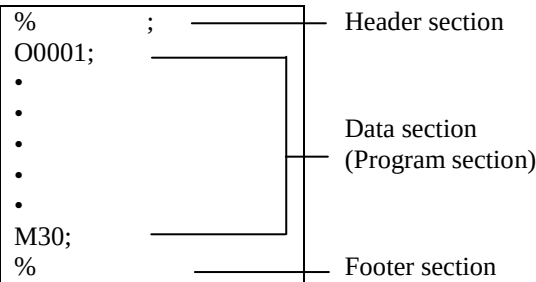
(In either case, an error state is generated if the the memory card does not contain the relevant program.

- * If no value is preset for the <Pallet program folder>, the operation program is switched to that having the program number in the root folder.
- * Do not set the following: <Pallet 1 program No.> and <Pallet 2 program No.>. when operating a program that is not a (O****.NC) program, except when the **[PALLET]** key is turned “OFF”.

4.2 Configuration of Tape Run Program

4.2.1 General Communication Device

For tape run (general COMM device), programs to be sent from the external input device should be formatted as shown below.

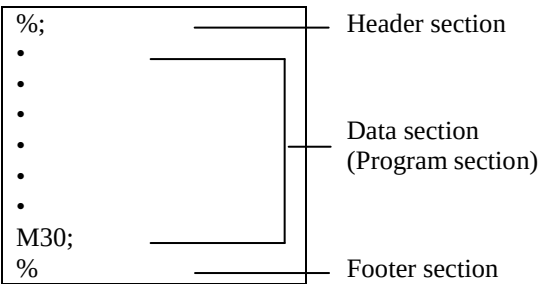


- Header section** The section from % to EOB is recognized as the header.
The header is not displayed on the screen. It is only read and then discarded.
- Data Section** The program number is included in the first block of the data.
The four-digit number after address O is recognized as the program number, and the program number is excluded when the program is loaded. When address O does not exist, the four-digit number after address N is recognized as the program number. When address O, address N, and “:” (colon) do not exist, it is considered that there is no program number. “O” or “N” in the second block or subsequent block in the data section is not recognized as the program number.
- Footer section** When % appears after the header is read, this section is regarded as the footer.

4.2.2 Memory Card

For tape run (memory card), the program file in the memory card should be created in the following format. The format varies depending on the settings for the <Header/footer> in the communication parameters (memory card).

- When <Header/footer> is set to <0: No>:
The entire data in the file are recognized as programs.
This section is regarded as the program.
- When <Header/footer> is set to <1: Yes>:



Header section	The section of %+EOB is recognized as the header. The header is not displayed on the screen. It is only read and then discarded.
Data Section	This section is regarded as the program. This section is regarded as the program.
Footer section	When % appears after the header is read, this section is regarded as the footer.

4.3 Notices

- In tape run mode, the pallet is loaded when operation is started although <Pallet loading> of “User parameter (quick table)” is set to <At 1st tool change>.
- Although <Program repeat> is set to <1: Repeat>, the operation stops at the point of program end (M30, M02) without repeating the operation.
- The files such as programs and data banks cannot be input or output during operation (including during pause).

Prohibited commands

The following code cannot be used during tape operation.

- Macro (WHILE DO to END)

Refer to “6.4 Control Function” in the Programming Manual (NC language) for further details.

Restricted commands

There are restrictions for the following codes during tape operation.

1. Sub program/macro program call (M98, M99, G65, G66)
 - Only those programs contained in the root folder in the machine can be called up.
 - Address P cannot be omitted when M98 is specified.
 - M99 cannot be specified in the main program.
 - Address P cannot be specified when M99 is specified.

Refer to “Chapter 8 Subprogram Function” in the Programming Manual (NC language) for further details.
2. Macros (GOTO to N)
 - If there is no specified sequence number in the data between GOTO and the end of program, a “no specified sequence” alarm is generated. (Sequence numbers contained in the part between the head of program and GOTO are not searched for.)
 - If there is no specified sequence number in the data within approx. 60 Kbyte from the block at which the execution pointer is located, a “no specified sequence” alarm is generated.

Refer to “6.4 Control Function” in the Programming Manual (NC language) for further details.

CHAPTER 5

COMMUNICATION

5

- 5.1 Outline**
- 5.2 Master Station Communication**
- 5.3 Slave Station Communication**
- 5.4 Maintenance Communication**
- 5.5 Communication Protocol**
- 5.6 Details of Communication Data**

5.1 Overview

5.1.1 Special Notes Before Using Communication Function

This product has communication functions, such as Ethernet. However it is not possible to connect directly to a telecommunication line (including public wireless LAN) for telecommunication carriers (such as mobile communication carriers, fixed-line communication carriers, Internet providers, etc.). Make sure you use a router or firewall when connecting your Brother machine to the Internet to protect it against unauthorized access from the Internet.

To avoid inadvertently losing important data, we recommend regularly backing up the data in this product. If there is an accident, such as data being deleted due to a mistaken operation, data being corrupted due to a failure, or data being deleted and/or falsified via remote operation from an external program on the network, the customer can shorten the recovery time and minimize the impact on work by using backup data.

A description of how to back up the internal data on this product is described later on.

In addition, we recommend for the customer to save any data in a separate data storage device at the appropriate time, for example, when creating or making changes to data, such as data found in NC programs, conversation-language programs, internal PLC programs or in data banks.

To customers using the FTP master station communication function:

The FTP master station communication function allows customers to use this product to access an FTP server that is prepared separately by the customer. When using an FTP server, be sure to limit access to files used by this product, for example, by minimizing the range of access that is granted under the login name used on this product.

Procedure to back up internal data for this product

Items needed: Formatted memory card (USB memory)

1. Turn the machine breaker ON, and then turn the power ON.
2. Insert the memory card into the memory card interface, which is under the machine's operation panel.
3. Change to maintenance mode.
 - 3-1. Change to <Program edit menu> screen.
Press the **[EDIT]** key.
 - 3-2. Change to <External I/O menu> screen.
Select <External I/O> on the menu, and press the **[ENT]** key.
 - 3-3. Change to <Maintenance menu> screen.
Select <Maintenance> on the menu, and press the **[ENT]** key.
 - 3-4. Make sure that <Connected> on the bottom-right corner of the screen reads <Memory card>.
If the connecting location is different, select <Change connection device> and change it to <Memory card>.
4. Output the internal machine data to the memory card.
 - 4-1. Change to <Maintenance – Output all data> screen.
Press the **[F0]** (Output all data) key.
 - 4-2. Output data to memory card.
Press the **[F2]** (Output) key.
After output is complete, it goes back to the screen before outputting.
5. Remove memory card.

Copy the internal machine data saved in the memory card to the computer or other device to store it. If data is corrupted, this procedure can be used to restore the data by transferring and inputting the backup data.

5.1.2 Master station and slave station communication

Programs, data banks, and maintenance data can be input/ output between NC and external I/O equipment.

Master station communication: External program I/O menu and the subsequent screens of NC are used to start data input/ output. External equipment is the responding equipment (server). Memory card is also the subject of master station communication.

Slave station communication: External equipment initiates communication with the NC, which is the responding equipment (server). Slave station communication is limited when NC is displaying the external program I/O menu or the subsequent screens.

Maintenance communication: Special inputs/outputs necessary for maintenance are performed (board exchange, etc.).

Connecting lines	Communication type	Master/ Slave	Maintenance	Description
Serial	General communication device	(Master)	×	Reader/ puncher protocol communication.
	NC communication device (Type 1)	Master	×	External I/O equipment must be compatible with NC communication device protocol type 1.
	NC communication device (Type 2)	Master/ slave	×	External I/O equipment must be compatible with NC communication device protocol type 2. Input/output between NCs is available.
Ethernet	NC communication device (Type 2)	Slave	○	External I/O equipment must be compatible with NC communication device protocol type 2. Input/output between NCs is available.
	FTP	Master	○	External I/O equipment must have the FTP server function.
	FTP	Slave	○	External I/O equipment must have the FTP client function.
Memory card		Master	○	Inputs/outputs files, etc. with a memory card connected to the USB port of the operation panel.

5.1.3 File Name

The programs, data banks output from NC are normally stored in the I/O equipment under the following file name:

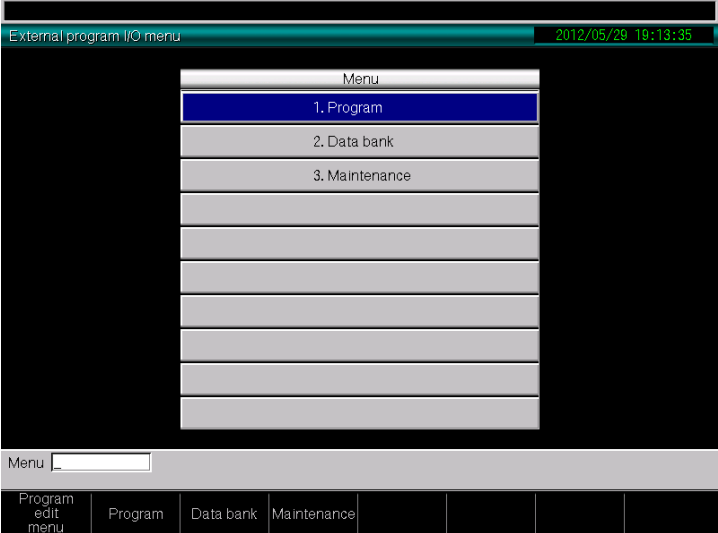
Type	File name
NC program	O****.NC (NC language) O****.TC (conversation) **** : 4 digits fixed
Conversation program	QM****.NC (NC language) QM****.TC (conversation) **** : 4 digits fixed
Data bank files, Maintenance data	*****.NC (NC language) *****.TC (conversation) ***** : by file See 5.6.1 Data Name List.
Ladder files	*****.ldr ***** : by file See 5.6.1 Data Name List.

5.2 Master station communication

5.2.1 Screen

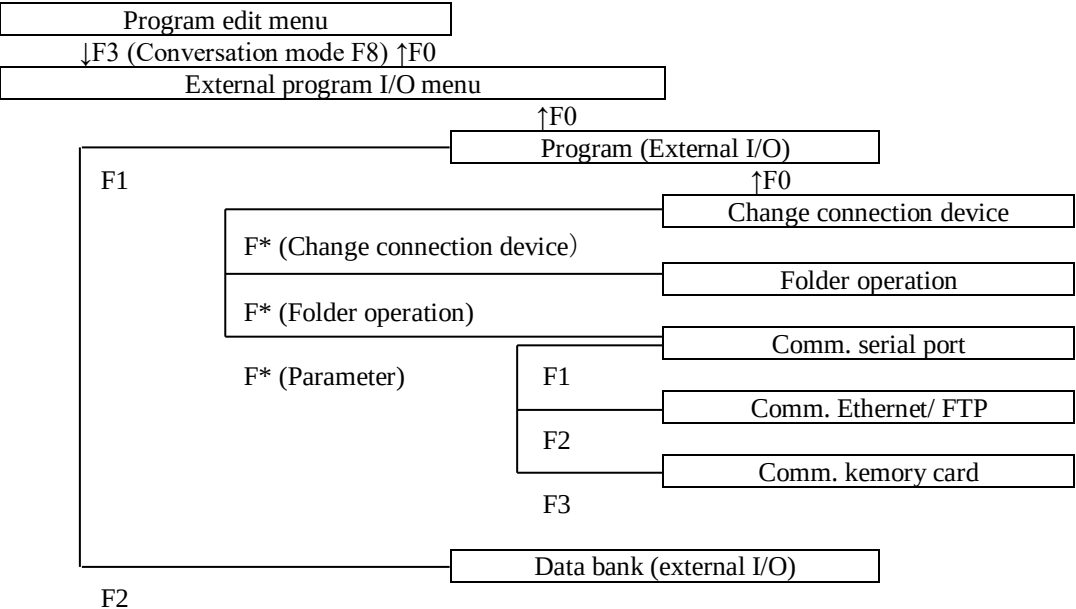
5.2.1.1 External program I/O menu

1. Display <Program edit menu> screen.
Press **[EDIT]** key.
Press **[PRGRM]** key.
2. Display <External program I/O menu> screen.
Press <External program I/O> key from the function keys.



3. Select input/output data.
[F1]: <Program>
[F2]: <Data bank>
[F3]: <Maintenance>

5.2.1.2 Screen flow



The layers below this are the same as the program (external I/O).

5.2.1.3 Program (external I/O)

NC language mode

Program details(external program I/O)					2012/05/29 19:18:32		
NC directory				Gen. COMM dir.			
No.	Size	Comment	Date				
MM/			12/05/29 18:58:29				
000/			12/05/25 20:39:02				
SSS/			12/04/26 18:53:48				
SSSPRP/			12/05/14 21:59:44				
VVV/			12/04/26 18:53:48				
0001	1.27K	(*****)	12/05/24 14:05:10	Directory of general communication device cannot be displayed.			
0002	144	(PRG2)	11/11/22 13:20:28				
0004	74	(PRG3)	11/11/22 13:20:28				
0005	4	M99	11/11/22 13:20:29				
0010	64	M00	11/11/22 13:20:29				
0011	24	M450	11/11/22 13:20:30				
0012	14	T901	11/11/22 13:20:30				
0013	9	M400	11/11/22 13:20:31				
0014	14	M450	11/11/22 13:20:39				
Program No.						Available 430MByte	Available -----Byte
No. 0001 1.27K Byte 12/05/24 14:05:10				Selected 0	Selected -----		
A-B:Change file A to B, and input or output. [HELP]							
A/B:Output files A to B.							
Data protection on:Check mode Data protection off:Input mode							
☒ Running ☐ Protected ☐ Selected Connected:General COMM device							
External program I/O menu	List	Sort	Select program	Input	Output	Folder operation	Change connection device

Conversation mode

Program details(external program I/O)					2012/05/29 19:18:41		
NC directory				Gen. COMM dir.			
No.	Size	Comment	Date				
LLL/			12/05/25 11:41:31				
MM/			12/05/29 12:05:36				
000/			12/05/25 20:39:29				
SSS/			12/05/25 11:41:31				
SSSPRP/			12/05/25 11:41:31				
VVV/			12/05/25 11:41:31				
0001	2.47K	(*****)	12/05/24 14:04:00	Directory of general communication device cannot be displayed.			
0002	12.1K	RENZOKU UNTEN 14T	11/12/22 11:43:37				
0003	1.82K	REAMER	12/01/23 14:34:05				
0004	1.31K	(*****)	12/04/23 22:15:48				
0005	1.22K	(*****)	12/01/23 17:16:04				
0010	4.96K	(*****)	11/11/22 13:20:30				
0011	4.83K	(*****)	11/11/22 13:20:30				
0012	1.89K	(*****)	12/01/30 15:10:25				
Program No.						Available 430MByte	Available -----Byte
No. 0001 2.47K Byte 12/05/24 14:04:00						Selected 0	Selected -----
A-B:Change file A to B, and input or output. [HELP]							
A/B:Output files A to B.							
Data protection on:Check mode Data protection off:Input mode							
☒ Running ☐ Protected ☐ Selected Connected:General COMM device							
External program I/O menu	List	Sort	Select program	Input	Output	Folder operation	Change connection device

<NC directory>

Displays the programs and sub-folders existing in the machine. The lower row shows the position of the current folder of the NC directory.

<Gen. COMM dir.>

(cannot be displayed)

<NC COMM directory>

Displays a program list in the connected device (NC COMM device).

<Memory card directory>

Displays a list of programs and subfolders that are in the connected media's directory (memory card).

The size that is displayed even for the same program varies because the saving format for data in the NC directory is different.

When the subfolder name is long, it is abbreviated.

The current folder's location in the directory for the connecting device is shown in the lower half.

<FTP server side>

Displays the programs and sub-folders existing in the connected device (FTP server).

Data storage format is different from the NC side so that the size is different even if the program is the same.

When the subfolder name is long, it is abbreviated.

The current folder's location in the directory for the connecting device is shown in the lower half.

<Available> (left)	Displays remaining storage capacity of the machine (bytes).
<Selected> (left)	Displays the total number of programs selected in the program list of the NC directory.
<Available> (right)	Displays the remaining system storage capacity of the connected device (bytes). Values are shown only when the connected device is set to <Memory card>.
<Selected> (right)	Displays the total number of programs selected in the program list of the directory of the connected device.
<Connected>	Displays the currently set connected device (general COMM device, NC COMM device, memory card, or FTP server). See “5.2.2 Change connection device” for the setting procedure.
<I>	This symbol is attached to inch programs when <Metric> is set for <Machine unit system>. Programs without this symbol are metric programs. This symbol is displayed only in the conversation mode.
<M>	This symbol is attached to metric programs when <Inch> is set for <Machine unit system>. Programs without this symbol are inch programs. This symbol is displayed only in the conversation mode.
[Details / List]	Changes the display. <Details>: Displays comments or the time of the last update. <List>: Increases the number of programs that can be displayed on the screen.
[Sort]	Sorts the programs according to the time of the last update, program number (in alphabetical order in the case of folders), and comment (in alphabetical order in the case of folders). The lists can be sorted in ascending or descending order.
[Select program]	Designates one or more programs to be input/output.
[Input]	Inputs the designated program on the connected device into the machine. Operate by enabling data protection switch, and the program is identified and checked.
[Output]	Outputs the designated program on the machine to the connected device.
[Folder operation]	Instructs various operations for folders. "Movement" and "preparation" can be performed for the folders on the NC side.
[Change connection device]	Changes the target device for communication.
[Move cursor right/left]	Switches cursor between machine side and connected device.
[Delete]	Deletes programs of the connected device.
[Input all programs]	Inputs the programs output by 'Output all programs'.
[Output all programs]	Outputs all programs in the current folder. Folders are not output.
[Parameter]	Changes the communication parameter.
[Input all]	Inputs a program on connected device by dividing it into multiple programs.
[Display all files / Display NC exclusive files]	
Display all files:	Displays all files existing in the connected device
Display NC exclusive files:	Displays only the files of a designated program name of the connected device. Part of a long file name is substituted by “_” as appropriate.

- * The function keys vary with the connected device.
Upper row **[F0]** to **[F7]**; press **[>>]** (next functions) to view the lower row **[F0]** to **[F7]**

General COMM device							
External program I/O menu	List/Details	Sort	Select program	Input	Output	Folder operation	Change connection device
External program I/O menu	Input all programs	Output all programs	Parameter	Input all			
NC COMM device							
External program I/O menu	List/Details	Sort	Select program	Input	Output	Folder operation	Move cursor right/left
External program I/O menu	Delete	Change connection device	Parameter	Input all			
Memory card							
External program I/O menu	List/Details	Sort	Select program	Input	Output	Folder operation	Move cursor right/left
External program I/O menu	Delete	Change connection device	Parameter	Input all	Display all files / Display NC exclusive files		
FTP server							
External program I/O menu	List/Details	Sort	Select program	Input	Output	Folder operation	Move cursor right/left
External program I/O menu	Delete	Change connection device	Parameter	Input all	Display all files / Display NC exclusive files		

- * <Input all> and <Display all files / Display NC exclusive files> are available only in the NC language mode.
- * To list the programs of the connected device, press the function key <Folder operation>, followed by Latest Directory. This is not valid, however, when a <general COMM device> is connected.
- * When the cursor is located on a folder on the <Program (external program I/O)> screen, the folder changes as you press **[ENT]** key (NC, memory card, and FTP server side).
- * Up to 4096 sub-folders and programs are displayed in the program list of the connected device. Sub-folders and programs exceeding this figure are not displayed.

Program(external program I/O)										2012/05/29 19:26:53			
NC directory						Memory card directory							
Current folder: /						Current folder: /							
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date		
AAA/		12/05/11	SSS/		12/04/26	AAA/		12/05/29	SSS/		12/05/29		
BBB/		12/04/26	SSSPPP/		12/05/14	BBB/		12/05/29	SSSPPP/		12/05/29		
CCC/		12/04/26	WWW/		12/04/26	CCC/		12/05/29	WWW/		12/05/29		
DDD/		12/04/26	0001	1.27K	12/05/24	DDD/		12/05/29	0001	1.33K	12/05/29		
EEE/		12/04/26	0002	144	11/11/22	EEE/		12/05/29	0002	165	12/05/29		
FFF/		12/04/26	0004	74	11/11/22	FFF/		12/05/29	0004	85	12/05/29		
GGG/		12/04/26	0005	4	11/11/22	GGG/		12/05/29	0005	5	12/05/29		
HHH/		12/05/23	0010	64	11/11/22	HHH/		12/05/29	0010	77	12/05/29		
III/		12/04/26	0011	24	11/11/22	III/		12/05/29	0011	29	12/05/29		
JJJ/		12/04/26	0012	14	11/11/22	JJJ/		12/05/29	0012	17	12/05/29		
KKK/		12/04/26	0013	9	11/11/22	KKK/		12/05/29	0013	11	12/05/29		
LLL/		12/04/26	0014	14	11/11/22	LLL/		12/05/29	0014	17	12/05/29		
MMM/		12/05/26	0016	21	11/11/22	MMM/		12/05/29	0016	25	12/05/29		
OOO/		12/05/25	0017	8	11/11/22	OOO/		12/05/29	0017	10	12/05/29		
No. 1			Available Selected 430Byte 0			Available Selected 932Byte 0							
No. 0001 1.27K Byte 12/05/24 14:05:10 (*****)													
A-B: Change file A to B, and input or output.													
A/B: Input or output files A to B.													
Data protection off: Input mode Data protection on: Check mode													
Destination: Memory card													
Running Protected Selected													
External program I/O menu	Details	Sort	Select program	Input	Output	Folder operation	Move cursor right/left						

5.2.1.4 Data bank (external program I/O)

NC language mode

Data bank(external program I/O)

2012/05/29 19:28:34

Items	Size	NC directory(Metric)					Gen. COMM dir.(Metric)
Workpc. coord. zero	17.1K	1	2	3	4	5	Directory of general communication device cannot be displayed.
Tool data	89.4K	1	2	3			
Macro variables	16.6K	1	2				
User parameter	10.7K	1		3	4		
External I/O signal	73.3K	1	2				
Commun. parameter	544	1					
Machine parameter	84.5K	1					
ATC tool data		1					

No.

Busy

Occupied 292KByte

Available 436MByte

Previous format

Occupied -----Byte

Available -----Byte

A-B:Change file A to B, and input or output.
A/B:Input or output files A to B.
Data protection off:Input mode Data protection on:Check mode

Connected:General COMM device (HELP)

External program I/O menu

Input

Output

Input all data banks

Output all data banks

Change connection device

Parameter

Conversation mode

Data bank(external program I/O)

2012/05/29 19:32:02

Items	Size	NC directory(Metric)					Gen. COMM dir.(Metric)
Tool list	52.2K	1	2	3	4	5	Directory of general communication device cannot be displayed.
Tool pattern	1.01K	1	2	3		9	
Tapping drill dia.	7.70K	1		3		10	
Cutting conditions	13.3K	1		3			
User parameter	10.7K	1		3	4		
External I/O signal	73.3K	1	2				
Commun. parameter	544	1					
Machine parameter	84.5K	1					

No.

Busy

Occupied 243KByte

Available 436MByte

Previous format

Occupied -----Byte

Available -----Byte

A-B:Change file A to B, and input or output.
A/B:Input or output files A to B.
Data protection off:Input mode Data protection on:Check mode

Connected:General COMM device (HELP)

External program I/O menu

Input

Output

Input all data banks

Output all data banks

Change connection device

Parameter

<Items>

Displays the type of data bank.

<Size>

Items that can be input and output vary with the connected device. Displays the total capacity (bytes) of data banks on the machine side.

<NC directory>

Displays the condition of registration of data banks of the machine. Data bank numbers currently in use are shown in green and those of previous formats in gray

<Gen. COMM dir.>

(cannot be displayed)

<NC COMM directory>

Displays a data bank list in the connected device (NC COMM device).

Data bank numbers of previous formats and those with different least input increments are shown in gray.

<Memory card directory>

Displays a data bank list in the connected device (Memory card). The lower row shows the position of the current folder of the connected device.

Data bank numbers of previous formats and those with different least input increments are shown in gray.

<FTP server side>	Displays a data bank list in the connected device (FTP server). The lower row shows the position of the current folder of the connected device. Data bank numbers of previous formats and those with different least input increments are shown in gray.
<Occupied> (left)	Displays the total capacity (byte) of all data banks on the machine side.
<Available> (left)	Displays remaining system storage capacity (bytes) of the machine.
<Occupied> (right)	Displays total capacity (bytes) of all data banks of the connected device.
<Available> (right)	Displays remaining system storage capacity (bytes) of the connected device. Values are displayed only when the connected device is set to <Memory card>.
<Connected>	Displays the currently set connected device (general COMM device, NC COMM device, memory card, or FTP server). See “5.2.2 Change connection device” for the setting procedure.

* The function keys vary with the connected device.

Upper row **[F0]** to **[F7]**; press **[>>]** (next functions) to view the lower row **[F0]** to **[F7]**

General COMM device							
External program I/O menu	Input	Output	Input all data banks	Output all data banks	Change connection device	Parameter	
NC COMM device							
External program I/O menu	Input	Output	Input all data banks	Output all data banks	Latest directory	Delete	Delete all data banks
External program I/O menu	Change connection device	Parameter					
Memory card							
External program I/O menu	Input	Output	Input all data banks	Output all data banks	Folder operation	Delete	Delete all data banks
External program I/O menu	Change connection device	Parameter					
FTP server							
External program I/O menu	Input	Output	Input all data banks	Output all data banks	Folder operation	Delete	Delete all data banks
External program I/O menu	Change connection device	Parameter					

* To display the data bank list of the connected device, press the function key <Latest directory>. If you do not see this key, first press the function key <Folder operation>, followed by <Latest directory>. This is not valid, however, when a <General COMM device> is connected.

5.2.2 Change connection device

5.2.2.1 Connection device changing procedure

The device to communicate is selected.

Specify the connected device correctly before starting master station communication.

1. Press <Change connection device> from the functions.

2. Specify the device to communicate.

Connected	Target device
General COMM device	PTP/PTR communication. Connect to the serial port.
NC COMM device	Communication with FACIT (Protocol type 1) Communication with devices available for command-response type communication (Protocol type 1, 2) Connect to the serial port. <NC COMM device> uses the Brother original communication system. The user must prepare software for the <NC COMM device> side when using communication device other than FACIT.
Memory card	Communication with a memory card. See "5.2.2.2 Memory card" for details.
FTP server	Communication with the <FTP server>. NC is the client of FTP. Connect NC to the network, and be ready with the <FTP server> on the network. NC is connected to the <FTP server> on the network in accordance the setting of <FTP server name (IP address)>, <FTP server port number>, <User name as client>, and <Password as client> of communication parameters. * The password for logging in the FTP server is not encrypted. You may not log in the FTP server that permits only the encrypted connection. Do not attempt to connect.

- * List of programs and data banks of the connected device cannot be displayed and folder operation for the connected device is not available depending on the connected device.
- * You cannot change the setting during tape operation (including pauses); the alarm <<Tape Operating mode>> occurs.
- * When not entering ASCII code data on a general communications device, we recommend performing a data check using ISO code with the communication parameter <Sending data code> set to <0: ISO>.
When inputting the data from a memory card, the data is automatically set to ASCII code.
- * When there is no RS232C option, the <General COMM device> and the <NC COMM device> may not display depending on the machine model being used.

Connected device	Display of lists of connected device	Folder operation for connected device
General COMM device	Display is not available.	Folder operation is not available
NC COMM device	Display is available	Folder operation is not available
Memory card	Display is available.	Folder operation is available
FTP server	Display is available.	Folder operation is available

* Files and folders existing in the FTP server may not be correctly displayed in the list if the name includes characters other than one-byte alphanumerals. When there is a file on the FTP server that has the same name as the input or output file but the file names differ only in letter case, then the file may not input or output correctly due to the FTP server configuration.

5.2.2.2 Memory card

NC has a USB interface for connecting memory cards. By inserting a USB memory or other adapter for USB-based recording media into the USB interface, these devices can be used as an external storage for NC.

Memory card for NC is a generic term for all these external storage devices including USB memory sticks.

* Precautions for use

1. Only FAT16 and FAT32 memory card file formats are supported (number of files to be created, etc. depends on format).
2. If write protect switch of a memory card is ON, input from the memory card to the machine is possible but output from the machine is impossible.
3. Security function of memory card must be “released”. The machine may not recognize the memory card.
4. The access lamp of a memory card lights up when the machine accesses the card. Remove the card only after the necessary operations are completed and the access lamp goes off. If you remove the card during an access, the machine may not recognize the card until power is turned off and then on next time.
5. Even if the empty space of a media exceeds 2G byte, the display on the screen remains 2G byte (maximum display value).
6. The alarm <<Multiple memory cards cannot be connected>> occurs if you connect more than one memory card.
7. The data cannot be read correctly if you use commercial memory cards without first logically formatting. Use them after logical formatting with the disc format of the machine.
8. Be sure to take steps to back up important data on your computer because some memory cards are not necessarily designed for long term storage of data.
9. Not all commercially available memory cards are guaranteed to function properly, and there are some cards with slower access speeds.
When actually using a memory card, carefully check the operation before use.
10. Memory cards that have not been physically formatted cannot be formatted on this machine. Format the card on a PC.

5.2.3 Folder operation

5.2.3.1 Folder operation screen

Various operations are performed for folders.

<Movement> and <Preparation> can be performed for the folders present on the NC side.

<Movement>, <Preparation>, <Deletion>, and <Change of Name> can be performed on the folders on the connected side.

Operations for the NC side folders are available only when the Folder Operation screen is opened from the <Program (external program I/O)> screen.

Operations for the connected side folders are not available depending on the connected device. See “5.2.2 Change connection device”.

1. Press the function key <Folder operation>.

Folder operation (external I/O) 2012/05/29 18:37:29			
NC directory		Memory card directory	
Current folder: /		Current folder: /AAA/	
Name	Date	Name	Date
MM/	12/05/29 18:38:29	../(master folder)	
000/	12/05/25 20:39:02	00001.NC	12/05/29 18:39:24
SSS/	12/04/26 18:53:48	00002.NC	12/05/29 18:39:24
SSSPPP/	12/05/14 21:59:44	00004.NC	12/05/29 18:39:26
000/	12/04/26 18:53:48	00005.NC	12/05/29 18:39:26
0001	12/05/24 14:05:10	00010.NC	12/05/29 18:39:26
0002	11/11/22 13:20:28	00011.NC	12/05/29 18:39:26
0004	11/11/22 13:20:28	00025.NC	12/05/29 18:39:26
0005	11/11/22 13:20:29	00045.NC	12/05/29 18:39:38
0010	11/11/22 13:20:29	00202.NC	12/05/29 18:39:38
0011	11/11/22 13:20:30	01114.NC	12/05/29 18:39:38
0012	11/11/22 13:20:30	GM0001.NC	12/05/29 18:39:24
0013	11/11/22 13:20:31	GM0002.NC	12/05/29 18:39:24
0014	11/11/22 13:20:39	GM0003.NC	12/05/29 18:39:24
[HELP]			
Destination: Memory card			
Back	Move	Create	Delete
Rename	Directory request	Sort	Move cursor right/left

<NC directory>

Displays the programs and sub-folders existing in the current folder of the machine. Folder names are identified by the suffix “/” and displayed in blue.

The lower row shows the position of the current folder of the NC directory.

This is displayed only when this screen is opened from the <Program (external program I/O)> screen.

<Gen. COMM dir.>

Not displayed.

<NC COMM directory>

Displays a program list in the connected device (NC COMM device). Extensions are not displayed.

<Memory card directory>

Displays a list of programs and subfolders in the current folder for the connected media’s directory (memory card). All files are displayed with their extensions regardless if the directory has been selected or not for communication. For folders, a slash “/” is inserted and displayed at the tail end of the name.

When the file or subfolder name is long, it is abbreviated.

The current folder’s location in the directory for the connecting device is shown in the lower half.

<FTP server side>

Displays the programs and sub-folders existing in the current folder of the connected device (FTP server). All files with extensions are listed irrespective of whether they are communicated or not. Folder names are identified by the suffix “/.”

When the file or subfolder name is long, it is abbreviated.

The current folder’s location in the directory for the connecting device is shown in the lower half.

<Connected>

Displays the currently set connected device (general COMM device, NC COMM device, memory card, or FTP server). See “5.2.2 Change connection device” for the setting procedure.

2. After completing folder operation, press **[F0]** (Back) key to quit the screen.

5.2.3.2 Moving the folder

1. Press the function key **[F7]** (Move cursor right/left) to select the NC or the connected device where you wish to move folders.
 2. Select the folder you wish to move with the cursor and press the function key **[F1]** (move).
 3. The current folder changes to the destination folder, and the programs and the sub-folder of the destination folder are displayed.
- * Alternatively, select a folder with the cursor and press **[ENT]** key, and you can move to the selected folder.

5.2.3.3 Creating a folder

1. Press the function key **[F7]** (Move cursor right/left) to select the NC or the connected device where you wish to create a folder.
2. Press the function key **[F2]** (Create).
The folder creation screen opens.

NC directory		Memory card directory	
Name	Date	Name	Date
000/	12/05/25 20:39:02	00001.NC	12/05/29 18:39:24
001	12/05/24 14:05:10	00002.NC	12/05/29 18:39:24
0002	11/11/22 13:20:28	00004.NC	12/05/29 18:39:26
0004	11/11/22 13:20:28	00005.NC	12/05/29 18:39:26
0005	11/11/22 13:20:29	00010.NC	12/05/29 18:39:26
0010	11/11/22 13:20:29	00011.NC	12/05/29 18:39:26
0011	11/11/22 13:20:30	00025.NC	12/05/29 18:39:26
0012	11/11/22 13:20:30	00045.NC	12/05/29 18:39:38
0013	11/11/22 13:20:31	00202.NC	12/05/29 18:39:38
0014	11/11/22 13:20:39	01114.NC	12/05/29 18:39:38
		GM0001.NC	12/05/29 18:39:24
		GM0002.NC	12/05/29 18:39:24
		GM0003.NC	12/05/29 18:39:24

Folder name:

Destination: Memory card

3. Type a new folder name in the entry field below the screen and press **[F1]** (Create) key.
A folder name consists of a maximum of 8 characters.
Up to 2 layers of folders including the root folder can be created on the NC device (folders cannot be created within a folder).
- * Names reserved in the NC (program file names and data bank file names) cannot be used as the folder name.

5.2.3.4 Deleting a folder

Folders on the connected device are deleted. Delete the folders on the NC directory on the directory display screen. Refer to “1.5.2.2 To delete a folder” for details.

1. Move the cursor to the connected device by pressing the **[F7]** (Move cursor right/left) key.
2. Select the folder you wish to delete with the cursor and press **[F3]** (Delete) key. Delete confirmation screen appears.

Folder operation (external I/O) 2012/05/29 19:45:59

NC directory		Memory card directory	
Current folder: /		Current folder: /AAA/	
Name	Date	Name	Date
MMM/	12/05/29 18:58:29	../(master folder)	
DDD/	12/05/25 20:39:02	MMM/	12/05/29 19:38:12
SSS/	12/04/26 18:53:48	00001.NC	12/05/29 18:39:24
SSSPHP/	12/05/14 21:59:44	00002.NC	12/05/29 18:39:24
VVV/	12/04/26 18:53:48	00004.NC	12/05/29 18:39:26
0001	12/05/24 14:05:10	00005.NC	12/05/29 18:39:26
0002	11/11/22 13:20:28	00010.NC	12/05/29 18:39:26
0004	11/11/22 13:20:28	00011.NC	12/05/29 18:39:26
0005	11/11/22 13:20:29	00025.NC	12/05/29 18:39:26
0010	11/11/22 13:20:29	00045.NC	12/05/29 18:39:38
0011	11/11/22 13:20:30	00202.NC	12/05/29 18:39:38
0012	11/11/22 13:20:30	01114.NC	12/05/29 18:39:38
0013	11/11/22 13:20:31	QM0001.NC	12/05/29 18:39:24
0014	11/11/22 13:20:39	QM0002.NC	12/05/29 18:39:24

FolderMMM
All files and sub folders in the folder are deleted.
Delete file?

Destination: Memory card ☐

Yes No

3. Confirm the deletion.
Press **[F0]** (Yes) key.

* “../(master folder)” cannot be deleted.

5.2.3.5 Changing the folder name

Folder names on the connected device are changed. Go to the Directory display screen to change the folder name on the NC side. Refer to “1.5.2.3 To change a folder name” for details.

1. Move the cursor to the connected device by pressing the **[F7]** (Move cursor right/left) key.
2. Press **[F4]** (Rename) key. The Rename screen appears.

Folder operation (external I/O) 2012/05/29 19:46:41

NC directory		Memory card directory	
Current folder: /		Current folder: /AAA/	
Name	Date	Name	Date
MMM/	12/05/29 18:58:29	../(master folder)	
DDD/	12/05/25 20:39:02	MMM/	12/05/29 19:38:12
SSS/	12/04/26 18:53:48	00001.NC	12/05/29 18:39:24
SSSPHP/	12/05/14 21:59:44	00002.NC	12/05/29 18:39:24
VVV/	12/04/26 18:53:48	00004.NC	12/05/29 18:39:26
0001	12/05/24 14:05:10	00005.NC	12/05/29 18:39:26
0002	11/11/22 13:20:28	00010.NC	12/05/29 18:39:26
0004	11/11/22 13:20:28	00011.NC	12/05/29 18:39:26
0005	11/11/22 13:20:29	00025.NC	12/05/29 18:39:26
0010	11/11/22 13:20:29	00045.NC	12/05/29 18:39:38
0011	11/11/22 13:20:30	00202.NC	12/05/29 18:39:38
0012	11/11/22 13:20:30	01114.NC	12/05/29 18:39:38
0013	11/11/22 13:20:31	QM0001.NC	12/05/29 18:39:24
0014	11/11/22 13:20:39	QM0002.NC	12/05/29 18:39:24

New folder name

Destination: Memory card ☐

Back Rename

3. Select the folder which name you wish to change. Type a new folder name in the entry field below the screen and press **[F1]** (Rename) key.
A folder name consists of a maximum of 8 characters.
- * Names reserved in the NC (program file names and data bank file names) cannot be used as the folder name.

The rename confirmation screen opens.

NC directory		Memory card directory	
Current folder: /		Current folder: /AAA/	
Name	Date	Name	Date
MM/	12/05/29 18:58:29	MM/	12/05/29 19:38:12
UU/	12/05/26 20:39:02	00001.NC	12/05/29 18:39:24
SS/	12/04/26 18:53:48	00002.NC	12/05/29 18:39:24
SSSP/	12/05/14 21:58:44	00004.NC	12/05/29 18:39:26
VW/	12/04/26 18:53:48	00005.NC	12/05/29 18:39:26
0001	12/05/24 14:05:10	00010.NC	12/05/29 18:39:26
0002	11/11/22 13:20:28	00011.NC	12/05/29 18:39:26
0004	11/11/22 13:20:28	00025.NC	12/05/29 18:39:26
0005	11/11/22 13:20:29	00045.NC	12/05/29 18:39:38
0010	11/11/22 13:20:29	00202.NC	12/05/29 18:39:38
0011	11/11/22 13:20:30	01114.NC	12/05/29 18:39:38
0012	11/11/22 13:20:30	GM0001.NC	12/05/29 18:39:24
0013	11/11/22 13:20:31	GM0002.NC	12/05/29 18:39:24
0014	11/11/22 13:20:39		

Folder MMM
Rename? → ZZZ

Destination: Memory card ☒

Yes No

4. Confirm renaming.
Press **[F0]** (Yes) key.
- * “../(master folder)” cannot be renamed.

5.2.3.6 Sort

1. Press **[F6]** (Sort) key.
The sort screen opens.
The lists can be sorted by date and name in ascending or descending order.

5.2.3.7 Latest directory

The list of programs and folders on the connected device is refreshed.
The list of programs is refreshed when the connected device is an <NC COMM device>.
This is invalid when the connected device is a <general COMM device>.

5.2.4 Parameters

Communication parameters are changed.

Settings and parameters of the connected device must be adjusted before starting master station communication.

See “2.7 Communication Parameters” for the details of setting.

1. Press the function key <Parameter>.

Parameter	Value
Reset(slave)[0:Off 1:On]	0
Data overwrite(slave)[0:No 1:Yes]	1
Remote operation[0:Off 1:On]	1
Display slave command alarm[0:No 1:Yes]	1
Program batch input system[0:System 1 1:System 2]	1
Add comment of file name[0:No 1:Yes]	0
Extended comment input[0:No 1:Yes]	1
Connected device	1234
NC communication device protocol(master)[0:Type1 1:Type2]	0
Baud rate[0:300 1:600 2:1200 3:2400 4:4800 5:9600 6:19200 7:38400 8:57600]	8
Parity[0:None 1:Odd 2:Even]	0
Stop bit[0:1 bit 1:2 bits]	0
Character length[0:7 bits 1:8 bits]	0
Communication mode[0:Line 1:Code 1 2:Code 2]	0
Response monitoring time	0 s
Invalid data[0:Convert 1>Error 2:Ignore]	0

Reset(slave)

[HELP]

Completion mode	Serial port	Ethernet/FTP	Memory card

2. Select the connected device with the function keys.
 [F1]: Serial port
 [F2]: Ethernet/FTP
 [F3]: Memory card
3. Adjust settings to the communication device.

5.2.5 Selecting programs to communicate

5.2.5.1 Program selection screen

Multiple programs can be specified for inputting/outputting.

Multiple programs are selected on the screen.

1. Press the function key <Select program>.

NC directory						Memory card directory					
Current folder: /						Current folder: /					
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
0001	1.27K	12/05/24	0021	8	11/11/22	0032	17	12/05/29	0046	1.99M	12/05/29
0002	144	11/11/22	0022	8	11/11/22	0033	53	12/05/29	0047	17	12/05/29
0004	74	11/11/22	0023	8	11/11/22	0034	29	12/05/29	0048	17	12/05/29
0005	4	11/11/22	0024	8	11/11/22	0035	73	12/05/29	0049	15	12/05/29
0010	64	11/11/22	0025	8	11/11/22	0036	11	12/05/29	0050	15	12/05/29
0011	24	11/11/22	0026	9	11/11/22	0037	11	12/05/29	0051	172	12/05/29
0012	14	11/11/22	0027	12	11/11/22	0038	11	12/05/29	0052	48	12/05/29
0013	9	11/11/22	0028	8	11/11/22	0039	11	12/05/29	0053	11	12/05/29
0014	14	11/11/22	0029	14	11/11/22	0040	27	12/05/29	0054	167	12/05/29
0016	21	11/11/22	0030	14	11/11/22	0041	19	12/05/29	0055	2.25M	12/05/29
0017	8	11/11/22	0031	14	11/11/22	0042	33	12/05/29	0056	23	12/05/29
0018	8	11/11/22	0032	14	11/11/22	0043	19	12/05/29	0057	64	12/05/29
0019	8	11/11/22	0033	44	11/11/22	0044	14	12/05/29	0058	11	12/05/29
0020	8	11/11/22	0034	24	11/11/22	0045	22	12/05/29	0059	46	12/05/29

Available 438MByte
Selected 0

Available 932MByte
Selected 0

No. 0001 1.27K Byte 12/05/24 14:05:10 (*****)

Move the cursor to the program to be input or output, and press [F1].

Destination: Memory card

Back	Select/ Deselect	Start selection	Select all	Deselect all	Move cursor right/left

- [Select / Cancel]** Select the desired number with the cursor and press the key. Unselected number is selected and the selected one deselected by pressing this key.
- [Start / End range selection]** Multiple programs are selected by specifying the range. Select the program number you wish to start selection, and press the key. Move the cursor to the ending program number, and press the key again.
- [Select all]** All programs are selected.
- [Cancel all]** All programs are deselected.
- [Move cursor right/left]** Switches cursor between machine side and connected device.

2. After completing program selection, press **[F0]** (Back) key to quit the screen.

5.2.5.2 Selecting programs to be input

Some programs may not be selected from the program list of the connected device depending on the connected device. See “5.2.2 Change connection device.”

Press the **[F7]** (Move cursor right/left) to move the cursor to the communication side (right side of the screen) and select the program.

Program(external program I/O)										2012/05/29 19:52:15			
NC directory						Memory card directory							
Current folder: /						Current folder: /							
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date		
0001	1.27K	12/05/24	0021	8	11/11/22	0032	17	12/05/29	0046	1.99M	12/05/29		
0002	144	11/11/22	0022	8	11/11/22	0033	53	12/05/29	0047	17	12/05/29		
0004	74	11/11/22	0023	8	11/11/22	0034	29	12/05/29	0048	17	12/05/29		
0005	4	11/11/22	0024	8	11/11/22	0035	73	12/05/29	0049	15	12/05/29		
0010	64	11/11/22	0025	8	11/11/22	0036	11	12/05/29	0050	15	12/05/29		
0011	24	11/11/22	0026	9	11/11/22	0037	11	12/05/29	0051	172	12/05/29		
0012	14	11/11/22	0027	12	11/11/22	0038	11	12/05/29	0052	48	12/05/29		
0013	9	11/11/22	0028	8	11/11/22	0039	11	12/05/29	0053	11	12/05/29		
0014	14	11/11/22	0029	14	11/11/22	0040	27	12/05/29	0054	167	12/05/29		
0016	21	11/11/22	0030	14	11/11/22	0041	19	12/05/29	0055	2.25M	12/05/29		
0017	8	11/11/22	0031	14	11/11/22	0042	33	12/05/29	0056	23	12/05/29		
0018	8	11/11/22	0032	14	11/11/22	0043	19	12/05/29	0057	64	12/05/29		
0019	8	11/11/22	0033	44	11/11/22	0044	14	12/05/29	0058	11	12/05/29		
0020	8	11/11/22	0034	24	11/11/22	0045	22	12/05/29	0059	46	12/05/29		
Available 430Byte Selected 0						Available 932Byte Selected 13							
No. 0058 11 Byte 12/05/29 18:41:34													
Move the cursor to the program to be input or output, and press [F1].													
[HELP]													
Destination: Memory card													
Back	Select/ Deselect	Start selection	Select all	Deselect all						Move cursor right/left			

- * When a <memory card> or <FTP server> is connected, you may select folders as well. When selecting a folder, the folder and all programs in the folder are input. If the folder name is not permitted on the NC side, the operator message <<This is a folder name that cannot be input>> appears and you cannot select. If the current folder on the NC side is not the root folder, the operator message <<No folder can enter the current folder position>> appears and your request for input is rejected. If sub-folders are included in the selected folder, the operator message <<There was a subfolder unable to input>> appears and the sub-folders will not be input.

In the check mode, when the folder name is not permitted on the NC side, the operator message <<This is a folder name that cannot be checked>> appears and the folder is not checked. If sub-folders are included in the selected folder, the operator message << There was a subfolder that cannot be checked>> appears and the sub-folders will not be checked.

- * All selected programs are deselected when you move the folder of the connected device or command Latest Directory.

5.2.5.3 Selecting programs to output

Press the **[F7]** (Move cursor right/left) to move the cursor to the NC side (left side of the screen) and select the program.

Program(external program I/O) 2012/05/29 19:53:51											
NC directory						Memory card directory					
Current folder: /						Current folder: /					
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
0036	9	11/11/22	0050	12	11/11/22	0202	6.63K	12/05/29			
0037	9	11/11/22	0051	159	11/11/22	0203	134	12/05/29			
0038	9	11/11/22	0052	41	11/11/22	0204	117	12/05/29			
0039	9	11/11/22	0053	9	11/11/22	0205	21.7K	12/05/29			
0040	25	11/11/22	0054	155	11/11/22	0206	302	12/05/29			
0041	16	11/11/22	0055	2.15M	11/11/22	0207	168	12/05/29			
0042	28	11/11/22	0056	19	11/11/22	0208	147	12/05/29			
0043	16	11/11/22	0057	57	11/11/22	0209	154	12/05/29			
0044	12	11/11/22	0058	9	11/11/22	0210	145	12/05/29			
0045	21	12/04/24	0059	42	11/11/22	0211	156	12/05/29			
0046	1.90M	11/11/22	0060	43	11/11/22	0212	142	12/05/29			
0047	14	11/11/22	0061	67	11/11/22						
0048	14	11/11/22	0068	58	11/11/22						
0049	12	11/11/22	0069	4	11/11/22						
Available 430Byte Selected 9						Available 932Byte Selected 0					
No. 0043 16 Byte 11/11/22 13:21:11 G04 X5 Move the cursor to the program to be input or output, and press [F1].						[HELP]					
						Destination: Memory card 					
											
Back	Select/ Deselect	Start selection	Select all	Deselect all						Move cursor right/left	

- * When a <memory card> or <FTP server> is connected, you may select folders as well. When selecting a folder, the folder and all programs in the folder are output.
- * All selected programs are deselected when you change the connected device.

5.2.5.4 Selecting programs to delete

The operation is the same as “5.2.5.2 Selecting programs to be input”.

- * When a <memory card> or <FTP server> is connected, you may select folders as well. When selecting a folder, the folder and all programs and folders in the selected folder are deleted. When a folder contains non-displayed files (programs, data banks, text files, etc. of the non-selected language mode), the folder and its non-displayed files are not deleted. To delete a folder including non-displayed files, go to the Folder operation screen to follow the necessary deletion procedure.
- * If the folder name is not permitted on the NC side, the operator message <<This is a folder name that cannot be input>> appears and you cannot select. To delete this type of folder, go to the Folder operation screen to follow the necessary deletion procedure.

5.2.6 Input/output of programs

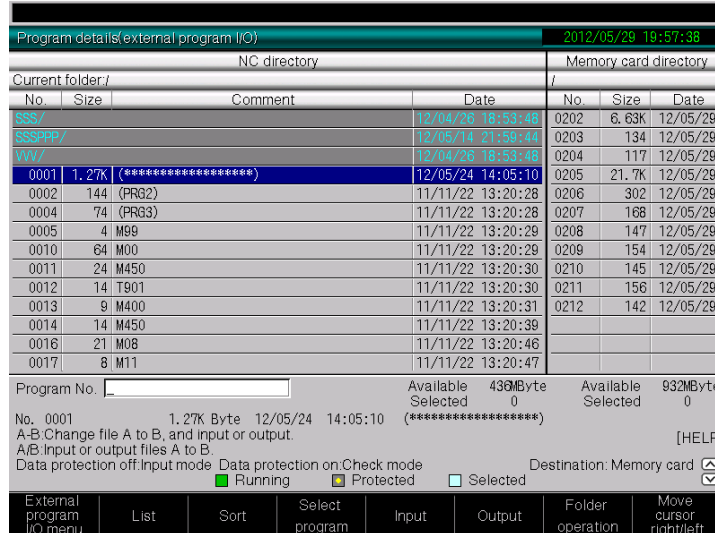
5.2.6.1 Program input method

(from connected device to machine)

Programs specified on the connected device are input into the machine. Programs in the current folder of the connected device are input into the current folder of the machine.

1. Display the <Program (external program I/O)> screen.
2. Connect the target device to the machine and check the display of <Connected> on the lower right corner of the screen.

See “5.2.2 Change connection device” for the connection changing procedure.



3. Check [DATA PROTECTION] switch.
Data protection off: input
Data protection on: check mode (compares files)
4. Display the directory of the connected device. (* For the connected devices other than <Gen. COMM device>)
Press [F6] (Folder operation) key, followed by [F5] (Directory request) key.
5. Specify the program number to be input.
(1) Select on the screen (* For the connected devices other than <Gen. COMM device>)
See “5.2.5 Selecting programs to communicate.”
(2) Enter the program number.

Enter the program number in the entry field below the screen.

Entry example	Communication operation
1234	Connected device 1234 ⇒ NC side 1234
1234-5678	Connected device 1234 ⇒ NC side 5678
1000/1999 (* For the connected devices other than gen. COMM device)	Connected device All of 1000 to 1999 ⇒ NC side All of 1000 to 1999
1000 2000 3000 (Separate by a space; up to 24 characters) (* For the connected devices other than gen. COMM device)	Connected device 1000, 2000, 3000 ⇒ NC side 1000, 2000, 3000

- * An alarm occurs when you use both procedures 1 and 2.
 - * Use procedure 1 to input by folder.
 - * The user may be unable to input a program from another company, for example, because the formatting is different. For example, when the beginning or end of the program statement has a % sign, delete the % sign and input the program again.
6. Start communication.
Press [F4] (Input) key.

7. Communication starts.

Program details(external program I/O)				2012/05/29 20:01:46		
NC directory				Memory card directory		
Current folder: /				/		
No.	Size	Comment	Date	No.	Size	Date
0202	6.32K	(RENZOKU UNTEN 21T)	11/11/22 13:22:10	0202	6.63K	12/05/29
0203	127	G90	11/11/22 13:22:11	0203	134	12/05/29
0204	110	G91	11/11/22 13:22:11	0204	117	12/05/29
0205	20.8K	(123456789123456789)	11/11/22 13:22:12	0205	21.7K	12/05/29
0206	288	N01 G90G54	12/05/29 20:01:25	0206	302	12/05/29
0207	159	T101H1D1	11/11/22 13:22:12	0207	168	12/05/29
0208	138	T101H1D1	11/11/22 13:22:12	0208	147	12/05/29
0209	145	T101H1D1	11/11/22 13:22:13	0209	154	12/05/29
0210	136	T101H1D1	11/11/22 13:22:13	0210	145	12/05/29
0211	147	T101H1D1	11/11/22 13:22:13	0211	156	12/05/29
0212	132	(0001)	11/11/22 13:22:13	0212	142	12/05/29
0213	132	(0002)	11/11/22 13:22:14			
0214	118	(0011)	11/11/22 13:22:14			
0215	122	(0012)	11/11/22 13:22:14			
Program						
Transferring						
Memory card 0206 → NC 0206 Destination: Memory card						

- * Overwrite confirmation screen opens when the same file name exists on the machine side.

[F0] (Yes) : The data are overwritten
[F1] (No) : Input of the relevant data is skipped
[F4] (Input all remaining) : All data is input hereafter without confirmation

8. When communication completes, the screen returns to the former screen.

- *1 In the NC language mode, when the connected device is a <general COMM device>, you can start inputting without specifying the program number. Press **[F4]** key without specifying a program number, and the number of the first block of the program is taken as the program number. If no program number exists in the first block, the system asks you to enter the program number.
- *2 In the NC language mode, a program can be input by dividing it into multiple programs.
- General COMM device
Press **[>>]** (Next functions) key, followed by **[F4]** (Input all) key. The program on the tape is divided into multiple programs.
 - Other than general COMM device
Select the file to input and press **[>>]** (Next functions) key, followed by **[F4]** (Input all) key. The program included in the file is divided into multiple programs. Just one file can be specified in the case of Input All.

Delimiters for dividing programs vary with the setting of communication parameter <Program batch input system>.

Setting of "Program Batch Input System"	Dividing method
Type 1	Programs are divided by M02, M30, and M99. Programs with data existing after M02, M30, or M99 and those without M02, M30, or M99 command are not correctly divided.
Type 2	Programs are divided by program number (O****). Programs without a program number in the first block are not correctly divided.

- *3 In the NC language mode, when the connected device is a <memory card> or an <FTP server>, you can input programs output from other than the machine (O****.NC).
- Press **[F5]** (Display all files) from the functions listed.
All files in the connected media's directory are displayed with their extensions.
 - Specify the program to input.
 - Select on the screen.
See "5.2.5 Selecting programs to communicate".
 - Enter the file name.
Enter the file name (including the extension) in the entry field below the screen.
Example: "ABC.TXT", "ABC.TXT-1234"

(Note) An alarm occurs when you use both procedures 1 and 2.

- Start communication.
Press **[F4]** (Input) key.
If the file to input is other than "O****.NC," the system asks you to enter a program number unless the program number is specified (ABC.TXT-1234). Enter the desired program number in the entry field below the screen and press **[F0]** (Input) key.
The file is stored on the NC side under the specified number.

By setting the communication parameter <Add comment of file name>, you can add the original file name to the first block as a comment line.

- *4 If the connecting destination is <Memory card> or <FTP server> for <Display NC exclusive files>, when a program with 3D smoothing data is input for the connecting destination, the 3D smoothing data is also input. If the program number is changed for a program, also change the program number for the 3D smoothing data. Press the **[F1]** (No) key on the <Overwrite confirmation> screen to skip the program input, and then the 3D smoothing data will also not be input.

Input example	Communication operation	
1234	Connecting dir.	NC directory
	O1234.NC	⇒ O1234
	SD1234.NC	SD1234
1234-5678	Connecting dir.	NC directory
	O1234.NC	⇒ O5678
	SD1234.NC	SD5678

- When using input all (multiple files or batch input), the 3D smoothing data will not input.
- When a program with 3D smoothing path data is output and edited on an external machine, the 3D smoothing path data is deleted when that program is input again. Therefore, the data must be created once more before operation. (The 3D smoothing parameter setting values are maintained.)
- If the communication parameter <Extended comment input> is set to <0: No>, when a program includes characters that are unavailable using the key operations, the comment for the program changes. It is the same as when editing data on an external machine, the 3D smoothing path data is deleted. When this happens, set the communication parameter <Extended comment input> to <1: Yes>.
- If the connecting destination is <Memory card> or <FTP server> for <Display of all files>, input the program and the 3D smoothing data separately. However, 3D smoothing data that has a file name other than "SD****.NC" cannot be input.

Input example	Communication operation	
O1234.NC	Connecting dir.	NC directory
	O1234.NC	⇒ O1234
	SD1234.NC	
SD1234.NC	Connecting dir.	NC directory
	O1234.NC	⇒
	SD1234.NC	SD1234
O1234.NC-5678	Connecting dir.	NC directory
	O1234.NC	⇒ O5678
	SD1234.NC	
SD1234.NC-5678	Connecting dir.	NC directory
	O1234.NC	⇒
	SD1234.NC	SD5678
ABC.TXT-1234	Connecting dir.	NC directory
	ABC.TXT	⇒ O1234

- The 3D smoothing data can be separately input only when the 3D smoothing data is distributed to separate systems that use the same program but the data is created in another system. Input the program data beforehand using a program that exists in the NC directory. When an attempt is made to only input 3D smoothing data without any programs present in the NC directory, the alarm <<3D smoothing data cannot be input>> is triggered.

5.2.6.2 Program output method (from machine to connecting directory)

Programs specified on the machine are output to the connected device. Programs in the current folder of the NC side are output to the current folder of the connected device.

1. Display <Program (external program I/O)> screen.
2. Connect the target device to the machine and check the display of <Connected> on the lower right corner of the screen.

See “5.2.2 Change connection device” for the connection changing procedure.

Program details(external program I/O) 2012/05/29 20:06:58

NC directory				Memory card directory		
No.	Size	Comment	Date	No.	Size	Date
2114	8.01K	(SUB PROGRAM)	11/11/22 13:22:37	0202	6.63K	12/05/29
2115	648	(MILL LINE)	11/11/22 13:22:37	0203	134	12/05/29
2116	536	(MILL CIRCLE)	11/11/22 13:22:38	0204	117	12/05/29
3210	147	(9876543210)	11/12/05 19:08:05	0205	21.7K	12/05/29
6030	485	N0001 (RUN-ED IT-DATA-NA IBU1)	12/04/16 16:02:44	0206	302	12/05/29
7000	495	N0001 (RUN-ED IT-DATA-NA IBU1)	12/04/18 11:04:54	0207	168	12/05/29
				0208	147	12/05/29
				0209	154	12/05/29
				0210	145	12/05/29
				0211	156	12/05/29
				0212	142	12/05/29

Program No. [] Available 430Byte Available 932MByte
 No. 6030 485 Byte 12/04/16 16:02:44 N0001 (RUN-ED IT-DATA-NA IBU1) Selected 0 Selected 0
 A-B: Change file A to B, and input or output. [HELP]
 A/B: Input or output files A to B.
 Data protection off: Input mode Data protection on: Check mode
☒ Running ☐ Protected ☐ Selected Destination: Memory card [v]
 External program I/O menu List Sort Select program Input Output Folder operation Move cursor right/left

3. Display the directory of the connected device. (* For the connected devices other than <gen. COMM device>)
Press [F6] (Folder operation) key, followed by [F5] (Directory request) key.
4. Specify the program number to be output.
 - (1) Select on the screen (* For the connected devices other than <gen. COMM device>)
See “5.2.5 Selecting programs to communicate”.
 - (2) Enter the program number.
Enter the program number in the entry field below the screen.

Entry example	Communication operation
1234	NC side 1234 ⇒ Connected device 1234
1234-5678	NC side 1234 ⇒ Connected device 5678
1000/1999 (* Batch output when <general COMM device> is connected.)	NC side All of 1000 to 1999 ⇒ Connected device All of 1000 to 1999
1000 2000 3000 (Separate by a space; up to 24 characters) (* Batch output when <general COMM device> is connected.)	NC side 1000, 2000, 3000 ⇒ Connected device 1000, 2000, 3000

- * An alarm occurs when you use both procedures (1) and (2).
- * Use procedure (1) to output by folder.

5. Start communication.
Press [F5] (Output) key.

7. When communication completes, the screen returns to the former screen.

Input example	Communication operation	
1234	NC directory O1234 SD1234	Connecting dir. ⇒ O1234.NC SD1234.NC
1234-5678	NC directory O1234 SD1234	Connecting dir. ⇒ O5678.NC SD5678.NC

- If the connecting destination is <Memory card> or <FTP server> for <Display of all files>, when a program with 3D smoothing data is output to the NC directory, the 3D smoothing data is also output. If the program number is changed for a program, also change the program number for the 3D smoothing data. However, if an attempt is made to change the file name (that is not “O****.NC”) for a program with 3D smoothing data in the NC directory, an alarm <<3D smoothing data cannot be output>> is triggered.

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- 5 - 23

eCOM3OPDT5

Input example	Communication operation	
1234	NC directory O1234 SD1234	Connecting dir. ⇒ O1234.NC SD1234.NC
1234-O5678.NC	NC directory O1234 SD1234	Connecting dir. ⇒ O5678.NC SD5678.NC

- If there is a file in the connecting directory with the same file name as the file being output, then the file deletion check screen is displayed before the program is output. If the program and 3D smoothing data are output at the same time, press the [F1] (No) key on the file deletion check screen to skip the program output, and the 3D smoothing data will not be output.

5.2.6.3 Program deletion method (from connected device)

Programs are deleted from the connected device. Programs on the current folder are deleted. No deletion is available when the connected device is a <general COMM device>.

1. Display <Program (external program I/O)> screen.
2. Connect the target device to the machine and check the display of <Connected> on the lower right corner of the screen.
See “5.2.2 Change connection device” for the changing procedure.
3. Display the directory of the connected device.
Press [F6] (Folder operation) key, followed by [F5] (Directory request) key.



4. Specify the program number to delete.
 - (1) Select on the screen.
See “5.2.5 Selecting programs to communicate”.
 - (2) Enter the program number.
Enter the program number in the entry field below the screen.

Entry example	Communication operation
1234	Delete 1234 on the connected device
1000/1999	Delete all of 1000 to 1999 on the connected device
1000 2000 3000 (Separate by a space)	Delete 1000, 2000, and 3000 on the connected device

- * An alarm occurs when you use both procedures (1) and (2).
 - * Use procedure (1) to delete by folder. If the folder name is not permitted on the NC side, the operator message <<This is a folder name that cannot be input>> appears and you cannot select. Go to the Folder Operation screen to delete a folder which name is not permitted on the machine side.
5. Start communication.
Press [>>] (Next functions) key, followed by [F1] (Delete).

6. Confirm the deletion.
Press **[F0]** (Yes) key.
7. Communication starts.

Program details(external program I/O)					2012/05/30 10:30:06		
NC directory					Memory card directory		
Current folder:/					/		
No.	Size	Comment	Date		No.	Size	Date
0001	1.27K	(*****)	12/05/24	14:05:10	0202	6.63K	12/05/29
0002	144	(PRG2)	11/11/22	13:20:28	0203	134	12/05/29
0004	74	(PRG3)	11/11/22	13:20:28	0204	117	12/05/29
0005	4	M99	11/11/22	13:20:29	0205	21.7K	12/05/29
0010	64	M00	11/11/22	13:20:29	0206	302	12/05/29
0011	24	M450	11/11/22	13:20:30	0207	168	12/05/29
0012	14	T901	11/11/22	13:20:30	0208	147	12/05/29
0013	9	M400	11/11/22	13:20:31	0209	154	12/05/29
0014	14	M450	11/11/22	13:20:39	0210	145	12/05/29
0016	21	M08	11/11/22	13:20:46	0211	156	12/05/29
0017	8	M11	11/11/22	13:20:47			
0018	8	M12	11/11/22	13:20:47			
0019	8	M13	11/11/22	13:20:48			
0020	8	M14	11/11/22	13:20:49			

Program 0211
Deleting

Destination: Memory card

* Deletion confirmation screen opens for the files to be deleted from the connected device. In the case of folders, you are asked if the folder is to be opened or not to specify files to be deleted.

[F0] (Yes) : The relevant data is deleted from the connected **device**.

[F1] (No) : Deletion of the relevant data is skipped. In the case of a folder, the entire folder is skipped.

[F4] (Delete all remaining) : All data is deleted hereafter without confirmation.

When selecting **[F0]** on the screen for confirming entry into the folder, the system asks you if the files and sub-folders in the folder should be deleted, or if the sub-folders should be opened before deletion.

When all files and folders in a folder are deleted, the folder itself is also deleted.

8. When communication completes, the screen returns to the former screen.

*1 In the NC language mode, when the connected device is a <memory card> or <FTP server>, you can also delete programs other than those output from the machine (O****.NC).

- Press **[F5]** (Display all files) from the functions listed.
All files in the connected media's directory are displayed with their extensions.
- Specify the program to delete.
 - (1) Select on the screen.
See "5.2.5 Selecting programs to communicate".
 - (2) Enter the file name.
Enter the file name (including the extension) in the entry field below the screen.
Example: "ABC.TXT"

(Note) An alarm occurs when you use both procedures (1) and (2).

- Start communication.
Press **[>>]** (Next functions) key, followed by **[F1]** (Delete) key.
The specified file is deleted.
- *2 When there is a file that does not display in the folder (i.e. program, data bank or text files created in a language mode that is not currently selected), that folder and the file(s) that do not display have not been deleted.
To delete the folder including the file(s) that do not display, delete the file(s) on the <Folder operation (external I/O)> screen.
- *3 If the connecting destination is <Memory card> or <FTP server> for <Display NC exclusive files>, when a program with 3D smoothing data is deleted for the connecting destination, the 3D smoothing data is also deleted.

Input example	Communication operation
1234	Delete O1234.NC and SD1234.NC in the connecting directory.

If the connecting destination is <Memory card> or <FTP server> for <Display of all files>, delete the program and the 3D smoothing data separately.

Input example	Communication operation
O1234.NC	Delete O1234.NC in the connecting directory.
SD1234.NC	Delete SD1234.NC in the connecting directory.

5.2.7 Data bank I/O

Data that can be input/output varies with the selected language and the connected device. Refer to “Details of communication data” for details.

- * When the connected device is a <general COMM device> or <NC COMM device>, ladder programs, and teaching pendant custom cannot be input/output.

5.2.7.1 Data bank input method (from connected device to machine)

Data banks of the connected device are input into the machine. Data banks in the current folder of the connected device are input into the root folder of the machine.

1. Display <Data bank (external program I/O)> screen.
2. Connect the target device to the machine and check the display of <Connected> on the lower right corner of the screen.

See “5.2.2 Change connection device” for the changing procedure.

Data bank(external program I/O) 2012/05/30 10:42:47

Items	Size	NC directory(Metric)					Memory card directory(Metric)		
		1	2	3	4	5	1	2	3
Workpc. coord. zero	17.1K	1					1		
Tool data	89.4K	1					1	2	
Macro variables	16.6K	1	2				1	2	
User parameter	10.7K	1			3	4	1		3
External I/O signal	73.3K	1	2				1	2	
Commun. parameter	544	1					1		
Machine parameter	84.5K	1					1		
ATC tool data		1					1		

No. ☒ Busy ☐ Previous format

Occupied 292KByte Available 436MByte Occupied 146KByte Available 932MByte

A-B: Change file A to B, and input or output.
A/B: Input or output files A to B.
Data protection off: Input mode Data protection on: Check mode

Destination: Memory card [HELP]

External program I/O menu Input Output Input all data banks Output all data banks Folder operation Delete Delete all data banks

- Items: Type of data bank file is shown. Figures to the right of Item are currently effective data number.

Items (I/O files)		Remarks
NC Language	Conversation	
Workpiece coordinates zero position		mm/inch
Tool data		mm/inch
Macro variables		mm/inch
	Tool List	mm/inch
	Tool pattern	mm/inch
	Setting Primary Hole Drilling Diameter for Tapping	mm/inch
	Cutting conditions:	mm/inch
User parameter		mm/inch
External I/O signal		
Communication parameter		
Machine parameter		Only one
G/M code macro		
Special settings		
Machining load monitor		
ATC tool data		Only one
Ladder program		Only one
Simplified/magazine panel custom		(Reserved)
Teaching pendant custom		Only one
Auto thermal distortion compensation (not displayed when the connected device is a <gen. COMM device>.		mm/inch; output only

Up to 10 respective data bank files corresponding to the current unit system can be registered on the data bank, except that auto thermal distortion compensation is limited to one file; machine parameter, ATC tool data, and teaching pendant custom are limited to one each file which is used commonly for metric and inch; and the ladder program registration is limited to one project.

When the connected device is a <general COMM device> or an <NC COMM device>, the ladder program, and teaching pendant custom cannot communicate (items and data numbers are shown but the function keys Input and Output do not appear).

- Size: The size of all data bank files
- NC directory: Data number stored in the NC side (machine).
Green background indicates the currently used data number.
- Connected device (memory card, NC COMM device, FTP server):
Data numbers stored in the connected device.

For the ladder program, one project contains 9 files (one each global comment, high speed 1, high speed 1 comment, standard 1, standard 1 comment, high speed 2, high speed 2 comment, standard 2, and standard 2 comment).

	Display of NC side / connected device
All ladder files are ready	"1" fixed 1
Files are not ready	"1" fixed <1>

3. Check **[DATA PROTECTION]** switch.
Protect OFF: Copy mode
Protect ON: Check mode (compares files).
4. Display the directory of the connected device. (* For the connected devices other than <gen. COMM device>)
 - NC COMM device
Press **[F5]** (Latest directory) key.
 - Memory card and FTP server
Press **[F6]** (Folder operation) key, followed by **[F5]** (Directory request) key.

5. Specify the data bank to be input.
Move the cursor to the data bank to input.
* You cannot enter thermal distortion compensation.
6. Specify the data bank number to input.
Enter the data bank number in the entry field below the screen.

Entry example	Communication operation	
1	Connected device 1	→ NC side 1
1-5	Connected device 1	→ NC side 5
1/10	Connected device All of 1 to 10	→ NC side All of 1 to 10
1 3 5 (Separate by a space)	Connected device 1, 3, 5	→ NC side 1, 3, 5

- * Only one machine parameter is allowed in the machine.
 - * Enter "1" for ATC tool data, ladder program, and teaching pendant custom.
7. Start communication.
Press **[F1]** (Input) key.
 8. Communication starts.

Data bank(external program I/O)									
2012/05/30 10:49:48									
Items	Size	NC directory(Metric)					Memory card directory(Metric)		
Workpc. coord. zero	17.1K	1	2	3	4	5	1	2	3
Tool data	89.4K	1	2	3			1	2	
Macro variables	16.6K	1	2				1	2	
User parameter	10.7K	1		3	4		1		3
External I/O signal	73.3K	1	2				1	2	
Commun. parameter	544	1					1		
Machine parameter	84.5K	1					1		
ATC tool data		1					1		
Workpc. coord. zero2(Metric)									
Transferring									
Memory card									
Workpc. coord. zero2 → NC Workpc. coord. zero2 Destination: Memory card									

- * Overwrite confirmation screen opens when the same data bank type or number exists on the machine side.

[F0] (Yes) : The data are overwritten
[F1] (No) : Input of the relevant data is skipped
[F4] (Input all remaining) : All data is input hereafter without confirmation

9. When communication completes, the screen returns to the former screen.
* Press **[F3]** (Input all data banks) key to input all data banks from the connected device. All data banks of the current unit system and language existing in the current folder of the connected device are input into the machine.

Remarks

- An alarm occurs depending on data bank file to be input.
 - The ladder program is input in the following order from the connecting destination: global comment, high-speed 1, high-speed 1 comment, standard 1, standard 1 comment, high-speed 2, high-speed 2 comment, standard 2 and standard 2 comment. The files are transferred even if there is only 1 file in the connecting destination, and if there are no files, input is automatically skipped.
- Refer to "5.2.7.4 I/O in the Unit of Record" to input by specifying files individually.

5.2.7.2 Data bank output method (from machine to connected device)

Data banks of the machine are output to the connected device. Data banks in the root folder of the NC side are out to the current folder of the connected device.

1. Display <Data bank (external program I/O)> screen.
2. Connect the target device to the machine and check the display of <Connected> on the lower right corner of the screen.

See “5.2.2 Change connection device” for the changing procedure.

Items	Size	NC directory(Metric)					Memory card directory(Metric)				
Workpc_coord_zero	17.1K	1	2	3	4	5	1	2	3		
Tool data	89.4K	1	2	3			1	2			
Macro variables	16.6K	1	2				1	2			
User parameter	10.7K	1		3	4		1		3		
External I/O signal	73.3K	1	2				1	2			
Commun. parameter	544	1					1				
Machine parameter	84.5K	1					1				
ATC tool data		1					1				

No. ☒ Busy ☐ Previous format
 Occupied 282KByte Available 438MByte
 Occupied 146KByte Available 932MByte

A-B:Change file A to B, and input or output.
 A/B:Input or output files A to B.
 Data protection off:Input mode Data protection on:Check mode Destination: Memory card [HELP] [v]

External program I/O menu Input Output Input all data banks Output all data banks Folder operation Delete Delete all data banks

3. Display the directory of the connected device. (*For the connected devices other than <gen. COMM device>)
 - NC COMM device
Press [F5] (Directory request) key.
 - Memory card and FTP server
Press [F6] (Folder operation) key, followed by [F5] (Directory request) key.
4. Specify the data bank to output.
Move the cursor to the output data banks position.
5. Specify the data bank number to output.
Enter the data bank number in the entry field below the screen.

Entry example	Communication operation	
1	NC side 1	→ Connected device 1
1-5	NC side 1	→ Connected device 5
1/10	NC side All of 1 to 10	→ Connected device All of 1 to 10
1 3 5 (Separate by a space)	NC side 1, 3, 5	→ Connected device 1, 3, 5

- * Only one machine parameter is allowed in the machine.
- * Auto thermal distortion compensation need not be specified.
- * Enter “1” for ATC tool data, ladder program, and teaching pendant custom.

6. Start communication.
Press [F2] (Output) key.

7. Communication starts.

Data bank(external program I/O)											2012/05/30 10:56:00			
Items		Size	NC directory(Metric)					Memory card directory(Metric)						
				2	3	4	5	/						
Workpc. coord. zero		17.1K	1	2	3	4	5	1	2	3				
Tool data		89.4K	1	2	3			1	2					
Macro variables		16.6K	1	2				1	2					
User parameter		10.7K	1		3	4		1		3				
External I/O signal		73.3K	1	2				1	2					
Commun. parameter		544	1					1						
Machine parameter		84.5K	1					1						
ATC tool data			1					1						
Workpc. coord. zero4(Metric)														
Transferring														
NC			Memory card					Destination: Memory card						
Workpc. coord. zero4			-> Workpc. coord. zero4											

- * Deletion confirmation screen opens when the same data bank type or number exists on the connected device.

[F0] (Yes) : The relevant data of the connected device is deleted before outputting

[F1] (No) : Data output is skipped

[F4] (Delete all remaining) : All data is deleted without confirmation before outputting

8. When communication completes, the screen returns to the former screen.

- * To output all data banks in a general COMM device, press **[F4]** (Output all data banks) key. All data banks of the current unit system and language existing in the machine are output to the current folder of the connected device.

- * It is possible to output workpiece coordinate zero point, tool data, and macro variables of type 1 format of the machine of type 2 least input increment.

Enter "B" before the number when specifying data bank number.

Example: Outputting No. 1 workpiece coordinate zero point using type 1 format:

Place the cursor on the workpiece coordinate zero point, type "B1" in the entry field, and press **[F2]** key.

Remarks

- The ladder program is output in the following order: global comment, high-speed 1, high-speed 1 comment, standard 1, standard 1 comment, high-speed 2, high-speed 2 comment, standard 2 and standard 2 comment. The files are transferred even if there is only 1 file, and if there are no files, output is automatically skipped.

5.2.7.3 Data bank deletion method (deleting data bank from the connected device)

Data banks are deleted from the connected device. Data banks on the current folder of the connected device are deleted.

No deletion is available when the connected device is a <gen. COMM device>.

1. Display <Data bank (external program I/O)> screen.
2. Connect the target device to the machine and check the display of <Connected> on the lower right corner of the screen.
See "5.2.2 Change connection device" for the changing procedure.
3. Display the directory of the connected device.
 - NC COMM device
Press **[F5]** (Latest directory) key.
 - Memory card and FTP server
Press **[F6]** (Folder operation) key, followed by **[F5]** (Directory request) key.
4. Specify the data bank to delete.
Move the cursor to the data bank position to be deleted.

5. Specify the data bank number to delete.
Enter the data bank number in the entry field below the screen.

Entry example	Communication operation
1	Delete 1 of the connected device
1/10	Delete all of 1 to 10 of the connected device
1 3 5 (Separate by a space)	Delete 1, 3, and 5 of the connected device

6. Start communication.
Press **[F6]** (Delete) key.
7. Confirm the deletion.
Press **[F0]** (Yes) key.
8. Communication starts.

Data bank(external program I/O)		2012/05/30 10:58:54									
Items	Size	NC directory(Metric)					Memory card directory(Metric)				
Workpc. coord. zero	17.1K	1	2	3	4	5	1	2	3	4	
Tool data	89.4K	1	2	3			1	2			
Macro variables	16.6K	1	2				1	2			
User parameter	10.7K	1		3	4		1		3		
External I/O signal	73.3K	1	2				1	2			
Commun. parameter	544	1					1				
Machine parameter	84.5K	1					1				
ATC tool data		1					1				
Workpc. coord. zero4(Metric)											
Deleting											
Destination: Memory card											

- * Deletion confirmation screen opens for each data bank type and number to be deleted from the connected device.
[F0] (Yes) : The relevant data is deleted from the connected device
[F1] (No) : Deletion of the relevant data is skipped
[F4] (Delete all remaining) : All data is deleted hereafter without confirmation
9. When communication completes, the screen returns to the former screen.
- * Press **[F7]** (Delete all data banks) key to delete all data banks from the connected device. All data banks of the current unit system and language existing in the current folder of the connected device are deleted.

5.2.7.4 I/O in the unit of record (User parameter, machine parameter, and ladder program)

For user parameters and machine parameters, you can select records to be input/output.

For ladder programs, you can select files for each ladder.

- Display <Data bank (external program I/O)> screen.
- Connect the target device to the machine and check the display of <Connected> on the lower right corner of the screen.
See “5.2.2 Change connection device” for the changing procedure.
- Display the directory of the connected device. (*For the connected devices other than general COMM device)
 - NC COMM device
Press **[F5]** (Latest directory) key.
 - Memory card and FTP server
Press **[F6]** (Folder operation) key, followed by **[F5]** (Directory request) key.
- Specify the data banks to be input/output.
Move the cursor to the position of data banks to be input/output.

5. The data bank numbers and record numbers (ladder program) that are input and output can be specified.

Enter the data bank number in the data input field on the bottom of the screen.

Input example	Communication operation
1.1	Inputs the 1 st record for data bank number 1.
1.1/3	Inputs the 1 st through 3 rd records for data bank number 1.
1.1 3 5 (Space inserted as delimiter)	Inputs the 1 st , 3 rd and 5 th records for data bank number 1.

* If a record is specified using "." and multiple data bank numbers are specified ("-") and then deleted with (<Delete>), an alarm is triggered.

Record	User parameter	Machine parameter
1	Switch 1	System 1
2	Switch 2	System 2
3	Switch 3	System 3
4	Reference position / conversation zero point	Pitch error compensation
5	User registered miscellaneous functions	Servo
6	Tool breakage detection	Auto thermal distortion compensation
7	Auto workpiece measurement	High accuracy
8	Auto thermal distortion compensation	PLC
9	Quick table	Special settings
10	High accuracy	
11	PLC	
12	PROFIBUS DP	
13	DeviceNet	
14	CC-Link Slave	
15	Rotary fixture offset	
16	3D smoothing	
17	Feature Coordinate Setting	
18	CC-Link Master	

Record	Ladder program
1	Global comment
2	High speed 1; High speed 1 comment
3	Standard 1; Standard 1 comment
4	High speed 2; High speed 2 comment
5	Standard 2; Standard 2 comment

For 2 through 5, communication is performed even if there is no comment file. No communication is performed if there is no ladder file.

6. Start communication.
Press **[F1]** (Input) or **[F2]** (Output) key.
7. Communication starts.
 - In the case of input, only the specified records (ladder program) are input. Other records become the record of the data in the machine.
 - In the case of output, only the specified records (ladder program) are output. Other records are not output.
8. When communication completes, the screen returns to the former screen.

5.3 Slave station communication

5.3.1 Outline

Communication is performed with the machine as the slave station using communication software having <NC communication device protocol>, FTP client software, or browser software.

Communication is possible even when the machine is running, subject to some restrictions depending on the type of communication.

- File operation for programs, data banks, etc.
- Operation of the machine

5.3.1.1 Slave station communication from “NC COMM software”

Serial port or Ethernet is used.

Software for the master station must be created because a communication system unique to Brother is used.

<Type2> is used in the slave station communication regardless of the <NC communication device protocol (master)> setting. Slave station communication is impossible with <Type1>.

In addition to file operation, operation of the machine can be controlled using commands. When using Ethernet, commands are transmitted as TCP/IP packet data.

The machine becomes the server, waiting for connection establishment request from the client at the port specified by the communication parameter <Port No.>.

When the communication parameter <Restrict Ethernet access> is set to <1: Yes>, the access becomes restricted. Use a temporary access command.

Even if <Restrict Ethernet access> is set to <1: Yes> when using the serial port, the access does not become restricted.

Refer to the “5.5 Communication protocol” for further details.

5.3.1.2 Slave station communication from FTP client

Ethernet is used. NC is the FTP server.

The master station need not create special software because FTP protocols are used. Use any desired FTP client software.

For the user name and password when connecting to the FTP, use the communication parameter settings: <Server user name>, <Password as server>, <Common user name> and <Common password>.

FTP allows file operation but cannot operate the machine by commands.

(Note 1) When creating a new folder for the NC (FTP server side), sometimes a character string (Ex: [New folder] (2 byte character or more than 8 characters), etc.) that is not allowed by the NC may be used to create the folder depending on the FTP client that is used.

In this situation, an error in the FTP protocol is triggered and the folder cannot be created.

When creating a new folder for the NC (FTP server side) using the slave station’s communication from the FTP client, create a folder on the FTP client side based on the rules (See below) for the folder level and name that are permitted by the NC. Then, copy the folder.

- Up to 8 characters (alphameric characters and period) can be used for the folder name.
- Up to 2 levels, including the route folder, can be created. (A folder cannot be created inside another folder that is created.)
- A name that already exists cannot be used for the folder name, such as another folder with the same name, a name being used inside the NC program (O0000 to O9999) or a name being used inside the conversation program.

(Note 2) Special response for delete (DELE) command

When inputting (STOR) data that does not follow the file format, the NC returns an error response. However, depending on the FTP client being used, a delete (DELE) response may be issued as a second reply to the error response. If a response is issued for this delete (DELE) command, the existing data in the NC may be deleted accidentally. Therefore, the following special response is used.

A normal response is issued back to the delete (DELE) command for the same file that receives from the error response in less than 1 second, in order to avoid deleting the existing data in the NC.

(Note 3) Set the number of simultaneous connections on the FTP client software to "1". If the number of simultaneous connections on the FTP client software is set to "2" or greater, then the NC (FTP server side) may reject the connection and not transfer data properly.**5.3.1.3 Slave station communication from browser**

Ethernet is used. NC is the web server.

The master station need not create special software because HTTP protocols are used. Use any desired browser software.

Use <Host name> or <IP address> of communication parameters to specify URL for HTTP connection.

http://host name (or IP address)

The machine's workpiece counter, operation time and alarm status can be checked using a browser. File I/O is also possible (FTP protocols are used).

5.3.2 Restrictions

Slave station communication is not available according to the setting or condition of the machine.

Setting/Condition	Serial port	Ethernet	FTP	Browser
Connected device: <Memory card> or <FTP server> In the master station mode	○	○	○	○
Connected device: <General COMM device> or <NC COMM device> In the master station mode	×	○	○	○
Connected device: <General COMM device> Not in the master station mode	×	○	○	○
Connected device: Other than <General COMM device> Not in the master station mode	○	○	○	○
In maintain and transfer mode	×	×	×	×
Same data is being transmitted through a certain port	×	×	×	×
Input of data existing in the machine is instructed when <Data overwrite (slave)> = <0: No>	×	×	-	-
A command in the operation control group is instructed when <Remote operation> = <0: Invalid>	×	×	×	×
Operation level or data input or deletion currently prohibited by data protection function of the machine is instructed	×	×	×	×
When issuing a command that is restricted from <Restrict Ethernet access> being set to <1: Yes>	○	×	-	-

* Master station mode: When the screen is under the directory of <External program I/O menu> screen.

* Maintain and transfer mode: Time while using the sub-screens under the <Maintenance menu> screen

* Data protection function: See "Chapter 6 Data Protection" for details.

- * Restrict Ethernet access: Refer to “5.5 Communication protocol” for details on commands and functions that are restricted.

When pressing the **[RST]** key during slave station communication, slave station communication stops according to the setting of <Reset (slave)>. When <1: Valid> is selected slave station communication stops, when <0: Invalid> it doesn't stop.

When slave station communication ends normally, an end of communication alarm appears on the screen if the <Display slave command alarm> is set to <1: Yes>. It does not appear when <0: No> is set. An alarm is shown regardless of the setting if the communication ends abnormally.

5.4 Maintenance communication

5.4.1 What is maintenance communication?

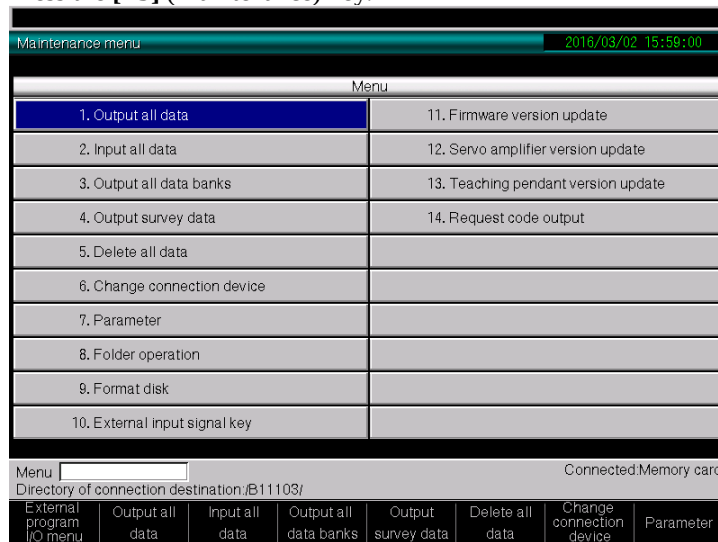
When performing maintenance on the control panel or similar tasks on the machine, there is a potential risk of losing machine data. This feature allows the user to back up and restore all internal data on the machine.

In addition, we recommend that the user regularly backs up the data to ensure the machine's internal data is preserved.

- * You cannot select <General COMM device> and <NC COMM device> as the connected device in the maintenance communication.

5.4.1.1 Maintenance menu

1. Display <Program edit menu> screen.
Press **[EDIT]** key.
Press the **[PRGRM]** key.
2. Display <External program I/O menu> screen.
Press <External program I/O> key from the function keys.
3. Display <Maintenance menu> screen.
Press the **[F3]** (Maintenance) key.



4. Select a function.

Function	Outline
Output all data	All data of the machine is backed up to the connected device.
Input all data	Data in the machine is recovered from the connected device.
Output all data banks	All data bank files of the machine are backed up to the connected device.
Output survey data	All data plus survey data is output. * This is not used in normal operation.
Delete all data	All back-up data stored in the connected device is deleted.
Change connection device	See “5.2 Master station communication”.
Parameter	See “5.2 Master station communication”.
Format disk	“Memory card” is logically formatted.
Folder operation	See “5.2 Master station communication”.
External input signal key	Memory card is specially formatted to create external input signal keys.
Version up of Firmware	System firmware is upgraded.
Servo amplifier version update	Updates the version for the servo amplifier firmware.
Teaching pendant version update	Updates the version of the teaching pendant that is connected.
Request code output	Outputs the request code for resetting the relocation detection status on the connecting device.
CM PCB version update	Updates the CM PCB version.

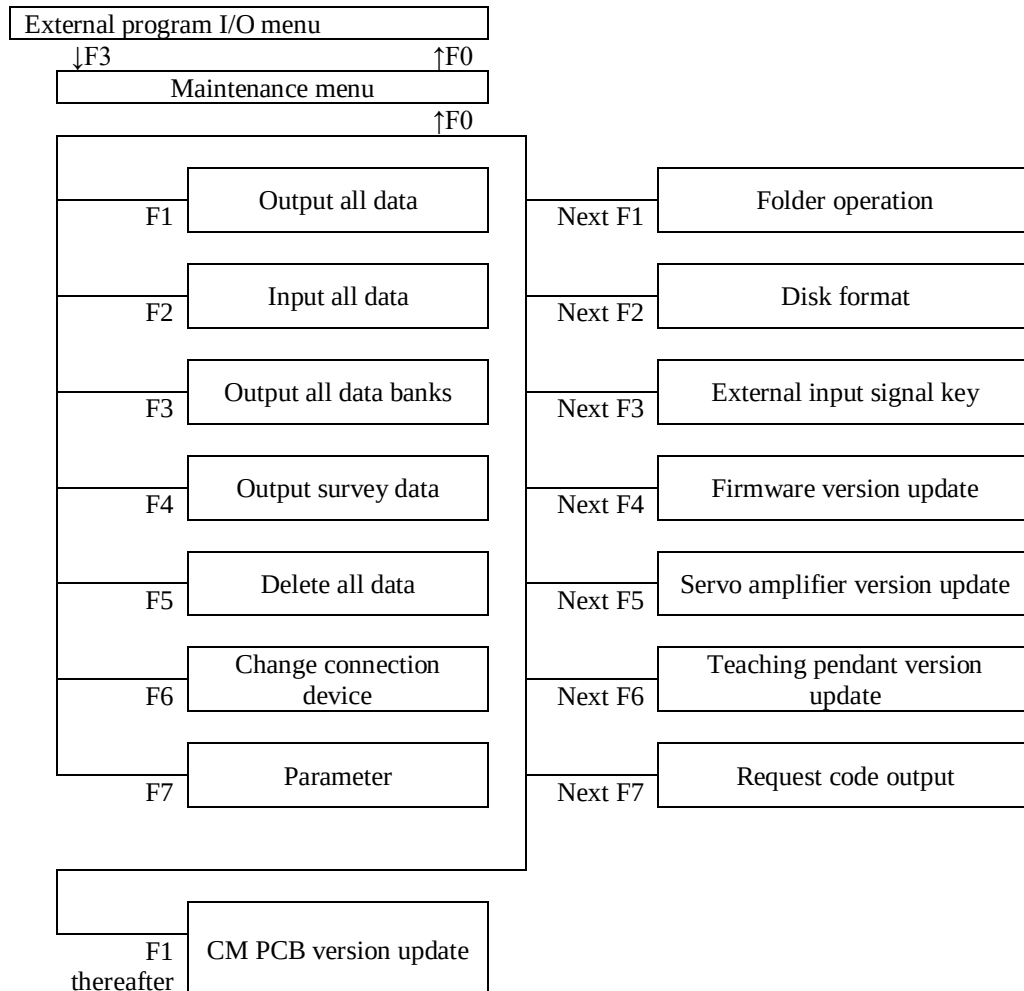
- * Menus and function keys vary with the connected device.

Upper row[F0]to[F7]; press[>>](next functions) to view the lower row[F0]to[F7]

Memory card							
External program I/O menu	Output all data	Input all data	Output all data bank	Output survey data	Delete all data	Change connection device	Parameter
External program I/O menu	Folder operation	Format disk	External input signal key	Version up of firmware	Servo amplifier version update	Teaching pendant version update	Request code output
External I/O menu	CM PCB version update						
FTP server							
External program I/O menu	Output all data	Input all data	Output all data bank	Output survey data	Delete all data	Change connection device	Parameter
External program I/O menu	Folder operation	Request code output					

- * When the connected device is a <General COMM device> or <NC COMM device>, only [F0] (External program I/O menu) and [F6] (Change connection device) are available.
- * An alarm <<The waveform data is being measured>> appears and the screen does not change if you press [F2] (Input all data) when the system is measuring waveform data.

5.4.1.2 Screen flow



5.4.2 Output all data

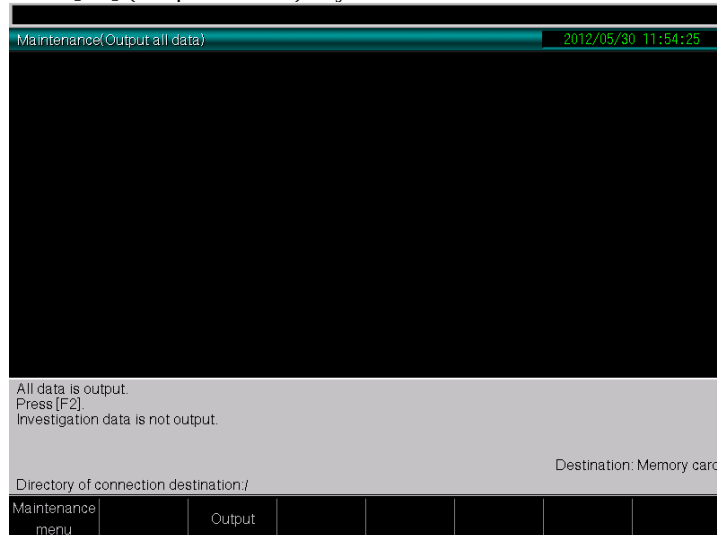
All data banks and programs of the machine are output to the connected device for backup. The folder structure of the machine is created on the current folder and below of the connected device. See “5.6 Details of communication data” for the data to be output.

1. Display <Maintenance menu> screen.
2. Connect the target device to the machine and check the display of <Connected> on the lower right corner of the screen.

See “5.2 Master station communication” for the connection changing procedure.

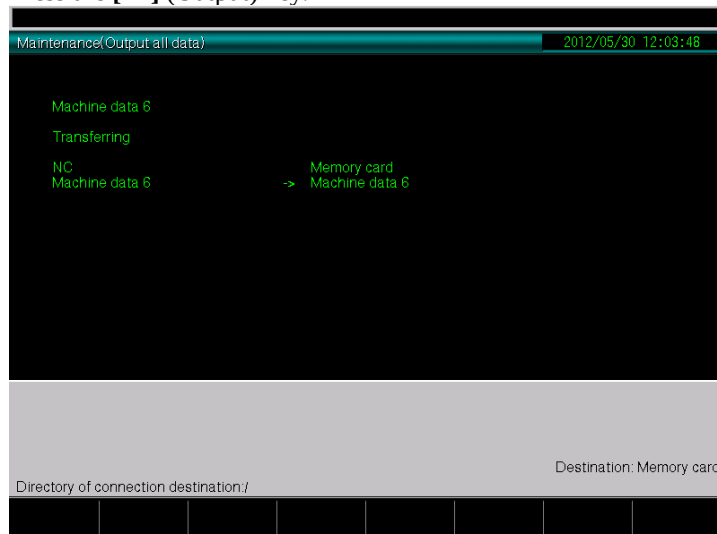
3. Display <Output all data> screen.

Press **[F1]** (Output all data) key.



4. Start communication.

Press the **[F2]** (Output) key.



5. When communication completes, the screen returns to the former screen.

* This operation is not available in the memory operation edit mode.

5.4.3 Input all data

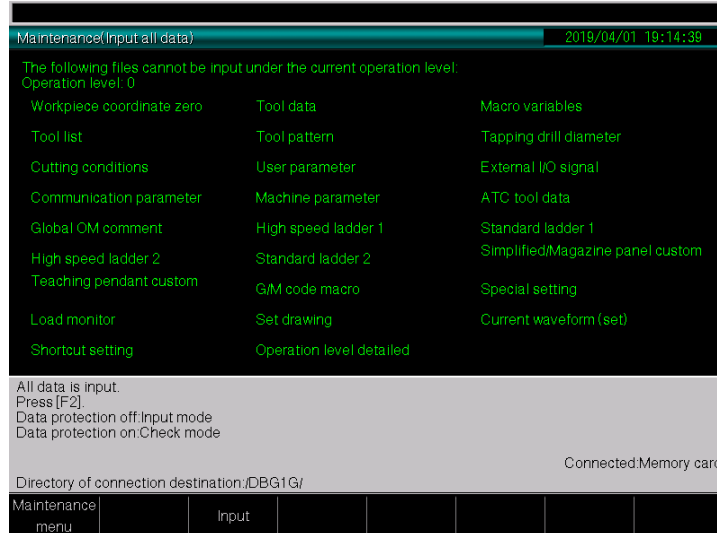
All data banks and programs backed up in the connected device are input into the machine for recovery. The folder structure of the current folder and below of the connected device is created on the root folder and below of the NC side. Note that up to 2 layers of folders including the root folder are created on the NC side; the 3rd and below layers, if any, are ignored (an operator message <<There was a subfolder unable to input>> occurs). If the folder name is not permitted on the NC side, the operator message <<This is a folder name that cannot be input>> appears and the folder is ignored.

See “5.6 Details of communication data” for the data to be input.

1. Check **[DATA PROTECTION]** switch and the operation level of the current data protection.

Data protection off: input

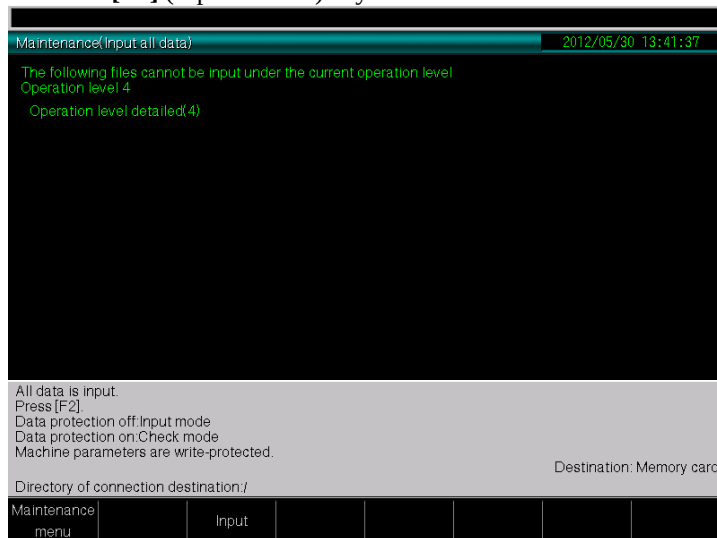
Data protection on: check mode (compares files)



2. When entering the machine parameter, high-speed ladder 2 and standard ladder 2, set <Parameter change> to <1: Writable>.
 - (1) Display the <I/O menu> screen.
Press the **[I/O]** key.
 - (2) Display the <I/O menu (version)> screen.
Press the **[F7]** (version) key.
 - (3) Set the <Parameter change> to <1: Writable>.
Enter 1 into the data input field, and press the **[ENT]** key.
 - (4) The screen returns to the <Program> screen.
Press the **[PRGRM]** key.
3. Display <Maintenance menu> screen.
4. Connect the target device to the machine and check the display of <Connected> on the lower right corner of the screen.

See “5.2 Master station communication” for the connection changing procedure.

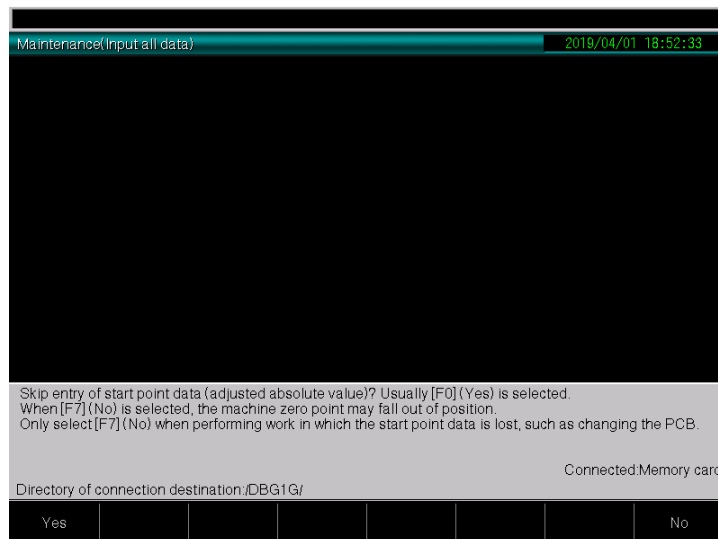
5. Display <Input all data> screen.
Press the **[F2]** (Input all data) key.



6. Start communication.
Press the **[F2]** (Input) key.



7. Select among the followings when <Overwrite check> screen is displayed.
[F0] (Yes) : Overwrite
[F1] (No) : Input of the relevant data is skipped
[F4] (Input all remaining) : All data is input hereafter without confirmation
8. When machine data 1 is included in the data to be input, this operation selects whether to the skip the input of start point data (adjusted absolute values).
[F0] (Yes): Skip the input of the start point data.
[F1] (No): Overwrite the start point data.
 Normally, select **[F0]** (Yes). When **[F7]** (No) is selected, the machine zero point may shift. Select **[F7]** (No) only when carrying out a work task, such as a PCB replacement, that causes the start point data to be lost.
 When **[F7]** (No) is selected, the alarm <<The machine coordinates may be out of position.>> may trigger after turning ON the power again. Check the machine status by following the recovery procedure for the alarm <<The machine coordinates may be out of position.>>.



9. When communication completes, the screen returns to the former screen.
10. Restart the power.

- * This operation is not available in the memory operation edit mode.
- * Files of the current language mode (NC/conversation) are input.
 - Files with an extension “.NC” are input in the NC language mode.
 - Files with an extension “.TC” are input in the conversation mode.

When no file corresponds to the current language mode, the other files will be input.
- * Files with a ladder program extension “.LDR” are input regardless of NC language or conversation mode.
- * After inputting all data, do not operate the built-in PLC until you turn power off and then on again.

5.4.4 Output all data banks

All data banks in the machine are output to the connected device for backup.

See “5.6 Details of communication data” for the data to be output.

1. Display <Maintenance menu> screen.
2. Connect the target device to the machine and check the display of <Connected> on the lower right corner of the screen.
See “5.2 Master station communication” for the connection changing procedure.
3. Display <Output all data banks> screen.
Press the **[F3]** (Output all data banks) key.
4. Start communication.
Press the **[F2]** (Output) key.
5. When communication completes, the screen returns to the former screen.

5.4.5 Output survey data

All data in the machine (data output by <Output all data> plus survey data) are output to the connected device for backup. The folder structure of the machine is created on the current folder and below of the connected device.

Survey data cannot be input.

See “5.6 Details of communication data” for the data to be output.

1. Display <Maintenance menu> screen.
2. Connect the target device to the machine and check the display of <Connected> on the lower right corner of the screen.
See “5.2 Master station communication” for the connection changing procedure.
3. Display <Output survey data> screen.
Press the **[F4]** (Output survey data) key.
4. Start communication.
Press the **[F2]** (Output) key.
5. When communication completes, the screen returns to the former screen.

5.4.6 Delete all data

All data stored on the connected device is deleted. All files in the current folder and below of the connected device are deleted together with the folders. Note that the files and folders other than those output from the machine are also deleted altogether.

See “5.6 Details of communication data” for the data to be deleted.

1. Display <Maintenance menu> screen.
2. Connect the target device to the machine and check the display of <Connected> on the lower right corner of the screen.
See “5.2 Master station communication” for the connection changing procedure.
3. Display <Delete all data> screen.
Press the **[F5]** (Delete all data) key.
4. Start communication.
Press **[F2]** (Delete) key.
5. When communication completes, the screen returns to the former screen.

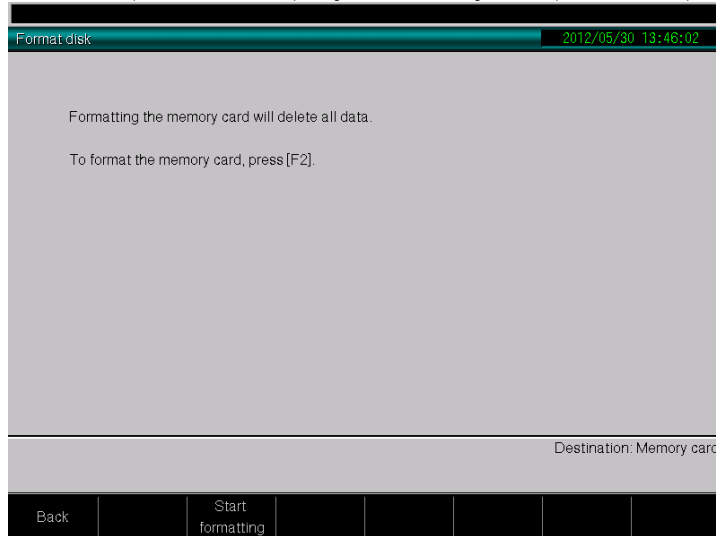
5.4.7 Disk format

A memory card is logically formatted (physical formatting is not performed).

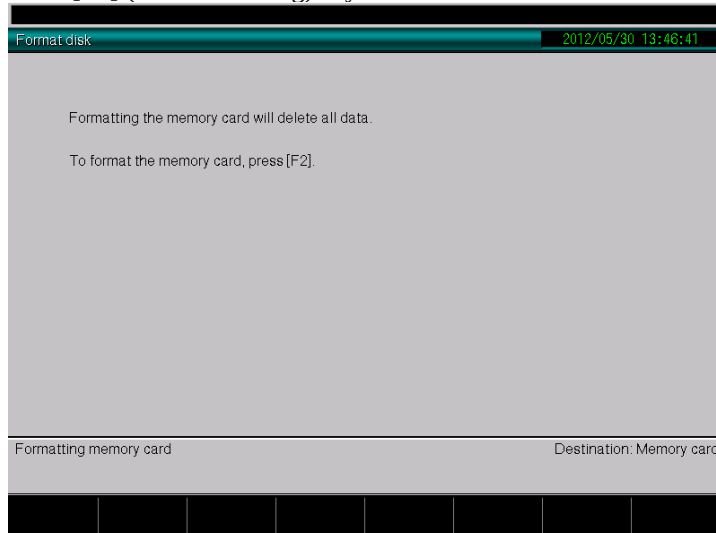
* You cannot use memory cards that are not physically formatted.

1. Display <Maintenance menu> screen.
2. Connect the memory card to the USB port and check the display of <Connected> on the lower right corner of the screen.
See “5.2 Master station communication” for the connection changing procedure.

3. Display <Format disk> screen.
Press [>>] (Next functions) key, followed by [F2] (Format disk) key.



4. Start formatting.
Press [F2] (Start formatting) key.



5. When formatting completes, the screen returns to the former screen.

5.4.8 External input signal key

Memory cards are specially formatted to create external input signal keys. The designated external input signal is turned ON as you insert the memory card into the USB port. External input signals can be turned on without adding switches, etc.

- * External input signals of the machine are turned on if one of the following external signal controls is selected to ON:
 - Actual external input signal input from the terminal
 - External input signal control by slave station commands
 - External input signal key
- * The external input signal key cannot be differentiated from an ordinary memory card if inserted into a PC. Handle with care.

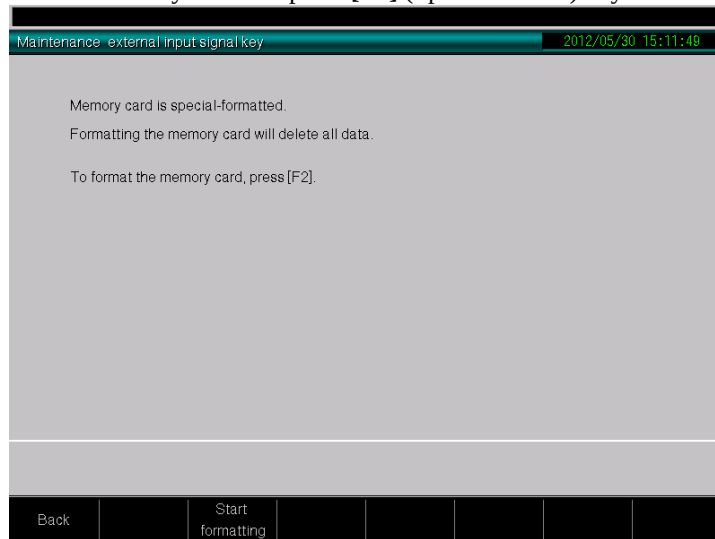
5.4.8.1 External input signal key screen

- * To display this screen, set the data protection operation level to <Level 5> beforehand.
1. Display <Maintenance menu> screen.
 2. Connect the memory card to the USB port and check the display of <Connected> on the lower right corner of the screen.
See “5.2 Master station communication” for the connection changing procedure.
 3. Display <Maintenance external input signal key> screen.
Press [>>] (Next functions) key, followed by [F3] (External input signal key) key.

Maintenance external input signal key			2013/10/17 13:40:17		
LEVEL4					
External input signal			Can be deleted		
1. PRO1	128. MPEDOK	138. LEVEL2	518. #1005	528. #1015	538. #1025
2. PRO2	129. WCEDOK	139. LEVEL3	519. #1006	529. #1016	539. #1026
3. PRO4	130. MGEDOK	140. LEVEL4	520. #1007	530. #1017	540. #1027
4. PRO8	131. MTEDOK	510. WPEDOK	521. #1008	531. #1018	541. #1028
5. PRO16	132. PREDOK	511. MCEDOK	522. #1009	532. #1019	542. #1029
6. PRO32	133. IOEDOK	513. #1000	523. #1010	533. #1020	543. #1030
7. PRO64	134. CPEDOK	514. #1001	524. #1011	534. #1021	544. #1031
124. EDLOCK	135. DIROK	515. #1002	525. #1012	535. #1022	545. #1200
126. TLEDOK	136. MODEOK	516. #1003	526. #1013	536. #1023	546. #1201
127. UPEDOK	137. LEVEL1	517. #1004	527. #1014	537. #1024	547. #1202
Back		Special format	Format cancel	Next menu	

5.4.8.2 Creation of external input signal key

1. Memory cards are specially formatted to use them as an external input signal key. Insert a memory card and press **[F3]** (Special format) key.



2. Confirm formatting.
Press **[F2]** (Start formatting) key.
3. When formatting completes, the screen returns to the former screen.

- * The memory card is physically formatted when specially formatted.
- * The memory card that is specially formatted as an external input signal key can also be used as a connection device for external I/O.

5.4.8.3 Adding external input signals

1. Signals are set to the external input signal key.
Type the number in the entry field below the screen and press **[ENT]** key.

Type in the order of “X” and “Number” to set a PLC input signal.

Example: Typing PLC input “X03FF”

[X] [3] [F] [F] [ENT]

- * Up to 64 types of signals can be set.
- * Some signals cannot be set.
- * The signal is immediately turned on when set.
- * Multiple signals can be set to a single external input signal key. In this case, all signals on the key are turned on when the key is inserted.

5.4.8.4 Deleting external input signals

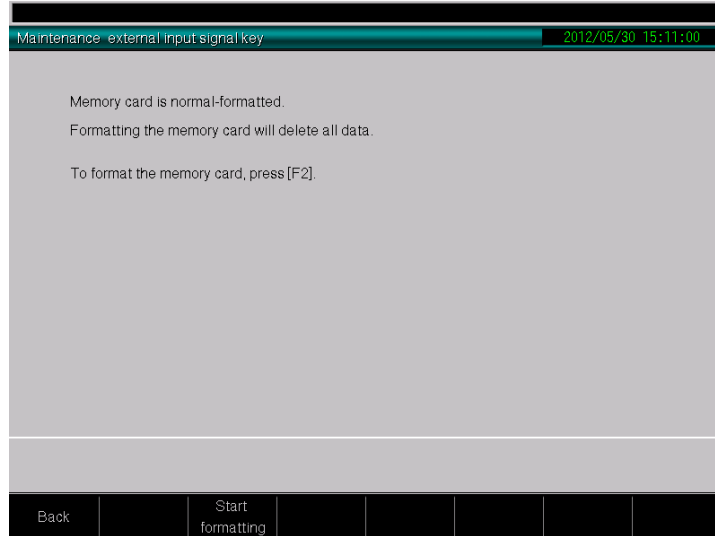
1. Signals are deleted from the external input signal key.
Move the cursor to the signal you wish to delete and press **[Delete]** key.

- * The signal is immediately turned off when it is deleted.

5.4.8.5 Cancellation of external input signal key

1. The memory card that has been used as an external input signal key is reformatted to ordinary card without the function of signal key.

Insert the memory card and press **[F4]** (Format cancel) key.



2. Confirm formatting.
Press **[F2]** (Start formatting) key.
3. When formatting completes, the screen returns to the former screen.

* The memory card is physically formatted when the special formatting is canceled.

5.4.9 Request code output

There are some machine models that do not require request code output. In this situation, the operator message <<There is no corresponding function.>> is triggered and the screen does not change to the request code output menu.

Refer to “Chapter 14 Relocation Detection Function” in the Installation Manual for further details on request code output.

5.5 Communication protocol

5.5.1 Character code table (NC language mode)

Character	ASCII								Character	ISO								Character	EIA								Processing	Meaning
	8	7	6	5	4	3	2	1		8	7	6	5	4	3	2	1		8	7	6	5	4	3	2	1		
0			○	○					0			○	○					0			○							Number0
1			○	○				○	1	○		○	○				○	1							○			Number1
2			○	○			○		2	○		○	○			○		2							○			Number2
3			○	○			○	○	3			○	○			○	○	3			○			○	○			Number3
4			○	○		○			4	○		○	○		○			4					○					Number4
5			○	○		○		○	5			○	○		○		○	5			○		○		○			Number5
6			○	○		○	○		6			○	○		○	○		6			○		○	○				Number6
7			○	○		○	○	○	7	○		○	○		○	○	○	7					○	○	○			Number7
8			○	○	○				8	○		○	○	○				8				○						Number8
9			○	○	○			○	9			○	○	○			○	9			○	○			○			Number9
A		○						○	A		○						○	A		○	○					○		AddressA
B		○					○		B		○					○		B		○	○				○			AddressB
C		○					○	○	C	○	○					○	○	C		○	○	○			○	○		AddressC
D		○				○			D		○				○			D		○	○			○				AddressD
E		○				○		○	E	○	○				○		○	E		○	○	○		○		○		AddressE
F		○				○	○		F	○	○				○	○		F		○	○	○		○	○			AddressF
G		○				○	○	○	G		○				○	○	○	G		○	○			○	○	○		AddressG
H		○			○				H		○			○				H		○	○		○					AddressH
I		○			○			○	I	○	○			○			○	I		○	○	○	○			○		AddressI
J		○			○		○		J	○	○			○		○		J		○		○				○		AddressJ
K		○			○		○	○	K		○			○		○	○	K		○		○			○			AddressK
L		○			○	○			L	○	○			○	○			L		○					○	○		AddressL

Character	ASCII							Character	ISO							Character	EIA							Processing	Meaning			
	8	7	6	5	4	3	2		1	8	7	6	5	4	3		2	1	8	7	6	5	4			3	2	1
M		○			○	○		○	M		○			○	○		○	M		○		○		○				AddressM
N		○			○	○	○		N		○			○	○	○		N		○				○		○		AddressN
O		○			○	○	○	○	O	○	○			○	○	○	○	O		○				○	○			AddressO
P		○		○					P		○		○					P		○		○		○	○	○		AddressP
Q		○		○				○	Q	○	○		○				○	Q		○		○	○					AddressQ
R		○		○			○		R	○	○		○			○		R		○			○			○		AddressR
S		○		○			○	○	S		○		○			○	○	S			○	○			○			AddressS
T		○		○		○			T	○	○		○		○			T			○				○	○		AddressT
U		○		○		○		○	U		○		○		○		○	U			○	○		○				AddressU
V		○		○		○	○		V		○		○		○	○		V			○			○		○		AddressV
W		○		○		○	○	○	W	○	○		○		○	○	○	W			○			○	○			AddressW
X		○		○	○				X	○	○		○	○				X			○	○		○	○	○		AddressX
Y		○		○	○			○	Y		○		○	○			○	Y			○	○	○					AddressY
Z		○		○	○		○		Z		○		○	○		○		Z			○		○			○		AddressZ
DEL		○	○	○	○	○	○	○	DEL	○	○	○	○	○	○	○	○	DEL		○	○	○	○	○	○	○	(1)	Delete
NUL									NUL									BLANK									(1)	No punched
BS					○				BS	○				○				BS			○		○		○		(1)	Back space
HT					○			○	HT					○			○	TAB			○	○	○	○	○	○	(5)	Tabulator
LF /NL					○		○		LF /NL					○		○		EOB /CR	○									End of block
CR					○	○		○	CR	○				○	○		○									(7)	Carriage return	
SP			○						SP	○		○						SP				○						Space
%			○			○		○	%	○		○			○		○	ER					○		○	○	(6)	End of record

Character	ASCII								Character	ISO								Character	EIA								Processing	Meaning
	8	7	6	5	4	3	2	1		8	7	6	5	4	3	2	1		8	7	6	5	4	3	2	1		
(			○		○				(			○		○				C/O (2+4+5)				○	○		○			Control Out
)			○		○			○	)	○		○		○			○	C/I (2+4+7)		○			○		○			Control In
+			○		○		○	○	+			○		○		○	○	+			○	○	○				Plus sign	
-			○		○	○		○	-			○		○	○		○	-			○							Minus sign
:			○	○	○			○	:			○	○	○			○									(4)	Colon	
/			○		○	○	○	○	/	○		○		○	○	○	○	/			○	○				○	Optional block skip	
.			○		○	○	○		.			○		○	○	○		.			○	○		○	○	○	Period (decimal point)	
#			○				○	○	#	○		○					○									(3)	Sharp	
\$			○			○			\$			○				○										(1)	Dollar mark	
&			○			○	○		&	○		○			○	○		&					○	○	○		(2)	Ampersand
'			○			○	○	○	'			○			○	○	○									(1)	Apostrophe	
*			○		○		○		*	○		○		○		○										(3)	Asterisk	
,			○		○	○			,	○		○		○	○			,			○	○	○		○	○	(2)	Comma
;			○	○	○		○	○	;	○		○	○	○		○	○									(1)	Semicolon	
<			○	○	○	○			<			○	○	○	○											(1)	Left angle bracket	
=			○	○	○	○		○	=	○		○	○	○	○		○									(3)	Equal sign	
>			○	○	○	○	○		>	○		○	○	○	○	○										(1)	Right angle bracket	
?			○	○	○	○	○	○	?			○	○	○	○	○										(2)	Question mark	
@		○							@	○	○															(1)	At mark	
"			○				○		"			○				○										(1)	Quotation mark	
[		○		○	○		○	○	[	○	○		○	○		○	○									(3)	Left square bracket	
]		○		○	○	○		○	]	○	○		○	○	○		○									(3)	Right square bracket	

- *1 Blanks in “Processing” indicate that the data is read normally.
- *2 Characters noted (1) in “Processing” and the codes not listed in this table, excluding label skip section, etc. (see *12) are processed as follows according to the setting of communication parameter <Invalid data>:
 <0: Convert> Read after converted into “?” (question mark).
 <1: Error> When inputted, the communication is stopped as an error.
 <2: Ignore> When inputted, it is ignored.
- *3 Characters noted (2) in “Processing” are read normally but will cause an error in operation if they are located in the significant section.
- *4 Characters noted (3) in “Processing” are converted into the values set to communication parameters <EIA character codes # through J> when inputted in EIA.
- *5 Processing for (4) is converted to “O” (Address O) for ISO / ASCII input. However, the communication parameter <Extended comment input> is set to “1: Yes”, the NC program content enclosed in parentheses () between (control out) and (control in) and the data bank or data to be maintained enclosed in apostrophes ‘ ’ is not converted.
- *6 Characters noted (5) in “Processing” are converted into “SP” (space) when inputted in ISO, EIA, and ASCII.
- *7 Characters noted (6) in “Processing” may not be used other than tape start and tape end code in ISO, EIA, and ASCII. They may not be used even between “(” (control out) and “)” (control in) of an NC program.
- *8 Characters noted (7) in “Processing” are always ignored when inputted in ISO or ASCII.
- *9 When communication parameter <Character length> is set to <0: 7 bits> in the ISO code, the 8th bit of the ISO code is always treated as mask (not punched) (same as ASCII).
- *10 Switching among ISO, EIA, and ASCII at inputting is automatically judged by “LF” / “EOB” / “CR” at the head of the tape.
- *11 “’” (apostrophe) as used in outputting a data capable of setting a character string such as <<External alarm 10 message>> of external I/O signal is processed as follows:
 - Same as the character code table in ISO and ASCII.
 - Converted into “&” (ampersand) in EIA.
 “’” (apostrophe) indicates the start and the end of a character string.
- *12 When communication parameter <Extended comment input> is set to <1:Yes> and communication is performed by other than serial port, characters between “(” (control out) and “)” (control in) in NC programs and those between two “’” (apostrophes) in data banks or maintenance data can be input if they are compatible with the following character codes:
 - Supported code display: When the <Display language> setting is Japanese → Shift-JIS
 - Supported code display: When the <Display language> setting is English, German, French, Portuguese, Italian or Spanish → ISO8859-1
 - Supported code display: When the <Display language> setting is Chinese → GB18030
 - Supported code display: When the <Display language> setting is Polish or Czech → ISO8859-2
 - Supported code display: When the <Display language> setting is Korean → EUC-KR

Codes other than the above cannot be input.

However, lower case English alphabets [“a” (0x61) to “z” (0x7A)] can be input in all communications excepting EIA.

(Note) There are some characters that cannot be displayed.

- *13 Data that is input/output as binary data is not necessarily as specified in this table.

5.5.2 Character code table (conversation mode)

Refer to “5.5.3 Internal data code list” for the codes used internally.

Character	code	Character	code
LF	0A H	:	3A H
CR	0D H	A	41 H
SP	20 H	B	42 H
%	25 H	C	43 H
0	30 H	D	44 H
1	31 H	E	45 H
2	32 H	F	46 H
3	33 H	G	47 H
4	34 H	H	48 H
5	35 H	I	49 H
6	36 H	J	4A H
7	37 H	K	4B H
8	38 H	L	4C H
9	39 H	M	4D H
		N	4E H
		O	4F H
		P	50 H
		Q	51 H
		R	52 H
		S	53 H
		T	54 H
		U	55 H
		V	56 H
		W	57 H
		X	58 H
		Y	59 H
		Z	5A H

* Data that is input/output as binary data is not necessarily as specified in this table.

5.5.3 Internal data code list

This machine uses the unique codes listed below. Make sure that all of the below-mentioned codes can be input to/output from your external equipment.

Higher order \ Lower order	0	1	2	3	4	5	6	7
0								
1								
2								
3								
4								
5								
6								
7								
8								
9								
A								
B								
C								
D								
E								
F								

(Internal data code list) “/” indicates the codes in use.

5.5.4 Ethernet Communication

The following methods are available to communicate with the machine over Ethernet:

- Slave station communication from “NC communication software”
- Master station communication to <FTP server>
- Slave station communication from FTP client
- Slave station communication from a browser

Network setting is required to communicate with the machine over Ethernet.

Set communication parameters <Host name>, <Use DHCP>, <IP address>, <Mask bit>, <Default gateway>, <Auto acquisition of DNS server address>, <DNS server address 1>, and <DNS server address 2> according to your network environment.

Contact your network administrator for any unknown points.

In addition, when an unrelated device is connected to the machine on the same network, it causes an additional load on communications. Therefore, take the appropriate steps with the configuration.

5.5.4.1 Slave station communication from “NC communication software”

- Protocol
TCP/IP protocol is used for Ethernet communication.
NC communication device protocol type 2 is encapsulated in the TCP/IP packet data. The machine runs as slave station and is a server in the TCP/IP protocol.
- TCP port number
The machine waits for connection with the client at the TCP port number set to communication parameter <Port NO.>. The client should ask establishment of connection at this TCP port number.
- Establishment and release of connection
Connection is established / released using transmission of one pair of command and response of NC communication device protocol, type 2, as a session, in principle.
When the client does not release the connection and no command arrives for a time period set by communication parameter <Response monitoring time>, the server (machine) releases the connection. If you wish not to release connection from the server, set the <Response monitoring time> to <0>.
- Restrict access
Restrict access to this machine using communication protocol type 2 for an NC communication device by setting the communication parameter <Restrict Ethernet access> to <1: Yes>.
 - All commands except a temporary access command
 When a temporary access command is issued, the access restriction is temporarily lifted. However, when the communication parameters < Common user name > and <Common password> are not set, the access restriction cannot be lifted.
Temporary access continues from the previous session until the communication parameter <Access timeout period> elapses.
If there are multiple release commands from the client side, access restriction is temporarily lifted only for the IP address on the client side when the temporary access command is finally issued.
Refer to “5.5.9.3. Command details” for further details on the temporary access command.

5.5.4.2 FTP communication

General protocols are used and the description is omitted here.

In the case of master station communication (the machine is an FTP client), connection is established with the “FTP server” on the network according to communication parameters <FTP server name (IP address)>, <User name as client>, <FTP server port number>, and <Password as client>.

For slave station communication (this machine is used as an FTP server), use the communication parameter settings: <Server user name>, <Password as server>, <Common user name> and <Common password> to connect on the client side.

(Note) <Server user name> and < Password as server > are output in a file using plain text. We recommend leaving the <Server user name> blank and using the <Common user name> and <Common password>. However, if the communication parameter <Restrict Ethernet access> is set to <1: Yes>, the settings <Server user name> and < Password as server > cannot be used.>

The FTP commands that support the FTP server are as follows:

USER, PASS, QUIT, PWD, XPWD, CWD, XCWD, CDUP, XCUP, TYPE, PORT, PASV, LIST, NLST, SIZE, MDTM, RETR, STOR, ABOR, DELE, MKD, XMKD, RMD, XRMD, RNFR(*), RNTD, NOOP

(*) It returns an error when the specified name is a file.

(Note) When this machine uses an FTP server, if data is sent to the machine while the data protection is enabled, a data send error will occur. However, an error may not occur depending on the client's software.

5.5.4.3 Slave station communication from browser

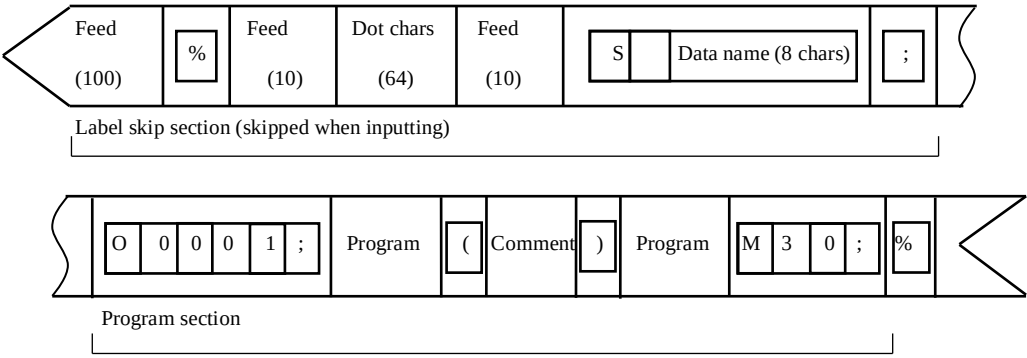
General protocols are used and the description is omitted here.

URL for connection is designated by communication parameter <Host name> or <IP address>. Enter as follows in the entry field: http://Host name (or IP address)/

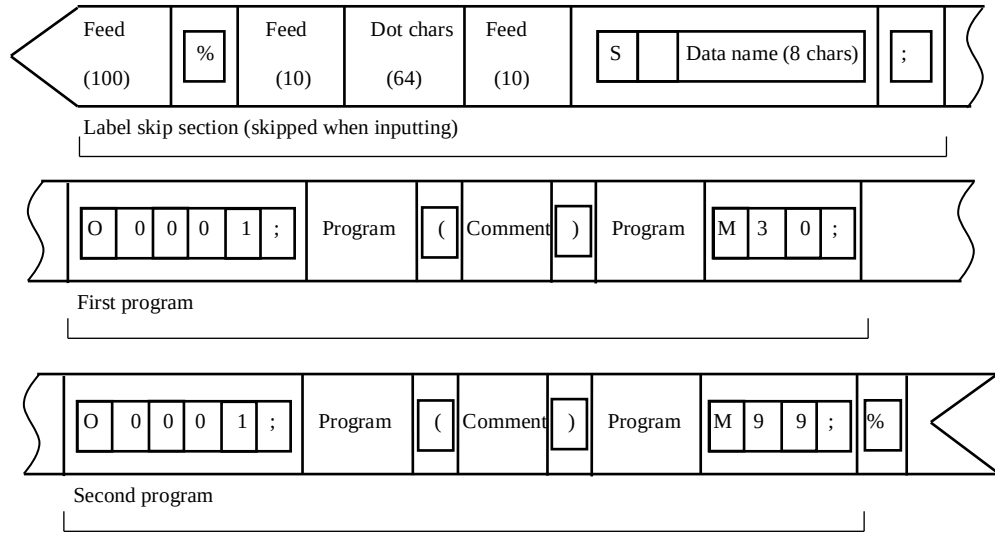
5.5.5 General communication device protocol

5.5.5.1 Tape format (NC language mode)

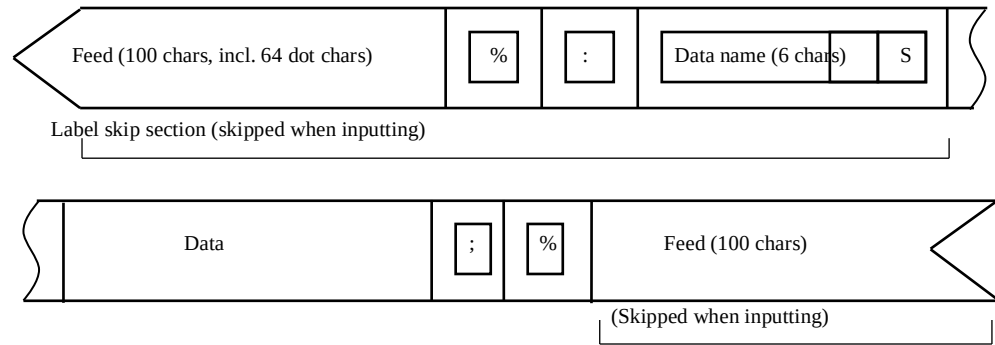
- 1. Just one program exists on the tape.



- 2. More than one program exists on the tape (input or output all programs)



5.5.5.2 Tape format (conversation mode)

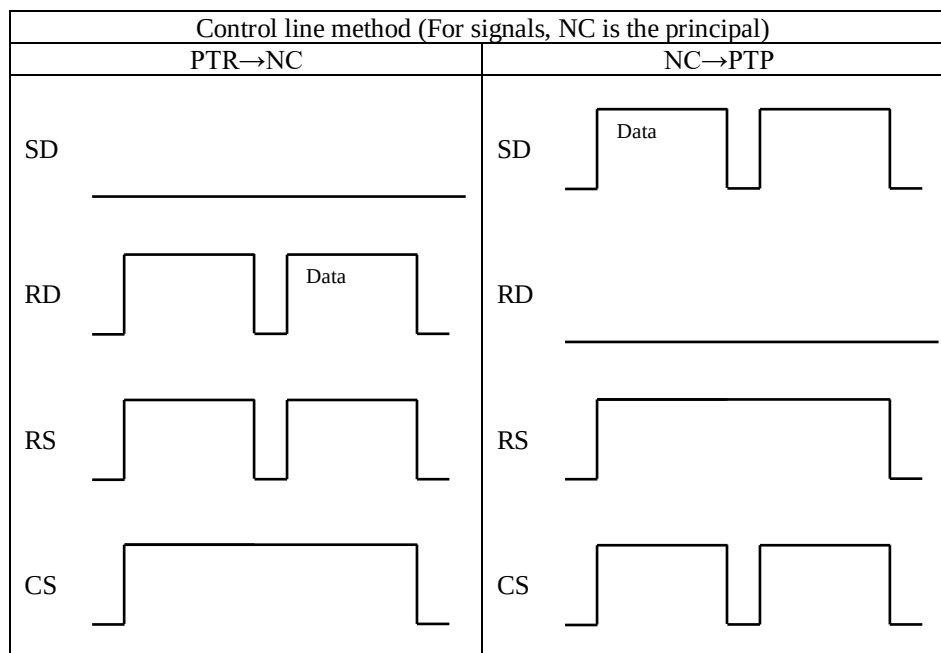


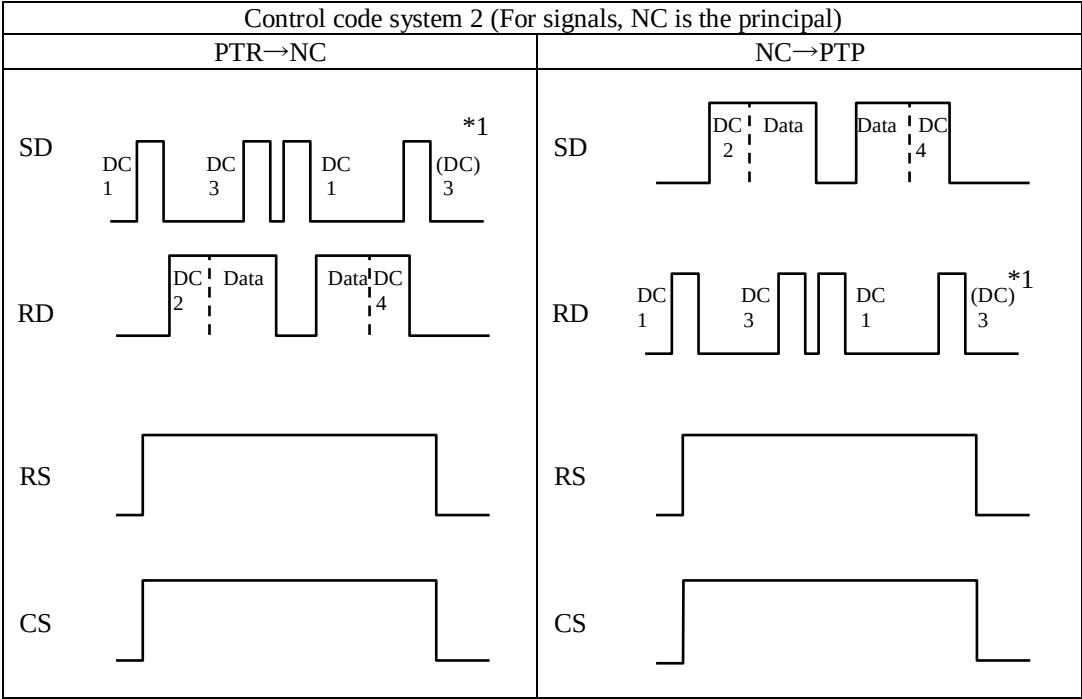
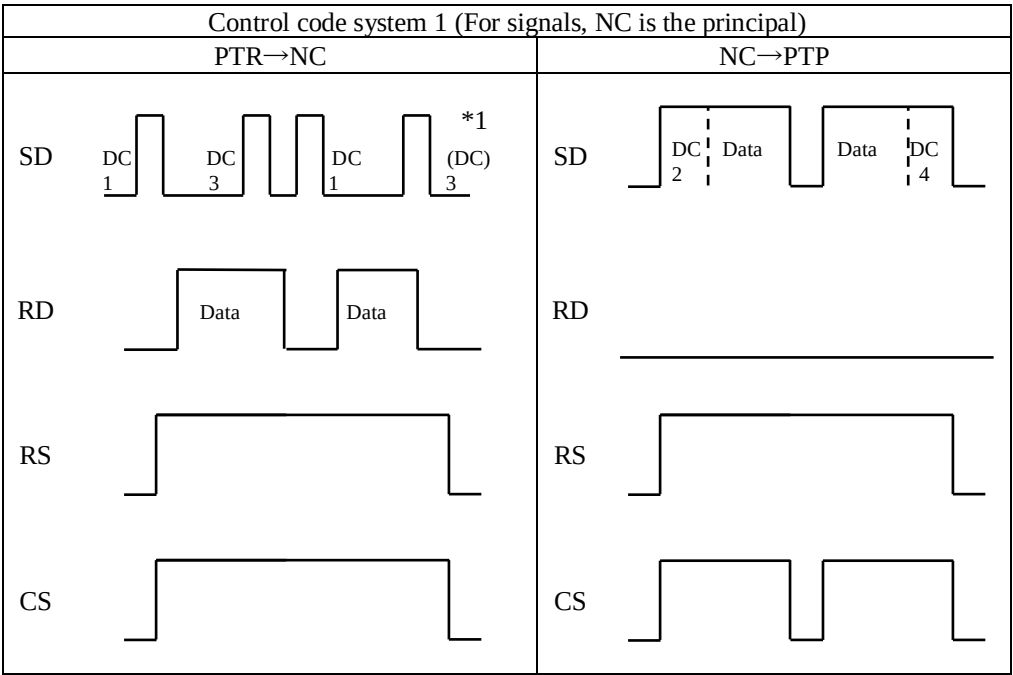
- *1 indicates tape start / end code (ISO/ASCII: %; EIA: ER).
- *2 indicates end of block code (ISO: LF; EIA: EOB; ASCII: CR + LF).
- *3 indicates space (ISO / EIA / ASCII: SP).
- *4 Feed is (ISO / ASCII: NULL; EIA: BLANK).
- *5 Dot characters are used as the heading of tapes. (Set communication parameter <Output dot character> to <0: No> or <1: Yes> as required.)
- *6 Data name and Data name indicate data transfer direction and data name.
- *7 The figures in parentheses () indicate the number of characters to be output.
- *8 Program number data at the program top
 (; ;) is automatically added according to the output program number when the program is output from NC.
 When input program number is missing as it is input into NC, the program is registered by the number of this data (this function is available only in the NC language mode).
- *9 Program number data at the program top is only used to register the program in NC when the input program number is missing. It is not registered in the memory.
 Accordingly, the block next to the program number data at the program top is the first block in the memory (available only in the NC language mode).

5.5.6 Communication methods of general communication device

You can specify one of the following 3 methods by setting communication parameter <Communication mode>.

1. Control line method
2. Control code system 1
3. Control code system 2





*1 DC3 code in parentheses are selectable between <0: No> and <1: Yes> according to the setting of communication parameter <Communication end DC3 code>.

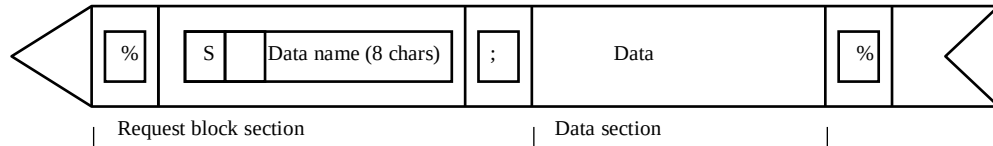
*2 Values of DC1 through DC4 are set by communication parameter <DC1 to 4 code>.

5.5.7 NC communication device protocol type 1 (NC language mode)

5.5.7.1 Data output request

Data is output from the master to the slave station.

1. The requested data is output from the master to the slave station by adding a request block before the data.

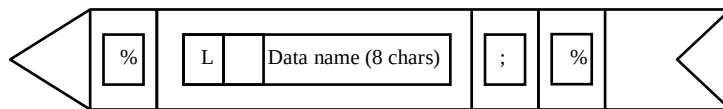


2. The slave station saves the data according to the information contained in the request block.

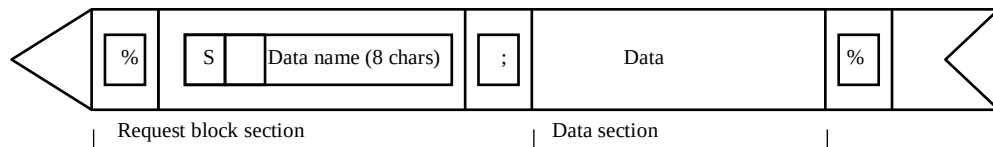
5.5.7.2 Data input request

Data is input from the slave station into the master station.

1. The master station outputs a request block for the data to be input to the master station.



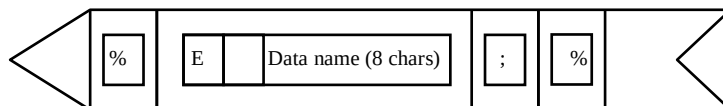
2. The slave station outputs the requested data to the master station by adding a request block at the top of the data.



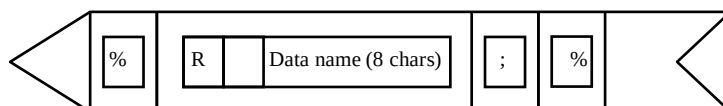
5.5.7.3 Data delete request

Data on the slave station is deleted.

1. The master station outputs a request block for the data to be deleted to the slave station.



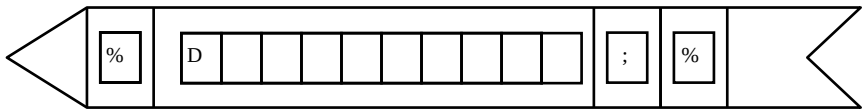
2. The slave station outputs a response block to the master station.



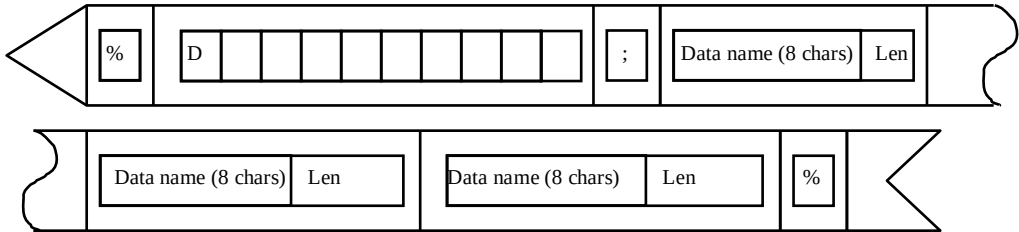
5.5.7.4 Request for inputting information on slave station directory

Directory information of the slave station is input into the master station.

1. The master station outputs the following request block to the slave station:



2. The slave station outputs the directory data to the master station by adding a request block at the top of the data.



- *1

%

 indicates tape start and tape end code (ISO/ASCII: %; EIA: ER).
- *2

;

 indicates end of block code (ISO: LF; EIA: EOB; ASCII: CR + LF).
- *3

--

 indicates end of block code (ISO: LF; EIA: EOB; ASCII: CR + LF).
- *4

S	
---	--

Data name (8 chars)

 and

L	
---	--

Data name (8 chars)

 indicate data transfer direction and data name (10 characters fixed).

S	
---	--

 indicates data output.

L	
---	--

 indicates data input.

E	
---	--

 indicates data deletion.

R	
---	--

 indicates response to data deletion.
- *5

Len

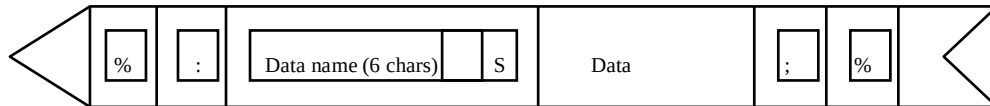
 indicates data size by 001 through 999 (3 characters fixed).
Data size is represented by 1 = 128 bytes.
Data name for requesting a directory is the same as *4.

5.5.8 NC communication device protocol type 1 (conversation mode)

5.5.8.1 Data output request

Data is output from the master to the slave station.

1. The requested data is output from the master to the slave station by adding a request block before the data.

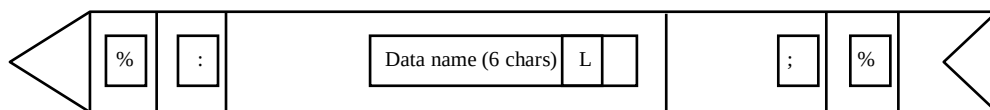


2. The slave station saves the data according to the information contained in the request block

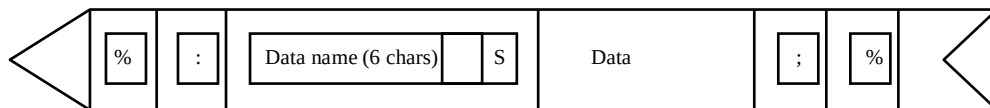
5.5.8.2 Data input request

The data of the slave station is input into the master station.

1. The master station outputs a request block for the data to be input to the master station.



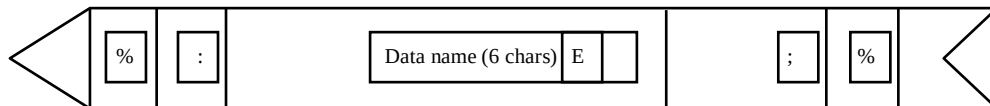
2. The slave station outputs the requested data to the master station by adding a request block at the top of the data.



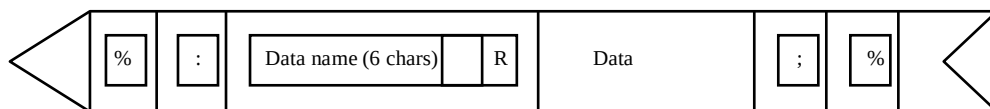
5.5.8.3 Data delete request

Data on the slave station is deleted.

1. The master station outputs a request block for the data to be deleted to the slave station.



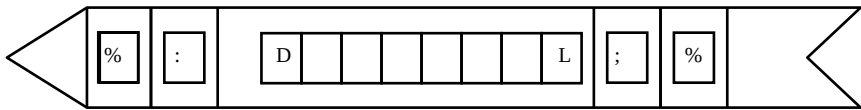
2. The slave station outputs a response block to the master station.



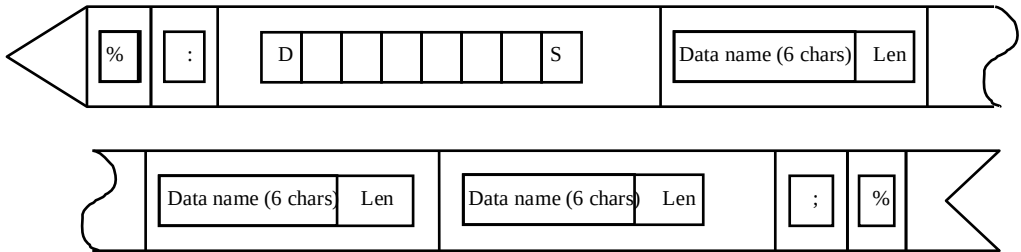
5.5.8.4 Request for slave station directory information

Directory information of the slave station is output to the master station.

1. Master station outputs the following request block to the slave station:



2. The directory data is output from the slave to the master station by adding a request block before the data.



- *1 “%” indicates tape start / end code (%).
- *2 ; indicates end of block code (CR + LF).
- *3 indicates space (SP).
- *4 Data name (6 chars) S and Data name (6 chars) L indicate data transfer direction and data name (8 characters fixed).
S indicated data output.
L indicates data input.
E indicates data deletion.
R indicates response to data deletion.
- *5 Len indicates data size by 001 through 999 (3 characters fixed).
Data size is represented by 1 = 128 bytes.
Data name for requesting a directory is the same as *4.

5.5.9 NC communication device protocol type 2

Once NC communication device protocol type 2 is set as Command-Response type, the command acceptance status can be externally checked, ensuring problem-free and precise control. This protocol applies to the case when type 2 is selected for the NC communication device at master station and to the NC communication software (serial or Ethernet) at slave station. Commands for not just data transfer but remote operations such as startup of memory operation for slave station and operation of external I/O signals are included.

5.5.9.1 Control format

1. Basic format

Command and response consist of:

Header + Data + Footer

Data is not necessarily included depending on cases.

2. Header (19 bytes)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
%	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2

% Start symbol

i1 Identifier – 1 byte

C: Command; R: response

One response of identifier R is always returned to each command of identifier C. The identifier of a response to a command is R, and the header of the same contents, excepting that a completion code is set, is echo-backed.

c1-c3 Command type – 3 bytes

f1-f4 Function – 4 bytes

s1-s8 Message – 8 bytes

r1-r2 Completion code – 2 bytes

00 :Normally completed: other than 00: Abnormally completed

Blank boxes appearing in the command header table hereafter indicate a space.

3. Data Section

The portion starting with “LF” until “LF” of the footer appears is defined as the data portion.

4. Footer (4 bytes)

Footer is indicated by LFss%.

“ss” indicates the checksum, which is represented by a 2-digit decimal number.

It is calculated by adding all values from % in the header to the character before LF in the footer in decimal expression, then dividing it by decimal 16, and keeping the remainder.

* % is the same as % in ISO and ASCII, and ER in EIA

* LF is the same as LF in ISO, EOB in EIA, and CR+LF in ASCII.

5.5.9.2 Completion code list

Completion code	Meaning
00	Normally ended
01	Invalid data is received
02	Illegal slave command header
04	Illegal slave command check sum
05	Currently in editing or operation mode, so processing is not possible The folder is currently in use; cannot delete.
06	Editing error occurred during file operation.
07	The specified data does not exist. Current or specified folder does not exist.
08	Slave command data name is incorrect. The specified folder name is abnormal.
09	The specified data cannot be saved or deleted.
10	Data protection enabled
11	Remote operation not permitted
13	The item of the received data is not within the allowed range or the number of items doesn't match. A symbol in the receiving data does not follow the record order in the file.
14	Data version error
15	During special startup
16	Cannot read the specified data.
17	Output of drawing data was attempted during drawing.
18	The folder already exists when creating the folder. The folder exists when deleting the folder; cannot delete.
19	Designation of data size is abnormal. Size of binary data is abnormal.
20	Binary data storage error.
21	Auto notification function reserve 1
22	Auto notification function reserve 2
23	Access is restricted
30	When changing the program, a program with the specified number does not exist. When changing the data bank, the specified number is not within the allowed range. When changing the data bank, the specified number is not a data bank. Changing the data bank is attempted while data protection is enabled. When changing the ATC tool, the specified number is not within the allowed range. An illegal signal name was specified via external input/output signal operation. The value is outside the specified range when writing individual data. Illegal signal name is designated in PLC signal operation.
31	Changing the program number was attempted in a mode other than memory operation mode. When changing the data bank, the machine parameter number was changed. Changing the data bank was attempted in a mode other than program edit mode. When changing the ATC tool, the specified tool is already registered. A signal that cannot be updated is specified for updating via external input/output signal operation. Write target is wrong in individual data writing. Attempted to update signals that cannot be updated in PLC signal setting.
32	Changing the program number was attempted during operation or editing. Changing the program number was attempted during tape operation. Changing the data bank was attempted during operation or editing. When changing the ATC tool, an attempt was made to set a large tool to a pot with a tool attached to the adjacent pot. Character strings other than ON/OFF were specified via external input/output signal operation. Attempted to set a value that cannot be set in PLC signal setting.
33	When changing the ATC tool, the tool is not registered in the specified group. A signal that cannot be turned off was attempted to be turned off via external input/output signal operation.
34	When changing the ATC tool, changing a magazine item (group/main tool/ drawing color) without a tool number assigned was attempted.
35	When changing the ATC tool, the pot adjacent to the specified pot contains a large tool.

Completion code	Meaning
36	Changing the ATC tool was attempted during memory operation.
37	Changing the ATC tool was attempted during MDI operation.
38	Unspecified error occurred during ATC tool change.
39	When changing the ATC tool, registering the unregistered tool in the tool list was attempted.
40	Conflict occurred due to communication using other port.
41	Check sum error occurred in the specified data.
42	Parity error occurred in the specified data.
43	The specified data is too large to be stored.
44	The specified data cannot be stored because programs #8000 to #8999 are write-protected.
45	Machine unit system is different.
46	The tool that is unable to change group/main tool/tool type/drawing color in ATC tool change is set.
60	Mode change not permitted
61	Mode change not permitted signal is on.
62	MDI operation mode
63	During tool change
64	During automatic centering
65	During automatic workpiece measurement
66	Automatic door operation not possible
67	Operation not possible
68	No program
69	Not in memory operation mode (or edit-during-operation mode)
70	The outer door is open.
71	The door is open.
72	The side door is open.
73	Resetting
74	Servo control is on.
75	[FEED HOLD] switch is held down.
76	Zero return was not conducted.
77	Restarting program Sequence search in progress
78	Pallet position error
79	Performing tool breakage detection
80	Program number error Different from pallet program
81	Outer pallet A and B-axes operating
82	No quick table
83	[PALLET] key is set to [OFF].
84	Workpiece counter end
85	Executing external output command
86	Memory operation mode
87	External input not available
88	In handle mode
89	XY-axes lock signal is on.
90	Z-axis lock signal is on.
91	*-axis lock signal is on.
92	Pot is not at the top end.
93	Zero return command error
94	Indexing not permitted signal is on.
95	Pallet start reversed
96	Outer pallet operating
97	Communicating
98	NC or conversation mode is not selected correctly.
99	Reservation

5.5.9.3 Details of commands

1. Data operation group

Restrict access

Temporary Ethernet access

The Ethernet access restriction, for using communication protocol type 2 for an NC communication device, is lifted temporarily. If the password is invalid, the completion code 23 is returned.

This is only needed for Ethernet access because there is no other access restriction. Even if a command is issued, there is no error completion code that is returned.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	S1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	E	T	H	L	G	I	N										

- Command data

Header	LF	Common user name	LF	Common password	LF	Footer
--------	----	------------------	----	-----------------	----	--------

Common user name...8 to 20 bytes

Specify the character string that is set in the communication parameter <Common user name>.

Common password...8 to 20 bytes

Specify the character string that is set in the communication parameter < Common password.>

End temporary Ethernet access

Temporary access ends and the Ethernet access restriction, for using communication protocol type 2 for an NC communication device, is applied.

This is only needed for Ethernet access because there is no other access restriction. Even if a command is issued, there is no error completion code that is returned.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	S1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	E	T	H	L	G	O	T										

2. Data operation group

Folder operation

Root folder is shown by “/” and the folder one layer up by “..”.

[Examples of folder name]

“ABC”: Moves to the folder named “ABC” existing in the current folder.

“/ABC” : Moves to the folder named “ABC” existing in the root folder.

“../ABC”: Moves to the folder named “ABC” existing in the master folder.

Request the current folder

Request is made for the path of the current folder of the other party.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	S1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	F	L	D	P	W	D											

- Response data

Header	LF	Path	LF	Footer
--------	----	------	----	--------

Path ... 1 to 255 bytes. Path, including the root folder, is returned.

Request the folder list

Request is made for a list of sub-folders existing in the current folder of the other party.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	F	L	D	D	R	Q											

Folder list is returned in the format shown below.

- Response data

Header	LF	Folder name	LF	Folder name	LF	Folder name	...	Footer
--------	----	-------------	----	-------------	----	-------------	-----	--------

Folder name ...1 to 255 bytes

The footer is output after all existing folders are output.

- * Folders in the next layer up (..) are also output if the current folder is not the root folder.

Move the current folder

Current folder of the other party to be used in slave station commands is moved to the specified folder.

Path may be included in the folder name to be specified.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	F	L	D	C	H	G											

- Command data

Header	LF	Folder name	LF	Footer
--------	----	-------------	----	--------

Folder name...1 to 255 bytes

- * Current folder is saved in memory at each port, and retained until power is turned off.

Create a folder

Folders are created on the other party.

Path may be included in the folder name to be specified.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	F	L	D	M	A	K	E										

- Command data

Header	LF	Folder name	LF	Footer
--------	----	-------------	----	--------

Folder name...1 to 255 bytes

Delete a folder

Folders of the other party are deleted.

Path may be included in the folder name to be specified.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	F	L	D	D	E	L											

- Command data

Header	LF	Folder name	LF	Footer
--------	----	-------------	----	--------

Folder name...1 to 255 bytes

- * Operation programs, edit programs, and the folder specified as the current folder of slave station of any port cannot be deleted.

Request directory

- * Directories that can be acquired are those in the current folder of the other party. First move the current folder by command and then execute Request directory.

Request the directories of all data

Request the master station to send the directory.

- Command header

	i1	c1	c2	c3	F1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	D	R	Q	A	L	A	L										

Directory information is returned in the format shown below.

- Response data

Header	LF	Data name	Size	Data name	Size	Data name	Size	...	Footer
--------	----	-----------	------	-----------	------	-----------	------	-----	--------

Data name...8 bytes

Size...10 bytes

Size is indicated by bytes. The footer is output after all existing data is output.

Request directory of all data (old type)

Ask Latest directory to the other party.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	D	R	Q	A	L	L											

Directory information is returned in the format shown below.

- Response data

Header	LF	Data name	Size	Data name	Size	Data name	Size	...	Footer
--------	----	-----------	------	-----------	------	-----------	------	-----	--------

Data name...8bytes

Size...3bytes

The data size is indicated by the number of blocks. One block contains 128 bytes. (Data exceeding 999 blocks is represented by 999.)

The footer is output after all existing data is output.

Request a directory by designating the data name

Request the master station to send the directory of the designated area.

This function is useful when confirming whether the designated data exists at the master station.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	D	R	Q	S	E	L		n1	n2	n3	n4	n5	n6	n7	n8		

n1n2n3n4n5n6n7n8...Specifies the data name.

- Response data

The data is returned in the same format as the case requesting directories of all data.

Ask detailed directory of registered programs only

Ask the other party for detailed directory of registered programs only.

- Command header

	i1	c1	c2	C3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	D	R	Q	P	R	A	L										

Directory information is returned in the format shown below.

- Response data

Header	LF	Data name	Size	Comment	Date	Data name	Size	...	Footer
--------	----	-----------	------	---------	------	-----------	------	-----	--------

Data name...8bytes

Size...10bytes

Comment...60bytes

Date...14bytes

Size is indicated by bytes. Date is shown by year (4 digits), month (2), day (2), hours (2), minutes (2), and seconds (2), from top.

Request detailed directory by specifying registered program

Request detailed directory to the other party by specifying registered programs

- Command header

	i1	c1	c2	C3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	D	R	Q	P	R	S	L	n1	n2	n3	n4	n5	n6	n7	n8		

n1n2n3n4n5n6n7n8•••Data name

Result of search is returned in the format shown below.

- Response data

Header	LF	Data name	Size	Comment	Date	Attribute	LF	Footer
--------	----	-----------	------	---------	------	-----------	----	--------

Data name•••8 bytes

Size•••10 bytes

Comment•••60 bytes

Date•••14 bytes

Attribute•••1 byte 0: Ordinary state (other than 1 and 2)

1: Program is being used.

2: Edit-prohibited program

Date is shown by year (4 digits), month (2), day (2), hours (2), minutes (2), and seconds (2), from top.

Save data

- * Programs are saved in the current folder of the other party.
First move the current folder by command and then request Save data.
- * Non-program data is saved only when the current folder is the root folder.

Save the specified data in the other party

Specified data is transferred to the other party.

- Command header

	i1	c1	c2	C3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	S	A	V					n1	n2	n3	n4	n5	n6	n7	n8		

n1n2n3n4n5n6n7n8...Specifies the data name.

- Command data

Header	LF	...Data...	LF	Footer
--------	----	------------	----	--------

Specified data comprises from LF immediately after the header to LF of the footer.

Save the specified data (binary) on the other party

Specified data is transferred to the other party by binary.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	S	A	V	B	I	N		n1	n2	n3	n4	n5	n6	n7	n8		

n1n2n3n4n5n6n7n8...Specifies the data name.

- Command data

Header	LF	Size	LF	...Data...	LF	Footer
--------	----	------	----	------------	----	--------

Size...1 to 10 bytes. Data size is represented by the number of bytes.

Data...Specified data with the length specified by size (binary data).

- * Data that can be transferred as binary data is limited to ladder programs and other certain files.
Refer to “5.6 Details of communication data” for details. No transmission is possible to the serial port.

Read data

- * In the case of programs, programs existing in the current folder of the other party are read.
First move the current folder by command and then execute Data reading.
- * Non-program data is read only when the current folder is the root folder.

Read specified data from other party

Ask the other party for the specified data, and receive reply data.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	L	O	D					n1	n2	n3	n4	n5	n6	n7	n8		

n1n2n3n4n5n6n7n8...Specifies the data name.

- Response data

The format is the same as the command data for “Save data”.

Read specified data (binary) from other party

Ask the other party for specified data and receive replay data.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	L	O	D	B	I	N		n1	n2	n3	n4	n5	n6	n7	n8		

n1n2n3n4n5n6n7n8...Specifies the data name.

- Response data

The format is the same as the command data for “Save data”.

- * Data that can be transferred as binary data is limited to ladder programs and other certain files.
Refer to “5.6 Details of communication data” for details. No transmission is possible to the serial port.

Read specified records in specified data from other party

Ask the other party for specified records in specified data, and receive the reply data.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	L	O	D	R	E	C		n1	n2	n3	n4	n5	n6	n7	n8		

n1n2n3n4n5n6n7n8...Specifies data name.

- Command data

Header	LF	Symbol 1	LF	...Symbol n...	LF	Footer
--------	----	----------	----	----------------	----	--------

Data...Symbol for record to be referenced

- Response data

The same format is used as the command data to <Save data>.

- * The specified symbol order must follow the same record order in the file.
- * Only some files can be specified. The following files can be specified.

Data name
Workpiece coordinate zero (Type 1)
Workpiece coordinate zero (Type 2)
Tool data (Type 1)
Tool data (Type 2)
Macro variables (Type 1)
Macro variables (Type 2)

Delete files

- * In the case of programs, programs existing in the current folder of the other party are deleted.
First move the current folder by command and then execute Delete data.
- * Non-program data is read only when the current folder is the root folder.

Delete specified data of other party

Ask the other party to delete the specified data.

The receiving station deletes the specified data and returns a response.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	D	E	L					n1	n2	n3	n4	n5	n6	n7	n8		

n1n2n3n4n5n6n7n8...Specifies the data name.

Delete all programs of the other party

Ask the other party to delete all NC programs, interactive machining data, and schedule programs. Programs in not only the current folder but also all other folders are deleted. The receiving station deletes all programs and returns a response.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	D	E	L	P	R	A	L										

Read individual dataAcquire file control data

File control data is acquired from the NC.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	F	I	L	E										

Machining program control data is returned in the format shown below.

- Response data

Header	LF	Number of registrations	Number of possible registrations	Memory usage	Remaining memory	LF	Footer
--------	----	-------------------------	----------------------------------	--------------	------------------	----	--------

Number of registrations ... 4 bytes; number of programs currently registered (total of NC and conversation)

Number of possible registrations...4 bytes; number of programs that can be registered (total of NC and conversation)

Memory usage ... 10 bytes; memory in use (bytes)

Remaining memory ... 10 bytes; remaining capacity of the memory (bytes)

- * Number of possible registrations, Memory usage, and Remaining memory include files other than programs such as data banks.

Acquire information on currently executed program

Information on the program currently selected in the memory operation is acquired.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	P	R	G	N										

Information on currently executed program is returned in the format shown below.

- Response data

Header	LF	Currently executed program number	Main program number	Currently executed block number	LF	Footer
--------	----	-----------------------------------	---------------------	---------------------------------	----	--------

Currently executed program number ... 4 bytes; currently selected program number

Main program number ... 4 bytes; Main program number

Currently executed block number ... 14 bytes; NC: currently executed block number
Conversation: process (3), tool (2), machining position (3), workpiece number (2), line number (2), infeed (2)

Currently executed program number is the same as Main program number when the system is running according to the main program.

Acquire currently executed programs by the number equivalent to the specified character number

That portion of the currently selected program running in the memory operation which is equivalent to the specified number of characters is acquired, starting from the block number currently being executed.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	P	R	G		c1	c2	c3	c4	c5	c6	c7	c8		

c1c2c3c4c5c6c7c8...Acquired character count: Between 1 and the screen display character count.
The required information on the program is returned in the format shown below.

- Response data

Header	LF	Program	LF	Footer
--------	----	---------	----	--------

Program ... That portion of the NC program which is equivalent to the specified number of characters, starting from the block currently being executed.

Those programs that have an execution point are output. Only sub-programs are output when the execution pointer is a sub-program. If a file ends before outputting the data equivalent to the specified number of characters, spaces are output thereafter.

An error occurs in the case of conversation mode.

Acquire date and time

Date and time of the other party are acquired.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	D	A	T	E										

Date and time are returned in the format shown below.

- Response data

Header	LF	Date	LF	Footer
--------	----	------	----	--------

Date... 14 bytes; shown by year (4 digits), month (2), day (2), hours (2), minutes (2), and seconds (2), from top.

Acquire PLC signal data

Acquires the specified PLC signal data.

- * You can instruct a command when the communication parameter <Remote operation> is set to <1: Valid>.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	P	L	C	D	k1	k2	k3	k4	n1	n2	n3	n4		

k1k2k3k4...Signal type

X : External input 000 to 3FF (hexadecimals)
 Y : External output 000 to 3FF (hexadecimals)
 BX : Internal input (bit) 000 to 4FF (hexadecimals)
 BY : Internal output (bit) 000 to 4FF (hexadecimals)
 BDY : Internal input (word) 0 to 303 (decimals)
 BDYL : Internal input (long word) 0 to 302 (Specify even hexadecimal numbers)

BDY : Internal output (word) 0 to 303 (decimals)
 BDYL : Internal output (long word) 0 to 302 (Specify even hexadecimal numbers)

M : Internal relay 0000 to 8191 (decimals)
 D : Data register(word) 0 to 2047 (decimals)
 DL : Data register (long word) 0 to 2046 (Specify even hexadecimal numbers)

LM0 to 3 : Local relay (High speed 1/2, Standard 1/2) 0000 to 2046 (decimals)
 (LM0: high speed 1; LM1: standard 1; LM2: high speed 2; LM3: standard 2)

LD0 to 3 : Local data register (high speed 1/2; Standard 1/2) 0 to 2047 (decimals)

LDL0 to 3: Local relay(long word) 0 to 1046 (Specify even hexadecimal numbers)
 (high speed 1/2; Standard 1/2)

LTV0 to 3: Local timer – current value (high speed 1/2; Standard 1/2) 0 to 255 (decimals)

LTP0 to 3: Local timer – set value (high speed 1/2; Standard 1/2) 0 to 255 (decimals)

LCV0 to 3: Local counter – current value (high speed 1/2; Standard 1/2) 0 to 255 (decimals)

LCP0 to 3: Local counter – set value (high speed 1/2; Standard 1/2) 0 to 255 (decimals)

- Response data

Header	LF	Set value	LF	Footer
--------	----	-----------	----	--------

Set value ... 1 byte (when k1k2k3k4 is X/Y/BX/BY/M/LM0 to 3) 0/1
 11 byte (when k1k2k3k4 is BDYL/BDYL/DL/LDL0 to 3) –2147483648 to 2147483647
 6 bytes (when k1k2k3k4 is other than the above) –32768~32767

Specify the range and acquire the PLC signal data

Acquires the specified range of the PLC signal data.

- * When the communication parameter <Remote operation> is set to <1:Valid>, this command can be issued.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	P	L	C	R	k1	k2	k3	k4	n1	n2	n3	n4		

k1k2k3k4...Signal type

n1n2n3n4...No.

The signal type and number are the same type and number used in “Acquire PLC signal data”.

- Command data

Header	LF	Data size	LF	Footer
--------	----	-----------	----	--------

Data size...4 bytes, 1 to 9999 (Specify using decimals regardless of the specified signal.)

The signals for the specified data size are referenced from the specified signals in the command header.

- * If the value has a data size that exceeds the range, it references the signals within the range.

- Response data

Header	LF	Data size	Set value	Set value	...	Set value	LF	Footer
--------	----	-----------	-----------	-----------	-----	-----------	----	--------

Data size...4 bytes, data size for signals that are referenced

Set value... 1 byte (when k1k2k3k4 is X/Y/BX/BY/M/LM0 to 3) 0/1

11 bytes (when k1k2k3k4 is BDXL/BDYL/DL/LDL0 to 3)-2147483648 to 2147483647

6 bytes (when k1k2k3k4 is data not specified above) -32768 to 32767

Acquire all current data bank names

All current data bank names are requested.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	C	D	B	N										

The current data bank names are issued in a response using the formats below.

- Response data

Header	LF	Data bank names	Data bank names	Data bank names	...	LF	Footer
--------	----	-----------------	-----------------	-----------------	-----	----	--------

Data bank names ...8 bytes

Acquire specific current data bank names

Specific current data bank names are requested.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	C	D	S	L	n1	n2	n3	n4	n5	n6	n7	n8		

n1n2 n3n4 n5n6 n7n8...Data bank names

* Specify the data bank names using a format without the data number. (Ex: UPRCM)

When the data number is included, the number is ignored. (Even if the name is entered as "UPRCM1" the name is processed as "UPRCM")

The current data bank names are issued in a response using the formats below.

- Response data

Header	LF	Data bank names	LF	Footer
--------	----	-----------------	----	--------

Data bank names ...8 bytes

Acquire tool compensation

The tool length offset, cutter compensation and tool position offset are acquired for a specific tool in the NC language.

- Command header

	i1	c1	c2	C3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	T	O	F	S	n1	n2	k1							

n1n2... Tool No.: 01 to 99

k1... Types → 0: Tool length offset, 1: T length wear offset, 2: Cutter compensation, 3: Cutter wear offset, 4: Tool position offset (X), 5: Tool pos. wear offset (X), 6: Tool position offset (Y) and 7: Tool pos. wear offset (Y).

The tool compensation is issued in a response using the formats below.

- Response data

Header	LF	Compensation	LF	Footer
--------	----	--------------	----	--------

Compensation... 9 bytes (when k1 is 0, 2, 4 and 6)

8 bytes (when k1 is 1, 3, 5 and 7)

(Commands are issued with decimal points for compensation, and when there is no decimal point, the integer part is processed.)

Acquire tool life

A specific tool life is acquired in the NC language.

- Command header

	i1	c1	c2	C3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	T	L	L	F	n1	n2	k1							

n1n2... Tool No.: 01 to 99

k1... Type → 0: Life unit, 1: Initial life, 2: Life warning, 3: Tool life

- Response data

Header	LF	Set value	LF	Footer
--------	----	-----------	----	--------

Set value...1 byte (when k1 is 0) for life unit → 1: Not counted, 2: Time (min), 3: Drilling (holes), 4: Program (cycles)

6 bytes (when k1 is 1, 2 or 3) for tool life value

Acquire H/D modal

The current H/D modal value is acquired.

- Command header

	i1	c1	c2	C3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	T	O	F	M										

The H/D modal value is issued in a response using the formats below.

- Response data

Header	LF	H modal	LF	D modal	Footer
--------	----	---------	----	---------	--------

H modal ...3bytes

D modal ...3bytes

Acquire macro variable

Acquires the value of the specified macro variable.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	M	C	N	M	n1	n2	n1							

n1n2n3... Tool No.: 500 to 999

The set value for the macro variable is issued in a response using the formats below.

- Response data

Header	LF	Set value	LF	Footer
--------	----	-----------	----	--------

Set value...12 byte

Specify the range and acquire the macro variables

Acquires the values for the macro variables in the specified range.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	R	E	D	M	C	N	M	n1	n2	n3							

n1n2n3... No.: 500 to 999

- Command data

Header	LF	Data size	LF	Footer
--------	----	-----------	----	--------

Data size...3 bytes, 1 to 999

The macro variables for the specified data size are referenced from the specified macro variables in the command header.

* If the value has a data size that exceeds the range, it references the macro variables within the range.

- Response data

Header	LF	Data size	Set value	Set value	...	Set value	LF	Footer
--------	----	-----------	-----------	-----------	-----	-----------	----	--------

Data size...3 bytes, data size for macro variables that are referenced

Set value...12 bytes

Write individual data

Preset (resets) offsets for relative coordinate positions

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	W	R	T	R	E	L		a1	a2								

a1a2...Axis number: 1 to 15

All axes: -1

- Command data

Header	LF	Offset 1	Offset 2	...	LF	Footer
--------	----	----------	----------	-----	----	--------

Offset 1... 10 bytes; offset for a single specified axis

Offset for the 1st axis when “all axes” is specified.

Offset 2 to 15... 10 bytes; Offset for the 2nd through 15th axes are added when “all axes” is specified. (Offsets are commanded with a decimal point. In the absence of a decimal point, the value is treated as an integer.)

Set tool offset

The tool length offset, cutter compensation and tool position offset are set for a specific tool in the NC language.

The setting cannot be changed while editing the tool data and while the initialize screen for tool life is displayed.

- Command header

	i1	c1	c2	C3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	W	R	T	T	O	F	S	n1	n2	k1							

n1n2...Tool number: 01 to 99

k1...Type. 0: tool length offset; 1: tool wear offset; 2: tool diameter offset; 3: tool diameter wear offset; 4: Tool position offset (X), 5: Tool pos. wear offset (X), 6: Tool position offset (Y) and 7: Tool pos. wear offset (Y).

- Command data

Header	LF	Offset	LF	Footer
--------	----	--------	----	--------

Offset ... 9 bytes (for k1 = 0,2,4,6)

8 bytes (for k1 = 1,3,5,7)

(Offsets are commanded with a decimal point. In the absence of a decimal point, the value is treated as an integer.)

Set tool life

Set the life of the tool of the specified NC language. No change is permitted during tool data editing or when the Initialize Life screen is displayed.

• Command header

	i1	c1	c2	C3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	W	R	T	T	L	L	F	n1	n2	k1							

n1n2... Tool number :01 to 99

k1... Type. 0: life unit; 1: initial life; 2: life warning; 3: life

• Command data

Header	LF	Set value	LF	Footer
--------	----	-----------	----	--------

Setting... 1 byte (for k1 = 0); life unit: 1: not counted; 2: time (min); 3: drilling (holes); 4: program (cycles)

6 bytes (for k1= 1 to 3): life value

Set tools to tool groups

Tools are added to the tool group of the specified NC language.

No change is permitted during tool data editing or when the Initialize Life screen is displayed.

• Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	W	R	T	T	G	R	P	n1	n2	i1	i2	t1	t2				

n1n2... Group number: 01 to 30

i1i2... Order of use: 01 to 30

t1t2... Tool number: 01 to 99

- * When specifying an existing order of use, the tool of that order of use is shifted to the next place.
- * Vacancy, if any, before the specified order of use is automatically closed up by advancing the tool of the next-occurring order of use.

Set date and time

Date and time of NC are set.

• Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	W	R	T	D	A	T	E										

• Command data

Header	LF	Date, time	LF	Footer
--------	----	------------	----	--------

Date...14 bytes; shown by year (4 digits), month (2), day (2), hours (2), minutes (2), and seconds (2), from top.

Set the operation time

Operation time is set.

• Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	W	R	T	O	P	T	M										

• Command data

Header	LF	Operation time	LF	Footer
--------	----	----------------	----	--------

Date...9 bytes; shown by hours (2 digits), minutes (2), and seconds (2), from top.

Set PLC signal data

The specified PLC signal data is set.

- * You can instruct commands when the communication parameter <Remote operation> is set to <1: Valid>.
- * “M: Internal relay” and “D (DL): Data register” are loaded from and written into the writing range when the ladder program is running. When using this function, set the location outside of the writing range before using the function.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	W	R	T	P	L	C	D	k1	k2	k3	k4	n1	n2	n3	n4		

k1k2k3k4	Signal type	n1n2n3n4	Number
Y	: External output	000 to 3FF	(hexadecimals)
M	: Internal relay	0000 to 8191	(decimals)
D	: Data register (word)	0 to 2047	(decimals)
DL	: Data register (long word)	0 to 2046	(Specify even hexadecimal numbers)
LM0 to 3	: Local relay (High speed 1/2, Standard 1/2)	0000 to 1023	(decimals)
	(LM0: high speed 1; LM1: standard 1; LM2: high speed 2; LM3: standard 2)		
LD0 to 3	: Local data register (high speed 1/2; Standard 1/2)	0 to 2047	(decimals)
LDL0 to 3	: Local data register(long word)	0 to 1046	(Specify even hexadecimal numbers)
	(high speed 1/2; Standard 1/2)		
LTV0 to 3	: Local timer – current value (high speed 1/2; Standard 1/2)	0 to 255	(decimals)
LTP0 to 3	: Local timer – set value (high speed 1/2; Standard 1/2)	0 to 255	(decimals)
LCV0 to 3	: Local counter – current value (high speed 1/2; Standard 1/2)	0 to 255	(decimals)
LCP0 to 3	: Local counter – set value (high speed 1/2; Standard 1/2)	0 to 255	(decimals)

- Command data

Header	LF	Set value	LF	Footer
--------	----	-----------	----	--------

Set value... 1 byte (When k1k2k3k4 is Y/M/LM0 - 3) 0/1
 6 bytes (When k1k2k3k4 is LTV0 - 3, LTP0 - 3, LCV0 - 3 or LCP0 - 3) 0 to 32767
 11 bytes (When k1k2k3k4 is DL/LDL0 - 3) -2147483648 to 2147483647
 6 bytes (When k1k2k3k4 is data not specified above) -32768 to 32767

Set macro variable

The value of the specific macro variable is set.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	W	R	T	M	C	N	M	n1	n2	n1							

n1n2n3... Tool No.: 500 to 999

- Command data

Header	LF	Set value	LF	Footer
--------	----	-----------	----	--------

Set value...12 bytes

(Commands are issued with decimal points for the set value, and when there is no decimal point, the integer part is processed.)

Clear individual dataInitialize tool life

Life of tool of the specified NC language is initialized.

No change is permitted during tool data editing or when the Initialize Life screen is displayed.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	C	L	R	T	L	L	F	n1	n2								

n1n2...Tool number : 1 to 99

Delete tools from tool group

Tools in the tool group of the specified NC language that are registered on the specified order of use are deleted from the tool group. No change is permitted during tool data editing or when the Initialize Life screen is displayed.

An alarm <<Tool changed during operation>> occurs and the machine stops when you delete a tool that is registered in the tool group currently used by the program. The tool group number to which the deleted tool belongs is also deleted from the ATC tool registration.

• Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	C	L	R	T	G	R	P	n1	n2	i1	i2						

n1n2...Tool group number : 1 to 30

i1i2... Order of use: 1 to 30

* Tools after the deleted order of use are moved forward.

3. Operation control group

- * The target of program number designation is the programs present in the current folder of the other party. Instruct commands only after moving the current folder by command.
- * You can instruct commands when the communication parameter <Remote operation> is set to <1: Valid>.

Operation commandStart the program operation

Memory operation of the specified program is started. Memory operation of the specified program starts when a program number is specified regardless of the input condition of program number select signal (PRO1, PRO2, PRO4, PRO8, PRO16, PRO32, PRO64, or PR9900) of the external input signal.

The program number may be omitted, in which case, the following program is selected:

- When any of the program select signals is input, the program number specified by that signal is used.
- When no program select signals are input, the program number selected in the current memory operation mode is used.

• Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	M	E	M	S	T	R	T	p1	p2	p3	p4						

p1p2p3p4...Specifies the program number to run (may be omitted).

- * The program cannot be re-opened. During startup, the following error appears: << Receive command abnormal end (0067)>>.

Stopping the program operation

Memory operation is stopped.

When ON is selected for this command, the [FEED HOLD] switch keeps the ON state until OFF is selected for this command. It is therefore necessary to select this command to OFF to resume operation.

The machine stops when either the external input signal (EXSTOP) or the state selected by this command is ON. Unless both are OFF, the [FEED HOLD] switch recognizes the ON state.

• Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	M	E	M	S	T	O	P	s1	s2	s3							

s1s2s3...ON: A state where [FEED HOLD] switch is pressed.

OFF: A state where [FEED HOLD] switch is not pressed.

Starting a program externally

The specified program is started externally. Memory operation of the specified program starts when a program number is specified regardless of the input condition of program number select signal (PRO1, PRO2, PRO4, PRO8, PRO16, PRO32, PRO64, or PR9900) of the external input signal.

The program number may be omitted, in which case, the following program is selected:

- When any of the program select signals is input, the program number specified by that signal is used.
- When no program select signals are input, the program number specified by the user parameter <Pallet 1/2 program No.>.

External startup reservation, if present, is released on instructing this command.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	M	E	M	Q	T	S	T	p1	p2	p3	p4						

p1p2p3p4... Specifies the program number to run (may be omitted).

Change commandChanging the program selection

Program of the memory operation is changed to the program of a specified program number. No change is available during operation.

This command is available only in the memory operation mode and in the edit mode during operation.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	C	H	G	P	R	O	G	p1	p2	p3	p4						

p1p2p3p4... Specifies the program number.

Changing the current data bank

Data bank number currently in use is changed.

Data name other than data bank name may be assigned to the data name.

No change is available during operation or when editing a specified data.

Commanding is only available in the edit mode.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	C	H	G	D	B	K		n1	n2	n3	n4	n5	n6	n7	n8		

n1n2n3n4n5n6n7n8... Specifies the data bank name.

Changing the mode

The mode can be changed to a designated mode.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	C	H	G	M	O	D	E	m1	m2	m3	m4						

m1m2m3m4... MNL : Manual mode

MDI : MDI mode

MEM : Memory operation mode

EDIT: Program edit mode

Changing the key status of memory operation mode

This command changes the ON/OFF status of each key used in memory operation mode.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	C	H	G	k1	k2	k3	k4	s1	s2	s3							

k1k2k3k4... DRYR : [DRY]key s1s2s3... ON : On

SNGL : [SINGL]key OFF : Off

OPTS : [OP.STP]key

BLKS : [B.SKIP]key

MACL : [M.LCK]key

Changing ATC tool

Tools set to the ATC tool are changed.

No change is available during operation.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	C	H	G	M	A	G	k1	m1	m2	t1	t2						

k1... M : Change tool number

S : Change group number (NC language mode)

Change main tool number (conversation mode)

K : Change tool type

C : Change color

D : Delete tool number

m1m2...Magazine number (00 when commanding spindle tool)

t1t2... Tool number, group number, spindle tool number (for k1 = M or S)

t1... 1: Standard tool; 2: large tool; 3: medium tool (for k1 = K)

t1... 0: No color; 1: blue; 2: red; 3: purple; 4: green; 5: light blue (for k1 = C)

Signal operation commandOperating External I/O signals

The functions of signals that are prepared as external I/O signals can be used using communication.

Input signals are available for operation and reference, and output signals for reference.

When referencing input signals, not only the status of signals due to communication commands but also the signals input into actual terminals (actual signals) are referenced and then the status of signals as recognized by the machine is returned.

Relation between actual signals and signal operation by communication command as well as the signal names that can be commanded are shown in the attached table.

- Command header

	i1	c1	c2	c3	f1	f2	f3	f4	s1	s2	s3	s4	s5	s6	s7	s8	r1	r2
%	C	I	O	C	c1	c2	c3	c4	s1	s2	s3	s4	s5	s6				

c1c2c3c4... REF : Reference

MOD : Operated the specified signal

s1s2s3s4s5s6...Signal name (see the attached table)

- Command data in operation and Response data in referencing

Header	LF	d1d2d3	LF	Footer
--------	----	--------	----	--------

d1d2d3...ON : Turns the specified signal ON (when operating); specified signal turns ON (when referencing)

OFF : Turns the specified signal OFF (when operation); specified signal turns OFF (when referencing)

External input signal (NC language mode)
Reference and change are available.

Signal name	Signal operation function	Signal name	Signal operation function	Signal name	Signal operation function
PRO1	ON/OFF	WCEDOK	ON/OFF	OVTR5P	ON/OFF
PRO2	ON/OFF	MGEDOK	ON/OFF	OVTR5M	ON/OFF
PRO4	ON/OFF	MTEDOK	ON/OFF	OVTR6P	ON/OFF
PRO8	ON/OFF	PREDOK	ON/OFF	OVTR6M	ON/OFF
PRO16	ON/OFF	IOEDOK	ON/OFF	OVTR7P	ON/OFF
PRO32	ON/OFF	CPEDOK	ON/OFF	OVTR7M	ON/OFF
PRO64	ON/OFF	DIROK	ON/OFF	OVTR8P	ON/OFF
EXSTRT	ON/OFF			OVTR8M	ON/OFF
OUTST	ON/OFF	MODEOK	ON/OFF	EXREF	ON/OFF
EXSTOP	ON/OFF	LEVEL1	ON/OFF	EXREF2	ON/OFF
		LEVEL2	ON/OFF	EXREF3	ON/OFF
EXORG	ON/OFF	LEVEL3	ON/OFF	EXREF4	ON/OFF
EXZORG	ON/OFF	LEVEL4	ON/OFF	EXREF5	ON/OFF
CTURN	ON/OFF	WKCLR1	ON/OFF	EXREF6	ON/OFF
MFIN	ON/OFF	WKCLR2	ON/OFF	OPSKIP	ON/OFF
EXER10	ON/OFF	WKCLR3	ON/OFF	OPSTOP	ON/OFF
:		WKCLR4	ON/OFF	EXWORK	ON/OFF
EXER99		DRLOCK	ON/OFF	WPEDOK	ON/OFF
ALMRLS	ON/OFF	ATDROP	ON/OFF	MCEDOK	ON/OFF
M460	ON/OFF	ATDRCL	ON/OFF	UINT	ON/OFF
M462	ON/OFF	OUTAP	ON/OFF	#1000	ON/OFF
M464	ON/OFF	OUTAM	ON/OFF	:	
M466	ON/OFF	OUTBP	ON/OFF	#1031	ON/OFF
M468	ON/OFF	OUTBM	ON/OFF	#1200	
SPLOCK	ON/OFF	OUTLCK	ON/OFF	:	
ATCLCK	ON/OFF	UNCLS5	ON/OFF	#1295	
XYLOCK	ON/OFF	UNCLS6	ON/OFF		
ZLOCK	ON/OFF	UNCLS7	ON/OFF	EXEMS	ON/OFF
4LOCK	ON/OFF	UNCLS8	ON/OFF	EXRESET	ON
5LOCK	ON/OFF	UNCSP1	ON/OFF		
6LOCK	ON/OFF	UNCSP2	ON/OFF		
7LOCK	ON/OFF	UNCSP3	ON/OFF		
8LOCK	ON/OFF	UNCSP4	ON/OFF		
MDLOCK	ON/OFF	CLPLS5	ON/OFF		
KYLOCK	ON/OFF	CLPLS6	ON/OFF		
PRLOCK	ON/OFF	CLPLS7	ON/OFF		
EDLOCK	ON/OFF	CLPLS8	ON/OFF		
MGPLCK	ON/OFF	CLPSP1	ON/OFF		
TLEDOK	ON/OFF	CLPSP2	ON/OFF		
UPEDOK	ON/OFF	CLPSP3	ON/OFF		
MPEDOK	ON/OFF	CLPSP4	ON/OFF		

Refer to “2.6.3 External Input Signals” for the function of each signal.

EXEMS: The alarm <<Emergency button activated>> or <<[MASTER ON] is off>> occurs depending on the value of machine parameter 1 <Emergency stop switch change>.

EXWORK: No changes can be made while in feature coordinate manufacturing mode (G68.2 modal in progress).

External output signal (NC language mode)
Only reference is available.

Signal name	Signal name	Signal name	Signal name
M00	TOOL	RPD100	HWMO
M30/1	TOLPRE	SPN100	RYCHKI
M30/2	ORGFIN	FED100	XPSSW2
M08	SPTURN	RELESE	YPSSW2
M11	ZPX	SAFETY	ZPSSW2
M12	ZPY		4PSSW2
M14	ZPZ	COOLSW	5PSSW2
M18	ZPSP	CHIPSW	6PSSW2
M21	ZP4	PULOFF	7PSSW2
M22	ZP5	QTSEL1	8PSSW2
M24	ZP6	QTSEL2	TLDHI
M28	ZP7	WAVDSP	TLDLO
MF	ZP8	XPOSSW	TLDON
M900	PFIN1	YPOSSW	
M400	PFIN2	ZPOSSW	REFIN
M402	RSTOUT	4POSSW	REFIN2
M404	TLBRKE	5POSSW	REFIN3
M406	RUNSTP	6POSSW	REFIN4
M408	BATALM	7POSSW	REFIN5
M480	MNTALM	8POSSW	REFIN6
M482	GRN	OUTAON	#1100
M484	YEL	OUTBON	:
M486	RED	UNCLP5	#1131
M450	CNTUP	UNCLP6	#1300
M451	CNTPRE	UNCLP7	:
M455	STPOUT	UNCLP8	#1395
M456	STPLED	UCLPP1	
AUTO	DROPEN	UCLPP2	LSTURN
STL	DRCLS	UCLPP3	LSSLCT
MEMOK	INDRCL	UCLPP4	
MEMMOD	SDDRCL	MGZ1	
EXPRUN	XDR1CL	MGZ2	
NCOK	XDR2CL	MGZ4	
	ATDRSL	MGZ8	
ALM	DRRDY	MGZ16	
ALM2	DROFIN	MGZ32	
ALMLV1	DRCFIN	UNLOCK	
ALMLV2	DRALM1	ALMA	
ASTLV1	DRALM2	MOUT	
ASTLV2	RDYLED	SPIALM	
ASTLV3	DRYRUN	DRHOLD	
ASTLV4	SINGL	DRORFN	
ASTLV5	RESTAT	RDBTAL	

Refer to “2.6.4 External Output Signals” for the function of each signal.

ALMA: The same signal as ALM.

MOUT: Auxiliary function numbers currently being output are set, in place of ON/OFF, in d1 through d3 of the response data. d1, d2, and d3 are decode values of M900, M21 through M28, and M11 through M18, respectively.

External input signal (Conversation mode)
Reference and change are available.

Signal name	Signal operation function	Signal name	Signal operation function	Signal name	Signal operation function
PRO1	ON/OFF	WCEDOK	ON/OFF	OVTR5P	ON/OFF
PRO2	ON/OFF	MGEDOK	ON/OFF	OVTR5M	ON/OFF
PRO4	ON/OFF	MTEDOK	ON/OFF	OVTR6P	ON/OFF
PRO8	ON/OFF	PREDOK	ON/OFF	OVTR6M	ON/OFF
PRO16	ON/OFF	IOEDOK	ON/OFF	OVTR7P	ON/OFF
PRO32	ON/OFF	CPEDOK	ON/OFF	OVTR7M	ON/OFF
PRO64	ON/OFF	DIROK	ON/OFF	OVTR8P	ON/OFF
EXSTRT	ON/OFF			OVTR8M	ON/OFF
OUTST	ON/OFF	MODEOK	ON/OFF		
EXSTOP	ON/OFF	LEVEL1	ON/OFF		
		LEVEL2	ON/OFF		
EXORG	ON/OFF	LEVEL3	ON/OFF		
EXZORG	ON/OFF	LEVEL4	ON/OFF		
CTURN	ON/OFF	WKCLR1	ON/OFF		
MFIN	ON/OFF	WKCLR2	ON/OFF	PR9900	ON/OFF
EXER10	ON/OFF	WKCLR3	ON/OFF	EXORG2	ON/OFF
:		WKCLR4	ON/OFF	EXORG3	ON/OFF
EXER99		DRLOCK	ON/OFF	TPEDOK	ON/OFF
ALMRLS	ON/OFF	ATDROP	ON/OFF	THEDOK	ON/OFF
M460	ON/OFF	ATDRCL	ON/OFF	CTEDOK	ON/OFF
M462	ON/OFF	OUTAP	ON/OFF		
M464	ON/OFF	OUTAM	ON/OFF	EXEMS	ON/OFF
M466	ON/OFF	OUTBP	ON/OFF	EXRESET	ON
M468	ON/OFF	OUTBM	ON/OFF		
SPLOCK	ON/OFF	OUTLCK	ON/OFF		
ATCLCK	ON/OFF	UNCLS5	ON/OFF		
XYLOCK	ON/OFF	UNCLS6	ON/OFF		
ZLOCK	ON/OFF	UNCLS7	ON/OFF		
4LOCK	ON/OFF	UNCLS8	ON/OFF		
5LOCK	ON/OFF	UNCSP1	ON/OFF		
6LOCK	ON/OFF	UNCSP2	ON/OFF		
7LOCK	ON/OFF	UNCSP3	ON/OFF		
8LOCK	ON/OFF	UNCSP4	ON/OFF		
MDLOCK	ON/OFF	CLPLS5	ON/OFF		
KYLOCK	ON/OFF	CLPLS6	ON/OFF		
PRLOCK	ON/OFF	CLPLS7	ON/OFF		
EDLOCK	ON/OFF	CLPLS8	ON/OFF		
MGPLCK	ON/OFF	CLPSP1	ON/OFF		
TLEDOK	ON/OFF	CLPSP2	ON/OFF		
UPEDOK	ON/OFF	CLPSP3	ON/OFF		
MPEDOK	ON/OFF	CLPSP4	ON/OFF		

Refer to “2.6.3 External Input Signals” for the function of each signal.

External output signal (Conversation mode)
Only reference is available.

Signal name	Signal name	Signal name	Signal name
M00	TOOL	RPD100	SPIALM
M30/1	TOLPRE	SPN100	DRHOLD
M30/2	ORGFIN	FED100	DRORFN
M08	SPTURN	RELESE	RDBTAL
M11	ZPX	SAFETY	HWMO
M12	ZPY		RYCHK1
M14	ZPZ		XPSSW2
M18	ZPSP	COOLSW	YPSSW2
M21	ZP4	CHIPSW	ZPSSW2
M22	ZP5	PULOFF	4PSSW2
M24	ZP6	QTSEL1	5PSSW2
M28	ZP7	QTSEL2	6PSSW2
MF	ZP8	WAVDSP	7PSSW2
M900	PFIN1	XPOSSW	8PSSW2
M400	PFIN2	YPOSSW	TLDHI
M402	RSTOUT	ZPOSSW	TLDLO
M404	TLBRKE	4POSSW	TLDON
M406	RUNSTP	5POSSW	
M408	BATALM	6POSSW	
M480	MNTALM	7POSSW	
M482	GRN	8POSSW	
M484	YEL	OUTAON	
M486	RED	OUTBON	
M450	CNTUP	UNCLP5	
M451	CNTPRE	UNCLP6	
M455	STPOUT	UNCLP7	
M456	STPLED	UNCLP8	
AUTO	DROPEN	UCLPP1	
STL	DRCLS	UCLPP2	
MEMOK	INDRCL	UCLPP3	
MEMMOD	SDDRCL	UCLPP4	
EXPRUN	XDR1CL	MGZ1	
NCOK	XDR2CL	MGZ2	
	ATDRSL	MGZ4	
ALM	DRRDY	MGZ8	
ALM2	DROFIN	MGZ16	
ALMLV1	DRCFIN	MGZ32	
ALMLV2	DRALM1	UNLOCK	
ASTLV1	DRALM2		
ASTLV2	RDYLED		
ASTLV3	DRYRUN		
ASTLV4	SINGL	ALMA	ORGFN2
ASTLV5	RESTAT	MOUT	ORGFN3

Refer to “2.6.4 External Output Signals” for the function of each signal.

- *1 Decode values of M900, M21 through M28, and M11 through M18 enter s1, s2, and s3, respectively.

4. Auto notification group

Control of auto notification function (NC ← Notification host)

When the communication parameter <Auto notification function> is set to <1: Enable>, the auto notification function can be controlled.

• Command header

	i1	c1	c2	c3	f1	f2	f3	f4	S1	s2	s3	S4	s5	S6	s7	s8	r1	r2
%	C	S	N	C	k1	k2	k3	k4										

k1k2k3k4...Types of controls

STRT: Start (only possible from the set host)

STOP: Stop (not just possible from the set host)

STAT: Acquire status (not just possible from the set host)

• Response data

header	LF	Return value	LF	Footer
--------	----	--------------	----	--------

Return value...6 bytes -32768 to 32767

0: End normally (during STRT, STOP)

1: Stopped (during STAT, STOP)

2: Operating for host issued command (during STAT, STRT)

3: Operating for another host (during STAT, STRT)

4: Send being set by another host (during STAT, STRT)

5: Disabled setting for auto notification function

6: Cannot start due to invalid parameter specification (during STRT)

7: Cannot stop due to auto notification being sent (during STOP)

Auto notification (NC → Notification host)

When the communication parameter <Auto notification function> is set to <1: Enable> and the function is activated, if the SND_REQ(BY1F0) is turned ON from a ladder program, then an auto notification is sent for the notification host from the NC.

• Command header

	i1	c1	c2	C3	f1	f2	f3	f4	s1	s2	s3	s4	s5	S6	s7	s8	r1	r2
%	C	S	N	D														

- When the command data communication parameter <Type of sending data> is set to <0: No data>

header	LF	LF	Footer
--------	----	----	--------

- When the command data communication parameter <Type of sending data> is not set to <0: No data>

header	LF	Type of data	Leading address	Data length	Data	Data	...	LF	Footer
--------	----	--------------	-----------------	-------------	------	------	-----	----	--------

Type of data ...8 bytes

The type of data is defined by the communication parameter <Method to specify sending data> and <Type of sending data> or SND_TYPE(BDY096).

PLCDM : PLC internal relay

PLCDD : PLC data register (word)

PLCDDL : PLC data register (long word)

Leading address...4 bytes

1 to 8191

The leading address is defined by the communication parameter <Method to specify sending data> and <Leading address for sending data> or SND_TOP(BDY097).

The following ranges apply according to the type of data.

0 to 8191 (specified using decimal numbers) for PLCDM

0 to 2047 (specified using decimal numbers) for PLCDD

0 to 2046 (specified using even decimal numbers) for PLCDDL

Data length ...4 bytes

1 to 600 (packets)

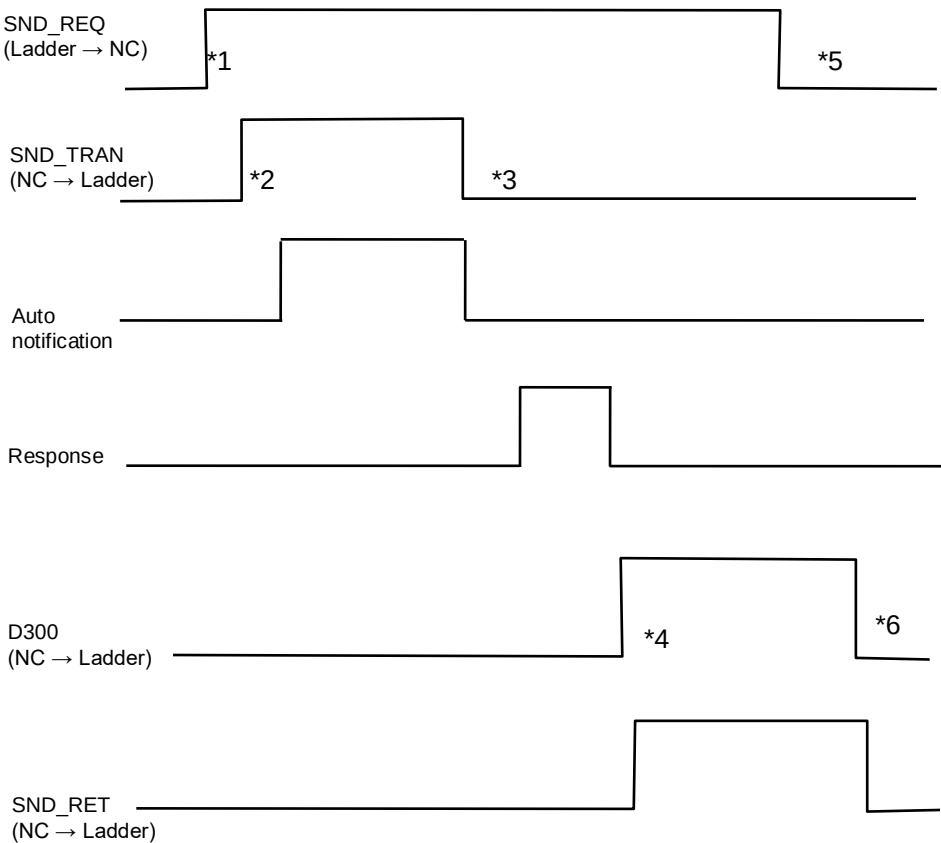
The data length is defined by the communication parameter <Method to specify sending data> and <Sending data length> or SND_LEN(BDY098). The following ranges apply according to the type of data.

1 to 600 (packets) for PLCDM

Data (multiple possible)	1 to 100 (packets) for PLCDD
	1 to 50 (packets) for PLCDDL
	The number alone for the data length continues. The number of bytes for 1 data is defined by the type of data.
	Every 1 byte (0/1) for PLCDM
	Every 6 bytes (-32768 to 32767) for PLCDD
	Every 11 bytes (-2147483648 to 2147483647) for PLCDDL

Control method for auto notification function (when standard)

Time chart example of PLC signal when the <Control method> for the auto notification function is set to <0: Standard>
This example shows when the <Data register address for storing error code> is set to <300>.



- *1 : When the SND_REQ is OFF, the ladder program prepares data for the auto notification and turns the SND_REQ to ON.
- *2 : The NC confirms that the SND_REQ is ON, and after the SND_TRAN turns ON, the auto notification starts to send.
- *3 : After the NC sends the auto notification, the SND_TRAN is turned OFF, and the NC waits for a response from the notification destination.
- *4 : Once the NC receives a response from the notification destination, the error code is stored in D300, and the SND_RET is turned ON.
- *5 : The ladder program confirms that the SND_RET is ON, and the SND_REQ is turned OFF. (Check D300 before SND-REQ is turned OFF.)
- *6 : The NC confirms that the SND_REQ is OFF, and after clearing D300, SND_RET is turned OFF.

Error code for auto notification function (Value entered in D300 in example above)

When between 0 and 101

0 to 99 : Completion code in the response received from the notification destination.

100 : No response has been received from the notification destination.

(Failed while SND command was sending. (Essentially, response was not received))

101 : A response was received from the notification destination but was not a number.

When greater than 1000

1100 : Failed TCP/IP connection with notification destination.

2100 : Timed out after waiting for response from notification destination.

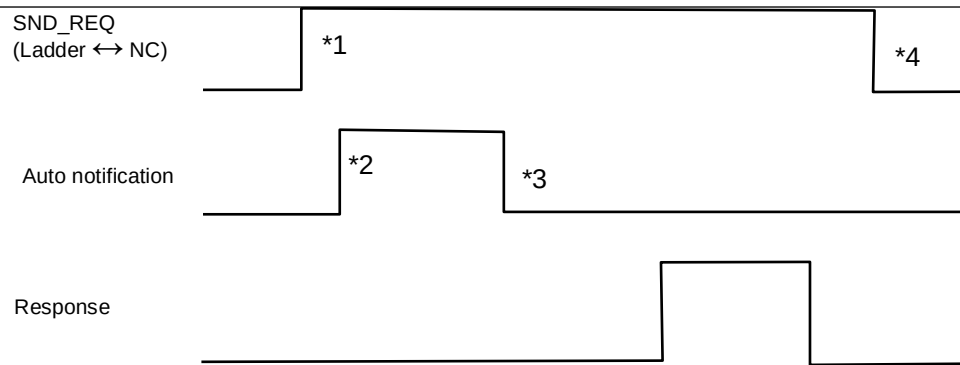
(After the SND command was sent, the time set in the parameter has elapsed)

3xxx : Check sum error, the low-order 3 digits are the same as when between 0 and 101.

4100 : It was not an auto notification response.

Control method for auto notification function (when set to simple)

Time chart of PLC signal when the <Control method> for the auto notification function is set to <1: Simple>



*1 : When the SND_REQ is OFF, the ladder program prepares data for the auto notification and turns the SND_REQ to ON.

*2: The NC confirms that the SND_REQ is ON, and the auto notification starts to send.

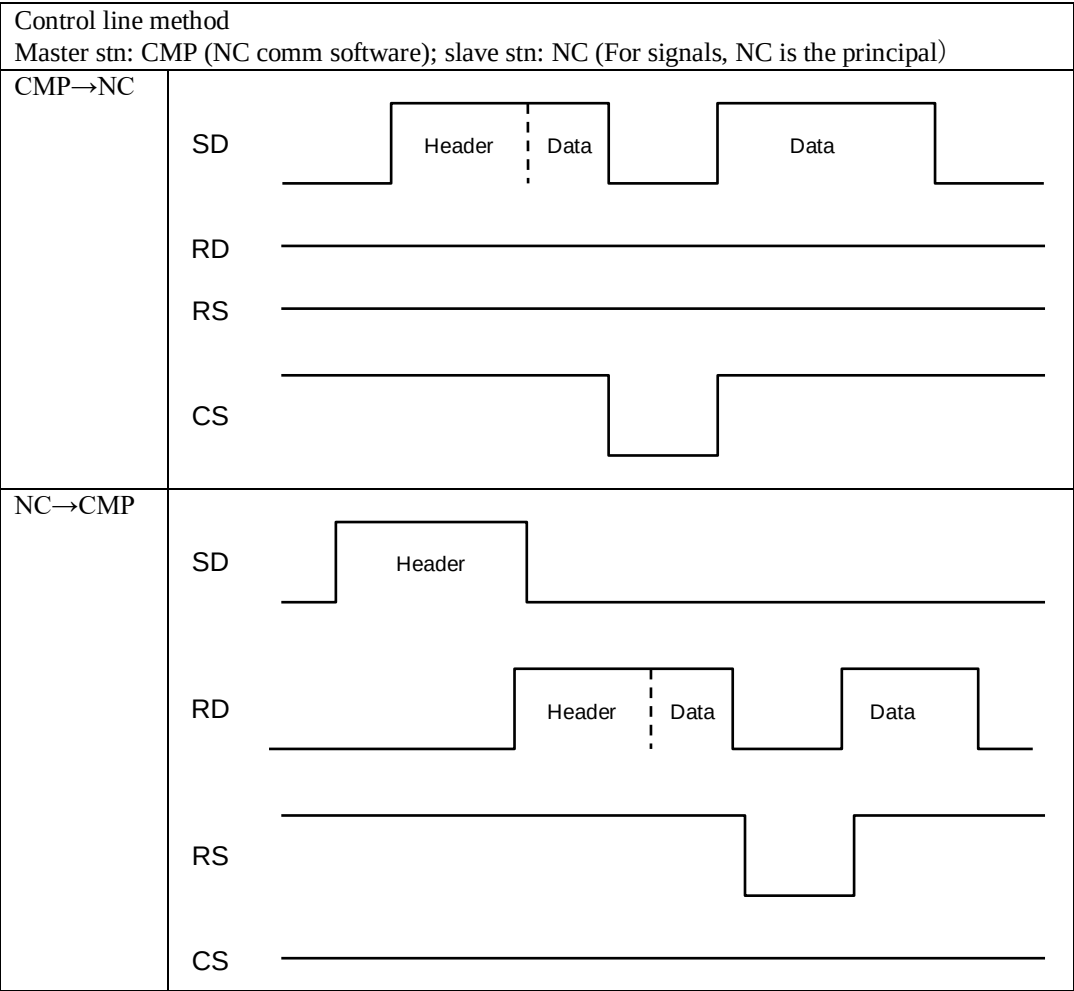
*3: After the NC sends the auto notification, the NC waits for a response from the notification destination.

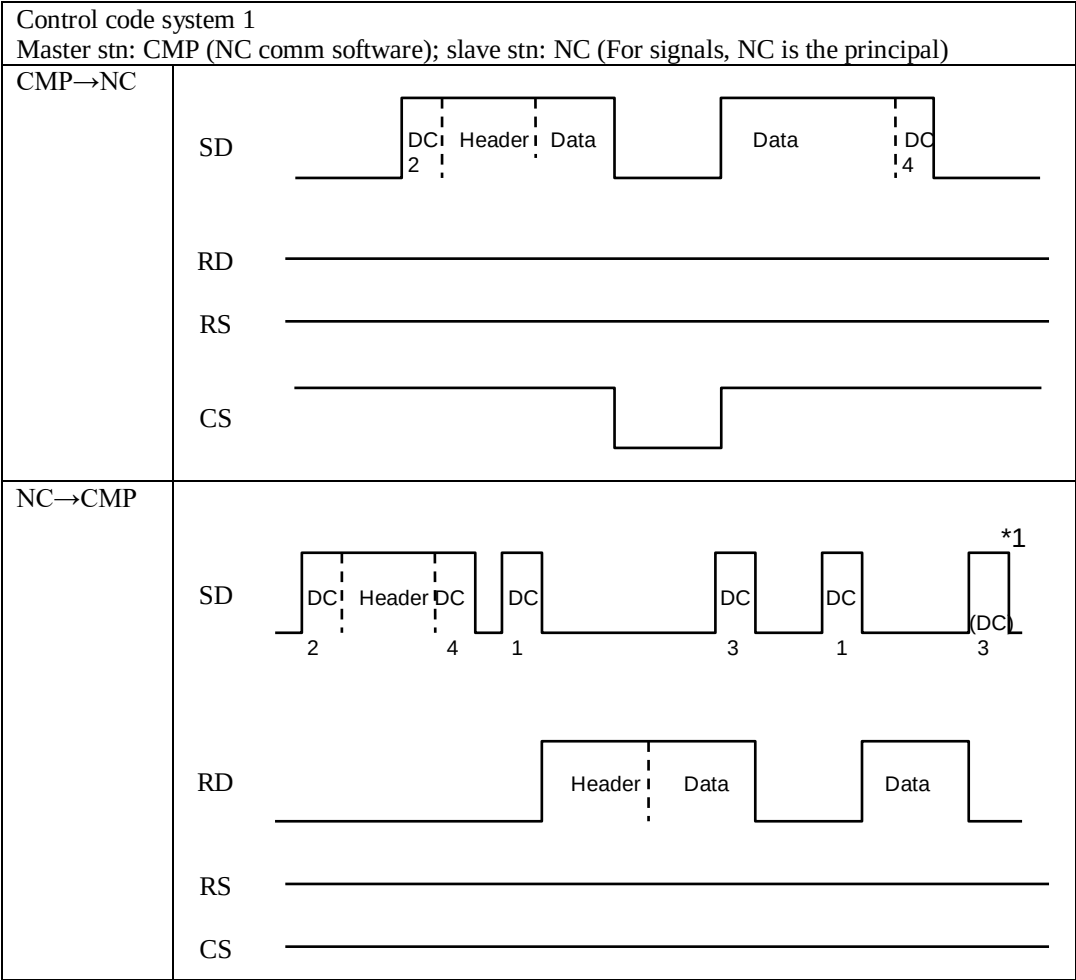
*4: After the NC receives a response, the SND_REQ is turned OFF.

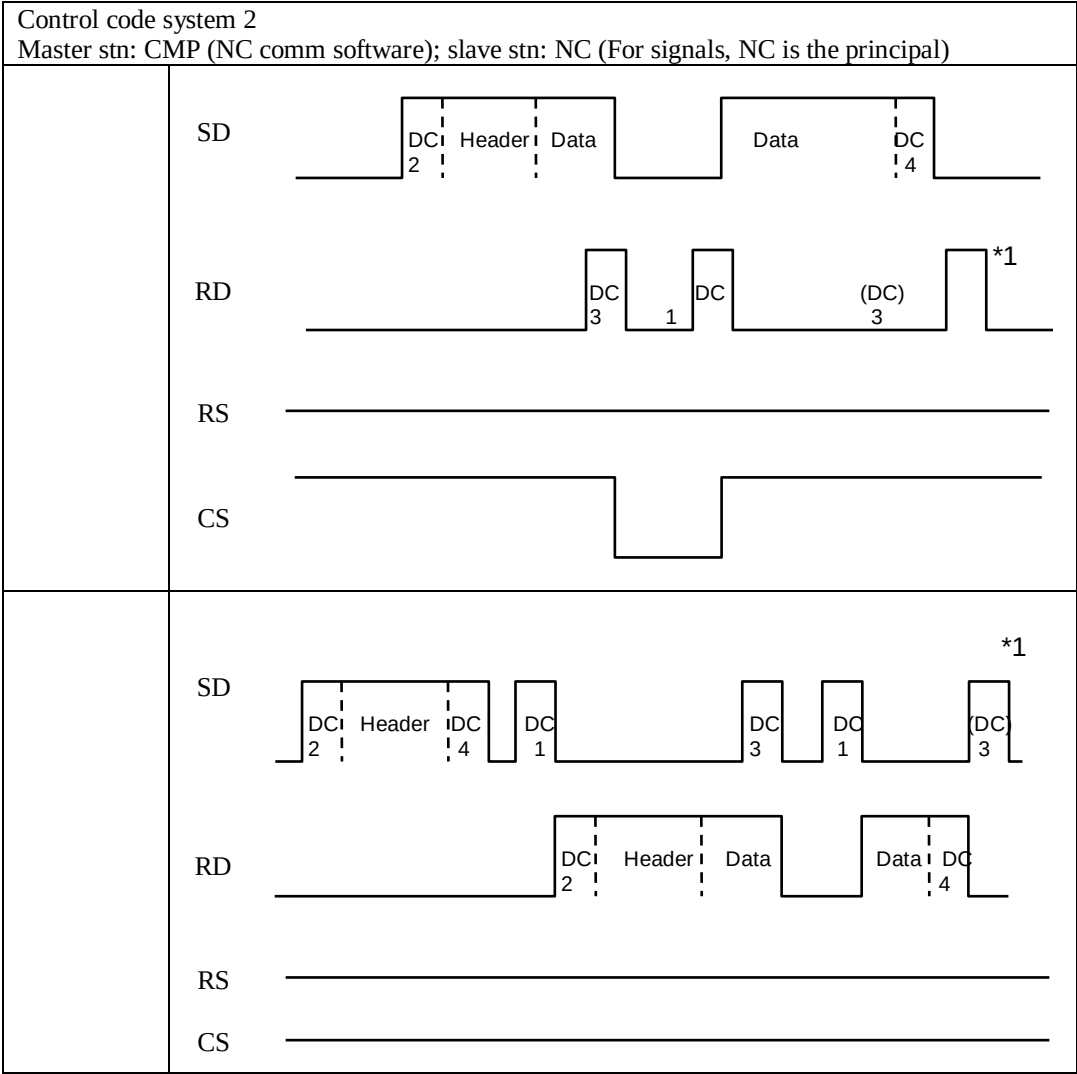
5.5.10 Communication methods of NC communication device

You can specify one of the following 3 methods by setting communication parameter <Communication mode>.

- 1. Control line method
- 2. Control code system 1
- 3. Control code system 2







- *1 DC3 code in parentheses is selectable between <0: No> and <1: Yes> according to the communication parameter <Communication end DC3 code>.
- *2 Values of DC1 through DC4 are set by communication parameter <DC1 to 4 code>.
- *3 For communication with NC as the master station, change NC and CMP with each other in the above table.

5.6 Details of communication data

5.6.1 Data name list

The followings are data names' list handled in external I/O.

Explanatory note

pppp: Program number

u: Unit system (M: Metric / I: Inch / D: Current unit system)

n: Data number (When data number is from 0 to 9 and D:0, it represents No.10. When unit system is set to D, it represents the current plane.)

Program

Data name	Data name					
Program (NC)	O	p	p	p	p	
Machining data (Conversation)	Q	u	p	p	p	p
Schedule program (Conversation)	J	p	p	p	p	
Old machining data (*1) (Conversation)	K	u	p	p	p	p
Old machining data (*2) (Conversation)	G	u	p	p	p	p
Old machining data (*3) (Conversation)	N	u	p	p	p	p
Old machining data (*4) (Conversation)	O	p	p	p	p	
3D smoothing data	S	D	p	p	p	p

Data Bank

Data name	Data name					
Workpiece zero point (Type 1)(NC)	P	O	S	N	u	n
Workpiece zero point (Type 2)(NC)	P	O	S	S	u	n
Tool data (Type 1)(NC)	T	O	L	N	u	n
Tool data (Type 2)(NC)	T	O	L	S	u	n
Macro variable (Type 1)(NC)	M	C	R	N	u	n
Macro variable (Type 2)(NC)	M	C	R	S	u	n
G/M code macro (NC)	G	M	M	C	n	
User parameter	U	P	R	C	u	n
External I/O signal	E	X	I	O	n	
Communication parameter	C	M	P	R	n	
Machine parameter	M	P	R	C	n	
Special settings	M	S	P	S	T	n
Machining load monitor	T	L	O	A	D	n
ATC tool data	A	T	C	T	L	
Automatic thermal distortion compensation data	H	E	A	C	u	
High speed ladder program 1 (*6)	H	T	Y	P	E	1
High speed ladder program 1 comment (*6)	H	T	Y	P	E	1
High speed ladder program 2 (*6)	H	T	Y	P	E	2
High speed ladder program 2 comment (*6)	H	T	Y	P	E	2
Standard ladder program 1 (*6)	S	T	Y	P	E	1
Standard ladder program 1 comment (*6)	S	T	Y	P	E	1
Standard ladder program 2 (*6)	S	T	Y	P	E	2
Standard ladder program 2 comment (*6)	S	T	Y	P	E	2
Global OM comment	G	C	O	M	T	
Teaching pendant custom	C	S	T	T	P	1
Tool list (Conversation)	T	O	L	C	u	n
Tool pattern (Conversation)	T	P	T	N	C	n
Tapping drill diameter (Conversation)	T	P	U	C	u	n
Cutting condition (Conversation)	C	N	D	C	u	n
Old tool list (*3) (Conversation)	L	u	n			
Old tool pattern (*3) (Conversation)	P		n			
Old cutting condition (*3) (Conversation)	C	u	n			

Maintenance data

Maintenance data	Data name		Data name			
Machine data 1	S	Y	S	C	9	9
Machine data 2	S	Y	S	C	9	8
Machine data 3	S	Y	S	C	9	7
Machine data 4	S	Y	S	C	9	6
Machine data 5	S	Y	S	C	9	5
Machine data 6	S	Y	S	C	9	4
Machine data 11	S	Y	S	C	8	9
Production data 1	P	R	D	1		
Production data 2	P	R	D	C	2	
Production data 3	P	R	D	3		
Data for machine monitor (Production data 4)	M	O	N	T	R	
Maintenance notice	M	A	I	N	T	C
Workpiece counter	W	K	C	N	T	R
Measurement result	M	S	R	R	S	C
Drawing data	P	A	I	N	T	
Waveform display parameters	W	V	P	R	M	
PLC signal data	P	L	C	D	A	T
PLC register monitor	P	L	C	M	O	N
Shortcut information	S	H	T	C	U	T
Input/ output data	I	O				
Memory operation data	M	E	M			
Operation panel data	P	A	N	E	L	
Position data	P	D	S	P		
Version data	V	E	R			
Alarm log data (up to item #100)	L	O	G			
Alarm log data (from item #101)	L	O	G	B	K	
Current alarm data	A	L	A	R	M	
Operation log	O	P	L	O	G	
Operation level data	P	R	T	C	T	C
Request code (Slave station)	R	Q	C			
Request code (Master station)	R	Q	C	nnnnnnnyymmdd		
	nnnnnn: Machine No. (User parameter), yy: Last two digits of year, mm: Month, dd: Day					

*1 TC32B, 31B, 22B, 20B, S2C, S2D, R2B, 32BN, S2DN data

*2 TC32A/ 31A/ 22A/ 20A/ S2A/ S2B/ R2A data

*3 TC201, 217/ 227/ 228, 218/ 229, 311/ 323, 312/ 324, 203 data

*4 TC211/ 212/ 221, 215/ 225 data

*5 When output from memory card or FTP, an extension is added to the above-mentioned file name. Extensions to be added vary depending on the current language mode: “.TC” for interactive mode and “.NC” for NC language mode. The extension for high speed ladder 1/2 and standard ladder 1/2 is fixed to “.LDR”, while that for high speed ladder 1/2 comment, standard ladder 1/2 comment and global OM comment is fixed to “.CMT”.

*6 High speed ladder 1/2 and standard ladder 1/2 are as shown below (Ethernet communication only.).

	File name in communication protocol method 2 of NC-dedicated communication device	File name after output/ file name in FTP protocol
High speed ladder program 1	HTYP1L	HTYPE1.LDR
High speed ladder program 1 comment	HTYP1C	HTYPE1.CMT
High speed ladder program 2	HTYP2L	HTYPE2.LDR
High speed ladder program 2 comment	HTYP2C	HTYPE2.CMT
Standard ladder program 1	STYP1L	STYPE1.LDR
Standard ladder program 1 comment	STYP1C	STYPE1.CMT
Standard ladder program 2	STYP2L	STYPE2.LDR
Standard ladder program 2 comment	STYP2C	STYPE2.CMT

5.6.2 Operation range

5.6.2.1 Operation from external devices to NC (NC:slave)

Output : Output from NC to external device

(Refer to “5.5 Communication Protocol” for files that can be output according to the record with the specified symbol.)

Input : Input from external device to NC

Deletion : NC file deleted according to request from external device

When the communication parameter <Restrict Ethernet access> is set to <1: Yes>, all output, input and delete operations based on the communication protocol type 2 for the NC communication device are restricted. First, issue a temporary access command and then make the operation request. When a serial port is used, there is no restriction applied. Refer to “Communication protocol” for further details.

○: Can be operated individually

▲: Can be operated in a batch

×: Cannot be operated

Data name	Output	Input	Deletion
Program(*1)	○	○	○(*2)
Machining data(*1)	○	○	○(*2)
Schedule program(*1)	○	○	○(*2)
Old machining data (K)(*6)	×	×	×
Old machining data (G) (*6)	×	×	×
Old machining data (N)(*6)	×	×	×
Old machining data (O)(*6)	×	×	×
3D smoothing data (*8)(*14)	○	○	○
Workpiece zero point (* 1)(* 10)	○	○(*16)	○(*3)
Workpiece zero point (* 2)(* 10)	○	○(*16)	○(*3)
Tool data (*1)(*10)	○	○(*11)	○(*3)
Tool data (*2)(*10)	○	○(*11)	○(*3)
Macro variable(*1)(*10)	○	○	○(*3)
Macro variable(*2)(*10)	○	○	○(*3)
User parameter (*1)	○	○	○(*3)
External I/O signal(*1)	○	○	○(*3)
Communication parameters(*1)	○	○	○(*3)
Machine parameters(*1)	○	○	○(*3)
G/M code macro	○	○	○(*3)
Special settings(*1)	○	○	○(*3)
Machining load monitor	○	○	○(*3)
ATC tool data	○	○(*4)	×
Automatic thermal distortion compensation data	○	×	×
High speed ladder program 1(*1)(*7)(*8)	○	○	×
High speed ladder program 1 comment (*1)(*7)(*8)	○	○	×
High speed ladder program 2(*1)(*7)(*8)	○	○	×
High speed ladder program 2 comment (*1)(*7)(*8)	○	○	×
Standard ladder program 1(*1)(*7)(*8)	○	○	×
Standard ladder program 1 comment(*1)(*7)(*8)	○	○	×
Standard ladder program 2(*1)(*7)(*8)	○	○	×
Standard ladder program 2 comment(*1)(*7)(*8)	○	○	×
Global OM comment(*1)(*7) (*8)	○	○	×
Teaching pendant custom(*8)	○	○	○
Tool list (*1)	○	○(*11)	○(*3)
Tool pattern (*1)	○	○	○(*3)
Tapping drill diameter(*1)	○	○	○(*3)
Cutting condition(*1)	○	○	○(*3)

Data name	Output	Input	Deletion
Old tool list (*6)	×	×	×
Old tool pattern (*6)	×	×	×
Old cutting condition (*6)	×	×	×
Machine data 1 (*17)	○(*4)	○(*4)	×
Machine data 2	○	○(*4)	×
Machine data 3	○	○(*4)	×
Machine data 4	○	○(*4)	×
Machine data 5	○	○	×
Machine data 6	○	○(*4)	×
Machine data 11(*8)	○	○(*4)	×
Production data 1	○(*4)	○(*4)	×
Production data 2	○	○(*4)	×
Production data 3	○	○	×
Data for machine monitor (Production data 4)	○	×	×
Maintenance notice	○	○	×
Workpiece counter	○	○(*4)	×
Measurement result	○	○(*4) (*13)	×
Drawing data	○(*5)	○(*5)	×
Waveform display parameters	○	○(*12)	×
PLC signal data(*8)	○	○	×
PLC register monitor	○	○	×
Shortcut information	○	○	×
Input/ output data	○	×	×
Memory operation data	○	×	×
Operation panel data	○	×	×
Position data	○	×	×
Version data	○	×	×
Alarm log data (up to item #100)	○	×	×
Alarm log data (from item #101)	○	×	×
Current alarm data	○	×	×
Operation log	○	×	×
Operation level data(*8)	○	○	×
Request code (*14)	○	×	×

- *1 Cannot be operated while editing the object file.
- *2 Cannot be operated while editing the object program.
- *3 Active data bank cannot be operated.
- *4 Cannot be operated during memory run.
- *5 Cannot be operated during check drawing.
- *6 Old formatted data cannot be operated.
- *7 Cannot be operated while displaying PLC monitor screen or PLC I/O screen or while communicating with PLC PROGRAMMING SOFTWARE.
- *8 Being binary data, they cannot be input/output in slave station communication through a serial port using "NC-dedicated communication software".
- *9 When a data is entered to the NC, the input data is checked. The data that includes the unavailable value cannot be entered.
- *10 Despite the difference in the file name and format of data between the machines of type 1 and type 2 in minimum setting unit, either type of data can be operated.
When requested for directory, however, only the file names in the current minimum setting unit are returned. (File names in other minimum setting unit are not returned.)
- *11 For the active data bank, the relevant tool is deleted from the ATC tools. During memory run or MDI run, an alarm <<Tool changed during operation>> is generated, which stops the operation.
- *12 Cannot be operated while measuring waveform data.
- *13 Data in other machine unit system cannot be input.
- *14 Operation is not available while 3D smoothing data is being generated, during communication, while running a source program for creating data and during editing.
- *15 Output is not possible when the machine model does not require a request code.

- *16 The workpiece coordinate zero cannot be changed while in feature coordinate manufacturing mode (G68.2 modal in progress)
- *17 When input, the alarm <<The machine coordinates may be out of position.>> may trigger after turning ON the power again.

5.6.2.2 Operation from NC to external devices (NC: master) (NC language mode)

Output: Output from NC to external device

Input: Input from external device to NC

Deletion: External device file deleted according to request from NC

○: Can be operated individually

▲: Can be operated in a batch

×: Cannot be operated

Data name	Program			Data Bank			Maintenance				
	Output	Input	Deletion	Output	Input	Deletion	Output all	Input all	Output all DB	Check output	Delete all
Program (*1)	○	○	○	×	×	×	▲	▲	×	▲	▲
Machining data (*1)	×	×	×	×	×	×	▲	▲	×	▲	▲
Schedule program (*1)	×	×	×	×	×	×	▲	▲	×	▲	▲
Old machining data (K)(*1)	×	×	×	×	×	×	×	▲	×	×	▲
Old machining data (G)(*1)	×	×	×	×	×	×	×	▲	×	×	▲
Old machining data (N)(*1)	×	×	×	×	×	×	×	▲	×	×	▲
Old machining data (O)(*1)	×	×	×	×	×	×	×	×	×	×	×
3D smoothing data(*5)(*13)	○	○	○	×	×	×	▲	▲	×	▲	▲
Workpiece zero point (Type1)(* 1)	×	×	×	○*8	○*9	○*9	▲*8	▲*9	▲*8	▲*8	▲*9
Workpiece zero point (Type1)(* 2)	×	×	×	○*8	○*9	○*9	▲*8	▲*9	▲*8	▲*8	▲*9
Tool data (Type 1)(*1)	×	×	×	○*8	○*9	○*9	▲*8	▲*9	▲*8	▲*8	▲*9
Tool data (Type 2)(*1)	×	×	×	○*8	○*9	○*9	▲*8	▲*9	▲*8	▲*8	▲*9
Macro variable (Type1)(*1)	×	×	×	○*8	○*9	○*9	▲*8	▲*9	▲*8	▲*8	▲*9
Macro variable (Type1)(*2)	×	×	×	○*8	○*9	○*9	▲*8	▲*9	▲*8	▲*8	▲*9
User parameter (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
External I/O signal (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Communication parameter (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Machine parameter (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
G/M code macro (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Special settings (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Machining load monitor(*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
ATC tool data (*3)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Automatic thermal distortion compensation data (*4)	×	×	×	○	×	○	×	×	×	×	×
High speed ladder program 1 (*1)(*2)(*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
High speed ladder program 1 comment (*1)(*2)(*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
High speed ladder program 2 (*1)(*2)(*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
High speed ladder program 2 comment (*1)(*2)(*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲

Data name	Program			Data Bank			Maintenance				
	Output	Input	Deletion	Output	Input	Deletion	Output all	Input all	Output all DB	Check output	Delete all
Standard ladder program 1 (*1)(*2)(*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
Standard ladder program 1 comment (*1)(*2)(*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
Standard ladder program 2 (*1)(*2)(*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
Standard ladder program 2 comment (*1)(*2)(*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
Global OM comment (*1)(*2)(*5)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Teaching pendant custom (*5)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Tool list (*1)	×	×	×	×	×	×	▲	▲	▲	▲	▲
Tool pattern (*1)	×	×	×	×	×	×	▲	▲	▲	▲	▲
Tapping drill diameter(*1)	×	×	×	×	×	×	▲	▲	▲	▲	▲
Cutting condition (*1)	×	×	×	×	×	×	▲	▲	▲	▲	▲
Old tool list (*1)	×	×	×	×	×	×	×	▲ *10	×	×	▲ *10
Old tool pattern (*1)	×	×	×	×	×	×	×	▲ *10	×	×	▲ *10
Old cutting condition (*1)	×	×	×	×	×	×	×	▲ *10	×	×	▲ *10
Machine data 1	×	×	×	×	×	×	▲	▲	▲	▲	▲
Machine data 2	×	×	×	×	×	×	▲	▲*7	▲	▲	▲
Machine data 3	×	×	×	×	×	×	▲	▲*7	▲	▲	▲
Machine data 4	×	×	×	×	×	×	▲	▲*7	▲	▲	▲
Machine data 5	×	×	×	×	×	×	▲	▲	▲	▲	▲
Machine data 6	×	×	×	×	×	×	▲	▲*7	▲	▲	▲
Machine data 11 (*5)	×	×	×	×	×	×	▲	▲	▲	▲	▲
Production data 1	×	×	×	×	×	×	▲	▲	▲	▲	▲
Production data 2	×	×	×	×	×	×	▲	▲	▲	▲	▲
Production data 3	×	×	×	×	×	×	▲	▲*7	▲	▲	▲
Data for machine monitor (Production data 4)	×	×	×	×	×	×	×	×	×	×	×
Maintenance notice	×	×	×	×	×	×	▲	▲	▲	▲	▲
Workpiece counter	×	×	×	×	×	×	▲	▲	▲	▲	▲
Measurement result	×	×	×	×	×	×	▲	▲ *12	▲	▲	▲
Drawing data	×	×	×	×	×	×	▲	▲	▲	▲	▲
Waveform display parameter	×	×	×	×	×	×	▲	▲ *11	▲	▲	▲
PLC signal data(*5)	×	×	×	×	×	×	▲	▲	▲	▲	▲
PLC register monitor	×	×	×	×	×	×	▲	▲	▲	▲	▲
Shortcut information	×	×	×	×	×	×	▲	▲	▲	▲	▲
Input/ output data	×	×	×	×	×	×	×	×	×	▲	▲
Memory operation data	×	×	×	×	×	×	×	×	×	▲	▲
Operation panel data	×	×	×	×	×	×	×	×	×	▲	▲
Position data	×	×	×	×	×	×	×	×	×	▲	▲
Version data	×	×	×	×	×	×	×	×	×	▲	▲
Alarm log data (up to item #100)	×	×	×	×	×	×	×	×	×	▲	▲
Alarm log data (from item #101)	×	×	×	×	×	×	×	×	×	▲	▲
Current alarm data	×	×	×	×	×	×	×	×	×	▲	▲

Data name	Program			Data Bank			Maintenance				
	Output	Input	Deletion	Output	Input	Deletion	Output all	Input all	Output all DB	Check output	Delete all
Operation log	×	×	×	×	×	×	×	×	×	▲	▲
Operation level data (*5)	×	×	×	×	×	×	▲	▲	▲	▲	▲
Request code (*14)	×	×	×	×	×	×	▲	×	×	▲	▲

- *1 Cannot be operated while editing the object file.
- *2 Cannot be operated while displaying PLC monitor screen or PLC I/O screen or while communicating with PLC PROGRAMMING SOFTWARE.
- *3 Entry is not possible during memory operation.
- *4 Cannot be operated when connected to <General COMM device.>
- *5 Being binary data, they cannot be input/output when connected to other than <Memory card> or <FTP server.>
- *6 High speed ladder programs 1 and 2 , high speed ladder program 1 and 2 comments, standard ladder programs 1 and 2, and standard ladder program 1 and 2 comments are operated in sets, respectively.
- *7 Check input is skipped.
- *8 The file name and format of data output differ between the machines of type 1 and type 2 in minimum setting unit.
- *9 Despite the difference in the file name and format of data between the machines of type 1 and type 2 in minimum setting unit, either type of data can be operated. If data with the same unit system and number exists in the connected device, the data with a file name which is supported by the machine is operated.
- *10 Can be operated only when there is no tool list, tool pattern or cutting condition with the same unit system and number in the connected device.
- *11 Cannot be operated while measuring waveform data.
- *12 Data in other machine unit are skipped.
- *13 Operation is not available while 3D smoothing data is being generated, during communication, while running a source program for creating data and during editing.
- *14 Output is not possible when the machine model does not require a request code.

5.6.2.3 Operation from NC to external devices (NC: master) (conversation mode)

Output : Output from NC to external device

Input : Input from external device to NC

Deletion : External device file deleted according to request from NC

○: Can be operated individually

▲: Can be operated in a batch

×: Cannot be operated

Data name	Program			Data Bank			Maintenance				
	Output	Input	Deletion	Output	Input	Deletion	Output all	Input all	Output all DB	Check output	Delete all
Program (*1)	×	×	×	×	×	×	▲	▲	×	▲	▲
Machining data (*1)	○	○	○	×	×	×	▲	▲	×	▲	▲
Schedule program (*1)	○	○	○	×	×	×	▲	▲	×	▲	▲
Old machining data (K)(*1)	×	○	○	×	×	×	×	▲	×	×	▲
Old machining data (G)(*1)	×	○	○	×	×	×	×	▲	×	×	▲
Old machining data (N)(*1)	×	○	○	×	×	×	×	▲	×	×	▲
Old machining data (O)(*1)	×	○ *11	×	×	×	×	×	×	×	×	×
3D smoothing data (*5)(*14)	×	×	×	×	×	×	▲	▲	×	▲	▲
Workpiece zero point (Type1)(* 1)	×	×	×	×	×	×	▲	▲	▲	▲	▲
Workpiece zero point (Type1)(* 2)	×	×	×	×	×	×	▲*8	▲*9	▲*8	▲*8	▲*9
Tool data (Type 1)(*1)	×	×	×	×	×	×	▲*8	▲*9	▲*8	▲*8	▲*9
Tool data (Type 2)(*1)	×	×	×	×	×	×	▲*8	▲*9	▲*8	▲*8	▲*9
Macro variable (Type1)(*1)	×	×	×	×	×	×	▲*8	▲*9	▲*8	▲*8	▲*9
Macro variable (Type1)(*2)	×	×	×	×	×	×	▲*8	▲*9	▲*8	▲*8	▲*9
User parameter (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
External I/O signal (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Communication parameter (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Machine parameter (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
G/M code macro	×	×	×	×	×	×	▲	▲	▲	▲	▲
Special settings (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Machining load monitor(*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
ATC tool data (*3)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Automatic thermal distortion compensation data (*4)	×	×	×	○	×	○	×	×	×	×	×
High speed ladder program 1(*1)(*2)(*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
High speed ladder program 1 comment (*1)(*2)(*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
High speed ladder program 2 (*1)(*2)(*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲

Data name	Program			Data Bank			Maintenance				
	Output	Input	Deletion	Output	Input	Deletion	Output all	Input all	Output all DB	Check output	Delete all
High speed ladder program 2 comment (*1) (*2) (*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
Standard ladder program 1 (*1) (*2) (*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
Standard ladder program 1 comment (*1) (*2) (*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
Standard ladder program 2 (*1) (*2) (*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
Standard ladder program 2 comment (*1) (*2) (*5)	×	×	×	○*6	○*6	○*6	▲	▲	▲	▲	▲
Global OM comment (*1) (*2) (*5)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Teaching pendant custom (*5)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Tool list (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Tool pattern (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Tapping drill diameter (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Cutting condition (*1)	×	×	×	○	○	○	▲	▲	▲	▲	▲
Old tool list (*1)	×	×	×	×	○*10	×	×	▲*10	×	×	▲*10
Old tool pattern (*1)	×	×	×	×	○*10	×	×	▲*10	×	×	▲*10
Old cutting condition (*1)	×	×	×	×	○*10	×	×	▲*10	×	×	▲*10
Machine data 1	×	×	×	×	×	×	▲	▲	▲	▲	▲
Machine data 2	×	×	×	×	×	×	▲	▲*7	▲	▲	▲
Machine data 3	×	×	×	×	×	×	▲	▲*7	▲	▲	▲
Machine data 4	×	×	×	×	×	×	▲	▲*7	▲	▲	▲
Machine data 5	×	×	×	×	×	×	▲	▲	▲	▲	▲
Machine data 6	×	×	×	×	×	×	▲	▲*7	▲	▲	▲
Machine data 11 (*5)	×	×	×	×	×	×	▲	▲	▲	▲	▲
Production data 1	×	×	×	×	×	×	▲	▲	▲	▲	▲
Production data 2	×	×	×	×	×	×	▲	▲	▲	▲	▲
Production data 3	×	×	×	×	×	×	▲	▲*7	▲	▲	▲
Data for machine monitor (Production data 4)	×	×	×	×	×	×	×	×	×	×	×
Maintenance notice	×	×	×	×	×	×	▲	▲	▲	▲	▲
Workpiece counter	×	×	×	×	×	×	▲	▲	▲	▲	▲
Measurement result	×	×	×	×	×	×	▲	▲*13	▲	▲	▲
Drawing data	×	×	×	×	×	×	▲	▲	▲	▲	▲
Waveform display parameters	×	×	×	×	×	×	▲	▲*12	▲	▲	▲
PLC signal data (*5)	×	×	×	×	×	×	▲	▲	▲	▲	▲
PLC register monitor	×	×	×	×	×	×	▲	▲	▲	▲	▲
Shortcut information	×	×	×	×	×	×	▲	▲	▲	▲	▲
I/O data	×	×	×	×	×	×	×	×	×	▲	▲
Memory run data	×	×	×	×	×	×	×	×	×	▲	▲
Operation panel data	×	×	×	×	×	×	×	×	×	▲	▲
Position data	×	×	×	×	×	×	×	×	×	▲	▲
Version data	×	×	×	×	×	×	×	×	×	▲	▲

Data name	Program			Data Bank			Maintenance				
	Output	Input	Deletion	Output	Input	Deletion	Output all	Input all	Output all DB	Check output	Delete all
Alarm log data (up to item #100)	×	×	×	×	×	×	×	×	×	▲	▲
Alarm log data (from item #101)	×	×	×	×	×	×	×	×	×	▲	▲
Current alarm data	×	×	×	×	×	×	×	×	×	▲	▲
Operation log	×	×	×	×	×	×	×	×	×	▲	▲
Operation level data (*5)	×	×	×	×	×	×	▲	▲	▲	▲	▲
Request code (*15)	×	×	×	×	×	×	▲	×	×	▲	▲

- *1 Cannot be operated while editing the object file.
- *2 Cannot be operated while displaying PLC monitor screen or PLC I/O screen or while communicating with PLC PROGRAMMING SOFTWARE.
- *3 Entry is not possible during memory operation.
- *4 Cannot be operated when connected to <General COMM device.>
- *5 Being binary data, they cannot be input/output when connected to other than <Memory card> or <FTP server.>
- *6 High speed ladder programs 1 and 2 , high speed ladder program 1 and 2 comments, standard ladder programs 1 and 2, and standard ladder program 1 and 2 comments are operated in sets, respectively.
- *7 Check input is skipped.
- *8 The file name and format of data output differ between the machines of type 1 and type 2 in minimum setting unit.
- *9 Despite the difference in the file name and format of data between the machines of type 1 and type 2 in minimum setting unit, either type of data can be operated. If data with the same unit system and number exists in the connected device, the data with a file name which is supported by the machine is operated.
- *10 Can be operated only when there is no tool list, tool pattern or cutting condition with the same unit system and number in the connected device.
- *11 Can be operated only when connected to general COMM device.
- *12 Cannot be operated while measuring waveform data.
- *13 Data in other machine unit are skipped.
- *14 Operation is not available while 3D smoothing data is being generated, during communication, while running a source program for creating data and during editing.
- *15 Output is not possible when the machine model does not require a request code.

5.6.3 Data compatibility with other models

5.6.3.1 Compatibility of NC language data

When using machining data created for other models, be sure to conduct test runs and check that there is no interference or other problems.

1. Compatibility with TC-217, 227, 218, 228, 229, 311, 312, 323, 324 and 731

Input

Program	Can be input.
Data Bank	Cannot be input.

Output

Program	Can be used. Note: Some codes cannot be used.
Data Bank	Cannot be used.

2. Compatibility with TC-20A, S2A, S2B, 22A, 31A, 32A and R2A

Input

Program	Can be input.
Data Bank	Workpiece coordinates zero position Can be input. Note: Extended coordinates are blank. Tool data Can be input. Note: Tool name, peripheral speed, feed per rotation, spindle speed (S), federate (F), Max. spindle speed, tool wash and CTs are blank. Macro variables Can be input. Note: Items #600 and after are blank. User parameter Cannot be input. Machine parameter Cannot be input.

Output

Program	Can be used. Note: Some codes cannot be used.
Data Bank	Workpiece coordinates zero position Can be used. Note: Extended coordinates are deleted. Tool data Can be used. Note: Tool name, peripheral speed, feed per rotation, spindle speed (S), federate (F), Max. spindle speed, tool wash and CTs are deleted. Macro variables Can be used. Note: Items #600 and after are deleted. User parameter Cannot be used. Machine parameter Cannot be used.

3. Compatibility with TC-20B, S2C, S2D, 22B, R2B, 31B, 32B, 32BN and S2DN

Input

Program	Can be input.
Data Bank	Workpiece coordinates zero position Can be input. Tool data Can be input. Macro variables Can be input. User parameter Cannot be input. Machine parameter Cannot be input.

Output

Program	Can be used. Note: Some codes cannot be used.
Data Bank	Workpiece coordinates zero position Can be used. Tool data Can be used. Macro variables Can be used. User parameter Cannot be used. Machine parameter Cannot be used.

5.6.3.2 Compatibility of conversational language data

When using machining data created for other models, be sure to conduct test runs and check that there is no interference or other problems.

(Note) Machining data that uses <10. NPT>, <11. Sewing machine thread,> or <12. Miniature screw thread> as screw type, if input to the machine whose machine parameter <Change screw type> is set to <0: Language set> and <Display language> is set to <Japanese> or whose <Change screw type> is set to <1: For Japan>, cannot be edited or used for operation.

1. Compatibility with TC-211, 212, 221, 215, and 225

Input

Machining data	Can be input. Note: Machining sequence, tool pattern, cutting conditions, and tool assignment data are automatically separated from the machining data. Set them again after input.
Data Bank	Cannot be input.

Output

No data can be used.

2. Compatibility with TC-321, 201, 203, 217, 218, 227, 228, 229, 311, 312, 323, and 324

Input

Machining data	Can be input. (Note) Although the machining data can be input, the machine may not operate correctly due to differences in programming specifications for some models. Modify the program while paying attention to the matters below. In addition, when initially performing memory operation, use functions such as single operation, dry run, machine lock, or override, to check that the machine operates correctly. 1. Some cutting conditions need to be reset. 2. Rotational transformations in the operation program are deleted. 3. The type of threads may vary for those other than metric coarse thread, metric fine pitch thread, unified coarse thread, and unified fine pitch thread. 4. The command method of the tool change M code varies for some models. Change it if necessary. 5. The Z-axis measurement system M code cannot be used. Delete it.												
Data Bank	Even when data banks can be input, the machine may not operate correctly because some items are added or changed. Set and change the data bank appropriately, according to the precautions regarding data banks described in the Operation Manual. In addition, when initially performing memory operation, use functions such as single operation, dry run, machine lock, or override, to check that the machine operates correctly. <table border="0"> <tr> <td>Tool List</td><td> Can be input. Tool wash and CTS are blank. (Note) 1. When the data bank is input from the TC-321, 201, 217, 227, or 228, set the tool life again. 2. The type of threads may vary for those other than metric coarse thread, metric fine pitch thread, unified coarse thread, and unified fine pitch thread. </td></tr> <tr> <td>Tool pattern</td><td>Can be input.</td></tr> <tr> <td>Setting Primary Hole Drilling Diameter for Tapping</td><td>Cannot be input.</td></tr> <tr> <td>Cutting conditions:</td><td>Can be input.</td></tr> <tr> <td></td><td>Note: Some data needs to be reset.</td></tr> <tr> <td>Parameter</td><td>Cannot be input.</td></tr> </table>	Tool List	Can be input. Tool wash and CTS are blank. (Note) 1. When the data bank is input from the TC-321, 201, 217, 227, or 228, set the tool life again. 2. The type of threads may vary for those other than metric coarse thread, metric fine pitch thread, unified coarse thread, and unified fine pitch thread.	Tool pattern	Can be input.	Setting Primary Hole Drilling Diameter for Tapping	Cannot be input.	Cutting conditions:	Can be input.		Note: Some data needs to be reset.	Parameter	Cannot be input.
Tool List	Can be input. Tool wash and CTS are blank. (Note) 1. When the data bank is input from the TC-321, 201, 217, 227, or 228, set the tool life again. 2. The type of threads may vary for those other than metric coarse thread, metric fine pitch thread, unified coarse thread, and unified fine pitch thread.												
Tool pattern	Can be input.												
Setting Primary Hole Drilling Diameter for Tapping	Cannot be input.												
Cutting conditions:	Can be input.												
	Note: Some data needs to be reset.												
Parameter	Cannot be input.												

Output

No data can be used.

3. Compatibility with TC-20A, S2A, S2B, 22A, 31A, 32A and R2A

Input

Machining data	Can be input. Notes: 1. Cutting condition (Others) requires partial resetting. 2. The Z-axis measurement system M code cannot be used. Delete it.	
SCHEDULE PROGRAM	Can be input.	
Data Bank	Tool List	Can be input. Tool wash and CTS are blank.
	Tool pattern	Can be input.
	Setting Primary Hole Drilling Diameter for Tapping	Can be input.
	Cutting conditions:	Can be input. Note: Cutting condition 3 (Others) requires partial resetting.
	User parameter	Cannot be input.
	Machine parameter	Cannot be input.

Output

Machining data	Cannot be used.	
SCHEDULE PROGRAM	Cannot be used.	
Data Bank	Tool List	Can be used. Tool wash and TCS are deleted.
	Tool pattern	Can be used.
	Setting Primary Hole Drilling Diameter for Tapping	Can be used.
	Cutting conditions:	Can be used. Note: Cutting condition 3 (Others) requires partial resetting.
	User parameter	Cannot be used.
	Machine parameter	Cannot be used.

4. Compatibility with TC-20B, S2C, S2D, 22B, R2B, 31B, 32B, 32BN and S2DN

Input

Machining data	Can be input.	
SCHEDULE PROGRAM	Can be input.	
Data Bank	Tool List	Can be input. Tool wash and CTS are blank.
	Tool pattern	Can be input.
	Setting Primary Hole Drilling Diameter for Tapping	Can be input.
	Cutting conditions:	Can be input.
	User parameter	Cannot be input.
	Machine parameter	Cannot be input.

Output

Machining data	Cannot be used.	
SCHEDULE PROGRAM	Cannot be used.	
Data Bank	Tool List	Can be used. Tool wash and CTS are blank.
	Tool pattern	Can be used.
	Setting Primary Hole Drilling Diameter for Tapping	Can be used.
	Cutting conditions:	Can be used.
	User parameter	Cannot be used.
	Machine parameter	Cannot be used.

5.6.4 File format

The data format is shown for each file.

A comma “,” is used as a delimiter between a symbol and item, or between each item. In addition, a “line break” code is used as a delimiter between each symbol.

- In the table, the delimiter between symbol and item is omitted.
- For the data length, only the data length section for the item is shown, and the symbol as well as the delimiter section is not included.

5.6.4.1 Workpiece counter

Filename: WKCNTN

Symbol	Item	Delimiter	Data length	Notes	NC/Conversation
A01	Workpiece counter 1 (Count)	,	3	0-999	NC
	Workpiece counter 1 (Current)	,	6	0-99999	NC
	Workpiece counter 1 (End)	,	6	0-99999	NC
	Workpiece counter 1 (End warning)	CR+LF	6	0-99999	NC
A02-A04	Workpiece counter 2-4	Same structure as A01			NC
B01	Workpiece counter 1 (Count)	,	3	0-999	Conversation
	Workpiece counter 1 (Current)	,	6	0-99999	Conversation
	Workpiece counter 1 (End)	,	6	0-99999	Conversation
	Workpiece counter 1 (End warning)	CR+LF	6	0-99999	Conversation
B02-B04	Workpiece counter 2-4	Same structure as B01			Conversation

A01-A04 and B01-B04 are supported by 1-4 in the “workpiece counter” screen.

5.6.4.2 Production data 2 (Time display)

Filename: PRDC2

Symbol	Item	Delimiter	Data length	Notes
L01	Cycle time (1) (Hrs./Min./0.1 Sec.)	,	9	hhhhmmss
	Cutting time (1) (Hrs./Min./0.1 Sec.)	,	9	hhhhmmss
	Cycle time (2) (Hrs./Min./0.1 Sec.)	,	9	hhhhmmss
	Cutting time (2) (Hrs./Min./0.1 Sec.)	,	9	hhhhmmss
	Program No. (1)	,	4	
	Program No. (2)	,	4	
	Program folder (1)	,	10	8 half-width characters (Characters that can be entered from the panel). The leading and last characters are single quotation marks.
	Program folder (2)	,	10	Same as above
	Memory operation type (1)	,	1	0: Internal memory operation, 1: Tape operation - General communications device, 2: Tape operation - Memory card
	Memory operation type (2)	CR+LF	1	Same as above
M01	Operation time status	,	1	
	Operation time index	CR+LF	2	
N01	Operation time log 1 (Date)	,	8	YYYYMMDD
	Operation time log 1 (Hours, minutes, seconds)	,	6	hhmmss
	Operation time log 1 (No. of operations)	,	5	

Symbol	Item	Delimiter	Data length	Notes
	Operation time log 1 (Power ON time)	CR+LF	6	
N02-20	Operation time log 2-20	Same structure as N01		
O01	Panel status (Coolant)	,	1	0: OFF, 1: ON
	Panel status (Chip shower)	,	1	0: OFF, 1: ON
	Panel status (Machine lights)	,	1	0: OFF, 1: ON
	Panel status (Pallet select key)	CR+LF	1	0: OFF, 1: 1, 2: 2, 3: 1-2
P01	Relative position offset (X-axis)	,	10	
	Relative position offset (Y-axis)	,	10	
	Relative position offset (Z-axis)	,	10	
	Relative position offset (No 4-axis)	,	10	
	Relative position offset (No 5-axis)	,	10	
	Relative position offset (No 6-axis)	,	10	
	Relative position offset (No 7-axis)	,	10	
	Relative position offset (No 8-axis)	,	10	
	Relative position offset (P1-axis)	,	9	
	Relative position offset (P2-axis)	,	9	
	Relative position offset (P3-axis)	,	9	
	Relative position offset (P4-axis)	CR+LF	9	
Q01	Cycle time log (1) (Hrs./Min./0.1 Sec.)	,	9	hhhhmmss
	Cutting time (1) (Hrs./Min./0.1 Sec.)	,	9	hhhhmmss
	Cycle time (2) (Hrs./Min./0.1 Sec.)	,	9	hhhhmmss
	Cutting time (2) (Hrs./Min./0.1 Sec.)	,	9	hhhhmmss
	Program No. (1)	,	4	
	Program No. (2)	,	4	
	Program folder (1)	,	10	8 half-width characters (Characters that can be entered from the panel). The leading and last characters are single quotation marks.
	Program folder (2)	,	10	Same as above
	Memory operation type (1)	,	1	0: Internal memory operation, 1: Tape operation - General communications device, 2: Tape operation - Memory card
	Memory operation type (2)		1	Same as above
	Operation end date and time (1)		14	YYYYMMDDhhmmss
	Operation end date and time (2)		14	YYYYMMDDhhmmss
	Operation end counter	CR+LF	4	0-9999

5.6.4.3 Measurement results

Filename: MSRRSC

Symbol	Item	Delimiter	Data length	Notes	NC/ Conversation
C01	Unit system	CR+LF	1	0: Metric, 1: Inch	NC
D01	Measurement result 1 (Measurement date and time)	,	14	YYYYMMDDhhmmss	NC
	Measurement result 1 (X)	,	10		NC
	Measurement result 1 (Y)	,	10		NC
	Measurement result 1 (Z)	,	10		NC
	Measurement result 1 (Rotation)	,	10		NC
	Measurement result 1 (G code 1)	,	1	1: G121, 2: G122, 3: G123 4: G124, 5: G125, 6: G126 7: G127, 9: G129, Only the G128 measurement result is 0.	NC
	Measurement result 1 (G code 2)	,	1	8: G128 The measurement result that does not include G128 is 0.	NC
	Measurement result 1 (Effective axis)	CR+LF	3	Bit 3: Rotation, 2: Z, 1: Y, 0: X	NC
D02-D09	Measurement result 1, Log 2-9	Same structure as D01			NC
E01-E09	Measurement result 2	Same structure as D01-D09			NC
F01-F09	Measurement result 3	Same structure as D01-D09			NC
G01-G09	Measurement result 4	Same structure as D01-D09			NC
H01	Measurement result 1 (Measurement date and time)	,	14	YYYYMMDDhhmmss	Conversation
	Measurement result 1 (Measurement type)	,	3	Bit 7: Always 1 5: Rotation Y 4: Rotation X 3: Z-axis height 2: Circle center 1: Parallel 0: Corner	Conversation
	Measurement result 1 (X)	,	9		Conversation
	Measurement result 1 (Y)	,	9		Conversation
	Measurement result 1 (Z)	,	9		Conversation
	(Reference position X)	,	9	Not used	Conversation
	(Reference position Y)	,	9	Not used	Conversation
	(Reference position Z)	,	9	Not used	Conversation
	Measurement result 1 (Center X)	,	9		Conversation
	Measurement result 1 (Center Y)	,	9		Conversation
	Measurement result 1 (Rotation)	,	9	0-999.999	Conversation
	Measurement result 1 (Effective axis)	,	3	Bit 5: Rotation 4: Center Y 3: Center X 2: Z 1: Y 0: X	Conversation
H02-H09	Measurement result 1, Log 2-9	Same structure as H01			Conversation
I01-I09	Measurement result 2	Same structure as H01-H09			Conversation

Symbol	Item	Delimiter	Data length	Notes	NC/ Conversation
J01-J09	Measurement result 3	Same structure as H01-H09			Conversation
K01-K09	Measurement result 4	Same structure as H01-H09			Conversation

5.6.4.4 Automatic thermal distortion compensation

Filename: HEACu, u: Unit system (M: Metric I: Inch)

Symbol	Item	Delimiter	Data length	Notes
A0001	Log 1 (No. of calculations)	,	5	
	Log 1 (Reserved)	,	1	Not used (0 fixed)
	Log 1 (Reserved)	,	11	Same as above
	Log 1 (Reserved)	,	10	Same as above
	Log 1 (Reserved)	,	9	Same as above
	Log 1 (Reserved)	,	9	Same as above
	Log 1 (Reserved)	,	7	Same as above
	Log 1 (Reserved)	,	7	Same as above
	Log 1 (Time)	,	14	YYYYMMDDhhmmss
	Log 1 (Average speed on Z-axis)	,	11	0.0000~429496.7295 0.00000~42949.67295*1
	Log 1 (Spindle rotation sum)	,	10	0~4294967295
	Log 1 (Ball screw displacement 1 (Z-axis))	,	8	-99.9999~99.9999 -9.99999~9.99999 *1
	Log 1 (Ball screw displacement 2 (Z-axis))	,	8	-99.9999~99.9999 -9.99999~9.99999 *1
	Log 1 (Spindle displacement (Z-axis))	,	8	-99.9999~99.9999 -9.99999~9.99999 *1
	Log 1 (Position 1 (Z-axis))	,	10	-9999.9999~9999.9999 -999.99999~999.99999 *1
	Log 1 (Position 2 (Z-axis))	,	10	-9999.9999~9999.9999 -999.99999~999.99999 *1
	Log 1 (Compensation 1 (Z-axis))	,	10	-9999.9999~9999.9999 -999.99999~999.99999 *1
	Log 1 (Compensation 2 (Z-axis))	,	10	-9999.9999~9999.9999 -999.99999~999.99999 *1
	Log 1 (Lathe spindle rotation sum)	,	10	0~4294967295
	Log 1 (Lathe spindle displacement (Z-axis))	CR+LF	8	-99.9999~99.9999 -9.99999~9.99999 *1
A0002-A10801	Same structure as A0001			
B0001	Log 1 (No. of calculations)	,	5	
	Log 1 (Reserved)	,	1	Not used (0 fixed)
	Log 1 (Reserved)	,	11	Same as above
	Log 1 (Reserved)	,	10	Same as above
	Log 1 (Reserved)	,	9	Same as above
	Log 1 (Reserved)	,	9	Same as above
	Log 1 (Reserved)	,	7	Same as above
	Log 1 (Reserved)	,	7	Same as above
	Log 1 (Time)	,	14	YYYYMMDDhhmmss
	Log 1 (Average speed on X-axis)	,	11	0.0000~429496.7295 0.00000~42949.67295*1
	Log 1 (Spindle rotation sum)	,	10	0~4294967295
	Log 1 (Ball screw displacement 1 (X-axis))	,	8	-99.9999~99.9999 -9.99999~9.99999 *1
	Log 1 (Ball screw displacement 2 (X-axis))	,	8	-99.9999~99.9999 -9.99999~9.99999 *1

Symbol	Item	Delimiter	Data length	Notes
	Log 1 (Spindle displacement (X-axis))	,	8	-99.9999~99.9999 -9.99999~9.99999*1
	Log 1 (Position 1 (X-axis))	,	10	-9999.9999~9999.9999 -999.99999~999.99999*1
	Log 1 (Position 2 (X-axis))	,	10	-9999.9999~9999.9999 -999.99999~999.99999*1 ¹
	Log 1 (Compensation 1 (X-axis))	,	10	-9999.9999~9999.9999 -999.99999~999.99999*1
	Log 1 (Compensation 2 (X-axis))	,	10	-9999.9999~9999.9999 -999.99999~999.99999*1
	Log 1 (Lathe spindle rotation sum)	,	10	0~4294967295
	Log 1 (Lathe spindle displacement (X-axis))	CR+LF	8	-99.9999~99.9999 -9.99999~9.99999*1
B0002-B1080	Same structure as B0001			
C0001	Log 1 (No. of calculations)	,	5	
	Log 1 (Reserved)	,	1	Not used (0 fixed)
	Log 1 (Reserved)	,	11	Same as above
	Log 1 (Reserved)	,	10	Same as above
	Log 1 (Reserved)	,	9	Same as above
	Log 1 (Reserved)	,	9	Same as above
	Log 1 (Reserved)	,	7	Same as above
	Log 1 (Reserved)	,	7	Same as above
	Log 1 (Time)	,	14	YYYYMMDDhhmmss
	Log 1 (Average speed on Y-axis)	,	11	0.0000~429496.7295 0.00000~42949.67295*1
	Log 1 (Spindle rotation sum)	,	10	0~4294967295
	Log 1 (Ball screw displacement 1 (Y-axis))	,	8	-99.9999~99.9999 -9.99999~9.99999*1
	Log 1 (Ball screw displacement 2 (Y-axis))	,	8	-99.9999~99.9999 -9.99999~9.99999*1
	Log 1 (Spindle displacement (Y-axis))	,	8	-99.9999~99.9999 -9.99999~9.99999*1
	Log 1 (Position 1 (Y-axis))	,	10	-9999.9999~9999.9999 -999.99999~999.99999*1
	Log 1 (Position 2 (Y-axis))	,	10	-9999.9999~9999.9999 -999.99999~999.99999*1
	Log 1 (Compensation 1 (Y-axis))	,	10	-9999.9999~9999.9999 -999.99999~999.99999*1
	Log 1 (Compensation 2 (Y-axis))	,	10	-9999.9999~9999.9999 -999.99999~999.99999*1
	Log 1 (Lathe spindle rotation sum)	,	10	0~4294967295
	Log 1 (Lathe spindle displacement (Y-axis))	CR+LF	8	-99.9999~99.9999 -9.99999~9.99999*1
C0002-C1080	Same structure as C0001			

*1: Minimum digit for micron is a space.

5.6.4.5 Maintenance notice

Filename: MAINTC

Symbol	Item	Delimiter	Data length	Notes
Q01	No.1 (Type)	,	2	0: None 1: Spindle speed (× 1000 revs.) 2: X-axis travel distance (m) [in. × 100 inch] 3: Y-axis travel distance (m) [in. × 100 inch] 4: Z-axis travel distance (m) [in. × 100 inch] 5-9: No 4-8-axes speed (Rotation) 10: Tool change times (No. of times) 11: Magazine turn pitches (Pitch) 12: Center-through-coolant ON time (Hours, minutes, seconds) 13: Center-through-coolant ON times (No. of times) 14: Outer/Front door closings (No. of times) 15: Side door closings (No. of times) 16: Inner door closings (No. of times) 17: Servo ON times (No. of times) 18: Power ON times (No. of times) 19: Power ON time (Hours, minutes, seconds) 20-23: P1-4-axes travel amount
	No.1 (Message)	,	22	8 half-width characters (Characters that can be entered from the panel). The leading and last characters are single quotation marks.
	No.1 (Function)	,	1	0: Disabled, 1: Enabled
	No.1 (Message notification)	,	1	0: Do not notify, 1: Notify
	No.1 (Current value)	,	9	If the set value for the “type” is displayed in a date and time format, 2008/1/1 is converted to 100000000 and any difference in the value is expressed in seconds. If the set value for the “type” is displayed as a time period, the unit is in seconds.
	No.1 (End value)	CR+LF	9	Same as above
Q02-Q16	No.2-No.16	Same structure as Q01		

(Note) To input into the NC, enter the value for the range that can be set on the NC screen.

5.6.4.6 Production data 3 (Status history)

Filename: PRD3

Symbol	Item	Delimiter	Data length	Notes
A01	Start pointer	,	4	
	End pointer	,	4	
	Status	CR+LF	1	
C01	Current (Start date and time)	,	14	YYYYMMDDhhmmss
	Current (Status)	,	1	1: Power OFF 2: Standby mode 3: Operating 4: Stopped 5: Error
	Current status (Language)	,	1	0: NC, 1: Conversation
	Current status (Program No. / Error No.)	,	6	Program No. when the "status" is "3: Operating" Error No. when the "status" is "5: Error"
	Current status (Folder name)	,	10	The leading and last characters are single quotation marks.
	Memory operation type	CR+LF	1	0: Internal memory operation, 1: Tape operation - General communications device, 2: Tape operation - Memory card
B0001-B3500	Status log 1-3500	Same structure as C01		

The start date and time is the start time from when the status was changed.

5.6.4.7 Position data

Filename: PDSP

Symbol	Item	Delimiter	Data length	Notes
L01	Language	CR+LF	1	0: NC, 1: Conversation
G01	G00 group	,	3	00-03, 02.2, 03.2, 102, 103, 202, 203, 33, 392 (Outputs 922 and 932 for G02.2 and G03.2)
	G17 group	,	3	17-19
	G22 group	,	3	22, 23
	G40 group	,	3	40-42, 141, 142
	G49 group	,	3	43, 44, 49, 143, 144
	G50 group	,	3	50, 51
	G50.1 group	,	3	501, 511
	G54 group	,	3	1-48, 54-59 (Outputs 1-48 for G54.1P1-G54.1P48)
	G64 group	,	3	61, 64
	G67 group	,	3	66, 67
	G69 group	,	3	68, 69, 168, 68.2 (Outputs 682 for G68.2)
	G80 group	,	3	73-189
	G90 group	,	3	90, 91
	G94 group	,	3	93, 94, 95
	G97 group	,	3	96, 97
	G98 group	,	3	98, 99
	G54.2 group	,	3	Displays P number (Ex: Outputs 1 when G54.2P1)
	G321 group	CR+LF	3	321-323
M01	M05 group	,	3	03-05, 19, 111
	M09 group	,	3	08, 09
	M97 group	,	3	96, 97
	M141 group	,	3	141, 142

Symbol	Item	Delimiter	Data length	Notes
	M221 group	,	3	211, 221
	M222 group	,	3	212, 222
	M223 group	,	3	213, 223
	M224 group	,	3	214, 224
	M230 group	,	3	230, 231
	Reserved	,	3	Reserved
	M252 group	,	3	241-254
	M269 group	,	3	260-267, 269, 280-287, 289
	M270 group	,	3	270-279
	M290 group	,	3	290-293
	M305 group	,	3	303-305
	M401 group	,	3	400, 401
	M403 group	,	3	402, 403
	M405 group	,	3	404, 405
	M407 group	,	3	406, 407
	M409 group	,	3	408, 409
	M419 group	,	3	418, 419
	M431 group	,	3	430, 431
	M435 group	,	3	435-437
	M441 group	,	3	440, 441
	M443 group	,	3	442, 443
	M445 group	,	3	444, 445
	M481 group	,	3	480, 481
	M483 group	,	3	482, 483
	M485 group	,	3	484, 485
	M487 group	,	3	486, 487
	M495 group	,	3	490-495
	Reserved	,	3	Reserved
	M322 group	,	3	322,323
	M389 group		3	380-387,389
	M299 group		3	298,299
	M474 group		3	474,475
	M420 group		3	420-423
	M340 group	CR+LF	3	340-343
P01	Machine coordinate position (X-axis)	,	10	-9999.9999~9999.9999 -999.99999~999.99999 (*1)
	- Machine coordinate position (Y to 8-axes)	,	10*7	Same as machine coordinate position (X-axis)
	Machine coordinate position (P1 to P4-axes)	, CR+LF	9*4	
P02	Relative coordinate position, X to P4-axes	Same structure as P01		
P03	Absolute coordinate position, X to P4-axes	Same structure as P01		
P04	Remaining distance, X to P4-axes	Same structure as P01		
X01	Feedrate	,	10	
	Spindle speed	,	6	
	Inner pallet	,	1	0: Not indexed/QT not mounted, 1: Inner side pallet 1, 2: Inner side pallet 2
	Spindle tool No.	,	2	
	Next tool No.	,	2	
	Pot No.	,	2	
	Door interlock/ mode of safe operation	,	1	0: Disable / Maintenance, 1: Enable / Machine setup, 2: Automatic operation
	Outer door	,	1	0: Close, 1: Open

Symbol	Item	Delimiter	Data length	Notes
	Inner door	,	1	0: Close, 1: Open
	Side door	,	1	0: Close, 1: Open
	Rapid traverse override	,	1	0: speed1, 1: speed2, 2: speed3, 3: speed4 , 4: 100%, 5: 0%
	Feedrate override	,	3	%
	Spindle override	,	3	%
	High accuracy		2	0: OFF 1: Standard 2: Rough 3: Medium rough 4: Medium rough S 5: Finishing 6: Finishing S 7: Adjustment A 8: Adjustment B 9: Adjustment C 21: Accuracy spec. A 22: Accuracy spec. B 23: Accuracy spec. C 60:M260 61:M261 62:M262 63:M263 64:M264 65:M265 66:M266 67:M267 80:M280 81:M281 82:M282 83:M283 84:M284 85:M285 86:M286 87:M287
	Feature coordinate setting status		1	0: Before feature coordinate index 1: During feature coordinate index
	Machining load monitor	CR+LF	2	0: OFF 1 to 99: Parameter number for machining load monitor being used
A01	4th-axis name	,	4	Axis name: QT, A, B, C, (blank) (*2) The leading and last characters are single quotation marks.
	5th-axis name	,	4	Axis name: A,A1,A2,B,B1,B2,C,C1,C2, (blank) (*2) The leading and last characters are single quotation marks.
	6th-axis name	,	4	Axis name: A,A1,A2,B,B1,B2,C,C1,C2, (blank) (*2) The leading and last characters are single quotation marks.
	7th-axis name	,	4	Axis name: A,A1,A2,B,B1,B2,C,C1,C2, (blank) (*2) The leading and last characters are single quotation marks.
	8th-axis name	,	4	Axis name: A,A1,A2,B,B1,B2,C,C1,C2, (blank) (*2) The leading and last characters are single quotation marks.
	P1-axis name	,	4	Axis name: P1, (blank) (*2) The leading and last characters are single quotation marks.
	P2-axis name	,	4	Axis name: P2, (blank) (*2) The leading and last characters are single quotation marks.
	P3-axis name	,	4	Axis name: P3, (blank) (*2) The leading and last characters are single quotation marks.
	P4-axis name	CR+LF,	4	Axis name: P4, (blank) (*2) The leading and last characters are single quotation marks.

(*1) The last digit is a blank space when the unit is a micron.

(*2) When <Optional *-axis> is set to <Invalid> (blank)

5.6.4.8 Memory operation

Filename: MEM

Symbol	Item	Delimiter	Data length	Notes
A01	Operation folder name	,	10	8 half-width characters (Characters that can be entered from the panel). The leading and last characters are single quotation marks.
	Program No.	,	4	0: NC, 1: Conversation
	Operation status	,	1	0: Reset, 1: Operation, 2: Temporary stop, 3: Block stop
	Inner pallet status	,	1	0: Not indexed, 1: Inner side No.1, 2: Inner side No.2
	Spare tool	,	1	0: Not used, 1: Being used
	Mode	,	1	
	Expansion	CR+LF	1	0: Normal, 1: Expanded

5.6.4.9 ATC tool

Filename: ATCTL

Symbol	Item	Delimiter	Data length	Notes
M01	Spindle (Tool No.)	,	3	0: Not set, 1-99: Tool No, 255: Cap setting.
	Spindle (Conversation/NC)	,	1	0:Conversation, 1:NC
	Spindle (Group No. (NC) / Main tool No. (Conversation))	,	2	Group No.: 0 (Not set), 1-30 Main tool No.: 0 (Not set), 1-99
	Spindle (Type)	,	1	1: Standard, 2: Large diameter, 3: Medium diameter
	Spindle (Graph color)	CR+LF	1	0: No color, 1: Blue, 2: Red, 3: Purple, 4: Green, 5: Light blue, 6: Yellow, 7: White
M02-M51	Pot 1-50	Same structure as M01		

5.6.4.10 Operation panel data

Filename: PANEL

Symbol	Item	Delimiter	Data length	Notes
D01	Outer door	,	1	0: Close, 1: Open
	Inner door	,	1	0: Close, 1: Open
	Side door	CR+LF	1	0: Close, 1: Open
K01	Mode	,	1	0: Manual, 1: MDI operation, 2: Memory operation, 3: Program edit, 4: MDI manual, 5: Operating edit
	Screen	,	1	0: Display OFF, 1: Alarm, 2: Data bank, 3: ATC tool, 4: Program, 5: Manual conditions, 6: Position, 7: I/O, 8: Monitor, 9: Graph
	Block skip	,	1	0: OFF, 1: ON
	OPT stop	,	1	0: OFF, 1: ON
	Single block	,	1	0: OFF, 1: ON
	Dry run	,	1	0: OFF, 1: ON
	Machine lock	,	1	0: OFF, 1: ON
	Coolant pump	,	1	0: OFF, 1: ON
	Chip shower	,	1	0: OFF, 1: ON
	Machine light	,	1	0: OFF, 1: ON
	Pallet select key	CR+LF	1	0: OFF, 1: 1, 2: 2, 3: 1-2

Symbol	Item	Delimiter	Data length	Notes
S01	Rapid traverse override	,	1	0: Speed 1, 1: Speed 2, 2: Speed 3, 3: Speed 4, 4: 100%, 5: 0%, 9: Override prohibited
	Feedrate override	,	3	% display, 999: Override prohibited
	Spindle override	,	3	% display, 999: Override prohibited
	Emergency stop	,	1	0: ON, 1: OFF
	Door interlock	,	1	0: OFF, 1: ON
	Data protection	,	1	0: Enabled, 1: Disabled
	[MASTER ON]	CR+LF	1	0: OFF, 1: ON

5.6.4.11 Version data

Filename: VER

Symbol	Item	Delimiter	Data length	Notes
M01	Model	CR+LF	12	The leading and last characters are single quotation marks.
V01	Version (Overall)	,	14	The leading and last characters are single quotation marks.
	Version (Main)	,	14	Same as above
	Version (Local)	,	14	Same as above
	Version (Reserved)	,	14	Same as above
	Version (Reserved)	,	14	Same as above
	Version (Teaching pendant)	,	14	Same as above
	Version (Help data)	,	14	Same as above
	Version (Main Boot)	,	14	Same as above
	Version (Reserved)	,	14	Same as above
	Version (Reserved)	,	14	Same as above
	Version (Teaching pendant Boot)	,	14	Same as above
	Function version	,	8	Same as above
	Version (Local Boot)		14	Same as above
	CM PCB		14	
	CM PCB(Boot)		14	
	Version(Load system)		14	
	Version(Load system Boot)	CR+LF	14	
S01	Machine I.D. No.		14	
	Reserved	,	1	
	Reserved	CR+LF	1	

5.6.4.12 Current alarm

Filename: ALARM

Symbol	Item	Delimiter	Data length	Notes
E01	Alarm/Operator message No.1	,	10	2 high-order digits: 01: EX, 02: EC, 03: SV, 04: NC, 05: IO, 06: SP, 07: SM, 08: SL, 09: CM, 10: ES, 11: FC, 90: OM Next 4 digits: No. 4 low-order digits: Auxiliary No.
	- Alarm/Operator messages No.2-36 are omitted.	, CR+LF	10*35	Same as alarm/operator message No.1
L01	Loading system alarm message No.1	,	11	First 3 digits: Category number Next 4 digits: Number Last 4 digits: Auxiliary number
	Loading system alarm message No.2-18	, CR+LF	11*17	<Same as alarm/operator message No.1>

5.6.4.13 Alarm history

Filename: LOG

Symbol	Item	Delimiter	Data length	Notes	
A01	Table status	,	1	Internal data (Unnecessary for reference)	
	Table index	,	3	Internal data (Unnecessary for reference)	
	Detailed write index	,	2	Internal data (Unnecessary for reference)	
	Detailed read index	CR+LF	2	Internal data (Unnecessary for reference)	
B0001	Log 1 (Date and time)	,	14	YYYYMMDDhhmmss	
	Log 1 (Alarm No.)	,	6	2 high-order digits: 01: EX, 02: EC, 03: SV, 04: NC, 05: IO, 06: SP, 07: SM, 08: SL, 09: CM, 10: ES, 11: FC, 90: OM, 4 low-order digits: No.	
	Log 1 (Auxiliary No.)	,	8		
	Log 1 (Attributes)	,	5		
	Log 1 (Operation program No.)	,	4		
	Log 1 (Edit program No.)	,	4		
	Log 1 (Block No./Job No.)	,	13	Block No. when using the NC language. When using conversation: Job (2)/Tool (2)/Machining position (3)/Workpiece No. (2)/Line No. (2)/Infeed (2).	
	Log 1 (Mode)	,	1	0: Manual, 1: MDI, 2: Memory, 3: Edit, 4: MDI manual, 5: Memory edit, 8: Memory operation (Tape operation), 9: Operating edit (Tape operation)	
	Log 1 (Screen)	,	1	0: Display OFF, 1: Alarm, 2: Data bank, 3: ATC tool, 4: Program, 5: Manual conditions, 6: Position, 7: I/O, 8: Monitor, 9: Graph	
	Log 1 (Expansion)	,	1	0: Normal, 1: Expanded	
	Log 1 (Operation program folder)	,	10	The leading and last characters are single quotation marks.	
	Log 1 (Edit program folder)	CR+LF	10	Same as above	
B0002-B0100	Log 2 – Log 100	Same structure as B0001			
C0001	Log 1 (Machine coordinates for X-axis)	,	10		
	- Log 1 (Machine coordinates for Y through P4-axes) omitted.	,	10*13	Same structure as log 001 (Machine coordinates for X-axis)	
	Log 1 (Remaining distance for X-axis)	,	10		
	- Log 1 (Remaining distance for Y through P4-axes) omitted.	, CR+LF	10*13	Same structure as log 001 (Remaining distance for X-axis)	
C0002-C0100	Log 2 – Log 100	Same structure as C0001			
D0001	Log 1 (Spindle tool)	,	2		

Symbol	Item	Delimiter	Data length	Notes	
	Log 1 (Spindle speed)	,	5		
	Log 1 (Feed type)	,	1	0: Not traveling, 1: Positioning, 2: Feed axis cutting, 3: Additional axis cutting and 4: Feed axis + Additional axis cutting	
	Log 1 (Feedrate)	,	10		
	Log 1 (Door interlock)	,	1	0: Disabled, 1: Enabled	
	Log 1 (Outer door)	,	1	0: Close, 1: Open	
	Log 1 (Inner door)	,	1	0: Close, 1: Open	
	Log 1 (Side door)	,	1	bit0: Right door 0: Close, 1: Open bit1: Left door 0: Close, 1: Open	
	Log 1 (Rapid traverse override)	,	1	0: Speed 1, 1: Speed 2, 2: Speed 3, 3: Speed 4, 4: 100%, 5: 0%	
	Log 1 (Feedrate override)	,	3	%	
	Log 1 (Spindle override)	,	3	%	
	Log 1 (Tool length offset H modal)	,	8	0: No offset, 1-99: Tool No.	
	Log 1 (Tool position compensation (X))	,	10	Tool position compensation (X) + Tool position wear offset (X)	
	Log 1 (Tool position compensation (Y))	,	10	Tool position compensation (Y) + Tool position wear offset (Y)	
	Log 1 (Tool length offset (Z))	,	10	Tool length offset (Z) + T length wear offset (Z)	
	Log 1 (Cutter compensation D modal)	,	8	0: No offset, 1-99: Tool No.	
	Log 1 (Cutter compensation)	CR+LF	10	Cutter compensation + Cutter wear offset	
D0002-D0100	Log 2 – Log 100	Same structure as D0001			
E0001-E0100	Log 1-100 (Input main 1-160)	, CR+LF	16*160* 100	Bit expression for 0 or 1	
F0001-F0100	Log 1-100 (Output main 1-41)	, CR+LF	16*41*100	Bit expression for 0 or 1	
G0001-G0100	Log 1-100 (Input local 1-5)	, CR+LF	16*5*100	Bit expression for 0 or 1	
H0001-H0100	Log 1-100 (Output local 1-4)	, CR+LF	16*4*100	Bit expression for 0 or 1	
I0001-I0100	Log 1-100 (Deviation for S to P4-axes)	, CR+LF	11*15*100		
J0001-J0100	Log 1-100 (Execution block)	CR+LF	128*100		
K0001	Log 1 (Current limit)	,	4		
	Log 1 (Current command)	CR+LF	5		
K0002-K0100	Log 2-100	Same structure as K0001			
L0001-L0100	Log 1-100 (PLC external input X0-64)	, CR+LF	16*64*100	Bit expression for 0 or 1	
M0001-M0100	Log 1-100 (PLX external output Y0-64)	, CR+LF	16*64*100	Bit expression for 0 or 1	
N0001-N0100	Log 1-100 (PLC internal input BX0-64)	, CR+LF	16*64*100	Bit expression for 0 or 1	
O0001-O0100	Log 1-100 (PLC internal output BY0-64)	, CR+LF	16*64*100	Bit expression for 0 or 1	

Filename: LOGBK

The LOG and file format is the same.

The data starting from the 101th item in the alarm history is recorded in this file (LOGBK).

5.6.4.14 Operation history

Filename: OPLOG

Symbol	Item	Delimiter	Data length	Notes
A001	Read index	,	3	Internal data
	Write index	CR+LF	3	Internal data
B001	Key log 1 (Epoch time)	,	10	Time period since operation (Epoch time)
	Key log 1 (Key type)	CR+LF	10	Refer below
B002-B132	Key log 2 – Key log 132	Same structure as B001		
C001	Key log 133 (Epoch time)	,	10	
	Key log 133 (Key type)	,	10	
	Key log 134 – Key log 182	, CR+LF	10*2*49	Repeat system for epoch time and key type
C002-C200	Operation before key log 138	Same structure as C00		

Key type list

32 - 127: Same as ASCII code

128: OPT stop	129: Block skip	130: Single operation
132: Machine lock	133: Dry run	135: Machine light
136: Coolant pump	137: Chip shower	138: Shift
144: Machine zero return	145: Magazine forward	146: ATC
147: Spindle normal rotation	148: Spindle stop	149: High-speed travel
150: Low-speed travel	151: Step travel	152: +
153: -	154: Pallet index	155: PALLET
156: Magazine reverse	157: L.SP	
160: +X	161: -X	162: +Y
163: -Y	164: +Z	165: -Z
166: +4	167: -4	
176: Previous page	177: Next page	178: Left
179: Right	180: Upper	181: Lower
182: Insert	182: Delete	183: Cancel
185: EOB/Setting	191: BS	192-196: F0-F4
197: Next function	198-200: F5-F7	201: Help
202: Shortcut menu	203-204: Shortcut 1/2	207: Reset
224: Display OFF	225: Alarm	226: Data bank
227: ATC tool	228: Program	229: Manual conditions
230: Position	231: I/O	232: Production monitor
233: Graph		
240: Manual operation	241: MDI operation	242: Memory operation
243: Edit	244: MDI + Manual operation	245: Memory operation + Edit
246: Shift + Left	247: Shift + Right	248: Shift + Up
249: Shift + Down	250: Shift + Insert	251: Shift + Delete
256: Power	257: Start	258: Feed hold
259: Data protection enabled	260: Data protection disabled	261: Emergency stop
262: Door interlock enabled	263: Door interlock disabled	264-268: Rapid traverse override
269-289: Feedrate override	290-306: Spindle override	307: Pallet start
308-311: Outside index	312-315: Automatic door	318: Magazine turn SW Fwd.
319-328: Handle operation	333: PENDANT	334: PENDANT O
335: Magazine turn SW Rev.	336-337: [MASTER ON]	338-341: Chip auger
342: SHIFT + RELSE		

(+32768 or +16384 is added to repeat key operations.)

(+65536 is added to operations used in combination with the delete key.)

5.6.4.15 Data for Machine Monitor (Production Data 4)

Filename: MONTR

Symbol	Item	Delimiter	Data length	Notes
P01	Operation program No.	,	4	
	Edit program No.	,	4	
	Operation folder name	,	258	256 half-width characters. The leading and last characters are single quotation marks.
	Edit folder name	CR+LF	10	8 half-width characters. The leading and last characters are single quotation marks.
T01	Total operation time	,	9	
	Power on time	,	9	
	Operation time	CR+LF	9	
C01	Workpiece counter 1 (Count)	,	3	
	Workpiece counter 1 (Current)	,	6	
	Workpiece counter 1 (End)	,	6	
	Workpiece counter 1 (End warning)	CR+LF	6	
C02-C04	Workpiece counter 2-4	Same structure as C01		

5

5.6.4.16 Request code

File name: RQCnnnnnnnyymmdd

nnnnnn: Machine No., yy: Year, mm: Month, dd: Day

Symbol	Item	Delimiter	Data length	Notes
A01	Request code	,	12	Upper case letters + Numbers
B01	Machine No.	,	6	

5.6.4.17 Workpiece coordinate zero (Type 1)

File name: POSNun

u: Unit system (M: Metric I: Inch)

n: Data bank number (0 to 9: "0" is used for "10")

Symbol	Item	Delimiter	Data length	Notes
G54	X-axis	,	9	-9999.9999~9999.9999 -999.99999~999.99999 ※1
	Y-axis	,	9	-9999.9999~9999.9999 -999.99999~999.99999 ※1
	Z-axis	,	9	-9999.9999~9999.9999 -999.99999~999.99999 ※1
	A-axis	,	9	-9999.999~9999.999 ※1
	B-axis	,	9	-9999.999~9999.999 ※1
	C-axis	CR+LF	9	-9999.999~9999.999 ※1
G55-G59	Same structure as G54			
X01-X48	Extended workpiece coordinate system Same structure as G54			
H01	External workpiece coordinate system Same structure as G54			
B01	Reference rotary fixture offset	,	9	-9999.999~9999.999 ※1
	Reference angle A-axis	,	9	-9999.999~9999.999 ※1
	Reference angle B-axis	,	9	-9999.999~9999.999 ※1
	Reference angle C-axis	,	9	-9999.999~9999.999 ※1
	Reference offset X-axis	,	9	-9999.9999~9999.9999 -999.99999~999.99999 ※1
	Reference offset Y-axis	,	9	-9999.9999~9999.9999 -999.99999~999.99999 ※1

Symbol	Item	Delimiter	Data length	Notes
	Reference offset Z-axis		9	-9999.9999~9999.9999 -999.99999~999.99999 ※1
	Axis for calculation	CR+LF		0:A 1:B 2:C 3:A,B 4:A,C 5:B,C

*1: When using the micron unit system without purchasing the smallest unit system option, the last digit for the sub micron place holder is a space.

(Example of reference offset on Z-axis: Metric → -99.999 to 99.999 and Inch → -9.9999 to 9.9999)

5.6.4.18 Workpiece coordinate zero (Type 2)

File name: POSSun

The same format as workpiece coordinate zero (type 1) is used, but following point is different.

*1: In micron mode as well, the smallest digit values are output as is in the interal data during communication.

5.6.4.19 Tool data (Type 1)

File name: TOLNun

u: Unit system (M: Metric I: Inch)

n: Data bank number (0 to 9: "0" is used for "10")

Symbol	Item	Delimiter	Data length	Notes
T01	Tool length offset	,	8	-999.999~999.999 -99.9999~99.9999 ※1
	T length wear offset	,	7	-99.999~99.999 -9.9999~9.9999 ※1
	Cutter compensation	,	8	-999.999~999.999 -99.9999~99.9999 ※1
	Cutter wear offset	,	7	-99.999~99.999 -9.9999~9.9999 ※1
	Tool life unit	,	1	1~4
	Initial tool life / End of tool life	,	6	0~999999
	Tool life warning		6	0~999999
	Tool life		6	0~999999
	Tool name	,	16	The first and last characters in 14 byte character string (2 byte characters possible) is a single quotation mark.
	Rotation feed		4	0.01 ~9.99 0.001~0.999 ※4
	S command value		5	1~99999
	F command value		9	0.01~999999.99 0.001~99999.999 ※4
	Maximum speed		5	0~999999 0: Rotation not possible
	Tool wash		1	0: Possible 1: Not possible
	CTS		1	0: Possible 1: Not possible
	Tool type number		8	0~99999999 0: Not set
	Tool position offset (X)		8	-999.999~999.999 -99.9999~99.9999 ※1
	Tool position wear offset (X)		7	-99.999~99.999 -9.9999~9.9999 ※1
	Tool position offset (Y)		8	-999.999~999.999 -99.9999~99.9999 ※1
	Tool position wear offset (Y)		7	-99.999~99.999 -9.9999~9.9999 ※1
	Virtual teeth direction	CR+LF	1	-999.999~999.999 -99.9999~99.9999 ※1
T02- T99	Same structure as T01			
Y01	Tool number 1-30 set in group number 1	CR+LF	2*30	1~99 ※2
Y02- Y99	Same structure as Y01			

Symbol	Item	Delimiter	Data length	Notes
M01	Minimum value for tool length offset		8	-999.999~999.999 -99.9999~99.9999 ※1
	Maximum value for tool length offset		8	-999.999~999.999 -99.9999~99.9999 ※1
	Minimum value for T length wear offset		7	-99.999~99.999 -9.9999~9.9999 ※1
	Maximum value for T length wear offset		7	-99.999~99.999 -9.9999~9.9999 ※1
	Minimum value for tool diameter offset		8	-999.999~999.999 -99.9999~99.9999 ※1
	Maximum value for tool diameter offset		8	-999.999~999.999 -99.9999~99.9999 ※1
	Minimum value for cutter wear offset		7	-99.999~99.999 -9.9999~9.9999 ※1
	Maximum value for cutter wear offset		7	-99.999~99.999 -9.9999~9.9999 ※1
	Minimum value for tool position offset (X)		8	-999.999~999.999 -99.9999~99.9999 ※1
	Maximum value for tool position offset (X)		8	-999.999~999.999 -99.9999~99.9999 ※1
	Minimum value for tool position wear offset (X)		7	-99.999~99.999 -9.9999~9.9999 ※1
	Maximum value for tool position wear offset (X)		7	-99.999~99.999 -9.9999~9.9999 ※1
	Minimum value for tool position offset (Y)		8	-999.999~999.999 -99.9999~99.9999 ※1
	Maximum value for tool position offset (Y)		8	-999.999~999.999 -99.9999~99.9999 ※1
	Minimum value for tool position wear offset (Y)		7	-99.999~99.999 -9.9999~9.9999 ※1
	Maximum value for tool position wear offset (Y)		7	-99.999~99.999 -9.9999~9.9999 ※1
M02-M99	Same structure as M01			

*1: The last digit is a blank space when the unit is a micron.

*2: When there is a blank space, the cell shifts up to fill in the gap.

*3: It does not output rounding off to the smallest digit value. However, if the value only occupies the last digit (0 when not that value), the value is rounded off when outputting in micron mode.

*4: The metric and inch sizes are different.

5.6.4.20 Tool data (Type 2)

File name: TOLSun

u: Unit system (M: Metric I: Inch)

n: Data bank number (0 to 9: "0" is used for "10")

The same format as tool data (type 1) is used, but following point is different.

*1: In micron mode as well, the smallest digit values are output as is in the internal data during communication.

5.6.4.21 Macro variables (Type 1)

File name: MCRNun

u: Unit system (M: Metric I: Inch)

n: Data bank number (0 to 9: "0" is used for "10")

Symbol	Item	Delimiter	Data length	Notes
C500	Variable	,	11	-999999.999~999999.999 -99999.9999~99999.9999 ※1
C501- C999	Same structure as C500			

*1: The last digit is a blank space when the unit is a micron.

5.6.4.22 Macro variables (Type 2)

File name: MCERSun

u: Unit system (M: Metric I: Inch)

n: Data bank number (0 to 9: "0" is used for "10")

The same format as macro variables (type 1) is used, but following point is different.

*1: In micron mode as well, the smallest digit values are output as is in the internal data during communication.

5.6.4.23 Machining load monitor

File name: TLOADn

n: Data bank number (0 to 9: "0" is used for "10")

Symbol	Item	Delimiter	Data length	Notes
A01	Load monitor method	,	1	0: Peak value 1: Average
	Time constant	,	3	1~999
	Stop level for max.	,	1	0~3
	Stop level for min.	,	1	0~3
	Max. machining load	,	7	0.000~999.999
	Min. machining load	CR+LF	7	0.000~999.999
A02- A99	Same structure as A01			
B01	Machining load monitor function	,	1	0: Disable 1: Enable
	Machining load monitor during override	CR+LF	1	0:No monitor 1:Monitor

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CHAPTER 6

DATA PROTECTION

- 6.1 Outline
- 6.2 Set operation level
- 6.3 Change password
- 6.4 Edit detailed settings

6.1 Outline

Six operation levels are provided, at each of which data to be prevented from being changed and operation to be prevented can be set. By making settings for these operation levels, the machine is protected from erroneous operations.

Operation level	Operator	Setting
Level 0 (low)	User	Neither password nor external input signal
Level 1	User	Password or external signal (LEVEL1)
Level 2	User	Password or external input signal (LEVEL2)
Level 3	User	Password or external input signal (LEVEL3)
Level 4	User	Password or external input signal (LEVEL4)
Level 5	User	Password

- Detailed settings can be registered for each data such as programs and data banks at each operation level, and only the data set to <Op. Enabled> at the current operation level are permitted to be edited.
 Detailed settings can also be registered for keys, switches and screen displays.
 In some cases as shown below, however, some data cannot be operated even if they are set to <Op. Enabled>.
 - (1) When [DATA PROTECTION] switch is enabled;
 - (2) When editing is prevented by the combination of external input signals (EDLOCK, PRLOCK, TLEDOK, UPEDOK, MPEDOK, ECEDOK, MGEDOK, MTEDOK, WPEDOK, MCEDOK, TPEDOK, THEDOK, CTEDOK and DIROK);
 - (3) When data protection is ON for editing of programs #8000-8999;
 - (4) When <Parameter change> is set to <0: Write-protected> on the I/O (version) screen for the editing of machine parameters;
 - (5) When operation is prevented by the combination of external input signals (MDLOCK, KYLOCK, MODEOK);
 - (6) When operation is prevented by PLC input signals (KYLOCK1 to 6); or
 - (7) When <Prevent rapid traverse override>, <Prevent cutting feedrateoverride> or <Prevent spindle override> is set to <1: Yes> in the user parameters (switch 1).

* Data can be changed through the operator using G10 or macro variables even if the data is set to <Op. disabled>
- If an operation set to <Op. disabled> is attempted, an operator message <<No operation is possible under the current level>> is generated.
- Although input through communication is subject to restriction depending on the operation level, check input is possible.
- For Input all data banks or for Maintenance (Input all data), data set to <Op. disabled> are skipped. (An operator message <<Operation disable data has been skipped>> is generated.)

6.2 Set operation level

6.2.1 Setting by password

1. Select <Operation level> from the I/O menu to display the Operation level screen.

Operation level		2012/05/30 15:29:20
Operation level	Password	External input signal
Level0		
Level1		On
Level2		On
Level3	Setting	Off
Level4		Off
Level5		
Password		
		Current operation level: Level3
I/O menu	Reset password	Change password
Detail setting		

- <Password> <ON> is displayed for the operation level when the password is successfully entered. When a password is successfully entered for another operation level, the <ON> display changes to that level.
- <External input signal> The current external input signal status is displayed. When multiple signals are turned ON, the color of all the operation levels that are turned ON become green.
- <Current operation level> The current, valid operation level is displayed.

2. Move the cursor to the operation level that you wish to set, enter the password into the data input field and press the **[ENT]** key. (The following is displayed: <****> at a level when the password setting is enabled.)
 - If the password is entered correctly, the <Password> function is turned <ON>.
 - If the password is entered incorrectly, the following operator message appears: <<The password is wrong.>> Re-enter the correct password.

The following table shows the factory settings for the password.

Operation level	Password
Level 1	(Not set)
Level 2	(Not set)
Level 3	(Not set)
Level 4	(Not set)
Level 5	LEVEL5

3. Press the **[F1]** (Reset password) key to turn <OFF> all <Password> settings and to change to operation level 0. Note, when the external input signal is turned ON, the operation level is enabled for the signal that is turned ON.
- * The operation level is maintained even if the power is turned OFF. In this situation, the operation level screen is displayed when the power is turned ON, and the password input is requested for the current operation level. The screen cannot be changed until the operation level setting has been completed using one of the following procedures. In addition, the following alarm occurs:
<<*Enter your password.>> (Stop level 5, Cancel level 2)

0506 *Enter your password.

Operation level 2012/05/30 15:32:26

Operation level	Password	External input signal
Level0		
Level1		On
Level2		On
Level3	Power turned off dur. Setting	Off
Level4		Off
Level5		

Password

Previously, when the power was turned off, the password of level 3 was set.
To continue the previous operation level, enter the password and press the [ENT] key.
The operation level becomes 0 if either of the following operations is performed:
 • Press [F1] (password reset) key.
 • Press [RST] key without entering password.

Current operation level: Level3

I/O menu Reset password Change password Detail setting

- Enter the password into the data input field and press the **[ENT]** key to set the current operation level.
- Press the **[F1]** (Reset password) key, or press the **[RST]** key, to turn <OFF> all <Password> settings.
- Turn ON the external input signal for a higher level than the current operation level.
- The only valid keys on this screen are the alphanumeric keys, the **[RST]** key and the **[F1]** (Reset password) key.
- If the password is entered incorrectly 5 times in a row, all <Password> settings are automatically turned <OFF>.
- When the password is not set (factory settings) for levels 1 through 4, the operation level screen does not display when the power is turned ON. It starts up at the level when the power was turned OFF. (Excluding level 5)

6.2.2 Setting by external input signals

As an alternative to entering a password for operation levels 1 through 4, the operation level can be set by entering the external input signals: “LEVEL1”, “LEVEL2”, “LEVEL3” and “LEVEL4”.

- Actual external input signal that is input from the terminal
- External input signal operation using the slave station command.
- External input signal key.
- PLC input signal.

Turn OFF the external input signal to return to the previous operation level.

- * When multiple external input signals are turned ON, the highest operation level is enabled for the signals that are turned ON.

6.2.3 Precautions

- When the operation levels for the password and the external input signal are both set, the highest valid operation level is enabled.
- When the operation levels for the password and the external input signal are both set, if the password is turned OFF, the operation level for the external input signal is enabled, or viceversa.
- When the operation level is changed, the following operator message notification appears: <<Operation level setting (*)>>.

6.3 Change password

Passwords can be changed. Please note that, however, only the passwords for the operation levels not higher than the current operation level can be changed. Thus, to change a password, frist set up the current operation level that is higher than the one for which the password is to be changed. 4 to 10 alphanumeric characters (A to Z, 0 to 9) can be used for a password.

6.3.1 Changing password

1. One the operation level screen, locate the cursor at the desired operation level for which the password is going to be changed, and press **[F2]** (Change password) key. This displays the Change password screen.

Operation level	Password	External input signal
Level0		
Level1		On
Level2		On
Level3	Setting	Off
Level4		Off
Level5		

The password of operation level 2 is changed.
Enter 4 to 10 alphanumeric characters (A-Z, 0-9).
New password
Confirmation and input of new password

Current operation level: Level3

Cancel Change Initialize password

2. Enter a new password in the upper input area and press **[ENT]**. For confirmation, enter the new password again in the lower input area and press **[ENT]**. Pressing the **[F1]** (Change) key displays the Change password confirmation screen.

Operation level	Password	External input signal
Level0		
Level1		On
Level2		On
Level3	Setting	Off
Level4		Off
Level5		

Operation level2
Password is changed.
Are you sure?

Current operation level: Level3

Yes No

3. Press **[F0]** (Yes) key. This changes the password and returns the screen to the Original one.

6.3.2 Changing password to no-set

Operation levels 1 to 4 can be changed to the password no-set state.

Pressing **[F1]** (Change) key with no password entered in the input area of the Change password screen displays the Change password no-set confirmation screen.

Operation level Change password		2012/05/30 15:45:27
Operation level	Password	External input signal
Level0		
Level1		On
Level2		On
Level3	Setting	Off
Level4		Off
Level5		

Operation level2
Change the password to no-set.
By the password, this operation level cannot be set.
Are you sure?

Current operation level: Level3

Yes No

- * If the password is not set, just pressing the **[ENT]** key performs the same operation as successfully entering the password. (The operator message <<The password is wrong,>> appears after pressing the **[ENT]** key with text entered into the data input field.)

6.3.3 Reset password to default value

Passwords can be returned to the default values.

On the Change password screen, press **[F2]** (Initialize password) key.

Operation level Change password		2012/05/30 15:46:38
Operation level	Password	External input signal
Level0		
Level1		On
Level2		On
Level3	Setting	Off
Level4		Off
Level5		

Operation level2
Return the password to default value.
Are you sure?

Current operation level: Level3

Yes No

6.4 Edit detailed settings

6.4.1 Changing settings for detailed settings

Detailed settings can be changed at each operation level.

1. On the Operation level screen, press **[F3]** (Detail setting) key.
This displays the Detailed setting screen.

Operation level detailed setting						
2012/05/30 16:02:15						
Item	Operation level					
	0	1	2	3	4	5
Program edit screen	OK	OK	OK	OK	OK	OK
Program edit operation	OK	OK	OK	OK	OK	OK
Directory display screen (program)	OK	OK	OK	OK	OK	OK
Directory display operation (Program)	OK	OK	OK	OK	OK	OK
Directory display screen (data bank)	OK	OK	OK	OK	OK	OK
Directory display operation (Data bank)	OK	OK	OK	OK	OK	OK
Work coordinate zero point screen	OK	OK	OK	OK	OK	OK
Workpc. coord. zero	OK	OK	OK	OK	OK	OK
Tool data screen(NC)	OK	OK	OK	OK	OK	OK
Tool data	OK	OK	OK	OK	OK	OK
Macro variable screen	OK	OK	OK	OK	OK	OK
Macro variables	OK	OK	OK	OK	OK	OK
User parameter screen	OK	OK	OK	OK	OK	OK
User parameter	OK	OK	OK	OK	OK	OK
Machine parameter screen	OK	OK	OK	OK	OK	OK
Machine parameter	OK	OK	OK	OK	OK	OK
Tool list screen(Conversation)	OK	OK	OK	OK	OK	OK
Current operation level: Level5						
OK Oper. Enabled - Oper. disabled						
Back	Operation enabled	Operation disabled				

2. Locate the cursor at the desired operation level for the item to be changed and press **[F1]** (Operation enabled) key or **[F2]** (Operation disabled) key.

* Detailed settings can be changed only when the current operation level is set to level 5. Only the items at the operation levels lower than the current operation level can be changed. The items at the current operation level cannot be changed. Only the items set to <Op. Enabled> at the current operation level can be changed.

6.4.2 Item List

List of detailed setting items are as shown below.

Items	Remarks
Program edit screen	Menu selection is unavailable.
Program edit operation	Editing programs (NC program, processing data, schedule program)
Directory display screen (program)	Menu selection is unavailable when both programs and data banks are prevented.
Directory display operation (program)	Directory operations (file operation, folder operation, G code conversion)
Directory display screen (Data bank)	Menu selection is unavailable when both programs and data banks are prevented.
Directory display operation (Data bank)	Directory operation (current plane change, file operation)
Workpiece coordinate zero point screen	Menu selection is unavailable.
Workpc. coord. zero	
Tool data screen(NC)	Menu selection is unavailable.
Tool data	
Macro variable screen	Menu selection is unavailable.
Macro variables	
User parameter screen	Menu selection is unavailable.
User parameter	
Machine parameter screen	Menu selection is unavailable.
Machine parameter	
Tool list screen(Conversation)	Menu selection is unavailable.
Tool list	
Tool pattern screen	Menu selection is unavailable.
Tool pattern	

Items	Remarks
Tap prepared hole drill diameter screen	Menu selection is unavailable.
Tapping drill dia.	
Cutting condition screen	Menu selection is unavailable.
Cutting conditions	
Simplified/magazine panel custom	Reserved
Teaching pendant custom	
External I/O signal screen	Menu selection is unavailable.
External I/O signal	
Communication parameter screen	Menu selection is unavailable.
Communication parameter	
ATC tool data	Initialize tool life excluded
Initialize ATC tool life	
Workpiece counter	Setting, editing (Clear current value excluded)
Clear work counter current value	Clear current value/ work counter clear signal only
Measurement results	Measurement results, batch deletion
Therm. distortion comp.	Operation state setting, data deletion
Status log	Data deletion
Production monitor - Others	Tool breakage detection - No. of operations remaining
Maintenance notice	
Current position counter reset	Counter reset
Output signal value	Main, slave, local
ATC maintenance mode	
Upgrade	Menu selection is unavailable.
PLC edit screen	Menu selection is unavailable.
Global OM comment	
High speed ladder program 1 screen	Menu selection is unavailable.
High speed ladder program 1	
High speed ladder program 2 screen	Menu selection is unavailable.
High speed ladder program 2	
Standard ladder program 1 screen	Menu selection is unavailable.
Standard ladder program 1	
Standard ladder program 2 screen	Menu selection is unavailable.
Standard ladder program 2	
PLC parameter setting screen	Menu selection is unavailable.
PLC parameter setting	
PLC power failure hold/ writing area setting screen	Menu selection is unavailable.
PLC power failure hold/ writing area setting	
PLC conversion mode setting screen	Menu selection is unavailable.
PLC conversion mode setting	
PLC input/output	
Set drawing	
Operation program number	Change through external program selection signal or pallet program is possible.
Program protect	Program #8000-8999 protect ON/OFF change
Waveform display	Menu selection is unavailable.
Current waveform (setting)	Enter operation is not possible.
Shortcut setting	Shortcut [MENU] key selection is unavailable. Setting & editing
[MANU] key	Cannot be changed to MANU + MDI mode.
[MDI] key	
[MEM] key	
[EDIT] key	Cannot be changed to MEM + EDIT mode.
[B.SKIP] key	
[OP. STOP] key	
[SINGL.] key	

Items	Remarks
[MLCK.] key	
[DRY] key	
[PALLET] key	
[CLT .P] key	
[CHP.F] key	
[LIGHT] key	
[Z.RTN] key	
Operation level password change screen	Screen change is unavailable.
Operation level detail setting screen	Screen change is unavailable.
External I/O program screen	Menu selection is unavailable. I/O, communication parameters
External I/O program	
External I/O data bank screen	Menu selection is unavailable. I/O, communication parameter I/O
External I/O data bank	
External I/O maintenance screen	Menu selection is unavailable. I/O, communication parameters, format I/O
External I/O maintenance	
Expired battery recovery	The following key operations cannot be executed on the expired battery recovery screen and the position adjustment (*axis) screen. • [F1] to [F7] • [RELSE] + [P] • [RELSE] + [Q] • [RELSE] + [R]
G/M code macro screen	Menu selection is not available.
G/M code macro	
Special setting selection screen	Screen selection is not available. The menu cannot be selected when the the screens below are all restricted. • Rapid feed adjustment screen • Rapid feed adjustment - Linear interpolation type screen • Special setting selection screen
Rapid feed adjustment screen	Screen selection is not available. The menu cannot be selected when the the screens below are all restricted. • Rapid feed adjustment screen • Rapid feed adjustment - Linear interpolation type screen • Special setting selection screen
Special settings	
Rapid feed adjustment - Linear interpolation type screen	Screen selection is not available. The menu cannot be selected when the the screens below are all restricted. • Rapid feed adjustment screen • Rapid feed adjustment - Linear interpolation type screen • Special setting selection screen
PLC expanded function screen	Menu selection is not available.
PLC expanded function	
3D smoothing screen for preparations	Screen selection is not available.
3D smoothing for preparations	
Mode for enabling and disabling servo	
Machining load monitor screen	Menu selection is not available.
Machining load monitor	

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CHAPTER 7

SUPPLEMENT

- 7.1 Help
- 7.2 Display in different languages
- 7.3 HTTP (Hyper Text Transfer Protocol)

7.1 Help

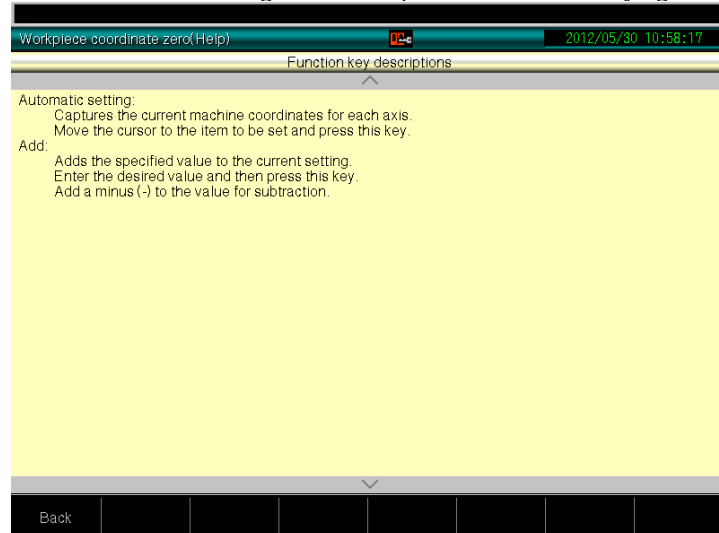
7.1.1 Help screen

7.1.1.1 Outline

Help screen displays information such as the operating method of the currently displayed screen and the contents of the relevant parameter.

Displayed state and contents of Help screen differ from key to key on each screen.

1. Press the **[HELP]** key.
2. To return to the original screen, press the **[HELP]** key again, or press the **[F0]** (Back) key.

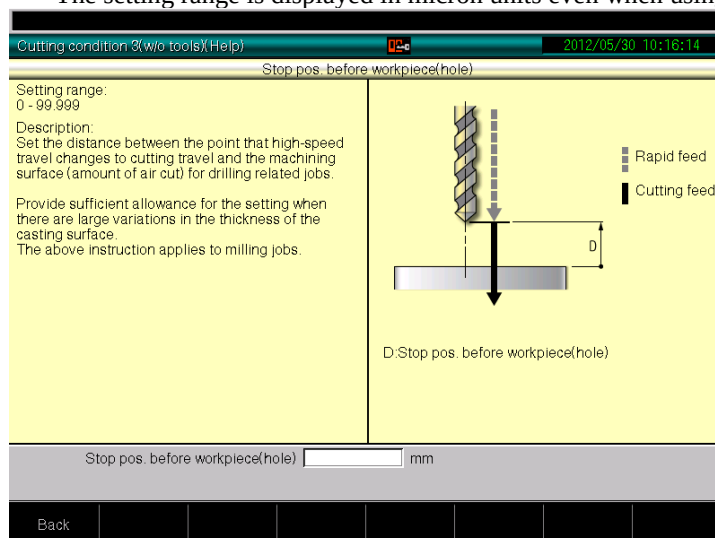


If the entire contents of the Help screen cannot be displayed within the frame, use the **[PAGE UP]** and **[PAGE DOWN]** keys to scroll up and down the page.
[F0] (Back)

Pressing this key closes the Help screen to return to the screen for displaying the Help.

When pressing the **[HELP]** key on a screen with the <Help> sign at the lower right of the teaching area, the Help screen dedicated to the relevant screen appears.

- Help screen at the entry of parameters
Pressing the **[HELP]** key in setting a value for parameters such as user parameters or conversation cutting conditions, the description of the relevant parameter appears.
A value can be entered in the input field provided in the teaching area.
* The setting range is displayed in micron units even when using submicron units.



7.2 Display in different languages

7.2.1 Changing display language

The text language on the machine unit display can be specified in the machine parameter <Display language>.

The following languages can be selected and displayed: Japanese, English, German, French, Chinese (simplified), Portuguese, Italian, Polish, Czech, Spanish and Korean.

When the <Display language> has been changed in the machine parameters, the display language is switched by restarting the machine.

When the machine parameters are not provided or if the machine parameter file is destroyed, the display is given in English. However, display language can be selected temporarily by a predetermined operation at power-ON.

For how to select temporary display language, refer to “Selecting temporary display language” described later.

In the data display area permitting entry of character string, selecting a display language other than Japanese may cause garbled characters or incorrect display of characters, if Japanese is set for the data of character string or if a file name or folder name in the connected device for external I/O is in Japanese.

The same situation occurs as with Chinese, German umlaut marks, and French accent marks.

Of the characters usable in comment sections, only the following characters can be displayed correctly even if the display language is changed.

Hexadecimal code	2*H	3*H	4*H	5*H	6*H	7*H
*0 H	(SP) (Note)	0	@	P	`	p
*1 H	!	1	A	Q	a	q
*2 H	"	2	B	R	b	r
*3 H	#	3	C	S	c	s
*4 H	\$	4	D	T	d	t
*5 H		5	E	U	e	u
*6 H	&	6	F	V	f	v
*7 H	' (Note)	7	G	W	g	w
*8 H	((Note)	8	H	X	h	x
*9 H	) (Note)	9	I	Y	i	y
*A H	*	:	J	Z	j	z
*B H	+	;	K	[	k	{
*C H	,	<	L	\ (Note)	l	
*D H	-	=	M	]	m	}
*E H	.	>	N		n	
*F H	/	?	O	_	o	

(Note) 20H (SP) refers to space.

27H “'” (apostrophe) is displayed only between “(” (control-out) and “)” (control-in) in NC programs.

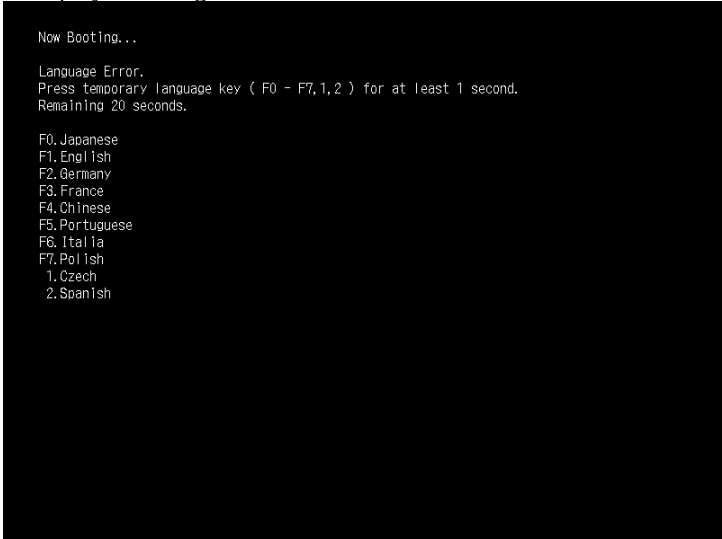
28 “(” and 29H “)” are displayed in the area allowing input of comments in other than NC programs. They are used as control-out and control-in in NC programs.

The 5CH “¥” (Yen symbol) is displayed as a “\ ” (backslash) in English, German, French, Chinese, Portuguese, Italian, Polish, Czech, Spanish and Korean.

7.2.2 Selecting temporary display language

When there are no machine parameters, or when the machine parameter file is corrupt, the display language is unknown and it is automatically set to English. Note, in this situation, the user can specify the display language temporarily using the startup operations.

1. When the system is started up and the <Now Booting> screen appears, the following message is displayed in English for 20 seconds.



```
Now Booting...

Language Error.
Press temporary language key ( F0 - F7,1,2 ) for at least 1 second.
Remaining 20 seconds.

F0. Japanese
F1. English
F2. German
F3. France
F4. Chinese
F5. Portuguese
F6. Italia
F7. Polish
1. Czech
2. Spanish
```

2. Within the 20 seconds, press the function key for the desired display language to be used temporarily.
3. Then the screen is changed to be displayed in the selected display language.

When no temporary display language has been selected within 20 seconds, the display is given in English.

The setting of the temporary display language is disabled by turning off the power supply.

7.3 HTTP (Hyper Text Transfer Protocol)

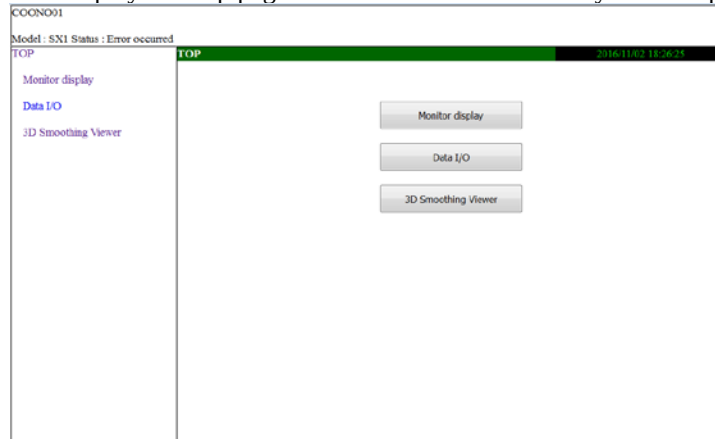
7.3.1 Outline

The machine can serve as a web server and you can browse the web pages provided by the machine through the PC. The machine can be monitored and data input/output can be performed through the browser of the PC.

- * It is required to connect the machine to the network and make settings for the network beforehand. For details, refer to “5.5 Communication protocol.”

Using the value set for the <Host name> or the <IP address> in the communication parameters, specify an URL (address) as shown below in the PC browser.
http://Host Name (or IP address)

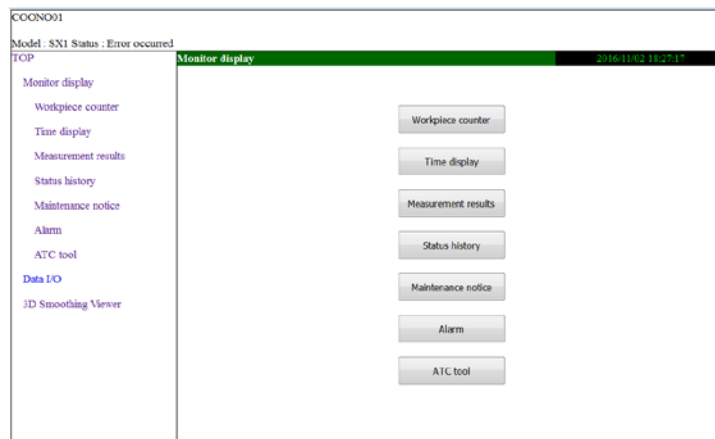
This displays the top page of the machine whose URL you have specified in the browser.



If you want to display another machine on the network, replace the <Host name> or <IP address> in the URL with that of the desired machine.

7.3.2 Monitor display

The status of the specified machine can be monitored.
Click the <Monitor display> on the <TOP> page.



- * Please note that the status of the machine displayed is not the real time status but the status when the web page was acquired.

7.3.2.1 Workpiece counter

Click the <Workpiece counter> on the <Monitor display> page.

COON001	
Model: SX1 Status: Error occurred	
TOP	
Monitor display	Workpiece counter 2016/11/02 18:27:35
Workpiece counter	
Time display	
Measurement results	
Status history	
Maintenance notice	
Alarm	
ATC tool	
Data I/O	
3D Smoothing Viewer	

	Counter 1	Counter 2	Counter 3	Counter 4
Count	0	0	0	0
Current value	0	0	0	0
Target value	0	0	0	0
End signal value	0	0	0	0
End / Signal				

7.3.2.2 Time display

Click the <Time display> on the <Monitor display> page.

NCX4	
Model: R650X2 Status: Error occurred	
TOP	
Monitor display	Time display 2019/02/22 16:23:33
Workpiece counter	
Time display	
Measurement results	
Status history	
Maintenance notice	
Alarm	
ATC tool	
Data I/O	

Machine data		
	Current	Previous value
Pallet 1 program	7777	7777
Cycle time	0000:00:00.0	0000:00:00.0
Cutting time	0000:00:00.0	0000:00:00.0
Non cutting time	0000:00:00.0	0000:00:00.0
Cutting time / cycle time	0 %	0 %
Operation end date and time		
Pallet 2 program	7777	7777
Cycle time	0000:00:00.0	0000:00:00.0
Cutting time	0000:00:00.0	0000:00:00.0
Non cutting time	0000:00:00.0	0000:00:00.0
Cutting time / cycle time	0 %	0 %
Operation end date and time		
Operation end counter		1580
Power on time	0556:09:10	
Total operation time	01208:32:00	
Date	2019/02/22 16:23:33	

Time display Operation time log

Clicking the appropriate tab switches between the <Time display> and the <Operation time log>.

COON001	
Model: SX1 Status: Error occurred	
TOP	
Monitor display	Operation time log 2016/11/02 18:26:14
Workpiece counter	
Time display	
Measurement results	
Status history	
Maintenance notice	
Alarm	
ATC tool	
Data I/O	
3D Smoothing Viewer	

Date	Operation time	Operation cycles	Operate rate (total operation time / power on time)
2016/09/05	0000:00	0	0 %
2016/10/01	0000:00	0	0 %
2016/10/04	0000:00	1	0 %
2016/10/05	0000:00	0	0 %
2016/10/06	0000:00	0	0 %
2016/10/12	0000:01	11	0 %
2016/10/13	0000:00	3	0 %
2016/10/14	0000:01	4	0 %
2016/10/17	0000:02	6	0 %
2016/10/18	0000:00	0	0 %
2016/10/19	0000:00	0	0 %
2016/10/20	0000:00	0	0 %
2016/10/21	0000:00	0	0 %
2016/10/22	0000:00	0	0 %
2016/10/26	0000:00	0	0 %
2016/10/27	0000:00	0	0 %
2016/10/28	0000:00	0	0 %
2016/10/31	0000:00	0	0 %
2016/11/01	0000:00	0	0 %
2016/11/02	0000:00	0	0 %

Time display Operation time log

7.3.2.3 Measurement results

Click the <Measurement results> on the <Monitor display> page.

Using the appropriate tab, select the desired measurement results from Measurement results 1 to 4.

COON001
Model : SX1 Status : Error occurred
TOP

Monitor display
Workpiece counter
Time display
Measurement results
Status history
Maintenance notice
Alarm
ATC tool
Data I/O
3D Smoothing Viewer

Measurement result 1			2016/11/02 11:28:09		
Latest	Previous 1	Previous 2	Previous 3	Previous 4	Previous 5
X	X	X	X	X	X
Y	Y	Y	Y	Y	Y
Z	Z	Z	Z	Z	Z
Rotation	Rotation	Rotation	Rotation	Rotation	Rotation
Time	Time	Time	Time	Time	Time
Previous 3	Previous 4	Previous 5	Previous 6	Previous 7	Previous 8
X	X	X	X	X	X
Y	Y	Y	Y	Y	Y
Z	Z	Z	Z	Z	Z
Rotation	Rotation	Rotation	Rotation	Rotation	Rotation
Time	Time	Time	Time	Time	Time
Previous 6	Previous 7	Previous 8	Previous 9	Previous 10	Previous 11
X	X	X	X	X	X
Y	Y	Y	Y	Y	Y
Z	Z	Z	Z	Z	Z
Rotation	Rotation	Rotation	Rotation	Rotation	Rotation
Time	Time	Time	Time	Time	Time

Measurement result 1 Measurement result 2 Measurement result 3 Measurement result 4

7.3.2.4 Status log

Click the <Status history> on the <Monitor display> page.

COON001
Model : SX1 Status : Error occurred
TOP

Monitor display
Workpiece counter
Time display
Measurement results
Status history
Maintenance notice
Alarm
ATC tool
Data I/O
3D Smoothing Viewer

Date	Off	Standby	Error	Operating	Stopped
11/02 (Today)	00:00	07:55	10:24	00:00	00:00
11/01 (Yesterday)	00:00	03:40	10:34	00:00	00:00
10/31 (2 days ago)	00:00	07:39	00:01	00:00	00:00
10/30 (3 days ago)	00:00	00:00	00:00	00:00	00:00
10/29 (4 days ago)	00:00	00:00	00:00	00:00	00:00
10/28 (5 days ago)	00:00	00:12	19:14	00:00	00:00
10/27 (6 days ago)	00:00	02:45	21:14	00:00	00:00

7

Clicking the desired date on the <Status history> page switches to the <Detailed display> page for the date.

COON001
Model : SX1 Status : Error occurred
TOP

Monitor display
Workpiece counter
Time display
Measurement results
Status history
Maintenance notice
Alarm
ATC tool
Data I/O
3D Smoothing Viewer

Date	11/02(Today)	
Status	Time	Error No./Program No.
Error occurred	00:00:00 - 09:42:39	100000 **
Standby mode	09:42:39 - 15:20:14	
Error occurred	15:20:14 - 15:25:04	100500 *
Standby mode	15:25:04 - 15:25:04	
Error occurred	15:25:04 - 15:31:28	100500 *
Standby mode	15:31:28 - 15:32:55	
Error occurred	15:32:55 - 15:33:25	1500 **
Standby mode	15:33:25 - 15:34:26	
Error occurred	15:34:26 - 15:34:51	1500 **
Standby mode	15:34:51 - 15:35:57	
Error occurred	15:35:57 - 15:36:31	1500 **
Standby mode	15:36:31 - 17:51:17	
Error occurred	17:51:17 - 17:51:21	SM4007 *
Standby mode	17:51:21 - 17:51:25	
Error occurred	17:51:25 - 18:20:46	SM2029 *
Standby mode	18:20:46 - 18:20:46	
Error occurred	18:20:46 -	100500 *

Back

COONO1			
Model: SX1 Status : Error occurred			
TOP			
Status history		2016/11/02 18:36:51	
Monitor display	Date	11/02(Today)	
Workpiece counter	Status	Time	Error No./Program No.
Time display	Power is off (00:00:00)		
Measurement results	Error occurred (10:24:59)	09:00:00 ~ 09:42:39	100008 **
Status history		15:20:14 ~ 15:25:04	100500 *
Maintenance notice		15:25:04 ~ 15:31:38	100590 *
		15:32:53 ~ 15:33:25	1500 **
Alarm		15:34:26 ~ 15:34:51	1503 **
ATC tool		15:35:57 ~ 15:36:31	1500 **
		17:51:17 ~ 17:51:21	SM46007 *
Data I/O		17:51:25 ~ 18:20:46	SM2628 *
		18:20:46 ~	100590 *
3D Smoothing View			
	Running (00:00:00)		
	Stopped (00:00:00)		
	Standby mode (07:55:47)	09:42:39 ~ 15:20:14	
		15:25:04 ~ 15:25:04	
		15:31:38 ~ 15:32:53	
		15:33:25 ~ 15:34:26	
		15:34:51 ~ 15:35:57	
		15:36:31 ~ 17:51:17	
		17:51:21 ~ 17:51:25	
		18:20:46 ~ 18:20:46	
	Back		

Click the <Maintenance notice> on the <Monitor display> page.

Click the **<Alarm>** on the **<Monitor display>** page.

Clicking the appropriate tab switches between the <Current alarm> and the <Alarm log list>.

COON001					
Model: SX1 Status: Error occurred					
TOP					
Monitor display Workpiece counter Time display Measurement results Status history Maintenance notice Alarm ATC tool Data I/O 3D Smoothing Viewer	Alarm		2016/11/02 18:33:07		
	Alarm	Message	Lang	Time	Date
	IO0550	*Emergency button activated	NC	18:32:41	2016/11/02
	SM2013	*Stroke exceeded (+X)	NC	18:32:37	2016/11/02
	SM2029	*Limit exceeded (+X)	NC	18:32:37	2016/11/02
	SM4014	*Feedrate error	NC	18:32:24	2016/11/02
	SM4001	*Specified G code cannot be used.	NC	18:32:09	2016/11/02
	SM4006	*Invalid data	NC	18:31:58	2016/11/02
	IO0550	*Emergency button activated	NC	18:20:45	2016/11/02
	IO0550	*Turn power off	NC	18:20:29	2016/11/02
	IO0550	*Emergency button activated	NC	17:51:28	2016/11/02
	SM2013	*Stroke exceeded (+X)	NC	17:51:25	2016/11/02
	SM2029	*Limit exceeded (+X)	NC	17:51:25	2016/11/02
	SM4007	*Command data range error	NC	17:51:17	2016/11/02
	IO0550	*Turn power off	NC	17:55:57	2016/11/02
	IO0550	*Turn power off	NC	17:54:25	2016/11/02
	IO0550	*Turn power off	NC	17:52:34	2016/11/02
	IO0550	*Emergency button activated	NC	17:25:04	2016/11/02
	IO0550	*Emergency button activated	NC	17:20:14	2016/11/02
	IO0007	*The 12-V DC power is overcurrent. (11)(KEY)	NC	19:58:58	2016/11/01
	IO0009	*The 12-V DC power is overcurrent. (31)(KEY)	NC	19:58:58	2016/11/01
	IO0008	*The 12-V DC power is overcurrent. (21)(KEY)	NC	19:58:58	2016/11/01
	IO000	*Press the [RST] key.	NC	19:02:09	2016/11/01
	SL2011.001	*User param. setting error (switch 1)	NC	19:01:38	2016/11/01
	SL2011.001	*User param. setting error (switch 1)	NC	19:01:34	2016/11/01
	IO000	*Press the [RST] key.	NC	19:01:33	2016/11/01
	IO005	*Version update has been completed.	NC	16:07:40	2016/11/01
	IO001	*Turn power off.	NC	16:04:45	2016/11/01
	IO001	*Turn power off.	NC	16:04:45	2016/11/01

Clicking the <Alarm> or <Date> on the title or <Sort by alarms> tab switches among the <Sort by number>, <Sort by time>, and <Sort by alarms>.

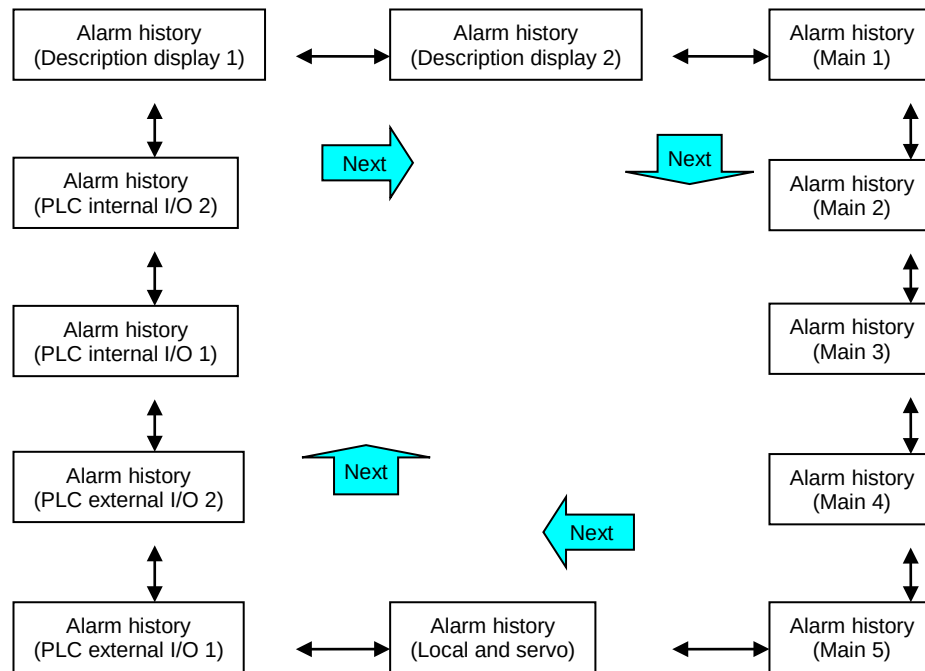
Up to 50 items can be displayed at a time. Click the <Previous 50 items> or <Next 50 items> tab to display previous or next 50 items.

Clicking any alarm cell displays its related detailed screen.

Model: R650X2 Status: Error occurred					
TOP					
Monitor display Workpiece counter Time display Measurement results Status history Maintenance notice Alarm ATC tool Data I/O	Alarm history (Detail display 1)		2019/02/22 16:26:22		
	Alarm	Message	Lang	Time	Date
	4000	*Press the [RST] key	NC	13:55:04	2019/02/22
	Mode	Screen	Edit program	Operation program	Block No.
	MANU	DATA BANK	KATOTA/0004	7777	
	Machine coordinate pos.		Distance left		
	X	0.000	X	0.000	
	Y	0.000	Y	0.000	
	Z	685.000	Z	0.000	
	QT	0.150	QT	0.000	
	B1	B1	Tool offset		Cutter compensation
	B2	B2			
	A1	A1	X		
	A2	A2	Y		
	Magazine	0.000	Z		
	P1	P1	Spindle tool		Spindle speed
	P2	P2			
	P3	P3			0 min-1
	P4	P4	Override		Feedrate
	AT	0.000	Rapid feed	100 %	
			Cutting feed	100 %	
			Spindle	100 %	
	Current alarm		Alarm history list		Previous
					Next

The page can be changed in the same way as the NC screen by using “Next” and “Previous.” The screen changes to the next page or the previous page based on the screen flow chart below.

Screen flow chart (Alarm description screen)



7.3.2.7 ATC tool

Click the <ATC tool> on the <Monitor display> page.

Model: SX1 Status: Error occurred

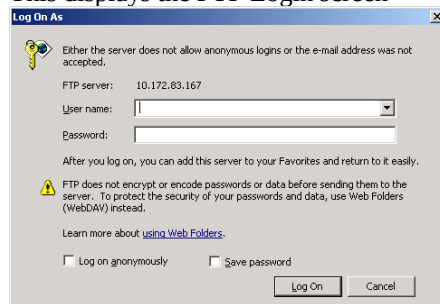
TOP

ATC tool							
Tool No.	Tool name	Tool data	Group	Tool life	Tool type	Color	
01	CENTER	10.0000x 1.0000		10/H 10	STD tool		
02	DRILL	45.0000x 8.0000		10/C 10	STD tool		
03					STD tool		
04	03	0.0000x 0.0000		10/m 10	Large tool		
05					STD tool		
96	99	160.0000x 10.0000		*****	STD tool		
07					STD tool		
08					STD tool		
09					STD tool		
10					STD tool		
11					STD tool		
12					STD tool		
13					STD tool		
14					STD tool		

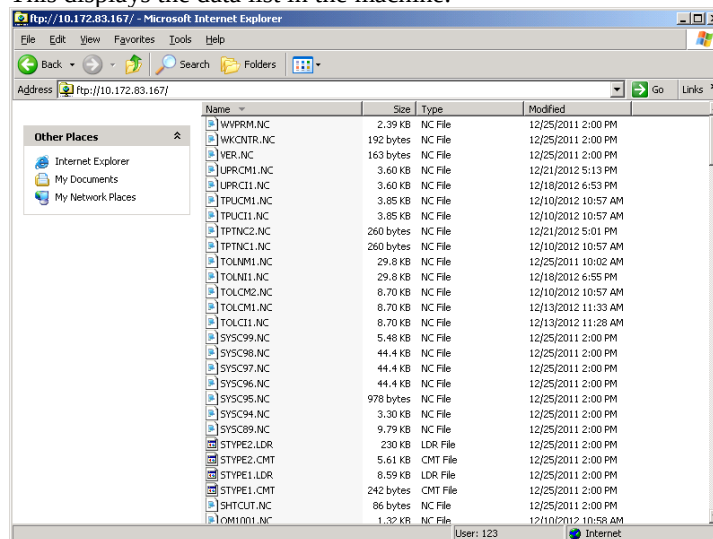
7.3.3 Data I/O

The selected machine data can be input/output through FTP.

1. Click the <Data I/O> on the <TOP> page.
This displays the FTP Login screen



2. Enter the user name and password that are set in the communication parameters: <Server user name> and <Password as server>, or <Common user name> and <Common password>, and then press the <Log On> button.
This displays the data list in the machine.



3. Data can be input/output or selected and deleted using drag and drop.
For data names, refer to “5.6 Details of Communication”.

7.3.4 3D smoothing viewer

Click on [3D smoothing viewer] on the <TOP> page to display the 3D smoothing viewer.

The smoothness (shape features) of a path that is corrected using the 3D smoothing function can be checked in the 3D smoothing viewer. Refer to “Chapter 12 (19) 3D Smoothing Viewer” in the Operation Manual for further details on the 3D smoothing viewer.

When not equipped with the 3D smoothing function option, the [3D smoothing viewer] does not display.

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